# Contents


**1. Introduction to Groups** **4**
1.1. Basic Axioms and Examples . . . . . 4
1.2. Dihedral Groups . . . . . 40
1.3. Symmetric Groups . . . . . 58
1.4. Matrix Groups . . . . . 78
1.5. The Quaternion Group . . . . . 89
1.6. Homomorphisms and Isomorphisms . . . . . 92
1.7. Group Actions . . . . . 118

**2. Subgroups** **141**
2.1. Definition and Examples . . . . . 141
2.2. Centralizers and Normalizers, Stabilizers and Kernels . . . . . 158
2.3. Cyclic Groups and Cyclic Subgroups . . . . . 172
2.4. Subgroups Generated by Subsets of a Group . . . . . 198
2.5. The Lattice of Subgroups of a Group . . . . . 218

**3. Quotient Groups and Homomorphisms** **238**
3.1. Definitions and Examples . . . . . 238
3.2. More on Cosets and Lagrange's Theorem . . . . . 281
3.3. The Isomorphism Theorems . . . . . 304
3.4. Composition Series and the Hölder Program . . . . . 314
3.5. Transpositions and the Alternating Group . . . . . 326

**4. Group Actions** **343**
4.1. Group Actions and Permutation Representations . . . . . 343
4.2. Groups Acting on Themselves by Left Multiplication—Cayley's Theorem . . . . . 353
4.3. Groups Acting on Themselves by Conjugation—The Class Equation . . . . . 367
4.4. Automorphisms . . . . . 403
4.5. The Sylow Theorems . . . . . 423
4.6. The Simplicity of $A_n$ . . . . . 479

**5. Direct and Semidirect Products and Abelian Groups** **487**
5.1. Direct Products . . . . . 487
5.2. The Fundamental Theorem of Finitely Generated Abelian Groups . . . . . 505
5.4. Recognizing Direct Products . . . . . 521
5.5. Semidirect Products . . . . . 541

**6. Further Topics in Group Theory** **566**
6.1. p-groups, Nilpotent Groups, and Solvable Groups . . . . . 566
6.2. Applications in Groups of Medium Order . . . . . 604
6.3. A Word on Free Groups . . . . . 634

**7. Introduction to Rings** **647**
7.1. Basic Definitions and Examples . . . . . 647
7.2. Examples: Polynomial Rings, Matrix Rings, and Group Rings . . . . . 677
7.3. Ring Homomorphisms and Quotient Rings . . . . . 690
7.4. Properties of Ideals . . . . . 727
7.5. Rings of Fractions . . . . . 768
7.6. The Chinese Remainder Theorem . . . . . 774

**8. Euclidean Domains, Principal Ideal Domains and Unique Factorization Domains** **785**
8.1. Euclidean Domains . . . . . 785
8.2. Principal Ideal Domains (P.I.D.s) . . . . . 797

8.3. Unique Factorization Domains (U.F.D.s) . . . 805

**9. Polynomial Rings** **816**
9.1. Definitions and Basic Properties . . . 816
9.2. Polynomial Rings over Fields I . . . 834
9.3. Polynomial Rings that are Unique Factorization Domains . . . 847
9.4. Irreducibility Criteria . . . 852
9.5. Polynomial Rings over Fields II . . . 872
9.6. Polynomials in Several Variables over a Field and Gröbner Bases . . . 879

**10. Introduction to Module Theory** **924**
10.1. Basic Definitions and Examples . . . 924
10.2. Quotient Modules and Module Homomorphisms . . . 947
10.3. Generation of Modules, Direct Sums, and Free Modules . . . 961
10.4. Tensor Products of Modules . . . 988
10.5. Exact Sequences—Projective, Injective, and Flat Modules . . . 1015

**11. Vector Spaces** **1043**
11.1. Definitions and Basic Theory . . . 1043
11.2. The Matrix of a Linear Transformation . . . 1057
11.3. Dual Vector Spaces . . . 1096
11.4. Determinants . . . 1101
11.5. Tensor Algebras, Symmetric and Exterior Algebras . . . 1107

**12. Modules over Principal Ideal Domains** **1121**
12.1. The Basic Theory . . . 1121
12.2. The Rational Canonical Form . . . 1143
12.3. The Jordan Canonical Form . . . 1168

**13. Field Theory** **1227**
13.1. Basic Theory of Field Extensions . . . 1227
13.2. Algebraic Extensions . . . 1235
13.3. Classical Straightedge and Compass Constructions . . . 1257
13.4. Splitting Fields and Algebraic Closures . . . 1262
13.5. Separable and Inseparable Extensions . . . 1268
13.6. Cyclotomic Polynomials and Extensions . . . 1279

**14. Galois Theory** **1296**
14.1. Basic Definitions . . . 1296
14.2. The Fundamental Theorem of Galois Theory . . . 1306
14.3. Finite Fields . . . 1337
14.4. Composite Extensions and Simple Extensions . . . 1354
14.5. Cyclotomic Extensions and Abelian Extensions over Q . . . 1362
14.6. Galois Groups of Polynomials . . . 1380
14.7. Solvable and Radical Extensions: Insolvability of the Quintic . . . 1431
14.8. Computation of Galois Groups over Q . . . 1452
14.9. Transcendental Extensions, Inseparable Extensions, Infinite Galois Groups . . . 1462

**15. Commutative Rings and Algebraic Geometry** **1481**
15.1. Noetherian Rings and Affine Algebraic Sets . . . 1481
15.2. Radicals and Affine Varieties . . . 1529
15.3. Integral Extensions and Hilbert's Nullstellensatz . . . 1583
15.4. Localization . . . 1611
15.5. The Prime Spectrum of a Ring . . . 1651

**16. Artinian Rings, Discrete Valuation Rings, and Dedekind Domains** **1682**
16.1. Artinian Rings . . . . . 1682
16.2. Discrete Valuation Rings . . . . . 1696
16.3. Dedekind Domains . . . . . 1705

**17. Introduction to Homological Algebra and Group Cohomology** **1730**
17.1. Introduction to Homological Algebra—Ext and Tor . . . . . 1730
17.2. The Cohomology of Groups . . . . . 1765
17.3. Crossed Homomorphisms and $H^1(G, A)$ . . . . . 1790
17.4. Group Extensions, Factor Sets and $H^2(G, A)$ . . . . . 1805

**18. Representation Theory and Character Theory** **1815**
18.1. Linear Actions and Modules over Group Rings . . . . . 1815
18.2. Wedderburn's Theorem and Some Consequences . . . . . 1839
18.3. Character Theory and the Orthogonality Relations . . . . . 1857

**19. Examples and Applications of Character Theory** **1885**
19.1. Characters of Groups of Small Order . . . . . 1885
19.2. Theorems of Burnside and Hall . . . . . 1902
19.3. Introduction to the Theory of Induced Characters . . . . . 1907

# Abstract Algebra, Third Edition

## 1. Introduction to Groups

### 1.1. Basic Axioms and Examples

Let $G$ be a group.

**1.** Determine which of the following binary operations are associative:

**(a)** the operation $\star$ on $\mathbb{Z}$ defined by $a \star b = a - b$

**(b)** the operation $\star$ on $\mathbb{R}$ defined by $a \star b = a + b + ab$

**(c)** the operation $\star$ on $\mathbb{Q}$ defined by $a \star b = \dfrac{a+b}{5}$

**(d)** the operation $\star$ on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) \star (c, d) = (ad + bc, bd)$

**(e)** the operation $\star$ on $\mathbb{Q} - \{0\}$ defined by $a \star b = \dfrac{a}{b}$.

Let $G$ be a group.

**2.** Decide which of the binary operations in the preceding exercise are commutative.

Let $G$ be a group.

**3.** Prove that addition of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative (you may assume it is well defined).

Let $G$ be a group.

**4.** Prove that multiplication of residue classes in $\mathbb{Z}/n\mathbb{Z}$ is associative (you may assume it is well defined).

Let $G$ be a group.

**5.** Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.

Let $G$ be a group.

**6.** Determine which of the following sets are groups under addition:

**(a)** the set of rational numbers (including 0 = 0/1) in lowest terms whose denominators are odd

**(b)** the set of rational numbers (including 0 = 0/1) in lowest terms whose denominators are even

**(c)** the set of rational numbers of absolute value $< 1$

**(d)** the set of rational numbers of absolute value $\geq 1$ together with 0

**(e)** the set of rational numbers with denominators equal to 1 or 2

**(f)** the set of rational numbers with denominators equal to 1, 2 or 3.

Let $G$ be a group.

**7.** Let $G = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$ and for $x, y \in G$ let $x \star y$ be the fractional part of $x + y$ (i.e., $x \star y = x + y - [x + y]$ where $[a]$ is the greatest integer less than or equal to $a$). Prove that $\star$ is a well defined binary operation on $G$ and that $G$ is an abelian group under $\star$ (called the *real numbers mod 1*).

Let $G$ be a group.

**8.** Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$.

**(a)** Prove that $G$ is a group under multiplication (called the group of *roots of unity* in $\mathbb{C}$).

**(b)** Prove that $G$ is not a group under addition.

Let $G$ be a group.

**9.** Let $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$.

**(a)** Prove that $G$ is a group under addition.

**(b)** Prove that the nonzero elements of $G$ are a group under multiplication. ["Rationalize the denominators" to find multiplicative inverses.]

Let $G$ be a group.

**10.** Prove that a finite group is abelian if and only if its group table is a symmetric matrix.

Let $G$ be a group.

**11.** Find the orders of each element of the additive group $\mathbb{Z}/12\mathbb{Z}$.

Let $G$ be a group.

**12.** Find the orders of the following elements of the multiplicative group $(\mathbb{Z}/12\mathbb{Z})^\times$: $\overline{1}$, $\overline{-1}$, $\overline{5}$, $\overline{7}$, $\overline{-7}$, $\overline{13}$.

Let $G$ be a group.

**13.** Find the orders of the following elements of the additive group $\mathbb{Z}/36\mathbb{Z}$: $\overline{1}$, $\overline{2}$, $\overline{6}$, $\overline{9}$, $\overline{10}$, $\overline{12}$, $\overline{-1}$, $\overline{-10}$, $\overline{-18}$.

Let $G$ be a group.

**14.** Find the orders of the following elements of the multiplicative group $(\mathbb{Z}/36\mathbb{Z})^\times$: $\overline{1}$, $\overline{-1}$, $\overline{5}$, $\overline{13}$, $\overline{-13}$, $\overline{17}$.

Let $G$ be a group.

**15.** Prove that $(a_1 a_2 \dots a_n)^{-1} = a_n^{-1} a_{n-1}^{-1} \dots a_1^{-1}$ for all $a_1, a_2, \dots, a_n \in G$.

Let $G$ be a group.

**16.** Let $x$ be an element of $G$. Prove that $x^2 = 1$ if and only if $|x|$ is either 1 or 2.

Let $G$ be a group.

**17.** Let $x$ be an element of $G$. Prove that if $|x| = n$ for some positive integer $n$ then $x^{-1} = x^{n-1}$.

Let $G$ be a group.

**18.** Let $x$ and $y$ be elements of $G$. Prove that $xy = yx$ if and only if $y^{-1}xy = x$ if and only if $x^{-1}y^{-1}xy = 1$.

Let $G$ be a group.

**19.** Let $x \in G$ and let $a, b \in \mathbb{Z}^+$.

**(a)** Prove that $x^{a+b} = x^a x^b$ and $(x^a)^b = x^{ab}$.

**(b)** Prove that $(x^a)^{-1} = x^{-a}$.

**(c)** Establish part (a) for arbitrary integers $a$ and $b$ (positive, negative or zero).

Let $G$ be a group.

**20.** For $x$ an element in $G$ show that $x$ and $x^{-1}$ have the same order.

Let $G$ be a group.

**21.** Let $G$ be a finite group and let $x$ be an element of $G$ of order $n$. Prove that if $n$ is odd, then $x = (x^2)^k$ for some k.

Let $G$ be a group.

**22.** If $x$ and $g$ are elements of the group $G$, prove that $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$.

Let $G$ be a group.

**23.** Suppose $x \in G$ and $|x| = n < \infty$. If $n = st$ for some positive integers $s$ and $t$, prove that $|x^s| = t$.

Let $G$ be a group.

**24.** If $a$ and $b$ are *commuting* elements of $G$, prove that $(ab)^n = a^n b^n$ for all $n \in \mathbb{Z}$. [Do this by induction for positive $n$ first.]

Let $G$ be a group.

**25.** Prove that if $x^2 = 1$ for all $x \in G$ then $G$ is abelian.

Let $G$ be a group.

**26.** Assume $H$ is a nonempty subset of $(G, \star)$ which is closed under the binary operation on $G$ and is closed under inverses, i.e., for all $h$ and $k \in H$, $hk$ and $h^{-1} \in H$. Prove that $H$ is a group under the operation $\star$ restricted to $H$ (such a subset $H$ is called a *subgroup* of $G$).

Let $G$ be a group.

**27.** Prove that if $x$ is an element of the group $G$ then $\{x^n \mid n \in \mathbb{Z}\}$ is a subgroup (cf. the preceding exercise) of $G$ (called the *cyclic subgroup* of $G$ generated by $x$).

Let $G$ be a group.

**28.** Let $(A, \star)$ and $(B, \diamond)$ be groups and let $A \times B$ be their direct product (as defined in Example 6). Verify all the group axioms for $A \times B$:

**(a)** prove that the associative law holds: for all $(a_i, b_i) \in A \times B$, $i = 1, 2, 3$
$(a_1, b_1)[(a_2, b_2)(a_3, b_3)] = [(a_1, b_1)(a_2, b_2)](a_3, b_3)$,

**(b)** prove that $(1, 1)$ is the identity of $A \times B$, and

**(c)** prove that the inverse of $(a, b)$ is $(a^{-1}, b^{-1})$.

Let $G$ be a group.

**29.** Prove that $A \times B$ is an abelian group if and only if both $A$ and $B$ are abelian.

Let $G$ be a group.

**30.** Prove that the elements $(a, 1)$ and $(1, b)$ of $A \times B$ commute and deduce that the order of $(a, b)$ is the least common multiple of $|a|$ and $|b|$.

Let $G$ be a group.

**31.** Prove that any finite group $G$ of even order contains an element of order 2. [Let $t(G)$ be the set $\{g \in G \mid g \neq g^{-1}\}$. Show that $t(G)$ has an even number of elements and every nonidentity element of $G - t(G)$ has order 2.]

Let $G$ be a group.

**32.** If $x$ is an element of finite order $n$ in $G$, prove that the elements $1, x, x^2, \ldots, x^{n-1}$ are all distinct. Deduce that $|x| \leq |G|$.

Let $G$ be a group.

**33.** Let $x$ be an element of finite order $n$ in $G$.

**(a)** Prove that if $n$ is odd then $x^i \neq x^{-i}$ for all $i = 1, 2, \ldots, n-1$.

**(b)** Prove that if $n = 2k$ and $1 \leq i < n$ then $x^i = x^{-i}$ if and only if $i = k$.

Let $G$ be a group.

**34.** If $x$ is an element of infinite order in $G$, prove that the elements $x^n$, $n \in \mathbb{Z}$ are all distinct.

Let $G$ be a group.

**35.** If $x$ is an element of finite order $n$ in $G$, use the Division Algorithm to show that *any* integral power of $x$ equals one of the elements in the set $\{1, x, x^2, \ldots, x^{n-1}\}$ (so these are all the distinct elements of the cyclic subgroup (cf. Exercise 27 above) of $G$ generated by $x$).

Let $G$ be a group.

**36.** Assume $G = \{1, a, b, c\}$ is a group of order 4 with identity 1. Assume also that $G$ has no elements of order 4 (so by Exercise 32, every element has order $\leq 3$). Use the cancellation laws to show that there is a unique group table for $G$. Deduce that $G$ is abelian.

## 1.2. Dihedral Groups

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**1.** Compute the order of each of the elements in the following groups:
**(a)** $D_6$ **(b)** $D_8$ **(c)** $D_{10}$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**2.** Use the generators and relations above to show that if $x$ is any element of $D_{2n}$ which is not a power of $r$, then $rx = xr^{-1}$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**3.** Use the generators and relations above to show that every element of $D_{2n}$ which is not a power of $r$ has order 2. Deduce that $D_{2n}$ is generated by the two elements $s$ and $sr$, both of which have order 2.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**4.** If $n = 2k$ is even and $n \geq 4$, show that $z = r^k$ is an element of order 2 which commutes with all elements of $D_{2n}$. Show also that $z$ is the only nonidentity element of $D_{2n}$ which commutes with all elements of $D_{2n}$. [cf. Exercise 33 of Section 1.]

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**5.** If $n$ is odd and $n \geq 3$, show that the identity is the only element of $D_{2n}$ which commutes with all elements of $D_{2n}$. [cf. Exercise 33 of Section 1.]

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**6.** Let $x$ and $y$ be elements of order 2 in any group $G$. Prove that if $t = xy$ then $tx = xt^{-1}$ (so that if $n = |xy| < \infty$ then $x$, $t$ satisfy the same relations in $G$ as $s$, $r$ do in $D_{2n}$).

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**7.** Show that $\langle a, b \mid a^2 = b^2 = (ab)^n = 1 \rangle$ gives a presentation for $D_{2n}$ in terms of the two generators $a = s$ and $b = sr$ of order 2 computed in Exercise 3 above. [Show that the relations for $r$ and $s$ follow from the relations for $a$ and $b$ and, conversely, the relations for $a$ and $b$ follow from those for $r$ and $s$.]

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**8.** Find the order of the cyclic subgroup of $D_{2n}$ generated by $r$ (cf. Exercise 27 of Section 1).

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

In each of Exercises 9 to 13 you can find the order of the group of rigid motions in $\mathbb{R}^3$ (also called the group of rotations) of the given Platonic solid by following the proof for the order of $D_{2n}$: find the number of positions to which an adjacent pair of vertices can be sent. Alternatively, you can find the number of places to which a given face may be sent and, once a face is fixed, the number of positions to which a vertex on that face may be sent.

**9.** Let $G$ be the group of rigid motions in $\mathbb{R}^3$ of a tetrahedron. Show that $|G| = 12$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

In each of Exercises 9 to 13 you can find the order of the group of rigid motions in $\mathbb{R}^3$ (also called the group of rotations) of the given Platonic solid by following the proof for the order of $D_{2n}$: find the number of positions to which an adjacent pair of vertices can be sent. Alternatively, you can find the number of places to which a given face may be sent and, once a face is fixed, the number of positions to which a vertex on that face may be sent.

**10.** Let $G$ be the group of rigid motions in $\mathbb{R}^3$ of a cube. Show that $|G| = 24$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

In each of Exercises 9 to 13 you can find the order of the group of rigid motions in $\mathbb{R}^3$ (also called the group of rotations) of the given Platonic solid by following the proof for the order of $D_{2n}$: find the number of positions to which an adjacent pair of vertices can be sent. Alternatively, you can find the number of places to which a given face may be sent and, once a face is fixed, the number of positions to which a vertex on that face may be sent.

**11.** Let $G$ be the group of rigid motions in $\mathbb{R}^3$ of an octahedron. Show that $|G| = 24$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

In each of Exercises 9 to 13 you can find the order of the group of rigid motions in $\mathbb{R}^3$ (also called the group of rotations) of the given Platonic solid by following the proof for the order of $D_{2n}$: find the number of positions to which an adjacent pair of vertices can be sent. Alternatively, you can find the number of places to which a given face may be sent and, once a face is fixed, the number of positions to which a vertex on that face may be sent.

**12.** Let $G$ be the group of rigid motions in $\mathbb{R}^3$ of a dodecahedron. Show that $|G| = 60$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

In each of Exercises 9 to 13 you can find the order of the group of rigid motions in $\mathbb{R}^3$ (also called the group of rotations) of the given Platonic solid by following the proof for the order of $D_{2n}$: find the number of positions to which an adjacent pair of vertices can be sent. Alternatively, you can find the number of places to which a given face may be sent and, once a face is fixed, the number of positions to which a vertex on that face may be sent.

**13.** Let $G$ be the group of rigid motions in $\mathbb{R}^3$ of an icosahedron. Show that $|G| = 60$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**14.** Find a set of generators for $\mathbb{Z}$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**15.** Find a set of generators and relations for $\mathbb{Z}/n\mathbb{Z}$.

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**16.** Show that the group $\langle x_1, y_1 \mid x_1^2 = y_1^2 = (x_1 y_1)^2 = 1 \rangle$ is the dihedral group $D_4$ (where $x_1$ may be replaced by the letter $r$ and $y_1$ by $s$). [Show that the last relation is the same as: $x_1 y_1 = y_1 x_1^{-1}$.]

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**17.** Let $X_{2n}$ be the group whose presentation is displayed in (1.2).

**(a)** Show that if $n = 3k$, then $X_{2n}$ has order 6, and it has the same generators and relations as $D_6$ when $x$ is replaced by $r$ and $y$ by $s$.

**(b)** Show that if $(3, n) = 1$, then $x$ satisfies the additional relation: $x = 1$. In this case deduce that $X_{2n}$ has order 2. [Use the facts that $x^n = 1$ and $x^3 = 1$.]

In these exercises, $D_{2n}$ has the usual presentation $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$.

**18.** Let $Y$ be the group whose presentation is displayed in (1.3).

**(a)** Show that $v^2 = v^{-1}$. [Use the relation: $v^3 = 1$.]

**(b)** Show that $v$ commutes with $u^3$. [Show that $v^2u^3v = u^3$ by writing the left hand side as $(v^2u^2)(uv)$ and using the relations to reduce this to the right hand side. Then use part (a).]

**(c)** Show that $v$ commutes with $u$. [Show that $u^9 = u$ and then use part (b).]

**(d)** Show that $uv = 1$. [Use part (c) and the last relation.]

**(e)** Show that $u = 1$, deduce that $v = 1$, and conclude that $Y = 1$. [Use part (d) and the equation $u^4v^3 = 1$.]

## 1.3. Symmetric Groups

**1.** Let $\sigma$ be the permutation

$$1 \mapsto 3 \qquad 2 \mapsto 4 \qquad 3 \mapsto 5 \qquad 4 \mapsto 2 \qquad 5 \mapsto 1$$

and let $\tau$ be the permutation

$$1 \mapsto 5 \qquad 2 \mapsto 3 \qquad 3 \mapsto 2 \qquad 4 \mapsto 4 \qquad 5 \mapsto 1.$$

Find the cycle decompositions of each of the following permutations: $\sigma$, $\tau$, $\sigma^2$, $\sigma\tau$, $\tau\sigma$, and $\tau^2\sigma$.

**2.** Let $\sigma$ be the permutation

$$\begin{array}{lllll} 1 \mapsto 13 & 2 \mapsto 2 & 3 \mapsto 15 & 4 \mapsto 14 & 5 \mapsto 10 \\ 6 \mapsto 6 & 7 \mapsto 12 & 8 \mapsto 3 & 9 \mapsto 4 & 10 \mapsto 1 \\ 11 \mapsto 7 & 12 \mapsto 9 & 13 \mapsto 5 & 14 \mapsto 11 & 15 \mapsto 8 \end{array}$$

and let $\tau$ be the permutation

$$\begin{array}{lllll} 1 \mapsto 14 & 2 \mapsto 9 & 3 \mapsto 10 & 4 \mapsto 2 & 5 \mapsto 12 \\ 6 \mapsto 6 & 7 \mapsto 5 & 8 \mapsto 11 & 9 \mapsto 15 & 10 \mapsto 3 \\ 11 \mapsto 8 & 12 \mapsto 7 & 13 \mapsto 4 & 14 \mapsto 1 & 15 \mapsto 13. \end{array}$$

Find the cycle decompositions of the following permutations: $\sigma$, $\tau$, $\sigma^2$, $\sigma\tau$, $\tau\sigma$, and $\tau^2\sigma$.

**3.** For each of the permutations whose cycle decompositions were computed in the preceding two exercises compute its order.

**4.** Compute the order of each of the elements in the following groups: **(a)** $S_3$ **(b)** $S_4$.

**5.** Find the order of (1 12 8 10 4)(2 13)(5 11 7)(6 9).

**6.** Write out the cycle decomposition of each element of order 4 in $S_4$.

**7.** Write out the cycle decomposition of each element of order 2 in $S_4$.

**8.** Prove that if $\Omega = \{1, 2, 3, \ldots\}$ then $S_\Omega$ is an infinite group (do not say $\infty! = \infty$).

**9.** **(a)** Let $\sigma$ be the 12-cycle (1 2 3 4 5 6 7 8 9 10 11 12). For which positive integers $i$ is $\sigma^i$ also a 12-cycle?

**(b)** Let $\tau$ be the 8-cycle (1 2 3 4 5 6 7 8). For which positive integers $i$ is $\tau^i$ also an 8-cycle?

**(c)** Let $\omega$ be the 14-cycle (1 2 3 4 5 6 7 8 9 10 11 12 13 14). For which positive integers $i$ is $\omega^i$ also a 14-cycle?

**10.** Prove that if $\sigma$ is the $m$-cycle $(a_1\ a_2\ \ldots\ a_m)$, then for all $i \in \{1, 2, \ldots, m\}$, $\sigma^i(a_k) = a_{k+i}$, where $k + i$ is replaced by its least residue mod $m$ when $k + i > m$. Deduce that $|\sigma| = m$.

**11.** Let $\sigma$ be the $m$-cycle $(1\ 2\ \ldots\ m)$. Show that $\sigma^i$ is also an $m$-cycle if and only if $i$ is relatively prime to $m$.

**12.** **(a)** If $\tau = (1\ 2)(3\ 4)(5\ 6)(7\ 8)(9\ 10)$ determine whether there is a $n$-cycle $\sigma$ $(n \geq 10)$ with $\tau = \sigma^k$ for some integer $k$.

**(b)** If $\tau = (1\ 2)(3\ 4\ 5)$ determine whether there is an $n$-cycle $\sigma$ $(n \geq 5)$ with $\tau = \sigma^k$ for some integer $k$.

**13.** Show that an element has order 2 in $S_n$ if and only if its cycle decomposition is a product of commuting 2-cycles.

**14.** Let $p$ be a prime. Show that an element has order $p$ in $S_n$ if and only if its cycle decomposition is a product of commuting $p$-cycles. Show by an explicit example that this need not be the case if $p$ is not prime.

**15.** Prove that the order of an element in $S_n$ equals the least common multiple of the lengths of the cycles in its cycle decomposition. [Use Exercise 10 and Exercise 24 of Section 1.]

**16.** Show that if $n \geq m$ then the number of $m$-cycles in $S_n$ is given by

$$\frac{n(n-1)(n-2)\ldots(n-m+1)}{m}.$$

[Count the number of ways of forming an $m$-cycle and divide by the number of representations of a particular $m$-cycle.]

**17.** Show that if $n \geq 4$ then the number of permutations in $S_n$ which are the product of two disjoint 2-cycles is $n(n-1)(n-2)(n-3)/8$.

**18.** Find all numbers $n$ such that $S_5$ contains an element of order $n$. [Use Exercise 15.]

**19.** Find all numbers $n$ such that $S_7$ contains an element of order $n$. [Use Exercise 15.]

**20.** Find a set of generators and relations for $S_3$.

## 1.4. Matrix Groups

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**1.** Prove that $|GL_2(\mathbb{F}_2)| = 6$.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**2.** Write out all the elements of $GL_2(\mathbb{F}_2)$ and compute the order of each element.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**3.** Show that $GL_2(\mathbb{F}_2)$ is non-abelian.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**4.** Show that if $n$ is not prime then $\mathbb{Z}/n\mathbb{Z}$ is not a field.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**5.** Show that $GL_n(F)$ is a finite group if and only if $F$ has a finite number of elements.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**6.** If $|F| = q$ is finite prove that $|GL_n(F)| < q^{n^2}$.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**7.** Let $p$ be a prime. Prove that the order of $GL_2(\mathbb{F}_p)$ is $p^4 - p^3 - p^2 + p$ (do not just quote the order formula in this section). [Subtract the number of $2 \times 2$ matrices which are *not* invertible from the total number of $2 \times 2$ matrices over $\mathbb{F}_p$. You may use the fact that a $2 \times 2$ matrix is not invertible if and only if one row is a multiple of the other.]

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**8.** Show that $GL_n(F)$ is non-abelian for any $n \geq 2$ and any $F$.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**9.** Prove that the binary operation of matrix multiplication of $2 \times 2$ matrices with real number entries is associative.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

**10.** Let $G = \{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{R},\ a \neq 0,\ c \neq 0\}$.

**(a)** Compute the product of $\begin{pmatrix} a_1 & b_1 \\ 0 & c_1 \end{pmatrix}$ and $\begin{pmatrix} a_2 & b_2 \\ 0 & c_2 \end{pmatrix}$ to show that $G$ is closed under matrix multiplication.

**(b)** Find the matrix inverse of $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ and deduce that $G$ is closed under inverses.

**(c)** Deduce that $G$ is a subgroup of $GL_2(\mathbb{R})$ (cf. Exercise 26, Section 1).

**(d)** Prove that the set of elements of $G$ whose two diagonal entries are equal (i.e., $a = c$) is also a subgroup of $GL_2(\mathbb{R})$.

Let $F$ be a field and let $n \in \mathbb{Z}^+$.

The next exercise introduces the *Heisenberg group* over the field $F$ and develops some of its basic properties. When $F = \mathbb{R}$ this group plays an important role in quantum mechanics and signal theory by giving a group theoretic interpretation (due to H. Weyl) of Heisenberg's Uncertainty Principle. Note also that the Heisenberg group may be defined more generally — for example, with entries in $\mathbb{Z}$.

**11.** Let $H(F) = \{\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \mid a, b, c \in F\}$ — called the *Heisenberg group* over $F$. Let $X = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$ and $Y = \begin{pmatrix} 1 & d & e \\ 0 & 1 & f \\ 0 & 0 & 1 \end{pmatrix}$ be elements of $H(F)$.

**(a)** Compute the matrix product $XY$ and deduce that $H(F)$ is closed under matrix multiplication. Exhibit explicit matrices such that $XY \neq YX$ (so that $H(F)$ is always non-abelian).

**(b)** Find an explicit formula for the matrix inverse $X^{-1}$ and deduce that $H(F)$ is closed under inverses.

**(c)** Prove the associative law for $H(F)$ and deduce that $H(F)$ is a group of order $|F|^3$. (Do not assume that matrix multiplication is associative.)

**(d)** Find the order of each element of the finite group $H(\mathbb{Z}/2\mathbb{Z})$.

**(e)** Prove that every nonidentity element of the group $H(\mathbb{R})$ has infinite order.

## 1.5. The Quaternion Group

**1.** Compute the order of each of the elements in $Q_8$.

**2.** Write out the group tables for $S_3$, $D_8$ and $Q_8$.

**3.** Find a set of generators and relations for $Q_8$.

## 1.6. Homomorphisms and Isomorphisms

Let $G$ and $H$ be groups.

**1.** Let $\varphi : G \to H$ be a homomorphism.
**(a)** Prove that $\varphi(x^n) = \varphi(x)^n$ for all $n \in \mathbb{Z}^+$.
**(b)** Do part (a) for $n = -1$ and deduce that $\varphi(x^n) = \varphi(x)^n$ for all $n \in \mathbb{Z}$.

Let $G$ and $H$ be groups.

**2.** If $\varphi : G \to H$ is an isomorphism, prove that $|\varphi(x)| = |x|$ for all $x \in G$. Deduce that any two isomorphic groups have the same number of elements of order $n$ for each $n \in \mathbb{Z}^+$. Is the result true if $\varphi$ is only assumed to be a homomorphism?

Let $G$ and $H$ be groups.

**3.** If $\varphi : G \to H$ is an isomorphism, prove that $G$ is abelian if and only if $H$ is abelian. If $\varphi : G \to H$ is a homomorphism, what additional conditions on $\varphi$ (if any) are sufficient to ensure that if $G$ is abelian, then so is $H$?

Let $G$ and $H$ be groups.

**4.** Prove that the multiplicative groups $\mathbb{R} - \{0\}$ and $\mathbb{C} - \{0\}$ are not isomorphic.

Let $G$ and $H$ be groups.

**5.** Prove that the additive groups $\mathbb{R}$ and $\mathbb{Q}$ are not isomorphic.

Let $G$ and $H$ be groups.

**6.** Prove that the additive groups $\mathbb{Z}$ and $\mathbb{Q}$ are not isomorphic.

Let $G$ and $H$ be groups.

**7.** Prove that $D_8$ and $Q_8$ are not isomorphic.

Let $G$ and $H$ be groups.

**8.** Prove that if $n \neq m$, $S_n$ and $S_m$ are not isomorphic.

Let $G$ and $H$ be groups.

**9.** Prove that $D_{24}$ and $S_4$ are not isomorphic.

Let $G$ and $H$ be groups.

**10.** Fill in the details of the proof that the symmetric groups $S_\Delta$ and $S_\Omega$ are isomorphic if $|\Delta| = |\Omega|$ as follows: let $\theta : \Delta \to \Omega$ be a bijection. Define

$$\varphi : S_\Delta \to S_\Omega \qquad \text{by} \qquad \varphi(\sigma) = \theta \circ \sigma \circ \theta^{-1} \quad \text{for all } \sigma \in S_\Delta$$

and prove the following:

**(a)** $\varphi$ is well defined, that is, if $\sigma$ is a permutation of $\Delta$ then $\theta \circ \sigma \circ \theta^{-1}$ is a permutation of $\Omega$.

**(b)** $\varphi$ is a bijection from $S_\Delta$ onto $S_\Omega$. [Find a 2-sided inverse for $\varphi$.]

**(c)** $\varphi$ is a homomorphism, that is, $\varphi(\sigma \circ \tau) = \varphi(\sigma) \circ \varphi(\tau)$.

Note the **similarity** to the *change of basis* or *similarity* transformations for matrices (we shall see the connections between these later in the text).

Let $G$ and $H$ be groups.

**11.** Let $A$ and $B$ be groups. Prove that $A \times B \cong B \times A$.

Let $G$ and $H$ be groups.

**12.** Let $A$, $B$, and $C$ be groups and let $G = A \times B$ and $H = B \times C$. Prove that $G \times C \cong A \times H$.

Let $G$ and $H$ be groups.

**13.** Let $G$ and $H$ be groups and let $\varphi : G \to H$ be a homomorphism. Prove that the image of $\varphi$, $\varphi(G)$, is a subgroup of $H$ (cf. Exercise 26 of Section 1). Prove that if $\varphi$ is injective then $G \cong \varphi(G)$.

Let $G$ and $H$ be groups.

**14.** Let $G$ and $H$ be groups and let $\varphi : G \to H$ be a homomorphism. Define the *kernel* of $\varphi$ to be $\{g \in G \mid \varphi(g) = 1_H\}$ (so the kernel is the set of elements in $G$ which map to the identity of $H$, i.e., is the fiber over the identity of $H$). Prove that the kernel of $\varphi$ is a subgroup (cf. Exercise 26 of Section 1) of $G$. Prove that $\varphi$ is injective if and only if the kernel of $\varphi$ is the identity subgroup of $G$.

Let $G$ and $H$ be groups.

**15.** Define a map $\pi : \mathbb{R}^2 \to \mathbb{R}$ by $\pi((x, y)) = x$. Prove that $\pi$ is a homomorphism and find the kernel of $\pi$ (cf. Exercise 14).

Let $G$ and $H$ be groups.

**16.** Let $A$ and $B$ be groups and let $G$ be their direct product, $A \times B$. Prove that the maps $\pi_1 : G \to A$ and $\pi_2 : G \to B$ defined by $\pi_1((a, b)) = a$ and $\pi_2((a, b)) = b$ are homomorphisms and find their kernels (cf. Exercise 14).

Let $G$ and $H$ be groups.

**17.** Let $G$ be any group. Prove that the map from $G$ to itself defined by $g \mapsto g^{-1}$ is a homomorphism if and only if $G$ is abelian.

Let $G$ and $H$ be groups.

**18.** Let $G$ be any group. Prove that the map from $G$ to itself defined by $g \mapsto g^2$ is a homomorphism if and only if $G$ is abelian.

Let $G$ and $H$ be groups.

**19.** Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$. Prove that for any fixed integer $k > 1$ the map from $G$ to itself defined by $z \mapsto z^k$ is a surjective homomorphism but is not an isomorphism.

Let $G$ and $H$ be groups.

**20.** Let $G$ be a group and let $\mathrm{Aut}(G)$ be the set of all isomorphisms from $G$ onto $G$. Prove that $\mathrm{Aut}(G)$ is a group under function composition (called the *automorphism group* of $G$ and the elements of $\mathrm{Aut}(G)$ are called *automorphisms* of $G$).

Let $G$ and $H$ be groups.

**21.** Prove that for each fixed nonzero $k \in \mathbb{Q}$ the map from $\mathbb{Q}$ to itself defined by $q \mapsto kq$ is an automorphism of $\mathbb{Q}$ (cf. Exercise 20).

Let $G$ and $H$ be groups.

**22.** Let $A$ be an abelian group and fix some $k \in \mathbb{Z}$. Prove that the map $a \mapsto a^k$ is a homomorphism from $A$ to itself. If $k = -1$ prove that this homomorphism is an isomorphism (i.e., is an automorphism of $A$).

Let $G$ and $H$ be groups.

**23.** Let $G$ be a finite group which possesses an automorphism $\sigma$ (cf. Exercise 20) such that $\sigma(g) = g$ if and only if $g = 1$. If $\sigma^2$ is the identity map from $G$ to $G$, prove that $G$ is abelian (such an automorphism $\sigma$ is called *fixed point free* of order 2). [Show that every element of $G$ can be written in the form $x^{-1}\sigma(x)$ and apply $\sigma$ to such an expression.]

Let $G$ and $H$ be groups.

**24.** Let $G$ be a finite group and let $x$ and $y$ be distinct elements of order 2 in $G$ that generate $G$. Prove that $G \cong D_{2n}$, where $n = |xy|$. [See Exercise 6 in Section 2.]

Let $G$ and $H$ be groups.

**25.** Let $n \in \mathbb{Z}^+$, let $r$ and $s$ be the usual generators of $D_{2n}$ and let $\theta = 2\pi/n$.

**(a)** Prove that the matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ is the matrix of the linear transformation which rotates the $x, y$ plane about the origin in a counterclockwise direction by $\theta$ radians.

**(b)** Prove that the map $\varphi : D_{2n} \to GL_2(\mathbb{R})$ defined on generators by

$$\varphi(r) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \qquad \text{and} \qquad \varphi(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism of $D_{2n}$ into $GL_2(\mathbb{R})$.

**(c)** Prove that the homomorphism $\varphi$ in part (b) is injective.

Let $G$ and $H$ be groups.

**26.** Let $i$ and $j$ be the generators of $Q_8$ described in Section 5. Prove that the map $\varphi$ from $Q_8$ to $GL_2(\mathbb{C})$ defined on generators by

$$\varphi(i) = \begin{pmatrix} \sqrt{-1} & 0 \\ 0 & -\sqrt{-1} \end{pmatrix} \qquad \text{and} \qquad \varphi(j) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

extends to a homomorphism. Prove that $\varphi$ is injective.

## 1.7. Group Actions

**1.** Let $F$ be a field. Show that the multiplicative group of nonzero elements of $F$ (denoted by $F^\times$) acts on the set $F$ by $g \cdot a = ga$, where $g \in F^\times$, $a \in F$ and $ga$ is the usual product in $F$ of the two field elements (state clearly which axioms in the definition of a field are used).

**2.** Show that the additive group $\mathbb{Z}$ acts on itself by $z \cdot a = z + a$ for all $z, a \in \mathbb{Z}$.

**3.** Show that the additive group $\mathbb{R}$ acts on the $x, y$ plane $\mathbb{R} \times \mathbb{R}$ by $r \cdot (x, y) = (x + ry, y)$.

**4.** Let $G$ be a group acting on a set $A$ and fix some $a \in A$. Show that the following sets are subgroups of $G$ (cf. Exercise 26 of Section 1):
**(a)** the kernel of the action,
**(b)** $\{g \in G \mid ga = a\}$ — this subgroup is called the *stabilizer* of $a$ in $G$.

**5.** Prove that the kernel of an action of the group $G$ on the set $A$ is the same as the kernel of the corresponding permutation representation $G \to S_A$ (cf. Exercise 14 in Section 6).

**6.** Prove that a group $G$ acts faithfully on a set $A$ if and only if the kernel of the action is the set consisting only of the identity.

**7.** Prove that in Example 2 in this section the action is faithful.

**8.** Let $A$ be a nonempty set and let $k$ be a positive integer with $k \leq |A|$. The symmetric group $S_A$ acts on the set $B$ consisting of all subsets of $A$ of cardinality $k$ by $\sigma \cdot \{a_1, \ldots, a_k\} = \{\sigma(a_1), \ldots, \sigma(a_k)\}$.

**(a)** Prove that this is a group action.

**(b)** Describe explicitly how the elements (1 2) and (1 2 3) act on the six 2-element subsets of {1, 2, 3, 4}.

**9.** Do both parts of the preceding exercise with "ordered $k$-tuples" in place of "$k$-element subsets," where the action on $k$-tuples is defined as above but with set braces replaced by parentheses (note that, for example, the 2-tuples (1,2) and (2,1) are different even though the sets $\{1, 2\}$ and $\{2, 1\}$ are the same, so the sets being acted upon are different).

**10.** With reference to the preceding two exercises determine:
**(a)** for which values of $k$ the action of $S_n$ on $k$-element subsets is faithful, and
**(b)** for which values of $k$ the action of $S_n$ on ordered $k$-tuples is faithful.

**11.** Write out the cycle decomposition of the eight permutations in $S_4$ corresponding to the elements of $D_8$ given by the action of $D_8$ on the vertices of a square (where the vertices of the square are labelled as in Section 2).

**12.** Assume $n$ is an even positive integer and show that $D_{2n}$ acts on the set consisting of pairs of opposite vertices of a regular $n$-gon. Find the kernel of this action (label vertices as usual).

**13.** Find the kernel of the left regular action.

**14.** Let $G$ be a group and let $A = G$. Show that if $G$ is non-abelian then the maps defined by $g \cdot a = ag$ for all $g, a \in G$ do *not* satisfy the axioms of a (left) group action of $G$ on itself.

**15.** Let $G$ be any group and let $A = G$. Show that the maps defined by $g \cdot a = ag^{-1}$ for all $g, a \in G$ *do* satisfy the axioms of a (left) group action of $G$ on itself.

**16.** Let $G$ be any group and let $A = G$. Show that the maps defined by $g \cdot a = gag^{-1}$ for all $g, a \in G$ *do* satisfy the axioms of a (left) group action (this action of $G$ on itself is called *conjugation*).

**17.** Let $G$ be a group and let $G$ act on itself by left conjugation, so each $g \in G$ maps $G$ to $G$ by

$$x \mapsto gxg^{-1}.$$

For fixed $g \in G$, prove that conjugation by $g$ is an isomorphism from $G$ onto itself (i.e., is an automorphism of $G$ — cf. Exercise 20, Section 6). Deduce that $x$ and $gxg^{-1}$ have the same order for all $x$ in $G$ and that for any subset $A$ of $G$, $|A| = |gAg^{-1}|$ (here $gAg^{-1} = \{gag^{-1} \mid a \in A\}$).

**18.** Let $H$ be a group acting on a set $A$. Prove that the relation $\sim$ on $A$ defined by

$$a \sim b \qquad \text{if and only if} \qquad a = hb \quad \text{for some } h \in H$$

is an equivalence relation. (For each $x \in A$ the equivalence class of $x$ under $\sim$ is called the *orbit* of $x$ under the action of $H$. The orbits under the action of $H$ partition the set $A$.)

**19.** Let $H$ be a subgroup (cf. Exercise 26 of Section 1) of the finite group $G$ and let $H$ act on $G$ (here $A = G$) by left multiplication. Let $x \in G$ and let $\mathcal{O}$ be the orbit of $x$ under the action of $H$. Prove that the map

$$H \to \mathcal{O} \qquad \text{defined by} \qquad h \mapsto hx$$

is a bijection (hence all orbits have cardinality $|H|$). From this and the preceding exercise deduce *Lagrange's Theorem*:

*if $G$ is a finite group and $H$ is a subgroup of $G$ then $|H|$ divides $|G|$.*

**20.** Show that the group of rigid motions of a tetrahedron is isomorphic to a subgroup (cf. Exercise 26 of Section 1) of $S_4$.

**21.** Show that the group of rigid motions of a cube is isomorphic to $S_4$. [This group acts on the set of four pairs of opposite vertices.]

**22.** Show that the group of rigid motions of an octahedron is isomorphic to a subgroup (cf. Exercise 26 of Section 1) of $S_4$. [This group acts on the set of four pairs of opposite faces.] Deduce that the groups of rigid motions of a cube and an octahedron are isomorphic. (These groups are isomorphic because these solids are "dual" — see *Introduction to Geometry* by H. Coxeter, Wiley, 1961. We shall see later that the groups of rigid motions of the dodecahedron and icosahedron are isomorphic as well — these solids are also dual.)

**23.** Explain why the action of the group of rigid motions of a cube on the set of three pairs of opposite faces is not faithful. Find the kernel of this action.

# 2. Subgroups

## 2.1. Definition and Examples

Let $G$ be a group.

**1.** In each of (a) – (e) prove that the specified subset is a subgroup of the given group:

**(a)** the set of complex numbers of the form $a + ai$, $a \in \mathbb{R}$ (under addition)

**(b)** the set of complex numbers of absolute value 1, i.e., the unit circle in the complex plane (under multiplication)

**(c)** for fixed $n \in \mathbb{Z}^+$ the set of rational numbers whose denominators divide $n$ (under addition)

**(d)** for fixed $n \in \mathbb{Z}^+$ the set of rational numbers whose denominators are relatively prime to $n$ (under addition)

**(e)** the set of nonzero real numbers whose square is a rational number (under multiplication).

Let $G$ be a group.

**2.** In each of (a) – (e) prove that the specified subset is *not* a subgroup of the given group:

**(a)** the set of 2-cycles in $S_n$ for $n \geq 3$

**(b)** the set of reflections in $D_{2n}$ for $n \geq 3$

**(c)** for $n$ a composite integer $> 1$ and $G$ a group containing an element of order $n$, the set $\{x \in G \mid |x| = n\} \cup \{1\}$

**(d)** the set of (positive and negative) odd integers in $\mathbb{Z}$ together with 0

**(e)** the set of real numbers whose square is a rational number (under addition).

Let $G$ be a group.

**3.** Show that the following subsets of the dihedral group $D_8$ are actually subgroups:
**(a)** $\{1, r^2, s, sr^2\}$, **(b)** $\{1, r^2, sr, sr^3\}$.

Let $G$ be a group.

**4.** Give an **explicit** example of a group $G$ and an infinite subset $H$ of $G$ that is closed under the group operation but is not a subgroup of $G$.

Let $G$ be a group.

**5.** Prove that $G$ cannot have a subgroup $H$ with $|H| = n - 1$, where $n = |G| > 2$.

Let $G$ be a group.

**6.** Let $G$ be an abelian group. Prove that $\{g \in G \mid |g| < \infty\}$ is a subgroup of $G$ (called the *torsion subgroup* of $G$). Give an explicit example where this set is not a subgroup when $G$ is non-abelian.

Let $G$ be a group.

**7.** Fix some $n \in \mathbb{Z}$ with $n > 1$. Find the torsion subgroup (cf. the previous exercise) of $\mathbb{Z} \times (\mathbb{Z}/n\mathbb{Z})$. Show that the set of elements of infinite order together with the identity is *not* a subgroup of this direct product.

Let $G$ be a group.

**8.** Let $H$ and $K$ be subgroups of $G$. Prove that $H \cup K$ is a subgroup if and only if either $H \subseteq K$ or $K \subseteq H$.

Let $G$ be a group.

**9.** Let $G = GL_n(F)$, where $F$ is any field. Define

$$SL_n(F) = \{A \in GL_n(F) \mid \det(A) = 1\}$$

(called the *special linear group*). Prove that $SL_n(F) \leq GL_n(F)$.

Let $G$ be a group.

**10.** **(a)** Prove that if $H$ and $K$ are subgroups of $G$ then so is their intersection $H \cap K$.
**(b)** Prove that the intersection of an arbitrary nonempty collection of subgroups of $G$ is again a subgroup of $G$ (do not assume the collection is countable).

Let $G$ be a group.

**11.** Let $A$ and $B$ be groups. Prove that the following sets are subgroups of the direct product $A \times B$:

**(a)** $\{(a, 1) \mid a \in A\}$
**(b)** $\{(1, b) \mid b \in B\}$
**(c)** $\{(a, a) \mid a \in A\}$, where here we assume $B = A$ (called the *diagonal subgroup*).

Let $G$ be a group.

**12.** Let $A$ be an abelian group and fix some $n \in \mathbb{Z}$. Prove that the following sets are subgroups of $A$:

**(a)** $\{a^n \mid a \in A\}$

**(b)** $\{a \in A \mid a^n = 1\}$.

Let $G$ be a group.

**13.** Let $H$ be a subgroup of the additive group of rational numbers with the property that $1/x \in H$ for every nonzero element $x$ of $H$. Prove that $H = 0$ or $\mathbb{Q}$.

Let $G$ be a group.

**14.** Show that $\{x \in D_{2n} \mid x^2 = 1\}$ is not a subgroup of $D_{2n}$ (here $n \geq 3$).

Let $G$ be a group.

**15.** Let $H_1 \leq H_2 \leq \cdots$ be an ascending chain of subgroups of $G$. Prove that $\cup_{i=1}^{\infty} H_i$ is a subgroup of $G$.

Let $G$ be a group.

**16.** Let $n \in \mathbb{Z}^+$ and let $F$ be a field. Prove that the set $\{(a_{ij}) \in GL_n(F) \mid a_{ij} = 0 \text{ for all } i > j\}$ is a subgroup of $GL_n(F)$ (called the group of *upper triangular* matrices). ∎

Let $G$ be a group.

**17.** Let $n \in \mathbb{Z}^+$ and let $F$ be a field. Prove that the set $\{(a_{ij}) \in GL_n(F) \mid a_{ij} = 0$ for all $i > j$, and $a_{ii} = 1$ for all $i\}$ is a subgroup of $GL_n(F)$.

## 2.2. Centralizers and Normalizers, Stabilizers and Kernels

**1.** Prove that $C_G(A) = \{g \in G \mid g^{-1}ag = a \text{ for all } a \in A\}$.

**2.** Prove that $C_G(Z(G)) = G$ and deduce that $N_G(Z(G)) = G$.

**3.** Prove that if $A$ and $B$ are subsets of $G$ with $A \subseteq B$ then $C_G(B)$ is a subgroup of $C_G(A)$.

**4.** For each of $S_3$, $D_8$, and $Q_8$ compute the centralizers of each element and find the center of each group. Does Lagrange's Theorem (Exercise 19 in Section 1.7) simplify your work?

**5.** In each of parts (a) to (c) show that for the specified group $G$ and subgroup $A$ of $G$, $C_G(A) = A$ and $N_G(A) = G$.

**(a)** $G = S_3$ and $A = \{1, (1\,2\,3), (1\,3\,2)\}$.

**(b)** $G = D_8$ and $A = \{1, s, r^2, sr^2\}$.

**(c)** $G = D_{10}$ and $A = \{1, r, r^2, r^3, r^4\}$.

**6.** Let $H$ be a subgroup of the group $G$.

**(a)** Show that $H \leq N_G(H)$. Give an example to show that this is not necessarily true if $H$ is not a subgroup.

**(b)** Show that $H \leq C_G(H)$ if and only if $H$ is abelian.

**7.** Let $n \in \mathbb{Z}$ with $n \geq 3$. Prove the following:

**(a)** $Z(D_{2n}) = 1$ if $n$ is odd

**(b)** $Z(D_{2n}) = \{1, r^k\}$ if $n = 2k$.

**8.** Let G = $S_n$, fix an $i \in \{1, 2, \ldots, n\}$ and let $G_i = \{\sigma \in G \mid \sigma(i) = i\}$ (the stabilizer of $i$ in $G$). Use group actions to prove that $G_i$ is a subgroup of $G$. Find $|G_i|$.

**9.** For any subgroup $H$ of $G$ and any nonempty subset $A$ of $G$ define $N_H(A)$ to be the set $\{h \in H \mid hAh^{-1} = A\}$. Show that $N_H(A) = N_G(A) \cap H$ and deduce that $N_H(A)$ is a subgroup of $H$ (note that $A$ need not be a subset of $H$).

**10.** Let $H$ be a subgroup of order 2 in $G$. Show that $N_G(H) = C_G(H)$. Deduce that if $N_G(H) = G$ then $H \leq Z(G)$.

**11.** Prove that $Z(G) \leq N_G(A)$ for any subset $A$ of $G$.

**12.** Let $R$ be the set of all polynomials with integer coefficients in the independent variables $x_1, x_2, x_3, x_4$ i.e., the members of $R$ are finite sums of elements of the form $ax_1^{r_1}x_2^{r_2}x_3^{r_3}x_4^{r_4}$, where $a$ is any integer and $r_1, \ldots, r_4$ are nonnegative integers. For example,

$$12x_1^5x_2^7x_4 - 18x_2^3x_3 + 11x_1^6x_2x_3^3x_4^{23} \qquad (*)$$

is a typical element of $R$. Each $\sigma \in S_4$ gives a permutation of $\{x_1, \ldots, x_4\}$ by defining $\sigma \cdot x_i = x_{\sigma(i)}$. This may be extended to a map from $R$ to $R$ by defining

$$\sigma \cdot p(x_1, x_2, x_3, x_4) = p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$$

for all $p(x_1, x_2, x_3, x_4) \in R$ (i.e., $\sigma$ simply permutes the indices of the variables). For example, if $\sigma = (1\ 2)(3\ 4)$ and $p(x_1, \ldots, x_4)$ is the polynomial in $(*)$ above, then

$$\begin{aligned}\sigma \cdot p(x_1, x_2, x_3, x_4) &= 12x_2^5x_1^7x_3 - 18x_1^3x_4 + 11x_2^6x_1x_4^3x_3^{23} \\ &= 12x_1^7x_2^5x_3 - 18x_1^3x_4 + 11x_1x_2^6x_3^{23}x_4^3.\end{aligned}$$

**(a)** Let $p = p(x_1, \ldots, x_4)$ be the polynomial in $(*)$ above, let $\sigma = (1\ 2\ 3\ 4)$ and let $\tau = (1\ 2\ 3)$. Compute $\sigma \cdot p$, $\tau \cdot (\sigma \cdot p)$, $(\tau \circ \sigma) \cdot p$, and $(\sigma \circ \tau) \cdot p$.

**(b)** Prove that these definitions give a (left) group action of $S_4$ on $R$.

**(c)** Exhibit all permutations in $S_4$ that stabilize $x_4$ and prove that they form a subgroup isomorphic to $S_3$.

**(d)** Exhibit all permutations in $S_4$ that stabilize the element $x_1 + x_2$ and prove that they form an abelian subgroup of order 4.

**(e)** Exhibit all permutations in $S_4$ that stabilize the element $x_1x_2 + x_3x_4$ and prove that they form a subgroup isomorphic to the dihedral group of order 8.

**(f)** Show that the permutations in $S_4$ that stabilize the element $(x_1 + x_2)(x_3 + x_4)$ are exactly the same as those found in part (e). (The two polynomials appearing in parts (e) and (f) and the subgroup that stabilizes them will play an important role in the study of roots of quartic equations in Section 14.6.)

**13.** Let $n$ be a positive integer and let $R$ be the set of all polynomials with integer coefficients in the independent variables $x_1, x_2, \ldots, x_n$, i.e., the members of $R$ are finite sums of elements of the form $ax_1^{r_1}x_2^{r_2}\cdots x_n^{r_n}$, where $a$ is any integer and $r_1, \ldots, r_n$ are nonnegative integers. For each $\sigma \in S_n$ define a map

$$\sigma : R \to R \qquad \text{by} \qquad \sigma \cdot p(x_1, x_2, \ldots, x_n) = p(x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(n)}).$$

Prove that this defines a (left) group action of $S_n$ on $R$.

**14.** Let $H(F)$ be the Heisenberg group over the field $F$ introduced in Exercise 11 of Section 1.4. Determine which matrices lie in the center of $H(F)$ and prove that $Z(H(F))$ is isomorphic to the additive group $F$.

## 2.3. Cyclic Groups and Cyclic Subgroups

**1.** Find all subgroups of $Z_{45} = \langle x \rangle$, giving a generator for each. Describe the containments between these subgroups.

**2.** If $x$ is an element of the finite group $G$ and $|x| = |G|$, prove that $G = \langle x \rangle$. Give an explicit example to show that this result need not be true if $G$ is an infinite group.

**3.** Find all generators for $\mathbb{Z}/48\mathbb{Z}$.

**4.** Find all generators for $\mathbb{Z}/202\mathbb{Z}$.

**5.** Find the number of generators for $\mathbb{Z}/49000\mathbb{Z}$.

**6.** In $\mathbb{Z}/48\mathbb{Z}$ write out all elements of $\langle \bar{a} \rangle$ for every $\bar{a}$. Find all inclusions between subgroups in $\mathbb{Z}/48\mathbb{Z}$.

**7.** Let $Z_{48} = \langle x \rangle$ and use the isomorphism $\mathbb{Z}/48\mathbb{Z} \cong Z_{48}$ given by $\bar{1} \mapsto x$ to list all subgroups of $Z_{48}$ as computed in the preceding exercise.

**8.** Let $Z_{48} = \langle x \rangle$. For which integers $a$ does the map $\varphi_a$ defined by $\varphi_a : \bar{1} \mapsto x^a$ extend to an *isomorphism* from $\mathbb{Z}/48\mathbb{Z}$ onto $Z_{48}$.

**9.** Let $Z_{36} = \langle x \rangle$. For which integers $a$ does the map $\psi_a$ defined by $\psi_a : \bar{1} \mapsto x^a$ extend to a *well defined homomorphism* from $\mathbb{Z}/48\mathbb{Z}$ into $Z_{36}$. Can $\psi_a$ ever be a surjective homomorphism?

**10.** What is the order of $\overline{30}$ in $\mathbb{Z}/54\mathbb{Z}$? Write out all of the elements and their orders in $\langle \overline{30} \rangle$.

**11.** Find all cyclic subgroups of $D_8$. Find a proper subgroup of $D_8$ which is not cyclic.

**12.** Prove that the following groups are *not* cyclic:

**(a)** $Z_2 \times Z_2$

**(b)** $Z_2 \times \mathbb{Z}$

**(c)** $\mathbb{Z} \times \mathbb{Z}$.

**13.** Prove that the following pairs of groups are *not* isomorphic:

**(a)** $\mathbb{Z} \times Z_2$ and $\mathbb{Z}$

**(b)** $\mathbb{Q} \times Z_2$ and $\mathbb{Q}$.

**14.** Let $\sigma = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12)$. For each of the following integers $a$ compute $\sigma^a$: $a = 13, 65, 626, 1195, -6, -81, -570$ and $-1211$.

**15. Prove that $\mathbb{Q} \times \mathbb{Q}$ is not cyclic.**

**16.** Assume $|x| = n$ and $|y| = m$. Suppose that $x$ and $y$ *commute*: $xy = yx$. Prove that $|xy|$ divides the least common multiple of $m$ and $n$. Need this be true if $x$ and $y$ do *not* commute? Give an example of commuting elements $x$, $y$ such that the order of $xy$ is not equal to the least common multiple of $|x|$ and $|y|$.

**17.** Find a presentation for $Z_n$ with one generator.

**18.** Show that if $H$ is any group and $h$ is an element of $H$ with $h^n = 1$, then there is a unique homomorphism from $Z_n = \langle x \rangle$ to $H$ such that $x \mapsto h$.

**19.** Show that if $H$ is any group and $h$ is an element of $H$, then there is a unique homomorphism from $\mathbb{Z}$ to $H$ such that $1 \mapsto h$.

**20.** Let $p$ be a prime and let $n$ be a positive integer. Show that if $x$ is an element of the group $G$ such that $x^{p^n} = 1$ then $|x| = p^m$ for some $m \leq n$.

**21.** Let $p$ be an odd prime and let $n$ be a positive integer. Use the Binomial Theorem to show that $(1 + p)^{p^{n-1}} \equiv 1 (\text{mod } p^n)$ but $(1 + p)^{p^{n-2}} \not\equiv 1 (\text{mod } p^n)$. Deduce that $1 + p$ is an element of order $p^{n-1}$ in the multiplicative group $(\mathbb{Z}/p^n\mathbb{Z})^\times$.

**22.** Let $n$ be an integer $\geq 3$. Use the Binomial Theorem to show that $(1+2^2)^{2^{n-2}} \equiv 1 (\text{mod } 2^n)$ but $(1+2^2)^{2^{n-3}} \not\equiv 1 (\text{mod } 2^n)$. Deduce that 5 is an element of order $2^{n-2}$ in the multiplicative group $(\mathbb{Z}/2^n\mathbb{Z})^\times$.

**23.** Show that $(\mathbb{Z}/2^n\mathbb{Z})^\times$ is not cyclic for any $n \geq 3$. [Find two distinct subgroups of order 2.]

**24.** Let $G$ be a finite group and let $x \in G$.

**(a)** Prove that if $g \in N_G(\langle x \rangle)$ then $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$.

**(b)** Prove conversely that if $gxg^{-1} = x^a$ for some $a \in \mathbb{Z}$ then $g \in N_G(\langle x \rangle)$. [Show first that $gx^kg^{-1} = (gxg^{-1})^k = x^{ak}$ for any integer $k$, so that $g\langle x \rangle g^{-1} \leq \langle x \rangle$. If $x$ has order $n$, show the elements $gx^ig^{-1}$, $i = 0, 1, \ldots, n-1$ are distinct, so that $|g\langle x \rangle g^{-1}| = |\langle x \rangle| = n$ and conclude that $g\langle x \rangle g^{-1} = \langle x \rangle$.]

Note that this cuts down some of the work in computing normalizers of cyclic subgroups since one does not have to check $ghg^{-1} \in \langle x \rangle$ for every $h \in \langle x \rangle$.

**25.** Let $G$ be a cyclic group of order $n$ and let $k$ be an integer relatively prime to $n$. Prove that the map $x \mapsto x^k$ is surjective. Use Lagrange's Theorem (Exercise 19, Section 1.7) to prove the same is true for any finite group of order $n$. (For such $k$ each element has a $k^{\text{th}}$ root in $G$. It follows from Cauchy's Theorem in Section 3.2 that if $k$ is not relatively prime to the order of $G$ then the map $x \mapsto x^k$ is not surjective.)

**26.** Let $Z_n$ be a cyclic group of order $n$ and for each integer $a$ let

$$\sigma_a : Z_n \to Z_n \qquad \text{by} \qquad \sigma_a(x) = x^a \text{ for all } x \in Z_n.$$

**(a)** Prove that $\sigma_a$ is an automorphism of $Z_n$ if and only if $a$ and $n$ are relatively prime (automorphisms were introduced in Exercise 20, Section 1.6).

**(b)** Prove that $\sigma_a = \sigma_b$ if and only if $a \equiv b \pmod{n}$.

**(c)** Prove that *every* automorphism of $Z_n$ is equal to $\sigma_a$ for some integer $a$.

**(d)** Prove that $\sigma_a \circ \sigma_b = \sigma_{ab}$. Deduce that the map $\bar{a} \mapsto \sigma_a$ is an isomorphism of $(\mathbb{Z}/n\mathbb{Z})^\times$ onto the automorphism group of $Z_n$ (so $\mathrm{Aut}(Z_n)$ is an abelian group of order $\varphi(n)$).

## 2.4. Subgroups Generated by Subsets of a Group

**1.** Prove that if $H$ is a subgroup of $G$ then $\langle H \rangle = H$.

**2.** Prove that if $A$ is a subset of $B$ then $\langle A \rangle \leq \langle B \rangle$. Give an example where $A \subseteq B$ with $A \neq B$ but $\langle A \rangle = \langle B \rangle$.

**3.** Prove that if $H$ is an abelian subgroup of a group $G$ then $\langle H, Z(G) \rangle$ is abelian. Give an explicit example of an abelian subgroup $H$ of a group $G$ such that $\langle H, C_G(H) \rangle$ is not abelian.

**4.** Prove that if $H$ is a subgroup of $G$ then $H$ is generated by the set $H - \{1\}$.

**5.** Prove that the subgroup generated by any two distinct elements of order 2 in $S_3$ is all of $S_3$.

**6.** Prove that the subgroup of $S_4$ generated by (1 2) and (1 2)(3 4) is a noncyclic group of order 4.

**7.** Prove that the subgroup of $S_4$ generated by (1 2) and (1 3)(2 4) is isomorphic to the dihedral group of order 8.

**8.** Prove that $S_4 = \langle (1\ 2\ 3\ 4),\ (1\ 2\ 4\ 3) \rangle$.

**9.** Prove that $SL_2(\mathbb{F}_3)$ is the subgroup of $GL_2(\mathbb{F}_3)$ generated by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. [Recall from Exercise 9 of Section 1 that $SL_2(\mathbb{F}_3)$ is the subgroup of matrices of determinant 1. You may assume this subgroup has order 24 — this will be an exercise in Section 3.2.]

**10.** Prove that the subgroup of $SL_2(\mathbb{F}_3)$ generated by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is isomorphic to the quaternion group of order 8. [Use a presentation for $Q_8$.]

**11.** Show that $SL_2(\mathbb{F}_3)$ and $S_4$ are two nonisomorphic groups of order 24.

**12.** Prove that the subgroup of upper triangular matrices in $GL_3(\mathbb{F}_2)$ is isomorphic to the dihedral group of order 8 (cf. Exercise 16, Section 1). [First find the order of this subgroup.]

**13.** Prove that the multiplicative group of positive rational numbers is generated by the set $\{\frac{1}{p} \mid p \text{ is a prime}\}$.

**14.** A group $H$ is called *finitely generated* if there is a finite set $A$ such that $H = \langle A \rangle$.

**(a)** Prove that every finite group is finitely generated.

**(b)** Prove that $\mathbb{Z}$ is finitely generated.

**(c)** Prove that every finitely generated subgroup of the additive group $\mathbb{Q}$ is cyclic. [If $H$ is a finitely generated subgroup of $\mathbb{Q}$, show that $H \leq \langle \frac{1}{k} \rangle$, where $k$ is the product of all the denominators which appear in a set of generators for $H$.]

**(d)** Prove that $\mathbb{Q}$ is not finitely generated.

**15.** Exhibit a proper subgroup of $\mathbb{Q}$ which is not cyclic.

**16.** A subgroup $M$ of a group $G$ is called a *maximal subgroup* if $M \neq G$ and the only subgroups of $G$ which contain $M$ are $M$ and $G$.

**(a)** Prove that if $H$ is a proper subgroup of the finite group $G$ then there is a maximal subgroup of $G$ containing $H$.

**(b)** Show that the subgroup of all rotations in a dihedral group is a maximal subgroup.

**(c)** Show that if $G = \langle x \rangle$ is a cyclic group of order $n \geq 1$ then a subgroup $H$ is maximal if and only $H = \langle x^p \rangle$ for some prime $p$ dividing $n$.

**17.** This is an exercise involving Zorn's Lemma (see Appendix I) to prove that every nontrivial finitely generated group possesses maximal subgroups. Let $G$ be a finitely generated group, say $G = \langle g_1, g_2, \ldots, g_n \rangle$, and let $\mathcal{S}$ be the set of all proper subgroups of $G$. Then $\mathcal{S}$ is partially ordered by inclusion. Let $\mathcal{C}$ be a chain in $\mathcal{S}$.

**(a)** Prove that the union, $H$, of all the subgroups in $\mathcal{C}$ is a subgroup of $G$.

**(b)** Prove that $H$ is a *proper* subgroup. [If not, each $g_i$ must lie in $H$ and so must lie in some element of the chain $\mathcal{C}$. Use the definition of a chain to arrive at a contradiction.]

**(c)** Use Zorn's Lemma to show that $\mathcal{S}$ has a maximal element (which is, by definition, a maximal subgroup).

**18.** Let $p$ be a prime and let $Z = \{z \in \mathbb{C} \mid z^{p^n} = 1 \text{ for some } n \in \mathbb{Z}^+\}$ (so $Z$ is the multiplicative group of all $p$-power roots of unity in $\mathbb{C}$). For each $k \in \mathbb{Z}^+$ let $H_k = \{z \in Z \mid z^{p^k} = 1\}$ (the group of $p^k$th roots of unity). Prove the following:

**(a)** $H_k \leq H_m$ if and only if $k \leq m$

**(b)** $H_k$ is cyclic for all $k$ (assume that for any $n \in \mathbb{Z}^+$, $\{e^{2\pi it/n} \mid t = 0, 1, \ldots, n-1\}$ is the set of all $n^{\text{th}}$ roots of 1 in $\mathbb{C}$)

**(c)** every proper subgroup of $Z$ equals $H_k$ for some $k \in \mathbb{Z}^+$ (in particular, every proper subgroup of $Z$ is finite and cyclic)

**(d)** $Z$ is not finitely generated.

**19.** A nontrivial abelian group $A$ (written multiplicatively) is called *divisible* if for each element $a \in A$ and each nonzero integer $k$ there is an element $x \in A$ such that $x^k = a$, i.e., each element has a $k^{\text{th}}$ root in $A$ (in additive notation, each element is the $k^{\text{th}}$ multiple of some element of $A$).

**(a)** Prove that the additive group of rational numbers, $\mathbb{Q}$, is divisible.

**(b)** Prove that no finite abelian group is divisible.

**20.** Prove that if $A$ and $B$ are nontrivial abelian groups, then $A \times B$ is divisible if and only if both $A$ and $B$ are divisible groups.

## 2.5. The Lattice of Subgroups of a Group

**1.** Let $H$ and $K$ be subgroups of $G$. Exhibit all possible sublattices which show only $G$, 1, $H$, $K$ and their joins and intersections. What distinguishes the different drawings?

**2.** In each of (a) to (d) list all subgroups of $D_{16}$ that satisfy the given condition.

**(a)** Subgroups that are contained in $\langle sr^2, r^4 \rangle$

**(b)** Subgroups that are contained in $\langle sr^7, r^4 \rangle$

**(c)** Subgroups that contain $\langle r^4 \rangle$

**(d)** Subgroups that contain $\langle s \rangle$.

**3.** Show that the subgroup $\langle s, r^2 \rangle$ of $D_8$ is isomorphic to $V_4$.

**4.** Use the given lattice to find all pairs of elements that generate $D_8$ (there are 12 pairs).

**5.** Use the given lattice to find all elements $x \in D_{16}$ such that $D_{16} = \langle x, s \rangle$ (there are 16 such elements $x$).

**6.** Use the given lattices to help find the centralizers of every element in the following groups:
**(a)** $D_8$ **(b)** $Q_8$ **(c)** $S_3$ **(d)** $D_{16}$.

**7.** Find the center of $D_{16}$.

**8.** In each of the following groups find the normalizer of each subgroup:
**(a)** $S_3$ **(b)** $Q_8$.

**9.** Draw the lattices of subgroups of the following groups:
**(a)** $\mathbb{Z}/16\mathbb{Z}$ **(b)** $\mathbb{Z}/24\mathbb{Z}$ **(c)** $\mathbb{Z}/48\mathbb{Z}$. [See Exercise 6 in Section 3.]

**10.** Classify groups of order 4 by proving that if $|G| = 4$ then $G \cong Z_4$ or $G \cong V_4$. [See Exercise 36, Section 1.1.]

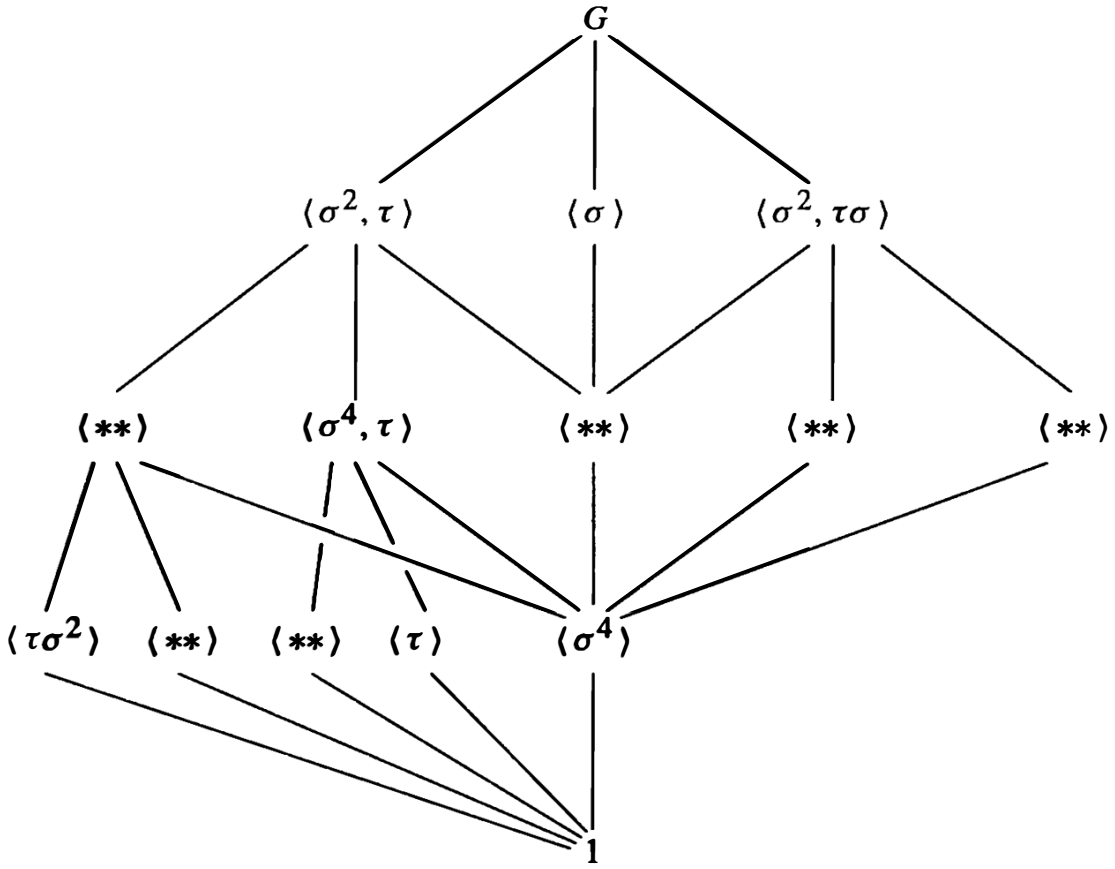
**11.** Consider the group of order 16 with the following presentation:

$$QD_{16} = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1,\ \sigma\tau = \tau\sigma^3 \rangle$$

(called the *quasidihedral* or *semidihedral* group of order 16). This group has three subgroups of order 8: $\langle \tau, \sigma^2 \rangle \cong D_8$, $\langle \sigma \rangle \cong Z_8$ and $\langle \sigma^2, \sigma\tau \rangle \cong Q_8$ and every proper subgroup is contained in one of these three subgroups. Fill in the missing subgroups in the lattice of all subgroups of the quasidihedral group on the following page, exhibiting each subgroup with at most two generators. (This is another example of a nonplanar lattice.)

The next three examples lead to two nonisomorphic groups that have the same lattice of subgroups.

**12.** The group $A = Z_2 \times Z_4 = \langle a, b \mid a^2 = b^4 = 1,\ ab = ba \rangle$ has order 8 and has three subgroups of order 4: $\langle a, b^2 \rangle \cong V_4$, $\langle b \rangle \cong Z_4$ and $\langle ab \rangle \cong Z_4$ and every proper




subgroup is contained in one of these three. Draw the lattice of all subgroups of $A$, giving each subgroup in terms of at most two generators.

The next three examples lead to two nonisomorphic groups that have the same lattice of subgroups.

**13.** The group $G = Z_2 \times Z_8 = \langle x, y \mid x^2 = y^8 = 1,\ xy = yx \rangle$ has order 16 and has three subgroups of order 8: $\langle x, y^2 \rangle \cong Z_2 \times Z_4$, $\langle y \rangle \cong Z_8$ and $\langle xy \rangle \cong Z_8$ and every proper subgroup is contained in one of these three. Draw the lattice of all subgroups of $G$, giving each subgroup in terms of at most two generators (cf. Exercise 12).

The next three examples lead to two nonisomorphic groups that have the same lattice of subgroups.

**14.** Let $M$ be the group of order 16 with the following presentation:

$$\langle u, v \mid u^2 = v^8 = 1,\ vu = uv^5 \rangle$$

(sometimes called the *modular* group of order 16). It has three subgroups of order 8: $\langle u, v^2 \rangle$, $\langle v \rangle$ and $\langle uv \rangle$ and every proper subgroup is contained in one of these three. Prove that $\langle u, v^2 \rangle \cong Z_2 \times Z_4$, $\langle v \rangle \cong Z_8$ and $\langle uv \rangle \cong Z_8$. Show that the lattice of subgroups of $M$ is the same as the lattice of subgroups of $Z_2 \times Z_8$ (cf. Exercise 13) but that these two groups are not isomorphic.

**15.** Describe the isomorphism type of each of the three subgroups of $D_{16}$ of order 8.

**16.** Use the lattice of subgroups of the quasidihedral group of order 16 to show that every element of order 2 is contained in the proper subgroup $\langle \tau, \sigma^2 \rangle$ (cf. Exercise 11).

**17.** Use the lattice of subgroups of the modular group $M$ of order 16 to show that the set $\{x \in M \mid x^2 = 1\}$ is a subgroup of $M$ isomorphic to the Klein 4-group (cf. Exercise 14).

**18.** Use the lattice to help find the centralizer of every element of $QD_{16}$ (cf. Exercise 11).

**19.** Use the lattice to help find $N_{D_{16}}(\langle s, r^4 \rangle)$.

**20.** Use the lattice of subgroups of $QD_{16}$ (cf. Exercise 11) to help find the normalizers
**(a)** $N_{QD_{16}}(\langle \tau\sigma \rangle)$ **(b)** $N_{QD_{16}}(\langle \tau, \sigma^4 \rangle)$.

# 3. Quotient Groups and Homomorphisms

## 3.1. Definitions and Examples

Let $G$ and $H$ be groups.

**1.** Let $\varphi : G \to H$ be a homomorphism and let $E$ be a subgroup of $H$. Prove that $\varphi^{-1}(E) \leq G$ (i.e., the preimage or pullback of a subgroup under a homomorphism is a subgroup). If $E \trianglelefteq H$ prove that $\varphi^{-1}(E) \trianglelefteq G$. Deduce that $\ker \varphi \trianglelefteq G$.

Let $G$ and $H$ be groups.

**2.** Let $\varphi : G \to H$ be a homomorphism of groups with kernel $K$ and let $a, b \in \varphi(G)$. Let $X \in G/K$ be the fiber above $a$ and let $Y$ be the fiber above $b$, i.e., $X = \varphi^{-1}(a)$, $Y = \varphi^{-1}(b)$. Fix an element $u$ of $X$ (so $\varphi(u) = a$). Prove that if $XY = Z$ in the quotient group $G/K$ and $w$ is any member of $Z$, then there is some $v \in Y$ such that $uv = w$. [Show $u^{-1}w \in Y$.]

Let $G$ and $H$ be groups.

**3.** Let $A$ be an abelian group and let $B$ be a subgroup of $A$. Prove that $A/B$ is abelian. Give an example of a non-abelian group $G$ containing a proper normal subgroup $N$ such that $G/N$ is abelian.

Let $G$ and $H$ be groups.

**4.** Prove that in the quotient group $G/N$, $(gN)^\alpha = g^\alpha N$ for all $\alpha \in \mathbb{Z}$.

Let $G$ and $H$ be groups.

**5.** Use the preceding exercise to prove that the order of the element $gN$ in $G/N$ is $n$, where $n$ is the smallest positive integer such that $g^n \in N$ (and $gN$ has infinite order if no such positive integer exists). Give an example to show that the order of $gN$ in $G/N$ may be strictly smaller than the order of $g$ in $G$.

Let $G$ and $H$ be groups.

**6.** Define $\varphi : \mathbb{R}^\times \to \{\pm 1\}$ by letting $\varphi(x)$ be $x$ divided by the absolute value of $x$. Describe the fibers of $\varphi$ and prove that $\varphi$ is a homomorphism.

Let $G$ and $H$ be groups.

**7.** Define $\pi : \mathbb{R}^2 \to \mathbb{R}$ by $\pi((x, y)) = x + y$. Prove that $\pi$ is a surjective homomorphism and describe the kernel and fibers of $\pi$ geometrically.

Let $G$ and $H$ be groups.

**8.** Let $\varphi : \mathbb{R}^\times \to \mathbb{R}^\times$ be the map sending $x$ to the absolute value of $x$. Prove that $\varphi$ is a homomorphism and find the image of $\varphi$. Describe the kernel and the fibers of $\varphi$.

Let $G$ and $H$ be groups.

**9.** Define $\varphi : \mathbb{C}^\times \to \mathbb{R}^\times$ by $\varphi(a+bi) = a^2 + b^2$. Prove that $\varphi$ is a homomorphism and find the image of $\varphi$. Describe the kernel and the fibers of $\varphi$ geometrically (as subsets of the plane).

Let $G$ and $H$ be groups.

**10.** Let $\varphi : \mathbb{Z}/8\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$ by $\varphi(\bar{a}) = \bar{a}$. Show that this is a well defined, surjective homomorphism and describe its fibers and kernel explicitly (showing that $\varphi$ is well defined involves the fact that $\bar{a}$ has a different meaning in the domain and range of $\varphi$).

Let $G$ and $H$ be groups.

**11.** Let $F$ be a field and let $G = \{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in F,\ ac \neq 0\} \leq GL_2(F)$.

**(a)** Prove that the map $\varphi : \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mapsto a$ is a surjective homomorphism from $G$ onto $F^\times$ (recall that $F^\times$ is the multiplicative group of nonzero elements in $F$). Describe the fibers and kernel of $\varphi$.

**(b)** Prove that the map $\psi : \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mapsto (a, c)$ is a surjective homomorphism from $G$ onto $F^\times \times F^\times$. Describe the fibers and kernel of $\psi$.

**(c)** Let $H = \{\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \mid b \in F\}$. Prove that $H$ is isomorphic to the additive group $F$.

Let $G$ and $H$ be groups.

**12.** Let $G$ be the additive group of real numbers, let $H$ be the multiplicative group of complex numbers of absolute value 1 (the unit circle $S^1$ in the complex plane) and let $\varphi : G \to H$ be the homomorphism $\varphi : r \mapsto e^{2\pi i r}$. Draw the points on a real line which lie in the kernel of $\varphi$. Describe similarly the elements in the fibers of $\varphi$ above the points $-1$, $i$, and $e^{4\pi i/3}$ of $H$. (Figure 1 of the text for this homomorphism $\varphi$ is usually depicted using the following diagram.)

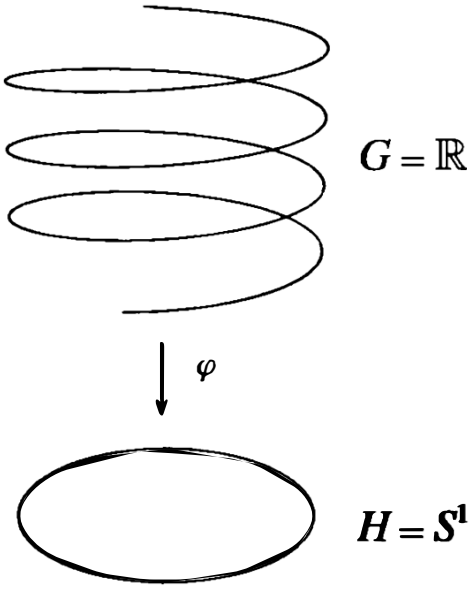



**Fig. 5**

Let $G$ and $H$ be groups.

**13.** Repeat the preceding exercise with the map $\varphi$ replaced by the map $\varphi : r \mapsto e^{4\pi i r}$.

Let $G$ and $H$ be groups.

**14.** Consider the additive quotient group $\mathbb{Q}/\mathbb{Z}$.

**(a)** Show that every coset of $\mathbb{Z}$ in $\mathbb{Q}$ contains exactly one representative $q \in \mathbb{Q}$ in the range $0 \leq q < 1$.

**(b)** Show that every element of $\mathbb{Q}/\mathbb{Z}$ has finite order but that there are elements of arbitrarily large order.

**(c)** Show that $\mathbb{Q}/\mathbb{Z}$ is the torsion subgroup of $\mathbb{R}/\mathbb{Z}$ (cf. Exercise 6, Section 2.1).

**(d)** Prove that $\mathbb{Q}/\mathbb{Z}$ is isomorphic to the multiplicative group of root of unity in $\mathbb{C}^\times$.

Let $G$ and $H$ be groups.

**15.** Prove that a quotient of a divisible abelian group by any proper subgroup is also divisible. Deduce that $\mathbb{Q}/\mathbb{Z}$ is divisible (cf. Exercise 19, Section 2.4).

Let $G$ and $H$ be groups.

**16.** Let $G$ be a group, let $N$ be a normal subgroup of $G$ and let $\overline{G} = G/N$. Prove that if $G = \langle x, y \rangle$ then $\overline{G} = \langle \overline{x}, \overline{y} \rangle$. Prove more generally that if $G = \langle S \rangle$ for any subset $S$ of $G$, then $\overline{G} = \langle \overline{S} \rangle$.

Let $G$ and $H$ be groups.

**17.** Let $G$ be the dihedral group of order 16 (whose lattice appears in Section 2.5):

$$G = \langle r, s \mid r^8 = s^2 = 1,\ rs = sr^{-1} \rangle$$

and let $\overline{G} = G/\langle r^4 \rangle$ be the quotient of $G$ by the subgroup generated by $r^4$ (this subgroup is the center of $G$, hence is normal).

**(a)** Show that the order of $\overline{G}$ is 8.

**(b)** Exhibit each element of $\overline{G}$ in the form $\overline{s}^a\overline{r}^b$, for some integers $a$ and $b$.

**(c)** Find the order of each of the elements of $\overline{G}$ exhibited in (b).

**(d)** Write each of the following elements of $\overline{G}$ in the form $\overline{s}^a\overline{r}^b$, for some integers $a$ and $b$ as in (b): $\overline{rs}$, $\overline{sr^{-2}s}$, $\overline{s^{-1}r^{-1}sr}$.

**(e)** Prove that $\overline{H} = \langle \overline{s}, \overline{r}^2 \rangle$ is a normal subgroup of $\overline{G}$ and $\overline{H}$ is isomorphic to the Klein 4-group. Describe the isomorphism type of the complete preimage of $\overline{H}$ in $G$.

**(f)** Find the center of $\overline{G}$ and describe the isomorphism type of $\overline{G}/Z(\overline{G})$.

Let $G$ and $H$ be groups.

**18.** Let $G$ be the quasidihedral group of order 16 (whose lattice was computed in Exercise 11 of Section 2.5):

$$G = \langle \sigma, \tau \mid \sigma^8 = \tau^2 = 1,\ \sigma\tau = \tau\sigma^3 \rangle$$

and let $\overline{G} = G/\langle \sigma^4 \rangle$ be the quotient of $G$ by the subgroup generated by $\sigma^4$ (this subgroup is the center of $G$, hence is normal).

**(a)** Show that the order of $\overline{G}$ is 8.

**(b)** Exhibit each element of $\overline{G}$ in the form $\overline{\tau}^a\overline{\sigma}^b$, for some integers $a$ and $b$.

**(c)** Find the order of each of the elements of $\overline{G}$ exhibited in (b).

**(d)** Write each of the following elements of $\overline{G}$ in the form $\overline{\tau}^a\overline{\sigma}^b$, for some integers $a$ and $b$ as in (b): $\overline{\sigma\tau}$, $\overline{\tau\sigma^{-2}\tau}$, $\overline{\tau^{-1}\sigma^{-1}\tau\sigma}$.

**(e)** Prove that $\overline{G} \cong D_8$.

Let $G$ and $H$ be groups.

**19.** Let $G$ be the modular group of order 16 (whose lattice was computed in Exercise 14 of Section 2.5):

$$G = \langle u, v \mid u^2 = v^8 = 1,\ vu = uv^5 \rangle$$

and let $\overline{G} = G/\langle v^4 \rangle$ be the quotient of $G$ by the subgroup generated by $v^4$ (this subgroup is contained in the center of $G$, hence is normal).

**(a)** Show that the order of $\overline{G}$ is 8.

**(b)** Exhibit each element of $\overline{G}$ in the form $\overline{u}^a\overline{v}^b$, for some integers $a$ and $b$.

**(c)** Find the order of each of the elements of $\overline{G}$ exhibited in (b).

**(d)** Write each of the following elements of $\overline{G}$ in the form $\overline{u}^a\overline{v}^b$, for some integers $a$ and $b$ as in (b): $\overline{vu}$, $\overline{uv^{-2}u}$, $\overline{u^{-1}v^{-1}uv}$.

**(e)** Prove that $\overline{G}$ is abelian and is isomorphic to $Z_2 \times Z_4$.

Let $G$ and $H$ be groups.

**20.** Let $G = \mathbb{Z}/24\mathbb{Z}$ and let $\widetilde{G} = G/\langle 12 \rangle$, where for each integer $a$ we simplify notation by writing $\widetilde{\overline{a}}$ as $\widetilde{a}$.

**(a)** Show that $\widetilde{G} = \{\widetilde{0}, \widetilde{1}, \ldots, \widetilde{11}\}$.

**(b)** Find the order of each element of $\widetilde{G}$.

**(c)** Prove that $\widetilde{G} \cong \mathbb{Z}/12\mathbb{Z}$. (Thus $(\mathbb{Z}/24\mathbb{Z})/(12\mathbb{Z}/24\mathbb{Z}) \cong \mathbb{Z}/12\mathbb{Z}$, just as if we inverted and cancelled the $24\mathbb{Z}$'s.)

Let $G$ and $H$ be groups.

**21.** Let $G = Z_4 \times Z_4$ be given in terms of the following generators and relations:

$$G = \langle x, y \mid x^4 = y^4 = 1,\ xy = yx \rangle.$$

Let $\overline{G} = G/\langle x^2y^2 \rangle$ (note that every subgroup of the abelian group $G$ is normal).

**(a)** Show that the order of $\overline{G}$ is 8.

**(b)** Exhibit each element of $\overline{G}$ in the form $\overline{x}^a\overline{y}^b$, for some integers $a$ and $b$.

**(c)** Find the order of each of the elements of $\overline{G}$ exhibited in (b).

**(d)** Prove that $\overline{G} \cong Z_4 \times Z_2$.

Let $G$ and $H$ be groups.

**22.** **(a)** Prove that if $H$ and $K$ are normal subgroups of a group $G$ then their intersection $H \cap K$ is also a normal subgroup of $G$.

**(b)** Prove that the intersection of an arbitrary nonempty collection of normal subgroups of a group is a normal subgroup (do not assume the collection is countable).

Let $G$ and $H$ be groups.

**23.** Prove that the join (cf. Section 2.5) of any nonempty collection of normal subgroups of a group is a normal subgroup.

Let $G$ and $H$ be groups.

**24.** Prove that if $N \trianglelefteq G$ and $H$ is any subgroup of $G$ then $N \cap H \trianglelefteq H$.

Let $G$ and $H$ be groups.

**25.** **(a)** Prove that a subgroup $N$ of $G$ is normal if and only if $gNg^{-1} \subseteq N$ for *all* $g \in G$.

**(b)** Let $G = GL_2(\mathbb{Q})$, let $N$ be the subgroup of upper triangular matrices with integer entries and 1's on the diagonal, and let $g$ be the diagonal matrix with entries 2,1. Show that $gNg^{-1} \subseteq N$ but $g$ does *not* normalize $N$.

Let $G$ and $H$ be groups.

**26.** Let $a, b \in G$.

**(a)** Prove that the conjugate of the product of $a$ and $b$ is the product of the conjugate of $a$ and the conjugate of $b$. Prove that the order of $a$ and the order of any conjugate of $a$ are the same.

**(b)** Prove that the conjugate of $a^{-1}$ is the inverse of the conjugate of $a$.

**(c)** Let $N = \langle S \rangle$ for some subset $S$ of $G$. Prove that $N \trianglelefteq G$ if $gSg^{-1} \subseteq N$ for all $g \in G$.

**(d)** Deduce that if $N$ is the cyclic group $\langle x \rangle$, then $N$ is normal in $G$ if and only if for each $g \in G$, $gxg^{-1} = x^k$ for some $k \in \mathbb{Z}$.

**(e)** Let $n$ be a positive integer. Prove that the subgroup $N$ of $G$ generated by all the elements of $G$ of order $n$ is a normal subgroup of $G$.

Let $G$ and $H$ be groups.

**27.** Let $N$ be a *finite* subgroup of a group $G$. Show that $gNg^{-1} \subseteq N$ if and only if $gNg^{-1} = N$. Deduce that $N_G(N) = \{g \in G \mid gNg^{-1} \subseteq N\}$.

Let $G$ and $H$ be groups.

**28.** Let $N$ be a *finite* subgroup of a group $G$ and assume $N = \langle S \rangle$ for some subset $S$ of $G$. Prove that an element $g \in G$ normalizes $N$ if and only if $gSg^{-1} \subseteq N$.

Let $G$ and $H$ be groups.

**29.** Let $N$ be a *finite* subgroup of $G$ and suppose $G = \langle T \rangle$ and $N = \langle S \rangle$ for some subsets $S$ and $T$ of $G$. Prove that $N$ is normal in $G$ if and only if $tSt^{-1} \subseteq N$ for all $t \in T$.

Let $G$ and $H$ be groups.

**30.** Let $N \leq G$ and let $g \in G$. Prove that $gN = Ng$ if and only if $g \in N_G(N)$.

Let $G$ and $H$ be groups.

**31.** Prove that if $H \leq G$ and $N$ is a normal subgroup of $H$ then $H \leq N_G(N)$. Deduce that $N_G(N)$ is the largest subgroup of $G$ in which $N$ is normal (i.e., is the join of all subgroups $H$ for which $N \trianglelefteq H$).

Let $G$ and $H$ be groups.

**32.** Prove that every subgroup of $Q_8$ is normal. For each subgroup find the isomorphism type of its corresponding quotient. [You may use the lattice of subgroups for $Q_8$ in Section 2.5.]

Let $G$ and $H$ be groups.

**33.** Find all normal subgroups of $D_8$ and for each of these find the isomorphism type of its corresponding quotient. [You may use the lattice of subgroups for $D_8$ in Section 2.5.]

Let $G$ and $H$ be groups.

**34.** Let $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$ be the usual presentation of the dihedral group of order $2n$ and let $k$ be a positive integer dividing $n$.

**(a)** Prove that $\langle r^k \rangle$ is a normal subgroup of $D_{2n}$.

**(b)** Prove that $D_{2n}/\langle r^k \rangle \cong D_{2k}$.

Let $G$ and $H$ be groups.

**35.** Prove that $SL_n(F) \trianglelefteq GL_n(F)$ and describe the isomorphism type of the quotient group (cf. Exercise 9, Section 2.1).

Let $G$ and $H$ be groups.

**36.** Prove that if $G/Z(G)$ is cyclic then $G$ is abelian. [If $G/Z(G)$ is cyclic with generator $xZ(G)$, show that every element of $G$ can be written in the form $x^a z$ for some integer $a \in \mathbb{Z}$ and some element $z \in Z(G)$.]

Let $G$ and $H$ be groups.

**37.** Let $A$ and $B$ be groups. Show that $\{(a, 1) \mid a \in A\}$ is a normal subgroup of $A \times B$ and the quotient of $A \times B$ by this subgroup is isomorphic to $B$.

Let $G$ and $H$ be groups.

**38.** Let $A$ be an abelian group and let $D$ be the (diagonal) subgroup $\{(a, a) \mid a \in A\}$ of $A \times A$. Prove that $D$ is a normal subgroup of $A \times A$ and $(A \times A)/D \cong A$.

Let $G$ and $H$ be groups.

**39.** Suppose $A$ is the non-abelian group $S_3$ and $D$ is the diagonal subgroup $\{(a, a) \mid a \in A\}$ of $A \times A$. Prove that $D$ is not normal in $A \times A$.

Let $G$ and $H$ be groups.

**40.** Let $G$ be a group, let $N$ be a normal subgroup of $G$ and let $\overline{G} = G/N$. Prove that $\bar{x}$ and $\bar{y}$ commute in $\overline{G}$ if and only if $x^{-1}y^{-1}xy \in N$. (The element $x^{-1}y^{-1}xy$ is called the *commutator* of $x$ and $y$ and is denoted by $[x, y]$.)

Let $G$ and $H$ be groups.

**41.** Let $G$ be a group. Prove that $N = \langle x^{-1}y^{-1}xy \mid x, y \in G \rangle$ is a normal subgroup of $G$ and $G/N$ is abelian ($N$ is called the *commutator subgroup* of $G$).

Let $G$ and $H$ be groups.

**42.** Assume both $H$ and $K$ are normal subgroups of $G$ with $H \cap K = 1$. Prove that $xy = yx$ for all $x \in H$ and $y \in K$. [Show $x^{-1}y^{-1}xy \in H \cap K$.]

Let $G$ and $H$ be groups.

**43.** Assume $\mathcal{P} = \{A_i \mid i \in I\}$ is any partition of $G$ with the property that $\mathcal{P}$ is a group under the "quotient operation" defined as follows: to compute the product of $A_i$ with $A_j$ take any element $a_i$ of $A_i$ and any element $a_j$ of $A_j$ and let $A_i A_j$ be the element of $\mathcal{P}$ containing $a_i a_j$ (this operation is assumed to be well defined). Prove that the element of $\mathcal{P}$ that contains the identity of $G$ is a normal subgroup of $G$ and the elements of $\mathcal{P}$ are the cosets of this subgroup (so $\mathcal{P}$ is just a quotient group of $G$ in the usual sense).

## 3.2. More on Cosets and Lagrange's Theorem

Let $G$ be a group.

**1.** Which of the following are permissible orders for subgroups of a group of order 120: 1, 2, 5, 7, 9, 15, 60, 240? For each permissible order give the corresponding index.

Let $G$ be a group.

**2.** Prove that the lattice of subgroups of $S_3$ in Section 2.5 is correct (i.e., prove that it contains all subgroups of $S_3$ and that their pairwise joins and intersections are correctly drawn).

Let $G$ be a group.

**3.** Prove that the lattice of subgroups of $Q_8$ in Section 2.5 is correct.

Let $G$ be a group.

**4.** Show that if $|G| = pq$ for some primes $p$ and $q$ (not necessarily distinct) then either $G$ is abelian or $Z(G) = 1$. [See Exercise 36 in Section 1.]

Let $G$ be a group.

**5.** Let $H$ be a subgroup of $G$ and fix some element $g \in G$.
**(a)** Prove that $gHg^{-1}$ is a subgroup of $G$ of the same order as $H$.
**(b)** Deduce that if $n \in \mathbb{Z}^+$ and $H$ is the unique subgroup of $G$ of order $n$ then $H \trianglelefteq G$.

Let $G$ be a group.

**6.** Let $H \leq G$ and let $g \in G$. Prove that if the right coset $Hg$ equals *some* left coset of $H$ in $G$ then it equals the left coset $gH$ and $g$ must be in $N_G(H)$.

Let $G$ be a group.

**7.** Let $H \leq G$ and define a relation $\sim$ on $G$ by $a \sim b$ if and only if $b^{-1}a \in H$. Prove that $\sim$ is an equivalence relation and describe the equivalence class of each $a \in G$. Use this to prove Proposition 4.

Let $G$ be a group.

**8.** Prove that if $H$ and $K$ are finite subgroups of $G$ whose orders are relatively prime then $H \cap K = 1$.

Let $G$ be a group.

**9.** This exercise outlines a proof of Cauchy's Theorem due to James McKay (*Another proof of Cauchy's group theorem*, Amer. Math. Monthly, 66(1959), p. 119). Let $G$ be a finite group and let $p$ be a prime dividing $|G|$. Let $\mathcal{S}$ denote the set of $p$-tuples of elements of $G$ the product of whose coordinates is 1:

$$\mathcal{S} = \{(x_1, x_2, \ldots, x_p) \mid x_i \in G \text{ and } x_1 x_2 \cdots x_p = 1\}.$$

**(a)** Show that $\mathcal{S}$ has $|G|^{p-1}$ elements, hence has order divisible by $p$.

Define the relation $\sim$ on $\mathcal{S}$ by letting $\alpha \sim \beta$ if $\beta$ is a cyclic permutation of $\alpha$.

**(b)** Show that a cyclic permutation of an element of $\mathcal{S}$ is again an element of $\mathcal{S}$.

**(c)** Prove that $\sim$ is an equivalence relation on $\mathcal{S}$.

**(d)** Prove that an equivalence class contains a single element if and only if it is of the form $(x, x, \ldots, x)$ with $x^p = 1$.

**(e)** Prove that every equivalence class has order 1 or $p$ (this uses the fact that $p$ is a *prime*). Deduce that $|G|^{p-1} = k + pd$, where $k$ is the number of classes of size 1 and $d$ is the number of classes of size $p$.

**(f)** Since $\{(1, 1, \ldots, 1)\}$ is an equivalence class of size 1, conclude from (e) that there must be a nonidentity element $x$ in $G$ with $x^p = 1$, i.e., $G$ contains an element of order $p$. [Show $p \mid k$ and so $k > 1$.]

Let $G$ be a group.

**10.** Suppose $H$ and $K$ are subgroups of finite index in the (possibly infinite) group $G$ with $|G : H| = m$ and $|G : K| = n$. Prove that l.c.m.$(m, n) \leq |G : H \cap K| \leq mn$. Deduce that if $m$ and $n$ are relatively prime then $|G : H \cap K| = |G : H| \cdot |G : K|$.

Let $G$ be a group.

**11.** Let $H \leq K \leq G$. Prove that $|G : H| = |G : K| \cdot |K : H|$ (do not assume $G$ is finite).

Let $G$ be a group.

**12.** Let $H \leq G$. Prove that the map $x \mapsto x^{-1}$ sends each left coset of $H$ in $G$ onto a right coset of $H$ and gives a bijection between the set of left cosets and the set of right cosets of $H$ in $G$ (hence the number of left cosets of $H$ in $G$ equals the number of right cosets).

Let $G$ be a group.

**13.** Fix any labelling of the vertices of a square and use this to identify $D_8$ as a subgroup of $S_4$. Prove that the elements of $D_8$ and $\langle (1\,2\,3) \rangle$ do not commute in $S_4$.

Let $G$ be a group.

**14.** Prove that $S_4$ does not have a normal subgroup of order 8 or a normal subgroup of order 3.

Let $G$ be a group.

**15.** Let $G = S_n$ and for fixed $i \in \{1, 2, \ldots, n\}$ let $G_i$ be the stabilizer of $i$. Prove that $G_i \cong S_{n-1}$.

Let $G$ be a group.

**16.** Use Lagrange's Theorem in the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ to prove ***Fermat's Little Theorem***: if $p$ is a prime then $a^p \equiv a \pmod{p}$ for all $a \in \mathbb{Z}$.

Let $G$ be a group.

**17.** Let $p$ be a prime and let $n$ be a positive integer. Find the order of $\bar{p}$ in $(\mathbb{Z}/(p^n-1)\mathbb{Z})^\times$ and deduce that $n \mid \varphi(p^n - 1)$ (here $\varphi$ is Euler's function).

Let $G$ be a group.

**18.** Let $G$ be a finite group, let $H$ be a subgroup of $G$ and let $N \trianglelefteq G$. Prove that if $|H|$ and $|G : N|$ are relatively prime then $H \leq N$.

Let $G$ be a group.

**19.** Prove that if $N$ is a normal subgroup of the finite group $G$ and $(|N|, |G : N|) = 1$ then $N$ is the unique subgroup of $G$ of order $|N|$.

Let $G$ be a group.

**20.** If $A$ is an abelian group with $A \trianglelefteq G$ and $B$ is any subgroup of $G$ prove that $A \cap B \trianglelefteq AB$.

Let $G$ be a group.

**21.** Prove that $\mathbb{Q}$ has no proper subgroups of finite index. Deduce that $\mathbb{Q}/\mathbb{Z}$ has no proper subgroups of finite index. [Recall Exercise 21, Section 1.6 and Exercise 15, Section 1.]

Let $G$ be a group.

**22.** Use Lagrange's Theorem in the multiplicative group $(\mathbb{Z}/n\mathbb{Z})^\times$ to prove *Euler's Theorem*: $a^{\varphi(n)} \equiv 1 \bmod n$ for every integer $a$ relatively prime to $n$, where $\varphi$ denotes Euler's $\varphi$-function.

Let $G$ be a group.

**23.** Determine the last two digits of $3^{3^{100}}$. [Determine $3^{100} \bmod \varphi(100)$ and use the previous exercise.]

## 3.3. The Isomorphism Theorems

Let $G$ be a group.

**1.** Let $F$ be a finite field of order $q$ and let $n \in \mathbb{Z}^+$. Prove that $|GL_n(F) : SL_n(F)| = q - 1$. [See Exercise 35, Section 1.]

Let $G$ be a group.

**2.** Prove all parts of the Lattice Isomorphism Theorem.

Let $G$ be a group.

**3.** Prove that if $H$ is a normal subgroup of $G$ of prime index $p$ then for all $K \leq G$ either
**(i)** $K \leq H$ or
**(ii)** $G = HK$ and $|K : K \cap H| = p$.

Let $G$ be a group.

**4.** Let $C$ be a normal subgroup of the group $A$ and let $D$ be a normal subgroup of the group $B$. Prove that $(C \times D) \trianglelefteq (A \times B)$ and $(A \times B)/(C \times D) \cong (A/C) \times (B/D)$.

Let $G$ be a group.

**5.** Let $QD_{16} = \langle \sigma, \tau \rangle$ be the quasidihedral group described in Exercise 11 of Section 2.5. Prove that $\langle \sigma^4 \rangle$ is normal in $QD_{16}$ and use the Lattice Isomorphism Theorem to draw the lattice of subgroups of $QD_{16}/\langle \sigma^4 \rangle$. Which group of order 8 has the same lattice as this quotient? Use generators and relations for $QD_{16}/\langle \sigma^4 \rangle$ to decide the isomorphism type of this group.

Let $G$ be a group.

**6.** Let $M = \langle v, u \rangle$ be the modular group of order 16 described in Exercise 14 of Section 2.5. Prove that $\langle v^4 \rangle$ is normal in $M$ and use the Lattice Isomorphism Theorem to draw the lattice of subgroups of $M/\langle v^4 \rangle$. Which group of order 8 has the same lattice as this quotient? Use generators and relations for $M/\langle v^4 \rangle$ to decide the isomorphism type of this group.

Let $G$ be a group.

**7.** Let $M$ and $N$ be normal subgroups of $G$ such that $G = MN$. Prove that $G/(M \cap N) \cong (G/M) \times (G/N)$. [Draw the lattice.]

Let $G$ be a group.

**8.** Let $p$ be a prime and let $G$ be the group of $p$-power roots of 1 in $\mathbb{C}$ (cf. Exercise 18, Section 2.4). Prove that the map $z \mapsto z^p$ is a surjective homomorphism. Deduce that $G$ is isomorphic to a proper quotient of itself.

Let $G$ be a group.

**9.** Let $p$ be a prime and let $G$ be a group of order $p^a m$, where $p$ does not divide $m$. Assume $P$ is a subgroup of $G$ of order $p^a$ and $N$ is a normal subgroup of $G$ of order $p^b n$, where $p$ does not divide $n$. Prove that $|P \cap N| = p^b$ and $|PN/N| = p^{a-b}$. (The subgroup $P$ of $G$ is called a *Sylow p-subgroup* of $G$. This exercise shows that the intersection of any Sylow $p$-subgroup of $G$ with a normal subgroup $N$ is a Sylow $p$-subgroup of $N$.)

Let $G$ be a group.

**10.** Generalize the preceding exercise as follows. A subgroup $H$ of a finite group $G$ is called a *Hall subgroup* of $G$ if its index in $G$ is relatively prime to its order: $(|G : H|, |H|) = 1$. Prove that if $H$ is a Hall subgroup of $G$ and $N \trianglelefteq G$, then $H \cap N$ is a Hall subgroup of $N$ and $HN/N$ is a Hall subgroup of $G/N$.

## 3.4. Composition Series and the Hölder Program

**1.** Prove that if $G$ is an abelian simple group then $G \cong Z_p$ for some prime $p$ (do not assume $G$ is a finite group).

**2.** Exhibit all 3 composition series for $Q_8$ and all 7 composition series for $D_8$. List the composition factors in each case.

**3.** Find a composition series for the quasidihedral group of order 16 (cf. Exercise 11, Section 2.5). Deduce that $QD_{16}$ is solvable.

**4.** Use Cauchy's Theorem and induction to show that a finite abelian group has a subgroup of order $n$ for each positive divisor $n$ of its order.

**5.** Prove that subgroups and quotient groups of a solvable group are solvable.

**6.** Prove part (1) of the Jordan–Hölder Theorem by induction on $|G|$.

**7.** If $G$ is a finite group and $H \trianglelefteq G$ prove that there is a composition series of $G$, one of whose terms is $H$.

**8.** Let $G$ be a *finite* group. Prove that the following are equivalent:

**(i)** $G$ is solvable

**(ii)** $G$ has a chain of subgroups: $1 = H_0 \trianglelefteq H_1 \trianglelefteq H_2 \trianglelefteq \ldots \trianglelefteq H_s = G$ such that $H_{i+1}/H_i$ is cyclic, $0 \leq i \leq s - 1$

**(iii)** all composition factors of $G$ are of prime order

**(iv)** $G$ has a chain of subgroups: $1 = N_0 \trianglelefteq N_1 \trianglelefteq N_2 \trianglelefteq \ldots \trianglelefteq N_t = G$ such that each $N_i$ is a normal subgroup of $G$ and $N_{i+1}/N_i$ is abelian, $0 \leq i \leq t - 1$.

[For (iv), prove that a minimal nontrivial normal subgroup $M$ of $G$ is necessarily abelian and then use induction. To see that $M$ is abelian, let $N \trianglelefteq M$ be of prime index (by (iii)) and show that $x^{-1}y^{-1}xy \in N$ for all $x, y \in M$ (cf. Exercise 40, Section 1). Apply the same argument to $gNg^{-1}$ to show that $x^{-1}y^{-1}xy$ lies in the intersection of all $G$-conjugates of $N$, and use the minimality of $M$ to conclude that $x^{-1}y^{-1}xy = 1$.]

**9.** Prove the following special case of part (2) of the Jordan–Hölder Theorem: assume the finite group $G$ has two composition series

$$1 = N_0 \trianglelefteq N_1 \trianglelefteq \ldots \trianglelefteq N_r = G \qquad \text{and} \qquad 1 = M_0 \trianglelefteq M_1 \trianglelefteq M_2 = G.$$

Show that $r = 2$ and that the list of composition factors is the same. [Use the Second Isomorphism Theorem.]

**10.** Prove part (2) of the Jordan–Hölder Theorem by induction on $\min\{r, s\}$. [Apply the inductive hypothesis to $H = N_{r-1} \cap M_{s-1}$ and use the preceding exercises.]

**11.** Prove that if $H$ is a nontrivial normal subgroup of the solvable group $G$ then there is a nontrivial subgroup $A$ of $H$ with $A \trianglelefteq G$ and $A$ abelian.

**12.** Prove (without using the Feit–Thompson Theorem) that the following are equivalent:
**(i)** every group of odd order is solvable
**(ii)** the only simple groups of odd order are those of prime order.

## 3.5. Transpositions and the Alternating Group

**1.** In Exercises 1 and 2 of Section 1.3 you were asked to find the cycle decomposition of some permutations. Write each of these permutations as a product of transpositions. Determine which of these is an even permutation and which is an odd permutation.

**2.** Prove that $\sigma^2$ is an even permutation for every permutation $\sigma$.

**3.** Prove that $S_n$ is generated by $\{(i \quad i+1) \mid 1 \leq i \leq n-1\}$. [Consider conjugates, viz. $(2\ 3)(1\ 2)(2\ 3)^{-1}$.]

**4.** Show that $S_n = \langle (1\ 2),\ (1\ 2\ 3 \ldots n) \rangle$ for all $n \geq 2$.

**5.** Show that if $p$ is prime, $S_p = \langle \sigma, \tau \rangle$ where $\sigma$ is any transposition and $\tau$ is any $p$-cycle.

**6.** Show that $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle$ is a proper subgroup of $S_4$. What is the isomorphism type of this subgroup?

**7.** Prove that the group of rigid motions of a tetrahedron is isomorphic to $A_4$. [Recall Exercise 20 in Section 1.7.]

**8.** Prove the lattice of subgroups of $A_4$ given in the text is correct. [By the preceding exercise and the comments following Lagrange's Theorem, $A_4$ has no subgroup of order 6.]

**9.** Prove that the (unique) subgroup of order 4 in $A_4$ is normal and is isomorphic to $V_4$.

**10.** Find a composition series for $A_4$. Deduce that $A_4$ is solvable.

**11.** Prove that $S_4$ has no subgroup isomorphic to $Q_8$.

**12.** Prove that $A_n$ contains a subgroup isomorphic to $S_{n-2}$ for each $n \geq 3$.

**13.** Prove that every element of order 2 in $A_n$ is the square of an element of order 4 in $S_n$. [An element of order 2 in $A_n$ is a product of $2k$ commuting transpositions.]

**14.** Prove that the subgroup of $A_4$ generated by any element of order 2 and any element of order 3 is all of $A_4$.

**15.** Prove that if $x$ and $y$ are distinct 3-cycles in $S_4$ with $x \neq y^{-1}$, then the subgroup of $S_4$ generated by $x$ and $y$ is $A_4$.

**16.** Let $x$ and $y$ be distinct 3-cycles in $S_5$ with $x \neq y^{-1}$.

**(a)** Prove that if $x$ and $y$ fix a common element of $\{1, \ldots, 5\}$, then $\langle x, y \rangle \cong A_4$.

**(b)** Prove that if $x$ and $y$ do not fix a common element of $\{1, \ldots, 5\}$, then $\langle x, y \rangle = A_5$.

**17.** If $x$ and $y$ are 3-cycles in $S_n$, prove that $\langle x, y \rangle$ is isomorphic to $Z_3$, $A_4$, $A_5$ or $Z_3 \times Z_3$.

# 4. Group Actions

## 4.1. Group Actions and Permutation Representations

Let $G$ be a group and let $A$ be a nonempty set.

**1.** Let $G$ act on the set $A$. Prove that if $a, b \in A$ and $b = g \cdot a$ for some $g \in G$, then $G_b = gG_ag^{-1}$ ($G_a$ is the stabilizer of $a$). Deduce that if $G$ acts transitively on $A$ then the kernel of the action is $\cap_{g \in G}\, gG_ag^{-1}$.

Let $G$ be a group and let $A$ be a nonempty set.

**2.** Let $G$ be a *permutation group* on the set $A$ (i.e., $G \leq S_A$), let $\sigma \in G$ and let $a \in A$. Prove that $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$. Deduce that if $G$ acts transitively on $A$ then

$$\bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1.$$

Let $G$ be a group and let $A$ be a nonempty set.

**3.** Assume that $G$ is an abelian, transitive subgroup of $S_A$. Show that $\sigma(a) \neq a$ for all $\sigma \in G - \{1\}$ and all $a \in A$. Deduce that $|G| = |A|$. [Use the preceding exercise.]

Let $G$ be a group and let $A$ be a nonempty set.

**4.** Let $S_3$ act on the set $\Omega$ of ordered pairs: $\{(i, j) \mid 1 \leq i, j \leq 3\}$ by $\sigma((i, j)) = (\sigma(i), \sigma(j))$. Find the orbits of $S_3$ on $\Omega$. For each $\sigma \in S_3$ find the cycle decomposition of $\sigma$ under this action (i.e., find its cycle decomposition when $\sigma$ is considered as an element of $S_9$ — first fix a labelling of these nine ordered pairs). For each orbit $\mathcal{O}$ of $S_3$ acting on these nine points pick some $a \in \mathcal{O}$ and find the stabilizer of $a$ in $S_3$.

Let $G$ be a group and let $A$ be a nonempty set.

**5.** For each of parts (a) and (b) repeat the preceding exercise but with $S_3$ acting on the specified set:

**(a)** the set of 27 triples $\{(i, j, k) \mid 1 \leq i, j, k \leq 3\}$

**(b)** the set $\mathcal{P}(\{1, 2, 3\}) - \{\emptyset\}$ of all 7 nonempty subsets of $\{1, 2, 3\}$.

Let $G$ be a group and let $A$ be a nonempty set.

**6.** As in Exercise 12 of Section 2.2 let $R$ be the set of all polynomials with integer coefficients in the independent variables $x_1, x_2, x_3, x_4$ and let $S_4$ act on $R$ by permuting the indices of the four variables:

$$\sigma \cdot p(x_1, x_2, x_3, x_4) = p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$$

for all $\sigma \in S_4$.

**(a)** Find the polynomials in the orbit of $S_4$ on $R$ containing $x_1 + x_2$ (recall from Exercise 12 in Section 2.2 that the stabilizer of this polynomial has order 4).

**(b)** Find the polynomials in the orbit of $S_4$ on $R$ containing $x_1x_2 + x_3x_4$ (recall from Exercise 12 in Section 2.2 that the stabilizer of this polynomial has order 8).

**(c)** Find the polynomials in the orbit of $S_4$ on $R$ containing $(x_1 + x_2)(x_3 + x_4)$.

Let $G$ be a group and let $A$ be a nonempty set.

**7.** Let $G$ be a transitive permutation group on the finite set $A$. A *block* is a nonempty subset $B$ of $A$ such that for all $\sigma \in G$ either $\sigma(B) = B$ or $\sigma(B) \cap B = \emptyset$ (here $\sigma(B)$ is the set $\{\sigma(b) \mid b \in B\}$).

**(a)** Prove that if $B$ is a block containing the element $a$ of $A$, then the set $G_B$ defined by $G_B = \{\sigma \in G \mid \sigma(B) = B\}$ is a subgroup of $G$ containing $G_a$.

**(b)** Show that if $B$ is a block and $\sigma_1(B), \sigma_2(B), \ldots, \sigma_n(B)$ are all the distinct images of $B$ under the elements of $G$, then these form a partition of $A$.

**(c)** A (transitive) group $G$ on a set $A$ is said to be *primitive* if the only blocks in $A$ are the trivial ones: the sets of size 1 and $A$ itself. Show that $S_4$ is primitive on $A = \{1, 2, 3, 4\}$. Show that $D_8$ is not primitive as a permutation group on the four vertices of a square.

**(d)** Prove that the transitive group $G$ is primitive on $A$ if and only if for each $a \in A$, the only subgroups of $G$ containing $G_a$ are $G_a$ and $G$ (i.e., $G_a$ is a *maximal* subgroup of $G$, cf. Exercise 16, Section 2.4). [Use part (a).]

Let $G$ be a group and let $A$ be a nonempty set.

**8.** A transitive permutation group $G$ on a set $A$ is called *doubly transitive* if for any (hence all) $a \in A$ the subgroup $G_a$ is transitive on the set $A - \{a\}$.

**(a)** Prove that $S_n$ is doubly transitive on $\{1, 2, \ldots, n\}$ for all $n \geq 2$.

**(b)** Prove that a doubly transitive group is primitive. Deduce that $D_8$ is not doubly transitive in its action on the 4 vertices of a square.

Let $G$ be a group and let $A$ be a nonempty set.

**9.** Assume $G$ acts transitively on the finite set $A$ and let $H$ be a normal subgroup of $G$. Let $\mathcal{O}_1, \mathcal{O}_2, \ldots, \mathcal{O}_r$ be the distinct orbits of $H$ on $A$.

**(a)** Prove that $G$ permutes the sets $\mathcal{O}_1, \mathcal{O}_2, \ldots, \mathcal{O}_r$ in the sense that for each $g \in G$ and each $i \in \{1, \ldots, r\}$ there is a $j$ such that $g\mathcal{O}_i = \mathcal{O}_j$, where $g\mathcal{O} = \{g \cdot a \mid a \in \mathcal{O}\}$ (i.e., in the notation of Exercise 7 the sets $\mathcal{O}_1, \ldots, \mathcal{O}_r$ are blocks). Prove that $G$ is transitive on $\{\mathcal{O}_1, \ldots, \mathcal{O}_r\}$. Deduce that all orbits of $H$ on $A$ have the same cardinality.

**(b)** Prove that if $a \in \mathcal{O}_1$ then $|\mathcal{O}_1| = |H : H \cap G_a|$ and prove that $r = |G : HG_a|$. [Draw the sublattice describing the Second Isomorphism Theorem for the subgroups $H$ and $G_a$ of $G$. Note that $H \cap G_a = H_a$.]

Let $G$ be a group and let $A$ be a nonempty set.

**10.** Let $H$ and $K$ be subgroups of the group $G$. For each $x \in G$ define the $HK$ *double coset* of $x$ in $G$ to be the set

$$HxK = \{hxk \mid h \in H,\ k \in K\}.$$

**(a)** Prove that $HxK$ is the union of the left cosets $x_1K, \ldots, x_nK$ where $\{x_1K, \ldots, x_nK\}$ is the orbit containing $xK$ of $H$ acting by left multiplication on the set of left cosets of $K$.

**(b)** Prove that $HxK$ is a union of right cosets of $H$.

**(c)** Show that $HxK$ and $HyK$ are either the same set or are disjoint for all $x, y \in G$. Show that the set of $HK$ double cosets partitions $G$.

**(d)** Prove that $|HxK| = |K| \cdot |H : H \cap xKx^{-1}|$.

**(e)** Prove that $|HxK| = |H| \cdot |K : K \cap x^{-1}Hx|$.

## 4.2. Groups Acting on Themselves by Left Multiplication—Cayley's Theorem

Let $G$ be a group and let $H$ be a subgroup of $G$.

**1.** Let $G = \{1, a, b, c\}$ be the Klein 4-group whose group table is written out in Section 2.5.

**(a)** Label $1, a, b, c$ with the integers 1,2,4,3, respectively, and prove that under the left regular representation of $G$ into $S_4$ the nonidentity elements are mapped as follows:

$$a \mapsto (1\ 2)(3\ 4) \qquad b \mapsto (1\ 4)(2\ 3) \qquad c \mapsto (1\ 3)(2\ 4).$$

**(b)** Relabel $1, a, b, c$ as 1,4,2,3, respectively, and compute the image of each element of $G$ under the left regular representation of $G$ into $S_4$. Show that the image of $G$ in $S_4$ under this labelling is the same *subgroup* as the image of $G$ in part (a) (even though the nonidentity elements individually map to different permutations under the two different labellings).

Let $G$ be a group and let $H$ be a subgroup of $G$.

**2.** List the elements of $S_3$ as 1, (1 2), (2 3), (1 3), (1 2 3), (1 3 2) and label these with the integers 1,2,3,4,5,6 respectively. Exhibit the image of each element of $S_3$ under the left regular representation of $S_3$ into $S_6$.

Let $G$ be a group and let $H$ be a subgroup of $G$.

**3.** Let $r$ and $s$ be the usual generators for the dihedral group of order 8.

**(a)** List the elements of $D_8$ as $1, r, r^2, r^3, s, sr, sr^2, sr^3$ and label these with the integers $1, 2, \ldots, 8$ respectively. Exhibit the image of each element of $D_8$ under the left regular representation of $D_8$ into $S_8$.

**(b)** Relabel this same list of elements of $D_8$ with the integers 1, 3, 5, 7, 2, 4, 6, 8 respectively and recompute the image of each element of $D_8$ under the left regular representation with respect to this new labelling. Show that the two subgroups of $S_8$ obtained in parts (a) and (b) are different.

Let $G$ be a group and let $H$ be a subgroup of $G$.

**4.** Use the left regular representation of $Q_8$ to produce two elements of $S_8$ which generate a subgroup of $S_8$ isomorphic to the quaternion group $Q_8$.

Let $G$ be a group and let $H$ be a subgroup of $G$.

**5.** Let $r$ and $s$ be the usual generators for the dihedral group of order 8 and let $H = \langle s \rangle$. List the left cosets of $H$ in $D_8$ as $1H, rH, r^2H$ and $r^3H$.

**(a)** Label these cosets with the integers 1,2,3,4, respectively. Exhibit the image of each element of $D_8$ under the representation $\pi_H$ of $D_8$ into $S_4$ obtained from the action of $D_8$ by left multiplication on the set of 4 left cosets of $H$ in $D_8$. Deduce that this representation is faithful (i.e., the elements of $S_4$ obtained form a subgroup isomorphic to $D_8$).

**(b)** Repeat part (a) with the list of cosets relabelled by the integers 1,3,2,4, respectively. Show that the permutations obtained from this labelling form a subgroup of $S_4$ that is different from the subgroup obtained in part (a).

**(c)** Let $K = \langle sr \rangle$, list the cosets of $K$ in $D_8$ as $1K, rK, r^2K$ and $r^3K$, and label these with the integers 1,2,3,4. Prove that, with respect to this labelling, the image of $D_8$ under the representation $\pi_K$ obtained from left multiplication on the cosets of $K$ is the same *subgroup* of $S_4$ as in part (a) (even though the subgroups $H$ and $K$ are different and some of the elements of $D_8$ map to different permutations under the two homomorphisms).

Let $G$ be a group and let $H$ be a subgroup of $G$.

**6.** Let $r$ and $s$ be the usual generators for the dihedral group of order 8 and let $N = \langle r^2 \rangle$. List the left cosets of $N$ in $D_8$ as $1N$, $rN$, $sN$ and $srN$. Label these cosets with the integers 1,2,3,4 respectively. Exhibit the image of each element of $D_8$ under the representation $\pi_N$ of $D_8$ into $S_4$ obtained from the action of $D_8$ by left multiplication on the set of 4 left cosets of $N$ in $D_8$. Deduce that this representation is not faithful and prove that $\pi_N(D_8)$ is isomorphic to the Klein 4-group.

Let $G$ be a group and let $H$ be a subgroup of $G$.

**7.** Let $Q_8$ be the quaternion group of order 8.
   **(a)** Prove that $Q_8$ is isomorphic to a subgroup of $S_8$.
   **(b)** Prove that $Q_8$ is not isomorphic to a subgroup of $S_n$ for any $n \leq 7$. [If $Q_8$ acts on any set $A$ of order $\leq 7$ show that the stabilizer of any point $a \in A$ must contain the subgroup $\langle -1 \rangle$.]

Let $G$ be a group and let $H$ be a subgroup of $G$.

**8.** Prove that if $H$ has finite index $n$ then there is a normal subgroup $K$ of $G$ with $K \leq H$ and $|G : K| \leq n!$.

Let $G$ be a group and let $H$ be a subgroup of $G$.

**9.** Prove that if $p$ is a prime and $G$ is a group of order $p^\alpha$ for some $\alpha \in \mathbb{Z}^+$, then every subgroup of index $p$ is normal in $G$. Deduce that every group of order $p^2$ has a normal subgroup of order $p$.

Let $G$ be a group and let $H$ be a subgroup of $G$.

**10.** Prove that every non-abelian group of order 6 has a nonnormal subgroup of order 2. Use this to classify groups of order 6. [Produce an injective homomorphism into $S_3$.]

Let $G$ be a group and let $H$ be a subgroup of $G$.

**11.** Let $G$ be a finite group and let $\pi : G \to S_G$ be the left regular representation. Prove that if $x$ is an element of $G$ of order $n$ and $|G| = mn$, then $\pi(x)$ is a product of $m$ $n$-cycles. Deduce that $\pi(x)$ is an odd permutation if and only if $|x|$ is even and $\dfrac{|G|}{|x|}$ is odd.

Let $G$ be a group and let $H$ be a subgroup of $G$.

**12.** Let $G$ and $\pi$ be as in the preceding exercise. Prove that if $\pi(G)$ contains an odd permutation then $G$ has a subgroup of index 2. [Use Exercise 3 in Section 3.3.]

Let $G$ be a group and let $H$ be a subgroup of $G$.

**13.** Prove that if $|G| = 2k$ where $k$ is odd then $G$ has a subgroup of index 2. [Use Cauchy's Theorem to produce an element of order 2 and then use the preceding two exercises.]

Let $G$ be a group and let $H$ be a subgroup of $G$.

**14.** Let $G$ be a finite group of composite order $n$ with the property that $G$ has a subgroup of order $k$ for each positive integer $k$ dividing $n$. Prove that $G$ is not simple.

## 4.3. Groups Acting on Themselves by Conjugation—The Class Equation

Let $G$ be a group.

**1.** Suppose $G$ has a left action on a set $A$, denoted by $g \cdot a$ for all $g \in G$ and $a \in A$. Denote the corresponding right action on $A$ by $a \cdot g$. Prove that the (equivalence) relations $\sim$ and $\sim'$ defined by

$$a \sim b \qquad \text{if and only if} \qquad a = g \cdot b \quad \text{for some } g \in G$$

and

$$a \sim' b \qquad \text{if and only if} \qquad a = b \cdot g \quad \text{for some } g \in G$$

are the same relation (i.e., $a \sim b$ if and only if $a \sim' b$).

Let $G$ be a group.

**2.** Find all conjugacy classes and their sizes in the following groups:
**(a)** $D_8$ **(b)** $Q_8$ **(c)** $A_4$.

Let $G$ be a group.

**3.** Find all the conjugacy classes and their sizes in the following groups:
**(a)** $Z_2 \times S_3$ **(b)** $S_3 \times S_3$ **(c)** $Z_3 \times A_4$.

Let $G$ be a group.

**4.** Prove that if $S \subseteq G$ and $g \in G$ then $gN_G(S)g^{-1} = N_G(gSg^{-1})$ and $gC_G(S)g^{-1} = C_G(gSg^{-1})$.

Let $G$ be a group.

**5.** If the center of $G$ is of index $n$, prove that every conjugacy class has at most $n$ elements.

Let $G$ be a group.

**6.** Assume $G$ is a non-abelian group of order 15. Prove that $Z(G) = 1$. Use the fact that $\langle g \rangle \leq C_G(g)$ for all $g \in G$ to show that there is at most one possible class equation for $G$. [Use Exercise 36, Section 3.1.]

Let $G$ be a group.

**7.** For $n = 3, 4, 6$ and 7 make lists of the partitions of $n$ and give representatives for the corresponding conjugacy classes of $S_n$.

**Let $G$ be a group.**

**8. Prove that $Z(S_n) = 1$ for all $n \geq 3$.**

Let $G$ be a group.

**9.** Show that $|C_{S_n}((1\,2)(3\,4))| = 8 \cdot (n-4)!$ for all $n \geq 4$. Determine the elements in this centralizer explicitly.

Let $G$ be a group.

**10.** Let $\sigma$ be the 5-cycle (1 2 3 4 5) in $S_5$. In each of (a) to (c) find an explicit element $\tau \in S_5$ which accomplishes the specified conjugation:

**(a)** $\tau\sigma\tau^{-1} = \sigma^2$

**(b)** $\tau\sigma\tau^{-1} = \sigma^{-1}$

**(c)** $\tau\sigma\tau^{-1} = \sigma^{-2}$.

Let $G$ be a group.

**11.** In each of (a) – (d) determine whether $\sigma_1$ and $\sigma_2$ are conjugate. If they are, give an explicit permutation $\tau$ such that $\tau\sigma_1\tau^{-1} = \sigma_2$.

**(a)** $\sigma_1 = (1\ 2)(3\ 4\ 5)$ and $\sigma_2 = (1\ 2\ 3)(4\ 5)$

**(b)** $\sigma_1 = (1\ 5)(3\ 7\ 2)(10\ 6\ 8\ 11)$ and $\sigma_2 = (3\ 7\ 5\ 10)(4\ 9)(13\ 11\ 2)$

**(c)** $\sigma_1 = (1\ 5)(3\ 7\ 2)(10\ 6\ 8\ 11)$ and $\sigma_2 = \sigma_1^3$

**(d)** $\sigma_1 = (1\ 3)(2\ 4\ 6)$ and $\sigma_2 = (3\ 5)(2\ 4)(5\ 6)$.

Let $G$ be a group.

**12.** Find a representative for each conjugacy class of elements of order 4 in $S_8$ and in $S_{12}$.

**Let $G$ be a group.**

**13. Find all finite groups which have exactly two conjugacy classes.**

Let $G$ be a group.

**14.** In Exercise 1 of Section 2 two labellings of the elements $\{1, a, b, c\}$ of the Klein 4-group $V$ were chosen to give two versions of the left regular representation of $V$ into $S_4$. Let $\pi_1$ be the version of regular representation obtained in part (a) of that exercise and let $\pi_2$ be the version obtained via the labelling in part (b). Let $\tau = (2\ 4)$. Show that $\tau \circ \pi_1(g) \circ \tau^{-1} = \pi_2(g)$ for each $g \in V$ (i.e., conjugation by $\tau$ sends the image of $\pi_1$ to the image of $\pi_2$ elementwise).

Let $G$ be a group.

**15.** Find an element of $S_8$ which conjugates the subgroup of $S_8$ obtained in part (a) of Exercise 3, Section 2 to the subgroup of $S_8$ obtained in part (b) of that same exercise (both of these subgroups are isomorphic to $D_8$).

Let $G$ be a group.

**16.** Find an element of $S_4$ which conjugates the subgroup of $S_4$ obtained in part (a) of Exercise 5, Section 2 to the subgroup of $S_4$ obtained in part (b) of that same exercise (both of these subgroups are isomorphic to $D_8$).

Let $G$ be a group.

**17.** Let $A$ be a nonempty set and let $X$ be any subset of $S_A$. Let

$$F(X) = \{a \in A \mid \sigma(a) = a \text{ for all } \sigma \in X\} \qquad \text{— the } \textit{fixed set} \text{ of } X.$$

Let $M(X) = A - F(X)$ be the elements which are *moved* by some element of $X$. Let $D = \{\sigma \in S_A \mid |M(\sigma)| < \infty\}$. Prove that $D$ is a normal subgroup of $S_A$.

Let $G$ be a group.

**18.** Let $A$ be a set, let $H$ be a subgroup of $S_A$ and let $F(H)$ be the fixed points of $H$ on $A$ as defined in the preceding exercise. Prove that if $\tau \in N_{S_A}(H)$ then $\tau$ stabilizes the set $F(H)$ and its complement $A - F(H)$.

Let $G$ be a group.

**19.** Assume $H$ is a normal subgroup of $G$, $\mathcal{K}$ is a conjugacy class of $G$ contained in $H$ and $x \in \mathcal{K}$. Prove that $\mathcal{K}$ is a union of $k$ conjugacy classes of equal size in $H$, where $k = |G : HC_G(x)|$. Deduce that a conjugacy class in $S_n$ which consists of even permutations is either a single conjugacy class under the action of $A_n$ or is a union of two classes of the same size in $A_n$. [Let $A = C_G(x)$ and $B = H$ so $A \cap B = C_H(x)$. Draw the lattice diagram associated to the Second Isomorphism Theorem and interpret the appropriate indices. See also Exercise 9, Section 1.]

Let $G$ be a group.

**20.** Let $\sigma \in A_n$. Show that all elements in the conjugacy class of $\sigma$ in $S_n$ (i.e., all elements of the same cycle type as $\sigma$) are conjugate in $A_n$ if and only if $\sigma$ commutes with an odd permutation. [Use the preceding exercise.]

Let $G$ be a group.

**21.** Let $\mathcal{K}$ be a conjugacy class in $S_n$ and assume that $\mathcal{K} \subseteq A_n$. Show $\sigma \in S_n$ does *not* commute with any odd permutation if and only if the cycle type of $\sigma$ consists of distinct odd integers. Deduce that $\mathcal{K}$ consists of two conjugacy classes in $A_n$ if and only if the cycle type of an element of $\mathcal{K}$ consists of distinct odd integers. [Assume first that $\sigma \in \mathcal{K}$ does not commute with any odd permutation. Observe that $\sigma$ commutes with each individual cycle in its cycle decomposition — use this to show that all its cycles must be of odd length. If two cycles have the same odd length, $k$, find a product of $k$ transpositions which interchanges them and commutes with $\sigma$. Conversely, if the cycle type of $\sigma$ consists of distinct integers, prove that $\sigma$ commutes *only* with the group generated by the cycles in its cycle decomposition.]

Let $G$ be a group.

**22.** Show that if $n$ is odd then the set of all $n$-cycles consists of two conjugacy classes of equal size in $A_n$.

Let $G$ be a group.

**23.** Recall (cf. Exercise 16, Section 2.4) that a proper subgroup $M$ of $G$ is called *maximal* if whenever $M \leq H \leq G$, either $H = M$ or $H = G$. Prove that if $M$ is a maximal subgroup of $G$ then either $N_G(M) = M$ or $N_G(M) = G$. Deduce that if $M$ is a maximal subgroup of $G$ that is not normal in $G$ then the number of nonidentity elements of $G$ that are contained in conjugates of $M$ is at most $(|M| - 1)|G : M|$.

Let $G$ be a group.

**24.** Assume $H$ is a proper subgroup of the finite group $G$. Prove $G \neq \cup_{g \in G} gHg^{-1}$, i.e., $G$ is not the union of the conjugates of any proper subgroup. [Put $H$ in some maximal subgroup and use the preceding exercise.]

Let $G$ be a group.

**25.** Let $G = GL_2(\mathbb{C})$ and let $H = \{\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{C},\ ac \neq 0\}$. Prove that every element of $G$ is conjugate to some element of the subgroup $H$ and deduce that $G$ is the union of conjugates of $H$. [Show that every element of $GL_2(\mathbb{C})$ has an eigenvector.]

Let $G$ be a group.

**26.** Let $G$ be a transitive permutation group on the finite set $A$ with $|A| > 1$. Show that there is some $\sigma \in G$ such that $\sigma(a) \neq a$ for all $a \in A$ (such an element $\sigma$ is called *fixed point free*).

Let $G$ be a group.

**27.** Let $g_1, g_2, \ldots, g_r$ be representatives of the conjugacy classes of the finite group $G$ and assume these elements pairwise commute. Prove that $G$ is abelian.

Let $G$ be a group.

**28.** Let $p$ and $q$ be primes with $p < q$. Prove that a non-abelian group $G$ of order $pq$ has a nonnormal subgroup of index $q$, so that there exists an injective homomorphism into $S_q$. Deduce that $G$ is isomorphic to a subgroup of the normalizer in $S_q$ of the cyclic group generated by the $q$-cycle $(1\,2\ldots q)$.

Let $G$ be a group.

**29.** Let $p$ be a prime and let $G$ be a group of order $p^\alpha$. Prove that $G$ has a subgroup of order $p^\beta$, for every $\beta$ with $0 \leq \beta \leq \alpha$. [Use Theorem 8 and induction on $\alpha$.]

Let $G$ be a group.

**30.** If $G$ is a group of odd order, prove for any nonidentity element $x \in G$ that $x$ and $x^{-1}$ are not conjugate in $G$.

Let $G$ be a group.

**31.** Using the usual generators and relations for the dihedral group $D_{2n}$ (cf. Section 1.2) show that for $n = 2k$ an even integer the conjugacy classes in $D_{2n}$ are the following: $\{1\}$, $\{r^k\}$, $\{r^{\pm 1}\}$, $\{r^{\pm 2}\}$, $\ldots$, $\{r^{\pm(k-1)}\}$, $\{sr^{2b} \mid b = 1, \ldots, k\}$ and $\{sr^{2b-1} \mid b = 1, \ldots, k\}$. Give the class equation for $D_{2n}$.

Let $G$ be a group.

**32.** For $n = 2k + 1$ an odd integer show that the conjugacy classes in $D_{2n}$ are $\{1\}$, $\{r^{\pm 1}\}$, $\{r^{\pm 2}\}$, $\ldots$, $\{r^{\pm k}\}$, $\{sr^b \mid b = 1, \ldots, n\}$. Give the class equation for $D_{2n}$.

Let $G$ be a group.

**33.** This exercise gives a formula for the size of each conjugacy class in $S_n$. Let $\sigma$ be a permutation in $S_n$ and let $m_1, m_2, \ldots, m_s$ be the *distinct* integers which appear in the cycle type of $\sigma$ (including 1-cycles). For each $i \in \{1, 2, \ldots, s\}$ assume $\sigma$ has $k_i$ cycles of length $m_i$ (so that $\Sigma_{i=1}^{s} k_i m_i = n$). Prove that the number of conjugates of $\sigma$ is

$$\frac{n!}{(k_1! m_1^{k_1})(k_2! m_2^{k_2}) \ldots (k_s! m_s^{k_s})}.$$

[See Exercises 6 and 7 in Section 1.3 where this formula was given in some special cases.]

Let $G$ be a group.

**34.** Prove that if $p$ is a prime and $P$ is a subgroup of $S_p$ of order $p$, then $|N_{S_p}(P)| = p(p-1)$. [Argue that every conjugate of $P$ contains exactly $p-1$ $p$-cycles and use the formula for the number of $p$-cycles to compute the index of $N_{S_p}(P)$ in $S_p$.]

Let $G$ be a group.

**35.** Let $p$ be a prime. Find a formula for the number of conjugacy classes of elements of order $p$ in $S_n$ (using the greatest integer function).

Let $G$ be a group.

**36.** Let $\pi : G \to S_G$ be the left regular representation afforded by the action of $G$ on itself by left multiplication. For each $g \in G$ denote the permutation $\pi(g)$ by $\sigma_g$, so that $\sigma_g(x) = gx$ for all $x \in G$. Let $\lambda : G \to S_G$ be the permutation representation afforded by the corresponding right action of $G$ on itself, and for each $h \in G$ denote the permutation $\lambda(h)$ by $\tau_h$. Thus $\tau_h(x) = xh^{-1}$ for all $x \in G$ ($\lambda$ is called the *right regular representation* of $G$).

**(a)** Prove that $\sigma_g$ and $\tau_h$ commute for all $g, h \in G$. (Thus the centralizer in $S_G$ of $\pi(G)$ contains the subgroup $\lambda(G)$, which is isomorphic to $G$).

**(b)** Prove that $\sigma_g = \tau_g$ if and only if $g$ is an element of order 1 or 2 in the center of $G$.

**(c)** Prove that $\sigma_g = \tau_h$ if and only if $g$ and $h$ lie in the center of $G$. Deduce that $\pi(G) \cap \lambda(G) = \pi(Z(G)) = \lambda(Z(G))$.

## 4.4. Automorphisms

Let $G$ be a group.

**1.** If $\sigma \in \mathrm{Aut}(G)$ and $\varphi_g$ is conjugation by $g$ prove $\sigma\varphi_g\sigma^{-1} = \varphi_{\sigma(g)}$. Deduce that $\mathrm{Inn}(G) \trianglelefteq \mathrm{Aut}(G)$. (The group $\mathrm{Aut}(G)/\mathrm{Inn}(G)$ is called the *outer automorphism group* of $G$.)

Let $G$ be a group.

**2.** Prove that if $G$ is an abelian group of order $pq$, where $p$ and $q$ are distinct primes, then $G$ is cyclic. [Use Cauchy's Theorem to produce elements of order $p$ and $q$ and consider the order of their product.]

Let $G$ be a group.

**3.** Prove that under any automorphism of $D_8$, $r$ has at most 2 possible images and $s$ has at most 4 possible images ($r$ and $s$ are the usual generators — cf. Section 1.2). Deduce that $|\mathrm{Aut}(D_8)| \leq 8$.

Let $G$ be a group.

**4.** Use arguments similar to those in the preceding exercise to show $|\mathrm{Aut}(Q_8)| \leq 24$.

Let $G$ be a group.

**5.** Use the fact that $D_8 \trianglelefteq D_{16}$ to prove that $\mathrm{Aut}(D_8) \cong D_8$.

Let $G$ be a group.

**6.** Prove that characteristic subgroups are normal. Give an example of a normal subgroup that is not characteristic.

Let $G$ be a group.

**7.** If $H$ is the unique subgroup of a given order in a group $G$ prove $H$ is characteristic in $G$.

Let $G$ be a group.

**8.** Let $G$ be a group with subgroups $H$ and $K$ with $H \leq K$.

**(a)** Prove that if $H$ is characteristic in $K$ and $K$ is normal in $G$ then $H$ is normal in $G$.

**(b)** Prove that if $H$ is characteristic in $K$ and $K$ is characteristic in $G$ then $H$ is characteristic in $G$. Use this to prove that the Klein 4-group $V_4$ is characteristic in $S_4$.

**(c)** Give an example to show that if $H$ is normal in $K$ and $K$ is characteristic in $G$ then $H$ need not be normal in $G$.

Let $G$ be a group.

**9.** If $r$, $s$ are the usual generators for the dihedral group $D_{2n}$, use the preceding two exercises to deduce that every subgroup of $\langle r \rangle$ is normal in $D_{2n}$.

Let $G$ be a group.

**10.** Let $G$ be a group, let $A$ be an abelian normal subgroup of $G$, and write $\overline{G} = G/A$. Show that $\overline{G}$ acts (on the left) by conjugation on $A$ by $\overline{g} \cdot a = gag^{-1}$, where $g$ is *any* representative of the coset $\overline{g}$ (in particular, show that this action is well defined). Give an explicit example to show that this action is not well defined if $A$ is non-abelian.

Let $G$ be a group.

**11.** If $p$ is a prime and $P$ is a subgroup of $S_p$ of order $p$, prove $N_{S_p}(P)/C_{S_p}(P) \cong \mathrm{Aut}(P)$. [Use Exercise 34, Section 3.]

Let $G$ be a group.

**12.** Let $G$ be a group of order 3825. Prove that if $H$ is a normal subgroup of order 17 in $G$ then $H \leq Z(G)$.

Let $G$ be a group.

**13.** Let $G$ be a group of order 203. Prove that if $H$ is a normal subgroup of order 7 in $G$ then $H \leq Z(G)$. Deduce that $G$ is abelian in this case.

Let $G$ be a group.

**14.** Let $G$ be a group of order 1575. Prove that if $H$ is a normal subgroup of order 9 in $G$ then $H \leq Z(G)$.

Let $G$ be a group.

**15.** Prove that each of the following (multiplicative) groups is cyclic: $(\mathbb{Z}/5\mathbb{Z})^\times$, $(\mathbb{Z}/9\mathbb{Z})^\times$ and $(\mathbb{Z}/18\mathbb{Z})^\times$.

Let $G$ be a group.

**16.** Prove that $(\mathbb{Z}/24\mathbb{Z})^\times$ is an elementary abelian group of order 8. (We shall see later that $(\mathbb{Z}/n\mathbb{Z})^\times$ is an elementary abelian group if and only if $n \mid 24$.)

Let $G$ be a group.

**17.** Let $G = \langle x \rangle$ be a cyclic group of order $n$. For $n = 2, 3, 4, 5, 6$ write out the elements of Aut($G$) explicitly (by Proposition 16 above we know Aut($G$) $\cong (\mathbb{Z}/n\mathbb{Z})^\times$, so for each element $a \in (\mathbb{Z}/n\mathbb{Z})^\times$, write out explicitly what the automorphism $\psi_a$ does to the elements $\{1, x, x^2, \ldots, x^{n-1}\}$ of $G$).

Let $G$ be a group.

**18.** This exercise shows that for $n \neq 6$ every automorphism of $S_n$ is inner. Fix an integer $n \geq 2$ with $n \neq 6$.

**(a)** Prove that the automorphism group of a group $G$ permutes the conjugacy classes of $G$, i.e., for each $\sigma \in \mathrm{Aut}(G)$ and each conjugacy class $\mathcal{K}$ of $G$ the set $\sigma(\mathcal{K})$ is also a conjugacy class of $G$.

**(b)** Let $\mathcal{K}$ be the conjugacy class of transpositions in $S_n$ and let $\mathcal{K}'$ be the conjugacy class of any element of order 2 in $S_n$ that is not a transposition. Prove that $|\mathcal{K}| \neq |\mathcal{K}'|$. Deduce that any automorphism of $S_n$ sends transpositions to transpositions. [See Exercise 33 in Section 3.]

**(c)** Prove that for each $\sigma \in \mathrm{Aut}(S_n)$

$$\sigma : (1\ 2) \mapsto (a\ b_2), \qquad \sigma : (1\ 3) \mapsto (a\ b_3), \quad \ldots, \quad \sigma : (1\ n) \mapsto (a\ b_n)$$

for some distinct integers $a, b_2, b_3, \ldots, b_n \in \{1, 2, \ldots, n\}$.

**(d)** Show that $(1\ 2)$, $(1\ 3)$, $\ldots$, $(1\ n)$ generate $S_n$ and deduce that any automorphism of $S_n$ is uniquely determined by its action on these elements. Use (c) to show that $S_n$ has at most $n!$ automorphisms and conclude that $\mathrm{Aut}(S_n) = \mathrm{Inn}(S_n)$ for $n \neq 6$.

Let $G$ be a group.

**19.** This exercise shows that $|\mathrm{Aut}(S_6) : \mathrm{Inn}(S_6)| \leq 2$ (Exercise 10 in Section 6.3 shows that equality holds by exhibiting an automorphism of $S_6$ that is not inner).

**(a)** Let $\mathcal{K}$ be the conjugacy class of transpositions in $S_6$ and let $\mathcal{K}'$ be the conjugacy class of any element of order 2 in $S_6$ that is not a transposition. Prove that $|\mathcal{K}| \neq |\mathcal{K}'|$ unless $\mathcal{K}'$ is the conjugacy class of products of three disjoint transpositions. Deduce that $\mathrm{Aut}(S_6)$ has a subgroup of index at most 2 which sends transpositions to transpositions.

**(b)** Prove that $|\mathrm{Aut}(S_6) : \mathrm{Inn}(S_6)| \leq 2$. [Follow the same steps as in (c) and (d) of the preceding exercise to show that any automorphism that sends transpositions to transpositions is inner.]

Let $G$ be a group.

The next exercise introduces a subgroup, $J(P)$, which (like the center of $P$) is defined for an arbitrary finite group $P$ (although in most applications $P$ is a group whose order is a power of a prime). This subgroup was defined by J. Thompson in 1964 and it now plays a pivotal role in the study of finite groups, in particular, in the classification of finite simple groups.

**20.** For any finite group $P$ let $d(P)$ be the minimum number of generators of $P$ (so, for example, $d(P) = 1$ if and only if $P$ is a nontrivial cyclic group and $d(Q_8) = 2$). Let $m(P)$ be the maximum of the integers $d(A)$ as $A$ runs over all *abelian* subgroups of $P$ (so, for example, $m(Q_8) = 1$ and $m(D_8) = 2$). Define

$$J(P) = \langle\, A \mid A \text{ is an abelian subgroup of } P \text{ with } d(A) = m(P) \,\rangle.$$

($J(P)$ is called the *Thompson subgroup* of $P$.)

**(a)** Prove that $J(P)$ is a characteristic subgroup of $P$.

**(b)** For each of the following groups $P$ list all abelian subgroups $A$ of $P$ that satisfy $d(A) = m(P)$: $Q_8$, $D_8$, $D_{16}$ and $QD_{16}$ (where $QD_{16}$ is the quasidihedral group of order 16 defined in Exercise 11 of Section 2.5). [Use the lattices of subgroups for these groups in Section 2.5.]

**(c)** Show that $J(Q_8) = Q_8$, $J(D_8) = D_8$, $J(D_{16}) = D_{16}$ and $J(QD_{16})$ is a dihedral subgroup of order 8 in $QD_{16}$.

**(d)** Prove that if $Q \leq P$ and $J(P)$ is a subgroup of $Q$, then $J(P) = J(Q)$. Deduce that if $P$ is a subgroup (not necessarily normal) of the finite group $G$ and $J(P)$ is contained in some subgroup $Q$ of $P$ such that $Q \trianglelefteq G$, then $J(P) \trianglelefteq G$.

## 4.5. The Sylow Theorems

Let $G$ be a finite group and let $p$ be a prime.

**1.** Prove that if $P \in Syl_p(G)$ and $H$ is a subgroup of $G$ containing $P$ then $P \in Syl_p(H)$. Give an example to show that, in general, a Sylow $p$-subgroup of a subgroup of $G$ need not be a Sylow $p$-subgroup of $G$.

Let $G$ be a finite group and let $p$ be a prime.

**2.** Prove that if $H$ is a subgroup of $G$ and $Q \in Syl_p(H)$ then $gQg^{-1} \in Syl_p(gHg^{-1})$ for all $g \in G$.

Let $G$ be a finite group and let $p$ be a prime.

**3.** Use Sylow's Theorem to prove Cauchy's Theorem. (Note that we only used Cauchy's Theorem for abelian groups — Proposition 3.21 — in the proof of Sylow's Theorem so this line of reasoning is not circular.)

Let $G$ be a finite group and let $p$ be a prime.

**4.** Exhibit all Sylow 2-subgroups and Sylow 3-subgroups of $D_{12}$ and $S_3 \times S_3$.

Let $G$ be a finite group and let $p$ be a prime.

**5.** Show that a Sylow $p$-subgroup of $D_{2n}$ is cyclic and normal for every odd prime $p$.

Let $G$ be a finite group and let $p$ be a prime.

**6.** Exhibit all Sylow 3-subgroups of $A_4$ and all Sylow 3-subgroups of $S_4$.

Let $G$ be a finite group and let $p$ be a prime.

**7.** Exhibit all Sylow 2-subgroups of $S_4$ and find elements of $S_4$ which conjugate one of these into each of the others.

Let $G$ be a finite group and let $p$ be a prime.

**8.** Exhibit two distinct Sylow 2-subgroups of $S_5$ and an element of $S_5$ that conjugates one into the other.

Let $G$ be a finite group and let $p$ be a prime.

**9.** Exhibit all Sylow 3-subgroups of $SL_2(\mathbb{F}_3)$ (cf. Exercise 9, Section 2.1).

Let $G$ be a finite group and let $p$ be a prime.

**10.** Prove that the subgroup of $SL_2(\mathbb{F}_3)$ generated by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is the unique Sylow 2-subgroup of $SL_2(\mathbb{F}_3)$ (cf. Exercise 10, Section 2.4).

Let $G$ be a finite group and let $p$ be a prime.

**11.** Show that the center of $SL_2(\mathbb{F}_3)$ is the group of order 2 consisting of $\pm I$, where $I$ is the identity matrix. Prove that $SL_2(\mathbb{F}_3)/Z(SL_2(\mathbb{F}_3)) \cong A_4$. [Use facts about groups of order 12.]

Let $G$ be a finite group and let $p$ be a prime.

**12.** Let $2n = 2^a k$ where $k$ is odd. Prove that the number of Sylow 2-subgroups of $D_{2n}$ is $k$. [Prove that if $P \in Syl_2(D_{2n})$ then $N_{D_{2n}}(P) = P$.]

Let $G$ be a finite group and let $p$ be a prime.

**13.** Prove that a group of order 56 has a normal Sylow $p$-subgroup for some prime $p$ dividing its order.

Let $G$ be a finite group and let $p$ be a prime.

**14.** Prove that a group of order 312 has a normal Sylow $p$-subgroup for some prime $p$ dividing its order.

Let $G$ be a finite group and let $p$ be a prime.

**15.** Prove that a group of order 351 has a normal Sylow $p$-subgroup for some prime $p$ dividing its order.

Let $G$ be a finite group and let $p$ be a prime.

**16.** Let $|G| = pqr$, where $p, q$ and $r$ are primes with $p < q < r$. Prove that $G$ has a normal Sylow subgroup for either $p$, $q$ or $r$.

Let $G$ be a finite group and let $p$ be a prime.

**17.** Prove that if $|G| = 105$ then $G$ has a normal Sylow 5-subgroup and a normal Sylow 7-subgroup.

Let $G$ be a finite group and let $p$ be a prime.

**18.** Prove that a group of order 200 has a normal Sylow 5-subgroup.

Let $G$ be a finite group and let $p$ be a prime.

**19.** Prove that if $|G| = 6545$ then $G$ is not simple.

Let $G$ be a finite group and let $p$ be a prime.

**20.** Prove that if $|G| = 1365$ then $G$ is not simple.

Let $G$ be a finite group and let $p$ be a prime.

**21.** Prove that if $|G| = 2907$ then $G$ is not simple.

Let $G$ be a finite group and let $p$ be a prime.

**22.** Prove that if $|G| = 132$ then $G$ is not simple.

Let $G$ be a finite group and let $p$ be a prime.

**23.** Prove that if $|G| = 462$ then $G$ is not simple.

Let $G$ be a finite group and let $p$ be a prime.

**24.** Prove that if $G$ is a group of order 231 then $Z(G)$ contains a Sylow 11-subgroup of $G$ and a Sylow 7-subgroup is normal in $G$.

Let $G$ be a finite group and let $p$ be a prime.

**25.** Prove that if $G$ is a group of order 385 then $Z(G)$ contains a Sylow 7-subgroup of $G$ and a Sylow 11-subgroup is normal in $G$.

Let $G$ be a finite group and let $p$ be a prime.

**26.** Let $G$ be a group of order 105. Prove that if a Sylow 3-subgroup of $G$ is normal then $G$ is abelian.

Let $G$ be a finite group and let $p$ be a prime.

**27.** Let $G$ be a group of order 315 which has a normal Sylow 3-subgroup. Prove that $Z(G)$ contains a Sylow 3-subgroup of $G$ and deduce that $G$ is abelian.

Let $G$ be a finite group and let $p$ be a prime.

**28.** Let $G$ be a group of order 1575. Prove that if a Sylow 3-subgroup of $G$ is normal then a Sylow 5-subgroup and a Sylow 7-subgroup are normal. In this situation prove that $G$ is abelian.

Let $G$ be a finite group and let $p$ be a prime.

**29.** If $G$ is a non-abelian simple group of order $< 100$, prove that $G \cong A_5$. [Eliminate all orders but 60.]

Let $G$ be a finite group and let $p$ be a prime.

**30.** How many elements of order 7 must there be in a simple group of order 168?

Let $G$ be a finite group and let $p$ be a prime.

**31.** For $p = 2$, 3 and 5 find $n_p(A_5)$ and $n_p(S_5)$. [Note that $A_4 \leq A_5$.]

Let $G$ be a finite group and let $p$ be a prime.

**32.** Let $P$ be a Sylow $p$-subgroup of $H$ and let $H$ be a subgroup of $K$. If $P \trianglelefteq H$ and $H \trianglelefteq K$, prove that $P$ is normal in $K$. Deduce that if $P \in Syl_p(G)$ and $H = N_G(P)$, then $N_G(H) = H$ (in words: *normalizers of Sylow p-subgroups are self-normalizing*).

Let $G$ be a finite group and let $p$ be a prime.

**33.** Let $P$ be a normal Sylow $p$-subgroup of $G$ and let $H$ be any subgroup of $G$. Prove that $P \cap H$ is the unique Sylow $p$-subgroup of $H$.

Let $G$ be a finite group and let $p$ be a prime.

**34.** Let $P \in Syl_p(G)$ and assume $N \trianglelefteq G$. Use the conjugacy part of Sylow's Theorem to prove that $P \cap N$ is a Sylow $p$-subgroup of $N$. Deduce that $PN/N$ is a Sylow $p$-subgroup of $G/N$ (note that this may also be done by the Second Isomorphism Theorem — cf. Exercise 9, Section 3.3).

Let $G$ be a finite group and let $p$ be a prime.

**35.** Let $P \in Syl_p(G)$ and let $H \leq G$. Prove that $gPg^{-1} \cap H$ is a Sylow $p$-subgroup of $H$ for some $g \in G$. Give an explicit example showing that $hPh^{-1} \cap H$ is not necessarily a Sylow $p$-subgroup of $H$ for any $h \in H$ (in particular, we cannot always take $g = 1$ in the first part of this problem, as we could when $H$ was normal in $G$).

Let $G$ be a finite group and let $p$ be a prime.

**36.** Prove that if $N$ is a normal subgroup of $G$ then $n_p(G/N) \leq n_p(G)$.

Let $G$ be a finite group and let $p$ be a prime.

**37.** Let $R$ be a normal $p$-subgroup of $G$ (not necessarily a Sylow subgroup).

**(a)** Prove that $R$ is contained in every Sylow $p$-subgroup of $G$.

**(b)** If $S$ is another normal $p$-subgroup of $G$, prove that $RS$ is also a normal $p$-subgroup of $G$.

**(c)** The subgroup $O_p(G)$ is defined to be the group generated by all normal $p$-subgroups of $G$. Prove that $O_p(G)$ is the unique largest normal $p$-subgroup of $G$ and $O_p(G)$ equals the intersection of all Sylow $p$-subgroups of $G$.

**(d)** Let $\overline{G} = G/O_p(G)$. Prove that $O_p(\overline{G}) = \overline{1}$ (i.e., $\overline{G}$ has no nontrivial normal $p$-subgroup).

Let $G$ be a finite group and let $p$ be a prime.

**38.** Use the method of proof in Sylow's Theorem to show that if $n_p$ is not congruent to $1 \pmod{p^2}$ then there are distinct Sylow $p$-subgroups $P$ and $Q$ of $G$ such that $|P : P \cap Q| = |Q : P \cap Q| = p$.

Let $G$ be a finite group and let $p$ be a prime.

**39.** Show that the subgroup of strictly upper triangular matrices in $GL_n(\mathbb{F}_p)$ (cf. Exercise 17, Section 2.1) is a Sylow $p$-subgroup of this finite group. [Use the order formula in Section 1.4 to find the order of a Sylow $p$-subgroup of $GL_n(\mathbb{F}_p)$.]

Let $G$ be a finite group and let $p$ be a prime.

**40.** Prove that the number of Sylow $p$-subgroups of $GL_2(\mathbb{F}_p)$ is $p + 1$. [Exhibit two distinct Sylow $p$-subgroups.]

Let $G$ be a finite group and let $p$ be a prime.

**41.** Prove that $SL_2(\mathbb{F}_4) \cong A_5$ (cf. Exercise 9, Section 2.1 for the definition of $SL_2(\mathbb{F}_4)$).

Let $G$ be a finite group and let $p$ be a prime.

**42.** Prove that the group of rigid motions in $\mathbb{R}^3$ of an icosahedron is isomorphic to $A_5$. [Recall that the order of this group is 60: Exercise 13, Section 1.2.]

Let $G$ be a finite group and let $p$ be a prime.

**43.** Prove that the group of rigid motions in $\mathbb{R}^3$ of a dodecahedron is isomorphic to $A_5$. (As with the cube and the tetrahedron, the icosahedron and the dodecahedron are dual solids.) [Recall that the order of this group is 60: Exercise 12, Section 1.2.]

Let $G$ be a finite group and let $p$ be a prime.

**44.** Let $p$ be the smallest prime dividing the order of the finite group $G$. If $P \in Syl_p(G)$ and $P$ is cyclic prove that $N_G(P) = C_G(P)$.

Let $G$ be a finite group and let $p$ be a prime.

**45.** Find generators for a Sylow $p$-subgroup of $S_{2p}$, where $p$ is an odd prime. Show that this is an abelian group of order $p^2$.

Let $G$ be a finite group and let $p$ be a prime.

**46.** Find generators for a Sylow $p$-subgroup of $S_{p^2}$, where $p$ is a prime. Show that this is a non-abelian group of order $p^{p+1}$.

Let $G$ be a finite group and let $p$ be a prime.

**47.** Write and execute a computer program which

**(i)** gives each odd number $n < 10,000$ that is not a power of a prime and that has some prime divisor $p$ such that $n_p$ is not forced to be 1 for all groups of order $n$ by the congruence condition of Sylow's Theorem, and

**(ii)** gives for each $n$ in (i) the factorization of $n$ into prime powers and gives the list of all permissible values of $n_p$ for all primes $p$ dividing $n$ (i.e., those values not ruled out by Part 3 of Sylow's Theorem).

Let $G$ be a finite group and let $p$ be a prime.

**48.** Carry out the same process as in the preceding exercise for all even numbers less than 1000. Explain the relative lengths of the lists versus the number of integers tested.

Let $G$ be a finite group and let $p$ be a prime.

**49.** Prove that if $|G| = 2^n m$ where $m$ is odd and $G$ has a cyclic Sylow 2-subgroup then $G$ has a normal subgroup of order $m$. [Use induction and Exercises 11 and 12 in Section 2.]

Let $G$ be a finite group and let $p$ be a prime.

**50.** Prove that if $U$ and $W$ are normal subsets of a Sylow $p$-subgroup $P$ of $G$ then $U$ is conjugate to $W$ in $G$ if and only if $U$ is conjugate to $W$ in $N_G(P)$. Deduce that two elements in the center of $P$ are conjugate in $G$ if and only if they are conjugate in $N_G(P)$. (A subset $U$ of $P$ is normal in $P$ if $N_P(U) = P$.)

Let $G$ be a finite group and let $p$ be a prime.

**51.** Let $P$ be a Sylow $p$-subgroup of $G$ and let $M$ be any subgroup of $G$ which contains $N_G(P)$. Prove that $|G : M| \equiv 1 (\text{mod } p)$.

Let $G$ be a finite group and let $p$ be a prime.

The following sequence of exercises leads to the classification of all numbers $n$ with the property that every group of order $n$ is cyclic (for example, $n = 15$ is such an integer). These arguments are a vastly simplified prototype for the proof that every group of odd order is solvable in the sense that they use the *structure* (commutativity) of the proper subgroups and their *embedding* in the whole group (we shall see that distinct maximal subgroups intersect in the identity) to obtain a contradiction by counting arguments. In the proof that groups of odd order are solvable one uses induction to reduce to the situation in which a minimal counterexample is a simple group — but here every proper subgroup is solvable (not abelian as in our situation). The analysis of the structure and embedding of the maximal subgroups in this situation is much more complicated and the counting arguments are (roughly speaking) replaced by character theory arguments (as will be discussed in Part VI).

**52.** Suppose $G$ is a finite simple group in which every proper subgroup is abelian. If $M$ and $N$ are distinct maximal subgroups of $G$ prove $M \cap N = 1$. [See Exercise 23 in Section 3.]

Let $G$ be a finite group and let $p$ be a prime.

The following sequence of exercises leads to the classification of all numbers $n$ with the property that every group of order $n$ is cyclic (for example, $n = 15$ is such an integer). These arguments are a vastly simplified prototype for the proof that every group of odd order is solvable in the sense that they use the *structure* (commutativity) of the proper subgroups and their *embedding* in the whole group (we shall see that distinct maximal subgroups intersect in the identity) to obtain a contradiction by counting arguments. In the proof that groups of odd order are solvable one uses induction to reduce to the situation in which a minimal counterexample is a simple group — but here every proper subgroup is solvable (not abelian as in our situation). The analysis of the structure and embedding of the maximal subgroups in this situation is much more complicated and the counting arguments are (roughly speaking) replaced by character theory arguments (as will be discussed in Part VI).

**53.** Use the preceding exercise to prove that if $G$ is any non-abelian group in which every proper subgroup is abelian then $G$ is not simple. [Let $G$ be a counterexample to this assertion and use Exercise 24 in Section 3 to show that $G$ has more than one conjugacy class of maximal subgroups. Use the method of Exercise 23 in Section 3 to count the elements which lie in all conjugates of $M$ and $N$, where $M$ and $N$ are nonconjugate maximal subgroups of $G$; show that this gives more than $|G|$ elements.]

Let $G$ be a finite group and let $p$ be a prime.

The following sequence of exercises leads to the classification of all numbers $n$ with the property that every group of order $n$ is cyclic (for example, $n = 15$ is such an integer). These arguments are a vastly simplified prototype for the proof that every group of odd order is solvable in the sense that they use the *structure* (commutativity) of the proper subgroups and their *embedding* in the whole group (we shall see that distinct maximal subgroups intersect in the identity) to obtain a contradiction by counting arguments. In the proof that groups of odd order are solvable one uses induction to reduce to the situation in which a minimal counterexample is a simple group — but here every proper subgroup is solvable (not abelian as in our situation). The analysis of the structure and embedding of the maximal subgroups in this situation is much more complicated and the counting arguments are (roughly speaking) replaced by character theory arguments (as will be discussed in Part VI).

**54.** Prove the following classification: if $G$ is a finite group of order $p_1p_2 \dots p_r$ where the $p_i$'s are distinct primes such that $p_i$ does not divide $p_j - 1$ for all $i$ and $j$, then $G$ is cyclic. [By induction, every proper subgroup of $G$ is cyclic, so $G$ is not simple by the preceding exercise. If $N$ is a nontrivial proper normal subgroup, $N$ is cyclic and $G/N$ acts as automorphisms of $N$. Use Proposition 16 to show that $N \leq Z(G)$ and use induction to show $G/Z(G)$ is cyclic, hence $G$ is abelian by Exercise 36 of Section 3.1.]

Let $G$ be a finite group and let $p$ be a prime.

The following sequence of exercises leads to the classification of all numbers $n$ with the property that every group of order $n$ is cyclic (for example, $n = 15$ is such an integer). These arguments are a vastly simplified prototype for the proof that every group of odd order is solvable in the sense that they use the *structure* (commutativity) of the proper subgroups and their *embedding* in the whole group (we shall see that distinct maximal subgroups intersect in the identity) to obtain a contradiction by counting arguments. In the proof that groups of odd order are solvable one uses induction to reduce to the situation in which a minimal counterexample is a simple group — but here every proper subgroup is solvable (not abelian as in our situation). The analysis of the structure and embedding of the maximal subgroups in this situation is much more complicated and the counting arguments are (roughly speaking) replaced by character theory arguments (as will be discussed in Part VI).

**55.** Prove the converse to the preceding exercise: if $n \geq 2$ is an integer such that every group of order $n$ is cyclic, then $n = p_1 p_2 \dots p_r$ is a product of distinct primes and $p_i$ does not divide $p_j - 1$ for all $i$, $j$. [If $n$ is not of this form, construct noncyclic groups of order $n$ using direct products of noncyclic groups of order $p^2$ and $pq$, where $p \mid q - 1$.]

Let $G$ be a finite group and let $p$ be a prime.

The following sequence of exercises leads to the classification of all numbers $n$ with the property that every group of order $n$ is cyclic (for example, $n = 15$ is such an integer). These arguments are a vastly simplified prototype for the proof that every group of odd order is solvable in the sense that they use the *structure* (commutativity) of the proper subgroups and their *embedding* in the whole group (we shall see that distinct maximal subgroups intersect in the identity) to obtain a contradiction by counting arguments. In the proof that groups of odd order are solvable one uses induction to reduce to the situation in which a minimal counterexample is a simple group — but here every proper subgroup is solvable (not abelian as in our situation). The analysis of the structure and embedding of the maximal subgroups in this situation is much more complicated and the counting arguments are (roughly speaking) replaced by character theory arguments (as will be discussed in Part VI).

**56.** If $G$ is a finite group in which every proper subgroup is abelian, show that $G$ is solvable.

## 4.6. The Simplicity of $A_n$

Let $G$ be a group and let $\Omega$ be an infinite set.

**1.** Prove that $A_n$ does not have a proper subgroup of index $< n$ for all $n \geq 5$.

Let $G$ be a group and let $\Omega$ be an infinite set.

**2.** Find all normal subgroups of $S_n$ for all $n \geq 5$.

Let $G$ be a group and let $\Omega$ be an infinite set.

**3.** Prove that $A_n$ is the only proper subgroup of index $< n$ in $S_n$ for all $n \geq 5$.

Let $G$ be a group and let $\Omega$ be an infinite set.

**4.** Prove that $A_n$ is generated by the set of all 3-cycles for each $n \geq 3$.

Let $G$ be a group and let $\Omega$ be an infinite set.

**5.** Prove that if there exists a chain of subgroups $G_1 \leq G_2 \leq \ldots \leq G$ such that $G = \cup_{i=1}^{\infty} G_i$ and each $G_i$ is simple then $G$ is simple.

Let $G$ be a group and let $\Omega$ be an infinite set.

**6.** Let $D$ be the subgroup of $S_\Omega$ consisting of permutations which move only a finite number of elements of $\Omega$ (described in Exercise 17 in Section 3) and let $A$ be the set of all elements $\sigma \in D$ such that $\sigma$ acts as an even permutation on the (finite) set of points it moves. Prove that $A$ is an infinite simple group. [Show that every pair of elements of $D$ lie in a finite simple subgroup of $D$.]

Let $G$ be a group and let $\Omega$ be an infinite set.

**7.** Under the notation of the preceding, exercise prove that if $H \trianglelefteq S_\Omega$ and $H \neq 1$ then $A \leq H$, i.e., $A$ is the unique (nontrivial) minimal normal subgroup of $S_\Omega$.

Let $G$ be a group and let $\Omega$ be an infinite set.

**8.** Under the notation of the preceding two exercises prove that $|D| = |A| = |\Omega|$. Deduce that

$$\text{if } S_\Omega \cong S_\Delta \text{ then } |\Omega| = |\Delta|.$$

[Use the fact that $D$ is generated by transpositions. You may assume that countable unions and finite direct products of sets of cardinality $|\Omega|$ also have cardinality $|\Omega|$.]

# 5. Direct and Semidirect Products and Abelian Groups

## 5.1. Direct Products

**1.** Show that the center of a direct product is the direct product of the centers:

$$Z(G_1 \times G_2 \times \cdots \times G_n) = Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n).$$

Deduce that a direct product of groups is abelian if and only if each of the factors is abelian.

**2.** Let $G_1, G_2, \ldots, G_n$ be groups and let $G = G_1 \times \cdots \times G_n$. Let $I$ be a proper, nonempty subset of $\{1, \ldots, n\}$ and let $J = \{1, \ldots, n\} - I$. Define $G_I$ to be the set of elements of $G$ that have the identity of $G_j$ in position $j$ for all $j \in J$.

**(a)** Prove that $G_I$ is isomorphic to the direct product of the groups $G_i$, $i \in I$.

**(b)** Prove that $G_I$ is a normal subgroup of $G$ and $G/G_I \cong G_J$.

**(c)** Prove that $G \cong G_I \times G_J$.

**3.** Under the notation of the preceding exercise let $I$ and $K$ be any disjoint nonempty subsets of $\{1, 2, \ldots, n\}$ and let $G_I$ and $G_K$ be the subgroups of $G$ defined above. Prove that $xy = yx$ for all $x \in G_I$ and all $y \in G_K$.

**4.** Let $A$ and $B$ be finite groups and let $p$ be a prime. Prove that any Sylow $p$-subgroup of $A \times B$ is of the form $P \times Q$, where $P \in Syl_p(A)$ and $Q \in Syl_p(B)$. Prove that $n_p(A \times B) = n_p(A)n_p(B)$. Generalize both of these results to a direct product of any finite number of finite groups (so that the number of Sylow $p$-subgroups of a direct product is the product of the numbers of Sylow $p$-subgroups of the factors).

**5.** Exhibit a nonnormal subgroup of $Q_8 \times Z_4$ (note that every subgroup of each factor is normal).

**6.** Show that all subgroups of $Q_8 \times E_{2^n}$ are normal.

**7.** Let $G_1, G_2, \ldots, G_n$ be groups and let $\pi$ be a fixed element of $S_n$. Prove that the map

$$\varphi_\pi : G_1 \times G_2 \times \cdots \times G_n \to G_{\pi^{-1}(1)} \times G_{\pi^{-1}(2)} \times \cdots \times G_{\pi^{-1}(n)}$$

defined by

$$\varphi_\pi(g_1, g_2, \ldots, g_n) = (g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \ldots, g_{\pi^{-1}(n)})$$

is an isomorphism (so that changing the order of the factors in a direct product does not change the isomorphism type).

**8.** Let $G_1 = G_2 = \cdots = G_n$ and let $G = G_1 \times \cdots \times G_n$. Under the notation of the preceding exercise show that $\varphi_\pi \in \mathrm{Aut}(G)$. Show also that the map $\pi \mapsto \varphi_\pi$ is an injective homomorphism of $S_n$ into $\mathrm{Aut}(G)$. (In particular, $\varphi_{\pi_1} \circ \varphi_{\pi_2} = \varphi_{\pi_1\pi_2}$. It is at this point that the $\pi^{-1}$'s in the definition of $\varphi_\pi$ are needed. The underlying reason for this is because if $e_i$ is the $n$-tuple with 1 in position $i$ and zeros elsewhere, $1 \leq i \leq n$, then $S_n$ acts on $\{e_1, \ldots, e_n\}$ by $\pi \cdot e_i = e_{\pi(i)}$; this is a left group action. If the $n$-tuple $(g_1, \ldots, g_n)$ is represented by $g_1e_1 + \cdots + g_ne_n$, then this left group action on $\{e_1, \ldots, e_n\}$ extends to a left group action on sums by

$$\pi \cdot (g_1e_1 + g_2e_2 + \cdots + g_ne_n) = g_1e_{\pi(1)} + g_2e_{\pi(2)} + \cdots + g_ne_{\pi(n)}.$$

The coefficient of $e_{\pi(i)}$ on the right hand side is $g_i$, so the coefficient of $e_i$ is $g_{\pi^{-1}(i)}$. Thus the right hand side may be rewritten as $g_{\pi^{-1}(1)}e_1 + g_{\pi^{-1}(2)}e_2 + \cdots + g_{\pi^{-1}(n)}e_n$, which is precisely the sum attached to the $n$-tuple $(g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \ldots, g_{\pi^{-1}(n)})$. In other words, any permutation of the "position vectors" $e_1, \ldots, e_n$ (which fixes their coefficients) is the same as the inverse permutation on the coefficients (fixing the $e_i$'s). If one uses $\pi$'s in place of $\pi^{-1}$'s in the definition of $\varphi_\pi$ then the map $\pi \mapsto \varphi_\pi$ is not necessarily a homomorphism — it corresponds to a *right* group action.)

**9.** Let $G_i$ be a field $F$ for all $i$ and use the preceding exercise to show that the set of $n \times n$ matrices with one 1 in each row and each column is a subgroup of $GL_n(F)$ isomorphic to $S_n$ (these matrices are called *permutation matrices* since they simply permute the standard basis $e_1, \ldots, e_n$ (as above) of the $n$-dimensional vector space $F \times F \times \cdots \times F$).

**10.** Let $p$ be a prime. Let $A$ and $B$ be two cyclic groups of order $p$ with generators $x$ and $y$, respectively. Set $E = A \times B$ so that $E$ is the elementary abelian group of order $p^2$: $E_{p^2}$. Prove that the distinct subgroups of $E$ of order $p$ are

$$\langle x \rangle, \quad \langle xy \rangle, \quad \langle xy^2 \rangle, \quad \ldots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

(note that there are $p + 1$ of them).

**11.** Let $p$ be a prime and let $n \in \mathbb{Z}^+$. Find a formula for the number of subgroups of order $p$ in the elementary abelian group $E_{p^n}$.

**12.** Let $A$ and $B$ be groups. Assume $Z(A)$ contains a subgroup $Z_1$ and $Z(B)$ contains a subgroup $Z_2$ with $Z_1 \cong Z_2$. Let this isomorphism be given by the map $x_i \mapsto y_i$ for all $x_i \in Z_1$. A *central product* of $A$ and $B$ is a quotient

$$(A \times B)/Z \quad \text{where} \quad Z = \{(x_i, y_i^{-1}) \mid x_i \in Z_1\}$$

and is denoted by $A * B$ — it is not unique since it depends on $Z_1$, $Z_2$ and the isomorphism between them. (Think of $A * B$ as the direct product of $A$ and $B$ "collapsed" by identifying each element $x_i \in Z_1$ with its corresponding element $y_i \in Z_2$.)

**(a)** Prove that the images of $A$ and $B$ in the quotient group $A * B$ are isomorphic to $A$ and $B$, respectively, and that these images intersect in a central subgroup isomorphic to $Z_1$. Find $|A * B|$.

**(b)** Let $Z_4 = \langle x \rangle$. Let $D_8 = \langle r, s \rangle$ and $Q_8 = \langle i, j \rangle$ be given by their usual generators and relations. Let $Z_4 * D_8$ be the central product of $Z_4$ and $D_8$ which identifies $x^2$ and $r^2$ (i.e., $Z_1 = \langle x^2 \rangle$, $Z_2 = \langle r^2 \rangle$ and the isomorphism is $x^2 \mapsto r^2$) and let $Z_4 * Q_8$ be the central product of $Z_4$ and $Q_8$ which identifies $x^2$ and $-1$. Prove that $Z_4 * D_8 \cong Z_4 * Q_8$.

**13.** Give presentations for the groups $Z_4 * D_8$ and $Z_4 * Q_8$ constructed in the preceding exercise.

**14.** Let $G = A_1 \times A_2 \times \cdots \times A_n$ and for each $i$ let $B_i$ be a normal subgroup of $A_i$. Prove that $B_1 \times B_2 \times \cdots \times B_n \trianglelefteq G$ and that

$$(A_1 \times A_2 \times \cdots \times A_n)/(B_1 \times B_2 \times \cdots \times B_n) \cong (A_1/B_1) \times (A_2/B_2) \times \cdots \times (A_n/B_n).$$

The following exercise describes the direct product of an arbitrary collection of groups. The terminology for the Cartesian product of an arbitrary collection of sets may be found in the Appendix.

**15.** Let $I$ be any nonempty index set and let $(G_i, \star_i)$ be a group for each $i \in I$. The *direct product* of the groups $G_i, i \in I$ is the set $G = \prod_{i \in I} G_i$ (the Cartesian product of the $G_i$'s) with a binary operation defined as follows: if $\prod a_i$ and $\prod b_i$ are elements of $G$, then their product in $G$ is given by

$$\left(\prod_{i \in I} a_i\right)\left(\prod_{i \in I} b_i\right) = \prod_{i \in I} (a_i \star_i b_i)$$

(i.e., the group operation in the direct product is defined componentwise).

**(a)** Show that this binary operation is well defined and associative.

**(b)** Show that the element $\prod 1_i$ satisfies the axiom for the identity of $G$, where $1_i$ is the identity of $G_i$ for all $i$.

**(c)** Show that the element $\prod a_i^{-1}$ is the inverse of $\prod a_i$, where the inverse of each component element $a_i$ is taken in the group $G_i$.

Conclude that the direct product is a group.
(Note that if $I = \{1, 2, \ldots, n\}$, this definition of the direct product is the same as the $n$-tuple definition in the text.)

**16.** State and prove the generalization of Proposition 2 to arbitrary direct products.

**17.** Let $I$ be any nonempty index set and let $G_i$ be a group for each $i \in I$. The *restricted direct product* or *direct sum* of the groups $G_i$ is the set of elements of the direct product which are the identity in all but finitely many components, that is, the set of all elements $\prod a_i \in \prod_{i \in I} G_i$ such that $a_i = 1_i$ for all but a finite number of $i \in I$.

**(a)** Prove that the restricted direct product is a subgroup of the direct product.

**(b)** Prove that the restricted direct product is normal in the direct product.

**(c)** Let $I = \mathbb{Z}^+$ and let $p_i$ be the $i^{\text{th}}$ integer prime. Show that if $G_i = \mathbb{Z}/p_i\mathbb{Z}$ for all $i \in \mathbb{Z}^+$, then every element of the restricted direct product of the $G_i$'s has finite order but $\prod_{i \in \mathbb{Z}^+} G_i$ has elements of infinite order. Show that in this example the restricted direct product is the torsion subgroup of the direct product (cf. Exercise 6, Section 2.1).

**18.** In each of (a) to (e) give an example of a group with the specified properties:

**(a)** an infinite group in which every element has order 1 or 2

**(b)** an infinite group in which every element has finite order but for each positive integer $n$ there is an element of order $n$

**(c)** a group with an element of infinite order and an element of order 2

**(d)** a group $G$ such that every finite group is isomorphic to some subgroup of $G$

**(e)** a nontrivial group $G$ such that $G \cong G \times G$.

## 5.2. The Fundamental Theorem of Finitely Generated Abelian Groups

**1.** In each of parts (a) to (e) give the number of nonisomorphic abelian groups of the specified order — do not list the groups: **(a)** order 100, **(b)** order 576, **(c)** order 1155, **(d)** order 42875, **(e)** order 2704.

**2.** In each of parts (a) to (e) give the lists of invariant factors for all abelian groups of the specified order:
**(a)** order 270, **(b)** order 9801, **(c)** order 320, **(d)** order 105, **(e)** order 44100.

**3.** In each of parts (a) to (e) give the lists of elementary divisors for all abelian groups of the specified order and then match each list with the corresponding list of invariant factors found in the preceding exercise:
**(a)** order 270, **(b)** order 9801, **(c)** order 320, **(d)** order 105, **(e)** order 44100.

**4.** In each of parts (a) to (d) determine which pairs of abelian groups listed are isomorphic (here the expression $\{a_1, a_2, \ldots, a_k\}$ denotes the abelian group $Z_{a_1} \times Z_{a_2} \times \cdots \times Z_{a_k}$).

**(a)** $\{4, 9\}$, $\{6, 6\}$, $\{8, 3\}$, $\{9, 4\}$, $\{6, 4\}$, $\{64\}$.

**(b)** $\{2^2, 2 \cdot 3^2\}$, $\{2^2 \cdot 3, 2 \cdot 3\}$, $\{2^3 \cdot 3^2\}$, $\{2^2 \cdot 3^2, 2\}$.

**(c)** $\{5^2 \cdot 7^2, 3^2 \cdot 5 \cdot 7\}$, $\{3^2 \cdot 5^2 \cdot 7, 5 \cdot 7^2\}$, $\{3 \cdot 5^2, 7^2, 3 \cdot 5 \cdot 7\}$, $\{5^2 \cdot 7, 3^2 \cdot 5, 7^2\}$.

**(d)** $\{2^2 \cdot 5 \cdot 7, 2^3 \cdot 5^3, 2 \cdot 5^2\}$, $\{2^3 \cdot 5^3 \cdot 7, 2^3 \cdot 5^3\}$, $\{2^2, 2 \cdot 7, 2^3, 5^3, 5^3\}$, $\{2 \cdot 5^3, 2^2 \cdot 5^3, 2^3, 7\}$.

**5.** Let $G$ be a finite abelian group of type $(n_1, n_2, \ldots, n_t)$. Prove that $G$ contains an element of order $m$ if and only if $m \mid n_1$. Deduce that $G$ is of exponent $n_1$.

**6.** Prove that any finite group has a finite exponent. Give an example of an infinite group with finite exponent. Does a finite group of exponent $m$ always contain an element of order $m$?

**7.** Let $p$ be a prime and let $A = \langle x_1 \rangle \times \langle x_2 \rangle \times \cdots \times \langle x_n \rangle$ be an abelian $p$-group, where $|x_i| = p^{\alpha_i} > 1$ for all $i$. Define the $p^{\text{th}}$-*power map*

$$\varphi : A \to A \qquad \text{by} \qquad \varphi : x \mapsto x^p.$$

**(a)** Prove that $\varphi$ is a homomorphism.
**(b)** Describe the image and kernel of $\varphi$ in terms of the given generators.
**(c)** Prove both $\ker \varphi$ and $A/\text{im}\, \varphi$ have rank $n$ (i.e., have the same rank as $A$) and prove these groups are both isomorphic to the elementary abelian group, $E_{p^n}$, of order $p^n$.

**8.** Let $A$ be a finite abelian group (written multiplicatively) and let $p$ be a prime. Let

$$A^p = \{a^p \mid a \in A\} \qquad \text{and} \qquad A_p = \{x \mid x^p = 1\}$$

(so $A^p$ and $A_p$ are the image and kernel of the $p^{\text{th}}$-power map, respectively).

**(a)** Prove that $A/A^p \cong A_p$. [Show that they are both elementary abelian and they have the same order.]

**(b)** Prove that the number of subgroups of $A$ of order $p$ equals the number of subgroups of $A$ of index $p$. [Reduce to the case where $A$ is an elementary abelian $p$-group.]

**9.** Let $A = Z_{60} \times Z_{45} \times Z_{12} \times Z_{36}$. Find the number of elements of order 2 and the number of subgroups of index 2 in $A$.

**10.** Let $n$ and $k$ be positive integers and let $A$ be the free abelian group of rank $n$ (written additively). Prove that $A/kA$ is isomorphic to the direct product of $n$ copies of $\mathbb{Z}/k\mathbb{Z}$ (here $kA = \{ka \mid a \in A\}$). [See Exercise 14, Section 1.]

**11.** Let $G$ be a nontrivial finite abelian group of rank $t$.

**(a)** Prove that the rank of $G$ equals the maximum of the ranks of its Sylow subgroups.

**(b)** Prove that $G$ can be generated by $t$ elements but no subset with fewer than $t$ elements generates $G$. [One way of doing this is by using part (a) together with Exercise 7.]

**12.** Let $n$ and $m$ be positive integers with $d = (n, m)$. Let $Z_n = \langle x \rangle$ and $Z_m = \langle y \rangle$. Let $A$ be the central product of $\langle x \rangle$ and $\langle y \rangle$ with an element of order $d$ identified, which has presentation $\langle x, y \mid x^n = y^m = 1,\ xy = yx,\ x^{\frac{n}{d}} = y^{\frac{m}{d}} \rangle$. Describe $A$ as a direct product of two cyclic groups.

**13.** Let $A = \langle x_1 \rangle \times \cdots \times \langle x_r \rangle$ be a finite abelian group with $|x_i| = n_i$ for $1 \leq i \leq r$. Find a presentation for $A$. Prove that if $G$ is any group containing commuting elements $g_1, \ldots, g_r$ such that $g_i^{n_i} = 1$ for $1 \leq i \leq r$, then there is a unique homomorphism from $A$ to $G$ which sends $x_i$ to $g_i$ for all $i$.

**14.** For any group $G$ define the *dual group* of $G$ (denoted $\widehat{G}$) to be the set of all homomorphisms from $G$ into the multiplicative group of roots of unity in $\mathbb{C}$. Define a group operation in $\widehat{G}$ by pointwise multiplication of functions: if $\chi, \psi$ are homomorphisms from $G$ into the group of roots of unity then $\chi\psi$ is the homomorphism given by $(\chi\psi)(g) = \chi(g)\psi(g)$ for all $g \in G$, where the latter multiplication takes place in $\mathbb{C}$.

**(a)** Show that this operation on $\widehat{G}$ makes $\widehat{G}$ into an abelian group. [Show that the identity is the map $g \mapsto 1$ for all $g \in G$ and the inverse of $\chi \in \widehat{G}$ is the map $g \mapsto \chi(g)^{-1}$.]

**(b)** If $G$ is a finite abelian group, prove that $\widehat{G} \cong G$. [Write $G$ as $\langle x_1 \rangle \times \cdots \times \langle x_r \rangle$ and if $n_i = |x_i|$ define $\chi_i$ to be the homomorphism which sends $x_i$ to $e^{2\pi i/n_i}$ and sends $x_j$ to 1, for all $j \neq i$. Prove $\chi_i$ has order $n_i$ in $\widehat{G}$ and $\widehat{G} = \langle \chi_1 \rangle \times \cdots \times \langle \chi_r \rangle$.]

(This result is often phrased: a finite abelian group is self-dual. It implies that the lattice diagram of a finite abelian group is the same when it is turned upside down. Note however that there is no *natural* isomorphism between $G$ and its dual (the isomorphism depends on a choice of a set of generators for $G$). This is frequently stated in the form: a finite abelian group is *noncanonically* isomorphic to its dual.)

**15.** Let $G = \langle x \rangle \times \langle y \rangle$ where $|x| = 8$ and $|y| = 4$.

**(a)** Find all pairs $a, b$ in $G$ such that $G = \langle a \rangle \times \langle b \rangle$ (where $a$ and $b$ are expressed in terms of $x$ and $y$).

**(b)** Let $H = \langle x^2y,\ y^2 \rangle \cong Z_4 \times Z_2$. Prove that there are no elements $a, b$ of $G$ such that $G = \langle a \rangle \times \langle b \rangle$ and $H = \langle a^2 \rangle \times \langle b^2 \rangle$ (i.e., one cannot pick direct product generators for $G$ in such a way that some powers of these are direct product generators for $H$).

**16.** Prove that no finitely generated abelian group is divisible (cf. Exercise 19, Section 2.4).

## 5.4. Recognizing Direct Products

Let $G$ be a group.

**1.** Prove that if $x, y \in G$ then $[y, x] = [x, y]^{-1}$. Deduce that for any subsets $A$ and $B$ of $G$, $[A, B] = [B, A]$ (recall that $[A, B]$ is the *subgroup* of $G$ generated by the commutators $[a, b]$).

Let $G$ be a group.

**2.** Prove that a subgroup $H$ of $G$ is normal if and only if $[G, H] \leq H$.

Let $G$ be a group.

**3.** Let $a, b, c \in G$. Prove that

**(a)** $[a, bc] = [a, c](c^{-1}[a, b]c)$

**(b)** $[ab, c] = (b^{-1}[a, c]b)[b, c]$.

Let $G$ be a group.

**4.** Find the commutator subgroups of $S_4$ and $A_4$.

Let $G$ be a group.

**5.** Prove that $A_n$ is the commutator subgroup of $S_n$ for all $n \geq 5$.

Let $G$ be a group.

**6.** Exhibit a representative of each cycle type of $A_5$ as a commutator in $S_5$.

Let $G$ be a group.

**7.** Prove that if $p$ is a prime and $P$ is a non-abelian group of order $p^3$ then $P' = Z(P)$.

Let $G$ be a group.

**8.** Assume $x, y \in G$ and both $x$ and $y$ commute with $[x, y]$. Prove that for all $n \in \mathbb{Z}^+$, $(xy)^n = x^n y^n [y, x]^{\frac{n(n-1)}{2}}$.

Let $G$ be a group.

**9.** Prove that if $p$ is an odd prime and $P$ is a group of order $p^3$ then the $p^{\text{th}}$ power map $x \mapsto x^p$ is a homomorphism of $P$ into $Z(P)$. If $P$ is not cyclic, show that the kernel of the $p^{\text{th}}$ power map has order $p^2$ or $p^3$. Is the squaring map a homomorphism in non-abelian groups of order 8? Where is the oddness of $p$ needed in the above proof? [Use Exercise 8.]

Let $G$ be a group.

**10.** Prove that a finite abelian group is the direct product of its Sylow subgroups.

Let $G$ be a group.

**11.** Prove that if $G = HK$ where $H$ and $K$ are characteristic subgroups of $G$ with $H \cap K = 1$ then $\mathrm{Aut}(G) \cong \mathrm{Aut}(H) \times \mathrm{Aut}(K)$. Deduce that if $G$ is an abelian group of finite order then $\mathrm{Aut}(G)$ is isomorphic to the direct product of the automorphism groups of its Sylow subgroups.

Let $G$ be a group.

**12.** Use Theorem 4.17 to describe the automorphism group of a finite cyclic group.

Let $G$ be a group.

**13.** Prove that $D_{8n}$ is not isomorphic to $D_{4n} \times Z_2$.

Let $G$ be a group.

**14.** Let $G = \{(a_{ij}) \in GL_n(F) \mid a_{ij} = 0 \text{ if } i > j, \text{ and } a_{11} = a_{22} = \cdots = a_{nn}\}$, where $F$ is a field, be the group of upper triangular matrices all of whose diagonal entries are equal. Prove that $G \cong D \times U$, where $D$ is the group of all nonzero multiples of the identity matrix and $U$ is the group of upper triangular matrices with 1's down the diagonal.

Let $G$ be a group.

**15.** If $A$ and $B$ are normal subgroups of $G$ such that $G/A$ and $G/B$ are both abelian, prove that $G/(A \cap B)$ is abelian.

Let $G$ be a group.

**16.** Prove that if $K$ is a normal subgroup of $G$ then $K' \trianglelefteq G$.

Let $G$ be a group.

**17.** If $K$ is a normal subgroup of $G$ and $K$ is cyclic, prove that $G' \leq C_G(K)$. [Recall that the automorphism group of a cyclic group is abelian.]

Let $G$ be a group.

**18.** Let $K_1, K_2, \ldots, K_n$ be non-abelian simple groups and let $G = K_1 \times K_2 \times \cdots \times K_n$. Prove that every normal subgroup of $G$ is of the form $G_I$ for some subset $I$ of $\{1, 2, \ldots, n\}$ (where $G_I$ is defined in Exercise 2 of Section 1). [If $N \trianglelefteq G$ and $x = (a_1, \ldots, a_n) \in N$ with some $a_i \neq 1$, then show that there is some $g_i \in G_i$ not commuting with $a_i$. Show $[(1, \ldots, g_i, \ldots, 1), x] \in K_i \cap N$ and deduce $K_i \leq N$.]

Let $G$ be a group.

**19.** A group $H$ is called *perfect* if $H' = H$ (i.e., $H$ equals its own commutator subgroup).

**(a)** Prove that every non-abelian simple group is perfect.

**(b)** Prove that if $H$ and $K$ are perfect subgroups of a group $G$ then $\langle H, K \rangle$ is also perfect. Extend this to show that the subgroup of $G$ generated by any collection of perfect subgroups is perfect.

**(c)** Prove that any conjugate of a perfect subgroup is perfect.

**(d)** Prove that any group $G$ has a unique maximal perfect subgroup and that this subgroup is normal.

Let $G$ be a group.

**20.** Let $H(F)$ be the Heisenberg group over the field $F$, cf. Exercise 11 of Section 1.4. Find an explicit formula for the commutator $[X, Y]$, where $X, Y \in H(F)$, and show that the commutator subgroup of $H(F)$ equals the center of $H(F)$ (cf. Section 2.2, Exercise 14).

## 5.5. Semidirect Products

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**1.** Prove that $C_K(H) = \ker \varphi$ (recall that $C_K(H) = C_G(H) \cap K$).

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**2.** Prove that $C_H(K) = N_H(K)$.

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**3.** In Example 1 following the proof of Proposition 11 prove that every element of $G - H$ has order 2. Prove that $G$ is abelian if and only if $h^2 = 1$ for all $h \in H$.

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**4.** Let $p = 2$ and check that the construction of the two non-abelian groups of order $p^3$ is valid in this case. Prove that *both* resulting groups are isomorphic to $D_8$.

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**5.** Let $G = \mathrm{Hol}(Z_2 \times Z_2)$.

**(a)** Prove that $G = H \rtimes K$ where $H = Z_2 \times Z_2$ and $K \cong S_3$. Deduce that $|G| = 24$.

**(b)** Prove that $G$ is isomorphic to $S_4$. [Obtain a homomorphism from $G$ into $S_4$ by letting $G$ act on the left cosets of $K$. Use Exercise 1 to show this representation is faithful.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**6.** Assume that $K$ is a cyclic group, $H$ is an arbitrary group and $\varphi_1$ and $\varphi_2$ are homomorphisms from $K$ into $\mathrm{Aut}(H)$ such that $\varphi_1(K)$ and $\varphi_2(K)$ are conjugate subgroups of $\mathrm{Aut}(H)$. If $K$ is infinite assume $\varphi_1$ and $\varphi_2$ are injective. Prove by constructing an explicit isomorphism that $H \rtimes_{\varphi_1} K \cong H \rtimes_{\varphi_2} K$ (in particular, if the subgroups $\varphi_1(K)$ and $\varphi_2(K)$ are equal in $\mathrm{Aut}(H)$, then the resulting semidirect products are isomorphic). [Suppose $\sigma\varphi_1(K)\sigma^{-1} = \varphi_2(K)$ so that for some $a \in \mathbb{Z}$ we have $\sigma\varphi_1(k)\sigma^{-1} = \varphi_2(k)^a$ for all $k \in K$. Show that the map

$\psi : H \rtimes_{\varphi_1} K \to H \rtimes_{\varphi_2} K$ defined by $\psi((h, k)) = (\sigma(h), k^a)$ is a homomorphism. Show $\psi$ is bijective by constructing a 2-sided inverse.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**7.** This exercise describes thirteen isomorphism types of groups of order 56. (It is not too difficult to show that every group of order 56 is isomorphic to one of these.)

**(a)** Prove that there are three abelian groups of order 56.

**(b)** Prove that every group of order 56 has either a normal Sylow 2-subgroup or a normal Sylow 7-subgroup.

**(c)** Construct the following non-abelian groups of order 56 which have a normal Sylow 7-subgroup and whose Sylow 2-subgroup $S$ is as specified:

one group when $S \cong Z_2 \times Z_2 \times Z_2$
two nonisomorphic groups when $S \cong Z_4 \times Z_2$
one group when $S \cong Z_8$
two nonisomorphic groups when $S \cong Q_8$
three nonisomorphic groups when $S \cong D_8$.

[For a particular $S$, two groups are not isomorphic if the kernels of the maps from $S$ into $\mathrm{Aut}(Z_7)$ are not isomorphic.]

**(d)** Let $G$ be a group of order 56 with a nonnormal Sylow 7-subgroup. Prove that if $S$ is the Sylow 2-subgroup of $G$ then $S \cong Z_2 \times Z_2 \times Z_2$. [Let an element of order 7 act by conjugation on the seven nonidentity elements of $S$ and deduce that they all have the same order.]

**(e)** Prove that there is a unique group of order 56 with a nonnormal Sylow 7-subgroup. [For existence use the fact that $|GL_3(\mathbb{F}_2)| = 168$; for uniqueness use Exercise 6.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**8.** Construct a non-abelian group of order 75. Classify all groups of order 75 (there are three of them). [Use Exercise 6 to show that the non-abelian group is unique.] (The classification of groups of order $pq^2$, where $p$ and $q$ are primes with $p < q$ and $p$ not dividing $q - 1$, is quite similar.)

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**9.** Show that the matrix $\begin{pmatrix} 0 & -1 \\ 1 & 4 \end{pmatrix}$ is an element of order 5 in $GL_2(\mathbb{F}_{19})$. Use this matrix to construct a non-abelian group of order 1805 and give a presentation of this group. Classify groups of order 1805 (there are three isomorphism types). [Use Exercise 6 to prove uniqueness of the non-abelian group.] (A general method for finding elements of prime order in $GL_n(\mathbb{F}_p)$ is described in the exercises in Section 12.2; this particular matrix of order 5 in $GL_2(\mathbb{F}_{19})$ appears in Exercise 16 of that section as an illustration of the method.)

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**10.** This exercise classifies the groups of order 147 (there are six isomorphism types).

**(a)** Prove that there are two abelian groups of order 147.

**(b)** Prove that every group of order 147 has a normal Sylow 7-subgroup.

**(c)** Prove that there is a unique non-abelian group whose Sylow 7-subgroup is cyclic.

**(d)** Let $t_1 = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ and $t_2 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ be elements of $GL_2(\mathbb{F}_7)$. Prove $P = \langle t_1, t_2 \rangle$ is a Sylow 3-subgroup of $GL_2(\mathbb{F}_7)$ and that $P \cong Z_3 \times Z_3$. Deduce that every subgroup of $GL_2(\mathbb{F}_7)$ of order 3 is conjugate in $GL_2(\mathbb{F}_7)$ to a subgroup of $P$.

**(e)** By Example 3 in Section 1 the group $P$ has four subgroups of order 3 and these are: $P_1 = \langle t_1 \rangle$, $P_2 = \langle t_2 \rangle$, $P_3 = \langle t_1 t_2 \rangle$, and $P_4 = \langle t_1 t_2^2 \rangle$. For $i = 1, 2, 3, 4$ let $G_i = (Z_7 \times Z_7) \rtimes_{\varphi_i} Z_3$, where $\varphi_i$ is an isomorphism of $Z_3$ with the subgroup $P_i$ of Aut($Z_7 \times Z_7$). For each $i$ describe $G_i$ in terms of generators and relations. Deduce that $G_1 \cong G_2$.

**(f)** Prove that $G_1$ is not isomorphic to either $G_3$ or $G_4$. [Show that the center of $G_1$ has order 7 whereas the centers of $G_3$ and $G_4$ are trivial.]

**(g)** Prove that $G_3$ is not isomorphic to $G_4$. [Show that every subgroup of order 7 in $G_3$ is normal in $G_3$ but that $G_4$ has subgroups of order 7 that are not normal.]

**(h)** Classify the groups of order 147 by showing that the six nonisomorphic groups described above (two from part (a), one from part (c) and $G_1$, $G_3$, and $G_4$) are all the groups of order 147. [Use Exercise 6 and part (d).] (The classification of groups of order $pq^2$, where $p$ and $q$ are primes with $p < q$ and $p \mid q - 1$, is quite similar.)

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**11.** Classify groups of order 28 (there are four isomorphism types).

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**12.** Classify the groups of order 20 (there are five isomorphism types).

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**13.** Classify groups of order $4p$, where $p$ is a prime greater than 3. [There are four isomorphism types when $p \equiv 3 \pmod 4$ and five isomorphism types when $p \equiv 1 \pmod 4$.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**14.** This exercise classifies the groups of order 60 (there are thirteen isomorphism types). Let $G$ be a group of order 60, let $P$ be a Sylow 5-subgroup of $G$ and let $Q$ be a Sylow 3-subgroup of $G$.

**(a)** Prove that if $P$ is not normal in $G$ then $G \cong A_5$. [See Section 4.5.]

**(b)** Prove that if $P \trianglelefteq G$ but $Q$ is not normal in $G$ then $G \cong A_4 \times Z_5$. [Show in this case that $P \leq Z(G)$, $G/P \cong A_4$, a Sylow 2-subgroup $T$ of $G$ is normal and $TQ \cong A_4$.]

**(c)** Prove that if both $P$ and $Q$ are normal in $G$ then $G \cong Z_{15} \rtimes T$ where $T \cong Z_4$ or $Z_2 \times Z_2$. Show in this case that there are six isomorphism types when $T$ is cyclic (one abelian) and there are five isomorphism types when $T$ is the Klein 4-group (one abelian). [Use the same ideas as in the classifications of groups of orders 30 and 20.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**15.** Let $p$ be an odd prime. Prove that every element of order 2 in $GL_2(\mathbb{F}_p)$ is conjugate to a diagonal matrix with $\pm 1$'s on the diagonal. Classify the groups of order $2p^2$. [If $A$ is a $2 \times 2$ matrix with $A^2 = I$ and $v_1, v_2$ is a basis for the underlying vector space, look at $A$ acting on the vectors $w_1 = v_1 + v_2$ and $w_2 = v_1 - v_2$.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**16.** Show that there are exactly 4 distinct homomorphisms from $Z_2$ into Aut($Z_8$). Prove that the resulting semidirect products are the groups: $Z_8 \times Z_2$, $D_{16}$, the quasidihedral group $QD_{16}$ and the modular group $M$ (cf. the exercises in Section 2.5).

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**17.** Show that for any $n \geq 3$ there are exactly 4 distinct homomorphisms from $Z_2$ into $\mathrm{Aut}(Z_{2^n})$. Prove that the resulting semidirect products give 4 nonisomorphic groups of order $2^{n+1}$. [Recall Exercises 21 to 23 in Section 2.3.] (These four groups together with the cyclic group and the generalized quaternion group, $Q_{2^{n+1}}$, are all the groups of order $2^{n+1}$ which possess a cyclic subgroup of index 2.)

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**18.** Show that if $H$ is any group then there is a group $G$ that contains $H$ as a normal subgroup with the property that for every automorphism $\sigma$ of $H$ there is an element $g \in G$ such that conjugation by $g$ when restricted to $H$ is the given automorphism $\sigma$, i.e., every automorphism of $H$ is obtained as an inner automorphism of $G$ restricted to $H$.

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**19.** Let $H$ be a group of order $n$, let $K = \mathrm{Aut}(H)$ and form $G = \mathrm{Hol}(H) = H \rtimes K$ (where $\varphi$ is the identity homomorphism). Let $G$ act by left multiplication on the left cosets of $K$ in $G$ and let $\pi$ be the associated permutation representation $\pi : G \to S_n$.

**(a)** Prove the elements of $H$ are coset representatives for the left cosets of $K$ in $G$ and with this choice of coset representatives $\pi$ restricted to $H$ is the regular representation of $H$.

**(b)** Prove $\pi(G)$ is the normalizer in $S_n$ of $\pi(H)$. Deduce that under the regular representation of any finite group $H$ of order $n$, the normalizer in $S_n$ of the image of $H$ is isomorphic to $\mathrm{Hol}(H)$. [Show $|G| = |N_{S_n}(\pi(H))|$ using Exercises 1 and 2 above.]

**(c)** Deduce that the normalizer of the group generated by an $n$-cycle in $S_n$ is isomorphic to $\mathrm{Hol}(Z_n)$ and has order $n\varphi(n)$.

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**20.** Let $p$ be an odd prime. Prove that if $P$ is a non-cyclic $p$-group then $P$ contains a normal subgroup $U$ with $U \cong Z_p \times Z_p$. Deduce that for odd primes $p$ a $p$-group that contains a unique subgroup of order $p$ is cyclic. (For $p = 2$ it is a theorem that the generalized quaternion groups $Q_{2^n}$ are the only non-cyclic 2-groups which contain a unique subgroup of order 2). [Proceed by induction on $|P|$. Let $Z$ be a subgroup of order $p$ in $Z(P)$ and let $\overline{P} = P/Z$. If $\overline{P}$ is cyclic then $P$ is abelian by Exercise 36 in Section 3.1 — show the result is true for abelian groups. When $\overline{P}$ is not cyclic use induction to produce a normal subgroup $\overline{H}$ of $\overline{P}$ with $\overline{H} \cong Z_p \times Z_p$. Let $H$ be the complete preimage of $\overline{H}$ in $P$, so $|H| = p^3$. Let $H_0 = \{x \in H \mid x^p = 1\}$ so that $H_0$ is a characteristic subgroup of $H$ of order $p^2$ or $p^3$ by Exercise 9 in Section 4. Show that a suitable subgroup of $H_0$ gives the desired normal subgroup $U$.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**21.** Let $p$ be an odd prime and let $P$ be a $p$-group. Prove that if every subgroup of $P$ is normal then $P$ is abelian. (Note that $Q_8$ is a non-abelian 2-group with this property, so the result is false for $p = 2$.) [Use the preceding exercises and Exercise 15 of Section 4.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**22.** Let $F$ be a field let $n$ be a positive integer and let $G$ be the group of upper triangular matrices in $GL_n(F)$ (cf. Exercise 16, Section 2.1)

**(a)** Prove that $G$ is the semidirect product $U \rtimes D$ where $U$ is the set of upper triangular matrices with 1's down the diagonal (cf. Exercise 17, Section 2.1) and $D$ is the set of diagonal matrices in $GL_n(F)$.

**(b)** Let $n$=2. Recall that $U \cong F$ and $D \cong F^\times \times F^\times$ (cf. Exercise 11 in Section 3.1). Describe the homomorphism from $D$ into Aut($U$) explicitly in terms of these isomorphisms (i.e., show how each element of $F^\times \times F^\times$ acts as an automorphism on $F$).

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**23.** Let $K$ and $L$ be groups, let $n$ be a positive integer, let $\rho : K \to S_n$ be a homomorphism and let $H$ be the direct product of $n$ copies of $L$. In Exercise 8 of Section 1 an injective homomorphism $\psi$ from $S_n$ into Aut($H$) was constructed by letting the elements of $S_n$ permute the $n$ factors of $H$. The composition $\psi \circ \rho$ is a homomorphism from $G$ into Aut($H$). The *wreath product* of $L$ by $K$ is the semidirect product $H \rtimes K$ with respect to this homomorphism and is denoted by $L \wr K$ (this wreath product depends on the choice of permutation representation $\rho$ of $K$ — if none is given explicitly, $\rho$ is assumed to be the left regular representation of $K$).

**(a)** Assume $K$ and $L$ are finite groups and $\rho$ is the left regular representation of $K$. Find $|L \wr K|$ in terms of $|K|$ and $|L|$.

**(b)** Let $p$ be a prime, let $K = L = Z_p$ and let $\rho$ be the left regular representation of $K$. Prove that $Z_p \wr Z_p$ is a non-abelian group of order $p^{p+1}$ and is isomorphic to a Sylow $p$-subgroup of $S_{p^2}$. [The $p$ copies of $Z_p$ whose direct product makes up $H$ may be represented by $p$ disjoint $p$-cycles; these are cyclically permuted by $K$.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into $\mathrm{Aut}(H)$ and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**24.** Let $n$ be an integer $> 1$. Prove the following classification: every group of order $n$ is abelian if and only if $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$, where $p_1, \dots, p_r$ are distinct primes, $\alpha_i = 1$ or 2 for all $i \in \{1, \dots, r\}$ and $p_i$ does not divide $p_j^{\alpha_j} - 1$ for all $i$ and $j$. [See Exercise 56 in Section 4.5.]

Let $H$ and $K$ be groups, let $\varphi$ be a homomorphism from $K$ into Aut($H$) and, as usual, identify $H$ and $K$ as subgroups of $G = H \rtimes_\varphi K$.

**25.** Let $H(\mathbb{F}_p)$ be the Heisenberg group over the finite field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ (cf. Exercise 20 in Section 4). Prove that $H(\mathbb{F}_2) \cong D_8$, and that $H(\mathbb{F}_p)$ has exponent $p$ and is isomorphic to the first non-abelian group in Example 7.

# 6. Further Topics in Group Theory

## 6.1. p-groups, Nilpotent Groups, and Solvable Groups

**1.** Prove that $Z_i(G)$ is a characteristic subgroup of $G$ for all $i$.

**2.** Prove Parts 2 and 4 of Theorem 1 for $G$ a finite nilpotent group, not necessarily a $p$-group.

**3.** If $G$ is finite prove that $G$ is nilpotent if and only if it has a normal subgroup of each order dividing $|G|$, and is cyclic if and only if it has a unique subgroup of each order dividing $|G|$.

**4. Prove that a maximal subgroup of a finite nilpotent group has prime index.**

**5.** Prove Parts 2 and 4 of Theorem 1 for $G$ an infinite nilpotent group.

**6. Show that if $G/Z(G)$ is nilpotent then $G$ is nilpotent.**

**7.** Prove that subgroups and quotient groups of nilpotent groups are nilpotent (your proof should work for infinite groups). Give an explicit example of a group $G$ which possesses a normal subgroup $H$ such that both $H$ and $G/H$ are nilpotent but $G$ is not nilpotent.

**8.** Prove that if $p$ is a prime and $P$ is a non-abelian group of order $p^3$ then $|Z(P)| = p$ and $P/Z(P) \cong Z_p \times Z_p$.

**9.** Prove that a finite group $G$ is nilpotent if and only if whenever $a,\ b \in G$ with $(|a|, |b|) = 1$ then $ab = ba$. [Use Part 4 of Theorem 3.]

**10.** Prove that $D_{2n}$ is nilpotent if and only if $n$ is a power of 2. [Use Exercise 9.]

**11.** Give another proof of Proposition 5 under the additional assumption that $G$ is abelian by invoking the Fundamental Theorem of Finite Abelian Groups.

**12.** Find the upper and lower central series for $A_4$ and $S_4$.

**13.** Find the upper and lower central series for $A_n$ and $S_n$, $n \geq 5$.

**14.** Prove that $G^i$ is a characteristic subgroup of $G$ for all $i$.

**15.** Prove that $Z_i(D_{2^n}) = D_{2^n}^{n-1-i}$.

**16.** Prove that $\mathbb{Q}$ has no maximal subgroups. [Recall Exercise 21, Section 3.2.]

**17.** Prove that $G^{(i)}$ is a characteristic subgroup of $G$ for all $i$.

**18.** Show that if $G'/G''$ and $G''/G'''$ are both cyclic then $G'' = 1$. [You may assume $G''' = 1$. Then $G/G''$ acts by conjugation on the cyclic group $G''$.]

**19.** Show that there is no group whose commutator subgroup is isomorphic to $S_4$. [Use the preceding exercise.]

**20.** Let $p$ be a prime, let $P$ be a $p$-subgroup of the finite group $G$, let $N$ be a normal subgroup of $G$ whose order is relatively prime to $p$ and let $\overline{G} = G/N$. Prove the following:

**(a)** $N_{\overline{G}}(\overline{P}) = \overline{N_G(P)}$ [Use Frattini's Argument.]

**(b)** $C_{\overline{G}}(\overline{P}) = \overline{C_G(P)}$. [Use part (a).]

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**21.** Prove that $\Phi(G)$ is a characteristic subgroup of $G$.

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**22.** Prove that if $N \trianglelefteq G$ then $\Phi(N) \leq \Phi(G)$. Give an explicit example where this containment does not hold if $N$ is not normal in $G$.

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**23.** Compute $\Phi(S_3)$, $\Phi(A_4)$, $\Phi(S_4)$, $\Phi(A_5)$ and $\Phi(S_5)$.

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**24.** Say an element $x$ of $G$ is a *nongenerator* if for every proper subgroup $H$ of $G$, $\langle x, H \rangle$ is also a proper subgroup of $G$. Prove that $\Phi(G)$ is the set of nongenerators of $G$ (here $|G| > 1$).

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**25.** Let $G$ be a finite group. Prove that $\Phi(G)$ is nilpotent. [Use Frattini's Argument to prove that every Sylow subgroup of $\Phi(G)$ is normal in $G$.]

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**26.** Let $p$ be a prime, let $P$ be a finite $p$-group and let $\overline{P} = P/\Phi(P)$.

- **(a)** Prove that $\overline{P}$ is an elementary abelian $p$-group. [Show that $P' \leq \Phi(P)$ and that $x^p \in \Phi(P)$ for all $x \in P$.]
- **(b)** Prove that if $N$ is any normal subgroup of $P$ such that $P/N$ is elementary abelian then $\Phi(P) \leq N$. State this (universal) property in terms of homomorphisms and commutative diagrams.
- **(c)** Let $\overline{P}$ be elementary abelian of order $p^r$ (by (a)). Deduce from Exercise 24 that if $\overline{x_1}, \overline{x_2}, \ldots, \overline{x_r}$ are any basis for the $r$-dimensional vector space $\overline{P}$ over $\mathbb{F}_p$ and if $x_i$ is any element of the coset $\overline{x_i}$, then $P = \langle x_1, x_2, \ldots, x_r \rangle$. Show conversely that if $y_1, y_2, \ldots, y_s$ is any set of generators for $P$, then $s \geq r$ (you may assume that every minimal generating set for an $r$-dimensional vector space has $r$ elements, i.e., every basis has $r$ elements). Deduce *Burnside's Basis Theorem*: a set $y_1, \ldots, y_s$ is a minimal generating set for $P$ if and only if $\overline{y_1}, \ldots, \overline{y_s}$ is a basis of $\overline{P} = P/\Phi(P)$. Deduce that any minimal generating set for $P$ has $r$ elements.
- **(d)** Prove that if $P/\Phi(P)$ is cyclic then $P$ is cyclic. Deduce that if $P/P'$ is cyclic then so is $P$.
- **(e)** Let $\sigma$ be any automorphism of $P$ of prime order $q$ with $q \neq p$. Show that if $\sigma$ fixes the coset $x\Phi(P)$ then $\sigma$ fixes some element of this coset (note that since $\Phi(P)$ is characteristic in $P$ every automorphism of $P$ induces an automorphism of $P/\Phi(P)$). [Use the observation that $\sigma$ acts a permutation of order 1 or $q$ on the $p^a$ elements in the coset $x\Phi(P)$.]
- **(f)** Use parts (e) and (c) to deduce that every nontrivial automorphism of $P$ of order prime to $p$ induces a nontrivial automorphism on $P/\Phi(P)$. Deduce that any group of automorphisms of $P$ which has order prime to $p$ is isomorphic to a subgroup of $\mathrm{Aut}(\overline{P}) = GL_r(\mathbb{F}_p)$.

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**27.** Generalize part (d) of the preceding exercise as follows: let $p$ be a prime, let $P$ be a $p$-group and let $\overline{P} = P/\Phi(P)$ be elementary abelian of order $p^r$. Prove that $P$ has exactly $\dfrac{p^r - 1}{p - 1}$ maximal subgroups. [Since every maximal subgroup of $P$ contains $\Phi(P)$, the maximal subgroups of $P$ are, by the Lattice Isomorphism Theorem, in bijective correspondence with the maximal subgroups of the elementary abelian group $\overline{P}$. It therefore suffices to show that the number of maximal subgroups of an elementary abelian $p$-group of order $p^r$ is as stated above. One way of doing this is to use the result that an abelian group is isomorphic to its dual group (cf. Exercise 14 in Section 5.2) so the number of subgroups of *index* $p$ equals the number of subgroups of *order* $p$.]

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**28.** Prove that if $p$ is a prime and $P = Z_p \times Z_{p^2}$ then $|\Phi(P)| = p$ and $P/\Phi(P) \cong Z_p \times Z_p$. Deduce that $P$ has $p + 1$ maximal subgroups.

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**29.** Prove that if $p$ is a prime and $P$ is a non-abelian group of order $p^3$ then $\Phi(P) = Z(P)$ and $P/\Phi(P) \cong Z_p \times Z_p$. Deduce that $P$ has $p + 1$ maximal subgroups.

For any group $G$ the *Frattini subgroup* of $G$ (denoted by $\Phi(G)$) is defined to be the intersection of all the maximal subgroups of $G$ (if $G$ has no maximal subgroups, set $\Phi(G) = G$). The next few exercises deal with this important subgroup.

**30.** Let $p$ be an odd prime, let $P_1 = Z_p \times Z_{p^2}$ and let $P_2$ be the non-abelian group of order $p^3$ which has an element of order $p^2$. Prove that $P_1$ and $P_2$ have the same lattice of subgroups.

**31.** For any group $G$ a ***minimal normal subgroup*** is a normal subgroup $M$ of $G$ such that the only normal subgroups of $G$ which are contained in $M$ are 1 and $M$. Prove that every minimal normal subgroup of a finite solvable group is an elementary abelian $p$-group for some prime $p$. [If $M$ is a minimal normal subgroup of $G$, consider its characteristic subgroups: $M'$ and $\langle x^p \mid x \in M \rangle$.]

**32.** Prove that every maximal subgroup of a finite solvable group has prime power index. [Let $H$ be a maximal subgroup of $G$ and let $M$ be a minimal normal subgroup of $G$ — cf. the preceding exercise. Apply induction to $G/M$ and consider separately the two cases: $M \leq H$ and $M \not\leq H$.]

**33.** Let $\pi$ be any set of primes. A subgroup $H$ of a finite group is called a *Hall $\pi$-subgroup* of $G$ if the only primes dividing $|H|$ are in the set $\pi$ and $|H|$ is relatively prime to $|G : H|$. (Note that if $\pi = \{p\}$, Hall $\pi$-subgroups are the same as Sylow $p$-subgroups. Hall subgroups were introduced in Exercise 10 of Section 3.3). Prove the following generalization of Sylow's Theorem for solvable groups: if $G$ is a finite solvable group then for every set $\pi$ of primes, $G$ has a Hall $\pi$-subgroup and any two Hall $\pi$-subgroups (for the same set $\pi$) are conjugate in $G$. [Fix $\pi$ and proceed by induction on $|G|$, proving both existence and conjugacy at once. Let $M$ be a minimal normal subgroup of $G$, so $M$ is a $p$-group for some prime $p$. If $p \in \pi$, apply induction to $G/M$. If $p \notin \pi$, reduce to the case $|G| = p^\alpha n$, where $p^\alpha = |M|$ and $n$ is the order of a Hall $\pi$-subgroup of $G$. In this case let $N/M$ be a minimal normal subgroup of $G/M$, so $N/M$ is a $q$-group for some prime $q \neq p$. Let $Q \in Syl_q(N)$. If $Q \trianglelefteq G$ argue as before with $Q$ in place of $M$. If $Q$ is not normal in $G$, use Frattini's Argument to show $N_G(Q)$ is a Hall $\pi$-subgroup of $G$ and establish conjugacy in this case too.]

The following result shows how to produce normal $p$-subgroups of some groups on which the elements of order prime to $p$ act faithfully by conjugation. Exercise 26(f) then applies to restrict these actions and give some information about the structure of the group.

**34.** Let $p$ be a prime dividing the order of the finite solvable group $G$. Assume $G$ has no nontrivial normal subgroups of order prime to $p$. Let $P$ be the largest normal $p$-subgroup of $G$ (cf. Exercise 37, Section 4.5). Note that Exercise 31 above shows that $P \neq 1$. Prove that $C_G(P) \leq P$, i.e., $C_G(P) = Z(P)$. [Let $N = C_G(P)$ and use the preceding exercise to show $N = Z(P) \times H$ for some Hall $\pi$-subgroup $H$ of $N$ — here $\pi$ is the set of all prime divisors of $|N|$ except for $p$. Show $H \trianglelefteq G$ to obtain the desired conclusion: $H = 1$.]

**35.** Prove that if $G$ is a finite group in which every proper subgroup is nilpotent, then $G$ is solvable. [Show that a minimal counterexample is simple. Let $M$ and $N$ be distinct maximal subgroups chosen with $|M \cap N|$ as large as possible and apply Part 2 of Theorem 3 to show that $M \cap N = 1$. Now apply the methods of Exercise 53 in Section 4.5.]

**36.** Let $p$ be a prime, let $V$ be a nonzero finite dimensional vector space over the field of $p$ elements and let $\varphi$ be an element of $GL(V)$ of order a power of $p$ (i.e., $V$ is a nontrivial elementary abelian $p$-group and $\varphi$ is an automorphism of $V$ of $p$-power order). Prove that there is some nonzero element $v \in V$ such that $\varphi(v) = v$, i.e., $\varphi$ has a nonzero fixed point on $V$.

**37.** Let $V$ be a finite dimensional vector space over the field of 2 elements and let $\varphi$ be an element of $GL(V)$ of order 2. (i.e., $V$ is a nontrivial elementary abelian 2-group and $\varphi$ is an automorphism of $V$ of order 2). Prove that the map $v \mapsto v+\varphi(v)$ is a homomorphism from $V$ to itself. Show that every element in the image of this map is fixed by $\varphi$. Deduce that the subspace of elements of $V$ which are fixed by $\varphi$ has dimension $\geq \frac{1}{2}$(dimension $V$). (Note that if $G$ is the semidirect product of $V$ with $\langle \varphi \rangle$, where $V \trianglelefteq G$ and $\varphi$ acts by conjugation on $V$ by sending each $v \in V$ to $\varphi(v)$, then the fixed points of $\varphi$ on $V$ are $C_V(\varphi)$ and the above map is simply the commutator map: $v \mapsto [v, \varphi]$. In this terminology the problem is to show that $|C_V(\varphi)|^2 \geq |V|$.)

**38.** Use the preceding exercise to prove that if $P$ is a 2-group which has a cyclic center and $M$ is a subgroup of index 2 in $P$, then the center of $M$ has rank $\leq 2$. [The group $G/M$ of order 2 acts by conjugation on the $\mathbb{F}_2$ vector space: $\{z \in Z(M) \mid z^2 = 1\}$ and the fixed points of this action are in the center of $P$.]

## 6.2. Applications in Groups of Medium Order

**1.** Prove that for fixed $P \in Syl_p(G)$ if $P \cap R = 1$ for all $R \in Syl_p(G) - \{P\}$, then $P_1 \cap P_2 = 1$ whenever $P_1$ and $P_2$ are distinct Sylow $p$-subgroups of $G$. Deduce in this case that the number of nonidentity elements of $p$-power order in $G$ is $(|P| - 1)|G : N_G(P)|$.

**2.** In the group $S_3 \times S_3$ exhibit a pair of Sylow 2-subgroups that intersect in the identity and exhibit another pair that intersect in a group of order 2.

**3.** Prove that if $|G| = 380$ then $G$ is not simple. [Just count elements of odd prime order.]

**4.** Prove that there are no simple groups of order 80, 351, 3875 or 5313.

**5.** Let $G$ be a solvable group of order $pm$, where $p$ is a prime not dividing $m$, and let $P \in Syl_p(G)$. If $N_G(P) = P$, prove that $G$ has a normal subgroup of order $m$. Where was the solvability of $G$ needed in the proof? (This result is true for nonsolvable groups as well — it is a special case of *Burnside's N/C-Theorem*.)

**6.** Prove that there are no simple groups of order 2205, 4125, 5103, 6545 or 6435.

**7.** Prove that there are no simple groups of order 1755 or 5265. [Use Sylow 3-subgroups to show $G \leq S_{13}$ and look at the normalizer of a Sylow 13-subgroup.]

**8.** Prove that there are no simple groups of order 792 or 918.

**9.** Prove that there are no simple groups of order 336.

**10.** Prove that there are no simple groups of order 4095, 4389, 5313 or 6669.

**11.** Prove that there are no simple groups of order 4851 or 5145.

**12.** Prove that there are no simple groups of order 9555. [Let $Q \in Syl_{13}(G)$ and let $P \in Syl_7(N_G(Q))$. Argue that $Q \trianglelefteq N_G(P)$ — why is this a contradiction?]

**13.** Let $G$ be a group with more than one Sylow $p$-subgroup. Over all pairs of distinct Sylow $p$-subgroups let $P$ and $Q$ be chosen so that $|P \cap Q|$ is maximal. Show that $N_G(P \cap Q)$ has more than one Sylow $p$-subgroup and that any two distinct Sylow $p$-subgroups of $N_G(P \cap Q)$ intersect in the subgroup $P \cap Q$. (Thus $|N_G(P \cap Q)|$ is divisible by $p \cdot |P \cap Q|$ and by some prime other than $p$. Note that Sylow $p$-subgroups of $N_G(P \cap Q)$ need not be Sylow in $G$.)

**14.** Prove that there are no simple groups of order 144, 525, 2025 or 3159.

**15.** Classify groups of order 105.

**16.** Prove that there are no non-abelian simple groups of odd order $< 10000$.

**17.** **(a)** Prove that there is no simple group of order 420.
**(b)** Prove that there are no simple groups of even order $< 500$ except for orders 2, 60, 168 and 360.

**18.** Prove that if $G$ is a group of order 36 then $G$ has either a normal Sylow 2-subgroup or a normal Sylow 3-subgroup.

**19.** Show that a group of order 12 with no subgroup of order 6 is isomorphic to $A_4$.

**20.** Show that a group of order 24 with no element of order 6 is isomorphic to $S_4$.

**21.** Generalize Lemma 13 by proving that if $n_p \not\equiv 1 (\text{mod } p^k)$ then there are distinct Sylow $p$-subgroups $P$ and $R$ of $G$ such that $P \cap R$ is of index $\leq p^{k-1}$ in both $P$ and $R$.

**22.** Suppose over all pairs of distinct Sylow $p$-subgroups of $G$, $P$ and $R$ are chosen with $|P \cap R|$ maximal. Prove that $N_G(P \cap R)$ is not a $p$-group.

**23.** Let $A$ and $B$ be normal subsets of a Sylow $p$-subgroup $P$ of $G$. Prove that if $A$ and $B$ are conjugate in $G$ then they are conjugate in $N_G(P)$.

**24.** Let $G$ be a group of order $pqr$ where $p$, $q$ and $r$ are primes with $p < q < r$. Prove that a Sylow $r$-subgroup of $G$ is normal.

**25.** Let $G$ be a simple group of order $p^2qr$ where $p$, $q$ and $r$ are primes. Prove that $|G| = 60$.

**26.** Prove or construct a counterexample to the assertion: if $G$ is a group of order 168 with more than one Sylow 7-subgroup then $G$ is simple.

**27.** Show that if $\mathcal{F}$ is any set of points and lines satisfying properties (11) to (13) in the subsection on simple groups of order 168 then the graph of incidences for $\mathcal{F}$ is uniquely determined and is the same as Figure 1 (up to relabeling points and lines). [Take a line and any point not on this line. Depict the line as the base of an equilateral triangle and the point as the vertex of this triangle not on the base. Use the axioms to show that the incidences of the remaining points and lines are then uniquely determined as in Figure 1.]

**28.** Let $G$ be a simple group of order $3^3 \cdot 7 \cdot 13 \cdot 409$. Compute all permissible values of $n_p$ for each $p \in \{3, 7, 13, 409\}$ and reduce to the case where there is a unique possible value for each $n_p$.

**29.** Given the information on the Sylow numbers for a hypothetical simple group of order $3^3 \cdot 7 \cdot 13 \cdot 409$, prove that there is no such group. [Work with the permutation representation of degree 819.]

**30.** Suppose $G$ is a simple group of order 720. Find as many properties of $G$ as you can (Sylow numbers, isomorphism type of Sylow subgroups, conjugacy classes, etc.). Is there such a group?

## 6.3. A Word on Free Groups

**1.** Let $F_1$ and $F_2$ be free groups of finite rank. Prove that $F_1 \cong F_2$ if and only if they have the same rank. What facts do you need in order to extend your proof to infinite ranks (where the result is also true)?

**2.** Prove that if $|S| > 1$ then $F(S)$ is non-abelian.

**3.** Prove that the commutator subgroup of the free group on 2 generators is not finitely generated (in particular, subgroups of finitely generated groups need not be finitely generated).

**4.** Prove that every nonidentity element of a free group is of infinite order.

**5.** Establish a finite presentation for $A_4$ using 2 generators.

**6.** Establish a finite presentation for $S_4$ using 2 generators.

**7.** Prove that the following is a presentation for the quaternion group of order 8:

$$Q_8 = \langle a, b \mid a^2 = b^2,\ a^{-1}ba = b^{-1} \rangle.$$

**8.** Use presentations to find the orders of the automorphism groups of the groups $Z_2 \times Z_4$ and $Z_4 \times Z_4$.

**9.** Prove that $\mathrm{Aut}(Q_8) \cong S_4$.

**10.** This exercise exhibits an automorphism of $S_6$ that is not inner (hence, together with Exercise 19 in Section 4.4 it shows that $|\mathrm{Aut}(S_6) : \mathrm{Inn}(S_6)| = 2$). Let $t_1' = (1\ 2)(3\ 4)(5\ 6)$, $t_2' = (1\ 4)(2\ 5)(3\ 6)$, $t_3' = (1\ 3)(2\ 4)(5\ 6)$, $t_4' = (1\ 2)(3\ 6)(4\ 5)$, and $t_5' = (1\ 4)(2\ 3)(5\ 6)$. Show that $t_1', \ldots, t_5'$ satisfy the following relations:

$(t_i')^2 = 1$ for all $i$,

$(t_i't_j')^2 = 1$ for all $i$ and $j$ with $|i - j| \geq 2$, and

$(t_i't_{i+1}')^3 = 1$ for all $i \in \{1, 2, 3, 4\}$.

Deduce that $S_6 = \langle t_1', \ldots, t_5' \rangle$ and that the map

$$(1\ 2) \mapsto t_1', \quad (2\ 3) \mapsto t_2', \quad (3\ 4) \mapsto t_3', \quad (4\ 5) \mapsto t_4', \quad (5\ 6) \mapsto t_5'$$

extends to an automorphism of $S_6$ (which is clearly not inner since it does not send transpositions to transpositions). [Use the presentation for $S_6$ described in Example 5.]

**11.** Let $S$ be a set. The group with presentation $(S, R)$, where $R = \{[s, t] \mid s, t \in S\}$ is called the *free abelian* group on $S$ — denote it by $A(S)$. Prove that $A(S)$ has the following universal property: if $G$ is any abelian group and $\varphi : S \to G$ is any set map, then there is a unique group homomorphism $\Phi : A(S) \to G$ such that $\Phi|_S = \varphi$. Deduce that if $A$ is a free abelian group on a set of cardinality $n$ then

$$A \cong \mathbb{Z} \times \mathbb{Z} \times \cdots \times \mathbb{Z} \quad (n \text{ factors}).$$

**12.** Let $S$ be a set and let $c$ be a positive integer. Formulate the notion of a *free nilpotent group* on $S$ of nilpotence class $c$ and prove it has the appropriate universal property with respect to nilpotent groups of class $\leq c$.

**13.** Prove that there cannot be a nilpotent group $N$ generated by two elements with the property that *every* nilpotent group which is generated by two elements is a homomorphic image of $N$ (i.e., the specification of the class $c$ in the preceding problem was necessary).

# 7. Introduction to Rings

## 7.1. Basic Definitions and Examples

Let $R$ be a ring with 1.

**1.** Show that $(-1)^2 = 1$ in $R$.

Let $R$ be a ring with 1.

**2.** Prove that if $u$ is a unit in $R$ then so is $-u$.

Let $R$ be a ring with 1.

**3.** Let $R$ be a ring with identity and let $S$ be a subring of $R$ containing the identity. Prove that if $u$ is a unit in $S$ then $u$ is a unit in $R$. Show by example that the converse is false.

Let $R$ be a ring with 1.

**4.** Prove that the intersection of any nonempty collection of subrings of a ring is also a subring.

Let $R$ be a ring with 1.

**5.** Decide which of the following (a) – (f) are subrings of $\mathbb{Q}$:

**(a)** the set of all rational numbers with odd denominators (when written in lowest terms)

**(b)** the set of all rational numbers with even denominators (when written in lowest terms)

**(c)** the set of nonnegative rational numbers

**(d)** the set of squares of rational numbers

**(e)** the set of all rational numbers with odd numerators (when written in lowest terms)

**(f)** the set of all rational numbers with even numerators (when written in lowest terms).

Let $R$ be a ring with 1.

**6.** Decide which of the following are subrings of the ring of all functions from the closed interval [0,1] to $\mathbb{R}$:

**(a)** the set of all functions $f(x)$ such that $f(q) = 0$ for all $q \in \mathbb{Q} \cap [0, 1]$

**(b)** the set of all polynomial functions

**(c)** the set of all functions which have only a finite number of zeros, together with the zero function

**(d)** the set of all functions which have an infinite number of zeros

**(e)** the set of all functions $f$ such that $\lim_{x \to 1^-} f(x) = 0$

**(f)** the set of all rational linear combinations of the functions $\sin nx$ and $\cos mx$, where $m, n \in \{0, 1, 2, \ldots\}$.

Let $R$ be a ring with 1.

**7.** The *center* of a ring $R$ is $\{z \in R \mid zr = rz \text{ for all } r \in R\}$ (i.e., is the set of all elements which commute with every element of $R$). Prove that the center of a ring is a subring that contains the identity. Prove that the center of a division ring is a field.

Let $R$ be a ring with 1.

**8.** Describe the center of the real Hamilton Quaternions $\mathbb{H}$. Prove that $\{a + bi \mid a, b \in \mathbb{R}\}$ is a subring of $\mathbb{H}$ which is a field but is not contained in the center of $\mathbb{H}$.

Let $R$ be a ring with 1.

**9.** For a fixed element $a \in R$ define $C(a) = \{r \in R \mid ra = ar\}$. Prove that $C(a)$ is a subring of $R$ containing $a$. Prove that the center of $R$ is the intersection of the subrings $C(a)$ over all $a \in R$.

Let $R$ be a ring with 1.

**10.** Prove that if $D$ is a division ring then $C(a)$ is a division ring for all $a \in D$ (cf. the preceding exercise).

Let $R$ be a ring with 1.

**11.** Prove that if $R$ is an integral domain and $x^2 = 1$ for some $x \in R$ then $x = \pm 1$.

Let $R$ be a ring with 1.

**12.** Prove that any subring of a field which contains the identity is an integral domain.

Let $R$ be a ring with 1.

**13.** An element $x$ in $R$ is called *nilpotent* if $x^m = 0$ for some $m \in \mathbb{Z}^+$.

**(a)** Show that if $n = a^k b$ for some integers $a$ and $b$ then $\overline{ab}$ is a nilpotent element of $\mathbb{Z}/n\mathbb{Z}$.

**(b)** If $a \in \mathbb{Z}$ is an integer, show that the element $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is nilpotent if and only if every prime divisor of $n$ is also a divisor of $a$. In particular, determine the nilpotent elements of $\mathbb{Z}/72\mathbb{Z}$ explicitly.

**(c)** Let $R$ be the ring of functions from a nonempty set $X$ to a field $F$. Prove that $R$ contains no nonzero nilpotent elements.

Let $R$ be a ring with 1.

**14.** Let $x$ be a nilpotent element of the commutative ring $R$ (cf. the preceding exercise).

**(a)** Prove that $x$ is either zero or a zero divisor.

**(b)** Prove that $rx$ is nilpotent for all $r \in R$.

**(c)** Prove that $1 + x$ is a unit in $R$.

**(d)** Deduce that the sum of a nilpotent element and a unit is a unit.

Let $R$ be a ring with 1.

**15.** A ring $R$ is called a *Boolean ring* if $a^2 = a$ for all $a \in R$. Prove that every Boolean ring is commutative.

Let $R$ be a ring with 1.

**16.** Prove that the only Boolean ring that is an integral domain is $\mathbb{Z}/2\mathbb{Z}$.

Let $R$ be a ring with 1.

**17.** Let $R$ and $S$ be rings. Prove that the direct product $R \times S$ is a ring under componentwise addition and multiplication. Prove that $R \times S$ is commutative if and only if both $R$ and $S$ are commutative. Prove that $R \times S$ has an identity if and only if both $R$ and $S$ have identities.

Let $R$ be a ring with 1.

**18.** Prove that $\{(r, r) \mid r \in R\}$ is a subring of $R \times R$.

Let $R$ be a ring with 1.

**19.** Let $I$ be any nonempty index set and let $R_i$ be a ring for each $i \in I$. Prove that the direct product $\prod_{i \in I} R_i$ is a ring under componentwise addition and multiplication.

Let $R$ be a ring with 1.

**20.** Let $R$ be the collection of sequences $(a_1, a_2, a_3, \ldots)$ of integers $a_1, a_2, a_3, \ldots$ where all but finitely many of the $a_i$ are 0 (called the *direct sum* of infinitely many copies of $\mathbb{Z}$). Prove that $R$ is a ring under componentwise addition and multiplication which does not have an identity.

Let $R$ be a ring with 1.

**21.** Let $X$ be any nonempty set and let $\mathcal{P}(X)$ be the set of all subsets of $X$ (the *power set* of $X$). Define addition and multiplication on $\mathcal{P}(X)$ by

$$A + B = (A - B) \cup (B - A) \qquad \text{and} \qquad A \times B = A \cap B$$

i.e., addition is symmetric difference and multiplication is intersection.

**(a)** Prove that $\mathcal{P}(X)$ is a ring under these operations ($\mathcal{P}(X)$ and its subrings are often referred to as *rings of sets*).

**(b)** Prove that this ring is commutative, has an identity and is a Boolean ring.

Let $R$ be a ring with 1.

**22.** Give an example of an infinite Boolean ring.

Let $R$ be a ring with 1.

**23.** Let $D$ be a squarefree integer, and let $\mathcal{O}$ be the ring of integers in the quadratic field $\mathbb{Q}(\sqrt{D})$. For any positive integer $f$ prove that the set $\mathcal{O}_f = \mathbb{Z}[f\omega] = \{a + bf\omega \mid a, b \in \mathbb{Z}\}$ is a subring of $\mathcal{O}$ containing the identity. Prove that $[\mathcal{O} : \mathcal{O}_f] = f$ (index as additive abelian groups). Prove conversely that a subring of $\mathcal{O}$ containing the identity and having finite index $f$ in $\mathcal{O}$ (as additive abelian group) is equal to $\mathcal{O}_f$. (The ring $\mathcal{O}_f$ is called the *order of conductor* $f$ in the field $\mathbb{Q}(\sqrt{D})$. The ring of integers $\mathcal{O}$ is called the *maximal order* in $\mathbb{Q}(\sqrt{D})$.)

Let $R$ be a ring with 1.

**24.** Show for $D = 3, 5, 6$, and 7 that the group of units $\mathcal{O}^\times$ of the quadratic integer ring $\mathcal{O}$ is infinite by exhibiting an explicit unit of infinite (multiplicative) order in each ring.

Let $R$ be a ring with 1.

**25.** Let $I$ be the ring of integral Hamilton Quaternions and define

$$N : I \to \mathbb{Z} \qquad \text{by} \qquad N(a + bi + cj + dk) = a^2 + b^2 + c^2 + d^2$$

(the map $N$ is called a *norm*).

**(a)** Prove that $N(\alpha) = \alpha\overline{\alpha}$ for all $\alpha \in I$, where if $\alpha = a + bi + cj + dk$ then $\overline{\alpha} = a - bi - cj - dk$.

**(b)** Prove that $N(\alpha\beta) = N(\alpha)N(\beta)$ for all $\alpha, \beta \in I$.

**(c)** Prove that an element of $I$ is a **unit** if and only if it has norm $+1$. Show that $I^\times$ is **isomorphic** to the quaternion group of order 8. [The inverse in the ring of rational quaternions of a nonzero element $\alpha$ is $\dfrac{\overline{\alpha}}{N(\alpha)}$. ]

Let $R$ be a ring with 1.

**26.** Let $K$ be a field. A *discrete valuation* on $K$ is a function $v : K^\times \to \mathbb{Z}$ satisfying

**(i)** $v(ab) = v(a) + v(b)$ (i.e., $v$ is a homomorphism from the multiplicative group of nonzero elements of $K$ to $\mathbb{Z}$),

**(ii)** $v$ is surjective, and

**(iii)** $v(x + y) \geq \min\{v(x), v(y)\}$ for all $x, y \in K^\times$ with $x + y \neq 0$.

The set $R = \{x \in K^\times \mid v(x) \geq 0\} \cup \{0\}$ is called the *valuation ring* of $v$.

**(a)** Prove that $R$ is a subring of $K$ which contains the identity. (In general, a ring $R$ is called a *discrete valuation ring* if there is some field $K$ and some discrete valuation $v$ on $K$ such that $R$ is the valuation ring of $v$.)

**(b)** Prove that for each nonzero element $x \in K$ either $x$ or $x^{-1}$ is in $R$.

**(c)** Prove that an element $x$ is a unit of $R$ if and only if $v(x) = 0$.

Let $R$ be a ring with 1.

**27.** A specific example of a discrete valuation ring (cf. the preceding exercise) is obtained when $p$ is a prime, $K = \mathbb{Q}$ and

$$v_p : \mathbb{Q}^\times \to \mathbb{Z} \qquad \text{by} \qquad v_p(\frac{a}{b}) = \alpha \quad \text{where } \frac{a}{b} = p^\alpha \frac{c}{d}, \quad p \nmid c \text{ and } p \nmid d.$$

Prove that the corresponding valuation ring $R$ is the ring of all rational numbers whose denominators are relatively prime to $p$. Describe the units of this valuation ring.

Let $R$ be a ring with 1.

**28.** Let $R$ be a ring with $1 \neq 0$. A nonzero element $a$ is called a *left zero divisor* in $R$ if there is a nonzero element $x \in R$ such that $ax = 0$. Symmetrically, $b \neq 0$ is a *right zero divisor* if there is a nonzero $y \in R$ such that $yb = 0$ (so a zero divisor is an element which is either a left or a right zero divisor). An element $u \in R$ has a *left inverse* in $R$ if there is some $s \in R$ such that $su = 1$. Symmetrically, $v$ has a *right inverse* if $vt = 1$ for some $t \in R$.

**(a)** Prove that $u$ is a unit if and only if it has both a right and a left inverse (i.e., $u$ must have a two-sided inverse).

**(b)** Prove that if $u$ has a right inverse then $u$ is not a right zero divisor.

**(c)** Prove that if $u$ has more than one right inverse then $u$ is a left zero divisor.

**(d)** Prove that if $R$ is a finite ring then every element that has a right inverse is a unit (i.e., has a two-sided inverse).

Let $R$ be a ring with 1.

**29.** Let $A$ be any commutative ring with identity $1 \neq 0$. Let $R$ be the set of all group homomorphisms of the additive group $A$ to itself with addition defined as pointwise addition of functions and multiplication defined as function composition. Prove that these operations make $R$ into a ring with identity. Prove that the units of $R$ are the group automorphisms of $A$ (cf. Exercise 20, Section 1.6).

Let $R$ be a ring with 1.

**30.** Let $A = \mathbb{Z} \times \mathbb{Z} \times \mathbb{Z} \times \cdots$ be the direct product of copies of $\mathbb{Z}$ indexed by the positive integers (so $A$ is a ring under componentwise addition and multiplication) and let $R$ be the ring of all group homomorphisms from $A$ to itself as described in the preceding exercise. Let $\varphi$ be the element of $R$ defined by $\varphi(a_1, a_2, a_3, \ldots) = (a_2, a_3, \ldots)$. Let $\psi$ be the element of $R$ defined by $\psi(a_1, a_2, a_3, \ldots) = (0, a_1, a_2, a_3, \ldots)$.

**(a)** Prove that $\varphi\psi$ is the identity of $R$ but $\psi\varphi$ is not the identity of $R$ (i.e., $\psi$ is a *right inverse* for $\varphi$ but not a left inverse).

**(b)** Exhibit infinitely many right inverses for $\varphi$.

**(c)** Find a nonzero element $\pi$ in $R$ such that $\varphi\pi = 0$ but $\pi\varphi \neq 0$.

**(d)** Prove that there is no nonzero element $\lambda \in R$ such that $\lambda\varphi = 0$ (i.e., $\varphi$ is a left zero divisor but not a right zero divisor).

## 7.2. Examples: Polynomial Rings, Matrix Rings, and Group Rings

Let $R$ be a commutative ring with 1.

**1.** Let $p(x) = 2x^3 - 3x^2 + 4x - 5$ and let $q(x) = 7x^3 + 33x - 4$. In each of parts (a), (b) and (c) compute $p(x) + q(x)$ and $p(x)q(x)$ under the assumption that the coefficients of the two given polynomials are taken from the specified ring (where the integer coefficients are taken mod $n$ in parts (b) and (c) ):
**(a)** $R = \mathbb{Z}$, **(b)** $R = \mathbb{Z}/2\mathbb{Z}$, **(c)** $R = \mathbb{Z}/3\mathbb{Z}$.

Let $R$ be a commutative ring with 1.

**2.** Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be an element of the polynomial ring $R[x]$. Prove that $p(x)$ is a zero divisor in $R[x]$ if and only if there is a nonzero $b \in R$ such that $bp(x) = 0$. [Let $g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_0$ be a nonzero polynomial of minimal degree such that $g(x)p(x) = 0$. Show that $b_m a_n = 0$ and so $a_n g(x)$ is a polynomial of degree less than $m$ that also gives 0 when multiplied by $p(x)$. Conclude that $a_n g(x) = 0$. Apply a similar argument to show by induction on $i$ that $a_{n-i}\, g(x) = 0$ for $i = 0, 1, \ldots, n$, and show that this implies $b_m p(x) = 0$.]

Let $R$ be a commutative ring with 1.

**3.** Define the set $R[[x]]$ of *formal power series* in the indeterminate $x$ with coefficients from $R$ to be all formal infinite sums

$$\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots.$$

Define addition and multiplication of power series in the same way as for power series with real or complex coefficients i.e., extend polynomial addition and multiplication to power series as though they were "polynomials of infinite degree":

$$\sum_{n=0}^{\infty} a_n x^n \quad + \quad \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (a_n + b_n) x^n$$

$$\sum_{n=0}^{\infty} a_n x^n \quad \times \quad \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} \Big( \sum_{k=0}^{n} a_k b_{n-k} \Big) x^n.$$

(The term "formal" is used here to indicate that convergence is not considered, so that formal power series need not represent functions on $R$.)

**(a)** Prove that $R[[x]]$ is a commutative ring with 1.

**(b)** Show that $1 - x$ is a unit in $R[[x]]$ with inverse $1 + x + x^2 + \cdots$.

**(c)** Prove that $\sum_{n=0}^{\infty} a_n x^n$ is a unit in $R[[x]]$ if and only if $a_0$ is a unit in $R$.

Let $R$ be a commutative ring with 1.

**4.** Prove that if $R$ is an integral domain then the ring of formal power series $R[[x]]$ is also an integral domain.

Let $R$ be a commutative ring with 1.

**5.** Let $F$ be a field and define the ring $F((x))$ of *formal Laurent series* with coefficients from $F$ by

$$F((x)) = \{\sum_{n \geq N}^{\infty} a_n x^n \mid a_n \in F \text{ and } N \in \mathbb{Z}\}.$$

(Every element of $F((x))$ is a power series in $x$ plus a polynomial in $1/x$, i.e., each element of $F((x))$ has only a finite number of terms with negative powers of $x$.)

**(a)** Prove that $F((x))$ is a field.

**(b)** Define the map

$$v : F((x))^{\times} \to \mathbb{Z} \qquad \text{by} \qquad v(\sum_{n \geq N}^{\infty} a_n x^n) = N$$

where $a_N$ is the first nonzero coefficient of the series (i.e., $N$ is the "order of zero or pole of the series at 0"). Prove that $v$ is a discrete valuation on $F((x))$ whose discrete valuation ring is $F[[x]]$, the ring of formal power series (cf. Exercise 26, Section 1).

Let $R$ be a commutative ring with 1.

**6.** Let $S$ be a ring with identity $1 \neq 0$. Let $n \in \mathbb{Z}^+$ and let $A$ be an $n \times n$ matrix with entries from $S$ whose $i, j$ entry is $a_{ij}$. Let $E_{ij}$ be the element of $M_n(S)$ whose $i, j$ entry is 1 and whose other entries are all 0.

**(a)** Prove that $E_{ij}A$ is the matrix whose $i^{\text{th}}$ row equals the $j^{\text{th}}$ row of $A$ and all other rows are zero.

**(b)** Prove that $AE_{ij}$ is the matrix whose $j^{\text{th}}$ column equals the $i^{\text{th}}$ column of $A$ and all other columns are zero.

**(c)** Deduce that $E_{pq}AE_{rs}$ is the matrix whose $p, s$ entry is $a_{qr}$ and all other entries are zero.

Let $R$ be a commutative ring with 1.

**7.** Prove that the center of the ring $M_n(R)$ is the set of scalar matrices (cf. Exercise 7, Section 1). [Use the preceding exercise.]

Let $R$ be a commutative ring with 1.

**8.** Let $S$ be any ring and let $n \geq 2$ be an integer. Prove that if $A$ is any strictly upper triangular matrix in $M_n(S)$ then $A^n = 0$ (a strictly upper triangular matrix is one whose entries on and below the main diagonal are all zero).

Let $R$ be a commutative ring with 1.

**9.** Let $\alpha = r + r^2 - 2s$ and $\beta = -3r^2 + rs$ be the two elements of the integral group ring $\mathbb{Z}D_8$ described in this section. Compute the following elements of $\mathbb{Z}D_8$:
**(a)** $\beta\alpha$, **(b)** $\alpha^2$, **(c)** $\alpha\beta - \beta\alpha$, **(d)** $\beta\alpha\beta$.

Let $R$ be a commutative ring with 1.

**10.** Consider the following elements of the integral group ring $\mathbb{Z}S_3$:

$$\alpha = 3(1\ 2) - 5(2\ 3) + 14(1\ 2\ 3) \qquad \text{and} \qquad \beta = 6(1) + 2(2\ 3) - 7(1\ 3\ 2)$$

(where (1) is the identity of $S_3$). Compute the following elements:
**(a)** $\alpha + \beta$, **(b)** $2\alpha - 3\beta$, **(c)** $\alpha\beta$, **(d)** $\beta\alpha$, **(e)** $\alpha^2$.

Let $R$ be a commutative ring with 1.

**11.** Repeat the preceding exercise under the assumption that the coefficients of $\alpha$ and $\beta$ are in $\mathbb{Z}/3\mathbb{Z}$ (i.e., $\alpha, \beta \in \mathbb{Z}/3\mathbb{Z}S_3$).

Let $R$ be a commutative ring with 1.

**12.** Let $G = \{g_1, \ldots, g_n\}$ be a finite group. Prove that the element $N = g_1 + g_2 + \ldots + g_n$ is in the center of the group ring $RG$ (cf. Exercise 7, Section 1).

Let $R$ be a commutative ring with 1.

**13.** Let $\mathcal{K} = \{k_1, \ldots, k_m\}$ be a conjugacy class in the finite group $G$.

**(a)** Prove that the element $K = k_1 + \ldots + k_m$ is in the center of the group ring $RG$ (cf. Exercise 7, Section 1). [Check that $g^{-1}Kg = K$ for all $g \in G$.]

**(b)** Let $\mathcal{K}_1, \ldots, \mathcal{K}_r$ be the conjugacy classes of $G$ and for each $\mathcal{K}_i$ let $K_i$ be the element of $RG$ that is the sum of the members of $\mathcal{K}_i$. Prove that an element $\alpha \in RG$ is in the center of $RG$ if and only if $\alpha = a_1K_1 + a_2K_2 + \cdots + a_rK_r$ for some $a_1, a_2, \ldots, a_r \in R$.

## 7.3. Ring Homomorphisms and Quotient Rings

Let $R$ be a ring with identity $1 \neq 0$.

**1.** Prove that the rings $2\mathbb{Z}$ and $3\mathbb{Z}$ are not isomorphic.

Let $R$ be a ring with identity $1 \neq 0$.

**2.** Prove that the rings $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ are not isomorphic.

Let $R$ be a ring with identity $1 \neq 0$.

**3.** Find all homomorphic images of $\mathbb{Z}$.

Let $R$ be a ring with identity $1 \neq 0$.

**4.** Find all ring homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}/30\mathbb{Z}$. In each case describe the kernel and the image.

Let $R$ be a ring with identity $1 \neq 0$.

**5.** Describe all ring homomorphisms from the ring $\mathbb{Z} \times \mathbb{Z}$ to $\mathbb{Z}$. In each case describe the kernel and the image.

Let $R$ be a ring with identity $1 \neq 0$.

**6.** Decide which of the following are ring homomorphisms from $M_2(\mathbb{Z})$ to $\mathbb{Z}$:

**(a)** $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a$ (projection onto the 1,1 entry)

**(b)** $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto a + d$ (the *trace* of the matrix)

**(c)** $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto ad - bc$ (the *determinant* of the matrix).

Let $R$ be a ring with identity $1 \neq 0$.

**7.** Let $R = \{\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mid a, b, d \in \mathbb{Z}\}$ be the subring of $M_2(\mathbb{Z})$ of upper triangular matrices. Prove that the map

$$\varphi : R \to \mathbb{Z} \times \mathbb{Z} \quad \text{defined by} \quad \varphi : \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mapsto (a, d)$$

is a surjective homomorphism and describe its kernel.

Let $R$ be a ring with identity $1 \neq 0$.

**8.** Decide which of the following are ideals of the ring $\mathbb{Z} \times \mathbb{Z}$:

**(a)** $\{(a, a) \mid a \in \mathbb{Z}\}$

**(b)** $\{(2a, 2b) \mid a, b \in \mathbb{Z}\}$

**(c)** $\{(2a, 0) \mid a \in \mathbb{Z}\}$

**(d)** $\{(a, -a) \mid a \in \mathbb{Z}\}$.

Let $R$ be a ring with identity $1 \neq 0$.

**9.** Decide which of the sets in Exercise 6 of Section 1 are ideals of the ring of all functions from $[0,1]$ to $\mathbb{R}$.

Let $R$ be a ring with identity $1 \neq 0$.

**10.** Decide which of the following are ideals of the ring $\mathbb{Z}[x]$:

- **(a)** the set of all polynomials whose constant term is a multiple of 3
- **(b)** the set of all polynomials whose coefficient of $x^2$ is a multiple of 3
- **(c)** the set of all polynomials whose constant term, coefficient of $x$ and coefficient of $x^2$ are zero
- **(d)** $\mathbb{Z}[x^2]$ (i.e., the polynomials in which only even powers of $x$ appear)
- **(e)** the set of polynomials whose coefficients sum to zero
- **(f)** the set of polynomials $p(x)$ such that $p'(0) = 0$, where $p'(x)$ is the usual first derivative of $p(x)$ with respect to $x$.

Let $R$ be a ring with identity $1 \neq 0$.

**11.** Let $R$ be the ring of all continuous real valued functions on the closed interval $[0, 1]$. Prove that the map $\varphi : R \to \mathbb{R}$ defined by $\varphi(f) = \int_0^1 f(t)dt$ is a homomorphism of additive groups but not a ring homomorphism.

Let $R$ be a ring with identity $1 \neq 0$.

**12.** Let $D$ be an integer that is not a perfect square in $\mathbb{Z}$ and let $S = \{\begin{pmatrix} a & b \\ Db & a \end{pmatrix} \mid a, b \in \mathbb{Z}\}$.

**(a)** Prove that $S$ is a subring of $M_2(\mathbb{Z})$.

**(b)** If $D$ is not a perfect square in $\mathbb{Z}$ prove that the map $\varphi : \mathbb{Z}[\sqrt{D}] \to S$ defined by $\varphi(a + b\sqrt{D}) = \begin{pmatrix} a & b \\ Db & a \end{pmatrix}$ is a ring isomorphism.

**(c)** If $D \equiv 1 \bmod 4$ is squarefree, prove that the set $\{\begin{pmatrix} a & b \\ (D-1)b/4 & a+b \end{pmatrix} \mid a, b \in \mathbb{Z}\}$ is a subring of $M_2(\mathbb{Z})$ and is isomorphic to the quadratic integer ring $\mathcal{O}$.

Let $R$ be a ring with identity $1 \neq 0$.

**13.** Prove that the ring $M_2(\mathbb{R})$ contains a subring that is isomorphic to $\mathbb{C}$.

Let $R$ be a ring with identity $1 \neq 0$.

**14.** Prove that the ring $M_4(\mathbb{R})$ contains a subring that is isomorphic to the real Hamilton Quaternions, $\mathbb{H}$.

Let $R$ be a ring with identity $1 \neq 0$.

**15.** Let $X$ be a nonempty set and let $\mathcal{P}(X)$ be the Boolean ring of all subsets of $X$ defined in Exercise 21 of Section 1. Let $R$ be the ring of all functions from $X$ into $\mathbb{Z}/2\mathbb{Z}$. For each $A \in \mathcal{P}(X)$ define the function

$$\chi_A : X \to \mathbb{Z}/2\mathbb{Z} \qquad \text{by} \qquad \chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

($\chi_A$ is called the *characteristic function of* $A$ with values in $\mathbb{Z}/2\mathbb{Z}$). Prove that the map $\mathcal{P}(X) \to R$ defined by $A \mapsto \chi_A$ is a ring isomorphism.

Let $R$ be a ring with identity $1 \neq 0$.

**16.** Let $\varphi : R \to S$ be a surjective homomorphism of rings. Prove that the image of the center of $R$ is contained in the center of $S$ (cf. Exercise 7 of Section 1).

Let $R$ be a ring with identity $1 \neq 0$.

**17.** Let $R$ and $S$ be nonzero rings with identity and denote their respective identities by $1_R$ and $1_S$. Let $\varphi : R \to S$ be a nonzero homomorphism of rings.

**(a)** Prove that if $\varphi(1_R) \neq 1_S$ then $\varphi(1_R)$ is a zero divisor in $S$. Deduce that if $S$ is an integral domain then every ring homomorphism from $R$ to $S$ sends the identity of $R$ to the identity of $S$.

**(b)** Prove that if $\varphi(1_R) = 1_S$ then $\varphi(u)$ is a unit in $S$ and that $\varphi(u^{-1}) = \varphi(u)^{-1}$ for each unit $u$ of $R$.

Let $R$ be a ring with identity $1 \neq 0$.

**18.** **(a)** If $I$ and $J$ are ideals of $R$ prove that their intersection $I \cap J$ is also an ideal of $R$.
**(b)** Prove that the intersection of an arbitrary nonempty collection of ideals is again an ideal (do not assume the collection is countable).

Let $R$ be a ring with identity $1 \neq 0$.

**19.** Prove that if $I_1 \subseteq I_2 \subseteq \cdots$ are ideals of $R$ then $\cup_{n=1}^{\infty} I_n$ is an ideal of $R$.

Let $R$ be a ring with identity $1 \neq 0$.

**20.** Let $I$ be an ideal of $R$ and let $S$ be a subring of $R$. Prove that $I \cap S$ is an ideal of $S$. Show by example that not every ideal of a subring $S$ of a ring $R$ need be of the form $I \cap S$ for some ideal $I$ of $R$.

Let $R$ be a ring with identity $1 \neq 0$.

**21.** Prove that every (two-sided) ideal of $M_n(R)$ is equal to $M_n(J)$ for some (two-sided) ideal $J$ of $R$. [Use Exercise 6(c) of Section 2 to show first that the set of entries of matrices in an ideal of $M_n(R)$ form an ideal in $R$.]

Let $R$ be a ring with identity $1 \neq 0$.

**22.** Let $a$ be an element of the ring $R$.

**(a)** Prove that $\{x \in R \mid ax = 0\}$ is a right ideal and $\{y \in R \mid ya = 0\}$ is a left ideal (called respectively the right and left *annihilators* of $a$ in $R$).

**(b)** Prove that if $L$ is a left ideal of $R$ then $\{x \in R \mid xa = 0 \text{ for all } a \in L\}$ is a two-sided ideal (called the left *annihilator* of $L$ in $R$).

Let $R$ be a ring with identity $1 \neq 0$.

**23.** Let $S$ be a subring of $R$ and let $I$ be an ideal of $R$. Prove that if $S \cap I = 0$ then $\overline{S} \cong S$, where the bar denotes passage to $R/I$.

Let $R$ be a ring with identity $1 \neq 0$.

**24.** Let $\varphi : R \to S$ be a ring homomorphism.

**(a)** Prove that if $J$ is an ideal of $S$ then $\varphi^{-1}(J)$ is an ideal of $R$. Apply this to the special case when $R$ is a subring of $S$ and $\varphi$ is the inclusion homomorphism to deduce that if $J$ is an ideal of $S$ then $J \cap R$ is an ideal of $R$.

**(b)** Prove that if $\varphi$ is surjective and $I$ is an ideal of $R$ then $\varphi(I)$ is an ideal of $S$. Give an example where this fails if $\varphi$ is not surjective.

Let $R$ be a ring with identity $1 \neq 0$.

**25.** Assume $R$ is a commutative ring with 1. Prove that the Binomial Theorem

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$$

holds in $R$, where the binomial coefficient $\binom{n}{k}$ is interpreted in $R$ as the sum $1+1+\cdots+1$ of the identity 1 in $R$ taken $\binom{n}{k}$ times.

Let $R$ be a ring with identity $1 \neq 0$.

**26.** The *characteristic* of a ring $R$ is the smallest positive integer $n$ such that $1+1+\cdots+1 = 0$ ($n$ times) in $R$; if no such integer exists the characteristic of $R$ is said to be 0. For example, $\mathbb{Z}/n\mathbb{Z}$ is a ring of characteristic $n$ for each positive integer $n$ and $\mathbb{Z}$ is a ring of characteristic 0.

**(a)** Prove that the map $\mathbb{Z} \to R$ defined by

$$k \mapsto \begin{cases} 1+1+\cdots+1 \ (k \text{ times}) & \text{if } k > 0 \\ 0 & \text{if } k = 0 \\ -1-1-\cdots-1 \ (-k \text{ times}) & \text{if } k < 0 \end{cases}$$

is a ring homomorphism whose kernel is $n\mathbb{Z}$, where $n$ is the characteristic of $R$ (this explains the use of the terminology "characteristic 0" instead of the archaic phrase "characteristic $\infty$" for rings in which no sum of 1's is zero).

**(b)** Determine the characteristics of the rings $\mathbb{Q}$, $\mathbb{Z}[x]$, $\mathbb{Z}/n\mathbb{Z}[x]$.

**(c)** Prove that if $p$ is a prime and if $R$ is a commutative ring of characteristic $p$, then $(a+b)^p = a^p + b^p$ for all $a, b \in R$.

Let $R$ be a ring with identity $1 \neq 0$.

**27.** Prove that a nonzero Boolean ring has characteristic 2 (cf. Exercise 15, Section 1).

Let $R$ be a ring with identity $1 \neq 0$.

**28.** Prove that an integral domain has characteristic $p$, where $p$ is either a prime or 0 (cf. Exercise 26).

Let $R$ be a ring with identity $1 \neq 0$.

**29.** Let $R$ be a commutative ring. Recall (cf. Exercise 13, Section 1) that an element $x \in R$ is nilpotent if $x^n = 0$ for some $n \in \mathbb{Z}^+$. Prove that the set of nilpotent elements form an ideal — called the *nilradical* of $R$ and denoted by $\mathfrak{N}(R)$. [Use the Binomial Theorem to show $\mathfrak{N}(R)$ is closed under addition.]

Let $R$ be a ring with identity $1 \neq 0$.

**30.** Prove that if $R$ is a commutative ring and $\mathfrak{N}(R)$ is its nilradical (cf. the preceding exercise) then zero is the only nilpotent element of $R/\mathfrak{N}(R)$ i.e., prove that $\mathfrak{N}(R/\mathfrak{N}(R)) = 0$.

Let $R$ be a ring with identity $1 \neq 0$.

**31.** Prove that the elements $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ are nilpotent elements of $M_2(\mathbb{Z})$ whose sum is not nilpotent (note that these two matrices do not commute). Deduce that the set of nilpotent elements in the noncommutative ring $M_2(\mathbb{Z})$ is not an ideal.

Let $R$ be a ring with identity $1 \neq 0$.

**32.** Let $\varphi : R \to S$ be a homomorphism of rings. Prove that if $x$ is a nilpotent element of $R$ then $\varphi(x)$ is nilpotent in $S$.

Let $R$ be a ring with identity $1 \neq 0$.

**33.** Assume $R$ is commutative. Let $p(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be an element of the polynomial ring $R[x]$.

**(a)** Prove that $p(x)$ is a unit in $R[x]$ if and only if $a_0$ is a unit and $a_1, a_2, \ldots, a_n$ are nilpotent in $R$. [ See Exercise 14 of Section 1.]

**(b)** Prove that $p(x)$ is nilpotent in $R[x]$ if and only if $a_0, a_1, \ldots, a_n$ are nilpotent elements of $R$.

Let $R$ be a ring with identity $1 \neq 0$.

**34.** Let $I$ and $J$ be ideals of $R$.

**(a)** Prove that $I + J$ is the smallest ideal of $R$ containing both $I$ and $J$.

**(b)** Prove that $IJ$ is an ideal contained in $I \cap J$.

**(c)** Give an example where $IJ \neq I \cap J$.

**(d)** Prove that if $R$ is commutative and if $I + J = R$ then $IJ = I \cap J$.

Let $R$ be a ring with identity $1 \neq 0$.

**35.** Let $I, J, K$ be ideals of $R$.

**(a)** Prove that $I(J + K) = IJ + IK$ and $(I + J)K = IK + JK$.

**(b)** Prove that if $J \subseteq I$ then $I \cap (J + K) = J + (I \cap K)$.

Let $R$ be a ring with identity $1 \neq 0$.

**36.** Show that if $I$ is the ideal of all polynomials in $\mathbb{Z}[x]$ with zero constant term then $I^n = \{a_n x^n + a_{n+1}x^{n+1} + \cdots + a_{n+m}x^{n+m} \mid a_i \in \mathbb{Z},\ m \geq 0\}$ is the set of polynomials whose first nonzero term has degree at least $n$.

Let $R$ be a ring with identity $1 \neq 0$.

**37.** An ideal $N$ is called *nilpotent* if $N^n$ is the zero ideal for some $n \geq 1$. Prove that the ideal $p\mathbb{Z}/p^m\mathbb{Z}$ is a nilpotent ideal in the ring $\mathbb{Z}/p^m\mathbb{Z}$.

## 7.4. Properties of Ideals

Let $R$ be a ring with identity $1 \neq 0$.

**1.** Let $L_j$ be the left ideal of $M_n(R)$ consisting of arbitrary entries in the $j^{\text{th}}$ column and zero in all other entries and let $E_{ij}$ be the element of $M_n(R)$ whose $i, j$ entry is 1 and whose other entries are all 0. Prove that $L_j = M_n(R)E_{ij}$ for any $i$. [See Exercise 6, Section 2.]

Let $R$ be a ring with identity $1 \neq 0$.

**2.** Assume $R$ is commutative. Prove that the augmentation ideal in the group ring $RG$ is generated by $\{g - 1 \mid g \in G\}$. Prove that if $G = \langle \sigma \rangle$ is cyclic then the augmentation ideal is generated by $\sigma - 1$.

Let $R$ be a ring with identity $1 \neq 0$.

**3.** **(a)** Let $p$ be a prime and let $G$ be an abelian group of order $p^n$. Prove that the nilradical of the group ring $\mathbb{F}_p G$ is the augmentation ideal (cf. Exercise 29, Section 3). [Use the preceding exercise.]

**(b)** Let $G = \{g_1, \ldots, g_n\}$ be a finite group and assume $R$ is commutative. Prove that if $r$ is any element of the augmentation ideal of $RG$ then $r(g_1 + \cdots + g_n) = 0$. [Use the preceding exercise.]

Let $R$ be a ring with identity $1 \neq 0$.

**4.** Assume $R$ is commutative. Prove that $R$ is a field if and only if 0 is a maximal ideal.

Let $R$ be a ring with identity $1 \neq 0$.

**5.** Prove that if $M$ is an ideal such that $R/M$ is a field then $M$ is a maximal ideal (do not assume $R$ is commutative).

Let $R$ be a ring with identity $1 \neq 0$.

**6.** Prove that $R$ is a division ring if and only if its only left ideals are $(0)$ and $R$. (The analogous result holds when "left" is replaced by "right.")

Let $R$ be a ring with identity $1 \neq 0$.

**7.** Let $R$ be a commutative ring with 1. Prove that the principal ideal generated by $x$ in the polynomial ring $R[x]$ is a prime ideal if and only if $R$ is an integral domain. Prove that $(x)$ is a maximal ideal if and only if $R$ is a field.

Let $R$ be a ring with identity $1 \neq 0$.

**8.** Let $R$ be an integral domain. Prove that $(a) = (b)$ for some elements $a, b \in R$, if and only if $a = ub$ for some unit $u$ of $R$.

Let $R$ be a ring with identity $1 \neq 0$.

**9.** Let $R$ be the ring of all continuous functions on $[0, 1]$ and let $I$ be the collection of functions $f(x)$ in $R$ with $f(1/3) = f(1/2) = 0$. Prove that $I$ is an ideal of $R$ but is not a prime ideal.

Let $R$ be a ring with identity $1 \neq 0$.

**10.** Assume $R$ is commutative. Prove that if $P$ is a prime ideal of $R$ and $P$ contains no zero divisors then $R$ is an integral domain.

Let $R$ be a ring with identity $1 \neq 0$.

**11.** Assume $R$ is commutative. Let $I$ and $J$ be ideals of $R$ and assume $P$ is a prime ideal of $R$ that contains $IJ$ (for example, if $P$ contains $I \cap J$). Prove either $I$ or $J$ is contained in $P$.

Let $R$ be a ring with identity $1 \neq 0$.

**12.** Assume $R$ is commutative and suppose $I = (a_1, a_2, \ldots, a_n)$ and $J = (b_1, b_2, \ldots, b_m)$ are two finitely generated ideals in $R$. Prove that the product ideal $IJ$ is finitely generated by the elements $a_i b_j$ for $i = 1, 2, \ldots, n$, and $j = 1, 2, \ldots, m$.

Let $R$ be a ring with identity $1 \neq 0$.

**13.** Let $\varphi : R \to S$ be a homomorphism of commutative rings.

**(a)** Prove that if $P$ is a prime ideal of $S$ then either $\varphi^{-1}(P) = R$ or $\varphi^{-1}(P)$ is a prime ideal of $R$. Apply this to the special case when $R$ is a subring of $S$ and $\varphi$ is the inclusion homomorphism to deduce that if $P$ is a prime ideal of $S$ then $P \cap R$ is either $R$ or a prime ideal of $R$.

**(b)** Prove that if $M$ is a maximal ideal of $S$ and $\varphi$ is surjective then $\varphi^{-1}(M)$ is a maximal ideal of $R$. Give an example to show that this need not be the case if $\varphi$ is not surjective.

Let $R$ be a ring with identity $1 \neq 0$.

**14.** Assume $R$ is commutative. Let $x$ be an indeterminate, let $f(x)$ be a monic polynomial in $R[x]$ of degree $n \geq 1$ and use the bar notation to denote passage to the quotient ring $R[x]/(f(x))$.

**(a)** Show that every element of $R[x]/(f(x))$ is of the form $\overline{p(x)}$ for some polynomial $p(x) \in R[x]$ of degree less than $n$, i.e.,

$$R[x]/(f(x)) = \{\overline{a_0} + \overline{a_1 x} + \cdots + \overline{a_{n-1}x^{n-1}} \mid a_0, a_1, \ldots, a_{n-1} \in R\}.$$

[If $f(x) = x^n + b_{n-1}x^{n-1} + \cdots + b_0$ then $\overline{x^n} = \overline{-(b_{n-1}x^{n-1} + \cdots + b_0)}$. Use this to reduce powers of $\bar{x}$ in the quotient ring.]

**(b)** Prove that if $p(x)$ and $q(x)$ are distinct polynomials in $R[x]$ which are both of degree less than $n$, then $\overline{p(x)} \neq \overline{q(x)}$. [Otherwise $p(x) - q(x)$ is an $R[x]$-multiple of the monic polynomial $f(x)$.]

**(c)** If $f(x) = a(x)b(x)$ where both $a(x)$ and $b(x)$ have degree less than $n$, prove that $\overline{a(x)}$ is a zero divisor in $R[x]/(f(x))$.

**(d)** If $f(x) = x^n - a$ for some nilpotent element $a \in R$, prove that $\bar{x}$ is nilpotent in $R[x]/(f(x))$.

**(e)** Let $p$ be a prime, assume $R = \mathbb{F}_p$ and $f(x) = x^p - a$ for some $a \in \mathbb{F}_p$. Prove that $\overline{x - a}$ is nilpotent in $R[x]/(f(x))$. [Use Exercise 26(c) of Section 3.]

Let $R$ be a ring with identity $1 \neq 0$.

**15.** Let $x^2 + x + 1$ be an element of the polynomial ring $E = \mathbb{F}_2[x]$ and use the bar notation to denote passage to the quotient ring $\mathbb{F}_2[x]/(x^2 + x + 1)$.

**(a)** Prove that $\overline{E}$ has 4 elements: $\overline{0}$, $\overline{1}$, $\overline{x}$ and $\overline{x+1}$.

**(b)** Write out the $4 \times 4$ addition table for $\overline{E}$ and deduce that the additive group $\overline{E}$ is isomorphic to the Klein 4-group.

**(c)** Write out the $4 \times 4$ multiplication table for $\overline{E}$ and prove that $\overline{E}^{\times}$ is isomorphic to the cyclic group of order 3. Deduce that $\overline{E}$ is a field.

Let $R$ be a ring with identity $1 \neq 0$.

**16.** Let $x^4 - 16$ be an element of the polynomial ring $E = \mathbb{Z}[x]$ and use the bar notation to denote passage to the quotient ring $\mathbb{Z}[x]/(x^4 - 16)$.

**(a)** Find a polynomial of degree $\leq 3$ that is congruent to $7x^{13} - 11x^9 + 5x^5 - 2x^3 + 3$ modulo $(x^4 - 16)$.

**(b)** Prove that $\overline{x - 2}$ and $\overline{x + 2}$ are zero divisors in $\overline{E}$.

Let $R$ be a ring with identity $1 \neq 0$.

**17.** Let $x^3 - 2x + 1$ be an element of the polynomial ring $E = \mathbb{Z}[x]$ and use the bar notation to denote passage to the quotient ring $\mathbb{Z}[x]/(x^3-2x+1)$. Let $p(x) = 2x^7 - 7x^5 + 4x^3 - 9x + 1$ and let $q(x) = (x-1)^4$.

**(a)** Express each of the following elements of $\overline{E}$ in the form $\overline{f(x)}$ for some polynomial $f(x)$ of degree $\leq 2$: $\overline{p(x)}$, $\overline{q(x)}$, $\overline{p(x)+q(x)}$ and $\overline{p(x)q(x)}$.
**(b)** Prove that $\overline{E}$ is not an integral domain.
**(c)** Prove that $\overline{x}$ is a unit in $\overline{E}$.

Let $R$ be a ring with identity $1 \neq 0$.

**18.** Prove that if $R$ is an integral domain and $R[[x]]$ is the ring of formal power series in the indeterminate $x$ then the principal ideal generated by $x$ is a prime ideal (cf. Exercise 3, Section 2). Prove that the principal ideal generated by $x$ is a maximal ideal if and only if $R$ is a field.

Let $R$ be a ring with identity $1 \neq 0$.

**19.** Let $R$ be a finite commutative ring with identity. Prove that every prime ideal of $R$ is a maximal ideal.

Let $R$ be a ring with identity $1 \neq 0$.

**20.** Prove that a nonzero finite commutative ring that has no zero divisors is a field (if the ring has an identity, this is Corollary 3, so do not assume the ring has a 1).

Let $R$ be a ring with identity $1 \neq 0$.

**21.** Prove that a finite ring with identity $1 \neq 0$ that has no zero divisors is a field (you may quote Wedderburn's Theorem).

Let $R$ be a ring with identity $1 \neq 0$.

**22.** Let $p \in \mathbb{Z}^+$ be a prime and let the $\mathbb{F}_p$ Quaternions be defined by

$$a + bi + cj + dk \qquad a, b, c, d \in \mathbb{Z}/p\mathbb{Z}$$

where addition is componentwise and multiplication is defined using the same relations on $i$, $j$, $k$ as for the real Quaternions.

**(a)** Prove that the $\mathbb{F}_p$ Quaternions are a homomorphic image of the integral Quaternions (cf. Section 1).

**(b)** Prove that the $\mathbb{F}_p$ Quaternions contain zero divisors (and so they cannot be a division ring). [Use the preceding exercise.]

Let $R$ be a ring with identity $1 \neq 0$.

**23.** Prove that in a Boolean ring (cf. Exercise 15, Section 1) every prime ideal is a maximal ideal.

Let $R$ be a ring with identity $1 \neq 0$.

**24.** Prove that in a Boolean ring every finitely generated ideal is principal.

Let $R$ be a ring with identity $1 \neq 0$.

**25.** Assume $R$ is commutative and for each $a \in R$ there is an integer $n > 1$ (depending on $a$) such that $a^n = a$. Prove that every prime ideal of $R$ is a maximal ideal.

Let $R$ be a ring with identity $1 \neq 0$.

**26.** Prove that a prime ideal in a commutative ring $R$ contains every nilpotent element (cf. Exercise 13, Section 1). Deduce that the **nilradical** of $R$ (cf. Exercise 29, Section 3) is contained in the intersection of all the prime ideals of $R$. (It is shown in Section 15.2 that the nilradical of $R$ is equal to the intersection of all prime ideals of $R$.)

Let $R$ be a ring with identity $1 \neq 0$.

**27.** Let $R$ be a commutative ring with $1 \neq 0$. Prove that if $a$ is a nilpotent element of $R$ then $1 - ab$ is a unit for all $b \in R$.

Let $R$ be a ring with identity $1 \neq 0$.

**28.** Prove that if $R$ is a commutative ring and $N = (a_1, a_2, \ldots, a_m)$ where each $a_i$ is a nilpotent element, then $N$ is a nilpotent ideal (cf. Exercise 37, Section 3). Deduce that if the nilradical of $R$ is finitely generated then it is a nilpotent ideal.

Let $R$ be a ring with identity $1 \neq 0$.

**29.** Let $p$ be a prime and let $G$ be a finite group of order a power of $p$ (i.e., a $p$-group). Prove that the augmentation ideal in the group ring $\mathbb{Z}/p\mathbb{Z}G$ is a nilpotent ideal. (Note that this ring may be noncommutative.) [Use Exercise 2.]

Let $R$ be a ring with identity $1 \neq 0$.

**30.** Let $I$ be an ideal of the commutative ring $R$ and define

$$\text{rad } I = \{r \in R \mid r^n \in I \text{ for some } n \in \mathbb{Z}^+\}$$

called the *radical* of $I$. Prove that rad $I$ is an ideal containing $I$ and that $(\text{rad } I)/I$ is the nilradical of the quotient ring $R/I$, i.e., $(\text{rad } I)/I = \mathfrak{N}(R/I)$ (cf. Exercise 29, Section 3).

Let $R$ be a ring with identity $1 \neq 0$.

**31.** An ideal $I$ of the commutative ring $R$ is called a *radical ideal* if $\text{rad } I = I$.

**(a)** Prove that every prime ideal of $R$ is a radical ideal.
**(b)** Let $n > 1$ be an integer. Prove that 0 is a radical ideal in $\mathbb{Z}/n\mathbb{Z}$ if and only if $n$ is a product of distinct primes to the first power (i.e., $n$ is square free). Deduce that $(n)$ is a radical ideal of $\mathbb{Z}$ if and only if $n$ is a product of distinct primes in $\mathbb{Z}$.

Let $R$ be a ring with identity $1 \neq 0$.

**32.** Let $I$ be an ideal of the commutative ring $R$ and define

$$\text{Jac } I \text{ to be the intersection of all maximal ideals of } R \text{ that contain } I$$

where the convention is that Jac $R = R$. (If $I$ is the zero ideal, Jac 0 is called the *Jacobson radical* of the ring $R$, so Jac $I$ is the preimage in $R$ of the Jacobson radical of $R/I$.)

**(a)** Prove that Jac $I$ is an ideal of $R$ containing $I$.

**(b)** Prove that rad $I \subseteq$ Jac $I$, where rad $I$ is the radical of $I$ defined in Exercise 30.

**(c)** Let $n > 1$ be an integer. Describe Jac $n\mathbb{Z}$ in terms of the prime factorization of $n$.

Let $R$ be a ring with identity $1 \neq 0$.

**33.** Let $R$ be the ring of all continuous functions from the closed interval [0,1] to $\mathbb{R}$ and for each $c \in [0, 1]$ let $M_c = \{f \in R \mid f(c) = 0\}$ (recall that $M_c$ was shown to be a maximal ideal of $R$).

**(a)** Prove that if $M$ is *any* maximal ideal of $R$ then there is a real number $c \in [0, 1]$ such that $M = M_c$.

**(b)** Prove that if $b$ and $c$ are distinct points in [0,1] then $M_b \neq M_c$.

**(c)** Prove that $M_c$ is not equal to the principal ideal generated by $x - c$.

**(d)** Prove that $M_c$ is not a finitely generated ideal.

Let $R$ be a ring with identity $1 \neq 0$.

The preceding exercise shows that there is a bijection between the *points* of the closed interval [0,1] and the set of *maximal ideals* in the ring $R$ of all of continuous functions on [0,1] given by $c \leftrightarrow M_c$. For any subset $X$ of $\mathbb{R}$ or, more generally, for any completely regular topological space $X$, the map $c \mapsto M_c$ is an *injection* from $X$ to the set of maximal ideals of $R$, where $R$ is the ring of all bounded continuous real valued functions on $X$ and $M_c$ is the maximal ideal of functions that vanish at $c$. Let $\beta(X)$ be the set of maximal ideals of $R$. One can put a topology on $\beta(X)$ in such a way that if we identify $X$ with its image in $\beta(X)$ then $X$ (in its given topology) becomes a subspace of $\beta(X)$. Moreover, $\beta(X)$ is a compact space under this topology and is called the *Stone-Čech compactification* of $X$.

**34.** Let $R$ be the ring of all continuous functions from $\mathbb{R}$ to $\mathbb{R}$ and for each $c \in \mathbb{R}$ let $M_c$ be the maximal ideal $\{f \in R \mid f(c) = 0\}$.

**(a)** Let $I$ be the collection of functions $f(x)$ in $R$ with *compact support* (i.e., $f(x) = 0$ for $|x|$ sufficiently large). Prove that $I$ is an ideal of $R$ that is not a prime ideal.

**(b)** Let $M$ be a maximal ideal of $R$ containing $I$ (properly, by (a)). Prove that $M \neq M_c$ for any $c \in \mathbb{R}$ (cf. the preceding exercise).

Let $R$ be a ring with identity $1 \neq 0$.

**35.** Let $A = (a_1, a_2, \ldots, a_n)$ be a nonzero finitely generated ideal of $R$. Prove that there is an ideal $B$ which is maximal with respect to the property that it does not contain $A$. [Use Zorn's Lemma.]

Let $R$ be a ring with identity $1 \neq 0$.

**36.** Assume $R$ is commutative. Prove that the set of prime ideals in $R$ has a minimal element with respect to inclusion (possibly the zero ideal). [Use Zorn's Lemma.]

Let $R$ be a ring with identity $1 \neq 0$.

**37.** A commutative ring $R$ is called a *local ring* if it has a unique maximal ideal. Prove that if $R$ is a local ring with maximal ideal $M$ then every element of $R - M$ is a unit. Prove conversely that if $R$ is a commutative ring with 1 in which the set of nonunits forms an ideal $M$, then $R$ is a local ring with unique maximal ideal $M$.

Let $R$ be a ring with identity $1 \neq 0$.

**38.** Prove that the ring of all rational numbers whose denominators is odd is a local ring whose unique maximal ideal is the principal ideal generated by 2.

Let $R$ be a ring with identity $1 \neq 0$.

**39.** Following the notation of Exercise 26 in Section 1, let $K$ be a field, let $v$ be a discrete valuation on $K$ and let $R$ be the valuation ring of $v$. For each integer $k \geq 0$ define $A_k = \{r \in R \mid v(r) \geq k\} \cup \{0\}$.

**(a)** Prove that $A_k$ is a principal ideal and that $A_0 \supseteq A_1 \supseteq A_2 \supseteq \cdots$.

**(b)** Prove that if $I$ is any nonzero ideal of $R$, then $I = A_k$ for some $k \geq 0$. Deduce that $R$ is a local ring with unique maximal ideal $A_1$.

Let $R$ be a ring with identity $1 \neq 0$.

**40.** Assume $R$ is commutative. Prove that the following are equivalent: (see also Exercises 13 and 14 in Section 1)

**(i)** $R$ has exactly one prime ideal

**(ii)** every element of $R$ is either nilpotent or a unit

**(iii)** $R/\eta(R)$ is a field (cf. Exercise 29, Section 3).

Let $R$ be a ring with identity $1 \neq 0$.

**41.** A proper ideal $Q$ of the commutative ring $R$ is called *primary* if whenever $ab \in Q$ and $a \notin Q$ then $b^n \in Q$ for some positive integer $n$. (Note that the symmetry between $a$ and $b$ in this definition implies that if $Q$ is a primary ideal and $ab \in Q$ with *neither* $a$ nor $b$ in $Q$, then a positive power of $a$ and a positive power of $b$ both lie in $Q$.) Establish the following facts about primary ideals.

**(a)** The primary ideals of $\mathbb{Z}$ are 0 and $(p^n)$, where $p$ is a prime and $n$ is a positive integer.

**(b)** Every prime ideal of $R$ is a primary ideal.

**(c)** An ideal $Q$ of $R$ is primary if and only if every zero divisor in $R/Q$ is a nilpotent element of $R/Q$.

**(d)** If $Q$ is a primary ideal then rad$(Q)$ is a prime ideal (cf. Exercise 30).

## 7.5. Rings of Fractions

Let $R$ be a commutative ring with identity $1 \neq 0$.

**1.** Fill in all the details in the proof of Theorem 15.

Let $R$ be a commutative ring with identity $1 \neq 0$.

**2.** Let $R$ be an integral domain and let $D$ be a nonempty subset of $R$ that is closed under multiplication. Prove that the ring of fractions $D^{-1}R$ is isomorphic to a subring of the quotient field of $R$ (hence is also an integral domain).

Let $R$ be a commutative ring with identity $1 \neq 0$.

**3.** Let $F$ be a field. Prove that $F$ contains a unique smallest subfield $F_0$ and that $F_0$ is isomorphic to either $\mathbb{Q}$ or $\mathbb{Z}/p\mathbb{Z}$ for some prime $p$ ($F_0$ is called the *prime subfield* of $F$). [See Exercise 26, Section 3.]

Let $R$ be a commutative ring with identity $1 \neq 0$.

**4.** Prove that any subfield of $\mathbb{R}$ must contain $\mathbb{Q}$.

Let $R$ be a commutative ring with identity $1 \neq 0$.

**5.** If $F$ is a field, prove that the field of fractions of $F[[x]]$ (the ring of formal power series in the indeterminate $x$ with coefficients in $F$) is the ring $F((x))$ of formal Laurent series (cf. Exercises 3 and 5 of Section 2). Show the field of fractions of the power series ring $\mathbb{Z}[[x]]$ is *properly* contained in the field of Laurent series $\mathbb{Q}((x))$. [Consider the series for $e^x$.]

Let $R$ be a commutative ring with identity $1 \neq 0$.

**6.** Prove that the real numbers, $\mathbb{R}$, contain a subring $A$ with $1 \in A$ and $A$ maximal (under inclusion) with respect to the property that $\frac{1}{2} \notin A$. [Use Zorn's Lemma.] (Exercise 13 in Section 15.3 shows $\mathbb{R}$ is the quotient field of $A$, so $\mathbb{R}$ is the quotient field of a proper subring.)

## 7.6. The Chinese Remainder Theorem

Let $R$ be a ring with identity $1 \neq 0$.

**1.** An element $e \in R$ is called an *idempotent* if $e^2 = e$. Assume $e$ is an idempotent in $R$ and $er = re$ for all $r \in R$. Prove that $Re$ and $R(1 - e)$ are two-sided ideals of $R$ and that $R \cong Re \times R(1 - e)$. Show that $e$ and $1 - e$ are identities for the subrings $Re$ and $R(1 - e)$ respectively.

Let $R$ be a ring with identity $1 \neq 0$.

**2.** Let $R$ be a finite Boolean ring with identity $1 \neq 0$ (cf. Exercise 15 of Section 1). Prove that $R \cong \mathbb{Z}/2\mathbb{Z} \times \cdots \times \mathbb{Z}/2\mathbb{Z}$. [Use the preceding exercise.]

Let $R$ be a ring with identity $1 \neq 0$.

**3.** Let $R$ and $S$ be rings with identities. Prove that every ideal of $R \times S$ is of the form $I \times J$ where $I$ is an ideal of $R$ and $J$ is an ideal of $S$.

Let $R$ be a ring with identity $1 \neq 0$.

**4.** Prove that if $R$ and $S$ are nonzero rings then $R \times S$ is never a field.

Let $R$ be a ring with identity $1 \neq 0$.

**5.** Let $n_1, n_2, \ldots, n_k$ be integers which are relatively prime in pairs: $(n_i, n_j) = 1$ for all $i \neq j$.

**(a)** Show that the Chinese Remainder Theorem implies that for any $a_1, \ldots, a_k \in \mathbb{Z}$ there is a solution $x \in \mathbb{Z}$ to the simultaneous congruences

$$x \equiv a_1 \bmod n_1 , \qquad x \equiv a_2 \bmod n_2 , \quad \ldots , \quad x \equiv a_k \bmod n_k$$

and that the solution $x$ is unique mod $n = n_1 n_2 \ldots n_k$.

**(b)** Let $n_i' = n/n_i$ be the quotient of $n$ by $n_i$, which is relatively prime to $n_i$ by assumption. Let $t_i$ be the inverse of $n_i'$ mod $n_i$. Prove that the solution $x$ in (a) is given by

$$x = a_1 t_1 n_1' + a_2 t_2 n_2' + \cdots + a_k t_k n_k' \bmod n.$$

Note that the elements $t_i$ can be quickly found by the Euclidean Algorithm as described in Section 2 of the Preliminaries chapter (writing $an_i + bn_i' = (n_i, n_i') = 1$ gives $t_i = b$) and that these then quickly give the solutions to the system of congruences above for any choice of $a_1, a_2, \ldots, a_k$.

**(c)** Solve the simultaneous system of congruences

$$x \equiv 1 \bmod 8 , \qquad x \equiv 2 \bmod 25 , \quad \text{and} \quad x \equiv 3 \bmod 81$$

and the simultaneous system

$$y \equiv 5 \bmod 8 , \qquad y \equiv 12 \bmod 25 , \quad \text{and} \quad y \equiv 47 \bmod 81.$$

Let $R$ be a ring with identity $1 \neq 0$.

**6.** Let $f_1(x), f_2(x), \ldots, f_k(x)$ be polynomials with integer coefficients of the same degree $d$. Let $n_1, n_2, \ldots, n_k$ be integers which are relatively prime in pairs (i.e., $(n_i, n_j) = 1$ for all $i \neq j$). Use the Chinese Remainder Theorem to prove there exists a polynomial $f(x)$ with integer coefficients and of degree $d$ with

$$f(x) \equiv f_1(x) \bmod n_1 , \qquad f(x) \equiv f_2(x) \bmod n_2 , \quad \ldots , \quad f(x) \equiv f_k(x) \bmod n_k$$

i.e., the coefficients of $f(x)$ agree with the coefficients of $f_i(x) \bmod n_i$. Show that if all the $f_i(x)$ are monic, then $f(x)$ may also be chosen monic. [Apply the Chinese Remainder Theorem in $\mathbb{Z}$ to each of the coefficients separately.]

Let $R$ be a ring with identity $1 \neq 0$.

**7.** Let $m$ and $n$ be positive integers with $n$ dividing $m$. Prove that the natural surjective ring projection $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ is also surjective on the units: $(\mathbb{Z}/m\mathbb{Z})^\times \to (\mathbb{Z}/n\mathbb{Z})^\times$.

Let $R$ be a ring with identity $1 \neq 0$.

The next four exercises develop the concept of *direct limits* and the "dual" notion of *inverse limits*. In these exercises $I$ is a nonempty index set with a partial order $\leq$ (cf. Appendix I). For each $i \in I$ let $A_i$ be an additive abelian group. In Exercise 8 assume also that $I$ is a *directed set*: for every $i, j \in I$ there is some $k \in I$ with $i \leq k$ and $j \leq k$.

**8.** Suppose for every pair of indices $i, j$ with $i \leq j$ there is a map $\rho_{ij} : A_i \to A_j$ such that the following hold:

**i.** $\rho_{jk} \circ \rho_{ij} = \rho_{ik}$ whenever $i \leq j \leq k$, and
**ii.** $\rho_{ii} = 1$ for all $i \in I$.

Let $B$ be the disjoint union of all the $A_i$. Define a relation $\sim$ on $B$ by

$$a \sim b \text{ if and only if there exists } k \text{ with } i, j \leq k \text{ and } \rho_{ik}(a) = \rho_{jk}(b),$$

for $a \in A_i$ and $b \in A_j$.

**(a)** Show that $\sim$ is an equivalence relation on $B$. (The set of equivalence classes is called the *direct* or *inductive limit* of the directed system $\{A_i\}$, and is denoted $\varinjlim A_i$. In the remaining parts of this exercise let $A = \varinjlim A_i$.)

**(b)** Let $\bar{x}$ denote the class of $x$ in $A$ and define $\rho_i : A_i \to A$ by $\rho_i(a) = \bar{a}$. Show that if each $\rho_{ij}$ is injective, then so is $\rho_i$ for all $i$ (so we may then identify each $A_i$ as a subset of $A$).

**(c)** Assume all $\rho_{ij}$ are group homomorphisms. For $a \in A_i, b \in A_j$ show that the operation

$$\bar{a} + \bar{b} = \overline{\rho_{ik}(a) + \rho_{jk}(b)}$$

where $k$ is any index with $i, j \leq k$, is well defined and makes $A$ into an abelian group. Deduce that the maps $\rho_i$ in (b) are group homomorphisms from $A_i$ to $A$.

**(d)** Show that if all $A_i$ are commutative rings with 1 and all $\rho_{ij}$ are ring homomorphisms that send 1 to 1, then $A$ may likewise be given the structure of a commutative ring with 1 such that all $\rho_i$ are ring homomorphisms.

**(e)** Under the hypotheses in (c) prove that the direct limit has the following *universal property*: if $C$ is any abelian group such that for each $i \in I$ there is a homomorphism $\varphi_i : A_i \to C$ with $\varphi_i = \varphi_j \circ \rho_{ij}$ whenever $i \leq j$, then there is a unique homomorphism $\varphi : A \to C$ such that $\varphi \circ \rho_i = \varphi_i$ for all $i$.

Let $R$ be a ring with identity $1 \neq 0$.

The next four exercises develop the concept of *direct limits* and the "dual" notion of *inverse limits*. In these exercises $I$ is a nonempty index set with a partial order $\leq$ (cf. Appendix I). For each $i \in I$ let $A_i$ be an additive abelian group. In Exercise 8 assume also that $I$ is a *directed set*: for every $i, j \in I$ there is some $k \in I$ with $i \leq k$ and $j \leq k$.

**9.** Let $I$ be the collection of open intervals $U = (a, b)$ on the real line containing a fixed real number $p$. Order these by reverse inclusion: $U \leq V$ if $V \subseteq U$ (note that $I$ is a directed set). For each $U$ let $A_U$ be the ring of continuous real valued functions on $U$. For $V \subseteq U$ define the *restriction maps* $\rho_{UV} : A_U \to A_V$ by $f \mapsto f|_V$, the usual restriction of a function on $U$ to a function on the subset $V$ (which is easily seen to be a ring homomorphism). Let $A = \varinjlim A_U$ be the direct limit. In the notation of the preceding exercise, show that the maps $\rho_U : A_U \to A$ are *not* injective but are all surjective ($A$ is called the ring of *germs of continuous functions* at $p$).

Let $R$ be a ring with identity $1 \neq 0$.

The next four exercises develop the concept of *direct limits* and the "dual" notion of *inverse limits*. In these exercises $I$ is a nonempty index set with a partial order $\leq$ (cf. Appendix I). For each $i \in I$ let $A_i$ be an additive abelian group. In Exercise 8 assume also that $I$ is a *directed set*: for every $i, j \in I$ there is some $k \in I$ with $i \leq k$ and $j \leq k$.

We now develop the notion of *inverse limits*. Continue to assume $I$ is a partially ordered set (but not necessarily directed), and $A_i$ is a group for all $i \in I$.

**10.** Suppose for every pair of indices $i, j$ with $i \leq j$ there is a map $\mu_{ji} : A_j \to A_i$ such that the following hold:

**i.** $\mu_{ji} \circ \mu_{kj} = \mu_{ki}$ whenever $i \leq j \leq k$, and
**ii.** $\mu_{ii} = 1$ for all $i \in I$.

Let $P$ be the subset of elements $(a_i)_{i \in I}$ in the direct product $\prod_{i \in I} A_i$ such that $\mu_{ji}(a_j) = a_i$ whenever $i \leq j$ (here $a_i$ and $a_j$ are the $i^{\text{th}}$ and $j^{\text{th}}$ components respectively of the element in the direct product). The set $P$ is called the *inverse* or *projective limit* of the system $\{A_i\}$, and is denoted $\varprojlim A_i$.)

**(a)** Assume all $\mu_{ji}$ are group homomorphisms. Show that $P$ is a subgroup of the direct product group (cf. Exercise 15, Section 5.1).

**(b)** Assume the hypotheses in (a), and let $I = \mathbb{Z}^+$ (usual ordering). For each $i \in I$ let $\mu_i : P \to A_i$ be the projection of $P$ onto its $i^{\text{th}}$ component. Show that if each $\mu_{ji}$ is surjective, then so is $\mu_i$ for all $i$ (so each $A_i$ is a quotient group of $P$).

**(c)** Show that if all $A_i$ are commutative rings with 1 and all $\mu_{ji}$ are ring homomorphisms that send 1 to 1, then $A$ may likewise be given the structure of a commutative ring with 1 such that all $\mu_i$ are ring homomorphisms.

**(d)** Under the hypotheses in (a) prove that the inverse limit has the following *universal property*: if $D$ is any group such that for each $i \in I$ there is a homomorphism $\pi_i : D \to A_i$ with $\pi_i = \mu_{ji} \circ \pi_j$ whenever $i \leq j$, then there is a unique homomorphism $\pi : D \to P$ such that $\mu_i \circ \pi = \pi_i$ for all $i$.

Let $R$ be a ring with identity $1 \neq 0$.

The next four exercises develop the concept of *direct limits* and the "dual" notion of *inverse limits*. In these exercises $I$ is a nonempty index set with a partial order $\leq$ (cf. Appendix I). For each $i \in I$ let $A_i$ be an additive abelian group. In Exercise 8 assume also that $I$ is a *directed set*: for every $i, j \in I$ there is some $k \in I$ with $i \leq k$ and $j \leq k$.

We now develop the notion of *inverse limits*. Continue to assume $I$ is a partially ordered set (but not necessarily directed), and $A_i$ is a group for all $i \in I$.

**11.** Let $p$ be a prime let $I = \mathbb{Z}^+$, let $A_i = \mathbb{Z}/p^i\mathbb{Z}$ and let $\mu_{ji}$ be the natural projection maps

$$\mu_{ji} : a \pmod{p^j} \longmapsto a \pmod{p^i}.$$

The inverse limit $\varprojlim \mathbb{Z}/p^i\mathbb{Z}$ is called the ring of *p-adic integers*, and is denoted by $\mathbb{Z}_p$.

**(a)** Show that every element of $\mathbb{Z}_p$ may be written uniquely as an infinite formal sum $b_0 + b_1 p + b_2 p^2 + b_3 p^3 + \cdots$ with each $b_i \in \{0, 1, \ldots, p-1\}$. Describe the rules for adding and multiplying such formal sums corresponding to addition and multiplication in the ring $\mathbb{Z}_p$. [Write a least residue in each $\mathbb{Z}/p^i\mathbb{Z}$ in its base $p$ expansion and then describe the maps $\mu_{ji}$.] (Note in particular that $\mathbb{Z}_p$ is uncountable.)

**(b)** Prove that $\mathbb{Z}_p$ is an integral domain that contains a copy of the integers.

**(c)** Prove that $b_0 + b_1 p + b_2 p^2 + b_3 p^3 + \cdots$ as in (a) is a unit in $\mathbb{Z}_p$ if and only if $b_0 \neq 0$.

**(d)** Prove that $p\mathbb{Z}_p$ is the unique maximal ideal of $\mathbb{Z}_p$ and $\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{Z}/p\mathbb{Z}$ (where $p = 0 + 1p + 0p^2 + 0p^3 + \cdots$). Prove that every ideal of $\mathbb{Z}_p$ is of the form $p^n\mathbb{Z}_p$ for some integer $n \geq 0$.

**(e)** Show that if $a_1 \not\equiv 0 \pmod{p}$ then there is an element $a = (a_i)$ in the direct limit $\mathbb{Z}_p$ satisfying $a_j^p \equiv 1 \pmod{p^j}$ and $\mu_{j1}(a_j) = a_1$ for all $j$. Deduce that $\mathbb{Z}_p$ contains $p - 1$ distinct $(p-1)^{\text{st}}$ roots of 1.

# 8. Euclidean Domains, Principal Ideal Domains and Unique Factorization Domains

## 8.1. Euclidean Domains

**1.** For each of the following five pairs of integers $a$ and $b$, determine their greatest common divisor $d$ and write $d$ as a linear combination $ax + by$ of $a$ and $b$.

**(a)** $a = 20, b = 13$.

**(b)** $a = 69, b = 372$.

**(c)** $a = 11391, b = 5673$.

**(d)** $a = 507885, b = 60808$.

**(e)** $a = 91442056588823$, $b = 779086434385541$ (the Euclidean Algorithm requires only 7 steps for these integers).

**2.** For each of the following pairs of integers $a$ and $n$, show that $a$ is relatively prime to $n$ and determine the inverse of $a$ mod $n$ (cf. Section 3 of the Preliminaries chapter).

**(a)** $a = 13, n = 20$.

**(b)** $a = 69, n = 89$.

**(c)** $a = 1891, n = 3797$.

**(d)** $a = 6003722857, n = 77695236973$ (the Euclidean Algorithm requires only 3 steps for these integers).

**3.** Let $R$ be a Euclidean Domain. Let $m$ be the minimum integer in the set of norms of nonzero elements of $R$. Prove that every nonzero element of $R$ of norm $m$ is a unit. Deduce that a nonzero element of norm zero (if such an element exists) is a unit.

**4.** Let $R$ be a Euclidean Domain.

**(a)** Prove that if $(a, b) = 1$ and $a$ divides $bc$, then $a$ divides $c$. More generally, show that if $a$ divides $bc$ with nonzero $a, b$ then $\dfrac{a}{(a, b)}$ divides $c$.

**(b)** Consider the Diophantine Equation $ax + by = N$ where $a, b$ and $N$ are integers and $a, b$ are nonzero. Suppose $x_0, y_0$ is a solution: $ax_0 + by_0 = N$. Prove that the full set of solutions to this equation is given by

$$x = x_0 + m\frac{b}{(a, b)}, \qquad y = y_0 - m\frac{a}{(a, b)}$$

as $m$ ranges over the integers. [If $x, y$ is a solution to $ax + by = N$, show that $a(x - x_0) = b(y_0 - y)$ and use (a).]

**5.** Determine all integer solutions of the following equations:

**(a)** $2x + 4y = 5$

**(b)** $17x + 29y = 31$

**(c)** $85x + 145y = 505.$

**6.** (*The Postage Stamp Problem*) Let $a$ and $b$ be two relatively prime positive integers. Prove that every sufficiently large positive integer $N$ can be written as a linear combination $ax+by$ of $a$ and $b$ where $x$ and $y$ are both *nonnegative*, i.e., there is an integer $N_0$ such that for all $N \geq N_0$ the equation $ax + by = N$ can be solved with both $x$ and $y$ nonnegative integers. Prove in fact that the integer $ab - a - b$ cannot be written as a positive linear combination of $a$ and $b$ but that every integer greater than $ab - a - b$ is a positive linear combination of $a$ and $b$ (so every "postage" greater than $ab - a - b$ can be obtained using only stamps in denominations $a$ and $b$).

**7.** Find a generator for the ideal $(85, 1+13i)$ in $\mathbb{Z}[i]$, i.e., a greatest common divisor for 85 and $1+13i$, by the Euclidean Algorithm. Do the same for the ideal $(47 - 13i, 53 + 56i)$.

It is known (but not so easy to prove) that $D = -1, -2, -3, -7, -11, -19, -43, -67$, and $-163$ are the only negative values of $D$ for which every ideal in $\mathcal{O}$ is principal (i.e., $\mathcal{O}$ is a P.I.D. in the terminology of the next section). The results of the next exercise determine precisely which quadratic integer rings with $D < 0$ are Euclidean.

**8.** Let $F = \mathbb{Q}(\sqrt{D})$ be a quadratic field with associated quadratic integer ring $\mathcal{O}$ and field norm $N$ as in Section 7.1.

**(a)** Suppose $D$ is $-1, -2, -3, -7$ or $-11$. Prove that $\mathcal{O}$ is a Euclidean Domain with respect to $N$. [Modify the proof for $\mathbb{Z}[i]$ ($D = -1$) in the text. For $D = -3, -7, -11$ prove that every element of $F$ differs from an element in $\mathcal{O}$ by an element whose norm is at most $(1 + |D|)^2/(16|D|)$, which is less than 1 for these values of $D$. Plotting the points of $\mathcal{O}$ in $\mathbb{C}$ may be helpful.]

**(b)** Suppose that $D = -43, -67$, or $-163$. Prove that $\mathcal{O}$ is not a Euclidean Domain with respect to any norm. [Apply the same proof as for $D = -19$ in the text.]

**9.** Prove that the ring of integers $\mathcal{O}$ in the quadratic integer ring $\mathbb{Q}(\sqrt{2})$ is a Euclidean Domain with respect to the norm given by the absolute value of the field norm $N$ in Section 7.1.

**10.** Prove that the quotient ring $\mathbb{Z}[i]/I$ is finite for any nonzero ideal $I$ of $\mathbb{Z}[i]$. [Use the fact that $I = (\alpha)$ for some nonzero $\alpha$ and then use the Division Algorithm in this Euclidean Domain to see that every coset of $I$ is represented by an element of norm less than $N(\alpha)$.]

**11.** Let $R$ be a commutative ring with 1 and let $a$ and $b$ be nonzero elements of $R$. A *least common multiple* of $a$ and $b$ is an element $e$ of $R$ such that

**(i)** $a \mid e$ and $b \mid e$, and

**(ii)** if $a \mid e'$ and $b \mid e'$ then $e \mid e'$.

**(a)** Prove that a least common multiple of $a$ and $b$ (if such exists) is a generator for the unique largest principal ideal contained in $(a) \cap (b)$.

**(b)** Deduce that any two nonzero elements in a Euclidean Domain have a least common multiple which is unique up to multiplication by a unit.

**(c)** Prove that in a Euclidean Domain the least common multiple of $a$ and $b$ is $\dfrac{ab}{(a, b)}$, where $(a, b)$ is the greatest common divisor of $a$ and $b$.

**12.** (*A Public Key Code*) Let $N$ be a positive integer. Let $M$ be an integer relatively prime to $N$ and let $d$ be an integer relatively prime to $\varphi(N)$, where $\varphi$ denotes Euler's $\varphi$-function. Prove that if $M_1 \equiv M^d \pmod{N}$ then $M \equiv M_1^{d'} \pmod{N}$ where $d'$ is the inverse of $d$ mod $\varphi(N)$: $dd' \equiv 1 \pmod{\varphi(N)}$.

*Remark:* This result is the basis for a standard *Public Key Code*. Suppose $N = pq$ is the product of two distinct large primes (each on the order of 100 digits, for example). If $M$ is a message, then $M_1 \equiv M^d \pmod{N}$ is a scrambled (*encoded*) version of $M$, which can be unscrambled (*decoded*) by computing $M_1^{d'} \pmod{N}$ (these powers can be computed quite easily even for large values of $M$ and $N$ by successive squarings). The values of $N$ and $d$ (but not $p$ and $q$) are made publicly known (hence the name) and then anyone with a message $M$ can send their encoded message $M^d \pmod{N}$. To decode the message it seems necessary to determine $d'$, which requires the determination of the value $\varphi(N) = \varphi(pq) = (p-1)(q-1)$ (no one has as yet *proved* that there is no other decoding scheme, however). The success of this method as a code rests on the necessity of determining the *factorization* of $N$ into primes, for which no sufficiently efficient algorithm exists (for example, the most naive method of checking all factors up to $\sqrt{N}$ would here require on the order of $10^{100}$ computations, or approximately 300 years even at 10 billion computations per second, and of course one can always increase the size of $p$ and $q$).

## 8.2. Principal Ideal Domains (P.I.D.s)

**1.** Prove that in a Principal Ideal Domain two ideals $(a)$ and $(b)$ are comaximal (cf. Section 7.6) if and only if a greatest common divisor of $a$ and $b$ is 1 (in which case $a$ and $b$ are said to be *coprime* or *relatively prime*).

**2.** Prove that any two nonzero elements of a P.I.D. have a least common multiple (cf. Exercise 11, Section 1).

**3.** Prove that a quotient of a P.I.D. by a prime ideal is again a P.I.D.

**4.** Let $R$ be an integral domain. Prove that if the following two conditions hold then $R$ is a Principal Ideal Domain:

**(i)** any two nonzero elements $a$ and $b$ in $R$ have a greatest common divisor which can be written in the form $ra + sb$ for some $r, s \in R$, and

**(ii)** if $a_1, a_2, a_3, \ldots$ are nonzero elements of $R$ such that $a_{i+1} \mid a_i$ for all $i$, then there is a positive integer $N$ such that $a_n$ is a unit times $a_N$ for all $n \geq N$.

**5.** Let $R$ be the quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$. Define the ideals $I_2 = (2, 1+\sqrt{-5})$, $I_3 = (3, 2+\sqrt{-5})$, and $I_3' = (3, 2-\sqrt{-5})$.

**(a)** Prove that $I_2$, $I_3$, and $I_3'$ are nonprincipal ideals in $R$. [Note that Example 2 following Proposition 1 proves this for $I_3$.]

**(b)** Prove that the product of two nonprincipal ideals can be principal by showing that $I_2^2$ is the principal ideal generated by 2, i.e., $I_2^2 = (2)$.

**(c)** Prove similarly that $I_2 I_3 = (1-\sqrt{-5})$ and $I_2 I_3' = (1+\sqrt{-5})$ areprincipal. Conclude that the principal ideal (6) is the product of 4 ideals: $(6) = I_2^2 I_3 I_3'$.

**6.** Let $R$ be an integral domain and suppose that every *prime* ideal in $R$ is principal. This exercise proves that every ideal of $R$ is principal, i.e., $R$ is a P.I.D.

**(a)** Assume that the set of ideals of $R$ that are not principal is nonempty and prove that this set has a maximal element under inclusion (which, by hypothesis, is not prime). [Use Zorn's Lemma.]

**(b)** Let $I$ be an ideal which is maximal with respect to being nonprincipal, and let $a, b \in R$ with $ab \in I$ but $a \notin I$ and $b \notin I$. Let $I_a = (I, a)$ be the ideal generated by $I$ and $a$, let $I_b = (I, b)$ be the ideal generated by $I$ and $b$, and define $J = \{r \in R \mid rI_a \subseteq I\}$. Prove that $I_a = (\alpha)$ and $J = (\beta)$ are principal ideals in $R$ with $I \subsetneq I_b \subseteq J$ and $I_a J = (\alpha\beta) \subseteq I$.

**(c)** If $x \in I$ show that $x = s\alpha$ for some $s \in J$. Deduce that $I = I_a J$ is principal, a contradiction, and conclude that $R$ is a P.I.D.

**7.** An integral domain $R$ in which every ideal generated by two elements is principal (i.e., for every $a, b \in R$, $(a, b) = (d)$ for some $d \in R$) is called a *Bezout Domain.* [cf. also Exercise 11 in Section 3.]

**(a)** Prove that the integral domain $R$ is a Bezout Domain if and only if every pair of elements $a, b$ of $R$ has a g.c.d. $d$ in $R$ that can be written as an $R$-linear combination of $a$ and $b$, i.e., $d = ax + by$ for some $x, y \in R$.

**(b)** Prove that every finitely generated ideal of a Bezout Domain is principal. [cf. the exercises in Sections 9.2 and 9.3 for Bezout Domains in which not every ideal is principal.]

**(c)** Let $F$ be the fraction field of the Bezout Domain $R$. Prove that every element of $F$ can be written in the form $a/b$ with $a, b \in R$ and $a$ and $b$ relatively prime (cf. Exercise 1).

**8.** Prove that if $R$ is a Principal Ideal Domain and $D$ is a multiplicatively closed subset of $R$, then $D^{-1}R$ is also a P.I.D. (cf. Section 7.5).

## 8.3. Unique Factorization Domains (U.F.D.s)

**1.** Let $G = \mathbb{Q}^\times$ be the multiplicative group of nonzero rational numbers. If $\alpha = p/q \in G$, where $p$ and $q$ are relatively prime integers, let $\varphi : G \to G$ be the map which interchanges the primes 2 and 3 in the prime power factorizations of $p$ and $q$ (so, for example, $\varphi(2^4 3^{11} 5^1 13^2) = 3^4 2^{11} 5^1 13^2$, $\varphi(3/16) = \varphi(3/2^4) = 2/3^4 = 2/81$, and $\varphi$ is the identity on all rational numbers with numerators and denominators relatively prime to 2 and to 3).

**(a)** Prove that $\varphi$ is a group isomorphism.

**(b)** Prove that there are infinitely many isomorphisms of the group $G$ to itself.

**(c)** Prove that none of the isomorphisms above can be extended to an isomorphism of the *ring* $\mathbb{Q}$ to itself. In fact prove that the identity map is the only ring isomorphism of $\mathbb{Q}$.

**2.** Let $a$ and $b$ be nonzero elements of the Unique Factorization Domain $R$. Prove that $a$ and $b$ have a least common multiple (cf. Exercise 11 of Section 1) and describe it in terms of the prime factorizations of $a$ and $b$ in the same fashion that Proposition 13 describes their greatest common divisor.

**3.** Determine all the representations of the integer $2130797 = 17^2 \cdot 73 \cdot 101$ as a sum of two squares.

**4.** Prove that if an integer is the sum of two rational squares, then it is the sum of two integer squares (for example, $13 = (1/5)^2 + (18/5)^2 = 2^2 + 3^2$).

**5.** Let $R = \mathbb{Z}[\sqrt{-n}]$ where $n$ is a squarefree integer greater than 3.

**(a)** Prove that 2, $\sqrt{-n}$ and $1 + \sqrt{-n}$ are irreducibles in $R$.

**(b)** Prove that $R$ is not a U.F.D. Conclude that the quadratic integer ring $\mathcal{O}$ is not a U.F.D. for $D \equiv 2, 3 \bmod 4$, $D < -3$ (so also not Euclidean and not a P.I.D.). [Show that either $\sqrt{-n}$ or $1 + \sqrt{-n}$ is not prime.]

**(c)** Give an explicit ideal in $R$ that is not principal. [Using (b) consider a maximal ideal containing the nonprime ideal $(\sqrt{-n})$ or $(1 + \sqrt{-n})$.]

**6.** **(a)** Prove that the quotient ring $\mathbb{Z}[i]/(1+i)$ is a field of order 2.

**(b)** Let $q \in \mathbb{Z}$ be a prime with $q \equiv 3 \bmod 4$. Prove that the quotient ring $\mathbb{Z}[i]/(q)$ is a field with $q^2$ elements.

**(c)** Let $p \in \mathbb{Z}$ be a prime with $p \equiv 1 \bmod 4$ and write $p = \pi\bar{\pi}$ as in Proposition 18. Show that the hypotheses for the Chinese Remainder Theorem (Theorem 17 in Section 7.6) are satisfied and that $\mathbb{Z}[i]/(p) \cong \mathbb{Z}[i]/(\pi) \times \mathbb{Z}[i]/(\bar{\pi})$ as rings. Show that the quotient ring $\mathbb{Z}[i]/(p)$ has order $p^2$ and conclude that $\mathbb{Z}[i]/(\pi)$ and $\mathbb{Z}[i]/(\bar{\pi})$ are both fields of order $p$.

**7.** Let $\pi$ be an irreducible element in $\mathbb{Z}[i]$.

**(a)** For any integer $n \geq 0$, prove that $(\pi^{n+1}) = \pi^{n+1}\mathbb{Z}[i]$ is an ideal in $(\pi^n) = \pi^n\mathbb{Z}[i]$ and that multiplication by $\pi^n$ induces an isomorphism $\mathbb{Z}[i]/(\pi) \cong (\pi^n)/(\pi^{n+1})$ as additive abelian groups.

**(b)** Prove that $|\mathbb{Z}[i]/(\pi^n)| = |\mathbb{Z}[i]/(\pi)|^n$.

**(c)** Prove for any nonzero $\alpha$ in $\mathbb{Z}[i]$ that the quotient ring $\mathbb{Z}[i]/(\alpha)$ has order equal to $N(\alpha)$. [Use (b) together with the Chinese Remainder Theorem and the results of the previous exercise.]

**8.** Let $R$ be the quadratic integer ring $\mathbb{Z}[\sqrt{-5}]$ and define the ideals $I_2 = (2, 1+\sqrt{-5})$, $I_3 = (3, 2+\sqrt{-5})$, and $I_3' = (3, 2-\sqrt{-5})$.

**(a)** Prove that 2, 3, $1+\sqrt{-5}$ and $1-\sqrt{-5}$ are irreducibles in $R$, no two of which are associate in $R$, and that $6 = 2 \cdot 3 = (1+\sqrt{-5}) \cdot (1-\sqrt{-5})$ are two distinct factorizations of 6 into irreducibles in $R$.

**(b)** Prove that $I_2$, $I_3$, and $I_3'$ are prime ideals in $R$. [One approach: for $I_3$, observe that $R/I_3 \cong (R/(3))/(I_3/(3))$ by the Third Isomorphism Theorem for Rings. Show that $R/(3)$ has 9 elements, $(I_3/(3))$ has 3 elements, and that $R/I_3 \cong \mathbb{Z}/3\mathbb{Z}$ as an additive abelian group. Conclude that $I_3$ is a maximal (hence prime) ideal and that $R/I_3 \cong \mathbb{Z}/3\mathbb{Z}$ as rings.]

**(c)** Show that the factorizations in (a) imply the equality of ideals $(6) = (2)(3)$ and $(6) = (1+\sqrt{-5})(1-\sqrt{-5})$. Show that these two ideal factorizations give the same factorization of the ideal (6) as the product of prime ideals (cf. Exercise 5 in Section 2).

**9.** Suppose that the quadratic integer ring $\mathcal{O}$ is a P.I.D. Prove that the absolute value of the field norm $N$ on $\mathcal{O}$ (cf. Section 7.1) is a Dedekind–Hasse norm on $\mathcal{O}$. Conclude that if the quadratic integer ring $\mathcal{O}$ possesses *any* Dedekind–Hasse norm, then in fact the absolute value of the field norm on $\mathcal{O}$ already provides a Dedekind–Hasse norm on $\mathcal{O}$. [If $\alpha, \beta \in \mathcal{O}$ then $(\alpha, \beta) = (\gamma)$ for some $\gamma \in \mathcal{O}$. Show that if $\beta$ does not divide $\alpha$ then $0 < |N(\gamma)| < |N(\beta)|$ — use the fact that the units in $\mathcal{O}$ are precisely the elements whose norm is $\pm 1$.]

*Remark:* If $\mathcal{O}$ is a Euclidean Domain with respect to some norm it is not necessarily true that it is a Euclidean Domain with respect to the absolute value of the field norm (although this is true for $D < 0$, cf. Exercise 8 in Section 1). An example is $D = 69$ (cf. D. Clark, *A quadratic field which is Euclidean but not norm-Euclidean*, Manuscripta Math., 83(1994), pp. 327–330).

**10.** (*k-stage Euclidean Domains*) Let $R$ be an integral domain and let $N : R \to \mathbb{Z}^+ \cup \{0\}$ be a norm on $R$. The ring $R$ is Euclidean with respect to $N$ if for any $a, b \in R$ with $b \neq 0$, there exist elements $q$ and $r$ in $R$ with

$$a = qb + r \qquad \text{with } r = 0 \text{ or } N(r) < N(b).$$

Suppose now that this condition is weakened, namely that for any $a, b \in R$ with $b \neq 0$, there exist elements $q, q'$ and $r, r'$ in $R$ with

$$a = qb + r, \quad b = q'r + r' \qquad \text{with } r' = 0 \text{ or } N(r') < N(b),$$

i.e., the remainder after two divisions is smaller. Call such a domain a *2-stage Euclidean Domain.*

**(a)** Prove that iterating the divisions in a 2-stage Euclidean Domain produces a greatest common divisor of $a$ and $b$ which is a linear combination of $a$ and $b$. Conclude that every *finitely generated* ideal of a 2-stage Euclidean Domain is principal. (There are 2-stage Euclidean Domains that are *not* P.I.D.s, however.) [Imitate the proof of Theorem 4.]

**(b)** Prove that a 2-stage Euclidean Domain in which every nonzero nonunit can be factored into a finite number of irreducibles is a Unique Factorization Domain. [Prove first that irreducible elements are prime, as follows. If $p$ is irreducible and $p \mid ab$ with $p$ not dividing $a$, use part (a) to write $px + ay = 1$ for some $x, y$. Multiply through by $b$ to conclude that $p \mid b$, so $p$ is prime. Now follow the proof of uniqueness in Theorem 14.]

**(c)** Make the obvious generalization to define the notion of a $k$-stage Euclidean Domain for any integer $k \geq 1$. Prove that statements (a) and (b) remain valid if "2-stage Euclidean" is replaced by "$k$-stage Euclidean."

*Remarks:* There are examples of rings which are 2-stage Euclidean but are not Euclidean. There are also examples of rings which are not Euclidean with respect to a given norm but which are $k$-stage Euclidean with respect to the norm (for example, the ring $\mathbb{Z}[\sqrt{14}]$ is not Euclidean with respect to the usual norm $N(a+b\sqrt{14}) = |a^2 - 14b^2|$, but is 2-stage Euclidean with respect to this norm). The $k$-stage Euclidean condition is also related to the question of whether the group $GL_n(R)$ of invertible $n \times n$ matrices with entries from $R$ is generated by elementary matrices (matrices with 1's along the main diagonal, a single 1 somewhere off the main diagonal, and 0's elsewhere).

**11.** (*Characterization of P.I.D.s*) Prove that $R$ is a P.I.D. if and only if $R$ is a U.F.D. that is also a Bezout Domain (cf. Exercise 7 in Section 2). [One direction is given by Theorem 14. For the converse, let $a$ be a nonzero element of the ideal $I$ with a minimal number of irreducible factors. Prove that $I = (a)$ by showing that if there is an element $b \in I$ that is not in $(a)$ then $(a, b) = (d)$ leads to a contradiction.]

# 9. Polynomial Rings

## 9.1. Definitions and Basic Properties

**1.** Let $p(x, y, z) = 2x^2y - 3xy^3z + 4y^2z^5$ and $q(x, y, z) = 7x^2 + 5x^2y^3z^4 - 3x^2z^3$ be polynomials in $\mathbb{Z}[x, y, z]$.

**(a)** Write each of $p$ and $q$ as a polynomial in $x$ with coefficients in $\mathbb{Z}[y, z]$.

**(b)** Find the degree of each of $p$ and $q$.

**(c)** Find the degree of $p$ and $q$ in each of the three variables $x$, $y$ and $z$.

**(d)** Compute $pq$ and find the degree of $pq$ in each of the three variables $x$, $y$ and $z$.

**(e)** Write $pq$ as a polynomial in the variable $z$ with coefficients in $\mathbb{Z}[x, y]$.

**2.** Repeat the preceding exercise under the assumption that the coefficients of $p$ and $q$ are in $\mathbb{Z}/3\mathbb{Z}$.

**3.** If $R$ is a commutative ring and $x_1, x_2, \ldots, x_n$ are independent variables over $R$, prove that $R[x_{\pi(1)}, x_{\pi(2)}, \ldots, x_{\pi(n)}]$ is isomorphic to $R[x_1, x_2, \ldots, x_n]$ for any permutation $\pi$ of $\{1, 2, \ldots, n\}$.

**4.** Prove that the ideals $(x)$ and $(x, y)$ are prime ideals in $\mathbb{Q}[x, y]$ but only the latter ideal is a maximal ideal.

**5.** Prove that $(x, y)$ and $(2, x, y)$ are prime ideals in $\mathbb{Z}[x, y]$ but only the latter ideal is a maximal ideal.

**6.** Prove that $(x, y)$ is not a principal ideal in $\mathbb{Q}[x, y]$.

**7.** Let $R$ be a commutative ring with 1. Prove that a polynomial ring in more than one variable over $R$ is not a Principal Ideal Domain.

**8.** Let $F$ be a field and let $R = F[x, x^2y, x^3y^2, \ldots, x^n y^{n-1}, \ldots]$ be a subring of the polynomial ring $F[x, y]$.

**(a)** Prove that the fields of fractions of $R$ and $F[x, y]$ are the same.

**(b)** Prove that $R$ contains an ideal that is not finitely generated.

**9.** Prove that a polynomial ring in infinitely many variables with coefficients in any commutative ring contains ideals that are not finitely generated.

**10.** Prove that the ring $\mathbb{Z}[x_1, x_2, x_3, \ldots]/(x_1x_2, x_3x_4, x_5x_6, \ldots)$ contains infinitely many minimal prime ideals (cf. Exercise 36 of Section 7.4).

**11.** Show that the radical of the ideal $I = (x, y^2)$ in $\mathbb{Q}[x, y]$ is $(x, y)$ (cf. Exercise 30, Section 7.4). Deduce that $I$ is a primary ideal that is not a power of a prime ideal (cf. Exercise 41, Section 7.4).

**12.** Let $R = \mathbb{Q}[x, y, z]$ and let bars denote passage to $\mathbb{Q}[x, y, z]/(xy - z^2)$. Prove that $\overline{P} = (\overline{x}, \overline{z})$ is a prime ideal. Show that $\overline{xy} \in \overline{P}^2$ but that no power of $\overline{y}$ lies in $\overline{P}^2$. (This shows $\overline{P}$ is a prime ideal whose square is *not* a primary ideal — cf. Exercise 41, Section 7.4).

**13.** Prove that the rings $F[x, y]/(y^2 - x)$ and $F[x, y]/(y^2 - x^2)$ are not isomorphic for any field $F$.

**14.** Let $R$ be an integral domain and let $i$, $j$ be relatively prime integers. Prove that the ideal $(x^i - y^j)$ is a prime ideal in $R[x, y]$. [Consider the ring homomorphism $\varphi$ from $R[x, y]$ to $R[t]$ defined by mapping $x$ to $t^j$ and mapping $y$ to $t^i$. Show that an element of $R[x, y]$ differs from an element in $(x^i - y^j)$ by a polynomial $f(x)$ of degree at most $j - 1$ in $y$ and observe that the exponents of $\varphi(x^r y^s)$ are distinct for $0 \leq s < j$.]

**15.** Let $p(x_1, x_2, \ldots, x_n)$ be a homogeneous polynomial of degree $k$ in $R[x_1, \ldots, x_n]$. Prove that for all $\lambda \in R$ we have $p(\lambda x_1, \lambda x_2, \ldots, \lambda x_n) = \lambda^k p(x_1, x_2, \ldots, x_n)$.

**16.** Prove that the product of two homogeneous polynomials is again homogeneous.

**17.** An ideal $I$ in $R[x_1, \ldots, x_n]$ is called a *homogeneous ideal* if whenever $p \in I$ then each homogeneous component of $p$ is also in $I$. Prove that an ideal is a homogeneous ideal if and only if it may be generated by homogeneous polynomials. [Use induction on degrees to show the "if" implication.]

The following exercise shows that some care must be taken when working with polynomials over noncommutative rings $R$ (the ring operations in $R[x]$ are defined in the same way as for commutative rings $R$), in particular when considering polynomials as functions.

**18.** Let $R$ be an arbitrary ring and let Func($R$) be the ring of all functions from $R$ to itself. If $p(x) \in R[x]$ is a polynomial, let $f_p \in$ Func($R$) be the function on $R$ defined by $f_p(r) = p(r)$ (the usual way of viewing a polynomial in $R[x]$ as defining a function on $R$ by "evaluating at $r$").

**(a)** For fixed $a \in R$, prove that "evaluation at $a$" is a ring homomorphism from Func($R$) to $R$ (cf. Example 4 following Theorem 7 in Section 7.3).

**(b)** Prove that the map $\varphi : R[x] \to$ Func($R$) defined by $\varphi(p(x)) = f_p$ is not a ring homomorphism in general. Deduce that polynomial identities need not give corresponding identities when the polynomials are viewed as functions. [If $R = \mathbb{H}$ is the ring of real Hamilton Quaternions show that $p(x) = x^2 + 1$ factors as $(x + i)(x - i)$, but that $p(j) = 0$ while $(j + i)(j - i) \neq 0$.]

**(c)** For fixed $a \in R$, prove that the composite "evaluation at $a$" of the maps in (a) and (b) mapping $R[x]$ to $R$ is a ring homomorphism if and only if $a$ is in the center of $R$.

## 9.2. Polynomial Rings over Fields I

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**1.** Let $f(x) \in F[x]$ be a polynomial of degree $n \geq 1$ and let bars denote passage to the quotient $F[x]/(f(x))$. **Prove that for each $\overline{g(x)}$** there is a unique polynomial $g_0(x)$ of degree $\leq n - 1$ such that $\overline{g(x)} = \overline{g_0(x)}$ (equivalently, the elements $\overline{1}, \overline{x}, \ldots, \overline{x^{n-1}}$ are a *basis* of the vector space $F[x]/(f(x))$ over $F$ — in particular, the dimension of this space is $n$). [Use the Division Algorithm.]

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**2.** Let $F$ be a finite field of order $q$ and let $f(x)$ be a polynomial in $F[x]$ of degree $n \geq 1$. Prove that $F[x]/(f(x))$ has $q^n$ elements. [Use the preceding exercise.]

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**3.** Let $f(x)$ be a polynomial in $F[x]$. Prove that $F[x]/(f(x))$ is a field if and only if $f(x)$ is irreducible. [Use Proposition 7, Section 8.2.]

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**4.** Let $F$ be a finite field. Prove that $F[x]$ contains infinitely many primes. (Note that over an infinite field the polynomials of degree 1 are an infinite set of primes in the ring of polynomials).

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**5.** Exhibit *all* the ideals in the ring $F[x]/(p(x))$, where $F$ is a field and $p(x)$ is a polynomial in $F[x]$ (describe them in terms of the factorization of $p(x)$).

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**6.** Describe (briefly) the ring structure of the following rings:
**(a)** $\mathbb{Z}[x]/(2)$, **(b)** $\mathbb{Z}[x]/(x)$, **(c)** $\mathbb{Z}[x]/(x^2)$, **(d)** $\mathbb{Z}[x, y]/(x^2, y^2, 2)$.
Show that $\alpha^2 = 0$ or 1 for every $\alpha$ in the last ring and determine those elements with $\alpha^2 = 0$. Determine the characteristics of each of these rings (cf. Exercise 26, Section 7.3).

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**7. Determine** all the ideals of the ring $\mathbb{Z}[x]/(2, x^3 + 1)$.

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**8.** Determine the greatest common divisor of $a(x) = x^3 - 2$ and $b(x) = x + 1$ in $\mathbb{Q}[x]$ and write it as a linear combination (in $\mathbb{Q}[x]$) of $a(x)$ and $b(x)$.

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**9.** Determine the greatest common divisor of $a(x) = x^5+2x^3+x^2+x+1$ and the polynomial $b(x) = x^5 + x^4 + 2x^3 + 2x^2 + 2x + 1$ in $\mathbb{Q}[x]$ and write it as a linear combination (in $\mathbb{Q}[x]$) of $a(x)$ and $b(x)$.

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**10.** Determine the greatest common divisor of $a(x) = x^3+4x^2+x-6$ and $b(x) = x^5-6x+5$ in $\mathbb{Q}[x]$ and write it as a linear combination (in $\mathbb{Q}[x]$) of $a(x)$ and $b(x)$.

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**11.** Suppose $f(x)$ and $g(x)$ are two nonzero polynomials in $\mathbb{Q}[x]$ with greatest common divisor $d(x)$.

**(a)** Given $h(x) \in \mathbb{Q}[x]$, show that there are polynomials $a(x), b(x) \in \mathbb{Q}[x]$ satisfying the equation $a(x)f(x) + b(x)g(x) = h(x)$ if and only if $h(x)$ is divisible by $d(x)$.

**(b)** If $a_0(x), b_0(x) \in \mathbb{Q}[x]$ are particular solutions to the equation in (a), show that the full set of solutions to this equation is given by

$$a(x) = a_0(x) + m(x)\frac{g(x)}{(x)d}$$

$$b(x) = b_0(x) - m(x)\frac{f(x)}{d(x)}$$

as $m(x)$ ranges over the polynomials in $\mathbb{Q}[x]$. [cf. Exercise 4 in Section 8.1]

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**12.** Let $F[x, y_1, y_2, \ldots]$ be the polynomial ring in the infinite set of variables $x, y_1, y_2, \ldots$ over the field $F$, and let $I$ be the ideal $(x - y_1^2, y_1 - y_2^2, \ldots, y_i - y_{i+1}^2, \ldots)$ in this ring. Define $R$ to be the ring $F[x, y_1, y_2, \ldots]/I$, so that in $R$ the square of each $y_{i+1}$ is $y_i$ and $y_1^2 = x$ modulo $I$, i.e., $x$ has a $2^i$th root, for every $i$. Denote the image of $y_i$ in $R$ as $x^{1/2^i}$. Let $R_n$ be the subring of $R$ generated by $F$ and $x^{1/2^n}$.

**(a)** Prove that $R_1 \subseteq R_2 \subseteq \cdots$ and that $R$ is the union of all $R_n$, i.e., $R = \cup_{n=1}^{\infty} R_n$.

**(b)** Prove that $R_n$ is isomorphic to a polynomial ring in one variable over $F$, so that $R_n$ is a P.I.D. Deduce that $R$ is a Bezout Domain (cf. Exercise 7 in Section 8.2). [First show that the ring $S_n = F[x, y_1, \ldots, y_n]/(x - y_1^2, y_1 - y_2^2, \ldots, y_{n-1} - y_n^2)$ is isomorphic to the polynomial ring $F[y_n]$. Then show any polynomial relation $y_n$ satisfies in $R_n$ gives a corresponding relation in $S_N$ for some $N \geq n$.]

**(c)** Prove that the ideal generated by $x, x^{1/2}, x^{1/4}, \ldots$ in $R$ is not finitely generated (so $R$ is not a P.I.D.).

Let $F$ be a field and let $x$ be an indeterminate over $F$.

**13.** This exercise introduces a noncommutative ring which is a "right" Euclidean Domain (and a "left" Principal Ideal Domain) but is not a "left" Euclidean Domain (and not a "right" Principal Ideal Domain). Let $F$ be a field of characteristic $p$ in which not every element is a $p^{\text{th}}$ power: $F \neq F^p$ (for example the field $F = \mathbb{F}_p(t)$ of rational functions in the variable $t$ with coefficients in $\mathbb{F}_p$ is such a field). Let $R = F\{x\}$ be the "twisted" polynomial ring of polynomials $\sum_{i=0}^{n} a_i x^i$ in $x$ with coefficients in $F$ with the usual (termwise) addition

$$\sum_{i=0}^{n} a_i x^i + \sum_{i=0}^{n} b_i x^i = \sum_{i=0}^{n} (a_i + b_i) x^i$$

but with a noncommutative multiplication defined by

$$\left(\sum_{i=0}^{n} a_i x^i\right)\left(\sum_{j=0}^{m} b_j x^j\right) = \sum_{k=0}^{n+m} \left(\sum_{i+j=k} a_i b_j^{p^i}\right) x^k .$$

This multiplication **arises from defining** $xa = a^p x$ **for every** $a \in F$ (so the powers of $x$ do not commute with the coefficients) and extending in a natural way. Let $N$ be the norm defined by taking the degree of a polynomial in $R$: $N(f) = \deg(f)$.

**(a)** Show that $x^k a = a^{p^k} x^k$ for every $a \in F$ and every integer $k \geq 0$ and that $R$ is a ring with this definition of multiplication. [Use the fact that $(a+b)^p = a^p + b^p$ for every $a, b \in F$ since $F$ has characteristic $p$, so also $(a+b)^{p^k} = a^{p^k} + b^{p^k}$ for every $a, b \in F$.]

**(b)** Prove that the degree of a product of two elements of $R$ is the sum of the degrees of the elements. Prove that $R$ has no zero divisors.

**(c)** Prove that $R$ is "right Euclidean" with respect to $N$, i.e., for any polynomials $f, g \in R$ with $g \neq 0$, there exist polynomials $q$ and $r$ in $R$ with

$$f = qg + r \qquad \text{with } r = 0 \text{ or } \deg(r) < \deg(g).$$

Use this to prove that every *left* ideal of $R$ is principal.

**(d)** Let $f = \theta x$ for some $\theta \in F$, $\theta \notin F^p$ and let $g = x$. Prove that there are no polynomials $q$ and $r$ in $R$ with

$$f = gq + r \qquad \text{with } r = 0 \text{ or } \deg(r) < \deg(g),$$

so in particular $R$ is not "left Euclidean" with respect to $N$. Prove that the right ideal of $R$ generated by $x$ and $\theta x$ is not principal. Conclude that $R$ is not "left Euclidean" with respect to *any* norm.

## 9.3. Polynomial Rings that are Unique Factorization Domains

**1.** Let $R$ be an integral domain with quotient field $F$ and let $p(x)$ be a monic polynomial in $R[x]$. Assume that $p(x) = a(x)b(x)$ where $a(x)$ and $b(x)$ are monic polynomials in $F[x]$ of smaller degree than $p(x)$. Prove that if $a(x) \notin R[x]$ then $R$ is not a Unique Factorization Domain. Deduce that $\mathbb{Z}[2\sqrt{2}]$ is not a U.F.D.

**2.** Prove that if $f(x)$ and $g(x)$ are polynomials with rational coefficients whose product $f(x)g(x)$ has integer coefficients, then the product of any coefficient of $g(x)$ with any coefficient of $f(x)$ is an integer.

**3.** Let $F$ be a field. Prove that the set $R$ of polynomials in $F[x]$ whose coefficient of $x$ is equal to 0 is a subring of $F[x]$ and that $R$ is not a U.F.D. [Show that $x^6 = (x^2)^3 = (x^3)^2$ gives two distinct factorizations of $x^6$ into irreducibles.]

**4.** Let $R = \mathbb{Z} + x\mathbb{Q}[x] \subset \mathbb{Q}[x]$ be the set of polynomials in $x$ with rational coefficients whose constant term is an integer.

**(a)** Prove that $R$ is an integral domain and its units are $\pm 1$.

**(b)** Show that the irreducibles in $R$ are $\pm p$ where $p$ is a prime in $\mathbb{Z}$ and the polynomials $f(x)$ that are irreducible in $\mathbb{Q}[x]$ and have constant term $\pm 1$. Prove that these irreducibles are prime in $R$.

**(c)** Show that $x$ cannot be written as the product of irreducibles in $R$ (in particular, $x$ is not irreducible) and conclude that $R$ is not a U.F.D.

**(d)** Show that $x$ is not a prime in $R$ and describe the quotient ring $R/(x)$.

**5.** Let $R = \mathbb{Z} + x\mathbb{Q}[x] \subset \mathbb{Q}[x]$ be the ring considered in the previous exercise.

**(a)** Suppose that $f(x), g(x) \in \mathbb{Q}[x]$ are two nonzero polynomials with rational coefficients and that $x^r$ is the largest power of $x$ dividing both $f(x)$ and $g(x)$ in $\mathbb{Q}[x]$, (i.e., $r$ is the degree of the lowest order term appearing in either $f(x)$ or $g(x)$). Let $f_r$ and $g_r$ be the coefficients of $x^r$ in $f(x)$ and $g(x)$, respectively (one of which is nonzero by definition of $r$). Then $\mathbb{Z}f_r + \mathbb{Z}g_r = \mathbb{Z}d_r$ for some nonzero $d_r \in \mathbb{Q}$ (cf. Exercise 14 in Section 2.4). Prove that there is a polynomial $d(x) \in \mathbb{Q}[x]$ that is a g.c.d. of $f(x)$ and $g(x)$ in $\mathbb{Q}[x]$ and whose term of minimal degree is $d_r x^r$.

**(b)** Prove that $f(x) = d(x)q_1(x)$ and $g(x) = d(x)q_2(x)$ where $q_1(x)$ and $q_2(x)$ are elements of the subring $R$ of $\mathbb{Q}[x]$.

**(c)** Prove that $d(x) = a(x)f(x) + b(x)g(x)$ for polynomials $a(x), b(x)$ in $R$. [The existence of $a(x)$, $b(x)$ in the Euclidean Domain $\mathbb{Q}[x]$ is immediate. Use Exercise 11 in Section 2 to show that $a(x)$ and $b(x)$ can be chosen to lie in $R$.]

**(d)** Conclude from (a) and (b) that $Rf(x) + Rg(x) = Rd(x)$ in $\mathbb{Q}[x]$ and use this to prove that $R$ is a Bezout Domain (cf. Exercise 7 in Section 8.2).

**(e)** Show that (d), the results of the previous exercise, and Exercise 11 of Section 8.3 imply that $R$ must contain ideals that are not principal (hence not finitely generated). Prove that in fact $I = x\mathbb{Q}[x]$ is an ideal of $R$ that is not finitely generated.

## 9.4. Irreducibility Criteria

**1.** Determine whether the following polynomials are irreducible in the rings indicated. For those that are reducible, determine their factorization into irreducibles. The notation $\mathbb{F}_p$ denotes the finite field $\mathbb{Z}/p\mathbb{Z}$, $p$ a prime.

**(a)** $x^2 + x + 1$ in $\mathbb{F}_2[x]$.

**(b)** $x^3 + x + 1$ in $\mathbb{F}_3[x]$.

**(c)** $x^4 + 1$ in $\mathbb{F}_5[x]$.

**(d)** $x^4 + 10x^2 + 1$ in $\mathbb{Z}[x]$.

**2.** Prove that the following polynomials are irreducible in $\mathbb{Z}[x]$:

**(a)** $x^4 - 4x^3 + 6$

**(b)** $x^6 + 30x^5 - 15x^3 + 6x - 120$

**(c)** $x^4 + 4x^3 + 6x^2 + 2x + 1$ [Substitute $x - 1$ for $x$.]

**(d)** $\dfrac{(x+2)^p - 2^p}{x}$, where $p$ is an odd prime.

**3.** Show that the polynomial $(x-1)(x-2)\cdots(x-n)-1$ is irreducible over $\mathbb{Z}$ for all $n \geq 1$. [If the polynomial factors consider the values of the factors at $x = 1, 2, \ldots, n$.]

**4.** Show that the polynomial $(x-1)(x-2)\cdots(x-n)+1$ is irreducible over $\mathbb{Z}$ for all $n \geq 1$, $n \neq 4$.

**5.** Find all the monic irreducible polynomials of degree $\leq 3$ in $\mathbb{F}_2[x]$, and the same in $\mathbb{F}_3[x]$.

**6.** Construct fields of each of the following orders: **(a)** 9, **(b)** 49, **(c)** 8, **(d)** 81 (you may exhibit these as $F[x]/(f(x))$ for some $F$ and $f$). [Use Exercises 2 and 3 in Section 2.]

**7.** Prove that $\mathbb{R}[x]/(x^2+1)$ is a field which is isomorphic to the complex numbers.

**8.** Prove that $K_1 = \mathbb{F}_{11}[x]/(x^2+1)$ and $K_2 = \mathbb{F}_{11}[y]/(y^2+2y+2)$ are both fields with 121 elements. Prove that the map which sends the element $p(\bar{x})$ of $K_1$ to the element $p(\bar{y}+1)$ of $K_2$ (where $p$ is any polynomial with coefficients in $\mathbb{F}_{11}$) is well defined and gives a ring (hence field) isomorphism from $K_1$ to $K_2$.

**9.** Prove that the polynomial $x^2 - \sqrt{2}$ is irreducible over $\mathbb{Z}[\sqrt{2}]$ (you may use the fact that $\mathbb{Z}[\sqrt{2}]$ is a U.F.D. — cf. Exercise 9 of Section 8.1).

**10.** Prove that the polynomial $p(x) = x^4 - 4x^2 + 8x + 2$ is irreducible over the quadratic field $F = \mathbb{Q}(\sqrt{-2}) = \{a + b\sqrt{-2} \mid a, b \in \mathbb{Q}\}$. [First use the method of Proposition 11 for the Unique Factorization Domain $\mathbb{Z}[\sqrt{-2}]$ (cf. Exercise 8, Section 8.1) to show that if $\alpha \in \mathbb{Z}[\sqrt{-2}]$ is a root of $p(x)$ then $\alpha$ is a divisor of 2 in $\mathbb{Z}[\sqrt{-2}]$. Conclude that $\alpha$ must be $\pm 1$, $\pm\sqrt{-2}$ or $\pm 2$, and hence show $p(x)$ has no linear factor over $F$. Show similarly that $p(x)$ is not the product of two quadratics with coefficients in $F$.]

**11.** Prove that $x^2 + y^2 - 1$ is irreducible in $\mathbb{Q}[x, y]$.

**12.** Prove that $x^{n-1} + x^{n-2} + \cdots + x + 1$ is irreducible over $\mathbb{Z}$ if and only if $n$ is a prime.

**13.** Prove that $x^3 + nx + 2$ is irreducible over $\mathbb{Z}$ for all integers $n \neq 1, -3, -5$.

**14.** Factor each of the two polynomials: $x^8 - 1$ and $x^6 - 1$ into irreducibles over each of the following rings: **(a)** $\mathbb{Z}$, **(b)** $\mathbb{Z}/2\mathbb{Z}$, **(c)** $\mathbb{Z}/3\mathbb{Z}$.

**15.** Prove that if $F$ is a field then the polynomial $X^n - x$ which has coefficients in the ring $F[[x]]$ of formal power series (cf. Exercise 3 of Section 7.2) is irreducible over $F[[x]]$. [Recall that $F[[x]]$ is a Euclidean Domain — cf. Exercise 5, Section 7.2 and Example 4, Section 8.1.]

**16.** Let $F$ be a field and let $f(x)$ be a polynomial of degree $n$ in $F[x]$. The polynomial $g(x) = x^n f(1/x)$ is called the *reverse* of $f(x)$.

**(a)** Describe the coefficients of $g$ in terms of the coefficients of $f$.

**(b)** Prove that $f$ is irreducible if and only if $g$ is irreducible.

**17.** Prove the following variant of Eisenstein's Criterion: let $P$ be a prime ideal in the Unique Factorization Domain $R$ and let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial in $R[x]$, $n \geq 1$. Suppose $a_n \notin P$, $a_{n-1}, \ldots, a_0 \in P$ and $a_0 \notin P^2$. Prove that $f(x)$ is irreducible in $F[x]$, where $F$ is the quotient field of $R$.

**18.** Show that $6x^5 + 14x^3 - 21x + 35$ and $18x^5 - 30x^2 + 120x + 360$ are irreducible in $\mathbb{Q}[x]$.

**19.** Let $F$ be a field and let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in F[x]$. The *derivative*, $D_x(f(x))$, of $f(x)$ is defined by

$$D_x(f(x)) = na_n x^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1$$

where, as usual, $na = a + a + \cdots + a$ ($n$ times). Note that $D_x(f(x))$ is again a polynomial with coefficients in $F$.

The polynomial $f(x)$ is said to have a *multiple root* if there is some field $E$ containing $F$ and some $\alpha \in E$ such that $(x - \alpha)^2$ divides $f(x)$ in $E[x]$. For example, the polynomial $f(x) = (x - 1)^2(x - 2) \in \mathbb{Q}[x]$ has $\alpha = 1$ as a multiple root and the polynomial $f(x) = x^4 + 2x^2 + 1 = (x^2 + 1)^2 \in \mathbb{R}[x]$ has $\alpha = \pm i \in \mathbb{C}$ as multiple roots. We shall prove in Section 13.5 that a nonconstant polynomial $f(x)$ has a multiple root if and only if $f(x)$ is not relatively prime to its derivative (which can be detected by the Euclidean Algorithm in $F[x]$). Use this criterion to determine whether the following polynomials have multiple roots:

**(a)** $x^3 - 3x - 2 \in \mathbb{Q}[x]$

**(b)** $x^3 + 3x + 2 \in \mathbb{Q}[x]$

**(c)** $x^6 - 4x^4 + 6x^3 + 4x^2 - 12x + 9 \in \mathbb{Q}[x]$

**(d)** Show for any prime $p$ and any $a \in \mathbb{F}_p$ that the polynomial $x^p - a$ has a multiple root.

**20.** Show that the polynomial $f(x) = x$ in $\mathbb{Z}/6\mathbb{Z}[x]$ factors as $(3x + 4)(4x + 3)$, hence is not an irreducible polynomial.

**(a)** Show that the reduction of $f(x)$ modulo both of the nontrivial ideals (2) and (3) of $\mathbb{Z}/6\mathbb{Z}$ is an irreducible polynomial, showing that the condition that $R$ be an integral domain in Proposition 12 is necessary.

**(b)** Show that in any factorization $f(x) = g(x)h(x)$ in $\mathbb{Z}/6\mathbb{Z}[x]$ the reduction of $g(x)$ modulo (2) is either 1 or $x$ and the reduction of $h(x)$ modulo (2) is then either $x$ or 1, and similarly for the reductions modulo (3). Determine all the factorizations of $f(x)$ in $\mathbb{Z}/6\mathbb{Z}[x]$. [Use the Chinese Remainder Theorem.]

**(c)** Show that the ideal $(3, x)$ is a principal ideal in $\mathbb{Z}/6\mathbb{Z}[x]$.

**(d)** Show that over the ring $\mathbb{Z}/30\mathbb{Z}[x]$ the polynomial $f(x) = x$ has the factorization $f(x) = (10x+21)(15x+16)(6x+25)$. Prove that the product of any of these factors is again of the same degree. Prove that the reduction of $f(x)$ modulo any prime in $\mathbb{Z}/30\mathbb{Z}$ is an irreducible polynomial. Determine all the factorizations of $f(x)$ in $\mathbb{Z}/30\mathbb{Z}[x]$. [Consider the reductions modulo (2), (3) and (5) and use the Chinese Remainder Theorem.]

**(e)** Generalize part (d) to $\mathbb{Z}/n\mathbb{Z}[x]$ where $n$ is the product of $k$ distinct primes.

## 9.5. Polynomial Rings over Fields II

**1.** Let $F$ be a field and let $f(x)$ be a nonconstant polynomial in $F[x]$. Describe the nilradical of $F[x]/(f(x))$ in terms of the factorization of $f(x)$ (cf. Exercise 29, Section 7.3).

**2.** For each of the fields constructed in Exercise 6 of Section 4 exhibit a generator for the (cyclic) multiplicative group of nonzero elements.

**3.** Let $p$ be an odd prime in $\mathbb{Z}$ and let $n$ be a positive integer. Prove that $x^n - p$ is irreducible over $\mathbb{Z}[i]$. [Use Proposition 18 in Chapter 8 and Eisenstein's Criterion.]

**4.** Prove that $x^3 + 12x^2 + 18x + 6$ is irreducible over $\mathbb{Z}[i]$. [Use Proposition 8.18 and Eisenstein's Criterion.]

**5.** Let $\varphi$ denote Euler's $\varphi$-function. Prove the identity $\sum_{d|n} \varphi(d) = n$, where the sum is extended over all the divisors $d$ of $n$. [First observe that the identity is valid when $n = p^m$ is the power of a prime $p$ since the sum telescopes. Write $n = p^m n'$ where $p$ does not divide $n'$. Prove that $\sum_{d|n} \varphi(d) = \sum_{d''|p^m} \varphi(d'') \sum_{d'|n'} \varphi(d')$ by multiplying out the right hand side and using the multiplicativity $\varphi(ab) = \varphi(a)\varphi(b)$ when $a$ and $b$ are relatively prime. Use induction to complete the proof. This problem may be done alternatively by letting $Z$ be the cyclic group of order $n$ and showing that since $Z$ contains a unique subgroup of order $d$ for each $d$ dividing $n$, the number of elements of $Z$ of order $d$ is $\varphi(d)$. Then $|Z|$ is the sum of $\varphi(d)$ as $d$ runs over all divisors of $n$.]

**6.** Let $G$ be a finite subgroup of order $n$ of the multiplicative group $F^\times$ of nonzero elements of the field $F$. Let $\varphi$ denote Euler's $\varphi$-function and let $\psi(d)$ denote the number of elements of $G$ of order $d$. Prove that $\psi(d) = \varphi(d)$ for every divisor $d$ of $n$. In particular conclude that $\psi(n) \geq 1$, so that $G$ is a cyclic group. [Observe that for any integer $N \geq 1$ the polynomial $x^N - 1$ has at most $N$ roots in $F$. Conclude that for any integer $N$ we have $\sum_{d|N} \psi(d) \leq N$. Since $\sum_{d|N} \varphi(d) = N$ by the previous exercise, show by induction that $\psi(d) \leq \varphi(d)$ for every divisor $d$ of $n$. Since $\sum_{d|n} \psi(d) = n = \sum_{d|n} \varphi(d)$ show that this implies $\psi(d) = \varphi(d)$ for every divisor $d$ of $n$.]

**7.** Prove that the additive and multiplicative groups of a field are never isomorphic. [Consider three cases: when $|F|$ is finite, when $-1 \neq 1$ in $F$, and when $-1 = 1$ in $F$.]

## 9.6. Polynomials in Several Variables over a Field and Gröbner Bases

**1.** Suppose $I$ is an ideal in $F[x_1, \ldots, x_n]$ generated by a (possibly infinite) set $\mathcal{S}$ of polynomials. Prove that a finite subset of the polynomials in $\mathcal{S}$ suffice to generate $I$. [Use Theorem 21 to write $I = (f_1, \ldots, f_m)$ and then write each $f_i \in I$ using polynomials in $\mathcal{S}$.]

**2.** Let $\geq$ be any monomial ordering.

**(a)** Prove that $LT(fg) = LT(f)LT(g)$ and $\partial(fg) = \partial(f) + \partial(g)$ for any nonzero polynomials $f$ and $g$.

**(b)** Prove that $\partial(f + g) \leq \max(\partial(f), \partial(g))$ with equality if $\partial(f) \neq \partial(g)$.

**(c)** Prove that $m \geq 1$ for every monomial $m$.

**(d)** Prove that if $m_1$ divides $m_2$ then $m_2 \geq m_1$. Deduce that the leading term of a polynomial does not divide any of its lower order terms.

**3.** Prove that if $\geq$ is any total or partial ordering on a nonempty set then the following are equivalent:

**(i)** Every nonempty subset contains a minimum element.

**(ii)** There is no infinite strictly decreasing sequence $a_1 > a_2 > a_3 > \cdots$ (this is called the *descending chain condition* or *D.C.C.*).

Deduce that General Polynomial Division always terminates in finitely many steps.

**4.** Let $\geq$ be a monomial ordering, and for monomials $m_1$, $m_2$ define $m_1 \geq_g m_2$ if either $\deg m_1 > \deg m_2$, or $\deg m_1 = \deg m_2$ and $m_1 \geq m_2$.

**(a)** Prove that $\geq_g$ is also a monomial ordering. (The relation $\geq_g$ is called the *grading* of $\geq$. An ordering in which the most important criterion for comparison is degree is sometimes called a *graded* or a *degree* ordering, so this exercise gives a method for constructing graded orderings.)

**(b)** The grading of the lexicographic ordering $x_1 > \cdots > x_n$ is called the *grlex* monomial ordering. Show that $x_2^4 > x_1^2 x_2 > x_1 x_2^2 > x_2^2 > x_1$ with respect to the grlex ordering and $x_1^2 x_2 > x_1 x_2^2 > x_1 > x_2^4 > x_2^2$ with respect to the lexicographic ordering.

**5.** The *grevlex* monomial ordering is defined by first choosing an ordering of the variables $\{x_1, x_2, \ldots, x_n\}$, then defining $m_1 \geq m_2$ for monomials $m_1, m_2$ if either $\deg m_1 > \deg m_2$ or $\deg m_1 = \deg m_2$ and the first exponent of $x_n, x_{n-1}, \ldots, x_1$ (in that order) where $m_1$ and $m_2$ differ is *smaller* in $m_1$.

**(a)** Prove that grevlex is a monomial ordering that satisfies $x_1 > x_2 > \cdots > x_n$.

**(b)** Prove that the grevlex ordering on $F[x_1, x_2]$ with respect to $\{x_1, x_2\}$ is the graded lexicographic ordering with $x_1 > x_2$, but that the grevlex ordering on $F[x_1, x_2, x_3]$ is not the grading of any lexicographic ordering.

**(c)** Show that $x_1x_2^2x_3 > x_1^2x_3^2 > x_2^2x_3^2 > x_2x_3^2 > x_1x_2 > x_2^2 > x_1x_3 > x_3^2 > x_1 > x_2$ for the grevlex monomial ordering with respect to $\{x_1, x_2, x_3\}$.

**6.** Show that $x^3y > x^3z^2 > x^3z > x^2y^2z > x^2y > xz^2 > y^2z^2 > y^2z$ with respect to the lexicographic monomial ordering $x > y > z$. Show that for the corresponding grlex monomial ordering $x^3z^2 > x^2y^2z > x^3y > x^3z > y^2z^2 > x^2y > xz^2 > y^2z$, and that $x^2y^2z > x^3z^2 > x^3y > x^3z > y^2z^2 > x^2y > y^2z > xz^2$ for the grevlex monomial ordering with respect to $\{x, y, z\}$.

**7.** Order the monomials $x^2z, x^2y^2z, xy^2z, x^3y, x^3z^2, x^2, x^2yz^2, x^2z^2$ for the lexicographic monomial ordering $x > y > z$, for the corresponding grlex monomial order, and for the grevlex monomial ordering with respect to $\{x, y, z\}$.

**8.** Show there are $n!$ distinct lexicographic monomial orderings on $F[x_1, \ldots, x_n]$. Show similarly that there are $n!$ distinct grlex and grevlex monomial orderings.

**9.** It can be shown that any monomial ordering on $F[x_1, \ldots, x_n]$ may be obtained as follows. For $k \leq n$ let $v_1, v_2, \ldots, v_k$ be nonzero vectors in Euclidean $n$-space, $\mathbb{R}^n$, that are pairwise orthogonal: $v_i \cdot v_j = 0$ for all $i \neq j$, where $\cdot$ is the usual dot product, and suppose also that all the coordinates of $v_1$ are nonnegative. Define an order, $\geq$, on monomials by $m_1 > m_2$ if and only if for some $t \leq k$ we have $v_i \cdot \partial(m_1) = v_i \cdot \partial(m_2)$ for all $i \in \{1, 2, \ldots, t-1\}$ and $v_t \cdot \partial(m_1) > v_t \cdot \partial(m_2)$.

**(a)** Let $k = n$ and let $v_i = (0, \ldots, 0, 1, 0, \ldots, 0)$ with 1 in the $i^{\text{th}}$ position. Show that $\geq$ defines the lexicographic order with $x_1 > x_2 > \cdots > x_n$.

**(b)** Let $k = n$ and define $v_1 = (1, 1, \ldots, 1)$ and $v_i = (1, 1, \ldots, 1, -n + i - 1, 0, \ldots, 0)$,

where there are $i - 2$ trailing zeros, $2 \leq i \leq n$. Show that $\geq$ defines the grlex order with respect to $\{x_1, \ldots, x_n\}$.

**10.** Suppose $I$ is a monomial ideal generated by monomials $m_1, \ldots, m_k$. Prove that the polynomial $f \in F[x_1, \ldots, x_n]$ is in $I$ if and only if every monomial term $f_i$ of $f$ is a multiple of one of the $m_j$. [For polynomials $a_1, \ldots, a_k \in F[x_1, \ldots, x_n]$ expand the polynomial $a_1 m_1 + \cdots + a_k m_k$ and note that every monomial term is a multiple of at least one of the $m_j$.] Show that $x^2yz + 3xy^2$ is an element of the ideal $I = (xyz, y^2) \subset F[x, y, z]$ but is not an element of the ideal $I' = (xz^2, y^2)$.

**11.** Fix a monomial ordering on $R = F[x_1, \ldots, x_n]$ and suppose $\{g_1, \ldots, g_m\}$ is a Gröbner basis for the ideal $I$ in $R$. Prove that $h \in LT(I)$ if and only if $h$ is a sum of monomial terms each divisible by some $LT(g_i)$, $1 \leq i \leq m$. [Use the previous exercise.]

**12.** Suppose $I$ is a monomial ideal with monomial generators $g_1, \ldots, g_m$. Use the previous exercise to prove directly that $\{g_1, \ldots, g_m\}$ is a Gröbner basis for $I$.

**13.** Suppose $I$ is a monomial ideal with monomial generators $g_1, \ldots, g_m$. Use Buchberger's Criterion to prove that $\{g_1, \ldots, g_m\}$ is a Gröbner basis for $I$.

**14.** Suppose $I$ is a monomial ideal in $R = F[x_1, \ldots, x_n]$ and suppose $\{m_1, \ldots, m_k\}$ is a minimal set of monomials generating $I$, i.e., each $m_i$ is a monomial and no proper subset of $\{m_1, \ldots, m_k\}$ generates $I$. Prove that the $m_i$, $1 \leq i \leq k$ are unique. [Use Exercise 10.]

**15.** Fix a monomial ordering on $R = F[x_1, \ldots, x_n]$.

**(a)** Prove that $\{g_1, \ldots, g_m\}$ is a minimal Gröbner basis for the ideal $I$ in $R$ if and only if $\{LT(g_1), \ldots, LT(g_m)\}$ is a minimal generating set for $LT(I)$.

**(b)** Prove that the leading terms of a minimal Gröbner basis for $I$ are uniquely determined and the number of elements in any two minimal Gröbner bases for $I$ is the same. [Use (a) and the previous exercise.]

**16.** Fix a monomial ordering on $F[x_1, \ldots, x_n]$ and suppose $G = \{g_1, \ldots, g_m\}$ is a set of generators for the nonzero ideal $I$. Show that if $S(g_i, g_j) \not\equiv 0 \bmod G$ then the ideal $(LT(g_1), \ldots, LT(g_m), LT(S(g_i, g_j))$ is strictly larger than the ideal $(LT(g_1), \ldots, LT(g_m))$. Conclude that the algorithm for computing a Gröbner basis described following Proposition 26 terminates after a finite number of steps. [Use Exercise 1.]

**17.** Fix the lexicographic ordering $x > y$ on $F[x, y]$. Use Buchberger's Criterion to show that $\{x^2y - y^2, x^3 - xy\}$ is a Gröbner basis for the ideal $I = (x^2y - y^2, x^3 - xy)$.

**18.** Show $\{x - y^3, y^5 - y^6\}$ is the reduced Gröbner basis for the ideal $I = (x - y^3, -x^2 + xy^2)$ with respect to the lexicographic ordering defined by $x > y$ in $F[x, y]$.

**19.** Fix the lexicographic ordering $x > y$ on $F[x, y]$.

**(a)** Show that $\{x^3 - y, x^2y - y^2, xy^2 - y^2, y^3 - y^2\}$ is the reduced Gröbner basis for the ideal $I = (-x^3 + y, x^2y - y^2)$.

**(b)** Determine whether the polynomial $f = x^6 - x^5y$ is an element of the ideal $I$.

**20.** Fix the lexicographic ordering $x > y > z$ on $F[x, y, z]$. Show that $\{x^2 + xy + z, xyz + z^2, xz^2, z^3\}$ is the reduced Gröbner basis for the ideal $I = (x^2 + xy + z, xyz + z^2)$ and in particular conclude that the leading term ideal $LT(I)$ requires four generators.

**21.** Fix the lexicographic ordering $x > y$ on $F[x, y]$. Use Buchberger's Criterion to show that $\{x^2y - y^2, x^3 - xy\}$ is a Gröbner basis for the ideal $I = (x^2y - y^2, x^3 - xy)$.

**22.** Let $I = (x^2 - y, x^2y - z)$ in $F[x, y, z]$.

**(a)** Show that $\{x^2 - y, y^2 - z\}$ is the reduced Gröbner basis for $I$ with respect to the lexicographic ordering defined by $x > y > z$.

**(b)** Show that $\{x^2 - y, z - y^2\}$ is the reduced Gröbner basis for $I$ with respect to the lexicographic ordering defined by $z > x > y$ (note these are essentially the same polynomials as in (a)).

**(c)** Show that $\{y - x^2, z - x^4\}$ is the reduced Gröbner basis for $I$ with respect to the lexicographic ordering defined by $z > y > x$.

**23.** Show that the ideals $I = (x^2y + xy^2 - 2y, x^2 + xy - x + y^2 - 2y, xy^2 - x - y + y^3)$ and $J = (x - y^2, xy - y, x^2 - y)$ in $F[x, y]$ are equal.

**24.** Use reduced Gröbner bases to show that the ideal $I = (x^3 - yz, yz + y)$ and the ideal $J = (x^3z + x^3, x^3 + y)$ in $F[x, y, z]$ are equal.

**25.** Show that the reduced Gröbner basis using the lexicographic ordering $x > y$ for the ideal $I = (x^2 + xy^2, x^2 - y^3, y^3 - y^2)$ is $\{x^2 - y^2, y^3 - y^2, xy^2 + y^2\}$.

**26.** Show that the reduced Gröbner basis for the ideal $I = (xy + y^2, x^2y + xy^2 + x^2)$ is $\{x^2, xy + y^2, y^3\}$ with respect to the lexicographic ordering $x > y$ and is $\{y^2 + yx, x^2\}$ with respect to the lexicographic ordering $y > x$.

There are generally substantial differences in computational complexity when using different monomial orders. The grevlex monomial ordering often provides the most efficient computation and produces simpler polynomials.

**27.** Show that $\{x^3-y^3, x^2+xy^2+y^4, x^2y+xy^3+y^2\}$ is a reduced Gröbner basis for the ideal $I$ in the example following Corollary 28 with respect to the grlex monomial ordering. (Note that while this gives three generators for $I$ rather than two for the lexicographic ordering as in the example, the degrees are smaller.)

**28.** Let $I = (x^4 - y^4 + z^3 - 1, x^3 + y^2 + z^2 - 1)$. Show that there are five elements in a reduced Gröbner basis for $I$ with respect to the lexicographic ordering with $x > y > z$ (the maximum degree among the five generators is 12 and the maximum number of monomial terms among the five generators is 35), that there are two elements for the lexicographic ordering $y > z > x$ (maximum degree is 6 and maximum number of terms is 8), and that $\{x^3 + y^2 + z^2 - 1, xy^2 + xz^2 - x + y^4 - z^3 + 1\}$ is the reduced Gröbner basis for the grevlex monomial ordering.

**29.** Solve the system of equations $x^2 - yz = 3$, $y^2 - xz = 4$, $z^2 - xy = 5$ over $\mathbb{C}$.

**30.** Find a Gröbner basis for the ideal $I = (x^2+xy+y^2-1, x^2+4y^2-4)$ for the lexicographic ordering $x > y$ and use it to find the four points of intersection of the ellipse $x^2+xy+y^2 = 1$ with the ellipse $x^2 + 4y^2 = 4$ in $\mathbb{R}^2$.

**31.** Use Gröbner bases to find all six solutions to the system of equations $2x^3+2x^2y^2+3y^3 = 0$ and $3x^5 + 2x^3y^3 + 2y^5 = 0$ over $\mathbb{C}$.

**32.** Use Gröbner bases to show that $(x, z) \cap (y^2, x - yz) = (xy, x - yz)$ in $F[x, y, z]$.

**33.** Use Gröbner bases to compute the intersection of the ideals $(x^3y - xy^2 + 1, x^2y^2 - y^3 - 1)$ and $(x^2 - y^2, x^3 + y^3)$ in $F[x, y]$.

The following four exercises deal with the *ideal quotient* of two ideals $I$ and $J$ in a ring $R$.

**Definition.** The *ideal quotient* $(I : J)$ of two ideals $I$, $J$ in a ring $R$ is the ideal

$$(I : J) = \{r \in R \mid rJ \in I\}.$$

**34.** **(a)** Suppose $R$ is an integral domain, $0 \neq f \in R$ and $I$ is an ideal in $R$. Show that if $\{g_1, \ldots, g_s\}$ are generators for the ideal $I \cap (f)$, then $\{g_1/f, \ldots, g_s/f\}$ are generators for the ideal quotient $(I : (f))$.

**(b)** If $I$ is an ideal in the commutative ring $R$ and $f_1, \ldots, f_s \in R$, show that the ideal quotient $(I : (f_1, \ldots f_s))$ is the ideal $\cap_{i=1}^{s} (I : (f_i))$.

The following four exercises deal with the *ideal quotient* of two ideals $I$ and $J$ in a ring $R$.

**Definition.** The *ideal quotient* $(I : J)$ of two ideals $I$, $J$ in a ring $R$ is the ideal

$$(I : J) = \{r \in R \mid rJ \in I\}.$$

**35.** If $I = (x^2y + z^3, x + y^3 - z, 2y^4z - yz^2 - z^3)$ and $J = (x^2y^5, x^3z^4, y^3z^7)$ in $\mathbb{Q}[x, y, z]$ show $(I : J)$ is the ideal $(z^2 + z\ x - z)$ [Use the revious exercise and Pro osition 30 ]

The following four exercises deal with the *ideal quotient* of two ideals $I$ and $J$ in a ring $R$.

**Definition.** The *ideal quotient* $(I : J)$ of two ideals $I$, $J$ in a ring $R$ is the ideal

$$(I : J) = \{r \in R \mid rJ \in I\}.$$

, y , . p p .

**36.** Suppose that $K$ is an ideal in $R$, that $I$ is an ideal containing $K$, and $J$ is any ideal. If $\overline{I}$ and $\overline{J}$ denote the images of $I$ and $J$ in the quotient ring $R/K$, show that $\overline{(I : J)} = (\overline{I} : \overline{J})$ where $\overline{(I : J)}$ is the image in $R/K$ of the ideal quotient $(I : J)$.

The following four exercises deal with the *ideal quotient* of two ideals $I$ and $J$ in a ring $R$.

**Definition.** The *ideal quotient* $(I : J)$ of two ideals $I$, $J$ in a ring $R$ is the ideal

$$(I : J) = \{r \in R \mid rJ \in I\}.$$

**37.** Let $K$ be the ideal $(y^5 - z^4)$ in $R = \mathbb{Q}[y, z]$. For each of the following pairs of ideals $I$ and $J$, use the previous two exercises together with Proposition 30 to verify the ideal quotients $(\overline{I} : \overline{J})$ in the ring $R/K$:

**i.** $I = (y^3, y^5 - z^4)$, $J = (z)$, $(\overline{I} : \overline{J}) = (\bar{y}^3, \bar{z}^3)$.

**ii.** $I = (y^3, z, y^5 - z^4)$, $J = (y)$, $(\overline{I} : \overline{J}) = (\bar{y}^2, \bar{z})$.

**iii.** $I = (y, y^3, z, y^5 - z^4)$, $J = (1)$, $(\overline{I} : \overline{J}) = (\bar{y}, \bar{z})$.

Exercises 38 to 44 develop some additional elementary properties of monomial ideals in $F[x_1, \ldots, x_n]$. It follows from Hilbert's Basis Theorem that ideals are finitely generated, however one need not assume this in these exercises—the arguments are the same for finitely or infinitely generated ideals. These exercises may be used to give an independent proof of Hilbert's Basis Theorem (Exercise 44). In these exercises, $M$ and $N$ are monomial ideals with monomial generators $\{m_i \mid i \in I\}$ and $\{n_j \mid j \in J\}$ for some index sets $I$ and $J$ respectively.

**38.** Prove that the sum and product of two monomial ideals is a monomial ideal by showing that $M + N = (m_i,\ n_j \mid i \in I,\ j \in J)$, and $MN = (m_i n_j \mid i \in I,\ j \in J)$.

Exercises 38 to 44 develop some additional elementary properties of monomial ideals in $F[x_1, \ldots, x_n]$. It follows from Hilbert's Basis Theorem that ideals are finitely generated, however one need not assume this in these exercises—the arguments are the same for finitely or infinitely generated ideals. These exercises may be used to give an independent proof of Hilbert's Basis Theorem (Exercise 44). In these exercises, $M$ and $N$ are monomial ideals with monomial generators $\{m_i \mid i \in I\}$ and $\{n_j \mid j \in J\}$ for some index sets $I$ and $J$ respectively.

**39.** Show that if $\{M_s \mid s \in S\}$ is any nonempty collection of monomial ideals that is totally ordered under inclusion then $\cup_{s \in S} M_s$ is a monomial ideal. (In particular, the union of any increasing sequence of monomial ideals is a monomial ideal, cf. Exercise 19, Section 7.3.)

Exercises 38 to 44 develop some additional elementary properties of monomial ideals in $F[x_1, \ldots, x_n]$. It follows from Hilbert's Basis Theorem that ideals are finitely generated, however one need not assume this in these exercises—the arguments are the same for finitely or infinitely generated ideals. These exercises may be used to give an independent proof of Hilbert's Basis Theorem (Exercise 44). In these exercises, $M$ and $N$ are monomial ideals with monomial generators $\{m_i \mid i \in I\}$ and $\{n_j \mid j \in J\}$ for some index sets $I$ and $J$ respectively.

**40.** Prove that the intersection of two monomial ideals is a monomial ideal by showing that $M \cap N = (e_{i,j} \mid i \in I,\ j \in J)$, where $e_{i,j}$ is the least common multiple of $m_i$ and $n_j$. [Use Exercise 10.]

Exercises 38 to 44 develop some additional elementary properties of monomial ideals in $F[x_1, \ldots, x_n]$. It follows from Hilbert's Basis Theorem that ideals are finitely generated, however one need not assume this in these exercises—the arguments are the same for finitely or infinitely generated ideals. These exercises may be used to give an independent proof of Hilbert's Basis Theorem (Exercise 44). In these exercises, $M$ and $N$ are monomial ideals with monomial generators $\{m_i \mid i \in I\}$ and $\{n_j \mid j \in J\}$ for some index sets $I$ and $J$ respectively.

**41.** Prove that for any monomial $n$, the ideal quotient $(M : (n))$ is $(m_i/d_i \mid i \in I)$, where $d_i$ is the greatest common divisor of $m_i$ and $n$ (cf. Exercise 34). Show that if $N$ is finitely generated, then the ideal quotient $(M : N)$ of two monomial ideals is a monomial ideal.

Exercises 38 to 44 develop some additional elementary properties of monomial ideals in $F[x_1, \ldots, x_n]$. It follows from Hilbert's Basis Theorem that ideals are finitely generated, however one need not assume this in these exercises—the arguments are the same for finitely or infinitely generated ideals. These exercises may be used to give an independent proof of Hilbert's Basis Theorem (Exercise 44). In these exercises, $M$ and $N$ are monomial ideals with monomial generators $\{m_i \mid i \in I\}$ and $\{n_j \mid j \in J\}$ for some index sets $I$ and $J$ respectively.

**42.** **(a)** Show that $M$ is a monomial prime ideal if and only if $M = (S)$ for some subset of $S$ of $\{x_1, x_2, \ldots, x_n\}$. (In particular, there are only finitely many monomial prime ideals, and each is finitely generated.)

**(b)** Show that $(x_1, \ldots, x_n)$ is the only monomial maximal ideal.

Exercises 38 to 44 develop some additional elementary properties of monomial ideals in $F[x_1, \ldots, x_n]$. It follows from Hilbert's Basis Theorem that ideals are finitely generated, however one need not assume this in these exercises—the arguments are the same for finitely or infinitely generated ideals. These exercises may be used to give an independent proof of Hilbert's Basis Theorem (Exercise 44). In these exercises, $M$ and $N$ are monomial ideals with monomial generators $\{m_i \mid i \in I\}$ and $\{n_j \mid j \in J\}$ for some index sets $I$ and $J$ respectively.

**43.** *(Dickson's Lemma*—a special case of Hilbert's Basis Theorem) Prove that every monomial ideal in $F[x_1, \ldots, x_n]$ is finitely generated as follows.
Let $\mathcal{S} = \{N \mid N \text{ is a monomial ideal that is not finitely generated}\}$, and assume by way of contradiction $\mathcal{S} \neq \emptyset$.

**(a)** Show that $\mathcal{S}$ contains a maximal element $M$. [Use Exercise 30 and Zorn's Lemma.]

**(b)** Show that there are monomials $x$, $y$ not in $M$ with $xy \in M$. [Use Exercise 33(a).]

**(c)** For $x$ as in (b), show that $M$ contains a finitely generated monomial ideal $M_0$ such that $M_0 + (x) = M + (x)$ and $M = M_0 + (x)(M : (x))$, where $(M : (x))$ is the (monomial) ideal defined in Exercise 32, and $(x)(M : (x))$ is the product of these two ideals. Deduce that $M$ is finitely generated, a contradiction which proves $\mathcal{S} = \emptyset$. [Use the maximality of $M$ and previous exercises.]

Exercises 38 to 44 develop some additional elementary properties of monomial ideals in $F[x_1, \ldots, x_n]$. It follows from Hilbert's Basis Theorem that ideals are finitely generated, however one need not assume this in these exercises—the arguments are the same for finitely or infinitely generated ideals. These exercises may be used to give an independent proof of Hilbert's Basis Theorem (Exercise 44). In these exercises, $M$ and $N$ are monomial ideals with monomial generators $\{m_i \mid i \in I\}$ and $\{n_j \mid j \in J\}$ for some index sets $I$ and $J$ respectively.

**44.** If $I$ is a nonzero ideal in $F[x_1, \ldots, x_n]$, use Dickson's Lemma to prove that $LT(I)$ is finitely generated. Conclude that $I$ has a Gröbner basis and deduce Hilbert's Basis Theorem. [cf. Proposition 24.]

**45.** *(n-colorings of graphs)* A finite *graph* $\mathcal{G}$ of size $N$ is a set of vertices $i \in \{1, 2, \ldots, N\}$ and a collection of edges $(i, j)$ connecting vertex $i$ with vertex $j$. An *n-coloring* of $\mathcal{G}$ is an assignment of one of $n$ colors to each vertex in such a way that vertices connected by an edge have distinct colors. Let $F$ be any field containing at least $n$ elements. If we introduce a variable $x_i$ for each vertex $i$ and represent the $n$ colors by choosing a set $S$ of $n$ distinct elements from $F$, then an $n$-coloring of $\mathcal{G}$ is equivalent to assigning a value $x_i = \alpha_i$ for each $i = 1, 2, \ldots, N$ where $\alpha_i \in S$ and $\alpha_i \neq \alpha_j$ if $(i, j)$ is an edge in $\mathcal{G}$. If $f(x) = \prod_{\alpha \in S}(x - \alpha)$ is the polynomial in $F[x]$ of degree $n$ whose roots are the elements in $S$, then $x_i = \alpha_i$ for some $\alpha_i \in S$ is equivalent to the statement that $x_i$ is a solution to the equation $f(x_i) = 0$. The statement $\alpha_i \neq \alpha_j$ is then the statement that $f(x_i) = f(x_j)$ but $x_i \neq x_j$, so $x_i$ and $x_j$ satisfy the equation $g(x_i, x_j) = 0$, where $g(x_i, x_j)$ is the polynomial $(f(x_i) - f(x_j))/(x_i - x_j)$ in $F[x_i, x_j]$. It follows that finding an *n-coloring* of $\mathcal{G}$ is equivalent to solving the system of equations

$$\begin{cases} f(x_i) = 0, & \text{for } i = 1, 2, \ldots, N, \\ g(x_i, x_j) = 0, & \text{for all edges } (i, j) \text{ in } \mathcal{G} \end{cases}$$

(note also we may use any polynomial $g$ satisfying $\alpha_i \neq \alpha_j$ if $g(\alpha_i, \alpha_j) = 0$). It follows by "Hilbert's Nullstellensatz" (cf. Corollary 33 in Section 15.3) that this system of equations has a solution, hence $\mathcal{G}$ has an $n$-coloring, unless the ideal $I$ in $F[x_1, x_2, \ldots, x_N]$ generated by the polynomials $f(x_i)$ for $i = 1, 2, \ldots, N$, together with the polynomials $g(x_i, x_j)$ for all the edges $(i, j)$ in the graph $\mathcal{G}$, is not a proper ideal. This in turn is equivalent to the statement that the reduced Gröbner basis for $I$ (with respect to any monomial ordering) is simply $\{1\}$. Further, when an $n$-coloring does exist, solving this system of equations as in the examples following Proposition 29 provides an explicit coloring for $\mathcal{G}$.

There are many possible choices of field $F$ and set $S$. For example, use any field $F$ containing a set $S$ of distinct $n^{\text{th}}$ roots of unity, in which case $f(x) = x^n - 1$ and we may take $g(x_i, x_j) = (x_i^n - x_j^n)/(x_i - x_j) = x_i^{n-1} + x_i^{n-2}x_j + \cdots + x_i x_j^{n-2} + x_j^{n-1}$, or use any subset $S$ of $F = \mathbb{F}_p$ with a prime $p \geq n$ (in the special case $n = p$, then, by Fermat's Little Theorem, we have $f(x) = x^p - x$ and $g(x_i, x_j) = (x_i - x_j)^{p-1} - 1$).

**(a)** Consider a possible 3-coloring of the graph $\mathcal{G}$ with eight vertices and 14 edges $(1, 3)$, $(1, 4)$, $(1, 5)$, $(2, 4)$, $(2, 7)$, $(2, 8)$, $(3, 4)$, $(3, 6)$, $(3, 8)$, $(4, 5)$, $(5, 6)$, $(6, 7)$, $(6, 8)$, $(7, 8)$. Take $F = \mathbb{F}_3$ with 'colors' $0, 1, 2 \in \mathbb{F}_3$ and suppose vertex 1 is colored by 0. In this case $f(x) = x(x - 1)(x - 2) = x^3 - x \in \mathbb{F}_3[x]$ and $g(x_i, x_j) = x_i^2 + x_i x_j + x_j^2 - 1$. If $I$ is the ideal generated by $x_1$, $x_i^3 - x_i$, $2 \leq i \leq 8$ and $g(x_i, x_j)$ for the edges $(i, j)$ in $\mathcal{G}$, show that the reduced Gröbner basis for $I$ with respect to the lexicographic monomial ordering $x_1 > x_2 > \cdots > x_8$ is $\{x_1, x_2, x_3 + x_8, x_4 + 2x_8, x_5 + x_8, x_6, x_7 + x_8, x_8^2 + 2\}$. Deduce that $\mathcal{G}$ has two distinct 3-colorings, determined by the coloring of vertex 8 (which must be colored by a nonzero element in $\mathbb{F}_3$), and exhibit the colorings of $\mathcal{G}$.

Show that if the edge $(3, 7)$ is added to $\mathcal{G}$ then the graph cannot be 3-colored.

**(b)** Take $F = \mathbb{F}_5$ with four 'colors' $1, 2, 3, 4 \in \mathbb{F}_5$, so $f(x) = x^4 - 1$ and we may use $g(x_i, x_j) = x_i^3 + x_i^2 x_j + x_i x_j^2 + x_j^3$. Show that the graph $\mathcal{G}$ with five vertices having 9 edges $(1, 3)$, $(1, 4)$, $(1, 5)$, $(2, 3)$, $(2, 4)$, $(2, 5)$, $(3, 4)$, $(3, 5)$, $(4, 5)$ (the "complete graph on five vertices" with one edge removed) can be 4-colored but cannot be 3-colored.

**(c)** Use Gröbner bases to show that the graph $\mathcal{G}$ with nine vertices and 22 edges $(1, 4)$, $(1, 6)$, $(1, 7)$, $(1, 8)$, $(2, 3)$, $(2, 4)$, $(2, 6)$, $(2, 7)$, $(3, 5)$, $(3, 7)$, $(3, 9)$, $(4, 5)$, $(4, 6)$, $(4, 7)$, $(4, 9)$, $(5, 6)$, $(5, 7)$, $(5, 8)$, $(5, 9)$, $(6, 7)$, $(6, 9)$, $(7, 8)$ has precisely four 4-colorings up to a permutation of the colors (so a total of 96 total 4-colorings). Show that if the edge $(1, 5)$ is added then $\mathcal{G}$ cannot be 4-colored.

# 10. Introduction to Module Theory

## 10.1. Basic Definitions and Examples

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**1.** Prove that $0m = 0$ and $(-1)m = -m$ for all $m \in M$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**2.** Prove that $R^\times$ and $M$ satisfy the two axioms in Section 1.7 for a *group action* of the multiplicative group $R^\times$ on the set $M$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**3.** Assume that $rm = 0$ for some $r \in R$ and some $m \in M$ with $m \neq 0$. Prove that $r$ does not have a left inverse (i.e., there is no $s \in R$ such that $sr = 1$).

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**4.** Let $M$ be the module $R^n$ described in Example 3 and let $I_1, I_2, \ldots, I_n$ be left ideals of $R$. Prove that the following are submodules of $M$:

**(a)** $\{(x_1, x_2, \ldots, x_n) \mid x_i \in I_i\}$

**(b)** $\{(x_1, x_2, \ldots, x_n) \mid x_i \in R \text{ and } x_1 + x_2 + \cdots + x_n = 0\}$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**5.** For any left ideal $I$ of $R$ define

$$IM = \{\sum_{\text{finite}} a_i m_i \mid a_i \in I,\ m_i \in M\}$$

to be the collection of all finite sums of elements of the form $am$ where $a \in I$ and $m \in M$. Prove that $IM$ is a submodule of $M$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**6.** Show that the intersection of any nonempty collection of submodules of an $R$-module is a submodule.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**7.** Let $N_1 \subseteq N_2 \subseteq \cdots$ be an ascending chain of submodules of $M$. Prove that $\cup_{i=1}^{\infty} N_i$ is a submodule of $M$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**8.** An element $m$ of the $R$-module $M$ is called a *torsion element* if $rm = 0$ for some nonzero element $r \in R$. The set of torsion elements is denoted

$$\text{Tor}(M) = \{m \in M \mid rm = 0 \text{ for some nonzero } r \in R\}.$$

**(a)** Prove that if $R$ is an integral domain then $\text{Tor}(M)$ is a submodule of $M$ (called the *torsion* submodule of $M$).
**(b)** Give an example of a ring $R$ and an $R$-module $M$ such that $\text{Tor}(M)$ is not a submodule. [Consider the torsion elements in the $R$-module $R$.]
**(c)** If $R$ has zero divisors show that every nonzero $R$-module has nonzero torsion elements.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**9.** If $N$ is a submodule of $M$, the *annihilator of $N$ in $R$* is defined to be $\{r \in R \mid rn = 0 \text{ for all } n \in N\}$. Prove that the annihilator of $N$ in $R$ is a 2-sided ideal of $R$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**10.** If $I$ is a right ideal of $R$, the *annihilator of $I$ in $M$* is defined to be $\{m \in M \mid am = 0 \text{ for all } a \in I\}$. Prove that the annihilator of $I$ in $M$ is a submodule of $M$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**11.** Let $M$ be the abelian group (i.e., $\mathbb{Z}$-module) $\mathbb{Z}/24\mathbb{Z} \times \mathbb{Z}/15\mathbb{Z} \times \mathbb{Z}/50\mathbb{Z}$.
**(a)** Find the annihilator of $M$ in $\mathbb{Z}$ (i.e., a generator for this principal ideal).
**(b)** Let $I = 2\mathbb{Z}$. Describe the annihilator of $I$ in $M$ as a direct product of cyclic groups.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**12.** In the notation of the preceding exercises prove the following facts about annihilators.

**(a)** Let $N$ be a submodule of $M$ and let $I$ be its annihilator in $R$. Prove that the annihilator of $I$ in $M$ contains $N$. Give an example where the annihilator of $I$ in $M$ does not equal $N$.

**(b)** Let $I$ be a right ideal of $R$ and let $N$ be its annihilator in $M$. Prove that the annihilator of $N$ in $R$ contains $I$. Give an example where the annihilator of $N$ in $R$ does not equal $I$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**13.** Let $I$ be an ideal of $R$. Let $M'$ be the subset of elements $a$ of $M$ that are annihilated by some power, $I^k$, of the ideal $I$, where the power may depend on $a$. Prove that $M'$ is a submodule of $M$. [Use Exercise 7.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**14.** Let $z$ be an element of the center of $R$, i.e., $zr = rz$ for all $r \in R$. Prove that $zM$ is a submodule of $M$, where $zM = \{zm \mid m \in M\}$. Show that if $R$ is the ring of $2 \times 2$ matrices over a field and $e$ is the matrix with a 1 in position 1,1 and zeros elsewhere then $eR$ is *not* a left $R$-submodule (where $M = R$ is considered as a left $R$-module as in Example 1) — in this case the matrix $e$ is not in the center of $R$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**15.** If $M$ is a finite abelian group then $M$ is naturally a $\mathbb{Z}$-module. Can this action be extended to make $M$ into a $\mathbb{Q}$-module?

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**16.** Prove that the submodules $U_k$ described in the example of $F[x]$-modules are all of the $F[x]$-submodules for the shift operator.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**17.** Let $T$ be the shift operator on the vector space $V$ and let $e_1, \ldots, e_n$ be the usual basis vectors described in the example of $F[x]$-modules. If $m \geq n$ find $(a_m x^m + a_{m-1} x^{m-1} + \cdots + a_0) e_n$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**18.** Let $F = \mathbb{R}$, let $V = \mathbb{R}^2$ and let $T$ be the linear transformation from $V$ to $V$ which is rotation clockwise about the origin by $\pi/2$ radians. Show that $V$ and 0 are the only $F[x]$-submodules for this $T$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**19.** Let $F = \mathbb{R}$, let $V = \mathbb{R}^2$ and let $T$ be the linear transformation from $V$ to $V$ which is projection onto the $y$-axis. Show that $V$, 0, the $x$-axis and the $y$-axis are the only $F[x]$-submodules for this $T$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**20.** Let $F = \mathbb{R}$, let $V = \mathbb{R}^2$ and let $T$ be the linear transformation from $V$ to $V$ which is rotation clockwise about the origin by $\pi$ radians. Show that *every* subspace of $V$ is an $F[x]$-submodule for this $T$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**21.** Let $n \in \mathbb{Z}^+, n > 1$ and let $R$ be the ring of $n \times n$ matrices with entries from a field $F$. Let $M$ be the set of $n \times n$ matrices with arbitrary elements of $F$ in the first column and zeros elsewhere. Show that $M$ is a submodule of $R$ when $R$ is considered as a left module over itself, but $M$ is not a submodule of $R$ when $R$ is considered as a right $R$-module.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**22.** Suppose that $A$ is a ring with identity $1_A$ that is a (unital) left $R$-module satisfying $r \cdot (ab) = (r \cdot a)b = a(r \cdot b)$ for all $r \in R$ and $a, b \in A$. Prove that the map $f : R \to A$ defined by $f(r) = r \cdot 1_A$ is a ring homomorphism mapping $1_R$ to $1_A$ and that $f(R)$ is contained in the center of $A$. Conclude that $A$ is an $R$-algebra and that the $R$-module structure on $A$ induced by its algebra structure is precisely the original $R$-module structure.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**23.** Let $A$ be the direct product ring $\mathbb{C} \times \mathbb{C}$ (cf. Section 7.6). Let $\tau_1$ denote the identity map on $\mathbb{C}$ and let $\tau_2$ denote complex conjugation. For any pair $p, q \in \{1, 2\}$ (not necessarily distinct) define

$$f_{p,q} : \mathbb{C} \to \mathbb{C} \times \mathbb{C} \qquad \text{by} \qquad f_{p,q}(z) = (\tau_p(z), \tau_q(z)).$$

So, for example, $f_{2,1} : z \mapsto (\bar{z}, z)$, where $\bar{z}$ is the complex conjugate of $z$, i.e., $\tau_2(z)$.

**(a)** Prove that each $f_{p,q}$ is an injective ring homomorphism, and that they all agree on the subfield $\mathbb{R}$ of $\mathbb{C}$. Deduce that $A$ has four distinct $\mathbb{C}$-algebra structures. Explicitly give the action $z \cdot (u, v)$ of a complex number $z$ on an ordered pair in $A$ in each case.

**(b)** Prove that if $f_{p,q} \neq f_{p',q'}$ then the identity map on $A$ is *not* a $\mathbb{C}$-algebra homomorphism from $A$ considered as a $\mathbb{C}$-algebra via $f_{p,q}$ to $A$ considered a $\mathbb{C}$-algebra via $f_{p',q'}$ (although the identity is an $\mathbb{R}$-algebra isomorphism).

**(c)** Prove that for any pair $p, q$ there is some ring isomorphism from $A$ to itself such that $A$ is isomorphic as a $\mathbb{C}$-algebra via $f_{p,q}$ to $A$ considered as $\mathbb{C}$-algebra via $f_{1,1}$ (the "natural" $\mathbb{C}$-algebra structure on $A$).

*Remark*: In the preceding exercise $A = \mathbb{C} \times \mathbb{C}$ is not a $\mathbb{C}$-algebra over either of the direct factor component copies of $\mathbb{C}$ (for example the subring $\mathbb{C} \times 0 \cong \mathbb{C}$) since it is not a unital module over these copies of $\mathbb{C}$ (the 1 of these subrings is not the same as the 1 of $A$).

## 10.2. Quotient Modules and Module Homomorphisms

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**1.** Use the submodule criterion to show that kernels and images of $R$-module homomorphisms are submodules.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**2.** Show that the relation "is $R$-module isomorphic to" is an equivalence relation on any set of $R$-modules.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**3.** Give an explicit example of a map from one $R$-module to another which is a group homomorphism but not an $R$-module homomorphism.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**4.** Let $A$ be any $\mathbb{Z}$-module, let $a$ be any element of $A$ and let $n$ be a positive integer. Prove that the map $\varphi_a : \mathbb{Z}/n\mathbb{Z} \to A$ given by $\varphi(\bar{k}) = ka$ is a well defined $\mathbb{Z}$-module homomorphism if and only if $na = 0$. Prove that $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong A_n$, where $A_n = \{a \in A \mid na = 0\}$ (so $A_n$ is the annihilator in $A$ of the ideal $(n)$ of $\mathbb{Z}$ — cf. Exercise 10, Section 1).

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**5.** Exhibit all $\mathbb{Z}$-module homomorphisms from $\mathbb{Z}/30\mathbb{Z}$ to $\mathbb{Z}/21\mathbb{Z}$.

**In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.**

**6.** Prove that $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/(n, m)\mathbb{Z}$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**7.** Let $z$ be a fixed element of the center of $R$. Prove that the map $m \mapsto zm$ is an $R$-module homomorphism from $M$ to itself. Show that for a commutative ring $R$ the map from $R$ to $\text{End}_R(M)$ given by $r \mapsto rI$ is a ring homomorphism (where $I$ is the identity endomorphism).

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**8.** Let $\varphi : M \to N$ be an $R$-module homomorphism. Prove that $\varphi(\text{Tor}(M)) \subseteq \text{Tor}(N)$ (cf. Exercise 8 in Section 1).

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**9.** Let $R$ be a commutative ring. Prove that $\mathrm{Hom}_R(R, M)$ and $M$ are isomorphic as left $R$-modules. [Show that each element of $\mathrm{Hom}_R(R, M)$ is determined by its value on the identity of $R$.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**10.** Let $R$ be a commutative ring. Prove that $\mathrm{Hom}_R(R, R)$ and $R$ are isomorphic as rings.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**11.** Let $A_1, A_2, \ldots, A_n$ be $R$-modules and let $B_i$ be a submodule of $A_i$ for each $i = 1, 2, \ldots, n$. Prove that

$$(A_1 \times \cdots \times A_n)/(B_1 \times \cdots \times B_n) \cong (A_1/B_1) \times \cdots \times (A_n/B_n).$$

[Recall Exercise 14 in Section 5.1.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**12.** Let $I$ be a left ideal of $R$ and let $n$ be a positive integer. Prove

$$R^n/IR^n \cong R/IR \times \cdots \times R/IR \quad (n \text{ times})$$

where $IR^n$ is defined as in Exercise 5 of Section 1. [Use the preceding exercise.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**13.** Let $I$ be a nilpotent ideal in a commutative ring $R$ (cf. Exercise 37, Section 7.3), let $M$ and $N$ be $R$-modules and let $\varphi : M \to N$ be an $R$-module homomorphism. Show that if the induced map $\overline{\varphi} : M/IM \to N/IN$ is surjective, then $\varphi$ is surjective.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**14.** Let $R = \mathbb{Z}[x]$ be the ring of polynomials in $x$ and let $A = \mathbb{Z}[t_1, t_2, \ldots]$ be the ring of polynomials in the independent indeterminates $t_1, t_2, \ldots$. Define an action of $R$ on $A$ as follows: 1) let $1 \in R$ act on $A$ as the identity, 2) for $n \geq 1$ let $x^n \circ 1 = t_n$, let $x^n \circ t_i = t_{n+i}$ for $i = 1, 2, \ldots$, and let $x^n$ act as 0 on monomials in $A$ of (total) degree at least two, and 3) extend $\mathbb{Z}$-linearly, i.e., so that the module axioms 2(a) and 2(c) are satisfied.

**(a)** Show that $x^{p+q} \circ t_i = x^p \circ (x^q \circ t_i) = t_{p+q+i}$ and use this to show that under this action the ring $A$ is a (unital) $R$-module.

**(b)** Show that the map $\varphi : R \to A$ defined by $\varphi(r) = r \circ 1_A$ is an $R$-module homomorphism of the ring $R$ into the ring $A$ mapping $1_R$ to $1_A$, but is not a ring homomorphism from $R$ to $A$.

## 10.3. Generation of Modules, Direct Sums, and Free Modules

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**1.** Prove that if $A$ and $B$ are sets of the same cardinality, then the free modules $F(A)$ and $F(B)$ are isomorphic.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**2.** Assume $R$ is commutative. Prove that $R^n \cong R^m$ if and only if $n = m$, i.e., two free $R$-modules of finite rank are isomorphic if and only if they have the same rank. [Apply Exercise 12 of Section 2 with $I$ a maximal ideal of $R$. You may assume that if $F$ is a field, then $F^n \cong F^m$ if and only if $n = m$, i.e., two finite dimensional vector spaces over $F$ are isomorphic if and only if they have the same dimension — this will be proved later in Section 11.1.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**3.** Show that the $F[x]$-modules in Exercises 18 and 19 of Section 1 are both cyclic.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**4.** An $R$-module $M$ is called a *torsion* module if for each $m \in M$ there is a nonzero element $r \in R$ such that $rm = 0$, where $r$ may depend on $m$ (i.e., $M = \mathrm{Tor}(M)$ in the notation of Exercise 8 of Section 1). Prove that every finite abelian group is a torsion $\mathbb{Z}$-module. Give an example of an infinite abelian group that is a torsion $\mathbb{Z}$-module.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**5.** Let $R$ be an integral domain. Prove that every finitely generated torsion $R$-module has a nonzero annihilator i.e., there is a nonzero element $r \in R$ such that $rm = 0$ for all $m \in M$ — here $r$ does not depend on $m$ (the annihilator of a module was defined in Exercise 9 of Section 1). Give an example of a torsion $R$-module whose annihilator is the zero ideal.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**6.** Prove that if $M$ is a finitely generated $R$-module that is generated by $n$ elements then every quotient of $M$ may be generated by $n$ (or fewer) elements. Deduce that quotients of cyclic modules are cyclic.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**7.** Let $N$ be a submodule of $M$. Prove that if both $M/N$ and $N$ are finitely generated then so is $M$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**8.** Let $S$ be the collection of sequences $(a_1, a_2, a_3, \ldots)$ of integers $a_1, a_2, a_3, \ldots$ where all but finitely many of the $a_i$ are 0 (called the *direct sum* of infinitely many copies of $\mathbb{Z}$). Recall that $S$ is a ring under componentwise addition and multiplication and $S$ does not have a multiplicative identity — cf. Exercise 20, Section 7.1. Prove that $S$ is not finitely generated as a module over itself.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**9.** An $R$-module $M$ is called *irreducible* if $M \neq 0$ and if 0 and $M$ are the only submodules of $M$. Show that $M$ is irreducible if and only if $M \neq 0$ and $M$ is a cyclic module with any nonzero element as generator. Determine all the irreducible $\mathbb{Z}$-modules.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**10.** Assume $R$ is commutative. Show that an $R$-module $M$ is irreducible if and only if $M$ is isomorphic (as an $R$-module) to $R/I$ where $I$ is a maximal ideal of $R$. [By the previous exercise, if $M$ is irreducible there is a natural map $R \to M$ defined by $r \mapsto rm$, where $m$ is any fixed nonzero element of $M$.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**11.** Show that if $M_1$ and $M_2$ are irreducible $R$-modules, then any nonzero $R$-module homomorphism from $M_1$ to $M_2$ is an isomorphism. Deduce that if $M$ is irreducible then $\mathrm{End}_R(M)$ is a division ring (this result is called *Schur's Lemma*). [Consider the kernel and the image.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**12.** Let $R$ be a commutative ring and let $A$, $B$ and $M$ be $R$-modules. Prove the following isomorphisms of $R$-modules:

**(a)** $\mathrm{Hom}_R(A \times B, M) \cong \mathrm{Hom}_R(A, M) \times \mathrm{Hom}_R(B, M)$

**(b)** $\mathrm{Hom}_R(M, A \times B) \cong \mathrm{Hom}_R(M, A) \times \mathrm{Hom}_R(M, B)$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**13.** Let $R$ be a commutative ring and let $F$ be a free $R$-module of finite rank. Prove the following isomorphism of $R$-modules: $\mathrm{Hom}_R(F, R) \cong F$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**14.** Let $R$ be a commutative ring and let $F$ be the free $R$-module of rank $n$. Prove that $\mathrm{Hom}_R(F, M) \cong M \times \cdots \times M$ ($n$ times). [Use Exercise 9 in Section 2 and Exercise 12.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**15.** An element $e \in R$ is called a *central idempotent* if $e^2 = e$ and $er = re$ for all $r \in R$. If $e$ is a central idempotent in $R$, prove that $M = eM \oplus (1-e)M$. [Recall Exercise 14 in Section 1.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

The next two exercises establish the Chinese Remainder Theorem for modules (cf. Section 7.6).

**16.** For any ideal $I$ of $R$ let $IM$ be the submodule defined in Exercise 5 of Section 1. Let $A_1, \ldots, A_k$ be any ideals in the ring $R$. Prove that the map

$$M \to M/A_1M \times \cdots \times M/A_kM \quad \text{defined by} \quad m \mapsto (m + A_1M, \ldots, m + A_kM)$$

is an $R$-module homomorphism with kernel $A_1M \cap A_2M \cap \cdots \cap A_kM$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

The next two exercises establish the Chinese Remainder Theorem for modules (cf. Section 7.6).

**17.** In the notation of the preceding exercise, assume further that the ideals $A_1, \ldots, A_k$ are pairwise comaximal (i.e., $A_i + A_j = R$ for all $i \neq j$). Prove that

$$M/(A_1 \cdots A_k)M \cong M/A_1M \times \cdots \times M/A_kM.$$

[See the proof of the Chinese Remainder Theorem for rings in Section 7.6.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**18.** Let $R$ be a Principal Ideal Domain and let $M$ be an $R$-module that is annihilated by the nonzero, proper ideal $(a)$. Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ be the unique factorization of $a$ into distinct prime powers in $R$. Let $M_i$ be the annihilator of $p_i^{\alpha_i}$ in $M$, i.e., $M_i$ is the set $\{m \in M \mid p_i^{\alpha_i} m = 0\}$ — called the *$p_i$-primary component of $M$*. Prove that

$$M = M_1 \oplus M_2 \oplus \cdots \oplus M_k.$$

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**19.** Show that if $M$ is a finite abelian group of order $a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ then, considered as a $\mathbb{Z}$-module, $M$ is annihilated by $(a)$, the $p_i$-primary component of $M$ is the unique Sylow $p_i$-subgroup of $M$ and $M$ is isomorphic to the direct product of its Sylow subgroups.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**20.** Let $I$ be a nonempty index set and for each $i \in I$ let $M_i$ be an $R$-module. The *direct product* of the modules $M_i$ is defined to be their direct product as abelian groups (cf. Exercise 15 in Section 5.1) with the action of $R$ componentwise multiplication. The *direct sum* of the modules $M_i$ is defined to be the restricted direct product of the abelian groups $M_i$ (cf. Exercise 17 in Section 5.1) with the action of $R$ componentwise multiplication. In other words, the direct sum of the $M_i$'s is the subset of the direct product, $\prod_{i\in I} M_i$, which consists of all elements $\prod_{i\in I} m_i$ such that only finitely many of the components $m_i$ are nonzero; the action of $R$ on the direct product or direct sum is given by $r\prod_{i\in I} m_i = \prod_{i\in I} rm_i$ (cf. Appendix I for the definition of Cartesian products of infinitely many sets). The direct sum will be denoted by $\oplus_{i\in I} M_i$.

**(a)** Prove that the direct product of the $M_i$'s is an $R$-module and the direct sum of the $M_i$'s is a submodule of their direct product.

**(b)** Show that if $R = \mathbb{Z}$, $I = \mathbb{Z}^+$ and $M_i$ is the cyclic group of order $i$ for each $i$, then the direct sum of the $M_i$'s is not isomorphic to their direct product. [Look at torsion.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**21.** Let $I$ be a nonempty index set and for each $i \in I$ let $N_i$ be a submodule of $M$. Prove that the following are equivalent:

**(i)** the submodule of $M$ generated by all the $N_i$'s is isomorphic to the direct sum of the $N_i$'s

**(ii)** if $\{i_1, i_2, \ldots, i_k\}$ is any finite subset of $I$ then $N_{i_1} \cap (N_{i_2} + \cdots + N_{i_k}) = 0$

**(iii)** if $\{i_1, i_2, \ldots, i_k\}$ is any finite subset of $I$ then $N_1 + \cdots + N_k = N_1 \oplus \cdots \oplus N_k$

**(iv)** for every element $x$ of the submodule of $M$ generated by the $N_i$'s there are unique elements $a_i \in N_i$ for all $i \in I$ such that all but a finite number of the $a_i$ are zero and $x$ is the (finite) sum of the $a_i$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**22.** Let $R$ be a Principal Ideal Domain, let $M$ be a torsion $R$-module (cf. Exercise 4) and let $p$ be a prime in $R$ (do not assume $M$ is finitely generated, hence it need not have a nonzero annihilator — cf. Exercise 5). The *$p$-primary component* of $M$ is the set of all elements of $M$ that are annihilated by some positive power of $p$.

**(a)** Prove that the $p$-primary component is a submodule. [See Exercise 13 in Section 1.]

**(b)** Prove that this definition of $p$-primary component agrees with the one given in Exercise 18 when $M$ has a nonzero annihilator.

**(c)** Prove that $M$ is the (possibly infinite) direct sum of its $p$-primary components, as $p$ runs over all primes of $R$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**23.** Show that any direct sum of free $R$-modules is free.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**24.** *(An arbitrary direct product of free modules need not be free)* For each positive integer $i$ let $M_i$ be the free $\mathbb{Z}$-module $\mathbb{Z}$, and let $M$ be the direct product $\prod_{i\in\mathbb{Z}^+} M_i$ (cf. Exercise 20). Each element of $M$ can be written uniquely in the form $(a_1, a_2, a_3, \dots)$ with $a_i \in \mathbb{Z}$ for all $i$. Let $N$ be the submodule of $M$ consisting of all such tuples with only finitely many nonzero $a_i$. Assume $M$ is a free $\mathbb{Z}$-module with basis $\mathcal{B}$.

**(a)** Show that $N$ is countable.

**(b)** Show that there is some countable subset $\mathcal{B}_1$ of $\mathcal{B}$ such that $N$ is contained in the submodule, $N_1$, generated by $\mathcal{B}_1$. Show also that $N_1$ is countable.

**(c)** Let $\overline{M} = M/N_1$. Show that $\overline{M}$ is a free $\mathbb{Z}$-module. Deduce that if $\overline{x}$ is any nonzero element of $\overline{M}$ then there are only finitely many distinct positive integers $k$ such that $\overline{x} = k\overline{m}$ for some $m \in M$ (depending on $k$).

**(d)** Let $\mathcal{S} = \{(b_1, b_2, b_3, \dots) \mid b_i = \pm i! \text{ for all } i\}$. Prove that $\mathcal{S}$ is uncountable. Deduce that there is some $s \in \mathcal{S}$ with $s \notin N_1$.

**(e)** Show that the assumption $M$ is free leads to a contradiction: By (d) we may choose $s \in \mathcal{S}$ with $s \notin N_1$. Show that for each positive integer $k$ there is some $m \in M$ with $\overline{s} = k\overline{m}$, contrary to (c). [Use the fact that $N \subseteq N_1$.]

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**25.** In the construction of direct limits, Exercise 8 of Section 7.6, show that if all $A_i$ are $R$-modules and the maps $\rho_{ij}$ are $R$-module homomorphisms, then the direct limit $A = \varinjlim A_i$ may be given the structure of an $R$-module in a natural way such that the maps $\rho_i : A_i \to A$ are all $R$-module homomorphisms. Verify the corresponding universal property (part (e)) for $R$-module homomorphisms $\varphi_i : A_i \to C$ commuting with the $\rho_{ij}$.

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**26.** Carry out the analysis of the preceding exercise corresponding to inverse limits to show that an inverse limit of $R$-modules is an $R$-module satisfying the appropriate universal property (cf. Exercise 10 of Section 7.6).

In these exercises $R$ is a ring with 1 and $M$ is a left $R$-module.

**27.** *(Free modules over noncommutative rings need not have a unique rank)* Let $M$ be the $\mathbb{Z}$-module $\mathbb{Z} \times \mathbb{Z} \times \cdots$ of Exercise 24 and let $R$ be its endomorphism ring, $R = \mathrm{End}_{\mathbb{Z}}(M)$ (cf. Exercises 29 and 30 in Section 7.1). Define $\varphi_1, \varphi_2 \in R$ by

$$\varphi_1(a_1, a_2, a_3, \ldots) = (a_1, a_3, a_5, \ldots)$$
$$\varphi_2(a_1, a_2, a_3, \ldots) = (a_2, a_4, a_6, \ldots)$$

**(a)** Prove that $\{\varphi_1, \varphi_2\}$ is a free basis of the left $R$-module $R$. [Define the maps $\psi_1$ and $\psi_2$ by $\psi_1(a_1, a_2, \ldots) = (a_1, 0, a_2, 0, \ldots)$ and $\psi_2(a_1, a_2, \ldots) = (0, a_1, 0, a_2, \ldots)$. Verify that $\varphi_i \psi_i = 1$, $\varphi_1 \psi_2 = 0 = \varphi_2 \psi_1$ and $\psi_1 \varphi_1 + \psi_2 \varphi_2 = 1$. Use these relations to prove that $\varphi_1, \varphi_2$ are independent and generate $R$ as a left $R$-module.]

**(b)** Use (a) to prove that $R \cong R^2$ and deduce that $R \cong R^n$ for all $n \in \mathbb{Z}^+$.

## 10.4. Tensor Products of Modules

Let $R$ be a ring with 1.

**1.** Let $f : R \to S$ be a ring homomorphism from the ring $R$ to the ring $S$ with $f(1_R) = 1_S$. Verify the details that $sr = sf(r)$ defines a right $R$-action on $S$ under which $S$ is an $(S, R)$-bimodule.

Let $R$ be a ring with 1.

**2.** Show that the element "$2 \otimes 1$" is 0 in $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ but is nonzero in $2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$.

Let $R$ be a ring with 1.

**3.** Show that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ and $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ are both left $\mathbb{R}$-modules but are not isomorphic as $\mathbb{R}$-modules.

Let $R$ be a ring with 1.

**4.** Show that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ and $\mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{Q}$ are isomorphic left $\mathbb{Q}$-modules. [Show they are both 1-dimensional vector spaces over $\mathbb{Q}$.]

Let $R$ be a ring with 1.

**5.** Let $A$ be a finite abelian group of order $n$ and let $p^k$ be the largest power of the prime $p$ dividing $n$. Prove that $\mathbb{Z}/p^k\mathbb{Z} \otimes_{\mathbb{Z}} A$ is isomorphic to the Sylow $p$-subgroup of $A$.

Let $R$ be a ring with 1.

**6.** If $R$ is any integral domain with quotient field $Q$, prove that $(Q/R) \otimes_R (Q/R) = 0$.

Let $R$ be a ring with 1.

**7.** If $R$ is any integral domain with quotient field $Q$ and $N$ is a left $R$-module, prove that every element of the tensor product $Q \otimes_R N$ can be written as a simple tensor of the form $(1/d) \otimes n$ for some nonzero $d \in R$ and some $n \in N$.

Let $R$ be a ring with 1.

**8.** Suppose $R$ is an integral domain with quotient field $Q$ and let $N$ be any $R$-module. Let $U = R^\times$ be the set of nonzero elements in $R$ and define $U^{-1}N$ to be the set of equivalence classes of ordered pairs of elements $(u, n)$ with $u \in U$ and $n \in N$ under the equivalence relation $(u, n) \sim (u', n)$ if and only if $u'n = un'$ in $N$.

**(a)** Prove that $U^{-1}N$ is an abelian group under the addition defined by $\overline{(u_1, n_1)} + \overline{(u_2, n_2)} = \overline{(u_1u_2, u_2n_1 + u_1n_2)}$. Prove that $r\overline{(u, n)} = \overline{(u, rn)}$ defines an action of $R$ on $U^{-1}N$ making it into an $R$-module. [This is an example of *localization* considered in general in Section 4 of Chapter 15, cf. also Section 5 in Chapter 7.]

**(b)** Show that the map from $Q \times N$ to $U^{-1}N$ defined by sending $(a/b, n)$ to $\overline{(b, an)}$ for $a \in R$, $b \in U$, $n \in N$, is an $R$-balanced map, so induces a homomorphism $f$ from $Q \otimes_R N$ to $U^{-1}N$. Show that the map $g$ from $U^{-1}N$ to $Q \otimes_R N$ defined by $g(\overline{(u, n)}) = (1/u) \otimes n$ is well defined and is an inverse homomorphism to $f$. Conclude that $Q \otimes_R N \cong U^{-1}N$ as $R$-modules.

**(c)** Conclude from (b) that $(1/d) \otimes n$ is 0 in $Q \otimes_R N$ if and only if $rn = 0$ for some nonzero $r \in R$.

**(d)** If $A$ is an abelian group, show that $\mathbb{Q} \otimes_{\mathbb{Z}} A = 0$ if and only if $A$ is a torsion abelian group (i.e., every element of $A$ has finite order).

Let $R$ be a ring with 1.

**9.** Suppose $R$ is an integral domain with quotient field $Q$ and let $N$ be any $R$-module. Let $Q \otimes_R N$ be the module obtained from $N$ by extension of scalars from $R$ to $Q$. Prove that the kernel of the $R$-module homomorphism $\iota : N \to Q \otimes_R N$ is the torsion submodule of $N$ (cf. Exercise 8 in Section 1). [Use the previous exercise.]

Let $R$ be a ring with 1.

**10.** Suppose $R$ is commutative and $N \cong R^n$ is a free $R$-module of rank $n$ with $R$-module basis $e_1, \ldots, e_n$.

**(a)** For any nonzero $R$-module $M$ show that every element of $M \otimes N$ can be written uniquely in the form $\sum_{i=1}^{n} m_i \otimes e_i$ where $m_i \in M$. Deduce that if $\sum_{i=1}^{n} m_i \otimes e_i = 0$ in $M \otimes N$ then $m_i = 0$ for $i = 1, \ldots, n$.

**(b)** Show that if $\sum m_i \otimes n_i = 0$ in $M \otimes N$ where the $n_i$ are merely assumed to be $R$-linearly independent then it is not necessarily true that all the $m_i$ are 0. [Consider $R = \mathbb{Z}$, $n = 1$, $M = \mathbb{Z}/2\mathbb{Z}$, and the element $1 \otimes 2$.]

Let $R$ be a ring with 1.

**11.** Let $\{e_1, e_2\}$ be a basis of $V = \mathbb{R}^2$. Show that the element $e_1 \otimes e_2 + e_2 \otimes e_1$ in $V \otimes_{\mathbb{R}} V$ cannot be written as a simple tensor $v \otimes w$ for any $v, w \in \mathbb{R}^2$.

Let $R$ be a ring with 1.

**12.** Let $V$ be a vector space over the field $F$ and let $v$, $v'$ be nonzero elements of $V$. Prove that $v \otimes v' = v' \otimes v$ in $V \otimes_F V$ if and only if $v = av'$ for some $a \in F$.

Let $R$ be a ring with 1.

**13.** Prove that the usual dot product of vectors defined by letting $(a_1, \ldots, a_n) \cdot (b_1, \ldots, b_n)$ be $a_1b_1 + \cdots + a_nb_n$ is a bilinear map from $\mathbb{R}^n \times \mathbb{R}^n$ to $\mathbb{R}$.

Let $R$ be a ring with 1.

**14.** Let $I$ be an arbitrary nonempty index set and for each $i \in I$ let $N_i$ be a left $R$-module. Let $M$ be a right $R$-module. Prove the group isomorphism: $M \otimes (\oplus_{i \in I} N_i) \cong \oplus_{i \in I} (M \otimes N_i)$, where the direct sum of an arbitrary collection of modules is defined in Exercise 20, Section 3. [Use the same argument as for the direct sum of two modules, taking care to note where the direct *sum* hypothesis is needed — cf. the next exercise.]

Let $R$ be a ring with 1.

**15.** Show that tensor products do not commute with direct products in general. [Consider the extension of scalars from $\mathbb{Z}$ to $\mathbb{Q}$ of the direct product of the modules $M_i = \mathbb{Z}/2^i\mathbb{Z}$, $i = 1, 2, \ldots$]

Let $R$ be a ring with 1.

**16.** Suppose $R$ is commutative and let $I$ and $J$ be ideals of $R$, so $R/I$ and $R/J$ are naturally $R$-modules.

**(a)** Prove that every element of $R/I \otimes_R R/J$ can be written as a simple tensor of the form $(1 \bmod I) \otimes (r \bmod J)$.

**(b)** Prove that there is an $R$-module isomorphism $R/I \otimes_R R/J \cong R/(I + J)$ mapping $(r \bmod I) \otimes (r' \bmod J)$ to $rr' \bmod (I + J)$.

Let $R$ be a ring with 1.

**17.** Let $I = (2, x)$ be the ideal generated by 2 and $x$ in the ring $R = \mathbb{Z}[x]$. The ring $\mathbb{Z}/2\mathbb{Z} = R/I$ is naturally an $R$-module annihilated by both 2 and $x$.

**(a)** Show that the map $\varphi : I \times I \to \mathbb{Z}/2\mathbb{Z}$ defined by

$$\varphi(a_0 + a_1x + \cdots + a_nx^n, b_0 + b_1x + \cdots + b_mx^m) = \frac{a_0}{2}b_1 \bmod 2$$

is $R$-bilinear.

**(b)** Show that there is an $R$-module homomorphism from $I \otimes_R I \to \mathbb{Z}/2\mathbb{Z}$ mapping $p(x) \otimes q(x)$ to $\frac{p(0)}{2}q'(0)$ where $q'$ denotes the usual polynomial derivative of $q$.

**(c)** Show that $2 \otimes x \neq x \otimes 2$ in $I \otimes_R I$.

Let $R$ be a ring with 1.

**18.** Suppose $I$ is a principal ideal in the integral domain $R$. Prove that the $R$-module $I \otimes_R I$ has no nonzero torsion elements (i.e., $rm = 0$ with $0 \neq r \in R$ and $m \in I \otimes_R I$ implies that $m = 0$).

Let $R$ be a ring with 1.

**19.** Let $I = (2, x)$ be the ideal generated by 2 and $x$ in the ring $R = \mathbb{Z}[x]$ as in Exercise 17. Show that the nonzero element $2 \otimes x - x \otimes 2$ in $I \otimes_R I$ is a torsion element. Show in fact that $2 \otimes x - x \otimes 2$ is annihilated by both 2 and $x$ and that the submodule of $I \otimes_R I$ generated by $2 \otimes x - x \otimes 2$ is isomorphic to $R/I$.

Let $R$ be a ring with 1.

**20.** Let $I = (2, x)$ be the ideal generated by 2 and $x$ in the ring $R = \mathbb{Z}[x]$. Show that the element $2 \otimes 2 + x \otimes x$ in $I \otimes_R I$ is not a simple tensor, i.e., cannot be written as $a \otimes b$ for some $a, b \in I$.

Let $R$ be a ring with 1.

**21.** Suppose $R$ is commutative and let $I$ and $J$ be ideals of $R$.

**(a)** Show there is a surjective $R$-module homomorphism from $I \otimes_R J$ to the product ideal $IJ$ mapping $i \otimes j$ to the element $ij$.

**(b)** Give an example to show that the map in (a) need not be injective (cf. Exercise 17).

Let $R$ be a ring with 1.

**22.** Suppose that $M$ is a left and a right $R$-module such that $rm = mr$ for all $r \in R$ and $m \in M$. Show that the elements $r_1r_2$ and $r_2r_1$ act the same on $M$ for every $r_1, r_2 \in R$. (This explains why the assumption that $R$ is commutative in the definition of an $R$-algebra is a fairly natural one.)

Let $R$ be a ring with 1.

**23.** Verify the details that the multiplication in Proposition 19 makes $A \otimes_R B$ into an $R$-algebra.

Let $R$ be a ring with 1.

**24.** Prove that the extension of scalars from $\mathbb{Z}$ to the Gaussian integers $\mathbb{Z}[i]$ of the ring $\mathbb{R}$ is isomorphic to $\mathbb{C}$ as a ring: $\mathbb{Z}[i] \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{C}$ as rings.

Let $R$ be a ring with 1.

**25.** Let $R$ be a subring of the commutative ring $S$ and let $x$ be an indeterminate over $S$. Prove that $S[x]$ and $S \otimes_R R[x]$ are isomorphic as $S$-algebras.

Let $R$ be a ring with 1.

**26.** Let $S$ be a commutative ring containing $R$ (with $1_S = 1_R$) and let $x_1, \ldots, x_n$ be independent indeterminates over the ring $S$. Show that for every ideal $I$ in the polynomial ring $R[x_1, \ldots, x_n]$ that $S \otimes_R (R[x_1, \ldots, x_n]/I) \cong S[x_1, \ldots, x_n]/IS[x_1, \ldots, x_n]$ as $S$-algebras.

Let $R$ be a ring with 1.

The next exercise shows the ring $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ introduced at the end of this section is isomorphic to $\mathbb{C} \times \mathbb{C}$. One may also prove this via Exercise 26 and Proposition 16 in Section 9.5, since $\mathbb{C} \cong \mathbb{R}[x]/(x^2 + 1)$. The ring $\mathbb{C} \times \mathbb{C}$ is also discussed in Exercise 23 of Section 1.

**27.** **(a)** Write down a formula for the multiplication of two elements $a \cdot 1 + b \cdot e_2 + c \cdot e_3 + d \cdot e_4$ and $a' \cdot 1 + b' \cdot e_2 + c' \cdot e_3 + d' \cdot e_4$ in the example $A = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ following Proposition 21 (where $1 = 1 \otimes 1$ is the identity of $A$).

**(b)** Let $\epsilon_1 = \frac{1}{2}(1 \otimes 1 + i \otimes i)$ and $\epsilon_2 = \frac{1}{2}(1 \otimes 1 - i \otimes i)$. Show that $\epsilon_1 \epsilon_2 = 0$, $\epsilon_1 + \epsilon_2 = 1$, and $\epsilon_j^2 = \epsilon_j$ for $j = 1, 2$ ($\epsilon_1$ and $\epsilon_2$ are called *orthogonal idempotents* in $A$). Deduce that $A$ is isomorphic as a ring to the direct product of two principal ideals: $A \cong A\epsilon_1 \times A\epsilon_2$ (cf. Exercise 1, Section 7.6).

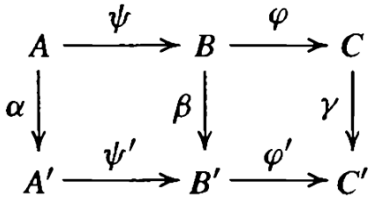
**(c)** Prove that the map $\varphi : \mathbb{C} \times \mathbb{C} \to \mathbb{C} \times \mathbb{C}$ by $\varphi(z_1, z_2) = (z_1 z_2, z_1 \overline{z_2})$, where $\overline{z_2}$ denotes the complex conjugate of $z_2$, is an $\mathbb{R}$-bilinear map.

**(d)** Let $\Phi$ be the $\mathbb{R}$-module homomorphism from $A$ to $\mathbb{C} \times \mathbb{C}$ obtained from $\varphi$ in (c). Show that $\Phi(\epsilon_1) = (0, 1)$ and $\Phi(\epsilon_2) = (1, 0)$. Show also that $\Phi$ is $\mathbb{C}$-linear, where the action of $\mathbb{C}$ is on the left tensor factor in $A$ and on both factors in $\mathbb{C} \times \mathbb{C}$. Deduce that $\Phi$ is surjective. Show that $\Phi$ is a $\mathbb{C}$-algebra isomorphism.

## 10.5. Exact Sequences—Projective, Injective, and Flat Modules

Let $R$ be a ring with 1.

**1.** Suppose that




is a commutative diagram of groups and that the rows are exact. Prove that

**(a)** if $\varphi$ and $\alpha$ are surjective, and $\beta$ is injective then $\gamma$ is injective. [If $c \in \ker \gamma$, show there is a $b \in B$ with $\varphi(b) = c$. Show that $\varphi'(\beta(b)) = 0$ and deduce that $\beta(b) = \psi'(a')$ for some $a' \in A'$. Show there is an $a \in A$ with $\alpha(a) = a'$ and that $\beta(\psi(a)) = \beta(b)$. Conclude that $b = \psi(a)$ and hence $c = \varphi(b) = 0$.]

**(b)** if $\psi'$, $\alpha$, and $\gamma$ are injective, then $\beta$ is injective,

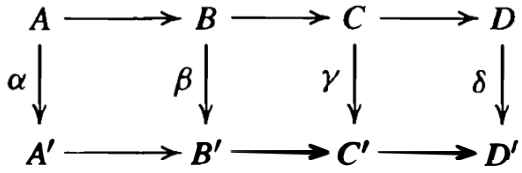
**(c)** if $\varphi$, $\alpha$, and $\gamma$ are surjective, then $\beta$ is surjective,

**(d)** if $\beta$ is injective, $\alpha$ and $\gamma$ are surjective, then $\gamma$ is injective,

**(e)** if $\beta$ is surjective, $\gamma$ and $\psi'$ are injective, then $\alpha$ is surjective.

Let $R$ be a ring with 1.

**2.** Suppose that




is a commutative diagram of groups, and that the rows are exact. Prove that

**(a)** if $\alpha$ is surjective, and $\beta$, $\delta$ are injective, then $\gamma$ is injective.

**(b)** if $\delta$ is injective, and $\alpha$, $\gamma$ are surjective, then $\beta$ is surjective.

Let $R$ be a ring with 1.

**3.** Let $P_1$ and $P_2$ be $R$-modules. Prove that $P_1 \oplus P_2$ is a projective $R$-module if and only if both $P_1$ and $P_2$ are projective.

Let $R$ be a ring with 1.

**4.** Let $Q_1$ and $Q_2$ be $R$-modules. Prove that $Q_1 \oplus Q_2$ is an injective $R$-module if and only if both $Q_1$ and $Q_2$ are injective.

Let $R$ be a ring with 1.

**5.** Let $A_1$ and $A_2$ be $R$-modules. Prove that $A_1 \oplus A_2$ is a flat $R$-module if and only if both $A_1$ and $A_2$ are flat. More generally, prove that an arbitrary direct sum $\sum A_i$ of $R$-modules is flat if and only if each $A_i$ is flat. [Use the fact that tensor product commutes with arbitrary direct sums.]

Let $R$ be a ring with 1.

**6.** Prove that the following are equivalent for a ring $R$:

**(i)** Every $R$-module is projective.

**(ii)** Every $R$-module is injective.

Let $R$ be a ring with 1.

**7.** Let $A$ be a nonzero finite abelian group.
**(a)** Prove that $A$ is not a projective $\mathbb{Z}$-module.
**(b)** Prove that $A$ is not an injective $\mathbb{Z}$-module.

Let $R$ be a ring with 1.

**8.** Let $Q$ be a nonzero divisible $\mathbb{Z}$-module. Prove that $Q$ is not a projective $\mathbb{Z}$-module. Deduce that the rational numbers $\mathbb{Q}$ is not a projective $\mathbb{Z}$-module. [Show first that if $F$ is any free module then $\cap_{n=1}^{\infty} nF = 0$ (use a basis of $F$ to prove this). Now suppose to the contrary that $Q$ is projective and derive a contradiction from Proposition 30(4).]

Let $R$ be a ring with 1.

**9.** Assume $R$ is commutative with 1.

**(a)** Prove that the tensor product of two free $R$-modules is free. [Use the fact that tensor products commute with direct sums.]

**(b)** Use (a) to prove that the tensor product of two projective $R$-modules is projective.

Let $R$ be a ring with 1.

**10.** Let $R$ and $S$ be rings with 1 and let $M$ and $N$ be left $R$-modules. Assume also that $M$ is an $(R, S)$-bimodule.

**(a)** For $s \in S$ and for $\varphi \in \mathrm{Hom}_R(M, N)$ define $(s\varphi) : M \to N$ by $(s\varphi)(m) = \varphi(ms)$. Prove that $s\varphi$ is a homomorphism of left $R$-modules, and that this action of $S$ on $\mathrm{Hom}_R(M, N)$ makes it into a *left* $S$-module.

**(b)** Let $S = R$ and let $M = R$ (considered as an $(R, R)$-bimodule by left and right ring multiplication on itself). For each $n \in N$ define $\varphi_n : R \to N$ by $\varphi_n(r) = rn$, i.e., $\varphi_n$ is the unique $R$-module homomorphism mapping $1_R$ to $n$. Show that $\varphi_n \in \mathrm{Hom}_R(R, N)$. Use part (a) to show that the map $n \mapsto \varphi_n$ is an isomorphism of left $R$-modules: $N \cong \mathrm{Hom}_R(R, N)$.

**(c)** Deduce that if $N$ is a free (respectively, projective, injective, flat) left $R$-module, then $\mathrm{Hom}_R(R, N)$ is also a free (respectively, projective, injective, flat) left $R$-module.

Let $R$ be a ring with 1.

**11.** Let $R$ and $S$ be rings with 1 and let $M$ and $N$ be left $R$-modules. Assume also that $N$ is an $(R, S)$-bimodule.

**(a)** For $s \in S$ and for $\varphi \in \mathrm{Hom}_R(M, N)$ define $(\varphi s) : M \to N$ by $(\varphi s)(m) = \varphi(m)s$. Prove that $\varphi s$ is a homomorphism of left $R$-modules, and that this action of $S$ on $\mathrm{Hom}_R(M, N)$ makes it into a *right* $S$-module. Deduce that $\mathrm{Hom}_R(M, R)$ is a right $R$-module, for any $R$-module $M$—called the *dual module* to $M$.

**(b)** Let $N = R$ be considered as an $(R, R)$-bimodule as usual. Under the action defined in part (a) show that the map $r \mapsto \varphi_r$ is an isomorphism of right $R$-modules: $\mathrm{Hom}_R(R, R) \cong R$, where $\varphi_r$ is the homomorphism that maps $1_R$ to $r$. Deduce that if $M$ is a finitely generated free left $R$-module, then $\mathrm{Hom}_R(M, R)$ is a free right $R$-module of the same rank. (cf. also Exercise 13.)

**(c)** Show that if $M$ is a finitely generated projective $R$-module then its dual module $\mathrm{Hom}_R(M, R)$ is also projective.

Let $R$ be a ring with 1.

**12.** Let $A$ be an $R$-module, let $I$ be any nonempty index set and for each $i \in I$ let $B_i$ be an $R$-module. Prove the following isomorphisms of abelian groups; when $R$ is commutative prove also that these are $R$-module isomorphisms. (Arbitrary direct sums and direct products of modules are introduced in Exercise 20 of Section 3.)

**(a)** $\mathrm{Hom}_R(\bigoplus_{i\in I} B_i, A) \cong \prod_{i\in I} \mathrm{Hom}_R(B_i, A)$

**(b)** $\mathrm{Hom}_R(A, \prod_{i\in I} B_i) \cong \prod_{i\in I} \mathrm{Hom}_R(A, B_i)$.

Let $R$ be a ring with 1.

**13.** **(a)** Show that the dual of the free $\mathbb{Z}$-module with countable basis is not free. [Use the preceding exercise and Exercise 24, Section 3.] (See also Exercise 5 in Section 11.3.)
**(b)** Show that the dual of the free $\mathbb{Z}$-module with countable basis is also not projective. [You may use the fact that any submodule of a free $\mathbb{Z}$-module is free.]

Let $R$ be a ring with 1.

**14.** Let $0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$ be a sequence of $R$-modules.

**(a)** Prove that the associated sequence

$$0 \longrightarrow \mathrm{Hom}_R(D, L) \xrightarrow{\psi'} \mathrm{Hom}_R(D, M) \xrightarrow{\varphi'} \mathrm{Hom}_R(D, N) \longrightarrow 0$$

is a short exact sequence of abelian groups for all $R$-modules $D$ if and only if the original sequence is a split short exact sequence. [To show the sequence splits, take $D = N$ and show the lift of the identity map in $\mathrm{Hom}_R(N, N)$ to $\mathrm{Hom}_R(N, M)$ is a splitting homomorphism for $\varphi$.]

**(b)** Prove that the associated sequence

$$0 \longrightarrow \mathrm{Hom}_R(N, D) \xrightarrow{\varphi'} \mathrm{Hom}_R(M, D) \xrightarrow{\psi'} \mathrm{Hom}_R(L, D) \longrightarrow 0$$

is a short exact sequence of abelian groups for all $R$-modules $D$ if and only if the original sequence is a split short exact sequence.

Let $R$ be a ring with 1.

**15.** Let $M$ be a left $R$-module where $R$ is a ring with 1.

**(a)** Show that $\mathrm{Hom}_{\mathbb{Z}}(R, M)$ is a left $R$-module under the action $(r\varphi)(r') = \varphi(r'r)$ (see Exercise 10).

**(b)** Suppose that $0 \to A \xrightarrow{\psi} B$ is an exact sequence of $R$-modules. Prove that if every homomorphism $f$ from $A$ to $M$ lifts to a homomorphism $F$ from $B$ to $M$ with $f = F \circ \psi$, then every homomorphism $f'$ from $A$ to $\mathrm{Hom}_{\mathbb{Z}}(R, M)$ lifts to a homomorphism $F'$ from $B$ to $\mathrm{Hom}_{\mathbb{Z}}(R, M)$ with $f' = F' \circ \psi$. [Given $f'$, show that $f(a) = f'(a)(1_R)$ defines a homomorphism of $A$ to $M$. If $F$ is the associated lift of $f$ to $B$, show that $F'(b)(r) = F(rb)$ defines a homomorphism from $B$ to $\mathrm{Hom}_{\mathbb{Z}}(R, M)$ that lifts $f'$.]

**(c)** Prove that if $Q$ is an injective $R$-module then $\mathrm{Hom}_{\mathbb{Z}}(R, Q)$ is also an injective $R$-module.

Let $R$ be a ring with 1.

**16.** This exercise proves Theorem 38 that every left $R$-module $M$ is contained in an injective left $R$-module.

**(a)** Show that $M$ is contained in an injective $\mathbb{Z}$-module $Q$. [$M$ is a $\mathbb{Z}$-module—use Corollary 37.]

**(b)** Show that $\mathrm{Hom}_R(R, M) \subseteq \mathrm{Hom}_{\mathbb{Z}}(R, M) \subseteq \mathrm{Hom}_{\mathbb{Z}}(R, Q)$.

**(c)** Use the $R$-module isomorphism $M \cong \mathrm{Hom}_R(R, M)$ (Exercise 10) and the previous exercise to conclude that $M$ is contained in an injective module.

Let $R$ be a ring with 1.

**17.** This exercise completes the proof of Proposition 34. Suppose that $Q$ is an $R$-module with the property that every short exact sequence $0 \to Q \to M_1 \to N \to 0$ splits and suppose that the sequence $0 \to L \xrightarrow{\psi} M$ is exact. Prove that every $R$-module homomorphism $f$ from $L$ to $Q$ can be lifted to an $R$-module homomorphism $F$ from $M$ to $Q$ with $f = F \circ \psi$. [By the previous exercise, $Q$ is contained in an injective $R$-module. Use the splitting property together with Exercise 4 (noting that Exercise 4 can be proved using (2) in Proposition 34 as the definition of an injective module).]

Let $R$ be a ring with 1.

**18.** Prove that the injective hull of the $\mathbb{Z}$-module $\mathbb{Z}$ is $\mathbb{Q}$. [Let $H$ be the injective hull of $\mathbf{Z}$ and argue that $\mathbb{Q}$ contains an isomorphic copy of $H$. Use the divisibility of $H$ to show $1/n \in H$ for all nonzero integers $n$, and deduce that $H = \mathbb{Q}$.]

Let $R$ be a ring with 1.

**19.** If $F$ is a field, prove that the injective hull of $F$ is $F$.

Let $R$ be a ring with 1.

**20.** Prove that the polynomial ring $R[x]$ in the indeterminate $x$ over the commutative ring $R$ is a flat $R$-module.

Let $R$ be a ring with 1.

**21.** Let $R$ and $S$ be rings with 1 and suppose $M$ is a right $R$-module, and $N$ is an $(R, S)$-bimodule. If $M$ is flat over $R$ and $N$ is flat as an $S$-module prove that $M \otimes_R N$ is flat as a right $S$-module.

Let $R$ be a ring with 1.

**22.** Suppose that $R$ is a commutative ring and that $M$ and $N$ are flat $R$-modules. Prove that $M \otimes_R N$ is a flat $R$-module. [Use the previous exercise.]

Let $R$ be a ring with 1.

**23.** Prove that the (right) module $M \otimes_R S$ obtained by changing the base from the ring $R$ to the ring $S$ (by some homomorphism $f : R \to S$ with $f(1_R) = 1_S$, cf. Example 6 following Corollary 12 in Section 4) of the flat (right) $R$-module $M$ is a flat $S$-module.

Let $R$ be a ring with 1.

**24.** Prove that $A$ is a flat $R$-module if and only if for any left $R$-modules $L$ and $M$ where $L$ is *finitely generated*, then $\psi : L \to M$ injective implies that also $1 \otimes \psi : A \otimes_R L \to A \otimes_R M$ is injective. [Use the techniques in the proof of Corollary 42.]

Let $R$ be a ring with 1.

**25.** *(A Flatness Criterion)* Parts (a)-(c) of this exercise prove that $A$ is a flat $R$-module if and only if for every finitely generated ideal $I$ of $R$, the map from $A \otimes_R I \to A \otimes_R R \cong A$ induced by the inclusion $I \subseteq R$ is again injective (or, equivalently, $A \otimes_R I \cong AI \subseteq A$).

**(a)** Prove that if $A$ is flat then $A \otimes_R I \to A \otimes_R R$ is injective.

**(b)** If $A \otimes_R I \to A \otimes_R R$ is injective for every finitely generated ideal $I$, prove that $A \otimes_R I \to A \otimes_R R$ is injective for every ideal $I$. Show that if $K$ is any submodule of a finitely generated free module $F$ then $A \otimes_R K \to A \otimes_R F$ is injective. Show that the same is true for any free module $F$. [Cf. the proof of Corollary 42.]

**(c)** Under the assumption in (b), suppose $L$ and $M$ are $R$-modules and $L \xrightarrow{\psi} M$ is injective. Prove that $A \otimes_R L \xrightarrow{1 \otimes \psi} A \otimes_R M$ is injective and conclude that $A$ is flat. [Write $M$ as a quotient of the free module $F$, giving a short exact sequence

$$0 \longrightarrow K \longrightarrow F \xrightarrow{f} M \longrightarrow 0.$$

Show that if $J = f^{-1}(\psi(L))$ and $\iota : J \to F$ is the natural injection, then the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & K & \longrightarrow & J & \longrightarrow & L & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle id} & & \downarrow{\scriptstyle \iota} & & \downarrow{\scriptstyle \psi} & & \\ 0 & \longrightarrow & K & \longrightarrow & F & \longrightarrow & M & \longrightarrow & 0 \end{array}$$

is commutative with exact rows. Show that the induced diagram

$$\begin{array}{ccccccccc} A \otimes_R K & \longrightarrow & A \otimes_R J & \longrightarrow & A \otimes_R L & \longrightarrow & 0 \\ \downarrow{\scriptstyle id} & & \downarrow{\scriptstyle 1 \otimes \iota} & & \downarrow{\scriptstyle 1 \otimes \psi} & & \\ A \otimes_R K & \longrightarrow & A \otimes_R F & \longrightarrow & A \otimes_R M & \longrightarrow & 0 \end{array}$$

is commutative with exact rows. Use (b) to show that $1 \otimes \iota$ is injective, then use Exercise 1 to conclude that $1 \otimes \psi$ is injective.]

**(d)** *(A Flatness Criterion for quotients)* Suppose $A = F/K$ where $F$ is flat (e.g., if $F$ is free) and $K$ is an $R$-submodule of $F$. Prove that $A$ is flat if and only if $FI \cap K = KI$ for every finitely generated ideal $I$ of $R$. [Use (a) to prove $F \otimes_R I \cong FI$ and observe the image of $K \otimes_R I$ is $KI$; tensor the exact sequence $0 \to K \to F \to A \to 0$ with $I$ to prove that $A \otimes_R I \cong FI/KI$, and apply the flatness criterion.]

Let $R$ be a ring with 1.

**26.** Suppose $R$ is a P.I.D. This exercise proves that $A$ is a flat $R$-module if and only if $A$ is torsion free $R$-module (i.e., if $a \in A$ is nonzero and $r \in R$, then $ra = 0$ implies $r = 0$).

**(a)** Suppose that $A$ is flat and for fixed $r \in R$ consider the map $\psi_r : R \to R$ defined by multiplication by $r$: $\psi_r(x) = rx$. If $r$ is nonzero show that $\psi_r$ is an injection. Conclude from the flatness of $A$ that the map from $A$ to $A$ defined by mapping $a$ to $ra$ is injective and that $A$ is torsion free.

**(b)** Suppose that $A$ is torsion free. If $I$ is a nonzero ideal of $R$, then $I = rR$ for some nonzero $r \in R$. Show that the map $\psi_r$ in (a) induces an isomorphism $R \cong I$ of $R$-modules and that the composite $R \xrightarrow{\psi} I \xrightarrow{\iota} R$ of $\psi_r$ with the inclusion $\iota : I \subseteq R$ is multiplication by $r$. Prove that the composite $A \otimes_R R \xrightarrow{1 \otimes \psi_r} A \otimes_R I \xrightarrow{1 \otimes \iota} A \otimes_R R$ corresponds to the map $a \mapsto ra$ under the identification $A \otimes_R R = A$ and that this composite is injective since $A$ is torsion free. Show that $1 \otimes \psi_r$ is an isomorphism and deduce that $1 \otimes \iota$ is injective. Use the previous exercise to conclude that $A$ is flat.

Let $R$ be a ring with 1.

**27.** Let $M$, $A$ and $B$ be $R$-modules.

**(a)** Suppose $f : A \to M$ and $g : B \to M$ are $R$-module homomorphisms. Prove that $X = \{(a, b) \mid a \in A,\ b \in B \text{ with } f(a) = g(b)\}$ is an $R$-submodule of the direct sum $A \oplus B$ (called the *pullback* or *fiber product* of $f$ and $g$) and that there is a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\pi_2} & B \\ {\scriptstyle \pi_1}\downarrow & & \downarrow{\scriptstyle g} \\ A & \xrightarrow{f} & M \end{array}$$

where $\pi_1$ and $\pi_2$ are the natural projections onto the first and second components.

**(b)** Suppose $f' : M \to A$ and $g' : M \to B$ are $R$-module homomorphisms. Prove that the quotient $Y$ of $A \oplus B$ by $\{(f'(m), -g'(m)) \mid m \in M\}$ is an $R$-module (called the *pushout* or *fiber sum* of $f'$ and $g'$) and that there is a commutative diagram

$$\begin{array}{ccc} M & \xrightarrow{g'} & B \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle \pi_2'} \\ A & \xrightarrow{\pi_1'} & Y \end{array}$$

where $\pi_1'$ and $\pi_2'$ are the natural maps to the quotient induced by the maps into the first and second components.

Let $R$ be a ring with 1.

**28.** **(a)** *(Schanuel's Lemma)* If $0 \to K \to P \xrightarrow{\varphi} M \to 0$ and $0 \to K' \to P' \xrightarrow{\varphi'} M \to 0$ are exact sequences of $R$-modules where $P$ and $P'$ are projective, prove $P \oplus K' \cong P' \oplus K$ as $R$-modules. [Show that there is an exact sequence $0 \to \ker \pi \to X \xrightarrow{\pi} P \to 0$ with $\ker \pi \cong K'$, where $X$ is the fiber product of $\varphi$ and $\varphi'$ as in the previous exercise. Deduce that $X \cong P \oplus K'$. Show similarly that $X \cong P' \oplus K$.]

**(b)** If $0 \to M \to Q \xrightarrow{\psi} L \to 0$ and $0 \to M \to Q' \xrightarrow{\psi'} L' \to 0$ are exact sequences of $R$-modules where $Q$ and $Q'$ are injective, prove $Q \oplus L' \cong Q' \oplus L$ as $R$-modules.

The $R$-modules $M$ and $N$ are said to be *projectively equivalent* if $M \oplus P \cong N \oplus P'$ for some projective modules $P$, $P'$. Similarly, $M$ and $N$ are *injectively equivalent* if $M \oplus Q \cong N \oplus Q'$ for some injective modules $Q$, $Q'$. The previous exercise shows $K$ and $K'$ are projectively equivalent and $L$ and $L'$ are injectively equivalent.

# 11. Vector Spaces

## 11.1. Definitions and Basic Theory

**1.** Let $V = \mathbb{R}^n$ and let $(a_1, a_2, \ldots, a_n)$ be a fixed vector in $V$. Prove that the collection of elements $(x_1, x_2, \ldots, x_n)$ of $V$ with $a_1x_1 + a_2x_2 + \ldots + a_nx_n = 0$ is a subspace of $V$. Determine the dimension of this subspace and find a basis.

**2.** Let $V$ be the collection of polynomials with coefficients in $\mathbb{Q}$ in the variable $x$ of degree at most 5. Prove that $V$ is a vector space over $\mathbb{Q}$ of dimension 6, with $1, x, x^2, \ldots, x^5$ as basis. Prove that $1, 1+x, 1+x+x^2, \ldots, 1+x+x^2+x^3+x^4+x^5$ is also a basis for $V$.

**3.** Let $\varphi$ be the linear transformation $\varphi : \mathbb{R}^4 \to \mathbb{R}^1$ such that

$$\varphi((1, 0, 0, 0)) = 1 \qquad \varphi((1, -1, 0, 0)) = 0$$
$$\varphi((1, -1, 1, 0)) = 1 \qquad \varphi((1, -1, 1, -1)) = 0.$$

Determine $\varphi((a, b, c, d))$.

**4.** Prove that the space of real-valued functions on the closed interval $[a, b]$ is an infinite dimensional vector space over $\mathbb{R}$, where $a < b$.

**5.** Prove that the space of continuous real-valued functions on the closed interval $[a, b]$ is an infinite dimensional vector space over $\mathbb{R}$, where $a < b$.

**6.** Let $V$ be a vector space of finite dimension. If $\varphi$ is any linear transformation from $V$ to $V$ prove there is an integer $m$ such that the intersection of the image of $\varphi^m$ and the kernel of $\varphi^m$ is $\{0\}$.

**7.** Let $\varphi$ be a linear transformation from a vector space $V$ of dimension $n$ to itself that satisfies $\varphi^2 = 0$. Prove that the image of $\varphi$ is contained in the kernel of $\varphi$ and hence that the rank of $\varphi$ is at most $n/2$.

**8.** Let $V$ be a vector space over $F$ and let $\varphi$ be a linear transformation of the vector space $V$ to itself. A nonzero element $v \in V$ satisfying $\varphi(v) = \lambda v$ for some $\lambda \in F$ is called an *eigenvector* of $\varphi$ with *eigenvalue* $\lambda$. Prove that for any fixed $\lambda \in F$ the collection of eigenvectors of $\varphi$ with eigenvalue $\lambda$ together with 0 forms a subspace of $V$.

**9.** Let $V$ be a vector space over $F$ and let $\varphi$ be a linear transformation of the vector space $V$ to itself. Suppose for $i = 1, 2, \ldots, k$ that $v_i \in V$ is an eigenvector for $\varphi$ with eigenvalue $\lambda_i \in F$ (cf. the preceding exercise) and that all the eigenvalues $\lambda_i$ are distinct. Prove that $v_1, v_2, \ldots, v_k$ are linearly independent. [Use induction on $k$: write a linear dependence relation among the $v_i$ and apply $\varphi$ to get another linear dependence relation among the $v_i$ involving the eigenvalues — now subtract a suitable multiple of the first linear relation to get a linear dependence relation on fewer elements.] Conclude that any linear transformation on an $n$-dimensional vector space has at most $n$ distinct eigenvalues.

In the following exercises let $V$ be a vector space of arbitrary dimension over a field $F$.

**10.** Prove that any vector space $V$ has a basis (by convention the null set is the basis for the zero space). [Let $\mathcal{S}$ be the set of subsets of $V$ consisting of linearly independent vectors, partially ordered under inclusion; apply Zorn's Lemma to $\mathcal{S}$ and show a maximal element of $\mathcal{S}$ is a basis.]

In the following exercises let $V$ be a vector space of arbitrary dimension over a field $F$.

**11.** Refine your argument in the preceding exercise to prove that any set of linearly independent vectors of $V$ is contained in a basis of $V$.

In the following exercises let $V$ be a vector space of arbitrary dimension over a field $F$.

**12.** If $F$ is a field with a finite or countable number of elements and $V$ is an infinite dimensional vector space over $F$ with basis $\mathcal{B}$, prove that the cardinality of $V$ equals the cardinality of $\mathcal{B}$. Deduce in this case that any two bases of $V$ have the same cardinality.

In the following exercises let $V$ be a vector space of arbitrary dimension over a field $F$.

**13.** Prove that as vector spaces over $\mathbb{Q}$, $\mathbb{R}^n \cong \mathbb{R}$, for all $n \in \mathbb{Z}^+$ (note that, in particular, this means $\mathbb{R}^n$ and $\mathbb{R}$ are isomorphic as additive abelian groups).

In the following exercises let $V$ be a vector space of arbitrary dimension over a field $F$.

**14.** Let $\mathcal{A}$ be a basis for the infinite dimensional space $V$. Prove that $V$ is isomorphic to the *direct sum* of copies of the field $F$ indexed by the set $\mathcal{A}$. Prove that the *direct product* of copies of $F$ indexed by $\mathcal{A}$ is a vector space over $F$ and it has strictly larger dimension than the dimension of $V$ (see the exercises in Section 10.3 for the definitions of direct sum and direct product of infinitely many modules).

## 11.2. The Matrix of a Linear Transformation

**1.** Let $V$ be the collection of polynomials with coefficients in $\mathbb{Q}$ in the variable $x$ of degree at most 5. Determine the transition matrix from the basis $1, x, x^2, \ldots, x^5$ for $V$ to the basis $1, 1+x, 1+x+x^2, \ldots, 1+x+x^2+x^3+x^4+x^5$ for $V$.

**2.** Let $V$ be the vector space of the preceding exercise. Let $\varphi = d/dx$ be the linear transformation of $V$ to itself given by usual differentiation of a polynomial with respect to $x$. Determine the matrix of $\varphi$ with respect to the two bases for $V$ in the previous exercise.

**3.** Let $V$ be the collection of polynomials with coefficients in $F$ in the variable $x$ of degree at most $n$. Determine the transition matrix from the basis $1, x, x^2, \ldots, x^n$ for $V$ to the elements

$$1,\ x-\lambda, \ldots,\ (x-\lambda)^{n-1},\ (x-\lambda)^n$$

where $\lambda$ is a fixed element of $F$. Conclude that these elements are a basis for $V$.

**4.** Let $\varphi$ be the linear transformation of $\mathbb{R}^2$ to itself given by rotation counterclockwise around the origin through an angle $\theta$. Show that the matrix of $\varphi$ with respect to the standard basis for $\mathbb{R}^2$ is $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

**5.** Show that the $m \times n$ matrix $A$ is nonsingular if and only if the linear transformation $\varphi$ is a nonsingular linear transformation from the $n$-dimensional space $V$ to the $m$-dimensional space $W$, where $A = M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$, regardless of the choice of bases $\mathcal{B}$ and $\mathcal{E}$.



**6.** Prove if $\varphi \in \mathrm{Hom}_F(F^n, F^m)$, and $\mathcal{B}, \mathcal{E}$ are the natural bases of $F^n$, $F^m$ respectively, then the range of $\varphi$ equals the span of the set of columns of $M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$. Deduce that the rank of $\varphi$ (as a linear transformation) equals the column rank of $M_{\mathcal{B}}^{\mathcal{E}}(\varphi)$.

**7.** Prove that any two similar matrices have the same row rank and the same column rank.

**8.** Let $V$ be an $n$-dimensional vector space over $F$ and let $\varphi$ be a linear transformation of the vector space $V$ to itself.

**(a)** Prove that if $V$ has a basis consisting of eigenvectors for $\varphi$ (cf. Exercise 8 of Section 1) then the matrix representing $\varphi$ with respect to this basis (for both domain and range) is diagonal with the eigenvalues as diagonal entries.

**(b)** If $A$ is the $n \times n$ matrix representing $\varphi$ with respect to a given basis for $V$ (for both domain and range) prove that $A$ is similar to a diagonal matrix if and only if $V$ has a basis of eigenvectors for $\varphi$.

**9.** If $W$ is a subspace of the vector space $V$ stable under the linear transformation $\varphi$ (i.e., $\varphi(W) \subseteq W$), show that $\varphi$ induces linear transformations $\varphi|_W$ on $W$ and $\bar{\varphi}$ on the quotient vector space $V/W$. If $\varphi|_W$ and $\bar{\varphi}$ are nonsingular prove $\varphi$ is nonsingular. Prove the converse holds if $V$ has finite dimension and give a counterexample with $V$ infinite dimensional.

**10.** Let $V$ be an $n$-dimensional vector space and let $\varphi$ be a linear transformation of $V$ to itself. Suppose $W$ is a subspace of $V$ of dimension $m$ that is stable under $\varphi$.

**(a)** Prove that there is a basis for $V$ with respect to which the matrix for $\varphi$ is of the form

$$\begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$$

where $A$ is an $m \times m$ matrix, $B$ is an $m \times (n-m)$ matrix and $C$ is an $(n-m) \times (n-m)$ matrix (such a matrix is called *block upper triangular*).

**(b)** Prove that if there is a subspace $W'$ invariant under $\varphi$ so that $V = W \oplus W'$ decomposes as a direct sum then the bases for $W$ and $W'$ give a basis for $V$ with respect to which the matrix for $\varphi$ is *block diagonal*:

$$\begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$$

where $A$ is an $m \times m$ matrix and $C$ is an $(n-m) \times (n-m)$ matrix.

**(c)** Prove conversely that if there is a basis for $V$ with respect to which $\varphi$ is block diagonal as in (b) then there are $\varphi$-invariant subspaces $W$ and $W'$ of dimensions $m$ and $n-m$, respectively, with $V = W \oplus W'$.

**11.** Let $\varphi$ be a linear transformation from the finite dimensional vector space $V$ to itself such that $\varphi^2 = \varphi$.

**(a)** Prove that image $\varphi \cap \ker \varphi = 0$.

**(b)** Prove that $V = \text{image } \varphi \oplus \ker \varphi$.

**(c)** Prove that there is a basis of $V$ such that the matrix of $\varphi$ with respect to this basis is a diagonal matrix whose entries are all 0 or 1.

A linear transformation $\varphi$ satisfying $\varphi^2 = \varphi$ is called an *idempotent* linear transformation. This exercise proves that idempotent linear transformations are simply projections onto some subspace.

**12.** Let $V = \mathbb{R}^2$, $v_1 = (1, 0)$, $v_2 = (0, 1)$, so that $v_1, v_2$ are a basis for $V$. Let $\varphi$ be the linear transformation of $V$ to itself whose matrix with respect to this basis is $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Prove that if $W$ is the subspace generated by $v_1$ then $W$ is stable under the action of $\varphi$. Prove that there is no subspace $W'$ invariant under $\varphi$ so that $V = W \oplus W'$.

**13.** Let $V$ be a vector space of dimension $n$ and let $W$ be a vector space of dimension $m$ over a field $F$. Suppose $A$ is the $m \times n$ matrix representing a linear transformation $\varphi$ from $V$ to $W$ with respect to the bases $\mathcal{B}_1$ for $V$ and $\mathcal{E}_1$ for $W$. Suppose similarly that $B$ is the $m \times n$ matrix representing $\varphi$ with respect to the bases $\mathcal{B}_2$ for $V$ and $\mathcal{E}_2$ for $W$. Let $P = M_{\mathcal{B}_2}^{\mathcal{B}_1}(I)$ where $I$ denotes the identity map from $V$ to $V$, and let $Q = M_{\mathcal{E}_2}^{\mathcal{E}_1}(I)$ where $I$ denotes the identity map from $W$ to $W$. Prove that $Q^{-1} = M_{\mathcal{E}_1}^{\mathcal{E}_2}(I)$ and that $Q^{-1}AP = B$, giving the general relation between matrices representing the same linear transformation but with respect to different choices of bases.

The following exercises recall the *Gauss–Jordan* elimination process. This is one of the fastest computational methods for the solution of a number of problems involving vector spaces — solving systems of linear equations, determining inverses of matrices, computing determinants, determining the span of a set of vectors, determining linear independence of a set of vectors etc.

Consider the system of $m$ linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= c_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= c_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= c_m \end{aligned} \tag{11.4}$$

in the $n$ unknowns $x_1, x_2, \ldots, x_n$ where $a_{ij}, c_i$, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n$ are elements of the field $F$. Associated to this system is the *coefficient matrix*:

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

and the *augmented matrix*:

$$(A \mid C) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & c_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & c_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & c_m \end{array}\right)$$

(the term *augmented* refers to the presence of the column matrix $C = (c_i)$ in addition to the coefficient matrix $A = (a_{ij})$ ). The set of solutions in $F$ of this system of equations is not altered if we perform any of the following three operations:

**(1)** interchange any two equations
**(2)** add a multiple of one equation to another
**(3)** multiply any equation by a nonzero element from $F$,

which correspond to the following three *elementary row operations* on the augmented matrix:

**(1)** interchange any two rows
**(2)** add a multiple of one row to another
**(3)** multiply any row by a unit in $F$, i.e., by any nonzero element in $F$.

If a matrix $A$ can be transformed into a matrix $C$ by a series of elementary row operations then $A$ is said to be *row reduced* to $C$.

**14.** Prove that if $A$ can be row reduced to $C$ then $C$ can be row reduced to $A$. Prove that the relation "$A \sim C$ if and only if $A$ can be row reduced to $C$" is an equivalence relation. [Observe that the elementary row operations are reversible.]

The following exercises recall the *Gauss–Jordan* elimination process. This is one of the fastest computational methods for the solution of a number of problems involving vector spaces — solving systems of linear equations, determining inverses of matrices, computing determinants, determining the span of a set of vectors, determining linear independence of a set of vectors etc.

Consider the system of $m$ linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= c_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= c_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= c_m \end{aligned} \tag{11.4}$$

in the $n$ unknowns $x_1, x_2, \ldots, x_n$ where $a_{ij}, c_i$, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n$ are elements of the field $F$. Associated to this system is the *coefficient matrix*:

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

and the *augmented matrix*:

$$(A \mid C) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & c_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & c_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & c_m \end{array}\right)$$

(the term *augmented* refers to the presence of the column matrix $C = (c_i)$ in addition to the coefficient matrix $A = (a_{ij})$ ). The set of solutions in $F$ of this system of equations is not altered if we perform any of the following three operations:

**(1)** interchange any two equations
**(2)** add a multiple of one equation to another
**(3)** multiply any equation by a nonzero element from $F$,

which correspond to the following three *elementary row operations* on the augmented matrix:

**(1)** interchange any two rows
**(2)** add a multiple of one row to another
**(3)** multiply any row by a unit in $F$, i.e., by any nonzero element in $F$.

If a matrix $A$ can be transformed into a matrix $C$ by a series of elementary row operations then $A$ is said to be *row reduced* to $C$.

Matrices lying in the same equivalence class under this equivalence relation are said to be *row equivalent*.

**15.** Prove that the row rank of two row equivalent matrices is the same. [It suffices to prove this for two matrices differing by an elementary row operation.]

An $m \times n$ matrix is said to be in *reduced row echelon form* if

**(a)** the first nonzero entry $a_{ij_i}$ in row $i$ is 1 and all other entries in the corresponding $j_i^{\text{th}}$ column are zero, and

**(b)** $j_1 < j_2 < \ldots < j_r$ where $r$ is the number of nonzero rows, i.e., the number of initial zeros in each row is strictly increasing (hence the term *echelon*).

An augmented matrix $(A \mid C)$ is said to be in reduced row echelon form if its coefficient matrix $A$ is in reduced row echelon form. For example, the following two matrices are in reduced row echelon form:

$$\left(\begin{array}{cccccc|c} \mathbf{1} & 0 & 5 & 7 & 0 & 3 & 0 \\ \mathbf{0} & 1 & -1 & 1 & 0 & -4 & -1 \\ \mathbf{0} & 0 & 0 & 0 & \mathbf{1} & 6 & 1 \\ \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \left(\begin{array}{cccc|c} 0 & \mathbf{1} & -1 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 2 \\ 0 & 0 & 0 & 0 & -3 \end{array}\right)$$

(with $j_1 = 1$, $j_2 = 2$, $j_3 = 5$ for the first matrix and $j_1 = 2$, $j_2 = 4$ for the second matrix). The first nonzero entry in any given row of the coefficient matrix of a reduced row echelon augmented matrix (in position $(i, j_i)$ by definition) is sometimes referred to as a *pivotal* element (so the pivotal elements in the first matrix are in positions (1,1), (2,2) and (3,5) and the pivotal elements in the second matrix are in positions (1,2) and (2,4)). The columns containing pivotal elements will be called *pivotal* columns and the columns of the coefficient matrix not containing pivotal elements will be called *nonpivotal*.

**16.** Prove by induction that any augmented matrix can be put in reduced row echelon form by a series of elementary row operations.

An $m \times n$ matrix is said to be in *reduced row echelon form* if

**(a)** the first nonzero entry $a_{ij_i}$ in row $i$ is 1 and all other entries in the corresponding $j_i^{\text{th}}$ column are zero, and

**(b)** $j_1 < j_2 < \ldots < j_r$ where $r$ is the number of nonzero rows, i.e., the number of initial zeros in each row is strictly increasing (hence the term *echelon*).

An augmented matrix $(A \mid C)$ is said to be in reduced row echelon form if its coefficient matrix $A$ is in reduced row echelon form. For example, the following two matrices are in reduced row echelon form:

$$\left(\begin{array}{cccccc|c} \mathbf{1} & 0 & 5 & 7 & 0 & 3 & 0 \\ \mathbf{0} & 1 & -1 & 1 & 0 & -4 & -1 \\ \mathbf{0} & 0 & 0 & 0 & \mathbf{1} & 6 & 1 \\ \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \left(\begin{array}{cccc|c} 0 & \mathbf{1} & -1 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 2 \\ 0 & 0 & 0 & 0 & -3 \end{array}\right)$$

(with $j_1 = 1$, $j_2 = 2$, $j_3 = 5$ for the first matrix and $j_1 = 2$, $j_2 = 4$ for the second matrix). The first nonzero entry in any given row of the coefficient matrix of a reduced row echelon augmented matrix (in position $(i, j_i)$ by definition) is sometimes referred to as a *pivotal* element (so the pivotal elements in the first matrix are in positions (1,1), (2,2) and (3,5) and the pivotal elements in the second matrix are in positions (1,2) and (2,4)). The columns containing pivotal elements will be called *pivotal* columns and the columns of the coefficient matrix not containing pivotal elements will be called *nonpivotal*.

**17.** Let $A$ and $C$ be two matrices in reduced row echelon form. Prove that if $A$ and $C$ are row equivalent then $A = C$.

An $m \times n$ matrix is said to be in *reduced row echelon form* if

**(a)** the first nonzero entry $a_{ij_i}$ in row $i$ is 1 and all other entries in the corresponding $j_i^{\text{th}}$ column are zero, and

**(b)** $j_1 < j_2 < \ldots < j_r$ where $r$ is the number of nonzero rows, i.e., the number of initial zeros in each row is strictly increasing (hence the term *echelon*).

An augmented matrix $(A \mid C)$ is said to be in reduced row echelon form if its coefficient matrix $A$ is in reduced row echelon form. For example, the following two matrices are in reduced row echelon form:

$$\left(\begin{array}{cccccc|c} \mathbf{1} & 0 & 5 & 7 & 0 & 3 & 0 \\ \mathbf{0} & 1 & -1 & 1 & 0 & -4 & -1 \\ \mathbf{0} & 0 & 0 & 0 & \mathbf{1} & 6 & 1 \\ \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \left(\begin{array}{cccc|c} 0 & \mathbf{1} & -1 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 2 \\ 0 & 0 & 0 & 0 & -3 \end{array}\right)$$

(with $j_1 = 1$, $j_2 = 2$, $j_3 = 5$ for the first matrix and $j_1 = 2$, $j_2 = 4$ for the second matrix). The first nonzero entry in any given row of the coefficient matrix of a reduced row echelon augmented matrix (in position $(i, j_i)$ by definition) is sometimes referred to as a *pivotal* element (so the pivotal elements in the first matrix are in positions (1,1), (2,2) and (3,5) and the pivotal elements in the second matrix are in positions (1,2) and (2,4)). The columns containing pivotal elements will be called *pivotal* columns and the columns of the coefficient matrix not containing pivotal elements will be called *nonpivotal*.

**18.** Prove that the row rank of a matrix in reduced row echelon form is the number of nonzero rows.

An $m \times n$ matrix is said to be in *reduced row echelon form* if

**(a)** the first nonzero entry $a_{ij_i}$ in row $i$ is 1 and all other entries in the corresponding $j_i^{\text{th}}$ column are zero, and

**(b)** $j_1 < j_2 < \ldots < j_r$ where $r$ is the number of nonzero rows, i.e., the number of initial zeros in each row is strictly increasing (hence the term *echelon*).

An augmented matrix $(A \mid C)$ is said to be in reduced row echelon form if its coefficient matrix $A$ is in reduced row echelon form. For example, the following two matrices are in reduced row echelon form:

$$\left(\begin{array}{cccccc|c} \mathbf{1} & 0 & 5 & 7 & 0 & 3 & 0 \\ \mathbf{0} & 1 & -1 & 1 & 0 & -4 & -1 \\ \mathbf{0} & 0 & 0 & 0 & \mathbf{1} & 6 & 1 \\ \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \left(\begin{array}{cccc|c} 0 & \mathbf{1} & -1 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{1} & 2 \\ 0 & 0 & 0 & 0 & -3 \end{array}\right)$$

(with $j_1 = 1$, $j_2 = 2$, $j_3 = 5$ for the first matrix and $j_1 = 2$, $j_2 = 4$ for the second matrix). The first nonzero entry in any given row of the coefficient matrix of a reduced row echelon augmented matrix (in position $(i, j_i)$ by definition) is sometimes referred to as a *pivotal* element (so the pivotal elements in the first matrix are in positions (1,1), (2,2) and (3,5) and the pivotal elements in the second matrix are in positions (1,2) and (2,4)). The columns containing pivotal elements will be called *pivotal* columns and the columns of the coefficient matrix not containing pivotal elements will be called *nonpivotal*.

**19.** Prove that the reduced row echelon forms of the matrices

$$\left(\begin{array}{cccccc|c} \mathbf{1} & 1 & 4 & 8 & 0 & -1 & -1 \\ \mathbf{1} & 2 & 3 & 9 & 0 & -5 & -2 \\ \mathbf{0} & -2 & 2 & -2 & \mathbf{1} & 14 & 3 \\ \mathbf{1} & 4 & 1 & 11 & 0 & -13 & -4 \end{array}\right) \qquad \left(\begin{array}{cccc|c} 0 & -3 & 3 & 1 & 5 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 2 & -2 & 0 & -3 \end{array}\right)$$

are the two matrices preceding Exercise 16.

The following exercises recall the *Gauss–Jordan* elimination process. This is one of the fastest computational methods for the solution of a number of problems involving vector spaces — solving systems of linear equations, determining inverses of matrices, computing determinants, determining the span of a set of vectors, **determining** linear independence of a set of vectors etc.

Consider the system of $m$ linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= c_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= c_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= c_m \end{aligned} \tag{11.4}$$

in the $n$ unknowns $x_1, x_2, \ldots, x_n$ where $a_{ij}, c_i, i = 1, 2, \ldots, m, j = 1, 2, \ldots, n$ are elements of the field $F$. Associated to this system is the *coefficient matrix*:

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

and the *augmented matrix*:

$$(A \mid C) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & c_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & c_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & c_m \end{array}\right)$$

(the term *augmented* refers to the presence of the column matrix $C = (c_i)$ in addition to the coefficient matrix $A = (a_{ij})$ ). The set of solutions in $F$ of this system of equations is not altered if we perform any of the following three operations:

**(1)** interchange any two equations
**(2)** add a multiple of one equation to another
**(3)** multiply any equation by a nonzero element from $F$,

which correspond to the following three *elementary row operations* on the augmented matrix:

**(1)** interchange any two rows
**(2)** add a multiple of one row to another
**(3)** multiply any row by a unit in $F$, i.e., by any nonzero element in $F$.

If a matrix $A$ can be transformed into a matrix $C$ by a series of elementary row operations then $A$ is said to be *row reduced* to $C$.

An $m \times n$ matrix is said to be in *reduced row echelon form* if

**(a)** the first nonzero entry $a_{ij_i}$ in row $i$ is 1 and all other entries in the corresponding $j_i^{\text{th}}$ column are zero, and

**(b)** $j_1 < j_2 < \ldots < j_r$ where $r$ is the number of nonzero rows, i.e., the number of initial zeros in each row is strictly increasing (hence the term *echelon*).

An augmented matrix $(A \mid C)$ is said to be in reduced row echelon form if its coefficient matrix $A$ is in reduced row echelon form. For example, the following two matrices are in reduced row echelon form:

$$\left(\begin{array}{cccccc|c} \mathbf{1} & 0 & 5 & 7 & 0 & 3 & 0 \\ \mathbf{0} & 1 & -1 & 1 & 0 & -4 & -1 \\ \mathbf{0} & 0 & 0 & 0 & 1 & 6 & 1 \\ \mathbf{0} & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \left(\begin{array}{cccc|c} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & -3 \end{array}\right)$$

(with $j_1 = 1$, $j_2 = 2$, $j_3 = 5$ for the first matrix and $j_1 = 2$, $j_2 = 4$ for the second matrix). The first nonzero entry in any given row of the coefficient matrix of a reduced row echelon augmented matrix (in position $(i, j_i)$ by definition) is sometimes referred to as a *pivotal* element (so the pivotal elements in the first matrix are in positions (1,1), (2,2) and (3,5) and the pivotal elements in the second matrix are in positions (1,2) and (2,4)). The columns containing pivotal elements will be called *pivotal* columns and the columns of the coefficient matrix not containing pivotal elements will be called *nonpivotal*.

The point of the reduced row echelon form is that the corresponding system of linear equations is in a particularly simple form, from which the solutions to the system $AX = C$ in (4) can be determined immediately:

**20.** (*Solving Systems of Linear Equations*) Let $(A' \mid C')$ be the reduced row echelon form of the augmented matrix $(A \mid C)$. The number of zero rows of $A'$ is clearly at least as great as the number of zero rows of $(A' \mid C')$.

**(a)** Prove that if the number of zero rows of $A'$ is strictly larger than the number of zero rows of $(A' \mid C')$ then there are no solutions to $AX = C$.

By (a) we may assume that $A'$ and $(A' \mid C')$ have the same number, $r$, of nonzero rows (so $n \geq r$).

**(b)** Prove that if $r = n$ then there is precisely one solution to the system of equations $AX = C$.

**(c)** Prove that if $r < n$ then there are infinitely many solutions to the system of equations $AX = C$. Prove in fact that the values of the $n - r$ variables corresponding to the nonpivotal columns of $(A' \mid C')$ can be chosen arbitrarily and that the remaining $r$ variables corresponding to the pivotal columns of $(A' \mid C')$ are then determined uniquely.

**21.** Determine the solutions of the following systems of equations:

**(a)**

$$\begin{aligned} -3x + 3y + z &= 5 \\ x - y &= 0 \\ 2x - 2y &= -3 \end{aligned}$$

**(b)**

$$\begin{aligned} x - 2y + z &= 5 \\ x - 4y - 6z &= 10 \\ 4x - 11y + 11z &= 12 \end{aligned}$$

**(c)**

$$\begin{aligned} x - 2y + z &= 5 \\ y - 2z &= 17 \\ 2x - 3y &= 27 \end{aligned}$$

**(d)**

$$\begin{aligned} x + y - 3z + 2u &= 2 \\ 3x - 2y + 5z + u &= 1 \\ 6x + y - 4z + 3u &= 7 \\ 2x + 2y - 6z &= 4 \end{aligned}$$

**(e)**

$$\begin{aligned} x + y + 4z + 8u - w &= -1 \\ x + 2y + 3z + 9u - 5w &= -2 \\ -2y + 2z - 2u + v + 14w &= 3 \\ x + 4y + z + 11u - 13w &= -4 \end{aligned}$$

The following exercises recall the *Gauss–Jordan* elimination process. This is one of the fastest computational methods for the solution of a number of problems involving vector spaces — solving systems of linear equations, determining inverses of matrices, computing determinants, determining the span of a set of vectors, determining linear independence of a set of vectors etc.

Consider the system of $m$ linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= c_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= c_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= c_m \end{aligned} \tag{11.4}$$

in the $n$ unknowns $x_1, x_2, \ldots, x_n$ where $a_{ij}, c_i$, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n$ are elements of the field $F$. Associated to this system is the *coefficient matrix*:

$$A = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

and the *augmented matrix*:

$$(A \mid C) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \ldots & a_{1n} & c_1 \\ a_{21} & a_{22} & \ldots & a_{2n} & c_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} & c_m \end{array}\right)$$

(the term *augmented* refers to the presence of the column matrix $C = (c_i)$ in addition to the coefficient matrix $A = (a_{ij})$ ). The set of solutions in $F$ of this system of equations is not altered if we perform any of the following three operations:

**(1)** interchange any two equations
**(2)** add a multiple of one equation to another
**(3)** multiply any equation by a nonzero element from $F$,

which correspond to the following three *elementary row operations* on the augmented matrix:

**(1)** interchange any two rows
**(2)** add a multiple of one row to another
**(3)** multiply any row by a unit in $F$, i.e., by any nonzero element in $F$.

If a matrix $A$ can be transformed into a matrix $C$ by a series of elementary row operations then $A$ is said to be *row reduced* to $C$.

Matrices lying in the same equivalence class under this equivalence relation are said to be *row equivalent*.

**22.** Suppose $A$ and $B$ are two row equivalent $m \times n$ matrices.

**(a)** Prove that the set

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

of solutions to the homogeneous linear equations $AX = 0$ as in equation (4) above are the same as the set of solutions to the homogeneous linear equations $BX = 0$. [It suffices to prove this for two matrices differing by an elementary row operation.]

**(b)** Prove that any linear dependence relation satisfied by the columns of $A$ viewed as vectors in $F^m$ is also satisfied by the columns of $B$.

**(c)** Conclude from (b) that the number of linearly independent columns of $A$ is the same as the number of linearly independent columns of $B$.

**23.** Let $A'$ be a matrix in reduced row echelon form.

**(a)** Prove that the nonzero rows of $A'$ are linearly independent. Prove that the pivotal columns of $A'$ are linearly independent and that the nonpivotal columns of $A'$ are linearly dependent on the pivotal columns. (Note the role the pivotal elements play.)

**(b)** Prove that the number of linearly independent columns of a matrix in reduced row echelon form is the same as the number of linearly independent rows, i.e., the row rank and the column rank of such a matrix are the same.

**24.** Use the previous two exercises and Exercise 15 above to prove in general that the row rank and the column rank of a matrix are the same.

**25.** (*Computing Inverses of Matrices*) Let $A$ be an $n \times n$ matrix.

**(a)** Show that $A$ has an inverse matrix $B$ with columns $B_1, B_2, \ldots, B_n$ if and only if the systems of equations:

$$AB_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad AB_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad \ldots, \quad AB_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

have solutions.

**(b)** Prove that $A$ has an inverse if and only if $A$ is row equivalent to the $n \times n$ identity matrix.

**(c)** Prove that $A$ has an inverse $B$ if and only if the augmented matrix $(A \mid I)$ can be row reduced to the augmented matrix $(I \mid B)$ where $I$ is the $n \times n$ identity matrix.

**26.** Determine the inverses of the following matrices using row reduction:

$$A = \begin{pmatrix} -7 & -1 & -4 \\ 7 & 1 & 3 \\ 1 & 0 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ 0 & 2 & 0 & 0 \\ -1 & 1 & 1 & 0 \end{pmatrix}.$$

**27.** (*Computing Spans, Linear Independence and Linear Dependencies in Vector Spaces*) Let $V$ be an $m$-dimensional vector space with basis $e_1, e_2, \ldots, e_m$ and let $v_1, v_2, \ldots, v_n$ be vectors in $V$. Let $A$ be the $m \times n$ matrix whose columns are the coordinates of the vectors $v_i$ (with respect to the basis $e_1, e_2, \ldots, e_m$) and let $A'$ be the reduced row echelon form of $A$.

**(a)** Let $B$ be any matrix row equivalent to $A$. Let $w_1, w_2, \ldots, w_n$ be the vectors whose coordinates (with respect to the basis $e_1, e_2, \ldots, e_m$) are the columns of $B$. Prove that any linear relation

$$x_1 v_1 + x_2 v_2 + \ldots + x_n v_n = 0 \tag{11.5}$$

satisfied by $v_1, v_2, \ldots, v_n$ is also satisfied when $v_i$ is replaced by $w_i$, $i = 1, 2, \ldots, n$.

**(b)** Prove that the vectors whose coordinates are given by the pivotal columns of $A'$ are linearly independent and that the vectors whose coordinates are given by the nonpivotal columns of $A'$ are linearly dependent on these.

**(c)** (*Determining Linear Independence of Vectors*) Prove that the vectors $v_1, v_2, \ldots, v_n$ are linearly independent if and only if $A'$ has $n$ nonzero rows (i.e., has rank $n$).

**(d)** (*Determining Linear Dependencies of Vectors*) By (c), the vectors $v_1, v_2, \ldots, v_n$ are linearly dependent if and only if $A'$ has nonpivotal columns. The solutions to (5) defining linear dependence relations among $v_1, v_2, \ldots, v_n$ are given by the linear equations defined by $A'$. Show that each of the variables $x_1, x_2, \ldots, x_n$ in (5) corresponding to the nonpivotal columns of $A'$ can be prescribed arbitrarily and the values of the remaining variables are then uniquely determined to give a linear dependence relation among $v_1, v_2, \ldots, v_n$ as in (5).

**(e)** (*Determining the Span of a Set of Vectors*) Prove that the subspace $W$ spanned by $v_1, v_2, \ldots, v_n$ has dimension $r$ where $r$ is the number of nonzero rows of $A'$ and that a basis for $W$ is given by the original vectors $v_{j_i}$ ($i = 1, 2, \ldots, r$) corresponding to the pivotal columns of $A'$.

**28.** Let $V = \mathbb{R}^5$ with the standard basis and consider the vectors

$$v_1 = (1, 1, 3, -2, 3)\ ,\ v_2 = (0, 1, 0, -1, 0)\ ,\ v_3 = (2, 3, 6, -5, 6)$$

$$v_4 = (0, 3, 1, -3, 1)\ ,\ v_5 = (2, -1, -1, -1, -1).$$

**(a)** Show that the reduced row echelon form of the matrix

$$A = \begin{pmatrix} 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & 3 & 3 & -1 \\ 3 & 0 & 6 & 1 & -1 \\ -2 & -1 & -5 & -3 & -1 \\ 3 & 0 & 6 & 1 & -1 \end{pmatrix}$$

whose columns are the coordinates of $v_1, v_2, v_3, v_4, v_5$ is the matrix

$$A' = \begin{pmatrix} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 1 & 0 & 18 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where the 1st, 2nd and 4th columns are pivotal and the remaining two are nonpivotal.

**(b)** Conclude that these vectors are linearly dependent, that the subspace $W$ spanned by $v_1, v_2, v_3, v_4, v_5$ is 3-dimensional and that the vectors

$$v_1 = (1, 1, 3, -2, 3), \quad v_2 = (0, 1, 0, -1, 0) \quad \text{and} \quad v_4 = (0, 3, 1, -3, 1)$$

are a basis for $W$.

**(c)** Conclude from (a) that the coefficients $x_1, x_2, x_3, x_4, x_5$ of any **linear** relation

$$x_1v_1 + x_2v_2 + x_3v_3 + x_4v_4 + x_5v_5 = 0$$

satisfied by $v_1, v_2, v_3, v_4, v_5$ are given by the equations

$$\begin{aligned} x_1 \quad + 2x_3 \quad + \ 2x_5 &= 0 \\ x_2 + \ x_3 \quad + 18x_5 &= 0 \\ x_4 - \ 7x_5 &= 0. \end{aligned}$$

Deduce that the 3rd and 5th variables, namely $x_3$ and $x_5$, corresponding to the non-pivotal columns of $A'$, can be prescribed arbitrarily and the remaining variables are then uniquely determined as:

$$\begin{aligned} x_1 &= -2x_3 - 2x_5 \\ x_2 &= -x_3 - 18x_5 \\ x_4 &= 7x_5 \end{aligned}$$

to give all the linear dependence relations satisfied by $v_1, v_2, v_3, v_4, v_5$. In particular show that

$$-2v_1 - v_2 + v_3 = 0$$

and

$$-2v_1 - 18v_2 + 7v_4 + v_5 = 0$$

corresponding to $(x_3 = 1, x_5 = 0)$ and $(x_3 = 0, x_5 = 1)$, respectively.

**29.** For each exercise below, determine whether the given vectors in $\mathbb{R}^4$ are linearly independent. If they are linearly dependent, determine an explicit linear dependence among them.

**(a)** $(1, -4, 3, 0)$, $(0, -1, 4, -3)$, $(1, -1, 1, -1)$, $(2, 2, -1, -3)$.

**(b)** $(1, -2, 4, 1)$, $(2, -3, 9, -1)$, $(1, 0, 6, -5)$, $(2, -5, 7, 5)$.

**(c)** $(1, -2, 0, 1)$, $(2, -2, 0, 0)$, $(-1, 3, 0, -2)$, $(-2, 1, 0, 1)$.

**(d)** $(0, 1, 1, 0)$, $(1, 0, 1, 1)$, $(2, 2, 2, 0)$, $(0, -1, 1, 1)$.

**30.** For each exercise below, determine the subspace spanned in $\mathbb{R}^4$ by the given vectors and give a basis for this subspace.

**(a)** $(1, -2, 5, 3), (2, 3, 1, -4), (3, 8, -3, -5)$.

**(b)** $(2, -5, 3, 0), (0, -2, 5, -3), (1, -1, 1, -1), (-3, 2, -1, 2)$.

**(c)** $(1, -2, 0, 1), (2, -2, 0, 0), (-1, 3, 0, -2), (-2, 1, 0, 1)$.

**(d)** $(1, 1, 0, -1), (1, 2, 3, 0), (2, 3, 3, -1), (1, 2, 2, -2), (2, 3, 2, -3), (1, 3, 4, -3)$.

**31.** (*Computing the Image and Kernel of a Linear Transformation*) Let $V$ be an $n$-dimensional vector space with basis $e_1, e_2, \ldots, e_n$ and let $W$ be an $m$-dimensional vector space with basis $f_1, f_2, \ldots, f_m$. Let $\varphi$ be a linear transformation from $V$ to $W$ and let $A$ be the corresponding $m \times n$ matrix with respect to these bases: $A = (a_{ij})$ where

$$\varphi(e_j) = \sum_{i=1}^{m} a_{ij} f_i , \qquad j = 1, 2, \ldots, n,$$

i.e., the columns of $A$ are the coordinates of the vectors $\varphi(e_1), \varphi(e_2), \ldots, \varphi(e_n)$ with respect to the basis $f_1, f_2, \ldots, f_m$ of $W$. Let $A'$ be the reduced row echelon form of $A$.

**(a)** (*Determining the Image of a Linear Transformation*) Prove that the image $\varphi(V)$ of $V$ under $\varphi$ has dimension $r$ where $r$ is the number of nonzero rows of $A'$ and that a basis for $\varphi(V)$ is given by the vectors $\varphi(e_{j_i})$ $(i = 1, 2, \ldots, r)$, i.e., the columns of $A$ corresponding to the pivotal columns of $A'$ give the coordinates of a basis for the image of $\varphi$.

**(b)** (*Determining the Kernel of a Linear Transformation*) The elements in the kernel of $\varphi$ are the vectors in $V$ whose coordinates $(x_1, x_2, \ldots, x_n)$ with respect to the basis $e_1, e_2, \ldots, e_n$ satisfy the equation

$$A \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = 0,$$

and the solutions $x_1, x_2, \ldots, x_n$ to this system of linear equations are determined by the matrix $A'$.

**(i)** Prove that $\varphi$ is injective if and only if $A'$ has $n$ nonzero rows (i.e., has rank $n$).

**(ii)** By (i), the kernel of $\varphi$ is nontrivial if and only if $A'$ has nonpivotal columns. Show that each of the variables $x_1, x_2, \ldots, x_n$ above corresponding to the nonpivotal columns of $A'$ can be prescribed arbitrarily and the values of the remaining variables are then uniquely determined to give an element $x_1e_1 + x_2e_2 + \ldots + x_ne_n$ in the kernel of $\varphi$. In particular, show that the coordinates of a basis for the kernel are obtained by successively setting one nonpivotal variable equal to 1 and all other nonpivotal variables to 0 and solving for the remaining pivotal variables. Conclude that the kernel of $\varphi$ has dimension $n - r$ where $r$ is the rank of $A$.

**32.** Let $V = \mathbb{R}^5$ and $W = \mathbb{R}^4$ with the standard bases. Let $\varphi$ be the linear transformation $\varphi : V \to W$ defined by

$$\varphi(x, y, z, u, v) = (x + 2y + 3z + 4u + 4v, -2x - 4y + 2v, x + 2y + u - 2v, x + 2y - v).$$

**(a)** Prove that the matrix $A$ corresponding to $\varphi$ and these bases is

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 & 4 \\ -2 & -4 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & -2 \\ 1 & 2 & 0 & 0 & -1 \end{pmatrix}$$

and that the reduced row echelon matrix $A'$ row equivalent to $A$ is

$$A' = \begin{pmatrix} 1 & 2 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

where the 1st, 3rd and 4th columns are pivotal and the remaining two are nonpivotal.

**(b)** Conclude that the image of $\varphi$ is 3-dimensional and that the image of the 1st, 3rd and 4th basis elements of $V$, namely, $(1, -2, 1, 1)$, $(3, 0, 0, 0)$ and $(4, 0, 1, 0)$ give a basis for the image $\varphi(V)$ of $V$.

**(c)** Conclude from (a) that the elements in the kernel of $\varphi$ are the vectors $(x, y, z, u, v)$ satisfying the equations

$$\begin{aligned} x + 2y \qquad - \ v &= 0 \\ z \ + 3v &= 0 \\ u - \ v &= 0. \end{aligned}$$

Deduce that the 2nd and 5th variables, namely $y$ and $v$, corresponding to the nonpivotal columns of $A'$ can be prescribed arbitrarily and the remaining variables are then uniquely determined as

$$\begin{aligned} x &= -2y + v \\ z &= -3v \\ u &= v. \end{aligned}$$

Show that $(-2, 1, 0, 0, 0)$ and $(1, 0, -3, 1, 1)$ give a basis for the 2-dimensional kernel of $\varphi$, corresponding to $(y = 1, v = 0)$ and $(y = 0, v = 1)$, respectively.

**33.** Let $\varphi$ be the linear transformation from $\mathbb{R}^4$ to itself defined by the matrix

$$A = \begin{pmatrix} 1 & -1 & 0 & 3 \\ -1 & 2 & 1 & -1 \\ -1 & 1 & 0 & -3 \\ 1 & -2 & -1 & 1 \end{pmatrix}$$

with respect to the standard basis for $\mathbb{R}^4$. Determine a basis for the image and for the kernel of $\varphi$.

**34.** Let $\varphi$ be the linear transformation $\varphi : \mathbb{R}^4 \to \mathbb{R}^2$ such that

$$\varphi((1,0,0,0)) = (1,-1) \qquad \varphi((1,-1,0,0)) = (0,0)$$
$$\varphi((1,-1,1,0)) = (1,-1) \qquad \varphi((1,-1,1,-1)) = (0,0).$$

Determine a basis for the image and for the kernel of $\varphi$.

**35.** Let $V$ be the set of all $2 \times 2$ matrices with real entries and let $\varphi : V \to \mathbb{R}$ be the map defined by sending a matrix $A \in V$ to the sum of the diagonal entries of $A$ (the *trace* of $A$).

**(a)** Show that

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

is a basis for $V$.

**(b)** Prove that $\varphi$ is a linear transformation and determine the matrix of $\varphi$ with respect to the basis in (a) for $V$. Determine the dimension of and a basis for the kernel of $\varphi$.

**36.** Let $V$ be the **6-dimensional** vector space over $\mathbb{Q}$ consisting of the polynomials in the variable $x$ of degree at most 5. Let $\varphi$ be the map of $V$ to itself defined by $\varphi(f) = x^2 f'' - 6xf' + 12f$, where $f''$ denotes the usual second derivative (with respect to $x$) of the polynomial $f \in V$ and $f'$ similarly denotes the usual first derivative.

**(a)** Prove that $\varphi$ is a linear transformation of $V$ to itself.

**(b)** Determine a basis for the image and for the kernel of $\varphi$.

**37.** Let $V$ be the 7-dimensional vector space over the field $F$ consisting of the polynomials in the variable $x$ of degree at most 6. Let $\varphi$ be the linear transformation of $V$ to itself defined by $\varphi(f) = f'$, where $f'$ denotes the usual derivative (with respect to $x$) of the polynomial $f \in V$. For each of the fields below, determine a basis for the image and for the kernel of $\varphi$:

**(a)** $F = \mathbb{R}$

**(b)** $F = \mathbb{F}_2$, the finite field of 2 elements (note that, for example, $(x^2)' = 2x = 0$ over this field)

**(c)** $F = \mathbb{F}_3$

**(d)** $F = \mathbb{F}_5$ .

**38.** Let $A$ and $B$ be square matrices. Prove that the trace of their Kronecker product is the product of their traces: $\operatorname{tr}(A \otimes B) = \operatorname{tr}(A)\operatorname{tr}(B)$. (Recall that the trace of a square matrix is the sum of its diagonal entries.)

**39.** Let $F$ be a subfield of $K$ and let $\psi : V \to W$ be a linear transformation of finite dimensional vector spaces over $F$.

**(a)** Prove that $1 \otimes \psi$ is a $K$–linear transformation from the vector spaces $K \otimes_F V$ to $K \otimes_F W$ over $K$. (Here 1 denotes the identity map from $K$ to itself.)

**(b)** Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ and $\mathcal{E} = \{w_1, \ldots, w_m\}$ be bases of $V$ and $W$ respectively. Prove that the matrix of $1 \otimes \psi$ with respect to the bases $\{1 \otimes v_1, \ldots, 1 \otimes v_n\}$ and $\{1 \otimes w_1, \ldots, 1 \otimes w_m\}$ is the same as the matrix of $\psi$ with respect to $\mathcal{B}$ and $\mathcal{E}$.

## 11.3. Dual Vector Spaces

**1.** Let $V$ be a finite dimensional vector space. Prove that the map $\varphi \mapsto \varphi^*$ in Theorem 20 gives a ring isomorphism of $\mathrm{End}(V)$ with $\mathrm{End}(V^*)$.

**2.** Let $V$ be the collection of polynomials with coefficients in $\mathbb{Q}$ in the variable $x$ of degree at most 5 with $1, x, x^2, \ldots, x^5$ as basis. Prove that the following are elements of the dual space of $V$ and express them as linear combinations of the dual basis:

**(a)** $E : V \to \mathbb{Q}$ defined by $E(p(x)) = p(3)$ (i.e., evaluation at $x = 3$).

**(b)** $\varphi : V \to \mathbb{Q}$ defined by $\varphi(p(x)) = \int_0^1 p(t)dt$.

**(c)** $\varphi : V \to \mathbb{Q}$ defined by $\varphi(p(x)) = \int_0^1 t^2 p(t)dt$.

**(d)** $\varphi : V \to \mathbb{Q}$ defined by $\varphi(p(x)) = p'(5)$ where $p'(x)$ denotes the usual derivative of the polynomial $p(x)$ with respect to $x$.

**3.** Let $S$ be any subset of $V^*$ for some finite dimensional space $V$. Define $\text{Ann}(S) = \{v \in V \mid f(v) = 0 \text{ for all } f \in S\}$. ($\text{Ann}(S)$ is called the *annihilator of $S$ in $V$*).

**(a)** Prove that $\text{Ann}(S)$ is a subspace of $V$.

**(b)** Let $W_1$ and $W_2$ be subspaces of $V^*$. Prove that $\text{Ann}(W_1 + W_2) = \text{Ann}(W_1) \cap \text{Ann}(W_2)$ and $\text{Ann}(W_1 \cap W_2) = \text{Ann}(W_1) + \text{Ann}(W_2)$.

**(c)** Let $W_1$ and $W_2$ be subspaces of $V^*$. Prove that $W_1 = W_2$ if and only if $\text{Ann}(W_1) = \text{Ann}(W_2)$.

**(d)** Prove that the annihilator of $S$ is the same as the annihilator of the subspace of $V^*$ spanned by $S$.

**(e)** Assume $V$ is finite dimensional with basis $v_1, \ldots, v_n$. Prove that if $S = \{v_1^*, \ldots, v_k^*\}$ for some $k \leq n$, then $\text{Ann}(S)$ is the subspace spanned by $\{v_{k+1}, \ldots, v_n\}$.

**(f)** Assume $V$ is finite dimensional. Prove that if $W^*$ is any subspace of $V^*$ then $\dim \text{Ann}(W^*) = \dim V - \dim W^*$.

**4.** If $V$ is infinite dimensional with basis $\mathcal{A}$, prove that $\mathcal{A}^* = \{v^* \mid v \in \mathcal{A}\}$ does *not* span $V^*$.

**5.** If $V$ is infinite dimensional with basis $\mathcal{A}$, prove that $V^*$ is isomorphic to the direct product of copies of $F$ indexed by $\mathcal{A}$. Deduce that $\dim V^* > \dim V$. [Use Exercise 14, Section 1.]

## 11.4. Determinants

**1.** Formulate and prove the cofactor expansion formula along the $j^{\text{th}}$ column of a square matrix $A$.

**2.** Let $F$ be a field and let $A_1, A_2, \ldots, A_n$ be (column) vectors in $F^n$. Form the matrix $A$ whose $i^{\text{th}}$ column is $A_i$. Prove that these vectors form a basis of $F^n$ if and only if $\det A \neq 0$.

**3.** Let $R$ be any commutative ring with 1, let $V$ be an $R$-module and let $x_1, x_2, \ldots, x_n \in V$. Assume that for some $A \in M_{n\times n}(R)$,

$$A\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0.$$

Prove that $(\det A)x_i = 0$, for all $i \in \{1, 2, \ldots, n\}$.

**4.** (*Computing Determinants of Matrices*) This exercise outlines the use of Gauss–Jordan elimination (cf. the exercises in Section 2) to compute determinants. This is the most efficient general procedure for computing large determinants. Let $A$ be an $n \times n$ matrix.

**(a)** Prove that the elementary row operations have the following effect on determinants:

**(i)** interchanging two rows changes the sign of the determinant

**(ii)** adding a multiple of one row to another does not alter the determinant

**(iii)** multiplying any row by a nonzero element $u$ from $F$ multiplies the determinant by $u$.

**(b)** Prove that $\det A$ is nonzero if and only if $A$ is row equivalent to the $n \times n$ identity matrix. Suppose $A$ can be row reduced to the identity matrix using a total of $s$ row interchanges as in (i) and by multiplying rows by the nonzero elements $u_1, u_2, \ldots, u_t$ as in (iii). Prove that $\det A = (-1)^s (u_1 u_2 \ldots u_t)^{-1}$.

**5.** Compute the determinants of the following matrices using row reduction:

$$A = \begin{pmatrix} 5 & 4 & -6 \\ -2 & 0 & 2 \\ 3 & 4 & -2 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 2 & -4 & 4 \\ 2 & -1 & 4 & -8 \\ 1 & 0 & 1 & -2 \\ 0 & 1 & -2 & 3 \end{pmatrix}.$$

**6.** (*Minkowski's Criterion*) Suppose $A$ is an $n \times n$ matrix with real entries such that the diagonal elements are all positive, the off-diagonal elements are all negative and the row sums are all positive. Prove that $\det A \neq 0$. [Consider the corresponding system of equations $AX = 0$ and suppose there is a nontrivial solution $(x_1, \ldots, x_n)$. If $x_i$ has the largest absolute value show that the $i^{\text{th}}$ equation leads to a contradiction.]

## 11.5. Tensor Algebras, Symmetric and Exterior Algebras

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**1.** Prove that if $M$ is a cyclic $R$-module then $\mathcal{T}(M) = \mathcal{S}(M)$, i.e., the tensor algebra $\mathcal{T}(M)$ is commutative.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**2.** Fill in the details for the proof of Proposition 33 that $S/I = \oplus_{k=0}^{\infty} S_k/I_k$. [Show first that $S_i I_j \subseteq I_{i+j}$. Use this to show that the multiplication $(S_i/I_i)(S_j/I_j) \subseteq S_{i+j}/I_{i+j}$ is well defined, and then check the ring axioms and verify the statements made in the proof of Proposition 33.]

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**3.** Show that the image of the map $Sym_2$ for the $\mathbb{Z}$-module $\mathbb{Z}$ consists of the 2-tensors $a(1 \otimes 1)$ where $a$ is an even integer. Conclude in particular that the symmetric tensor $1 \otimes 1$ in $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}$ is not contained in the image of the map $Sym$.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**4.** Prove that $m \wedge n_1 \wedge n_2 \wedge \cdots \wedge n_k = (-1)^k(n_1 \wedge n_2 \wedge \cdots \wedge n_k \wedge m)$. In particular, $x \wedge (y \wedge z) = (y \wedge z) \wedge x$ for all $x, y, z \in M$.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**5.** Prove that if $M$ is a free $R$-module of rank $n$ then $\bigwedge^i(M)$ is a free $R$-module of rank $\binom{n}{i}$ for $i = 0, 1, 2, \ldots$.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**6.** If $A$ is any $R$-algebra in which $a^2 = 0$ for all $a \in A$ and $\varphi : M \to A$ is an $R$-module homomorphism, prove there is a unique $R$-algebra homomorphism $\Phi : \bigwedge(M) \to A$ such that $\Phi|_M = \varphi$.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**7.** Let $R = \mathbb{Z}[x, y]$ and $I = (x, y)$.

**(a)** Prove that if $ax + by = a'x + b'y$ in $R$ then $a' = a + yf$ and $b' = b - xf$ for some polynomial $f(x, y) \in R$.

**(b)** Prove that the map $\varphi(ax+by, cx+dy) = ad-bc \bmod (x, y)$ in the example following Corollary 37 is a well defined alternating $R$-bilinear map from $I \times I$ to $\mathbb{Z} = R/I$.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**8.** Let $R$ be an integral domain and let $F$ be its field of fractions.

**(a)** Considering $F$ as an $R$-module, prove that $\bigwedge^2 F = 0$.

**(b)** Let $I$ be any $R$-submodule of $F$ (for example, any ideal in $R$). Prove that $\bigwedge^i I$ is a torsion $R$-module for $i \geq 2$ (i.e., for every $x \in \bigwedge^i I$ there is some nonzero $r \in R$ with $rx = 0$).

**(c)** Give an example of an integral domain $R$ and an $R$-module $I$ in $F$ with $\bigwedge^i I \neq 0$ for every $i \geq 0$ (cf. the example following Corollary 37).

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**9.** Let $R = \mathbb{Z}[G]$ be the group ring of the group $G = \{1, \sigma\}$ of order 2. Let $M = \mathbb{Z}e_1 + \mathbb{Z}e_2$ be a free $\mathbb{Z}$-module of rank 2 with basis $e_1$ and $e_2$. Define $\sigma(e_1) = e_1 + 2e_2$ and $\sigma(e_2) = -e_2$. Prove that this makes $M$ into an $R$-module and that the $R$-module $\bigwedge^2 M$ is a group of order 2 with $e_1 \wedge e_2$ as generator.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**10.** Prove that $z - (1/k!)Alt(z) = (1/k!)\sum_{\sigma \in S_k}(z - \epsilon(\sigma)\sigma z)$ for any $k$-tensor $z$ and use this to prove that the kernel of the $R$-module homomorphism $(1/k!)Alt$ in Proposition 40 is $\mathcal{A}^k(M)$.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**11.** Prove that the image of $Alt_k$ is the unique largest subspace of $\mathcal{T}^k(V)$ on which each permutation $\sigma$ in the symmetric group $S_k$ acts as multiplication by the scalar $\epsilon(\sigma)$.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**12.** **(a)** Prove that if $f(x, y)$ is an alternating bilinear map on $V$ (i.e., $f(x, x) = 0$ for all $x \in V$) then $f(x, y) = -f(y, x)$ for all $x, y \in V$.

**(b)** Suppose that $-1 \neq 1$ in $F$. Prove that $f(x, y)$ is an alternating bilinear map on $V$ (i.e., $f(x, x) = 0$ for all $x \in V$) if and only if $f(x, y) = -f(y, x)$ for all $x, y \in V$.

**(c)** Suppose that $-1 = 1$ in $F$. Prove that every alternating bilinear form $f(x, y)$ on $V$ is symmetric (i.e., $f(x, y) = f(y, x)$ for all $x, y \in V$). Prove that there is a symmetric bilinear map on $V$ that is not alternating. [One approach: show that $C^2(V) \subset \mathcal{A}^2(V)$ and $C^2(V) \neq \mathcal{A}^2(V)$ by counting dimensions. Alternatively, construct an explicit symmetric map that is not alternating.]

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**13.** Let $F$ be any field in which $-1 \neq 1$ and let $V$ be a vector space over $F$. Prove that $V \otimes_F V = \mathcal{S}^2(V) \oplus \bigwedge^2(V)$ i.e., that every 2-tensor may be written uniquely as a sum of a symmetric and an alternating tensor.

In these exercises $R$ is a commutative ring with 1 and $M$ is an $R$-module; $F$ is a field and $V$ is a finite dimensional vector space over $F$.

**14.** Prove that if $M$ is an $R$-module *direct factor* of the $R$-module $N$ then $\mathcal{T}(M)$ (respectively, $\mathcal{S}(M)$ and $\bigwedge(M)$) is an $R$-subalgebra of $\mathcal{T}(N)$ (respectively, $\mathcal{S}(N)$ and $\bigwedge(N)$).

# 12. Modules over Principal Ideal Domains

## 12.1. The Basic Theory

**1.** Let $M$ be a module over the integral domain $R$.

**(a)** Suppose $x$ is a nonzero torsion element in $M$. Show that $x$ and 0 are "linearly dependent." Conclude that the rank of Tor($M$) is 0, so that in particular any torsion $R$-module has rank 0.

**(b)** Show that the rank of $M$ is the same as the rank of the (torsion free) quotient $M/\mathrm{Tor}M$.

**2.** Let $M$ be a module over the integral domain $R$.

**(a)** Suppose that $M$ has rank $n$ and that $x_1, x_2, \ldots, x_n$ is any maximal set of linearly independent elements of $M$. Let $N = Rx_1 + \ldots + Rx_n$ be the submodule generated by $x_1, x_2, \ldots, x_n$. Prove that $N$ is isomorphic to $R^n$ and that the quotient $M/N$ is a torsion $R$-module (equivalently, the elements $x_1, \ldots, x_n$ are linearly independent and for any $y \in M$ there is a nonzero element $r \in R$ such that $ry$ can be written as a linear combination $r_1x_1 + \ldots + r_nx_n$ of the $x_i$).

**(b)** Prove conversely that if $M$ contains a submodule $N$ that is free of rank $n$ (i.e., $N \cong R^n$) such that the quotient $M/N$ is a torsion $R$-module then $M$ has rank $n$. [Let $y_1, y_2, \ldots, y_{n+1}$ be any $n + 1$ elements of $M$. Use the fact that $M/N$ is torsion to write $r_iy_i$ as a linear combination of a basis for $N$ for some nonzero elements $r_1, \ldots, r_{n+1}$ of $R$. Use an argument as in the proof of Proposition 3 to see that the $r_iy_i$, and hence also the $y_i$, are linearly dependent.]

**3.** Let $R$ be an integral domain and let $A$ and $B$ be $R$-modules of ranks $m$ and $n$, respectively. Prove that the rank of $A \oplus B$ is $m + n$. [Use the previous exercise.]

**4.** Let $R$ be an integral domain, let $M$ be an $R$-module and let $N$ be a submodule of $M$. Suppose $M$ has rank $n$, $N$ has rank $r$ and the quotient $M/N$ has rank $s$. Prove that $n = r + s$. [Let $x_1, x_2, \ldots, x_s$ be elements of $M$ whose images in $M/N$ are a maximal set of independent elements and let $x_{s+1}, x_{s+2}, \ldots, x_{s+r}$ be a maximal set of independent elements in $N$. Prove that $x_1, x_2, \ldots, x_{s+r}$ are linearly independent in $M$ and that for any element $y \in M$ there is a nonzero element $r \in R$ such that $ry$ is a linear combination of these elements. Then use Exercise 2.]

**5.** Let $R = \mathbb{Z}[x]$ and let $M = (2, x)$ be the ideal generated by 2 and $x$, considered as a submodule of $R$. Show that $\{2, x\}$ is not a basis of $M$. [Find a nontrivial $R$-linear dependence between these two elements.] Show that the rank of $M$ is 1 but that $M$ is not free of rank 1 (cf. Exercise 2).

**6.** Show that if $R$ is an integral domain and $M$ is any nonprincipal ideal of $R$ then $M$ is torsion free of rank 1 but is not a free $R$-module.

**7.** Let $R$ be any ring, let $A_1, A_2, \ldots, A_m$ be $R$-modules and let $B_i$ be a submodule of $A_i$, $1 \leq i \leq m$. Prove that

$$(A_1 \oplus A_2 \oplus \cdots \oplus A_m)/(B_1 \oplus B_2 \oplus \cdots \oplus B_m) \cong (A_1/B_1) \oplus (A_2/B_2) \oplus \cdots \oplus (A_m/B_m).$$

**8.** Let $R$ be a P.I.D., let $B$ be a torsion $R$-module and let $p$ be a prime in $R$. Prove that if $pb = 0$ for some nonzero $b \in B$, then $\text{Ann}(B) \subseteq (p)$.

**9.** Give an example of an integral domain $R$ and a nonzero torsion $R$-module $M$ such that $\text{Ann}(M) = 0$. Prove that if $N$ is a finitely generated torsion $R$-module then $\text{Ann}(N) \neq 0$.

**10.** For $p$ a prime in the P.I.D. $R$ and $N$ an $R$-module prove that the $p$-primary component of $N$ is a submodule of $N$ and prove that $N$ is the direct sum of its $p$-primary components (there need not be finitely many of them).

**11.** Let $R$ be a P.I.D., let $a$ be a nonzero element of $R$ and let $M = R/(a)$. For any prime $p$ of $R$ prove that

$$p^{k-1}M/p^kM \cong \begin{cases} R/(p) & \text{if } k \leq n \\ 0 & \text{if } k > n, \end{cases}$$

where $n$ is the power of $p$ dividing $a$ in $R$.

**12.** Let $R$ be a P.I.D. and let $p$ be a prime in $R$.

**(a)** Let $M$ be a finitely generated torsion $R$-module. Use the previous exercise to prove that $p^{k-1}M/p^kM \cong F^{n_k}$ where $F$ is the field $R/(p)$ and $n_k$ is the number of elementary divisors of $M$ which are powers $p^\alpha$ with $\alpha \geq k$.

**(b)** Suppose $M_1$ and $M_2$ are isomorphic finitely generated torsion $R$-modules. Use (a) to prove that, for every $k \geq 0$, $M_1$ and $M_2$ have the same number of elementary divisors $p^\alpha$ with $\alpha \geq k$. Prove that this implies $M_1$ and $M_2$ have the same set of elementary divisors.

**13.** If $M$ is a finitely generated module over the P.I.D. $R$, describe the structure of $M/\mathrm{Tor}(M)$.

**14.** Let $R$ be a P.I.D. and let $M$ be a torsion $R$-module. Prove that $M$ is irreducible (cf. Exercises 9 to 11 of Section 10.3) if and only if $M = Rm$ for any nonzero element $m \in M$ where the annihilator of $m$ is a nonzero prime ideal $(p)$.

**15.** Prove that if $R$ is a Noetherian ring then $R^n$ is a Noetherian $R$-module. [Fix a basis of $R^n$. If $M$ is a submodule of $R^n$ show that the collection of first coordinates of elements of $M$ is a submodule of $R$ hence is finitely generated. Let $m_1, m_2, \ldots, m_k$ be elements of $M$ whose first coordinates generate this submodule of $R$. Show that any element of $M$ can be written as an $R$-linear combination of $m_1, m_2, \ldots, m_k$ plus an element of $M$ whose first coordinate is 0. Prove that $M \cap R^{n-1}$ is a submodule of $R^{n-1}$ where $R^{n-1}$ is the set of elements of $R^n$ with first coordinate 0 and then use induction on $n$.

The following set of exercises outlines a proof of Theorem 5 in the special case where $R$ is a Euclidean Domain using a matrix argument involving row and column operations. This applies in particular to the cases $R = \mathbb{Z}$ and $R = F[x]$ of interest in the applications and is computationally useful.

Let $R$ be a Euclidean Domain and let $M$ be an $R$-module.

**16.** Prove that $M$ is finitely generated if and only if there is a surjective $R$-homomorphism $\varphi : R^n \to M$ for some integer $n$ (this is true for any ring $R$).

The following set of exercises outlines a proof of Theorem 5 in the special case where $R$ is a Euclidean Domain using a matrix argument involving row and column operations. This applies in particular to the cases $R = \mathbb{Z}$ and $R = F[x]$ of interest in the applications and is computationally useful.

Let $R$ be a Euclidean Domain and let $M$ be an $R$-module.

Suppose $\varphi : R^n \to M$ is a surjective $R$-module homomorphism. By Exercise 15, $\ker \varphi$ is finitely generated. If $x_1, x_2, \ldots, x_n$ is a basis for $R^n$ and $y_1, \ldots, y_m$ are generators for $\ker \varphi$ we have

$$y_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \quad i = 1, 2, \ldots, m$$

with coefficients $a_{ij} \in R$. It follows that the homomorphism $\varphi$ (hence the module structure of $M$) is determined by the choice of generators for $R^n$ and the matrix $A = (a_{ij})$. Such a matrix $A$ will be called a *relations matrix*.

**17.** **(a)** Show that interchanging $x_i$ and $x_j$ in the basis for $R^n$ interchanges the $i^{\text{th}}$ column with the $j^{\text{th}}$ column in the corresponding relations matrix.

**(b)** Show that, for any $a \in R$, replacing the element $x_j$ by $x_j - ax_i$ in the basis for $R^n$ gives another basis for $R^n$ and that the corresponding relations matrix for this basis is the same as the original relations matrix except that $a$ times the $j^{\text{th}}$ column has been added to the $i^{\text{th}}$ column. [Note that $\cdots + a_i x_i + \cdots + a_j x_j + \cdots = \cdots + (a_i + aa_j)x_i + \cdots + a_j(x_j - ax_i) + \ldots$ .]

The following set of exercises outlines a proof of Theorem 5 in the special case where $R$ is a Euclidean Domain using a matrix argument involving row and column operations. This applies in particular to the cases $R = \mathbb{Z}$ and $R = F[x]$ of interest in the applications and is computationally useful.

Let $R$ be a Euclidean Domain and let $M$ be an $R$-module.

Suppose $\varphi : R^n \to M$ is a surjective $R$-module homomorphism. By Exercise 15, $\ker \varphi$ is finitely generated. If $x_1, x_2, \ldots, x_n$ is a basis for $R^n$ and $y_1, \ldots, y_m$ are generators for $\ker \varphi$ we have

$$y_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \quad i = 1, 2, \ldots, m$$

with coefficients $a_{ij} \in R$. It follows that the homomorphism $\varphi$ (hence the module structure of $M$) is determined by the choice of generators for $R^n$ and the matrix $A = (a_{ij})$. Such a matrix $A$ will be called a *relations matrix*.

**18.** **(a)** Show that interchanging the generators $y_i$ and $y_j$ interchanges the $i^{\text{th}}$ row with the $j^{\text{th}}$ row in the relations matrix.

**(b)** Show that, for any $a \in R$, replacing the element $y_j$ by $y_j - ay_i$ gives another set of generators for $\ker \varphi$ and that the corresponding relations matrix for this choice of generators is the same as the original relations matrix except that $-a$ times the $i^{\text{th}}$ row has been added to the $j^{\text{th}}$ row.

The following set of exercises outlines a proof of Theorem 5 in the special case where $R$ is a Euclidean Domain using a matrix argument involving row and column operations. This applies in particular to the cases $R = \mathbb{Z}$ and $R = F[x]$ of interest in the applications and is computationally useful.

Let $R$ be a Euclidean Domain and let $M$ be an $R$-module.

Suppose $\varphi : R^n \to M$ is a surjective $R$-module homomorphism. By Exercise 15, $\ker \varphi$ is finitely generated. If $x_1, x_2, \ldots, x_n$ is a basis for $R^n$ and $y_1, \ldots, y_m$ are generators for $\ker \varphi$ we have

$$y_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \quad i = 1, 2, \ldots, m$$

with coefficients $a_{ij} \in R$. It follows that the homomorphism $\varphi$ (hence the module structure of $M$) is determined by the choice of generators for $R^n$ and the matrix $A = (a_{ij})$. Such a matrix $A$ will be called a *relations matrix*.

**19.** By the previous two exercises we may perform elementary row and column operations on a given relations matrix by choosing different generators for $R^n$ and $\ker \varphi$. If all relation matrices are the zero matrix then $\ker \varphi = 0$ and $M \cong R^n$. Otherwise let $a_1$ be the (nonzero) g.c.d. (recall $R$ is a Euclidean Domain) of all the entries in a fixed initial relations matrix for $M$.

**(a)** Prove that by elementary row and column operations we may assume $a_1$ occurs in a relations matrix of the form

$$\begin{pmatrix} a_1 & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

where $a_1$ divides $a_{ij}$, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n$.

**(b)** Prove that there is a relations matrix of the form

$$\begin{pmatrix} a_1 & 0 & \ldots & 0 \\ 0 & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

where $a_1$ divides all the entries.

**(c)** Let $a_2$ be a g.c.d. of all the entries except the element $a_1$ in the relations matrix in (b). Prove that there is a relations matrix of the form

$$\begin{pmatrix} a_1 & 0 & 0 & \ldots & 0 \\ 0 & a_2 & 0 & \ldots & 0 \\ 0 & 0 & a_{33} & \ldots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & a_{m3} & \ldots & a_{mn} \end{pmatrix}$$

where $a_1$ divides $a_2$ and $a_2$ divides all the other entries of the matrix.

**(d)** Prove that there is a relations matrix of the form $\begin{pmatrix} D & 0 \\ 0 & 0 \end{pmatrix}$ where $D$ is a diagonal matrix with nonzero entries $a_1, a_2, \ldots, a_k$, $k \leq n$, satisfying

$$a_1 \mid a_2 \mid \cdots \mid a_k.$$

Conclude that

$$M \cong R/(a_1) \oplus R/(a_2) \oplus \cdots \oplus R/(a_k) \oplus R^{n-k}.$$

If $n$ is not the minimal number of generators required for $M$ then some of the initial elements $a_1, a_2, \ldots$ above will be units, so the corresponding direct summands above will be 0. If we remove these irrelevant factors we have produced the invariant factors of the module $M$. Further, the image of the new generators for $R^n$ corresponding to the direct summands above will then be a set of $R$-generators for the cyclic submodules of $M$ in its invariant factor decomposition (note that the image in $M$ of the generators corresponding to factors with $a_i$ a unit will be 0). The *column* operations performed in the relations matrix reduction correspond to changing the basis used for $R^n$ as described in Exercise 17:

**(a)** Interchanging the $i^{\text{th}}$ column with the $j^{\text{th}}$ column corresponds to interchanging the $i^{\text{th}}$ and $j^{\text{th}}$ elements in the basis for $R^n$.

**(b)** For any $a \in R$, adding $a$ times the $j^{\text{th}}$ column to the $i^{\text{th}}$ column corresponds to subtracting $a$ times the $i^{\text{th}}$ basis element from the $j^{\text{th}}$ basis element.

Keeping track of the column operations performed and changing the initial choice of generators for $M$ in the same way therefore gives a set of $R$-generators for the cyclic submodules of $M$ in its invariant factor decomposition.

This process is quite fast computationally once an initial set of generators for $M$ and initial relations matrix are determined. The element $a_1$ is determined using the Euclidean Algorithm as the g.c.d. of the elements in the initial relations matrix. Using the row and column operations we can obtain the appropriate linear combination of the entries to produce this g.c.d. in the (1,1)-position of a new relations matrix. One then subtracts the appropriate multiple of the first column and first row to obtain a matrix as in Exercise 19(b), then iterates this process. Some examples of this procedure in a special case are given at the end of the following section.

**20.** Let $R$ be an integral domain with quotient field $F$ and let $M$ be any $R$-module. Prove that the rank of $M$ equals the dimension of the vector space $F \otimes_R M$ over $F$.

**21.** Prove that a finitely generated module over a P.I.D. is projective if and only if it is free.

**22.** Let $R$ be a P.I.D. that is not a field. Prove that no finitely generated $R$-module is injective. [Use Exercise 4, Section 10.5 to consider torsion and free modules separately.]

## 12.2. The Rational Canonical Form

**1.** Prove that similar linear transformations of $V$ (or $n \times n$ matrices) have the same characteristic and the same minimal polynomial.

**2.** Let $M$ be as in Lemma 19. Prove that the minimal polynomial of $M$ is the least common multiple of the minimal polynomials of $A_1, \ldots, A_k$.

**3.** Prove that two $2 \times 2$ matrices over $F$ which are not scalar matrices are similar if and only if they have the same characteristic polynomial.

**4.** Prove that two $3 \times 3$ matrices are similar if and only if they have the same characteristic and same minimal polynomials. Give an explicit counterexample to this assertion for $4 \times 4$ matrices.

**5.** Prove directly from the fact that the collection of *all* linear transformations of an $n$ dimensional vector space $V$ over $F$ to itself form a vector space over $F$ of dimension $n^2$ that the minimal polynomial of a linear transformation $T$ has degree at most $n^2$.

**6.** Prove that the constant term in the characteristic polynomial of the $n \times n$ matrix $A$ is $(-1)^n \det A$ and that the coefficient of $x^{n-1}$ is the negative of the sum of the diagonal entries of $A$ (the sum of the diagonal entries of $A$ is called the *trace* of $A$). Prove that $\det A$ is the product of the eigenvalues of $A$ and that the trace of $A$ is the sum of the eigenvalues of $A$.

**7.** Determine the eigenvalues of the matrix

$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

**8.** Verify that the characteristic polynomial of the companion matrix

$$\begin{pmatrix} 0 & 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & 0 & \dots & 0 & -a_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & -a_{n-1} \end{pmatrix}$$

is

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0.$$

**9.** Find the rational canonical forms of

$$\begin{pmatrix} 0 & -1 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} c & 0 & -1 \\ 0 & c & 1 \\ -1 & 1 & c \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 422 & 465 & 15 & -30 \\ -420 & -463 & -15 & 30 \\ 840 & 930 & 32 & -60 \\ -140 & -155 & -5 & 12 \end{pmatrix}.$$

**10.** Find all similarity classes of $6 \times 6$ matrices over $\mathbb{Q}$ with minimal polynomial $(x+2)^2(x-1)$ (it suffices to give all lists of invariant factors and write out some of their corresponding matrices).

**11.** Find all similarity classes of $6 \times 6$ matrices over $\mathbb{C}$ with characteristic polynomial $(x^4 - 1)(x^2 - 1)$.

**12.** Find all similarity classes of $3 \times 3$ matrices $A$ over $\mathbb{F}_2$ satisfying $A^6 = I$ (compare with the answer we computed over $\mathbb{Q}$). Do the same for $4 \times 4$ matrices $B$ satisfying $B^{20} = I$.

**13.** Prove that the number of similarity classes of $3 \times 3$ matrices over $\mathbb{Q}$ with a given characteristic polynomial in $\mathbb{Q}[x]$ is the same as the number of similarity classes over any extension field of $\mathbb{Q}$. Give an example to show that this is not true in general for $4 \times 4$ matrices.

**14.** Determine all possible rational canonical forms for a linear transformation with characteristic polynomial $x^2(x^2 + 1)^2$.

**15.** Determine up to similarity all $2 \times 2$ rational matrices (i.e., $\in M_2(\mathbb{Q})$) of precise order 4 (multiplicatively, of course). Do the same if the matrix has entries from $\mathbb{C}$.

**16.** Show that $x^5 - 1 = (x-1)(x^2 - 4x + 1)(x^2 + 5x + 1)$ in $\mathbb{F}_{19}[x]$. Use this to determine up to similarity all $2 \times 2$ matrices with entries from $\mathbb{F}_{19}$ of (multiplicative) order 5.

**17.** Determine representatives for the conjugacy classes for $GL_3(\mathbb{F}_2)$. [Compare your answer with Theorem 15 and Proposition 14 of Chapter 6.]

**18.** Let $V$ be a finite dimensional vector space over $\mathbb{Q}$ and suppose $T$ is a nonsingular linear transformation of $V$ such that $T^{-1} = T^2 + T$. Prove that the dimension of $V$ is divisible by 3. If the dimension of $V$ is precisely 3 prove that all such transformations $T$ are similar.

**19.** Let $V$ be the infinite dimensional real vector space

$$\mathbb{R}^{\infty} = \{(a_0, a_1, a_2, \ldots) \mid a_0, a_1, a_2, \cdots \in \mathbb{R}\}.$$

Define the map $T : V \to V$ by $T(a_0, a_1, a_2, \ldots) = (0, a_0, a_1, a_2, \ldots)$. Prove that $T$ has no eigenvectors.

**20.** Let $\ell$ be a prime and let $\Phi_\ell(x) = \frac{x^\ell - 1}{x-1} = x^{\ell-1} + x^{\ell-2} + \ldots + x + 1 \in \mathbb{Z}[x]$ be the $\ell^{\text{th}}$ cyclotomic polynomial, which is irreducible over $\mathbb{Q}$ (Example 4 following Corollary 9.14). This exercise determines the smallest degree of a factor of $\Phi_\ell(x)$ modulo $p$ for any prime $p$ and so in particular determines when $\Phi_\ell(x)$ is irreducible modulo $p$. (This actually determines the complete factorization of $\Phi_\ell(x)$ modulo $p$ — cf. Exercise 8 of Section 13.6.)

**(a)** Show that if $p = \ell$ then $\Phi_\ell(x)$ is divisible by $x - 1$ in $\mathbb{F}_\ell[x]$.

**(b)** Suppose $p \neq \ell$ and let $f$ denote the order of $p$ in $\mathbb{F}_\ell^\times$, i.e., $f$ is the smallest power of $p$ with $p^f \equiv 1 \bmod \ell$. Show that $m = f$ is the first value of $m$ for which the group $GL_m(\mathbb{F}_p)$ contains an element $A$ of order $\ell$. [Use the formula for the order of this group at the end of Section 11.1.]

**(c)** Show that $\Phi_\ell(x)$ is not divisible by any polynomial of degree smaller than $f$ in $\mathbb{F}_p[x]$ [consider the companion matrix for such a divisor and use (b)]. Let $m_A(x) \in \mathbb{F}_p[x]$ denote the minimal polynomial for the matrix $A$ in (b) and conclude that $m_A(x)$ is irreducible of degree $f$ and divides $\Phi_\ell(x)$ in $\mathbb{F}_p[x]$.

**(d)** In particular, prove that $\Phi_\ell(x)$ is irreducible modulo $p$ if and only if $l - 1$ is the smallest power of $p$ which is congruent to 1 modulo $\ell$, i.e., $p$ is a primitive root modulo $\ell$.

**21.** Prove that the first two elementary row and column operations described before Theorem 21 do not change the determinant of the matrix and the third elementary operation multiplies the determinant by a unit. Conclude from Theorem 21 that the characteristic polynomial of $A$ differs by a unit from the product of the invariant factors of $A$. Since both these polynomials are monic by definition, conclude that they are equal (this gives an alternate proof of Proposition 20).

The following exercises outline the proof of Theorem 21. They carry out explicitly the construction described in Exercises 16 to 19 of the previous section for the Euclidean Domain $F[x]$. Let $V$ be an $n$-dimensional vector space with basis $v_1, v_2, \ldots, v_n$ and let $T$ be the linear transformation of $V$ defined by the matrix $A$ and this choice of basis, i.e., $T$ is the linear transformation with

$$T(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \qquad j = 1, 2, \ldots, n$$

where $A = (a_{ij})$. Let $F[x]^n$ be the free module of rank $n$ over $F[x]$ and let $\xi_1, \xi_2, \ldots, \xi_n$ denote a basis. Then we have a natural surjective $F[x]$-module homomorphism

$$\varphi : F[x]^n \to V$$

defined by mapping $\xi_i$ to $v_i$, $i = 1, 2, \ldots, n$. As indicated in the exercises of the previous section the invariant factors for the $F[x]$-module $V$ can be determined once we have determined a set of generators and the corresponding relations matrix for $\ker \varphi$. Since by definition $x$ acts on $V$ by the linear transformation $T$, we have

$$x(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \quad j = 1, 2, \ldots, n.$$

**22.** Show that the elements

$$v_j = -a_{1j}\xi_1 - \cdots - a_{j-1\,j}\xi_{j-1} + (x - a_{jj})\xi_j - a_{j+1\,j}\xi_{j+1} - \cdots - a_{nj}\xi_n$$

for $j = 1, 2, \ldots, n$ are elements of the kernel of $\varphi$.

The following exercises outline the proof of Theorem 21. They carry out explicitly the construction described in Exercises 16 to 19 of the previous section for the Euclidean Domain $F[x]$. Let $V$ be an $n$-dimensional vector space with basis $v_1, v_2, \ldots, v_n$ and let $T$ be the linear transformation of $V$ defined by the matrix $A$ and this choice of basis, i.e., $T$ is the linear transformation with

$$T(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \qquad j = 1, 2, \ldots, n$$

where $A = (a_{ij})$. Let $F[x]^n$ be the free module of rank $n$ over $F[x]$ and let $\xi_1, \xi_2, \ldots, \xi_n$ denote a basis. Then we have a natural surjective $F[x]$-module homomorphism

$$\varphi : F[x]^n \to V$$

defined by mapping $\xi_i$ to $v_i$, $i = 1, 2, \ldots, n$. As indicated in the exercises of the previous section the invariant factors for the $F[x]$-module $V$ can be determined once we have determined a set of generators and the corresponding relations matrix for $\ker \varphi$. Since by definition $x$ acts on $V$ by the linear transformation $T$, we have

$$x(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \quad j = 1, 2, \ldots, n.$$

**23.** **(a)** Show that $x\xi_j = v_j + f_j$ where $f_j \in F\xi_1 + \cdots + F\xi_n$ is an element in the $F$-vector space spanned by $\xi_1, \ldots, \xi_n$.

**(b)** Show that

$$F[x]\xi_1 + \cdots + F[x]\xi_n = (F[x]v_1 + \cdots + F[x]v_n) + (F\xi_1 + \cdots + F\xi_n).$$

The following exercises outline the proof of Theorem 21. They carry out explicitly the construction described in Exercises 16 to 19 of the previous section for the Euclidean Domain $F[x]$. Let $V$ be an $n$-dimensional vector space with basis $v_1, v_2, \ldots, v_n$ and let $T$ be the linear transformation of $V$ defined by the matrix $A$ and this choice of basis, i.e., $T$ is the linear transformation with

$$T(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \qquad j = 1, 2, \ldots, n$$

where $A = (a_{ij})$. Let $F[x]^n$ be the free module of rank $n$ over $F[x]$ and let $\xi_1, \xi_2, \ldots, \xi_n$ denote a basis. Then we have a natural surjective $F[x]$-module homomorphism

$$\varphi : F[x]^n \to V$$

defined by mapping $\xi_i$ to $v_i$, $i = 1, 2, \ldots, n$. As indicated in the exercises of the previous section the invariant factors for the $F[x]$-module $V$ can be determined once we have determined a set of generators and the corresponding relations matrix for $\ker \varphi$. Since by definition $x$ acts on $V$ by the linear transformation $T$, we have

$$x(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \quad j = 1, 2, \ldots, n.$$

**24.** Show that $v_1, v_2, \ldots, v_n$ generate the kernel of $\varphi$. [Use the previous result to show that any element of $\ker \varphi$ is the sum of an element in the module generated by $v_1, v_2, \ldots, v_n$ and an element of the form $b_1\xi_1 + \cdots + b_n\xi_n$ where the $b_i$ are elements of $F$. Then show that such an element is in $\ker \varphi$ if and only if all the $b_i$ are 0 since $v_1, \ldots, v_n$ are a basis for $V$ over $F$.]

The following exercises outline the proof of Theorem 21. They carry out explicitly the construction described in Exercises 16 to 19 of the previous section for the Euclidean Domain $F[x]$. Let $V$ be an $n$-dimensional vector space with basis $v_1, v_2, \ldots, v_n$ and let $T$ be the linear transformation of $V$ defined by the matrix $A$ and this choice of basis, i.e., $T$ is the linear transformation with

$$T(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \qquad j = 1, 2, \ldots, n$$

where $A = (a_{ij})$. Let $F[x]^n$ be the free module of rank $n$ over $F[x]$ and let $\xi_1, \xi_2, \ldots, \xi_n$ denote a basis. Then we have a natural surjective $F[x]$-module homomorphism

$$\varphi : F[x]^n \to V$$

defined by mapping $\xi_i$ to $v_i, i = 1, 2, \ldots, n$. As indicated in the exercises of the previous section the invariant factors for the $F[x]$-module $V$ can be determined once we have determined a set of generators and the corresponding relations matrix for $\ker \varphi$. Since by definition $x$ acts on $V$ by the linear transformation $T$, we have

$$x(v_j) = \sum_{i=1}^{n} a_{ij} v_i, \quad j = 1, 2, \ldots, n.$$

**25.** Show that the generators $v_1, v_2, \ldots, v_n$ of $\ker \varphi$ have corresponding relations matrix

$$\begin{pmatrix} x - a_{11} & -a_{21} & \ldots & -a_{n1} \\ -a_{12} & x - a_{22} & \ldots & -a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{1n} & -a_{2n} & \ldots & x - a_{nn} \end{pmatrix} = xI - A^t,$$

where $A^t$ is the transpose of $A$. Conclude that Theorem 21 and the algorithm for determining the invariant factors of $A$ follows by Exercises 16 to 19 in the previous section (note that the row and column operations necessary to diagonalize this relations matrix are the column and row operations necessary to diagonalize the matrix in Theorem 21, which explains why the invariant factor algorithm keeps track of the *row* operations used).

## 12.3. The Jordan Canonical Form

**1.** Suppose the vector space $V$ is the direct sum of cyclic $F[x]$-modules whose annihilators are $(x+1)^2$, $(x-1)(x^2+1)^2$, $(x^4-1)$ and $(x+1)(x^2-1)$. Determine the invariant factors and elementary divisors for $V$.

**2.** Prove that if $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of the $n \times n$ matrix $A$ then $\lambda_1^k, \ldots, \lambda_n^k$ are the eigenvalues of $A^k$ for any $k \geq 0$.

**3.** Use the method of Example 2 above to determine explicit matrices $P_1$ and $P_2$ with $P_1^{-1} B P_1$ and $P_2^{-1} C P_2$ in Jordan canonical form. Use this to explicitly construct a matrix $Q$ which conjugates $B$ into $C$ (proving directly that these matrices are similar).

**4.** Prove that the Jordan canonical form for the matrix

$$\begin{pmatrix} 9 & 4 & 5 \\ -4 & 0 & -3 \\ -6 & -4 & -2 \end{pmatrix}$$

is that stated at the beginning of this chapter. Explicitly determine a matrix $P$ which conjugates this matrix to its Jordan canonical form. Explain why this matrix cannot be diagonalized.

**5.** Compute the Jordan canonical form for the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -2 \\ 0 & 1 & 3 \end{pmatrix}.$$

**6.** Determine which of the following matrices are similar:

$$\begin{pmatrix} -1 & 4 & -4 \\ 2 & -1 & 3 \\ 0 & -4 & 3 \end{pmatrix} \quad \begin{pmatrix} -3 & -4 & 0 \\ 2 & 3 & 0 \\ 8 & 8 & 1 \end{pmatrix} \quad \begin{pmatrix} -3 & 2 & -4 \\ 2 & 1 & 0 \\ 3 & -1 & 3 \end{pmatrix} \quad \begin{pmatrix} -1 & 4 & -4 \\ 0 & -3 & 2 \\ 0 & -4 & 3 \end{pmatrix}.$$

**7.** Determine the Jordan canonical forms for the following matrices:

$$\begin{pmatrix} 5 & 4 & 1 \\ -1 & 0 & 0 \\ -3 & -4 & 1 \end{pmatrix} \quad \begin{pmatrix} 3 & 4 & 2 \\ -2 & -3 & -1 \\ -4 & -4 & -3 \end{pmatrix}.$$

**8.** Prove that the matrices

$$A = \begin{pmatrix} 5 & 6 & 0 \\ -3 & -4 & 0 \\ -2 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 3 & -1 & 2 \\ -10 & 6 & -14 \\ -6 & 3 & -7 \end{pmatrix}$$

are similar. Prove that both $A$ and $B$ can be diagonalized and determine explicit matrices $P_1$ and $P_2$ with $P_1^{-1}AP_1$ and $P_2^{-1}BP_2$ in diagonal form.

**9.** Prove that the matrices

$$A = \begin{pmatrix} -8 & -10 & -1 \\ 7 & 9 & 1 \\ 3 & 2 & 0 \end{pmatrix} \quad B = \begin{pmatrix} -3 & 2 & -4 \\ 4 & -1 & 4 \\ 4 & -2 & 5 \end{pmatrix}$$

both have $(x - 1)^2(x + 1)$ as characteristic polynomial but that one can be diagonalized and the other cannot. Determine the Jordan canonical form for both matrices.

**10.** Find all Jordan canonical forms of $2 \times 2$, $3 \times 3$ and $4 \times 4$ matrices over $\mathbb{C}$.

**11.** Verify that the characteristic polynomial of

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -2 & 0 & 1 \\ -2 & 0 & -1 & -2 \end{pmatrix}$$

is a product of linear factors over $\mathbb{Q}$. Determine the rational and Jordan canonical forms for $A$ over $\mathbb{Q}$.

**12.** Determine the Jordan canonical form for the matrix

$$\begin{pmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

**13.** Determine the Jordan canonical form for the matrix

$$\begin{pmatrix} 3 & 0 & -2 & -3 \\ 4 & -8 & 14 & -15 \\ 2 & -4 & 7 & -7 \\ 0 & 2 & -4 & 3 \end{pmatrix}.$$

**14.** Prove that the matrices

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ -4 & -1 & -4 & 0 \\ 2 & 1 & 3 & 0 \\ -2 & 4 & 9 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 5 & 0 & -4 & -7 \\ 3 & -8 & 15 & -13 \\ 2 & -4 & 7 & -7 \\ 1 & 2 & -5 & 1 \end{pmatrix}$$

are similar.

**15.** Prove that the matrices

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 5 & 2 & -8 & -8 \\ -6 & -3 & 8 & 8 \\ -3 & -1 & 3 & 4 \\ 3 & 1 & -4 & -5 \end{pmatrix}$$

both have characteristic polynomial $(x - 3)(x + 1)^3$. Determine whether they are similar and determine the Jordan canonical form for each matrix.

**16.** Determine the Jordan canonical form for the matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and determine a matrix $P$ which conjugates this matrix into its Jordan canonical form.

**17.** Prove that any matrix $A$ is similar to its transpose $A^t$.

**18.** Determine all possible Jordan canonical forms for a linear transformation with characteristic polynomial $(x-2)^3(x-3)^2$.

**19.** Prove that all $n \times n$ matrices with characteristic polynomial $f(x)$ are similar if and only if $f(x)$ has no repeated factors in its unique factorization in $F[x]$.

**20.** Show that the following matrices are similar in $M_p(\mathbb{F}_p)$ ($p \times p$ matrices with entries from $\mathbb{F}_p$):

$$\begin{pmatrix} 0 & 0 & 0 & \dots & 0 & 1 \\ 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & 1 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 1 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{pmatrix}.$$

**21.** Show that if $A^2 = A$ then $A$ is similar to a diagonal matrix which has only 0's and 1's along the diagonal.

**22.** Prove that an $n \times n$ matrix $A$ with entries from $\mathbb{C}$ satisfying $A^3 = A$ can be diagonalized. Is the same statement true over *any* field $F$?

**23.** Suppose $A$ is a $2 \times 2$ matrix with entries from $\mathbb{Q}$ for which $A^3 = I$ but $A \neq I$. Write $A$ in rational canonical form and in Jordan canonical form viewed as a matrix over $\mathbb{C}$.

**24.** Prove there are no $3 \times 3$ matrices $A$ over $\mathbb{Q}$ with $A^8 = I$ but $A^4 \neq I$.

**25.** Determine the Jordan canonical form for the $n \times n$ matrix over $\mathbb{Q}$ whose entries are all equal to 1.

**26.** Determine the Jordan canonical form for the $n \times n$ matrix over $\mathbb{F}_p$ whose entries are all equal to 1 (the answer depends on whether or not $p$ divides $n$).

**27.** Determine the Jordan canonical form for the $n \times n$ matrix over $\mathbb{Q}$ whose entries are all equal to 1 except that the entries along the main diagonal are all equal to 0.

**28.** Determine the Jordan canonical form for the $n \times n$ matrix over $\mathbb{F}_p$ whose entries are all equal to 1 except that the entries along the main diagonal are all equal to 0.

The direct sum of the cyclic submodules of $V$ corresponding to all the elementary divisors of $V$ which are powers of the same $x - \lambda$ is called the *generalized eigenspace of* $T$ corresponding to the eigenvalue $\lambda$. Note that this is the $p$-primary component of $V$ for the prime $p = x - \lambda$ of $F[x]$ and consists of the elements of $V$ which are annihilated by some power of the linear transformation $T - \lambda$. The matrix for $T$ on the generalized eigenspace for $\lambda$ is the block diagonal matrix of all Jordan blocks for $T$ with the same eigenvalue $\lambda$.

**29.** Suppose $V_i$ is the generalized eigenspace of $T$ corresponding to eigenvalue $\lambda_i$. For any $k \geq 0$, prove that the nullity of $T - \lambda_i$ on the subspace $(T - \lambda_i)^k V_i$ is the same as the nullity of $T - \lambda_i$ on $(T - \lambda_i)^k V$ and equals the number of Jordan blocks of $T$ having eigenvalue $\lambda_i$ and size greater than $k$ (so for $k = 0$ this gives the number of Jordan blocks).

The direct sum of the cyclic submodules of $V$ corresponding to all the elementary divisors of $V$ which are powers of the same $x-\lambda$ is called the *generalized eigenspace of* $T$ corresponding to the eigenvalue $\lambda$. Note that this is the $p$-primary component of $V$ for the prime $p = x-\lambda$ of $F[x]$ and consists of the elements of $V$ which are annihilated by some power of the linear transformation $T-\lambda$. The matrix for $T$ on the generalized eigenspace for $\lambda$ is the block diagonal matrix of all Jordan blocks for $T$ with the same eigenvalue $\lambda$.

**30.** Let $\lambda$ be an eigenvalue of the linear transformation $T$ on the finite dimensional vector space $V$ over the field $F$. Let $r_k = \dim_F (T-\lambda)^k V$ be the rank of the linear transformation $(T-\lambda)^k$ on $V$. For any $k \geq 1$, prove that $r_{k-1} - 2r_k + r_{k+1}$ is the number of Jordan blocks of $T$ corresponding to $\lambda$ of size $k$ [use Exercise 12 in Section 1]. (This gives an efficient method for determining the Jordan canonical form for $T$ by computing the ranks of the matrices $(A-\lambda I)^k$ for a matrix $A$ representing $T$, cf. Exercise 31(a) in Section 11.2.)

**31.** Let $N$ be an $n \times n$ matrix with coefficients in the field $F$. The matrix $N$ is said to be *nilpotent* if some power of $N$ is the zero matrix, i.e., $N^k = 0$ for some $k$. Prove that any nilpotent matrix is similar to a block diagonal matrix whose blocks are matrices with 1's along the first superdiagonal and 0's elsewhere.

**32.** Prove that if $N$ is an $n \times n$ nilpotent matrix then in fact $N^n = 0$.

**33.** Let $A$ be a strictly upper triangular $n \times n$ matrix (all entries on and below the main diagonal are zero). Prove that $A$ is nilpotent.

**34.** Prove that the trace of a nilpotent $n \times n$ matrix is 0 (recall the trace of a matrix is the sum of the diagonal elements).

**35.** For $0 \leq i \leq n$, let $d_i$ be the g.c.d. of the determinants of all the $i \times i$ minors of $xI - A$, for $A$ as in Theorem 21 (take the $0 \times 0$ minor to be 1). Prove that the $i^{\text{th}}$ element along the diagonal of the Smith Normal Form for $A$ is $d_i/d_{i-1}$. This gives the invariant factors for $A$. [Show these g.c.d.s do not change under elementary row and column operations.]

**36.** Let $V = \mathbb{C}^n$ be the usual $n$-dimensional vector space of $n$-tuples $(\alpha_1, \alpha_2, \ldots, \alpha_n)$ of complex numbers. Let $T$ be the linear transformation defined by setting $T(\alpha_1, \alpha_2, \ldots, \alpha_n)$ equal to $(0, \alpha_1, \alpha_2, \ldots, \alpha_{n-1})$. Determine the Jordan canonical form for $T$.

**37.** Let $J$ be a Jordan block of size $n$ with eigenvalue $\lambda$ over $\mathbb{C}$.

**(a)** Prove that the Jordan canonical form for the matrix $J^2$ is the Jordan block of size $n$ with eigenvalue $\lambda^2$ if $\lambda \neq 0$.

**(b)** If $\lambda = 0$ prove that the Jordan canonical form for $J^2$ has two blocks (with eigenvalues 0) of size $\frac{n}{2}, \frac{n}{2}$ if $n$ is even and of size $\frac{n-1}{2}, \frac{n+1}{2}$ if $n$ is odd.

**38.** Determine necessary and sufficient conditions for a matrix $A \in M_n(\mathbb{C})$ to have a square root, i.e., for there to exist another matrix $B \in M_n(\mathbb{C})$ such that $A = B^2$. [Suppose $B$ is in Jordan canonical form and consider the Jordan canonical form for $B^2$ using the previous exercise.]

**39.** Let $J$ be a Jordan block of size $n$ with eigenvalue $\lambda$ over a field $F$ of characteristic 2. Determine the Jordan canonical form for the matrix $J^2$. Determine necessary and sufficient conditions for a matrix $A \in M_n(F)$ to have a square root, i.e., for there to exist another matrix $B \in M_n(F)$ such that $A = B^2$.

The remaining exercises explore functions (power series) of a matrix and introduce some applications of the Jordan canonical form to the theory of differential equations.

Throughout these exercises the matrices are assumed to be $n \times n$ matrices with entries from the field $K$, where $K$ is either the real or complex numbers. Let

$$G(x) = \sum_{k=0}^{\infty} \alpha_k x^k$$

be a power series with coefficients from $K$. Let $G_N(x) = \sum_{k=0}^{N} \alpha_k x^k$ be the $N^{\text{th}}$ partial sum of $G(x)$ and for each $A \in M_n(K)$ let $G_N(A)$ be the element of $M_n(K)$ obtained (as usual) by substituting $A$ in this polynomial. For each fixed $i, j$ we obtain a sequence of real or complex numbers $c_{ij}^N$, $N = 0, 1, 2, \ldots$ by taking $c_{ij}^N$ to be the $i, j$ entry of the matrix $G_N(A)$. The series

$$G(A) = \sum_{k=0}^{\infty} \alpha_k A^k$$

is said to *converge* to the matrix $C$ in $M_n(K)$ if for each $i, j \in \{1, 2, \ldots, n\}$ the sequence $c_{ij}^N$, $N = 0, 1, 2, \ldots$ converges to the $i, j$ entry of $C$ (in which case we write $G(A) = C$). Say $G(A)$ *converges* if there is some $C \in M_n(K)$ such that $G(A) = C$. If $A$ is a $1 \times 1$ matrix, this is the usual notion of convergence of a series in $K$.

For $A = (a_{ij}) \in M_n(K)$ define

$$\| A \| = \sum_{i,j=1}^{n} |a_{ij}|$$

i.e., $\| A \|$ is the sum of the absolute values of all the entries of $A$.

**40.** Prove that for all $A, B \in M_n(K)$ and all $\alpha \in K$

**(a)** $\| A + B \| \leq \| A \| + \| B \|$

**(b)** $\| AB \| \leq \| A \| \cdot \| B \|$

**(c)** $\| \alpha A \| = |\alpha| \cdot \| A \|$.

The remaining exercises explore functions (power series) of a matrix and introduce some applications of the Jordan canonical form to the theory of differential equations.

Throughout these exercises the matrices are assumed to be $n \times n$ matrices with entries from the field $K$, where $K$ is either the real or complex numbers. Let

$$G(x) = \sum_{k=0}^{\infty} \alpha_k x^k$$

be a power series with coefficients from $K$. Let $G_N(x) = \sum_{k=0}^{N} \alpha_k x^k$ be the $N^{\text{th}}$ partial sum of $G(x)$ and for each $A \in M_n(K)$ let $G_N(A)$ be the element of $M_n(K)$ obtained (as usual) by substituting $A$ in this polynomial. For each fixed $i, j$ we obtain a sequence of real or complex numbers $c_{ij}^N$, $N = 0, 1, 2, \ldots$ by taking $c_{ij}^N$ to be the $i, j$ entry of the matrix $G_N(A)$. The series

$$G(A) = \sum_{k=0}^{\infty} \alpha_k A^k$$

is said to *converge* to the matrix $C$ in $M_n(K)$ if for each $i, j \in \{1, 2, \ldots, n\}$ the sequence $c_{ij}^N$, $N = 0, 1, 2, \ldots$ converges to the $i, j$ entry of $C$ (in which case we write $G(A) = C$). Say $G(A)$ *converges* if there is some $C \in M_n(K)$ such that $G(A) = C$. If $A$ is a $1 \times 1$ matrix, this is the usual notion of convergence of a series in $K$.

For $A = (a_{ij}) \in M_n(K)$ define

$$\| A \| = \sum_{i,j=1}^{n} |a_{ij}|$$

i.e., $\| A \|$ is the sum of the absolute values of all the entries of $A$.

**41.** Let $R$ be the radius of convergence of the real or complex power series $G(x)$ (where $R = \infty$ if $G(x)$ converges for all $x \in K$).

**(a)** Prove that if $\| A \| < R$ then $G(A)$ converges.

**(b)** Deduce that for *all* matrices $A$ the following power series converge:

$$\sin(A) = A - \frac{A^3}{3!} + \frac{A^5}{5!} + \cdots + (-1)^k \frac{A^{2k+1}}{(2k+1)!} + \cdots$$

$$\cos(A) = I - \frac{A^2}{2!} + \frac{A^4}{4!} + \cdots + (-1)^k \frac{A^{2k}}{(2k)!} + \cdots$$

$$\exp(A) = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots + \frac{A^k}{k!} + \cdots$$

where $I$ is the $n \times n$ identity matrix.

The remaining exercises explore functions (power series) of a matrix and introduce some applications of the Jordan canonical form to the theory of differential equations.

Throughout these exercises the matrices are assumed to be $n \times n$ matrices with entries from the field $K$, where $K$ is either the real or complex numbers. Let

$$G(x) = \sum_{k=0}^{\infty} \alpha_k x^k$$

be a power series with coefficients from $K$. Let $G_N(x) = \sum_{k=0}^{N} \alpha_k x^k$ be the $N^{\text{th}}$ partial sum of $G(x)$ and for each $A \in M_n(K)$ let $G_N(A)$ be the element of $M_n(K)$ obtained (as usual) by substituting $A$ in this polynomial. For each fixed $i, j$ we obtain a sequence of real or complex numbers $c_{ij}^N$, $N = 0, 1, 2, \ldots$ by taking $c_{ij}^N$ to be the $i, j$ entry of the matrix $G_N(A)$. The series

$$G(A) = \sum_{k=0}^{\infty} \alpha_k A^k$$

is said to *converge* to the matrix $C$ in $M_n(K)$ if for each $i, j \in \{1, 2, \ldots, n\}$ the sequence $c_{ij}^N$, $N = 0, 1, 2, \ldots$ converges to the $i, j$ entry of $C$ (in which case we write $G(A) = C$). Say $G(A)$ *converges* if there is some $C \in M_n(K)$ such that $G(A) = C$. If $A$ is a $1 \times 1$ matrix, this is the usual notion of convergence of a series in $K$.

For $A = (a_{ij}) \in M_n(K)$ define

$$\| A \| = \sum_{i,j=1}^{n} |a_{ij}|$$

i.e., $\| A \|$ is the sum of the absolute values of all the entries of $A$.

In view of applications to the theory of differential equations we introduce a variable $t$ at this point, so that for $A \in M_n(K)$ the matrix $At$ is obtained from $A$ by multiplying each entry by $t$ (which is the same as multiplying $A$ by the "scalar" matrix $tI$). We obtain a function from a subset of $K$ into $M_n(K)$ defined by $t \mapsto G(At)$ at all points $t$ where the series $G(At)$ converges. In particular, $\sin(At)$, $\cos(At)$ and $\exp(At)$ converge for all $t \in K$.

**42.** Let $P$ be a nonsingular $n \times n$ matrix.

**(a)** Prove that $PG(At)P^{-1} = G(PAtP^{-1}) = G(PAP^{-1}t)$. (This implies that, up to a change of basis, it suffices to compute $G(At)$ for matrices $A$ in canonical form). [Take limits of partial sums to get the first equality. The second equality is immediate because the matrix $tI$ commutes with every matrix.]

**(b)** Prove that if $A$ is the direct sum of matrices $A_1, A_2, \ldots, A_m$, then $G(At)$ is the direct sum of the matrices $G(A_1t), G(A_2t), \ldots, G(A_mt)$.

**(c)** Show that if $Z$ is the diagonal matrix with entries $z_1, z_2, \ldots, z_n$ then $G(Zt)$ is the diagonal matrix with entries $G(z_1t), G(z_2t), \ldots, G(z_nt)$.

The matrix $\exp(A)$ defined in Exercise 41(b) is called the *exponential* of $A$ and is often denoted by $e^A$. The next three exercises lead to a formula for the matrix $\exp(Jt)$, where $J$ is an elementary Jordan matrix.

**43.** Prove that if $A$ and $B$ are *commuting* matrices then $\exp(A + B) = \exp(A)\exp(B)$. [Treat $A$ and $B$ as commuting indeterminates and deduce this by comparing the power series on the left hand side with the product of the two power series on the right hand side.]

The matrix $\exp(A)$ defined in Exercise 41(b) is called the *exponential* of $A$ and is often denoted by $e^A$. The next three exercises lead to a formula for the matrix $\exp(Jt)$, where $J$ is an elementary Jordan matrix.

**44.** Use the preceding exercise to show that if $M$ is any matrix and $\lambda$ is any element of $K$ then

$$\exp(\lambda It + M) = e^{\lambda t} \exp(M).$$

The matrix $\exp(A)$ defined in Exercise 41(b) is called the *exponential* of $A$ and is often denoted by $e^A$. The next three exercises lead to a formula for the matrix $\exp(Jt)$, where $J$ is an elementary Jordan matrix.

**45.** Let $N$ be the $r \times r$ matrix with 1's on the first superdiagonal and zeros elsewhere. Compute the exponential of the following nilpotent $r \times r$ matrix:

$$\text{if } Nt = \begin{pmatrix} 0 & t & & & \\ & 0 & t & & \\ & & \ddots & \ddots & \\ & & & 0 & t \\ & & & & 0 \end{pmatrix} \quad \text{then } \exp(Nt) = \begin{pmatrix} 1 & t & \frac{t^2}{2!} & \cdots & \cdots & \frac{t^{r-1}}{(r-1)!} \\ & 1 & t & \frac{t^2}{2!} & & \vdots \\ & & \ddots & \ddots & \ddots & \vdots \\ & & & \ddots & t & \frac{t^2}{2!} \\ & & & & 1 & t \\ & & & & & 1 \end{pmatrix}.$$

Deduce that if $J$ is the $r \times r$ elementary Jordan matrix with eigenvalue $\lambda$ then

$$\exp(Jt) = \begin{pmatrix} e^{\lambda t} & te^{\lambda t} & \frac{t^2}{2!}e^{\lambda t} & \cdots & \cdots & \frac{t^{r-1}}{(r-1)!}e^{\lambda t} \\ & e^{\lambda t} & te^{\lambda t} & \frac{t^2}{2!}e^{\lambda t} & & \vdots \\ & & \ddots & \ddots & \ddots & \vdots \\ & & & \ddots & te^{\lambda t} & \frac{t^2}{2!}e^{\lambda t} \\ & & & & e^{\lambda t} & te^{\lambda t} \\ & & & & & e^{\lambda t} \end{pmatrix}.$$

[To do the first part use the observation that since $Nt$ is a nilpotent matrix, $\exp(Nt)$ is a *polynomial* in $Nt$, i.e., all but a finite number of the terms in the power series are zero. To compute the exponential of $Jt$ write $Jt$ as $\lambda It + Nt$ and use Exercise 44 with $M = Nt$.]

The matrix $\exp(A)$ defined in Exercise 41(b) is called the *exponential* of $A$ and is often denoted by $e^A$. The next three exercises lead to a formula for the matrix $\exp(Jt)$, where $J$ is an elementary Jordan matrix.

Let $A \in M_n(K)$ and let $P$ be a change of basis matrix such that $P^{-1}AP$ is in Jordan canonical form. Suppose $P^{-1}AP$ is the sum of elementary Jordan matrices $J_1, \ldots, J_m$. The preceding exercises (with $t = 1$) show that $\exp(A)$ can easily be found by writing $E = \exp(P^{-1}AP)$ as the direct sum of the matrices $\exp(J_1), \ldots, \exp(J_m)$ and then changing the basis back again to obtain $\exp(A) = PEP^{-1}$.

**46.** For the $4 \times 4$ matrices $D$ and $P$ given in Example 3 of this section:

$$D = \begin{pmatrix} 1 & 2 & -4 & 4 \\ 2 & -1 & 4 & -8 \\ 1 & 0 & 1 & -2 \\ 0 & 1 & -2 & 3 \end{pmatrix} \qquad P = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 2 & 0 & -2 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

show that

$$E = \begin{pmatrix} e & e & 0 & 0 \\ 0 & e & 0 & 0 \\ 0 & 0 & e & e \\ 0 & 0 & 0 & e \end{pmatrix} \qquad \text{and} \qquad \exp(D) = \begin{pmatrix} e & 2e & -4e & 4e \\ 2e & -e & 4e & -8e \\ e & 0 & e & -2e \\ 0 & e & -2e & 3e \end{pmatrix}.$$

The matrix $\exp(A)$ defined in Exercise 41(b) is called the *exponential* of $A$ and is often denoted by $e^A$. The next three exercises lead to a formula for the matrix $\exp(Jt)$, where $J$ is an elementary Jordan matrix.

Let $A \in M_n(K)$ and let $P$ be a change of basis matrix such that $P^{-1}AP$ is in Jordan canonical form. Suppose $P^{-1}AP$ is the sum of elementary Jordan matrices $J_1, \ldots, J_m$. The preceding exercises (with $t = 1$) show that $\exp(A)$ can easily be found by writing $E = \exp(P^{-1}AP)$ as the direct sum of the matrices $\exp(J_1), \ldots, \exp(J_m)$ and then changing the basis back again to obtain $\exp(A) = PEP^{-1}$.

**47.** Compute the exponential of each of the following matrices:

**(a)** the matrix $A$ in Example 2 of this section

**(b)** the matrix in Exercise 4 (where you computed the Jordan canonical form and a change of basis matrix)

**(c)** the matrix in Exercise 16.

The matrix $\exp(A)$ defined in Exercise 41(b) is called the *exponential* of $A$ and is often denoted by $e^A$. The next three exercises lead to a formula for the matrix $\exp(Jt)$, where $J$ is an elementary Jordan matrix.

**48.** Show that $\exp(0) = I$ (here 0 is the zero matrix and $I$ is the identity matrix). Deduce that $\exp(A)$ is nonsingular with inverse $\exp(-A)$ for all matrices $A \in M_n(K)$.

The matrix $\exp(A)$ defined in Exercise 41(b) is called the *exponential* of $A$ and is often denoted by $e^A$. The next three exercises lead to a formula for the matrix $\exp(Jt)$, where $J$ is an elementary Jordan matrix.

**49.** Prove that $\det(\exp(A)) = e^{\mathrm{tr}(A)}$, where $\mathrm{tr}(A)$ is the trace of $A$ (the sum of the diagonal entries of $A$).

The matrix $\exp(A)$ defined in Exercise 41(b) is called the *exponential* of $A$ and is often denoted by $e^A$. The next three exercises lead to a formula for the matrix $\exp(Jt)$, where $J$ is an elementary Jordan matrix.

**50.** Fix any $A \in M_n(K)$. Prove that the map

$$K \to GL_n(K) \qquad \text{defined by} \qquad t \mapsto \exp(At)$$

is a group homomorphism (here $K$ is the additive group of the field). (Note how this generalizes the familiar exponential map from $K$ to $K^\times$, which is the $n = 1$ case. The subgroup $\{\exp(At) \mid t \in K\}$ is called a *1-parameter subgroup* of $GL_n(K)$. These subgroups and the exponential map play an important role in the theory of *Lie groups* — $GL_n(K)$ being a particular example of a Lie group.).

Let $G(x)$ be a power series having an infinite radius of convergence and fix a matrix $A \in M_n(K)$. The entries of the matrix $G(At)$ are $K$-valued functions of the variable $t$ that are defined for all $t$. Let $c_{ij}(t)$ be the function of $t$ in the $i, j$ entry of $G(At)$. The *derivative* of $G(At)$ with respect to $t$, denoted by $\frac{d}{dt}G(At)$, is the matrix whose $i, j$ entry is $\frac{d}{dt}c_{ij}(t)$ obtained by differentiating each of the entries of $G(At)$. In other words, if we identify $M_n(K)$ with $K^{n^2}$ by considering each $n \times n$ matrix as an $n^2$-tuple, then $t \mapsto G(At)$ is a map from $K$ to $K^{n^2}$ (i.e., is a vector valued function of $t$) whose derivative is just the usual (componentwise) derivative of this vector valued function.

**51.** Establish the following properties of derivatives:

**(a)** If $G(x) = \sum_{k=0}^{\infty} \alpha_k x^k$ then $\frac{d}{dt}G(At) = A \sum_{k=1}^{\infty} k\alpha_k (At)^{k-1}$.

**(b)** If $v$ is an $n \times 1$ matrix with (constant) entries from $K$ then

$$\frac{d}{dt}(G(At)v) = \left(\frac{d}{dt}G(At)\right)v.$$

Let $G(x)$ be a power series having an infinite radius of convergence and fix a matrix $A \in M_n(K)$. The entries of the matrix $G(At)$ are $K$-valued functions of the variable $t$ that are defined for all $t$. Let $c_{ij}(t)$ be the function of $t$ in the $i, j$ entry of $G(At)$. The *derivative* of $G(At)$ with respect to $t$, denoted by $\frac{d}{dt}G(At)$, is the matrix whose $i, j$ entry is $\frac{d}{dt}c_{ij}(t)$ obtained by differentiating each of the entries of $G(At)$. In other words, if we identify $M_n(K)$ with $K^{n^2}$ by considering each $n \times n$ matrix as an $n^2$-tuple, then $t \mapsto G(At)$ is a map from $K$ to $K^{n^2}$ (i.e., is a vector valued function of $t$) whose derivative is just the usual (componentwise) derivative of this vector valued function.

**52.** Deduce from part (a) of the preceding exercise that

$$\frac{d}{dt}\exp(At) = A\exp(At).$$

Now let $y_1(t), \ldots, y_n(t)$ be differentiable functions of the real variable $t$ that are related by the following linear system of first order differential equations with constant coefficients $a_{ij} \in K$:

$$\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + \ldots + a_{1n}y_n \\ y_2' &= a_{21}y_1 + a_{22}y_2 + \ldots + a_{2n}y_n \\ &\vdots \\ y_n' &= a_{n1}y_1 + a_{n2}y_2 + \ldots + a_{nn}y_n \end{aligned} \qquad (*)$$

(here the primes denote derivatives with respect to $t$). Let $A$ be the matrix whose $i, j$ entry is $a_{ij}$, so that $(*)$ may be written as

$$\begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{pmatrix} = A \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

or, more succinctly, as $y' = Ay$, where $y$ is the column vector of functions $y_1(t), \ldots, y_n(t)$.

An $n \times n$ matrix whose entries are functions of $t$ and whose columns are independent solutions to the system $(*)$ is called a *fundamental matrix* of $(*)$. By the theory of differential equations, the set of vectors $y$ that are solutions to the system $(*)$ form an $n$-dimensional vector space over $K$ and so the columns of a fundamental matrix are a *basis for the vector space of all solutions to* $(*)$.

**53.** Prove that $\exp(At)$ is a fundamental matrix of $(*)$. Show also that if $C$ is the $n \times 1$ constant vector whose entries are $y_1(0), \ldots, y_n(0)$ then $y(t) = \exp(At)C$ is the particular solution to the system $(*)$ satisfying the initial condition $y(0) = C$. (Note how this generalizes the 1-dimensional result that the single differential equation $y' = ay$ has $e^{at}$ as a basis for the 1-dimensional space of solutions and the unique solution to this differential equation satisfying the initial condition $y(0) = c$ is $y = ce^{at}$.) [Use the preceding exercises.]

Now let $y_1(t), \ldots, y_n(t)$ be differentiable functions of the real variable $t$ that are related by the following linear system of first order differential equations with constant coefficients $a_{ij} \in K$:

$$\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + \ldots + a_{1n}y_n \\ y_2' &= a_{21}y_1 + a_{22}y_2 + \ldots + a_{2n}y_n \\ &\vdots \\ y_n' &= a_{n1}y_1 + a_{n2}y_2 + \ldots + a_{nn}y_n \end{aligned} \tag{$*$}$$

(here the primes denote derivatives with respect to $t$). Let $A$ be the matrix whose $i, j$ entry is $a_{ij}$, so that $(*)$ may be written as

$$\begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{pmatrix} = A \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

or, more succinctly, as $y' = Ay$, where $y$ is the column vector of functions $y_1(t), \ldots, y_n(t)$.

An $n \times n$ matrix whose entries are functions of $t$ and whose columns are independent solutions to the system $(*)$ is called a *fundamental matrix* of $(*)$. By the theory of differential equations, the set of vectors $y$ that are solutions to the system $(*)$ form an $n$-dimensional vector space over $K$ and so the columns of a fundamental matrix are a *basis for the vector space of all solutions to* $(*)$.

**54.** Prove that if $M$ is a fundamental matrix of $(*)$ and if $Q$ is a nonsingular matrix in $M_n(K)$, then $MQ$ is also a fundamental matrix of $(*)$. [The columns of $MQ$ are linear combinations of the columns of $M$.]

Now let $y_1(t), \ldots, y_n(t)$ be differentiable functions of the real variable $t$ that are related by the following linear system of first order differential equations with constant coefficients $a_{ij} \in K$:

$$\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + \ldots + a_{1n}y_n \\ y_2' &= a_{21}y_1 + a_{22}y_2 + \ldots + a_{2n}y_n \\ &\vdots \\ y_n' &= a_{n1}y_1 + a_{n2}y_2 + \ldots + a_{nn}y_n \end{aligned} \qquad (*)$$

(here the primes denote derivatives with respect to $t$). Let $A$ be the matrix whose $i, j$ entry is $a_{ij}$, so that $(*)$ may be written as

$$\begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{pmatrix} = A \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

or, more succinctly, as $y' = Ay$, where $y$ is the column vector of functions $y_1(t), \ldots, y_n(t)$.

An $n \times n$ matrix whose entries are functions of $t$ and whose columns are independent solutions to the system $(*)$ is called a *fundamental matrix* of $(*)$. By the theory of differential equations, the set of vectors $y$ that are solutions to the system $(*)$ form an $n$-dimensional vector space over $K$ and so the columns of a fundamental matrix are a *basis for the vector space of all solutions to* $(*)$.

Now apply the preceding two exercises to solve some specific systems of differential equations as follows: given the matrix $A$ in a system $(*)$, calculate a change of basis matrix $P$ such that $B = P^{-1}AP$ is in Jordan canonical form. Then $\exp(At) = P\exp(Bt)P^{-1}$ is a fundamental matrix for $(*)$. By the preceding exercise, $P\exp(Bt)$ is also a fundamental matrix for $(*)$ and $\exp(Bt)$ can be calculated by the method described in the discussion following Exercise 45 (in particular, one does not have to find the inverse of the matrix $P$ to obtain a fundamental matrix for $(*)$). Thus, for example, if $A = D$ and $P$ are the matrices given in Exercise 46, then we saw that the Jordan canonical form for $A$ is the matrix $B = P^{-1}AP$ consisting of two $2 \times 2$ Jordan blocks with eigenvalues 1. A fundamental matrix for the system $y' = Ay$ is therefore

$$P\exp(B) = \begin{pmatrix} 0 & 1 & 2 & \mathbf{0} \\ 2 & 0 & -2 & \mathbf{1} \\ 1 & 0 & 0 & \mathbf{0} \\ 0 & 0 & 1 & \mathbf{0} \end{pmatrix} \begin{pmatrix} e^t & te^t & 0 & \mathbf{0} \\ 0 & e^t & 0 & \mathbf{0} \\ 0 & 0 & e^t & \boldsymbol{te^t} \\ 0 & 0 & 0 & \boldsymbol{e^t} \end{pmatrix} = \begin{pmatrix} 0 & e^t & 2e^t & 2te^t \\ 2e^t & 2te^t & -2e^t & e^t(1-\mathbf{2t}) \\ e^t & te^t & 0 & 0 \\ 0 & 0 & e^t & te^t \end{pmatrix}.$$

Writing this out more explicitly, this shows that the general solution to the system of differential equations

$$\begin{aligned} y_1' &= y_1 + 2y_2 - 4y_3 + 4y_4 \\ y_2' &= 2y_1 - y_2 + 4y_3 - 8y_4 \\ y_3' &= y_1 + y_3 - 2y_4 \\ y_4' &= y_2 - 2y_3 + 3y_4 \end{aligned}$$

is given by

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \alpha_1 \begin{pmatrix} 0 \\ 2e^t \\ e^t \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} e^t \\ 2te^t \\ te^t \\ 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 2e^t \\ -2e^t \\ 0 \\ e^t \end{pmatrix} + \alpha_4 \begin{pmatrix} 2te^t \\ e^t(1-2t) \\ 0 \\ te^t \end{pmatrix}$$

where $\alpha_1, \ldots, \alpha_4$ are arbitrary elements of the field $K$ (this describes the 4-dimensional vector space of solutions).

**55.** In each of Parts (a) to (c) find a fundamental matrix for the system $(*)$, where the coefficient matrix $A$ of $(*)$ is specified.

**(a)** $A$ is the matrix in Part (a) of Exercise 47.

**(b)** $A$ is the matrix in Part (b) of Exercise 47.

**(c)** $A$ is the matrix in Part (c) of Exercise 47.

Now let $y_1(t), \ldots, y_n(t)$ be differentiable functions of the real variable $t$ that are related by the following linear system of first order differential equations with constant coefficients $a_{ij} \in K$:

$$\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + \ldots + a_{1n}y_n \\ y_2' &= a_{21}y_1 + a_{22}y_2 + \ldots + a_{2n}y_n \\ &\vdots \\ y_n' &= a_{n1}y_1 + a_{n2}y_2 + \ldots + a_{nn}y_n \end{aligned} \qquad (*)$$

(here the primes denote derivatives with respect to $t$). Let $A$ be the matrix whose $i, j$ entry is $a_{ij}$, so that $(*)$ may be written as

$$\begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{pmatrix} = A \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

or, more succinctly, as $y' = Ay$, where $y$ is the column vector of functions $y_1(t), \ldots, y_n(t)$.

An $n \times n$ matrix whose entries are functions of $t$ and whose columns are independent solutions to the system $(*)$ is called a *fundamental matrix* of $(*)$. By the theory of differential equations, the set of vectors $y$ that are solutions to the system $(*)$ form an $n$-dimensional vector space over $K$ and so the columns of a fundamental matrix are a *basis for the vector space of all solutions to* $(*)$.

Now apply the preceding two exercises to solve some specific systems of differential equations as follows: given the matrix $A$ in a system $(*)$, calculate a change of basis matrix $P$ such that $B = P^{-1}AP$ is in Jordan canonical form. Then $\exp(At) = P\exp(Bt)P^{-1}$ is a fundamental matrix for $(*)$. By the preceding exercise, $P\exp(Bt)$ is also a fundamental matrix for $(*)$ and $\exp(Bt)$ can be calculated by the method described in the discussion following Exercise 45 (in particular, one does not have to find the inverse of the matrix $P$ to obtain a fundamental matrix for $(*)$). Thus, for example, if $A = D$ and $P$ are the matrices given in Exercise 46, then we saw that the Jordan canonical form for $A$ is the matrix $B = P^{-1}AP$ consisting of two $2 \times 2$ Jordan blocks with eigenvalues 1. A fundamental matrix for the system $y' = Ay$ is therefore

$$P\exp(B) = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 2 & 0 & -2 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} e^t & te^t & 0 & 0 \\ 0 & e^t & 0 & 0 \\ 0 & 0 & e^t & te^t \\ 0 & 0 & 0 & e^t \end{pmatrix} = \begin{pmatrix} 0 & e^t & 2e^t & 2te^t \\ 2e^t & 2te^t & -2e^t & e^t(1-2t) \\ e^t & te^t & 0 & 0 \\ 0 & 0 & e^t & te^t \end{pmatrix}.$$

Writing this out more explicitly, this shows that the general solution to the system of differential equations

$$\begin{aligned} y_1' &= y_1 + 2y_2 - 4y_3 + 4y_4 \\ y_2' &= 2y_1 - y_2 + 4y_3 - 8y_4 \\ y_3' &= y_1 + y_3 - 2y_4 \\ y_4' &= y_2 - 2y_3 + 3y_4 \end{aligned}$$

is given by

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix} = \alpha_1 \begin{pmatrix} 0 \\ 2e^t \\ e^t \\ 0 \end{pmatrix} + \alpha_2 \begin{pmatrix} e^t \\ 2te^t \\ te^t \\ 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} 2e^t \\ -2e^t \\ 0 \\ e^t \end{pmatrix} + \alpha_4 \begin{pmatrix} 2te^t \\ e^t(1-2t) \\ 0 \\ te^t \end{pmatrix}$$

where $\alpha_1, \ldots, \alpha_4$ are arbitrary elements of the field $K$ (this describes the 4-dimensional vector space of solutions).

**56.** Consider the system $(*)$ whose coefficient matrix $A$ is the matrix $D$ listed in Exercise 46 and whose fundamental matrix was computed just before the preceding exercise. Find the particular solution to $(*)$ that satisfies the initial condition $y_i(0) = 1$ for $i = 1, 2, 3, 4$.

Next we explore a special case of (∗). Given the linear $n^{\text{th}}$ *order* differential equation with constant coefficients

$$y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1y' + a_0y = 0 \qquad (**)$$

(where $y^{(k)}$ is the $k^{\text{th}}$ derivative of $y$ and $y^{(0)} = y$) one can form a *system* of linear *first order* differential equations by letting $y_i = y^{(i-1)}$ for $1 \leq i \leq n$ (the coefficient matrix of this system is described in the next exercise). A basis for the $n$-dimensional vector space of solutions to the $n^{\text{th}}$ order equation (∗∗) may then obtained from a fundamental matrix for the linear system. Specifically, in each of the $n \times 1$ columns of functions in a fundamental matrix for the system, the 1, 1 entry is a solution to (∗∗) and so the $n$ functions in the first row of the fundamental matrix for the system form a basis for the solutions to (∗∗).

**57.** Prove that the matrix, $A$, of coefficients of the system of $n$ first order equations obtained from (∗∗) is the transpose of the companion matrix of the polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$.

Next we explore a special case of (∗). Given the linear $n^{\text{th}}$ *order* differential equation with constant coefficients

$$y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1y' + a_0y = 0 \qquad (**)$$

(where $y^{(k)}$ is the $k^{\text{th}}$ derivative of $y$ and $y^{(0)} = y$) one can form a *system* of linear *first order* differential equations by letting $y_i = y^{(i-1)}$ for $1 \leq i \leq n$ (the coefficient matrix of this system is described in the next exercise). A basis for the $n$-dimensional vector space of solutions to the $n^{\text{th}}$ order equation (∗∗) may then obtained from a fundamental matrix for the linear system. Specifically, in each of the $n \times 1$ columns of functions in a fundamental matrix for the system, the 1, 1 entry is a solution to (∗∗) and so the $n$ functions in the first row of the fundamental matrix for the system form a basis for the solutions to (∗∗).

**58.** Use the above methods to find a basis for the vector space of solutions to the following differential equations

**(a)** $y''' - 3y' + 2y = 0$

**(b)** $y'''' + 4y''' + 6y'' + 4y' + y = 0.$

Now let $y_1(t), \ldots, y_n(t)$ be differentiable functions of the real variable $t$ that are related by the following linear system of first order differential equations with constant coefficients $a_{ij} \in K$:

$$\begin{aligned} y_1' &= a_{11}y_1 + a_{12}y_2 + \ldots + a_{1n}y_n \\ y_2' &= a_{21}y_1 + a_{22}y_2 + \ldots + a_{2n}y_n \\ &\vdots \\ y_n' &= a_{n1}y_1 + a_{n2}y_2 + \ldots + a_{nn}y_n \end{aligned} \qquad (*)$$

(here the primes denote derivatives with respect to $t$). Let $A$ be the matrix whose $i, j$ entry is $a_{ij}$, so that $(*)$ may be written as

$$\begin{pmatrix} y_1' \\ y_2' \\ \vdots \\ y_n' \end{pmatrix} = A \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

or, more succinctly, as $y' = Ay$, where $y$ is the column vector of functions $y_1(t), \ldots, y_n(t)$.

An $n \times n$ matrix whose entries are functions of $t$ and whose columns are independent solutions to the system $(*)$ is called a *fundamental matrix* of $(*)$. By the theory of differential equations, the set of vectors $y$ that are solutions to the system $(*)$ form an $n$-dimensional vector space over $K$ and so the columns of a fundamental matrix are a *basis for the vector space of all solutions to* $(*)$.

A system of differential equations

$$\begin{aligned} y_1' &= F_1(y_1, y_2, \ldots, y_n) \\ y_2' &= F_2(y_1, y_2, \ldots, y_n) \\ &\vdots \\ y_n' &= F_n(y_1, y_2, \ldots, y_n) \end{aligned}$$

where $F_1, F_2, \ldots, F_n$ are functions of $n$ variables, is called an *autonomous* system and it will be written more succinctly as $y' = F(y)$, where $F = (F_1, \ldots, F_n)$. (The expression autonomous means "independent of time" and it indicates that the variable $t$ — which may be thought of as a time variable — does not appear explicitly on the right hand side.) The system $(*)$ is the special type of autonomous system in which each $F_i$ is a linear function. In many instances it is desirable to analyze the behavior of solutions to an autonomous system of differential equations without explicitly finding these solutions (indeed, it is unlikely that it will be possible to find explicit solutions for a given nonlinear system). This investigation falls under the rubric "qualitative analysis" of autonomous differential equations and the rudiments of this study are often treated in basic calculus courses for $1 \times 1$ systems. The first step in a qualitative analysis of an $n \times n$ autonomous system is to find the *steady states*, namely the constant solutions (these are called steady states since they do not change with $t$). Note that a constant function $y = c$, where $c$ is the $n \times 1$ constant vector with entries $c_1, \ldots, c_n$, is a solution to $y' = F(y)$ if and only if

$$c_i' = 0 = F_i(c_1, \ldots, c_n) \quad \text{for } i = 1, 2, \ldots, n,$$

so the steady states are found by computing the zeros of $F$ (in the case of a nonlinear system this may require numerical methods). Next, given the initial value of some solution, one wishes to analyze the behavior of this solution as $t \to \infty$. This is called the *asymptotic behavior* of the solution. Again, it may not be possible to find the solution explicitly, although by the general theory of differential equations a solution to the initial value problem is unique provided the functions $F_i$ are differentiable. A steady state $y = c$ is called *globally asymptotically stable* if every solution tends to $c$ as $t \to \infty$, i.e., for any solution $y(t)$ we have $\lim_{t\to\infty} y_i(t) = c_i$ for all $i = 1, 2, \ldots, n$.

In the case of the linear autonomous system $(*)$ the solutions form a vector space, so the only constant solution is the zero solution. The next exercise gives a *sufficient* condition for zero to be globally asymptotically stable and it gives one example of how the behavior of a linear system may be analyzed in terms of the eigenvalues of its coefficient matrix. Nonlinear systems can be approximated by linear systems in some neighborhood of a steady state by considering $y' = Ty$, where $T = \left(\dfrac{\partial F_i}{\partial y_j}\right)$ is the $n \times n$ Jacobian matrix of $F$ evaluated at the steady state point. In this way the analysis of linear systems plays an important role in the local analysis of general autonomous systems.

**59.** Prove that the solution of $(*)$ given by $y_i(t) = 0$ for all $i \in \{1, \ldots, n\}$ (i.e., the zero solution) is globally asymptotically stable if all the eigenvalues of $A$ have negative real parts. [For those unfamiliar with the behavior of the complex exponential function, assume all eigenvalues are real (hence are negative real numbers). Use the explicit nature of the solutions to show that they all tend to zero as $t \to \infty$.]

# 13. Field Theory

## 13.1. Basic Theory of Field Extensions

**1.** Show that $p(x) = x^3 + 9x + 6$ is irreducible in $\mathbb{Q}[x]$. Let $\theta$ be a root of $p(x)$. Find the inverse of $1 + \theta$ in $\mathbb{Q}(\theta)$.

**2.** Show that $x^3 - 2x - 2$ is irreducible over $\mathbb{Q}$ and let $\theta$ be a root. Compute $(1+\theta)(1+\theta+\theta^2)$ and $\dfrac{1+\theta}{1+\theta+\theta^2}$ in $\mathbb{Q}(\theta)$.

**3.** Show that $x^3 + x + 1$ is irreducible over $\mathbb{F}_2$ and let $\theta$ be a root. Compute the powers of $\theta$ in $\mathbb{F}_2(\theta)$.

**4.** Prove directly that the map $a + b\sqrt{2} \mapsto a - b\sqrt{2}$ is an isomorphism of $\mathbb{Q}(\sqrt{2})$ with itself.

**5.** Suppose $\alpha$ is a rational root of a monic polynomial in $\mathbb{Z}[x]$. Prove that $\alpha$ is an integer.

**6.** Show that if $\alpha$ is a root of $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ then $a_n\alpha$ is a root of the monic polynomial $x^n + a_{n-1}x^{n-1} + a_n a_{n-2}x^{n-2} + \cdots + a_n^{n-2}a_1 x + a_n^{n-1}a_0$.

**7.** Prove that $x^3 - nx + 2$ is irreducible for $n \neq -1, 3, 5$.

**8.** Prove that $x^5 - ax - 1 \in \mathbb{Z}[x]$ is irreducible unless $a = 0$, 2 or $-1$. The first two correspond to linear factors, the third corresponds to the factorization $(x^2 - x + 1)(x^3 + x^2 - 1)$.

## 13.2. Algebraic Extensions

**1.** Let $\mathbb{F}$ be a finite field of characteristic $p$. Prove that $|\mathbb{F}| = p^n$ for some positive integer $n$.

**2.** Let $g(x) = x^2 + x - 1$ and let $h(x) = x^3 - x + 1$. Obtain fields of 4, 8, 9 and 27 elements by adjoining a root of $f(x)$ to the field $F$ where $f(x) = g(x)$ or $h(x)$ and $F = \mathbb{F}_2$ or $\mathbb{F}_3$. Write down the multiplication tables for the fields with 4 and 9 elements and show that the nonzero elements form a cyclic group.

**3.** Determine the minimal polynomial over $\mathbb{Q}$ for the element $1 + i$.

**4.** Determine the degree over $\mathbb{Q}$ of $2 + \sqrt{3}$ and of $1 + \sqrt[3]{2} + \sqrt[3]{4}$.

**5.** Let $F = \mathbb{Q}(i)$. Prove that $x^3 - 2$ and $x^3 - 3$ are irreducible over $F$.

**6.** Prove directly from the definitions that the field $F(\alpha_1, \alpha_2, \ldots, \alpha_n)$ is the composite of the fields $F(\alpha_1), F(\alpha_2), \ldots, F(\alpha_n)$.

**7.** Prove that $\mathbb{Q}(\sqrt{2}+\sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ [one inclusion is obvious, for the other consider $(\sqrt{2}+\sqrt{3})^2$, etc.]. Conclude that $[\mathbb{Q}(\sqrt{2}+\sqrt{3}) : \mathbb{Q}] = 4$. Find an irreducible polynomial satisfied by $\sqrt{2}+\sqrt{3}$.

**8.** Let $F$ be a field of characteristic $\neq 2$. Let $D_1$ and $D_2$ be elements of $F$, neither of which is a square in $F$. Prove that $F(\sqrt{D_1}, \sqrt{D_2})$ is of degree 4 over $F$ if $D_1 D_2$ is not a square in $F$ and is of degree 2 over $F$ otherwise. When $F(\sqrt{D_1}, \sqrt{D_2})$ is of degree 4 over $F$ the field is called a *biquadratic extension of* $F$.

**9.** Let $F$ be a field of characteristic $\neq 2$. Let $a, b$ be elements of the field $F$ with $b$ not a square in $F$. Prove that a necessary and sufficient condition for $\sqrt{a+\sqrt{b}} = \sqrt{m} + \sqrt{n}$ for some $m$ and $n$ in $F$ is that $a^2 - b$ is a square in $F$. Use this to determine when the field $\mathbb{Q}(\sqrt{a+\sqrt{b}})$ $(a, b \in \mathbb{Q})$ is biquadratic over $\mathbb{Q}$.

**10.** Determine the degree of the extension $\mathbb{Q}($ $+2\sqrt{2}$ over $\mathbb{Q}$.

ıe extension $\mathbb{Q}(\sqrt{3+2\sqrt{2}})$ over $\mathbb{Q}$.

**11.** **(a)** Let $\sqrt{3+4i}$ denote the square root of the complex number $3+4i$ that lies in the first quadrant and let $\sqrt{3-4i}$ denote the square root of $3-4i$ that lies in the fourth quadrant. Prove that $[\mathbb{Q}(\sqrt{3+4i}+\sqrt{3-4i}):\mathbb{Q}]=1$.

**(b)** Determine the degree of the extension $\mathbb{Q}(\sqrt{1+\sqrt{-3}}+\sqrt{1-\sqrt{-3}})$ over $\mathbb{Q}$.

**12.** Suppose the degree of the extension $K/F$ is a prime $p$. Show that any subfield $E$ of $K$ containing $F$ is either $K$ or $F$.

**13.** Suppose $F = \mathbb{Q}(\alpha_1, \alpha_2, \ldots, \alpha_n)$ where $\alpha_i^2 \in \mathbb{Q}$ for $i = 1, 2, \ldots, n$. Prove that $\sqrt[3]{2} \notin F$.

**14.** Prove that if $[F(\alpha) : F]$ is odd then $F(\alpha) = F(\alpha^2)$.

**15.** A field $F$ is said to be *formally real* if $-1$ is not expressible as a sum of squares in $F$. Let $F$ be a formally real field, let $f(x) \in F[x]$ be an irreducible polynomial of odd degree and let $\alpha$ be a root of $f(x)$. Prove that $F(\alpha)$ is also formally real. [Pick $\alpha$ a counterexample of minimal degree. Show that $-1 + f(x)g(x) = (p_1(x))^2 + \cdots + (p_m(x))^2$ for some $p_i(x), g(x) \in F[x]$ where $g(x)$ has odd degree $<$ deg $f$. Show that some root $\beta$ of $g$ has odd degree over $F$ and $F(\beta)$ is not formally real, violating the minimality of $\alpha$.]

**16.** Let $K/F$ be an algebraic extension and let $R$ be a *ring* contained in $K$ and containing $F$. Show that $R$ is a subfield of $K$ containing $F$.

**17.** Let $f(x)$ be an irreducible polynomial of degree $n$ over a field $F$. Let $g(x)$ be any polynomial in $F[x]$. Prove that every irreducible factor of the composite polynomial $f(g(x))$ has degree divisible by $n$.

**18.** Let $k$ be a field and let $k(x)$ be the field of rational functions in $x$ with coefficients from $k$. Let $t \in k(x)$ be the rational function $\dfrac{P(x)}{Q(x)}$ with relatively prime polynomials $P(x)$, $Q(x) \in k[x]$, with $Q(x) \neq 0$. Then $k(x)$ is an extension of $k(t)$ and to compute its degree it is necessary to compute the minimal polynomial with coefficients in $k(t)$ satisfied by $x$.

**(a)** Show that the polynomial $P(X) - tQ(X)$ in the variable $X$ and coefficients in $k(t)$ is irreducible over $k(t)$ and has $x$ as a root. [By Gauss' Lemma this polynomial is irreducible in $(k(t))[X]$ if and only if it is irreducible in $(k[t])[X]$. Then note that $(k[t])[X] = (k[X])[t]$.]

**(b)** Show that the degree of $P(X) - tQ(X)$ as a polynomial in $X$ with coefficients in $k(t)$ is the maximum of the degrees of $P(x)$ and $Q(x)$.

**(c)** Show that $[k(x) : k(t)] = [k(x) : k(\dfrac{P(x)}{Q(x)})] = \max(\deg P(x), \deg Q(x))$.

**19.** Let $K$ be an extension of $F$ of degree $n$.

**(a)** For any $\alpha \in K$ prove that $\alpha$ acting by left multiplication on $K$ is an $F$-linear transformation of $K$.

**(b)** Prove that $K$ is isomorphic to a subfield of the ring of $n \times n$ matrices over $F$, so the ring of $n \times n$ matrices over $F$ contains an isomorphic copy of *every* extension of $F$ of degree $\leq n$.

**20.** Show that if the matrix of the linear transformation "multiplication by $\alpha$" considered in the previous exercise is $A$ then $\alpha$ is a root of the characteristic polynomial for $A$. This gives an effective procedure for determining an equation of degree $n$ satisfied by an element $\alpha$ in an extension of $F$ of degree $n$. Use this procedure to obtain the monic polynomial of degree 3 satisfied by $\sqrt[3]{2}$ and by $1 + \sqrt[3]{2} + \sqrt[3]{4}$.

**21.** Let $K = \mathbb{Q}(\sqrt{D})$ for some squarefree integer $D$. Let $\alpha = a + b\sqrt{D}$ be an element of $K$. Use the basis $1, \sqrt{D}$ for $K$ as a vector space over $\mathbb{Q}$ and show that the matrix of the linear transformation "multiplication by $\alpha$" on $K$ considered in the previous exercises has the matrix $\begin{pmatrix} a & bD \\ b & a \end{pmatrix}$. Prove directly that the map $a + b\sqrt{D} \mapsto \begin{pmatrix} a & bD \\ b & a \end{pmatrix}$ is an isomorphism of the field $K$ with a subfield of the ring of $2 \times 2$ matrices with coefficients in $\mathbb{Q}$.

**22.** Let $K_1$ and $K_2$ be two finite extensions of a field $F$ contained in the field $K$. Prove that the $F$-algebra $K_1 \otimes_F K_2$ is a field if and only if $[K_1K_2 : F] = [K_1 : F][K_2 : F]$.

## 13.3. Classical Straightedge and Compass Constructions

**1.** Prove that it is impossible to construct the regular 9-gon.

**2.** Prove that Archimedes' construction actually trisects the angle $\theta$. [Note the isosceles triangles in Figure 5 to prove that $\beta = \gamma = 2\alpha$.]

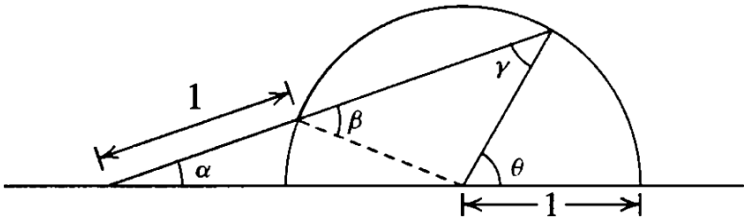



**Fig. 5**

**3.** Prove that Conway's construction indicated in the text actually constructs $2k^{1/3}$ and $2k^{2/3}$. [One method: let $(x, y)$ be the coordinates of the point $C$, $a$ the distance from $B$ to $C$ and $b$ the distance from $A$ to $D$; use similar triangles to prove (a) $\frac{y}{1} = \frac{\sqrt{1-k^2}}{1+a}$, (b) $\frac{x}{a} = \frac{b+k}{1+a}$, (c) $\frac{y}{x-k} = \frac{\sqrt{1-k^2}}{3k}$, and also show that (d) $(1-k^2)+(b+k)^2 = (1+a)^2$; solve these equations for $a$ and $b$.]

**4.** The construction of the regular 7-gon amounts to the constructibility of $\cos(2\pi/7)$. We shall see later (Section 14.5 and Exercise 2 of Section 14.7) that $\alpha = 2\cos(2\pi/7)$ satisfies the equation $x^3 + x^2 - 2x - 1 = 0$. Use this to prove that the regular 7-gon is not constructible by straightedge and compass.

**5.** Use the fact that $\alpha = 2\cos(2\pi/5)$ satisfies the equation $x^2 + x - 1 = 0$ to conclude that the regular 5-gon is constructible by straightedge and compass.

## 13.4. Splitting Fields and Algebraic Closures

**1.** Determine the splitting field and its degree over $\mathbb{Q}$ for $x^4 - 2$.

**2.** Determine the splitting field and its degree over $\mathbb{Q}$ for $x^4 + 2$.

**3.** Determine the splitting field and its degree over $\mathbb{Q}$ for $x^4 + x^2 + 1$.

Determine the sp

4. **Determine** the splitting field and its degree over $\mathbb{Q}$ for $x^6 - 4$.

Let $K$ be a finite

**5.** Let $K$ be a finite extension of $F$. Prove that $K$ is a splitting field over $F$ if and only if every irreducible polynomial in $F[x]$ that has a root in $K$ splits completely in $K[x]$. [Use Theorems 8 and 27.]

**6.** Let $K_1$ and $K_2$ be finite extensions of $F$ contained in the field $K$, and assume both are splitting fields over $F$.

**(a)** Prove that their composite $K_1K_2$ is a splitting field over $F$.

**(b)** Prove that $K_1 \cap K_2$ is a splitting field over $F$. [Use the preceding exercise.]

## 13.5. Separable and Inseparable Extensions

**1.** Prove that the derivative $D_x$ of a polynomial satisfies $D_x(f(x) + g(x)) = D_x(f(x)) + D_x(g(x))$ and $D_x(f(x)g(x)) = D_x(f(x))g(x) + D_x(g(x))f(x)$ for any two polynomials $f(x)$ and $g(x)$.

**2.** Find all irreducible polynomials of degrees 1, 2 and 4 over $\mathbb{F}_2$ and prove that their product is $x^{16} - x$.

**3.** Prove that $d$ divides $n$ if and only if $x^d - 1$ divides $x^n - 1$. [Note that if $n = qd + r$ then $x^n - 1 = (x^{qd+r} - x^r) + (x^r - 1)$.]

**4.** Let $a > 1$ be an integer. Prove for any positive integers $n, d$ that $d$ divides $n$ if and only if $a^d - 1$ divides $a^n - 1$ (cf. the previous exercise). Conclude in particular that $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n}$ if and only if $d$ divides $n$.

**5.** For any prime $p$ and any nonzero $a \in \mathbb{F}_p$ prove that $x^p - x + a$ is irreducible and separable over $\mathbb{F}_p$. [For the irreducibility: One approach — prove first that if $\alpha$ is a root then $\alpha + 1$ is also a root. Another approach — suppose it's reducible and compute derivatives.]

**6.** Prove that $x^{p^n-1}-1=\prod_{\alpha\in\mathbb{F}_{p^n}^\times}(x-\alpha)$. Conclude that $\prod_{\alpha\in\mathbb{F}_{p^n}^\times}\alpha=(-1)^{p^n}$ so the product of the nonzero elements of a finite field is $+1$ if $p=2$ and $-1$ if $p$ is odd. For $p$ odd and $n=1$ derive *Wilson's Theorem*: $(p-1)!\equiv -1 \pmod{p}$.

**7.** Suppose $K$ is a field of characteristic $p$ which is not a perfect field: $K \neq K^p$. Prove there exist irreducible inseparable polynomials over $K$. Conclude that there exist inseparable finite extensions of $K$.

**8.** Prove that $f(x)^p = f(x^p)$ for any polynomial $f(x) \in \mathbb{F}_p[x]$.

**9.** Show that the binomial coefficient $\binom{pn}{pi}$ is the coefficient of $x^{pi}$ in the expansion of $(1+x)^{pn}$. Working over $\mathbb{F}_p$ show that this is the coefficient of $(x^p)^i$ in $(1+x^p)^n$ and hence prove that $\binom{pn}{pi} \equiv \binom{n}{i} \pmod{p}$.

**10.** Let $f(x_1, x_2, \ldots, x_n) \in \mathbb{Z}[x_1, x_2, \ldots, x_n]$ be a polynomial in the variables $x_1, x_2, \ldots, x_n$ with integer coefficients. For any prime $p$ prove that the polynomial

$$f(x_1, x_2, \ldots, x_n)^p - f(x_1^p, x_2^p, \ldots, x_n^p) \in \mathbb{Z}[x_1, x_2, \ldots, x_n]$$

has all its coefficients divisible by $p$.

**11.** Suppose $K[x]$ is a polynomial ring over the field $K$ and $F$ is a subfield of $K$. If $F$ is a perfect field and $f(x) \in F[x]$ has no repeated irreducible factors in $F[x]$, prove that $f(x)$ has no repeated irreducible factors in $K[x]$.

## 13.6. Cyclotomic Polynomials and Extensions

**1.** Suppose $m$ and $n$ are relatively prime positive integers. Let $\zeta_m$ be a primitive $m^{\text{th}}$ root of unity and let $\zeta_n$ be a primitive $n^{\text{th}}$ root of unity. Prove that $\zeta_m \zeta_n$ is a primitive $mn^{\text{th}}$ root of unity.

**2.** Let $\zeta_n$ be a primitive $n^{\text{th}}$ root of unity and let $d$ be a divisor of $n$. Prove that $\zeta_n^d$ is a primitive $(n/d)^{\text{th}}$ root of unity.

**3.** Prove that if a field contains the $n^{\text{th}}$ roots of unity for $n$ odd then it also contains the $2n^{\text{th}}$ roots of unity.

**4.** Prove that if $n = p^k m$ where $p$ is a prime and $m$ is relatively prime to $p$ then there are precisely $m$ distinct $n^{\text{th}}$ roots of unity over a field of characteristic $p$.

**5.** Prove there are only a finite number of roots of unity in any finite extension $K$ of $\mathbb{Q}$.

**6.** Prove that for $n$ odd, $n > 1$, $\Phi_{2n}(x) = \Phi_n(-x)$.

**7.** Use the Möbius Inversion formula indicated in Section 14.3 to prove

$$\Phi_m(x) = \prod_{d|n} (x^d - 1)^{\mu(m/d)}.$$

**8.** Let $\ell$ be a prime and let $\Phi_\ell(x) = \frac{x^\ell - 1}{x-1} = x^{\ell-1} + x^{\ell-2} + \ldots + x + 1 \in \mathbb{Z}[x]$ be the $\ell^{\text{th}}$ cyclotomic polynomial, which is irreducible over $\mathbb{Z}$ by Theorem 41. This exercise determines the factorization of $\Phi_\ell(x)$ modulo $p$ for any prime $p$. Let $\zeta$ denote any fixed primitive $\ell^{\text{th}}$ root of unity.

**(a)** Show that if $p = l$ then $\Phi_\ell(x) = (x-1)^{\ell-1} \in \mathbb{F}_\ell[x]$.

**(b)** Suppose $p \neq \ell$ and let $f$ denote the order of $p$ mod $\ell$, i.e., $f$ is the smallest power of $p$ with $p^f \equiv 1 \bmod \ell$. Use the fact that $\mathbb{F}_{p^n}^\times$ is a cyclic group to show that $n = f$ is the smallest power $p^n$ of $p$ with $\zeta \in \mathbb{F}_{p^n}$. Conclude that the minimal polynomial of $\zeta$ over $\mathbb{F}_p$ has degree $f$.

**(c)** Show that $\mathbb{F}_p(\zeta) = \mathbb{F}_p(\zeta^a)$ for any integer $a$ not divisible by $\ell$. [One inclusion is obvious. For the other, note that $\zeta = (\zeta^a)^b$ where $b$ is the multiplicative inverse of $a$ mod $\ell$.] Conclude using (b) that, in $\mathbb{F}_p[x]$, $\Phi_\ell(x)$ is the product of $\frac{\ell-1}{f}$ distinct irreducible polynomials of degree $f$.

**(d)** In particular, prove that, viewed in $\mathbb{F}_p[x]$, $\Phi_7(x) = x^6 + x^5 + \ldots + x + 1$ is $(x-1)^6$ for $p = 7$, a product of distinct linear factors for $p \equiv 1 \bmod 7$, a product of 3 irreducible quadratics for $p \equiv 6 \bmod 7$, a product of 2 irreducible cubics for $p \equiv 2, 4 \bmod 7$, and is irreducible for $p \equiv 3, 5 \bmod 7$.

**9.** Suppose $A$ is an $n \times n$ matrix over $\mathbb{C}$ for which $A^k = I$ for some integer $k \geq 1$. Show that $A$ can be diagonalized. Show that the matrix $A = \begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$ where $\alpha$ is an element of a field of characteristic $p$ satisfies $A^p = I$ and cannot be diagonalized if $\alpha \neq 0$.

**10.** Let $\varphi$ denote the Frobenius map $x \mapsto x^p$ on the finite field $\mathbb{F}_{p^n}$. Prove that $\varphi$ gives an isomorphism of $\mathbb{F}_{p^n}$ to itself (such an isomorphism is called an *automorphism*). Prove that $\varphi^n$ is the identity map and that no lower power of $\varphi$ is the identity.

**11.** Let $\varphi$ denote the Frobenius map $x \mapsto x^p$ on the finite field $\mathbb{F}_{p^n}$ as in the previous exercise. Determine the rational canonical form over $\mathbb{F}_p$ for $\varphi$ considered as an $\mathbb{F}_p$-linear transformation of the $n$-dimensional $\mathbb{F}_p$-vector space $\mathbb{F}_{p^n}$.

**12.** Let $\varphi$ denote the Frobenius map $x \mapsto x^p$ on the finite field $\mathbb{F}_{p^n}$ as in the previous exercise. Determine the Jordan canonical form (over a field containing all the eigenvalues) for $\varphi$ considered as an $\mathbb{F}_p$-linear transformation of the $n$-dimensional $\mathbb{F}_p$-vector space $\mathbb{F}_{p^n}$.

**13.** (*Wedderburn's Theorem on Finite Division Rings*) This exercise outlines a proof (following Witt) of Wedderburn's Theorem that a finite division ring $D$ is a field (i.e., is commutative).

**(a)** Let $Z$ denote the center of $D$ (i.e., the elements of $D$ which commute with every element of $D$). Prove that $Z$ is a field containing $\mathbb{F}_p$ for some prime $p$. If $Z = \mathbb{F}_q$ prove that $D$ has order $q^n$ for some integer $n$ [$D$ is a vector space over $Z$].

**(b)** The nonzero elements $D^\times$ of $D$ form a multiplicative group. For any $x \in D^\times$ show that the elements of $D$ which commute with $x$ form a division ring which contains $Z$. Show that this division ring is of order $q^m$ for some integer $m$ and that $m < n$ if $x$ is not an element of $Z$.

**(c)** Show that the class equation (Theorem 4.7) for the group $D^\times$ is

$$q^n - 1 = (q-1) + \sum_{i=1}^{r} \frac{q^n - 1}{|C_{D^\times}(x_i)|}$$

where $x_1, x_2, \ldots, x_r$ are representatives of the distinct conjugacy classes in $D^\times$ not contained in the center of $D^\times$. Conclude from (b) that for each $i$, $|C_{D^\times}(x_i)| = q^{m_i} - 1$ for some $m_i < n$.

**(d)** Prove that since $\dfrac{q^n - 1}{q^{m_i} - 1}$ is an integer (namely, the index $|D^\times : C_{D^\times}(x_i)|$ ) then $m_i$ divides $n$ (cf. Exercise 4 of Section 5). Conclude that $\Phi_n(x)$ divides $(x^n - 1)/(x^{m_i} - 1)$ and hence that the integer $\Phi_n(q)$ divides $(q^n - 1)/(q^{m_i} - 1)$ for $i = 1, 2, \ldots, r$.

**(e)** Prove that (c) and (d) imply that $\Phi_n(q) = \prod_{\zeta \text{ primitive}} (q - \zeta)$ divides $q - 1$. Prove that $|q - \zeta| > q - 1$ (complex absolute value) for any root of unity $\zeta \neq 1$ [note that 1 is the closest point on the unit circle in $\mathbb{C}$ to the point $q$ on the real line]. Conclude that $n = 1$, i.e., that $D = Z$ is a field.

The following exercises provide a proof that for any positive integer $m$ there are infinitely many primes $p$ with $p \equiv 1 \pmod{m}$. This is a special case of *Dirichlet's Theorem on Primes in Arithmetic Progressions* which states more generally that there are infinitely many primes $p$ with $p \equiv a \pmod{m}$ for any $a$ relatively prime to $m$.

**14.** Given any monic polynomial $P(x) \in \mathbb{Z}[x]$ of degree at least one show that there are infinitely many distinct prime divisors of the integers

$$P(1), P(2), P(3), \ldots, P(n), \ldots.$$

[Suppose $p_1, p_2, \ldots, p_k$ are the only primes dividing the values $P(n)$, $n = 1, 2, \ldots$. Let $N$ be an integer with $P(N) = a \neq 0$. Show that $Q(x) = a^{-1}P(N + a\, p_1 p_2 \ldots p_k\, x)$ is an element of $\mathbb{Z}[x]$ and that $Q(n) \equiv 1 \pmod{p_1 p_2 \ldots p_k}$ for $n = 1, 2, \ldots$. Conclude that there is some integer $M$ such that $Q(M)$ has a prime factor different from $p_1, p_2, \ldots, p_k$ and hence that $P(N + ap_1 p_2 \cdots p_k M)$ has a prime factor different from $p_1, p_2, \ldots, p_k$.]

The following exercises provide a proof that for any positive integer $m$ there are infinitely many primes $p$ with $p \equiv 1 \pmod{m}$. This is a special case of *Dirichlet's Theorem on Primes in Arithmetic Progressions* which states more generally that there are infinitely many primes $p$ with $p \equiv a \pmod{m}$ for any $a$ relatively prime to $m$.

**15.** Let $p$ be an odd prime not dividing $m$ and let $\Phi_m(x)$ be the $m^{\text{th}}$ cyclotomic polynomial. Suppose $a \in \mathbb{Z}$ satisfies $\Phi_m(a) \equiv 0 \pmod{p}$. Prove that $a$ is relatively prime to $p$ and that the order of $a$ in $(\mathbb{Z}/p\mathbb{Z})^\times$ is precisely $m$. [Since

$$x^m - 1 = \prod_{d|m} \Phi_d(x) = \Phi_m(x) \prod_{\substack{d|m \\ d<m}} \Phi_d(x)$$

we see first that $a^m - 1 \equiv 0 \pmod{p}$ i.e., $a^m \equiv 1 \pmod{p}$. If the order of $a$ mod $p$ were less than $m$, then $a^d \equiv 1 \pmod{p}$ for some $d$ dividing $m$, so then $\Phi_d(a) \equiv 0 \pmod{p}$ for some $d < m$. But then $x^m - 1$ would have $a$ as a multiple root mod $p$, a contradiction.]

The following exercises provide a proof that for any positive integer $m$ there are infinitely many primes $p$ with $p \equiv 1 \pmod{m}$. This is a special case of *Dirichlet's Theorem on Primes in Arithmetic Progressions* which states more generally that there are infinitely many primes $p$ with $p \equiv a \pmod{m}$ for any $a$ relatively prime to $m$.

**16.** Let $a \in \mathbb{Z}$. Show that if $p$ is an odd prime dividing $\Phi_m(a)$ then either $p$ divides $m$ or $p \equiv 1 \pmod{m}$.

The following exercises provide a proof that for any positive integer $m$ there are infinitely many primes $p$ with $p \equiv 1 \pmod{m}$. This is a special case of *Dirichlet's Theorem on Primes in Arithmetic Progressions* which states more generally that there are infinitely many primes $p$ with $p \equiv a \pmod{m}$ for any $a$ relatively prime to $m$.

**17.** Prove there are infinitely many primes $p$ with $p \equiv 1 \pmod{m}$.

# 14. Galois Theory

## 14.1. Basic Definitions

**1.** **(a)** Show that if the field $K$ is generated over $F$ by the elements $\alpha_1, \ldots, \alpha_n$ then an automorphism $\sigma$ of $K$ fixing $F$ is uniquely determined by $\sigma(\alpha_1), \ldots, \sigma(\alpha_n)$. In particular show that an automorphism fixes $K$ if and only if it fixes a set of generators for $K$.

**(b)** Let $G \leq \mathrm{Gal}(K/F)$ be a subgroup of the Galois group of the extension $K/F$ and suppose $\sigma_1, \ldots, \sigma_k$ are generators for $G$. Show that the subfield $E/F$ is fixed by $G$ if and only if it is fixed by the generators $\sigma_1, \ldots, \sigma_k$.

**2.** Let $\tau$ be the map $\tau : \mathbb{C} \to \mathbb{C}$ defined by $\tau(a + bi) = a - bi$ (*complex conjugation*). Prove that $\tau$ is an automorphism of $\mathbb{C}$.

**3.** Determine the fixed field of complex conjugation on $\mathbb{C}$.

**4.** Prove that $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{3})$ are not isomorphic.

**5.** Determine the automorphisms of the extension $\mathbb{Q}(\sqrt[4]{2})/\mathbb{Q}(\sqrt{2})$ explicitly.

**6.** Let $k$ be a field.

**(a)** Show that the mapping $\varphi : k[t] \to k[t]$ defined by $\varphi(f(t)) = f(at + b)$ for fixed $a, b \in k$, $a \neq 0$ is an automorphism of $k[t]$ which is the identity on $k$.

**(b)** Conversely, let $\varphi$ be an automorphism of $k[t]$ which is the identity on $k$. Prove that there exist $a, b \in k$ with $a \neq 0$ such that $\varphi(f(t)) = f(at + b)$ as in (a).

**7.** This exercise determines $\mathrm{Aut}(\mathbb{R}/\mathbb{Q})$.

**(a)** Prove that any $\sigma \in \mathrm{Aut}(\mathbb{R}/\mathbb{Q})$ takes squares to squares and takes positive reals to positive reals. Conclude that $a < b$ implies $\sigma a < \sigma b$ for every $a, b \in \mathbb{R}$.

**(b)** Prove that $-\frac{1}{m} < a-b < \frac{1}{m}$ implies $-\frac{1}{m} < \sigma a - \sigma b < \frac{1}{m}$ for every positive integer $m$. Conclude that $\sigma$ is a continuous map on $\mathbb{R}$.

**(c)** Prove that any continuous map on $\mathbb{R}$ which is the identity on $\mathbb{Q}$ is the identity map, hence $\mathrm{Aut}(\mathbb{R}/\mathbb{Q}) = 1$.

**8.** Prove that the automorphisms of the rational function field $k(t)$ which fix $k$ are precisely the *fractional linear transformations* determined by $t \mapsto \dfrac{at+b}{ct+d}$ for $a, b, c, d \in k, ad-bc \neq 0$ (so $f(t) \in k(t)$ maps to $f(\dfrac{at+b}{ct+d})$ ) (cf. Exercise 18 of Section 13.2).

**9.** Determine the fixed field of the automorphism $t \mapsto t + 1$ of $k(t)$.

**10.** Let $K$ be an extension of the field $F$. Let $\varphi : K \to K'$ be an isomorphism of $K$ with a field $K'$ which maps $F$ to the subfield $F'$ of $K'$. Prove that the map $\sigma \mapsto \varphi\sigma\varphi^{-1}$ defines a group isomorphism $\mathrm{Aut}(K/F) \xrightarrow{\sim} \mathrm{Aut}(K'/F')$.

## 14.2. The Fundamental Theorem of Galois Theory

**1.** Determine the minimal polynomial over $\mathbb{Q}$ for the element $\sqrt{2} + \sqrt{5}$.

**2.** Determine the minimal polynomial over $\mathbb{Q}$ for the element $1 + \sqrt[3]{2} + \sqrt[3]{4}$.

**3.** Determine the Galois group of $(x^2 - 2)(x^2 - 3)(x^2 - 5)$. Determine *all* the subfields of the splitting field of this polynomial.

**4.** Let $p$ be a prime. Determine the elements of the Galois group of $x^p - 2$.

**5.** Prove that the Galois group of $x^p - 2$ for $p$ a prime is isomorphic to the group of matrices $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$ where $a, b \in \mathbb{F}_p$, $a \neq 0$.

**6.** Let $K = \mathbb{Q}(\sqrt[8]{2}, i)$ and let $F_1 = \mathbb{Q}(i)$, $F_2 = \mathbb{Q}(\sqrt{2})$, $F_3 = \mathbb{Q}(\sqrt{-2})$. Prove that $\mathrm{Gal}(K/F_1) \cong Z_8$, $\mathrm{Gal}(K/F_2) \cong D_8$, $\mathrm{Gal}(K/F_3) \cong Q_8$.

**7. Determine all the subfields of the splitting field of $x^8 - 2$ which are Galois over $\mathbb{Q}$.**

**8.** Suppose $K$ is a Galois extension of $F$ of degree $p^n$ for some prime $p$ and some $n \geq 1$. Show there are Galois extensions of $F$ contained in $K$ of degrees $p$ and $p^{n-1}$.

**9.** Give an example of fields $F_1$, $F_2$, $F_3$ with $\mathbb{Q} \subset F_1 \subset F_2 \subset F_3$, $[F_3 : \mathbb{Q}] = 8$ and each field is Galois over all its subfields with the exception that $F_2$ is not Galois over $\mathbb{Q}$.

**10.** Determine the Galois group of the splitting field over $\mathbb{Q}$ of $x^8 - 3$.

**11.** Suppose $f(x) \in \mathbb{Z}[x]$ is an irreducible quartic whose splitting field has Galois group $S_4$ over $\mathbb{Q}$ (there are many such quartics, cf. Section 6). Let $\theta$ be a root of $f(x)$ and set $K = \mathbb{Q}(\theta)$. Prove that $K$ is an extension of $\mathbb{Q}$ of degree 4 which has no proper subfields. Are there any Galois extensions of $\mathbb{Q}$ of degree 4 with no proper subfields?

**12.** Determine the Galois group of the splitting field over $\mathbb{Q}$ of $x^4 - 14x^2 + 9$.

**13.** Prove that if the Galois group of the splitting field of a cubic over $\mathbb{Q}$ is the cyclic group of order 3 then all the roots of the cubic are real.

**14.** Show that $\mathbb{Q}(\sqrt{2+\sqrt{2}})$ is a cyclic quartic field, i.e., is a Galois extension of degree 4 with cyclic Galois group.

**15.** (*Biquadratic Extensions*) Let $F$ be a field of characteristic $\neq 2$.

**(a)** If $K = F(\sqrt{D_1}, \sqrt{D_2})$ where $D_1, D_2 \in F$ have the property that none of $D_1$, $D_2$ or $D_1 D_2$ is a square in $F$, prove that $K/F$ is a Galois extension with $\mathrm{Gal}(K/F)$ isomorphic to the Klein 4-group.

**(b)** Conversely, suppose $K/F$ is a Galois extension with $\mathrm{Gal}(K/F)$ isomorphic to the Klein 4-group. Prove that $K = F(\sqrt{D_1}, \sqrt{D_2})$ where $D_1, D_2 \in F$ have the property that none of $D_1$, $D_2$ or $D_1 D_2$ is a square in $F$.

**16.** **(a)** Prove that $x^4 - 2x^2 - 2$ is irreducible over $\mathbb{Q}$.

**(b)** Show the roots of this quartic are

$$\alpha_1 = \sqrt{1+\sqrt{3}} \qquad \alpha_3 = -\sqrt{1+\sqrt{3}}$$
$$\alpha_2 = \sqrt{1-\sqrt{3}} \qquad \alpha_4 = -\sqrt{1-\sqrt{3}}.$$

**(c)** Let $K_1 = \mathbb{Q}(\alpha_1)$ and $K_2 = \mathbb{Q}(\alpha_2)$. Show that $K_1 \neq K_2$, and $K_1 \cap K_2 = \mathbb{Q}(\sqrt{3}) = F$.

**(d)** Prove that $K_1$, $K_2$ and $K_1K_2$ are Galois over $F$ with $\mathrm{Gal}(K_1K_2/F)$ the Klein 4-group. Write out the elements of $\mathrm{Gal}(K_1K_2/F)$ explicitly. Determine all the subgroups of the Galois group and give their corresponding fixed subfields of $K_1K_2$ containing $F$.

**(e)** Prove that the splitting field of $x^4 - 2x^2 - 2$ over $\mathbb{Q}$ is of degree 8 with dihedral Galois group.

The following two exercises indicate one method for constructing elements in subfields of a given field and are quite useful in many computations.

**17.** Let $K/F$ be any finite extension and let $\alpha \in K$. Let $L$ be a Galois extension of $F$ containing $K$ and let $H \leq \mathrm{Gal}(L/F)$ be the subgroup corresponding to $K$. Define the *norm* of $\alpha$ from $K$ to $F$ to be

$$\mathrm{N}_{K/F}(\alpha) = \prod_{\sigma} \sigma(\alpha),$$

where the product is taken over all the embeddings of $K$ into an algebraic closure of $F$ (so over a set of coset representatives for $H$ in $\mathrm{Gal}(L/F)$ by the Fundamental Theorem of Galois Theory). This is a product of Galois conjugates of $\alpha$. In particular, if $K/F$ is Galois this is $\prod_{\sigma \in \mathrm{Gal}(K/F)} \sigma(\alpha)$.

**(a)** Prove that $\mathrm{N}_{K/F}(\alpha) \in F$.

**(b)** Prove that $\mathrm{N}_{K/F}(\alpha\beta) = \mathrm{N}_{K/F}(\alpha)\mathrm{N}_{K/F}(\beta)$, so that the norm is a multiplicative map from $K$ to $F$.

**(c)** Let $K = F(\sqrt{D})$ be a quadratic extension of $F$. Show that $\mathrm{N}_{K/F}(a + b\sqrt{D}) = a^2 - Db^2$.

**(d)** Let $m_\alpha(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0 \in F[x]$ be the minimal polynomial for $\alpha \in K$ over $F$. Let $n = [K : F]$. Prove that $d$ divides $n$, that there are $d$ distinct Galois conjugates of $\alpha$ which are all repeated $n/d$ times in the product above and conclude that $\mathrm{N}_{K/F}(\alpha) = (-1)^n a_0^{n/d}$.

The following two exercises indicate one method for constructing elements in subfields of a given field and are quite useful in many computations.

**18.** With notation as in the previous problem, define the *trace* of $\alpha$ from $K$ to $F$ to be

$$\mathrm{Tr}_{K/F}(\alpha) = \sum_{\sigma} \sigma(\alpha),$$

a sum of Galois conjugates of $\alpha$.

**(a)** Prove that $\mathrm{Tr}_{K/F}(\alpha) \in F$.

**(b)** Prove that $\mathrm{Tr}_{K/F}(\alpha+\beta) = \mathrm{Tr}_{K/F}(\alpha) + \mathrm{Tr}_{K/F}(\beta)$, so that the trace is an additive map from $K$ to $F$.

**(c)** Let $K = F(\sqrt{D})$ be a quadratic extension of $F$. Show that $\mathrm{Tr}_{K/F}(a + b\sqrt{D}) = 2a$.

**(d)** Let $m_\alpha(x)$ be as in the previous problem. Prove that $\mathrm{Tr}_{K/F}(\alpha) = -\frac{n}{d} a_{d-1}$.

**19.** With notation as in the previous problems show that $\mathrm{N}_{K/F}(a\alpha) = a^n \mathrm{N}_{K/F}(\alpha)$ and $\mathrm{Tr}_{K/F}(a\alpha) = a\mathrm{Tr}_{K/F}(\alpha)$ for all $a$ in the base field $F$. In particular show that $\mathrm{N}_{K/F}(a) = a^n$ and $\mathrm{Tr}_{K/F}(a) = na$ for all $a \in F$.

**20.** With notation as in the previous problems show more generally that $\prod_\sigma (x - \sigma(\alpha)) = (m_\alpha(x))^{n/d}$.

**21.** Use the linear independence of characters to show that for any Galois extension $K$ of $F$ there is an element $\alpha \in K$ with $\mathrm{Tr}_{K/F}(\alpha) \neq 0$.

**22.** Suppose $K/F$ is a Galois extension and let $\sigma$ be an element of the Galois group.

**(a)** Suppose $\alpha \in K$ is of the form $\alpha = \dfrac{\beta}{\sigma\beta}$ for some nonzero $\beta \in K$. Prove that $\mathrm{N}_{K/F}(\alpha) = 1$.

**(b)** Suppose $\alpha \in K$ is of the form $\alpha = \beta - \sigma\beta$ for some $\beta \in K$. Prove that $\mathrm{Tr}_{K/F}(\alpha) = 0$.

The next exercise and Exercise 26 following establish the multiplicative and additive forms of Hilbert's Theorem 90. These are instances of the vanishing of a first cohomology group, as will be discussed in Section 17.3.

**23.** (*Hilbert's Theorem 90*) Let $K$ be a Galois extension of $F$ with cyclic Galois group of order $n$ generated by $\sigma$. Suppose $\alpha \in K$ has $N_{K/F}(\alpha) = 1$. Prove that $\alpha$ is of the form $\alpha = \dfrac{\beta}{\sigma\beta}$ for some nonzero $\beta \in K$. [By the linear independence of characters show there exists some $\theta \in K$ such that

$$\beta = \theta + \alpha\sigma(\theta) + (\alpha\,\sigma\alpha)\sigma^2(\theta) + \cdots + (\alpha\,\sigma\alpha \ldots \sigma^{n-2}\alpha)\sigma^{n-1}(\theta)$$

is nonzero. Compute $\dfrac{\beta}{\sigma\beta}$ using the fact that $\alpha$ has norm 1 to $F$.]

The next exercise and Exercise 26 following establish the multiplicative and additive forms of Hilbert's Theorem 90. These are instances of the vanishing of a first cohomology group, as will be discussed in Section 17.3.

**24.** Prove that the rational solutions $a, b \in \mathbb{Q}$ of Pythagoras' equation $a^2 + b^2 = 1$ are of the form $a = \dfrac{s^2 - t^2}{s^2 + t^2}$ and $b = \dfrac{2st}{s^2 + t^2}$ for some $s, t \in \mathbb{Q}$ and hence show that any right triangle with integer sides has sides of lengths $(m^2 - n^2, 2mn, m^2 + n^2)$ for some integers $m, n$. [Note that $a^2 + b^2 = 1$ is equivalent to $\mathrm{N}_{\mathbb{Q}(i)/\mathbb{Q}}(a + ib) = 1$, then use Hilbert's Theorem 90 above with $\beta = s + it$.]

The next exercise and Exercise 26 following establish the multiplicative and additive forms of Hilbert's Theorem 90. These are instances of the vanishing of a first cohomology group, as will be discussed in Section 17.3.

**25.** Generalize the previous problem to determine all the rational solutions of the equation $a^2 + Db^2 = 1$ for $D \in \mathbb{Z}$, $D > 0$, $D$ not a perfect square in $\mathbb{Z}$.

The next exercise and Exercise 26 following establish the multiplicative and additive forms of Hilbert's Theorem 90. These are instances of the vanishing of a first cohomology group, as will be discussed in Section 17.3.

**26.** (*Additive Hilbert's Theorem 90*) Let $K$ be a Galois extension of $F$ with cyclic Galois group of order $n$ generated by $\sigma$. Suppose $\alpha \in K$ has $\mathrm{Tr}_{K/F}(\alpha) = 0$. Prove that $\alpha$ is of the form $\alpha = \beta - \sigma\beta$ for some $\beta \in K$. [Let $\theta \in K$ be an element with $\mathrm{Tr}_{K/F}(\theta) \neq 0$ by a previous exercise, let

$$\beta = \frac{1}{\mathrm{Tr}_{K/F}(\theta)}[\alpha\sigma(\theta) + (\alpha + \sigma\alpha)\sigma^2(\theta) + \cdots + (\alpha + \sigma\alpha + \cdots + \sigma^{n-2}\alpha)\sigma^{n-1}(\theta)]$$

and compute $\beta - \sigma\beta$.]

**27.** Let $\alpha = \sqrt{(2+\sqrt{2})(3+\sqrt{3})}$ (positive real square roots for concreteness) and consider the extension $E = \mathbb{Q}(\alpha)$.

**(a)** Show that $a = (2+\sqrt{2})(3+\sqrt{3})$ is not a square in $F = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. [If $a = c^2$, $c \in F$, then $a\varphi(a) = (2+\sqrt{2})^2(6) = (c\varphi c)^2$ for the automorphism $\varphi \in \mathrm{Gal}(F/\mathbb{Q})$ fixing $\mathbb{Q}(\sqrt{2})$. Since $c\varphi c = \mathrm{N}_{F/\mathbb{Q}(\sqrt{2})}(c) \in \mathbb{Q}(\sqrt{2})$ conclude that this implies $\sqrt{6} \in \mathbb{Q}(\sqrt{2})$, a contradiction.]

**(b)** Conclude from (a) that $[E : \mathbb{Q}] = 8$. Prove that the roots of the minimal polynomial over $\mathbb{Q}$ for $\alpha$ are the 8 elements $\pm\sqrt{(2 \pm \sqrt{2})(3 \pm \sqrt{3})}$.

**(c)** Let $\beta = \sqrt{(2-\sqrt{2})(3+\sqrt{3})}$. Show that $\alpha\beta = \sqrt{2}(3+\sqrt{3}) \in F$ so that $\beta \in E$. Show similarly that the other roots are also elements of $E$ so that $E$ is a Galois extension of $\mathbb{Q}$. Show that the elements of the Galois group are precisely the maps determined by mapping $\alpha$ to one of the eight elements in (b).

**(d)** Let $\sigma \in \mathrm{Gal}(E/\mathbb{Q})$ be the automorphism which maps $\alpha$ to $\beta$. Show that since $\sigma(\alpha^2) = \beta^2$ that $\sigma(\sqrt{2}) = -\sqrt{2}$ and $\sigma(\sqrt{3}) = \sqrt{3}$. From $\alpha\beta = \sqrt{2}(3+\sqrt{3})$ conclude that $\sigma(\alpha\beta) = -\alpha\beta$ and hence $\sigma(\beta) = -\alpha$. Show that $\sigma$ is an element of order 4 in $\mathrm{Gal}(E/\mathbb{Q})$.

**(e)** Show similarly that the map $\tau$ defined by $\tau(\alpha) = \sqrt{(2+\sqrt{2})(3-\sqrt{3})}$ is an element of order 4 in $\mathrm{Gal}(E/\mathbb{Q})$. Prove that $\sigma$ and $\tau$ generate the Galois group, $\sigma^4 = \tau^4 = 1$, $\sigma^2 = \tau^2$ and that $\sigma\tau = \tau\sigma^3$.

**(f)** Conclude that $\mathrm{Gal}(E/\mathbb{Q}) \cong Q_8$, the quaternion group of order 8.

**28.** Let $f(x) \in F[x]$ be an irreducible polynomial of degree $n$ over the field $F$, let $L$ be the splitting field of $f(x)$ over $F$ and let $\alpha$ be a root of $f(x)$ in $L$. If $K$ is any Galois extension of $F$ contained in $L$, show that the polynomial $f(x)$ splits into a product of $m$ irreducible polynomials each of degree $d$ over $K$, where $m = [F(\alpha) \cap K : F]$ and $d = [K(\alpha) : K]$ (cf. also the generalization in Exercise 4 of Section 4). [If $H$ is the subgroup of the Galois group of $L$ over $F$ corresponding to $K$ then the factors of $f(x)$ over $K$ correspond to the orbits of $H$ on the roots of $f(x)$. Then use Exercise 9 of Section 4.1.]

**29.** Let $k$ be a field and let $k(t)$ be the field of rational functions in the variable $t$. Define the maps $\sigma$ and $\tau$ of $k(t)$ to itself by $\sigma f(t) = f(\frac{1}{1-t})$ and $\tau f(t) = f(\frac{1}{t})$ for $f(t) \in k(t)$.

**(a)** Prove that $\sigma$ and $\tau$ are automorphisms of $k(t)$ (cf. Exercise 8 of Section 1) and that the group $G = \langle \sigma, \tau \rangle$ they generate is isomorphic to $S_3$.

**(b)** Prove that the element $t = \dfrac{(t^2 - t + 1)^3}{t^2(t-1)^2}$ is fixed by all the elements of $G$.

**(c)** Prove that $k(t)$ is precisely the fixed field of $G$ in $k(t)$ [compute the degree of the extension].

**30.** Prove that the fixed field of the subgroup of automorphisms generated by $\tau$ in the previous problem is $k(t + \frac{1}{t})$. Prove that the fixed field of the subgroup generated by the automorphism $\tau\sigma^2$ (which maps $t$ to $1 - t$) is $k(t(1 - t))$. Determine the fixed field of the subgroup generated by $\tau\sigma$ and the fixed field of the subgroup generated by $\sigma$.

**31.** Let $K$ be a finite extension of $F$ of degree $n$. Let $\alpha$ be an element of $K$.

**(a)** Prove that $\alpha$ acting by left multiplication on $K$ is an $F$-linear transformation $T_\alpha$ of $K$.

**(b)** Prove that the minimal polynomial for $\alpha$ over $F$ is the same as the minimal polynomial for the linear transformation $T_\alpha$.

**(c)** Prove that the trace $\mathrm{Tr}_{K/F}(\alpha)$ is the trace of the $n \times n$ matrix defined by $T_\alpha$ (which justifies these two uses of the same word "trace"). Prove that the norm $\mathrm{N}_{K/F}(\alpha)$ is the determinant of $T_\alpha$.

## 14.3. Finite Fields

**1.** Factor $x^8 - x$ into irreducibles in $\mathbb{Z}[x]$ and in $\mathbb{F}_2[x]$.

**2.** Write out the multiplication table for $\mathbb{F}_4$ and $\mathbb{F}_8$.

**3.** Prove that an algebraically closed field must be infinite.

**4.** Construct the finite field of 16 elements and find a generator for the multiplicative group. How many generators are there?

**5.** Exhibit an explicit isomorphism between the splitting fields of $x^3 - x + 1$ and $x^3 - x - 1$ over $\mathbb{F}_3$.

**6.** Suppose $K = \mathbb{Q}(\theta) = \mathbb{Q}(\sqrt{D_1}\ , \sqrt{D_2}\ )$ with $D_1, D_2 \in \mathbb{Z}$, is a biquadratic extension and that $\theta = a + b\sqrt{D_1} + c\sqrt{D_2} + d\sqrt{D_1 D_2}$ where $a, b, c, d \in \mathbb{Z}$ are integers. Prove that the minimal polynomial $m_\theta(x)$ for $\theta$ over $\mathbb{Q}$ is irreducible of degree 4 over $\mathbb{Q}$ but is reducible modulo every prime $p$. In particular show that the polynomial $x^4 - 10x^2 + 1$ is irreducible in $\mathbb{Z}[x]$ but is reducible modulo every prime. [Use the fact that there are no biquadratic extensions over finite fields.]

**7.** Prove that one of $2, 3$ or $6$ is a square in $\mathbb{F}_p$ for every prime $p$. Conclude that the polynomial

$$x^6 - 11x^4 + 36x^2 - 36 = (x^2 - 2)(x^2 - 3)(x^2 - 6)$$

has a root modulo $p$ for every prime $p$ but has no root in $\mathbb{Z}$.

**8.** Determine the splitting field of the polynomial $x^p - x - a$ over $\mathbb{F}_p$ where $a \neq 0$, $a \in \mathbb{F}_p$. Show explicitly that the Galois group is cyclic. [Show $\alpha \mapsto \alpha + 1$ is an automorphism.] Such an extension is called an *Artin–Schreier extension* (cf. Exercise 9 of Section 7).

**9.** Let $q = p^m$ be a power of the prime $p$ and let $\mathbb{F}_q = \mathbb{F}_{p^m}$ be the finite field with $q$ elements. Let $\sigma_q = \sigma_p^m$ be the $m^{\text{th}}$ power of the Frobenius automorphism $\sigma_p$, called the $q$-Frobenius automorphism.

**(a)** Prove that $\sigma_q$ fixes $\mathbb{F}_q$.

**(b)** Prove that every finite extension of $\mathbb{F}_q$ of degree $n$ is the splitting field of $x^{q^n} - x$ over $\mathbb{F}_q$, hence is unique.

**(c)** Prove that every finite extension of $\mathbb{F}_q$ of degree $n$ is cyclic with $\sigma_q$ as generator.

**(d)** Prove that the subfields of the unique extension of $\mathbb{F}_q$ of degree $n$ are in bijective correspondence with the divisors $d$ of $n$.

**10.** Prove that $n$ divides $\varphi(p^n - 1)$. [Observe that $\varphi(p^n - 1)$ is the order of the group of automorphisms of a cyclic group of order $p^n - 1$.]

**11.** Prove that $x^{p^n} - x + 1$ is irreducible over $\mathbb{F}_p$ only when $n = 1$ or $n = p = 2$. [Note that if $\alpha$ is a root, then so is $\alpha + a$ for any $a \in \mathbb{F}_{p^n}$. Show that this implies $\mathbb{F}_p(\alpha)$ contains $\mathbb{F}_{p^n}$ and that $[\mathbb{F}_p(\alpha) : \mathbb{F}_{p^n}] = p$.]

(*Berlekamp's Factorization Algorithm*) The following exercises outline the Berlekamp factorization algorithm for factoring polynomials in $\mathbb{F}_p[x]$. The efficiency of this algorithm is based on the efficiency of computing greatest common divisors in $\mathbb{F}_p[x]$ by the Euclidean Algorithm and on the efficiency of row-reduction matrix algorithms for solving systems of linear equations.

Let $f(x) \in \mathbb{F}_p[x]$ be a monic polynomial of degree $n$ and let $f(x) = p_1(x)p_2(x)\dots p_k(x)$ where $p_1(x), p_2(x), \dots, p_k(x)$ are powers of distinct monic irreducibles in $\mathbb{F}_p[x]$.

**12.** Show that in order to write $f(x)$ as a product of irreducible polynomials in $\mathbb{F}_p[x]$ it suffices to determine the factors $p_1(x), \ldots, p_k(x)$. [If $p(x) = q(x)^N \in \mathbb{F}_p[x]$ with $q(x)$ monic and irreducible, show that $q(x)$ can be determined from $p(x)$ by checking for $p^{\text{th}}$ powers and by computing greatest common divisors with derivatives.]

**13.** Let $g(x) \in \mathbb{F}_p[x]$ be any polynomial of degree $< n$. Denote by $R(h(x))$ the remainder of $h(x)$ after division by $f(x)$. Prove the following are equivalent:

**(a)** $R(g(x^p)) = g(x)$.

**(b)** $f(x)$ divides $[g(x)-0][g(x)-1]\ldots[g(x)-(p-1)]$. [Use the fact that $g(x^p) = g(x)^p$ together with the factorization of $x^p - x$ in $\mathbb{F}_p[x]$.]

**(c)** $p_i(x)$ divides the product in (b) for $i = 1, 2, \ldots, k$.

**(d)** For each $i$, $i = 1, 2, \ldots, k$ there is an $s_i \in \mathbb{F}_p$ such that $p_i(x)$ divides $g(x) - s_i$, i.e., $g(x) \equiv s_i \pmod{p_i(x)}$.

**14.** Prove that the polynomials $g(x)$ of degree $< n$ satisfying the equivalent conditions of the previous exercise form a vector space $V$ over $\mathbb{F}_p$ of dimension $k$. [Use the Chinese Remainder Theorem applied to the $p^k$ possible choices for the $s_i$ in 13(d)].

**15.** Let $g(x) = b_0 + b_1x + \cdots + b_{n-1}x^{n-1} \in V$. For $j = 0, 1, \ldots, n-1$ let

$$R(x^{pj}) = a_{0,j} + a_{1,j}x + \cdots + a_{n-1,j}x^{n-1}$$

and let $A$ be the $n \times n$ matrix

$$A = \begin{pmatrix} a_{0,0} & a_{0,1} & \cdots & a_{0.n-1} \\ a_{1,0} & a_{1,1} & \cdots & a_{1.n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & a_{n-1,1} & \cdots & a_{n-1,n-1} \end{pmatrix}. \qquad (*)$$

Show that condition (a) of Exercise 13 for $g(x) \in V$ is equivalent to

$$(A - I)B = 0 \qquad (**)$$

where $B$ is the column matrix with entries $b_0, b_1, \ldots, b_{n-1}$. Conclude that the rank of the matrix $A - I$ is $n - k$. Note that this already suffices to determine if $f(x)$ is irreducible, without actually determining the factors.

**16.** Let $g_1(x), g_2(x), \ldots, g_k(x)$ be a basis of solutions to $(**)$ (so a basis for $V$), where we may take $g_1(x) = 1$. Beginning with $w(x) = f(x)$, compute the greatest common divisor $(w(x), g_i(x) - s)$ for $i = 2, 3, \ldots, k$ and $s \in \mathbb{F}_p$ for every factor of $f(x)$ already computed. Note by Exercise 13(d) that every factor $p_i(x)$ of $f(x)$ divides such a g.c.d. The process terminates when $k$ relatively prime factors have been determined.

Prove that this procedure actually gives all the factors $p_1(x), p_2(x), \ldots, p_k(x)$, i.e., one can separate the individual factors $p_1(x), p_2(x), \ldots, p_k(x)$ by this procedure, as follows:

If this were not the case, then for two of the factors, say $p_1(x)$ and $p_2(x)$, for each $i = 1, 2, \ldots, k$ there would exist $s_i \in \mathbb{F}_p$ such that $g_i(x) - s_i$ is divisible by both $p_1(x)$ and $p_2(x)$. By the Chinese Remainder Theorem, choose a $g(x) \in V$ satisfying $g(x) \equiv 0 \pmod{p_1(x)}$ and $g(x) \equiv 1 \pmod{p_2(x)}$. Write $g(x) = \sum_{i=1}^{k} c_i g_i(x)$ in terms of the basis for $V$ and let $s = \sum_{i=1}^{k} c_i s_i(x) \in \mathbb{F}_p$. Show that $s \equiv 0 \pmod{p_1(x)}$ so that $s = 0$ and $s \equiv 1 \pmod{p_2(x)}$ so that $s = 1$, a contradiction.

**17.** This exercise follows Berlekamp's Factorization Algorithm outlined in the previous exercises to determine the factorization of $f(x) = x^5 + x^2 + 4x + 6$ in $\mathbb{F}_7[x]$.

**(a)** Show that $x^7 \equiv x^2 + 3x^3 + 6x^4 \pmod{f(x)}$. Similarly compute $x^{14}$, $x^{21}$, and $x^{28}$ modulo $f(x)$ (note that $x^{14}$ can most easily be computed by squaring the result for $x^7$ and then reducing, etc.) to show that in this case the matrix $A$ in Exercise 15 is

$$\begin{pmatrix} 1 & 0 & 5 & 1 & 4 \\ 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 3 & 3 & 3 \\ 0 & 3 & 4 & 2 & 2 \\ 0 & 6 & 3 & 1 & 1 \end{pmatrix}.$$

**(b)** Show that the reduced row echelon form for $A - I$ is the matrix

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 6 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Conclude that $k = 2$ (so $f(x)$ is the product of precisely two factors which are powers of irreducible polynomials) and that $g_1(x) = 1$ and $g_2(x) = x^4 + 5x^3 + x^2 + x$ give a basis for the solutions to $(**)$ in Exercise 15.

**(c)** Following the procedure in Exercise 16, show that $(f(x), g_2(x) - 1) = x^2 + 3x + 5 = p_1(x)$, with $f(x)/p_1(x) = x^3 + 4x^2 + 4x + 4 = p_2(x)$, giving the powers of the irreducible polynomials dividing $f(x)$ in $\mathbb{F}_7[x]$. Show that neither factor is a $7^{\text{th}}$ power in $\mathbb{F}_7[x]$ and that each is relatively prime to its derivative to conclude that both factors are irreducible polynomials, giving the complete factorization of $f(x)$ into irreducible polynomials:

$$f(x) = (x^2 + 3x + 5)(x^3 + 4x^2 + 4x + 4) \in \mathbb{F}_7[x].$$

## 14.4. Composite Extensions and Simple Extensions

**1.** Determine the Galois closure of the field $\mathbb{Q}(\sqrt{1+\sqrt{2}})$ over $\mathbb{Q}$.

**2.** Find a primitive generator for $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$ over $\mathbb{Q}$.

**3.** Let $F$ be a field contained in the ring of $n \times n$ matrices over $\mathbb{Q}$. Prove that $[F : \mathbb{Q}] \leq n$. (Note that, by Exercise 19 of Section 13.2, the ring of $n \times n$ matrices over $\mathbb{Q}$ does contain fields of degree $n$ over $\mathbb{Q}$.)

**4.** Let $f(x) \in F[x]$ be an irreducible polynomial of degree $n$ over the field $F$, let $L$ be the splitting field of $f(x)$ over $F$ and let $\alpha$ be a root of $f(x)$ in $L$. If $K$ is any Galois extension of $F$, show that the polynomial $f(x)$ splits into a product of $m$ irreducible polynomials each of degree $d$ over $K$, where $d = [K(\alpha) : K] = [(L \cap K)(\alpha) : L \cap K]$ and $m = n/d = [F(\alpha) \cap K : F]$. [Show first that the factorization of $f(x)$ over $K$ is the same as its factorization over $L \cap K$. Then if $H$ is the subgroup of the Galois group of $L$ over $F$ corresponding to $L \cap K$ the factors of $f(x)$ over $L \cap K$ correspond to the orbits of $H$ on the roots of $f(x)$. Use Exercise 9 of Section 4.1.]

**5.** Let $p$ be a prime and let $F$ be a field. Let $K$ be a Galois extension of $F$ whose Galois group is a $p$-group (i.e., the degree $[K : F]$ is a power of $p$). Such an extension is called a *$p$-extension* (note that $p$-extensions are Galois by definition).

**(a)** Let $L$ be a $p$-extension of $K$. Prove that the Galois closure of $L$ over $F$ is a $p$-extension of $F$.

**(b)** Give an example to show that (a) need not hold if $[K : F]$ is a power of $p$ but $K/F$ is not Galois.

**6.** Prove that $\mathbb{F}_p(x, y)/\mathbb{F}_p(x^p, y^p)$ is not a simple extension by explicitly exhibiting an infinite number of intermediate subfields.

**7.** Let $F \subseteq K \subseteq L$ and let $\theta \in L$ with $p(x) = m_{\theta,F}(x)$. Prove that $K \otimes_F F(\theta) \cong K[x]/(p(x))$ as $K$-algebras.

**8.** Let $K_1$ and $K_2$ be two algebraic extensions of a field $F$ contained in the field $L$ of characteristic zero. Prove that the $F$-algebra $K_1 \otimes_F K_2$ has no nonzero nilpotent elements. [Use the preceding exercise.]

## 14.5. Cyclotomic Extensions and Abelian Extensions over Q

**1.** Determine the minimal polynomials satisfied by the primitive generators given in the text for the subfields of $\mathbb{Q}(\zeta_{13})$.

**2.** Determine the subfields of $\mathbb{Q}(\zeta_8)$ generated by the periods of $\zeta_8$ and in particular show that not every subfield has such a period as primitive element.

**3.** Determine the quadratic equation satisfied by the period $\alpha = \zeta_5 + \zeta_5^{-1}$ of the $5^{\text{th}}$ root of unity $\zeta_5$. Determine the quadratic equation satisfied by $\zeta_5$ over $\mathbb{Q}(\alpha)$ and use this to explicitly solve for the $5^{\text{th}}$ root of unity.

**4.** Let $\sigma_a \in \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ denote the automorphism of the cyclotomic field of $n^{\text{th}}$ roots of unity which maps $\zeta_n$ to $\zeta_n^a$ where $a$ is relatively prime to $n$ and $\zeta_n$ is a primitive $n^{\text{th}}$ root of unity. Show that $\sigma_a(\zeta) = \zeta^a$ for *every* $n^{\text{th}}$ root of unity.

**5.** Let $p$ be a prime and let $\epsilon_1, \epsilon_2, \ldots, \epsilon_{p-1}$ denote the primitive $p^{\text{th}}$ roots of unity. Set $p_n = \epsilon_1^n + \epsilon_2^n + \cdots + \epsilon_{p-1}^n$, the sum of the $n^{\text{th}}$ powers of the $\epsilon_i$. Prove that $p_n = -1$ if $p$ does not divide $n$ and that $p_n = p - 1$ if $p$ does divide $n$. [One approach: $p_1 = -1$ from $\Phi_p(x)$; show that $p_n$ is a Galois conjugate of $p_1$ for $p$ not dividing $n$, hence is also $-1$.]

**6.** Let $\zeta_n$ denote a primitive $n^{\text{th}}$ root of unity and let $K = \mathbb{Q}(\zeta_n)$ be the associated cyclotomic field. Let $a$ denote the trace of $\zeta_n$ from $K$ to $\mathbb{Q}$ (cf. Exercise 18 of Section 2). Prove that $a = 1$ if $n = 1$, $a = 0$ if $n$ is divisible by the square of a prime, and $a = (-1)^r$ if $n$ is the product of $r$ distinct primes.

**7.** Show that complex conjugation restricts to the automorphism $\sigma_{-1} \in \mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ of the cyclotomic field of $n^{\text{th}}$ roots of unity. Show that the field $K^+ = \mathbb{Q}(\zeta_n + \zeta_n^{-1})$ is the subfield of real elements in $K = \mathbb{Q}(\zeta_n)$, called the *maximal real subfield of* $K$.

**8.** Let $K_n = \mathbb{Q}(\zeta_{2^{n+2}})$ be the cyclotomic field of $2^{n+2}$-th roots of unity, $n \geq 0$. Set $\alpha_n = \zeta_{2^{n+2}} + \zeta_{2^{n+2}}^{-1}$ and $K_n^+ = \mathbb{Q}(\alpha_n)$, the maximal real subfield of $K_n$.

**(a)** Show that for all $n \geq 0$, $[K_n : \mathbb{Q}] = 2^{n+1}$, $[K_n : K_n^+] = 2$, $[K_n^+ : \mathbb{Q}] = 2^n$, and $[K_{n+1}^+ : K_n^+] = 2$.

**(b)** Determine the quadratic equation satisfied by $\zeta_{2^{n+2}}$ over $K_n^+$ in terms of $\alpha_n$.

**(c)** Show that for $n \geq 0$, $\alpha_{n+1}^2 = 2 + \alpha_n$ and hence show that

$$\alpha_n = \sqrt{2+\sqrt{2+\sqrt{\cdots+\sqrt{2}}}} \qquad (n \text{ times}),$$

giving an explicit formula for the (constructible) $2^{n+2}$-th roots of unity.

**9.** Notation as in the previous exercise.

**(a)** Prove that $K_n^+$ is a cyclic extension of $\mathbb{Q}$ of degree $2^n$. [Use an explicit isomorphism $(\mathbb{Z}/2^{n+2}\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^n\mathbb{Z}$ as abelian groups (i.e., $(\mathbb{Z}/2^{n+2}\mathbb{Z})^\times$ is isomorphic to a cyclic group of order 2 and a cyclic group of order $2^n$ — cf. Exercises 22 and 23 of Section 2.3]

**(b)** Prove that $K_n$ is a biquadratic extension of $K_{n-1}^+$ and that two of the three intermediate subfields are $K_n^+$ and $K_{n-1}$. Prove that the remaining field intermediate between $K_{n-1}^+$ and $K_n$ is a cyclic extension of $\mathbb{Q}$ of degree $2^n$.

**10.** Prove that $\mathbb{Q}(\sqrt[3]{2})$ is not a subfield of any cyclotomic field over $\mathbb{Q}$.

**11.** Prove that the primitive $n^{\text{th}}$ roots of unity form a basis over $\mathbb{Q}$ for the cyclotomic field of $n^{\text{th}}$ roots of unity if and only if $n$ is squarefree (i.e., $n$ is not divisible by the square of any prime).

**12.** Let $\sigma_p$ denote the Frobenius automorphism $x \mapsto x^p$ of the finite field $\mathbb{F}_q$ of $q = p^n$ elements. Viewing $\mathbb{F}_q$ as a vector space $V$ of dimension $n$ over $\mathbb{F}_p$ we can consider $\sigma_p$ as a linear transformation of $V$ to $V$. Determine the characteristic polynomial of $\sigma_p$ and prove that the linear transformation $\sigma_p$ is diagonalizable over $\mathbb{F}_p$ if and only if $n$ divides $p - 1$, and is diagonalizable over the algebraic closure of $\mathbb{F}_p$ if and only if $(n, p) = 1$.

**13.** Let $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ be the prime factorization of $n$ and let $\zeta_n$ be a primitive $n^{\text{th}}$ root of unity. For each $i = 1, 2, \dots, k$ define $d_i$ by $n = p_i^{a_i} d_i$ and let $\zeta_{p_i^{a_i}} = \zeta_n^{d_i}$, so that $\zeta_{p_i^{a_i}}$ is a particular primitive $p_i^{a_i}$-th root of unity. Let $\sigma_a \in \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ be the automorphism mapping $\zeta_n$ to $\zeta_n^a$ for $a$ relatively prime to $n$.

**(a)** Prove that for $i = 1, 2, \dots, k$, $\sigma_a$ maps $\zeta_{p_i^{a_i}}$ to $\zeta_{p_i^{a_i}}^a$ and gives an automorphism of $\mathbb{Q}(\zeta_{p_i^{a_i}})/\mathbb{Q}$ which depends only on $a \pmod{p_i^{a_i}}$, which we may denote $\sigma_{a \pmod{p_i^{a_i}}}$.

**(b)** Prove that the map $\sigma_a \mapsto (\sigma_{a \pmod{p_1^{a_1}}}, \dots, \sigma_{a \pmod{p_k^{a_k}}})$ is the isomorphism of Corollary 27 corresponding to the Chinese Remainder Theorem for $(\mathbb{Z}/n\mathbb{Z})^\times$.

The following Exercises 14 to 18 determine the periods associated to a primitive $17^{\text{th}}$ root of unity and provide a proof for the simple geometric construction indicated in Exercise 17 for the regular 17-gon. Let $\zeta = \zeta_{17} = \cos\dfrac{2\pi}{17} + i\sin\dfrac{2\pi}{17}$ be a fixed primitive $17^{\text{th}}$ root of unity in $\mathbb{C}$.

**14.** Define the *periods* of $\zeta$ as follows:

$$\eta_1 = \zeta+\zeta^2+\zeta^4+\zeta^8+\zeta^9+\zeta^{13}+\zeta^{15}+\zeta^{16} \qquad \eta_3' = \zeta^6+\zeta^7+\zeta^{10}+\zeta^{11}$$
$$\eta_2 = \zeta^3+\zeta^5+\zeta^6+\zeta^7+\zeta^{10}+\zeta^{11}+\zeta^{12}+\zeta^{14} \qquad \eta_4' = \zeta^3+\zeta^5+\zeta^{12}+\zeta^{14}$$
$$\eta_1' = \zeta+\zeta^4+\zeta^{13}+\zeta^{16} \qquad \eta_1'' = \zeta+\zeta^{16}$$
$$\eta_2' = \zeta^2+\zeta^8+\zeta^9+\zeta^{15} \qquad \eta_2'' = \zeta^4+\zeta^{13}.$$

**(a)** Show that all of these periods are real numbers and that $\eta_1'' = 2\cos\dfrac{2\pi}{17}$. Show that as real numbers these periods are approximately

$$\eta_1 \sim 1.562 \qquad \eta_1' \sim 2.049 \qquad \eta_3' \sim -2.906 \qquad \eta_1'' \sim 1.865$$
$$\eta_2 \sim -2.562 \qquad \eta_2' \sim -0.488 \qquad \eta_4' \sim 0.344 \qquad \eta_2'' \sim 0.185.$$

**(b)** Prove that $\eta_1$ and $\eta_2$ are roots of the equation $x^2 + x - 4 = 0$.

**(c)** Prove that $\eta_1'$ and $\eta_2'$ are roots of the equation $x^2 - \eta_1 x - 1 = 0$ and that $\eta_3'$ and $\eta_4'$ are roots of the equation $x^2 - \eta_2 x - 1 = 0$.

**(d)** Prove that $\eta_1''$ and $\eta_2''$ are roots of the equation $x^2 - \eta_1' x + \eta_4' = 0$.

The following Exercises 14 to 18 determine the periods associated to a primitive $17^{\text{th}}$ root of unity and provide a proof for the simple geometric construction indicated in Exercise 17 for the regular 17-gon. Let $\zeta = \zeta_{17} = \cos \dfrac{2\pi}{17} + i \sin \dfrac{2\pi}{17}$ be a fixed primitive $17^{\text{th}}$ root of unity in $\mathbb{C}$.

**15.** Prove that if $\tan 2\theta = a$ $(0 < 2\theta < \dfrac{\pi}{2})$ then $\tan \theta$ satisfies the equation $x^2 - \dfrac{2}{a}x - 1 = 0$.

The following Exercises 14 to 18 determine the periods associated to a primitive $17^{\text{th}}$ root of unity and provide a proof for the simple geometric construction indicated in Exercise 17 for the regular 17-gon. Let $\zeta = \zeta_{17} = \cos \dfrac{2\pi}{17} + i \sin \dfrac{2\pi}{17}$ be a fixed primitive $17^{\text{th}}$ root of unity in $\mathbb{C}$.

**16.** Let $C$ be the circle in $\mathbb{R}^2$ having the points $(h, k)$ and $(0, 1)$ as a diameter. Prove that this circle intersects the $x$-axis if and only if $h^2 - 4k \geq 0$ and in this case the two intercepts are the roots of the equation $x^2 - hx + k = 0$.

The following Exercises 14 to 18 determine the periods associated to a primitive $17^{\text{th}}$ root of unity and provide a proof for the simple geometric construction indicated in Exercise 17 for the regular 17-gon. Let $\zeta = \zeta_{17} = \cos\dfrac{2\pi}{17} + i\sin\dfrac{2\pi}{17}$ be a fixed primitive $17^{\text{th}}$ root of unity in $\mathbb{C}$.

**17.** (*Construction of the Regular 17-gon*) Draw a circle of radius 2 centered at the origin (0, 0).

**(a)** Join the point (4, 0) to the point (0, 1) and construct the line $\ell_1$ bisecting the angle between this line and the $y$-axis. Construct the line $\ell_2$ perpendicular to $\ell_1$ in Figure 2.

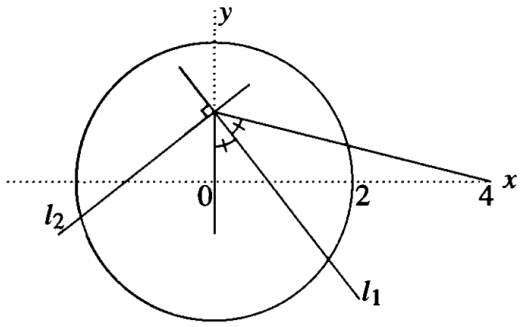



**Fig. 2**

**(b)** Using the intersection of $\ell_1$ and the $x$-axis as center and radius equal to the distance to (0, 1), construct the circle $C_1$ and let $A = (s, 0)$ be the right-hand point of intersection of $C_1$ with the $x$-axis. Similarly, let $B = (t, 0)$ denote the right-hand point of intersection of the $x$-axis and the circle $C_2$ whose center is the intersection of $\ell_2$ and the $x$-axis and whose radius is equal to the distance to (0, 1) as in Figure 3.

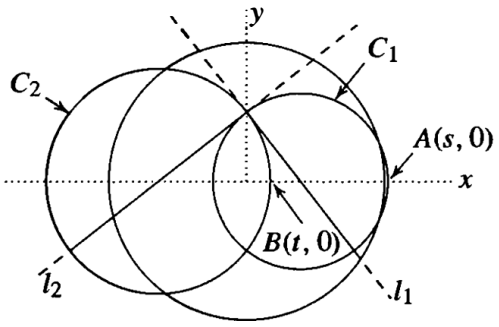



**Fig. 3**

**(c)** Construct a perpendicular to the $x$-axis at the point $A$ and mark off the distance $t$ from (0, 0) to $B$ to construct the point $(s, t)$. Construct the circle with $(s, t)$ and (0, 1) as a diameter and let $P$ denote the right-hand point of intersection of this circle with the $x$-axis. The perpendicular to the $x$-axis at $P$ intersects the circle of radius 2 at the second vertex of a regular 17-gon whose first vertex is at (2,0), hence constructs the regular 17-gon by straightedge and compass as in Figure 4.

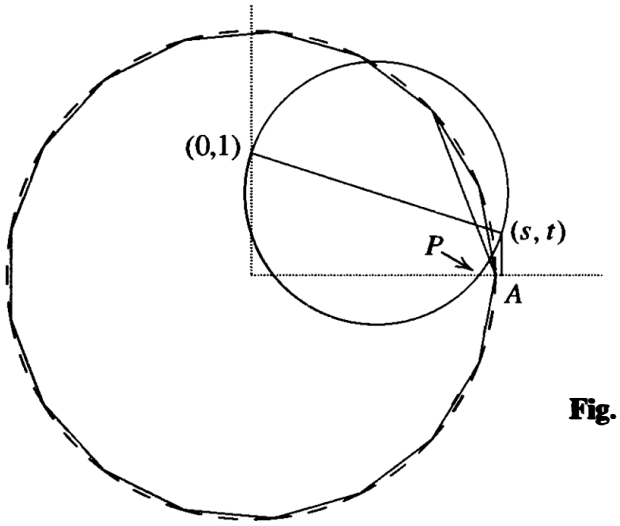



**Fig. 4**

The following Exercises 14 to 18 determine the periods associated to a primitive $17^{\text{th}}$ root of unity and provide a proof for the simple geometric construction indicated in Exercise 17 for the regular 17-gon. Let $\zeta = \zeta_{17} = \cos \dfrac{2\pi}{17} + i \sin \dfrac{2\pi}{17}$ be a fixed primitive $17^{\text{th}}$ root of unity in $\mathbb{C}$.

**18.** Notation as in the previous exercises.

**(a)** Prove that $\ell_1$ intersects the $x$-axis in the point $(\eta_1/2, 0)$ and that $\ell_2$ intersects the $x$-axis in the point $(\eta_2/2, 0)$.

**(b)** Prove that $C_1$ is the circle having the points $(\eta_1, -1)$ and $(0, 1)$ as diameter. Prove that $s = \eta_1'$. Similarly prove that $C_2$ is the circle having the points $(\eta_2, -1)$ and $(0, 1)$ as diameter and that $t = \eta_4'$.

**(c)** Prove that $P$ has coordinates $(\eta_1'', 0)$ and hence that the construction in the previous problem constructs the regular 17-gon by straightedge and compass.

## 14.6. Galois Groups of Polynomials

**1.** Show that a cubic with a multiple root has a linear factor. Is the same true for quartics?

**2.** Determine the Galois groups of the following polynomials:

**(a)** $x^3 - x^2 - 4$

**(b)** $x^3 - 2x + 4$

**(c)** $x^3 - x + 1$

**(d)** $x^3 + x^2 - 2x - 1$.

**3.** Prove for any $a, b \in \mathbb{F}_{p^n}$ that if $x^3 + ax + b$ is irreducible then $-4a^3 - 27b^2$ is a square in $\mathbb{F}_{p^n}$.

**4.** Determine the Galois group of $x^4 - 25$.

**5.** Determine the Galois group of $x^4 + 4$.

**6.** Determine the Galois group of $x^4 + 3x^3 - 3x - 2$.

**7.** Determine the Galois group of $x^4 + 2x^2 + x + 3$.

**8.** Determine the Galois group of $x^4 + 8x + 12$.

**9.** Determine the Galois group of $x^4 + 4x - 1$ (cf. Exercise 19).

**10.** Determine the Galois group of $x^5 + x - 1$.

**11.** Let $F$ be an extension of $\mathbb{Q}$ of degree 4 that is not Galois over $\mathbb{Q}$. Prove that the Galois closure of $F$ has Galois group either $S_4$, $A_4$ or the dihedral group $D_8$ of order 8. Prove that the Galois group is dihedral if and only if $F$ contains a quadratic extension of $\mathbb{Q}$.

**12.** Prove that an extension $F$ of $\mathbb{Q}$ of degree 4 can be generated by the root of an irreducible biquadratic $x^4 + ax^2 + b$ over $\mathbb{Q}$ if and only if $F$ contains a quadratic extension of $\mathbb{Q}$.

**13.** **(a)** Let $\pm\alpha, \pm\beta$ denote the roots of the polynomial $f(x) = x^4 + ax^2 + b \in \mathbb{Z}[x]$. Prove that $f(x)$ is irreducible if and only if $\alpha^2, \alpha \pm \beta$ are not elements of $\mathbb{Q}$.[3]

**(b)** Suppose $f(x)$ is irreducible and let $G$ be the Galois group of $f(x)$. Prove that

**(i)** $G \cong V$, the Klein 4-group, if and only if $b$ is a square in $\mathbb{Q}$ if and only if $\alpha\beta \in \mathbb{Q}$ is rational.

**(ii)** $G \cong C$, the cyclic group of order 4, if and only if $b(a^2 - 4b)$ is a square in $\mathbb{Q}$ if and only if $\mathbb{Q}(\alpha\beta) = \mathbb{Q}(\alpha^2)$.

**(iii)** $G \cong D_8$, the dihedral group of order 8, if and only if $b$ and $b(a^2 - 4b)$ are not squares in $\mathbb{Q}$ if and only if $\alpha\beta \notin \mathbb{Q}(\alpha^2)$.

**14.** Prove the polynomial $x^4 - px^2 + q \in \mathbb{Q}[x]$ is irreducible for any distinct odd primes $p$ and $q$ and has as Galois group the dihedral group of order 8.[4]

**15.** Prove the polynomial $x^4 + px + p \in \mathbb{Q}[x]$ is irreducible for every prime $p$ and for $p \neq 3, 5$ has Galois group $S_4$. Prove the Galois group for $p = 3$ is dihedral of order 8 and for $p = 5$ is cyclic of order 4.[5]

**16.** Determine the Galois group over $\mathbb{Q}$ of the polynomial $x^4+8x^2+8x+4$. Determine which of the subfields of this field are Galois over $\mathbb{Q}$ and for those which are Galois determine a polynomial $f(x) \in \mathbb{Q}[x]$ for which they are the splitting field over $\mathbb{Q}$.

**17.** Find the Galois group of $x^4 - 7$ over $\mathbb{Q}$ explicitly as a permutation group on the roots.

**18.** Let $\theta$ be a root of $x^3 - 3x + 1$. Prove that the splitting field of this polynomial is $\mathbb{Q}(\theta)$ and that the Galois group is cyclic of order 3. In particular the other roots of this polynomial can be written in the form $a + b\theta + c\theta^2$ for some $a, b, c \in \mathbb{Q}$. Determine the other roots explicitly in terms of $\theta$.

**19.** Let $f(x)$ be an irreducible polynomial of degree 4 in $\mathbb{Q}[x]$ with discriminant $D$. Let $K$ denote the splitting field of $f(x)$, viewed as a subfield of the complex numbers $\mathbb{C}$.

**(a)** Prove that $\mathbb{Q}(\sqrt{D}) \subset K$.

**(b)** Let $\tau$ denote complex conjugation and let $\tau_K$ denote the restriction of complex conjugation to $K$. Prove that $\tau_K$ is an element of $\mathrm{Gal}(K/\mathbb{Q})$ of order 1 or 2 depending on whether every element of $K$ is real or not.

**(c)** Prove that if $D < 0$ then $K$ cannot be cyclic of degree 4 over $\mathbb{Q}$ (i.e., $\mathrm{Gal}(K/\mathbb{Q})$ cannot be a cyclic group of order 4).

**(d)** Prove generally that $\mathbb{Q}(\sqrt{D})$ for squarefree $D < 0$ is not a subfield of a cyclic quartic field (cf. also Exercise 19 of Section 7).

**20.** Determine the Galois group of $(x^3 - 2)(x^3 - 3)$ over $\mathbb{Q}$. Determine all the subfields which contain $\mathbb{Q}(\rho)$ where $\rho$ is a primitive $3^{\text{rd}}$ root of unity.

**21.** Let $G \leq S_n$ be a subgroup of the symmetric group and suppose $\sigma_1, \ldots, \sigma_k$ are generators for $G$. If the function $f(x_1, x_2, \ldots, x_n)$ is fixed by the generators $\sigma_i$ show it is fixed by $G$.

**22.** (*Newton's Formulas*) Let $f(x)$ be a monic polynomial of degree $n$ with roots $\alpha_1, \ldots, \alpha_n$. Let $s_i$ be the elementary symmetric function of degree $i$ in the roots and define $s_i = 0$ for $i > n$. Let $p_i = \alpha_1^i + \cdots + \alpha_n^i$, $i \geq 0$, be the sum of the $i^{\text{th}}$ powers of the roots of $f(x)$.

---

[3]cf. the note *An Elementary Test for the Galois Group of a Quartic Polynomial*, Luise-Charlotte Kappe and Bette Warren, Amer. Math. Monthly, 96(1989), pp. 133–137.

[4]Ibid.

[5]Ibid.

Prove *Newton's Formulas*:

$$p_1 - s_1 = 0$$
$$p_2 - s_1 p_1 + 2s_2 = 0$$
$$p_3 - s_1 p_2 + s_2 p_1 - 3s_3 = 0$$
$$\vdots$$
$$p_i - s_1 p_{i-1} + s_2 p_{i-2} - \cdots + (-1)^{i-1} s_{i-1} p_1 + (-1)^i i s_i = 0$$

**23.** **(a)** If $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$ and $x^3 + y^3 + z^3 = 3$, determine $x^4 + y^4 + z^4$.
**(b)** Prove generally that $x$, $y$, $z$ are not rational but that $x^n + y^n + z^n$ is rational for every positive integer $n$.

**24.** Prove that an $n \times n$ matrix $A$ over a field of characteristic 0 is nilpotent if and only if the trace of $A^k$ is 0 for all $k \geq 0$.

**25.** Prove that two $n \times n$ matrices $A$ and $B$ over a field of characteristic 0 have the same characteristic polynomial if and only if the trace of $A^k$ equals the trace of $B^k$ for all $k \geq 0$.

**26.** Use the fact that the trace of $AB$ is the same as the trace of $BA$ for any two $n \times n$ matrices $A$ and $B$ to show that $AB$ and $BA$ have the same characteristic polynomial over a field of characteristic 0 (the same result is true over a field of arbitrary characteristic).

**27.** Let $f(x)$ be a monic polynomial of degree $n$ with roots $\alpha_1, \alpha_2, \ldots, \alpha_n$.

**(a)** Show that the discriminant $D$ of $f(x)$ is the square of the Vandermonde determinant

$$\begin{vmatrix} 1 & \alpha_1 & \alpha_1^2 & \ldots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \ldots & \alpha_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \ldots & \alpha_n^{n-1} \end{vmatrix} = \prod_{i>j} (\alpha_i - \alpha_j).$$

**(b)** Taking the Vandermonde matrix above, multiplying on the left by its transpose and taking the determinant show that one obtains

$$D = \begin{vmatrix} p_0 & p_1 & p_2 & \cdots & p_{n-1} \\ p_1 & p_2 & p_3 & \cdots & p_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ p_{n-1} & p_n & p_{n+1} & \cdots & p_{2n-2} \end{vmatrix}$$

where $p_i = \alpha_1^i + \cdots + \alpha_n^i$ is the sum of the $i^{\text{th}}$ powers of the roots of $f(x)$, which can be computed in terms of the coefficients of $f(x)$ using Newton's formulas above. This gives an efficient procedure for calculating the discriminant of a polynomial.

**28.** Let $\alpha$ be a root of the irreducible polynomial $f(x) \in F[x]$ and let $K = F(\alpha)$. Let $D$ be the discriminant of $f(x)$. Prove that $D = (-1)^{n(n-1)/2} \mathrm{N}_{K/F}(f'(\alpha))$, where $f'(x) = D_x f(x)$ is the derivative of $f(x)$.

The following exercises describe the *resultant* of two polynomials and in particular provide another efficient method for calculating the discriminant of a polynomial.

**29.** Let $F$ be a field and let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ and $g(x) = b_m x^m + b_{m-1}x^{m-1} + \cdots + b_1 x + b_0$ be two polynomials in $F[x]$.

**(a)** Prove that a necessary and sufficient condition for $f(x)$ and $g(x)$ to have a common root (or, equivalently, a common divisor in $F[x]$) is the existence of a polynomial $a(x) \in F[x]$ of degree at most $m-1$ and a polynomial $b(x) \in F[x]$ of degree at most $n-1$ with $a(x)f(x) = b(x)g(x)$.

**(b)** Writing $a(x)$ and $b(x)$ explicitly as polynomials show that equating coefficients in the equation $a(x)f(x) = b(x)g(x)$ gives a system of $n+m$ linear equations for the coefficients of $a(x)$ and $b(x)$. Prove that this system has a nontrivial solution (hence $f(x)$ and $g(x)$ have a common zero) if and only if the determinant

$$R(f,g) = \begin{vmatrix} a_n & a_{n-1} & \cdots & a_0 & & & & \\ & a_n & a_{n-1} & \cdots & a_0 & & & \\ & & a_n & a_{n-1} & \cdots & a_0 & & \\ & & & \ddots & & & & \\ & & & & a_n & a_{n-1} & \cdots & a_0 \\ b_m & b_{m-1} & \cdots & b_0 & & & & \\ & b_m & b_{m-1} & \cdots & b_0 & & & \\ & & b_m & b_{m-1} & \cdots & b_0 & & \\ & & & \ddots & & & & \\ & & & & b_m & b_{m-1} & \cdots & b_0 \end{vmatrix}$$

is zero. Here $R(f,g)$, called the *resultant* of the two polynomials, is the determinant of an $(n+m) \times (n+m)$ matrix $R$ with $m$ rows involving the coefficients of $f(x)$ and $n$ rows involving the coefficients of $g(x)$.

The following exercises describe the *resultant* of two polynomials and in particular provide another efficient method for calculating the discriminant of a polynomial.

**30.** **(a)** With notations as in the previous problem, show that we have the matrix equation

$$R\begin{pmatrix} x^{n+m-1} \\ x^{n+m-2} \\ \vdots \\ x \\ 1 \end{pmatrix} = \begin{pmatrix} x^{m-1}f(x) \\ x^{m-2}f(x) \\ \vdots \\ f(x) \\ x^{n-1}g(x) \\ x^{n-2}g(x) \\ \vdots \\ g(x) \end{pmatrix}.$$

**(b)** Let $R'$ denote the matrix of cofactors of $R$ as in Theorem 30 of Section 11.4, so $R'R = R(f, g)I$, where $I$ is the identity matrix. Multiply both sides of the matrix equation above by $R'$ and equate the bottom entry of the resulting column matrices to prove that there are polynomials $r(x), s(x) \in F[x]$ such that $R(f, g)$ is equal to $r(x)f(x) + s(x)g(x)$, i.e., the resultant of two polynomials is a linear combination (in $F[x]$) of the polynomials.

The following exercises describe the *resultant* of two polynomials and in particular provide another efficient method for calculating the discriminant of a polynomial.

**31.** Consider $f(x)$ and $g(x)$ as general polynomials and suppose the roots of $f(x)$ are $x_1, \ldots, x_n$ and the roots of $g(x)$ are $y_1, \ldots, y_m$. The coefficients of $f(x)$ are powers of $a_n$ times the elementary symmetric functions in $x_1, x_2, \ldots, x_n$ and the coefficients of $g(x)$ are powers of $b_m$ times the elementary symmetric functions in $y_1, y_2, \ldots, y_m$.

**(a)** By expanding the determinant show that $R(f, g)$ is homogeneous of degree $m$ in the coefficients $a_i$ and homogeneous of degree $n$ in the coefficients $b_j$.

**(b)** Show that $R(f, g)$ is $a_n^m b_m^n$ times a symmetric function in $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$.

**(c)** Since $R(f, g)$ is 0 if $f(x)$ and $g(x)$ have a common root, say $x_i = y_j$, show that $R(f, g)$ is divisible by $x_i - y_j$ for $i = 1, 2, \ldots, n$, $j = 1, 2, \ldots, m$. Conclude by degree considerations that

$$R = a_n^m b_m^n \prod_{i=1}^{n} \prod_{j=1}^{m} (x_i - y_j).$$

**(d)** Show that the product in (c) can be also be written

$$R(f, g) = a_n^m \prod_{i=1}^{n} g(x_i) = (-1)^{nm} b_m^n \prod_{j=1}^{m} f(y_j).$$

This gives an interesting *reciprocity* between the product of $g$ evaluated at the roots of $f$ and the product of $f$ evaluated at the roots of $g$.

The following exercises describe the *resultant* of two polynomials and in particular provide another efficient method for calculating the discriminant of a polynomial.

**32.** Consider now the special case where $g(x) = f'(x)$ is the derivative of the polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ and suppose the roots of $f(x)$ are $\alpha_1, \alpha_2, \ldots, \alpha_n$. Using the formula

$$R(f, f') = \prod_{i=1}^{n} f'(\alpha_i)$$

of the previous exercise, prove that

$$D = (-1)^{n(n-1)/2} R(f, f')$$

where $D$ is the discriminant of $f(x)$.

The following exercises describe the *resultant* of two polynomials and in particular provide another efficient method for calculating the discriminant of a polynomial.

**33.** **(a)** Prove that the discriminant of the cyclotomic polynomial $\Phi_p(x)$ of the $p^{\text{th}}$ roots of unity for an odd prime $p$ is $(-1)^{(p-1)/2}p^{p-2}$ [One approach: use Exercise 5 of the previous section together with the determinant form for the discriminant in terms of the power sums $p_i$.]

**(b)** Prove that $\mathbb{Q}(\sqrt{(-1)^{(p-1)/2}p}) \subset \mathbb{Q}(\zeta_p)$ for $p$ an odd prime. (Cf. also Exercise 11 of Section 7.)

The following exercises describe the *resultant* of two polynomials and in particular provide another efficient method for calculating the discriminant of a polynomial.

**34.** Use the previous exercise to prove that every quadratic extension of $\mathbb{Q}$ is contained in a cyclotomic extension (a special case of the Kronecker–Weber Theorem).

The following exercises describe the *resultant* of two polynomials and in particular provide another efficient method for calculating the discriminant of a polynomial.

**35.** Prove that the discriminant $D$ of the polynomial $x^n + px + q$ is given by the formula $(-1)^{n(n-1)/2} n^n q^{n-1} + (-1)^{(n-1)(n-2)/2} (n-1)^{n-1} p^n$.

The following exercises describe the *resultant* of two polynomials and in particular provide another efficient method for calculating the discriminant of a polynomial.

**36.** Prove that the discriminant of $x^n + nx^{n-1} + n(n-1)x^{n-2} + \cdots + n(n-1)\ldots(3)(2)x + n!$ is $(-1)^{n(n-1)/2}(n!)^n$.

The following exercises 37 to 43 outline two procedures for writing a symmetric function in terms of the elementary symmetric functions. Let $f(x_1, \ldots, x_n)$ be a polynomial which is symmetric in $x_1, \ldots, x_n$. Recall that the degree (sometimes called the *weight*) of the monomial $Ax_1^{a_1} x_2^{a_2} \ldots x_n^{a_n}$ $(a_i \geq 0)$ is $a_1 + a_2 + \cdots + a_n$ and that a polynomial is *homogeneous (of degree m)* if every monomial has the same degree $(m)$.

**37.** **(a)** Show that every polynomial $f(x_1, \ldots, x_n)$ can be written as a sum of homogeneous polynomials. Show that if $f(x_1, \ldots, x_n)$ is symmetric then each of these homogeneous polynomials is also symmetric.

**(b)** Show that the monomial $Bs_1^{a_1} s_2^{a_2} \ldots s_n^{a_n}$ in the elementary symmetric functions is a homogeneous polynomial in $x_1, x_2, \ldots, x_n$ of degree $a_1 + 2a_2 + \cdots + na_n$.

The following exercises 37 to 43 outline two procedures for writing a symmetric function in terms of the elementary symmetric functions. Let $f(x_1, \ldots, x_n)$ be a polynomial which is symmetric in $x_1, \ldots, x_n$. Recall that the degree (sometimes called the *weight*) of the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ $(a_i \geq 0)$ is $a_1 + a_2 + \cdots + a_n$ and that a polynomial is *homogeneous (of degree m)* if every monomial has the same degree $(m)$.

In writing $f(x_1, \ldots, x_n)$ as a polynomial in the symmetric functions it therefore suffices to assume that $f(x_1, \ldots, x_n)$ is homogeneous.

Recall the *lexicographic monomial order* with $x_1 > x_2 > \cdots > x_n$ defined in Section 9.6, where the nonzero monomial term with exponents $(a_1, a_2, \ldots, a_n)$ comes before the nonzero monomial term with exponents $(b_1, b_2, \ldots, b_n)$ if the initial components of the two $n$-tuples of exponents are equal and the first component where they differ has $a_i > b_i$. If $f(x_1, \ldots, x_n)$ contains the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ then since $f(x_1, \ldots, x_n)$ is symmetric it also contains all the permuted monomials. Among these choose the lexicographically largest monomial, which therefore satisfies $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$.

**38.** **(a)** Show that the monomial $As_1^{a_1-a_2}s_2^{a_2-a_3}\cdots s_n^{a_n}$ in the elementary symmetric functions has the same lexicographic initial term.

**(b)** Show that subtracting $As_1^{a_1-a_2}s_2^{a_2-a_3}\cdots s_n^{a_n}$ from $f(x)$ yields either 0 or a symmetric polynomial of the same degree whose terms are lexicographically smaller than the terms in $f(x_1, \ldots, x_n)$.

**(c)** Show that the iteration of this procedure (lexicographic ordering, choosing the lexicographically largest term, subtracting the associated monomial in the elementary symmetric functions) terminates, expressing $f(x_1, \ldots, x_n)$ as a polynomial in the elementary symmetric functions.

The following exercises 37 to 43 outline two procedures for writing a symmetric function in terms of the elementary symmetric functions. Let $f(x_1, \ldots, x_n)$ be a polynomial which is symmetric in $x_1, \ldots, x_n$. Recall that the degree (sometimes called the *weight*) of the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ $(a_i \geq 0)$ is $a_1 + a_2 + \cdots + a_n$ and that a polynomial is *homogeneous (of degree m)* if every monomial has the same degree $(m)$.

In writing $f(x_1, \ldots, x_n)$ as a polynomial in the symmetric functions it therefore suffices to assume that $f(x_1, \ldots, x_n)$ is homogeneous.

Recall the *lexicographic monomial order* with $x_1 > x_2 > \cdots > x_n$ defined in Section 9.6, where the nonzero monomial term with exponents $(a_1, a_2, \ldots, a_n)$ comes before the nonzero monomial term with exponents $(b_1, b_2, \ldots, b_n)$ if the initial components of the two $n$-tuples of exponents are equal and the first component where they differ has $a_i > b_i$. If $f(x_1, \ldots, x_n)$ contains the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ then since $f(x_1, \ldots, x_n)$ is symmetric it also contains all the permuted monomials. Among these choose the lexicographically largest monomial, which therefore satisfies $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$.

**39.** Use the algorithm described in Exercise 38 to prove that a polynomial $f(x_1, \ldots, x_n)$ that is symmetric in $x_1, \ldots, x_n$ can be expressed *uniquely* as a polynomial in the elementary symmetric functions.

The following exercises 37 to 43 outline two procedures for writing a symmetric function in terms of the elementary symmetric functions. Let $f(x_1, \ldots, x_n)$ be a polynomial which is symmetric in $x_1, \ldots, x_n$. Recall that the degree (sometimes called the *weight*) of the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ $(a_i \geq 0)$ is $a_1 + a_2 + \cdots + a_n$ and that a polynomial is *homogeneous (of degree m)* if every monomial has the same degree $(m)$.

In writing $f(x_1, \ldots, x_n)$ as a polynomial in the symmetric functions it therefore suffices to assume that $f(x_1, \ldots, x_n)$ is homogeneous.

Recall the *lexicographic monomial order* with $x_1 > x_2 > \cdots > x_n$ defined in Section 9.6, where the nonzero monomial term with exponents $(a_1, a_2, \ldots, a_n)$ comes before the nonzero monomial term with exponents $(b_1, b_2, \ldots, b_n)$ if the initial components of the two $n$-tuples of exponents are equal and the first component where they differ has $a_i > b_i$. If $f(x_1, \ldots, x_n)$ contains the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ then since $f(x_1, \ldots, x_n)$ is symmetric it also contains all the permuted monomials. Among these choose the lexicographically largest monomial, which therefore satisfies $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$.

**40.** Use the procedure in Exercise 38 to express each of the following symmetric functions as a polynomial in the elementary symmetric functions:

**(a)** $(x_1 - x_2)^2$

**(b)** $x_1^2 + x_2^2 + x_3^2$

**(c)** $x_1^2x_2^2 + x_1^2x_3^2 + x_2^2x_3^2$.

The following exercises 37 to 43 outline two procedures for writing a symmetric function in terms of the elementary symmetric functions. Let $f(x_1, \ldots, x_n)$ be a polynomial which is symmetric in $x_1, \ldots, x_n$. Recall that the degree (sometimes called the *weight*) of the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ $(a_i \geq 0)$ is $a_1 + a_2 + \cdots + a_n$ and that a polynomial is *homogeneous (of degree m)* if every monomial has the same degree $(m)$.

In writing $f(x_1, \ldots, x_n)$ as a polynomial in the symmetric functions it therefore suffices to assume that $f(x_1, \ldots, x_n)$ is homogeneous.

Recall the *lexicographic monomial order* with $x_1 > x_2 > \cdots > x_n$ defined in Section 9.6, where the nonzero monomial term with exponents $(a_1, a_2, \ldots, a_n)$ comes before the nonzero monomial term with exponents $(b_1, b_2, \ldots, b_n)$ if the initial components of the two $n$-tuples of exponents are equal and the first component where they differ has $a_i > b_i$. If $f(x_1, \ldots, x_n)$ contains the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ then since $f(x_1, \ldots, x_n)$ is symmetric it also contains all the permuted monomials. Among these choose the lexicographically largest monomial, which therefore satisfies $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$.

**41.** Use the procedure in Exercise 38 to express $\sum_{i \neq j} x_i^2 x_j$ as a polynomial in the elementary symmetric functions.

The following exercises 37 to 43 outline two procedures for writing a symmetric function in terms of the elementary symmetric functions. Let $f(x_1, \ldots, x_n)$ be a polynomial which is symmetric in $x_1, \ldots, x_n$. Recall that the degree (sometimes called the *weight*) of the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ $(a_i \geq 0)$ is $a_1 + a_2 + \cdots + a_n$ and that a polynomial is *homogeneous (of degree m)* if every monomial has the same degree $(m)$.

In writing $f(x_1, \ldots, x_n)$ as a polynomial in the symmetric functions it therefore suffices to assume that $f(x_1, \ldots, x_n)$ is homogeneous.

Recall the *lexicographic monomial order* with $x_1 > x_2 > \cdots > x_n$ defined in Section 9.6, where the nonzero monomial term with exponents $(a_1, a_2, \ldots, a_n)$ comes before the nonzero monomial term with exponents $(b_1, b_2, \ldots, b_n)$ if the initial components of the two $n$-tuples of exponents are equal and the first component where they differ has $a_i > b_i$. If $f(x_1, \ldots, x_n)$ contains the **monomial** $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ then since $f(x_1, \ldots, x_n)$ is symmetric it also contains all the permuted monomials. Among these choose the lexicographically largest monomial, which therefore satisfies $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$.

We now know that a symmetric polynomial $f(x_1, \ldots, x_n)$ can be written uniquely as a polynomial in the elementary symmetric functions. Using this existence and uniqueness we can describe an alternate and computationally useful method for determining the coefficients of the elementary symmetric functions in this polynomial. As in Exercise 37 we may assume that $f(x_1, \ldots, x_n)$ is homogeneous of degree $M$. Let $N$ be the maximum degree of any of the variables $x_1, \ldots, x_n$ in $f(x_1, \ldots, x_n)$.

**(a)** Determine all of the possible **monomials** $A_i s_1^{a_1} s_2^{a_2} \cdots s_n^{a_n}$ appearing in $f(x_1, \ldots, x_n)$ from the constraints

$$a_1 + 2a_2 + \cdots + na_n = M$$
$$a_1 + a_2 + \cdots + a_n \leq N.$$

**(b)** Since $f(x_1, \ldots, x_n) = \sum A_i s_1^{a_1} s_2^{a_2} \cdots s_n^{a_n}$ is a polynomial *identity*, it is valid for any substitution of values for $x_1, \ldots, x_n$. Each substitution into this equation gives a linear relation on the coefficients $A_i$ and so a sufficient number of substitutions will determine the $A_i$.

**42.** Show that the function $(x_1 + x_2 - x_3 - x_4)(x_1 + x_3 - x_2 - x_4)(x_1 + x_4 - x_2 - x_3)$ is symmetric in $x_1, x_2, x_3, x_4$ and use the preceding procedure to prove it can be expressed as a polynomial in the elementary symmetric functions as $s_1^3 - 4s_1s_2 + 8s_3$.

The following exercises 37 to 43 outline two procedures for writing a symmetric function in terms of the elementary symmetric functions. Let $f(x_1, \ldots, x_n)$ be a polynomial which is symmetric in $x_1, \ldots, x_n$. Recall that the degree (sometimes called the *weight*) of the monomial $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ $(a_i \geq 0)$ is $a_1 + a_2 + \cdots + a_n$ and that a polynomial is *homogeneous (of degree m)* if every monomial has the same degree $(m)$.

In writing $f(x_1, \ldots, x_n)$ as a polynomial in the symmetric functions it therefore suffices to assume that $f(x_1, \ldots, x_n)$ is homogeneous.

Recall the *lexicographic monomial order* with $x_1 > x_2 > \cdots > x_n$ defined in Section 9.6, where the nonzero monomial term with exponents $(a_1, a_2, \ldots, a_n)$ comes before the nonzero monomial term with exponents $(b_1, b_2, \ldots, b_n)$ if the initial components of the two $n$-tuples of exponents are equal and the first component where they differ has $a_i > b_i$. If $f(x_1, \ldots, x_n)$ contains the **monomial** $Ax_1^{a_1}x_2^{a_2}\ldots x_n^{a_n}$ then since $f(x_1, \ldots, x_n)$ is symmetric it also contains all the permuted monomials. Among these choose the lexicographically largest monomial, which therefore satisfies $a_1 \geq a_2 \geq \cdots \geq a_n \geq 0$.

We now know that a symmetric polynomial $f(x_1, \ldots, x_n)$ can be written uniquely as a polynomial in the elementary symmetric functions. Using this existence and uniqueness we can describe an alternate and computationally useful method for determining the coefficients of the elementary symmetric functions in this polynomial. As in Exercise 37 we may assume that $f(x_1, \ldots, x_n)$ is homogeneous of degree $M$. Let $N$ be the maximum degree of any of the variables $x_1, \ldots, x_n$ in $f(x_1, \ldots, x_n)$.

**(a)** Determine all of the possible **monomials** $A_i s_1^{a_1} s_2^{a_2} \cdots s_n^{a_n}$ appearing in $f(x_1, \ldots, x_n)$ from the constraints

$$a_1 + 2a_2 + \cdots + na_n = M$$
$$a_1 + a_2 + \cdots + a_n \leq N.$$

**(b)** Since $f(x_1, \ldots, x_n) = \sum A_i s_1^{a_1} s_2^{a_2} \cdots s_n^{a_n}$ is a polynomial *identity*, it is valid for any substitution of values for $x_1, \ldots, x_n$. Each substitution into this equation gives a linear relation on the coefficients $A_i$ and so a sufficient number of substitutions will determine the $A_i$.

**43.** Express each of the following in terms of the elementary symmetric functions:
**(a)** $\sum_{i \neq j} x_i^2 x_j$ **(b)** $\sum_{i,j,k \text{ distinct}} x_i^2 x_j x_k$ **(c)** $\sum_{i,j,k \text{ distinct}} x_i^2 x_j^2 x_k^2$.

**44.** Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be the roots of a quartic polynomial $f(x)$ over $\mathbb{Q}$. Show that the quantities $\alpha_1\alpha_2 + \alpha_3\alpha_4$, $\alpha_1\alpha_3 + \alpha_2\alpha_4$, and $\alpha_1\alpha_4 + \alpha_2\alpha_3$ are permuted by the Galois group of $f(x)$. Conclude that these elements are the roots of a cubic polynomial with coefficients in $\mathbb{Q}$ (also sometimes referred to as the *resolvent cubic* of $f(x)$).

**45.** If $f(x) = x^3 + px + q \in \mathbb{Z}[x]$ is irreducible, prove that its discriminant $D = -4p^3 - 27q^2$ is an integer not equal to $0, \pm 1$.

**46.** Prove that every finite group occurs as the Galois group of a field extension of the form $F(x_1, x_2, \ldots, x_n)/E$.

**47.** Let $F$ be a field of characteristic 0 in which every cubic polynomial has a root. Let $f(x)$ be an irreducible quartic polynomial over $F$ whose discriminant is a square in $F$. Determine the Galois group of $f(x)$.

**48.** This exercise determines the splitting field $K$ for the polynomial $f(x) = x^6 - 2x^3 - 2$ over $\mathbb{Q}$ (cf. also Exercise 2 of Section 8).

**(a)** Prove that $f(x)$ is irreducible over $\mathbb{Q}$ with roots the three cube roots of $1 \pm \sqrt{3}$.

**(b)** Prove that $K$ contains the field $\mathbb{Q}(\sqrt{-3})$ of 3rd roots of unity and contains $\mathbb{Q}(\sqrt{3})$, hence contains the biquadratic field $F = \mathbb{Q}(i, \sqrt{3})$. Take the product of two of the roots in (a) to prove that $K$ contains $\sqrt[3]{2}$ and conclude that $K$ is an extension of the field $L = \mathbb{Q}(\sqrt[3]{2}, i, \sqrt{3})$.

**(c)** Prove that $[L : \mathbb{Q}] = 12$ and that $K$ is obtained from $L$ by adjoining the cube root of an element in $L$, so that $[K : \mathbb{Q}] = 12$ or 36.

**(d)** Prove that if $[K : \mathbb{Q}] = 12$ then $K = \mathbb{Q}(\sqrt[3]{2}, i, \sqrt{3})$ and that $\mathrm{Gal}(K/\mathbb{Q})$ is isomorphic to the direct product of the cyclic group of order 2 and $S_3$. Prove that if $[K : \mathbb{Q}] = 12$ then there is a unique real cubic subfield in $K$, namely $\mathbb{Q}(\sqrt[3]{2})$.

**(e)** Take the quotient of the two real roots in (a) to show that $\sqrt[3]{2+\sqrt{3}}$ and $\sqrt[3]{2-\sqrt{3}}$ (real roots) are both elements of $K$. Show that $\alpha = \sqrt[3]{2+\sqrt{3}} + \sqrt[3]{2-\sqrt{3}}$ is a real root of the irreducible cubic equation $x^3 - 3x - 4$ whose discriminant is $-2^2 3^4$. Conclude that the Galois closure of $\mathbb{Q}(\alpha)$ contains $\mathbb{Q}(i)$ so in particular $\mathbb{Q}(\alpha) \neq \mathbb{Q}(\sqrt[3]{2})$.

**(f)** Conclude from (e) that $G = \mathrm{Gal}(K/\mathbb{Q})$ is of order 36. Determine all the elements of $G$ explicitly and in particular show that $G$ is isomorphic to $S_3 \times S_3$.

**49.** Prove that the Galois group over $\mathbb{Q}$ of $x^6 - 4x^3 + 1$ is isomorphic to the dihedral group of order 12. [Observe that the two real roots are inverses of each other.]

**50.** *(Criterion for the Galois Group of an Irreducible Cubic over an Arbitrary Field)* Suppose $K$ is a field and $f(x) = x^3 + ax^2 + bx + c \in K[x]$ is irreducible, so the Galois group of $f(x)$ over $K$ is either $S_3$ or $A_3$.

**(a)** Show that the Galois group of $f(x)$ is $A_3$ if and only if the resultant quadratic polynomial $g(x) = x^2+(ab-3c)x+(b^3+a^3c-6abc+9c^2)$ has a root in $K$. [If $\alpha$, $\beta$, $\gamma$ are the roots of $f(x)$ show that the Galois group is $A_3$ if and only if the element $\theta = \alpha\beta^2 + \beta\gamma^2 + \gamma\alpha^2$ is an element of $K$ and that $\theta$ is a root of $g(x)$.] Show that the discriminant of $g(x)$ is the same as the discriminant of $f(x)$.

**(b)** $(\mathrm{ch}(K) \neq 2)$ If $K$ has characteristic different from 2 show either from (a) or directly from the definition of the discriminant that the Galois group of $f(x)$ is $A_3$ if and only if the discriminant of $f(x)$ is a square in $K$.

**(c)** $(\mathrm{ch}(K) = 2)$ If $K$ has characteristic 2 show that the discriminant of $f(x)$ is always a square. Show that $f(x)$ can be taken to be of the form $x^3 + px + q$ and that the Galois group of $f(x)$ is $A_3$ if and only if the quadratic $x^2 + qx + (p^3 + q^2)$ has a root in $K$ (equivalently, if $(p^3 + q^2)/q^2 \in K$ is in the image of the *Artin–Schreier map* $x \mapsto x^2 - x$ mapping $K$ to $K$).

**(d)** If $K = \mathbb{F}_2(t)$ where $t$ is transcendental over $\mathbb{F}_2$. Prove that the polynomials $x^3 + t^2x + t^3$, $x^3 + (t^2 + t + 1)x + (t^2 + t + 1)$, and $x^3 + (t^2 + t + 1)x + (t^3 + t^2 + t)$ have $A_3$ as Galois group while $x^3 + t^2x + t$ and $x^3 = x + t$ have $S_3$ as Galois group.

**51.** This exercise proves *Sturm's Theorem* determining the number of real roots of a polynomial $f(x) \in \mathbb{R}[x]$ in an interval $[a, b]$. The multiple roots of $f(x)$ are zeros of the g.c.d. of $f(x)$ and its derivative $f'(x)$, and it follows that to determine the real roots of $f(x)$ in $[a, b]$ we may assume that the roots of $f(x)$ are *simple*.

Apply the Euclidean algorithm to $f_0(x) = f(x)$ and its derivative $f_1(x) = f'(x)$ using the *negative* of the remainder at each stage to find a sequence of polynomials $f(x), f'(x), f_2(x), \ldots, f_n(x)$ with

$$f_{i-1}(x) = q_i(x) f_i(x) - f_{i+1}(x) \qquad i = 0, 1, \ldots, n-1$$

where $f_n(x) \in \mathbb{R}$ is a nonzero constant.

**(a)** Prove that consecutive polynomials $f_i(x), f_{i+1}(x)$ for $i = 0, 1, \ldots, n-1$ have no common zeros. [Show that otherwise $f_{i+2}(c) = f_{i+3}(c) = \cdots = 0$, and derive a contradiction.]

**(b)** If $f_i(c) = 0$ for some $i = 0, 1, \ldots, n-1$, prove that one of the two values $f_{i-1}(c)$, $f_{i+1}(c)$ is strictly negative and the other is strictly positive.

For any real number $\alpha$, let $V(\alpha)$ denote the number of sign changes in the *Sturm sequence* of real numbers

$$f(\alpha), f'(\alpha), f_2(\alpha), \ldots, f_n(\alpha),$$

ignoring any 0's that appear (for example $-1, -2, 0, +3, -4$ has signs $--+-$ disregarding the 0, so there are 2 sign changes, the first from $-2$ to $+3$, the second from $+3$ to $-4$).

**(c)** Suppose $\alpha < \beta$ and that all the elements in the Sturm sequences for $\alpha$ and for $\beta$ are nonzero. Prove that unless $f_i(c) = 0$ for some $\alpha < c < \beta$ and some $i = 0, 1, \ldots, n-1$, then the signs of all the elements in these two Sturm sequences are the same, so in particular $V(\alpha) = V(\beta)$.

**(d)** If $f_j(c) = 0$ prove that there is a sufficiently small interval $(\alpha, \beta)$ containing $c$ so that $f_j(x)$ has no zero other than $c$ for $\alpha < x < \beta$.

**(e)** If $j \geq 1$ in (d), prove that the number of sign changes in $f_{j-1}(\alpha), f_j(\alpha), f_{j+1}(\alpha)$ and in $f_{j-1}(\beta), f_j(\beta), f_{j+1}(\beta)$ are the same. [Observe that $f_{j-1}(c)$ and $f_{j+1}(c)$ have opposite signs by (b) and $f_{j-1}(x)$ and $f_{j+1}(x)$ do not change sign in $(\alpha, \beta)$.]

**(f)** If $j = 0$ in (d) show that the number of sign changes in $f(\alpha), f'(\alpha)$ is one more than the number of sign changes in $f(\beta), f'(\beta)$. [If $f'(c) > 0$ then $f(x)$ is increasing at $c$, so that $f(\alpha) < 0$, $f(\beta) > 0$, and $f'(x)$ does not change sign in $(\alpha, \beta)$, so the signs change from $-+$ to $++$. Similarly if $f'(c) < 0$.]

**(g)** Prove *Sturm's Theorem*: if $f(x)$ is a polynomial with real coefficients all of whose real roots are simple then the number of real zeros of $f(x)$ in an interval $[a, b]$ where $f(a)$ and $f(b)$ are both nonzero is given by $V(a) - V(b)$. [Use (c), (e) and (f) to see that as $\alpha$ runs from $a$ to $b$ the number $V(\alpha)$ of sign changes is constant unless $\alpha$ passes through a zero of $f(x)$, in which case it decreases by precisely 1.]

**(h)** Suppose $f(x) = x^5 + px + q \in \mathbb{R}[x]$ has simple roots. Show that the sequence of polynomials above is given by $f(x)$, $5x^4 + p$, $(-4p/5)x + q$, and $-D/(256p^4)$ where $D = 256p^5 + 3125q^4$ is the discriminant of $f(x)$. Conclude for $p > 0$ that $f(x)$ has precisely one real root and for $p < 0$ that $f(x)$ has precisely 1 or 3 real roots depending on whether $D > 0$ or $D < 0$, respectively. [E.g., if $p < 0$ and $D < 0$ then at $-\infty$ the signs are $- + -+$ with 3 sign changes and at $+\infty$ the signs are $+ + ++$ with no sign changes.]

## 14.7. Solvable and Radical Extensions: Insolvability of the Quintic

**1.** Use Cardano's Formulas to solve the equation $x^3 + x^2 - 2 = 0$. In particular show that the equation has the real root

$$\frac{1}{3}(\sqrt[3]{26 + 15\sqrt{3}} + \sqrt[3]{26 - 15\sqrt{3}} - 1).$$

Show directly that the roots of this cubic are $1, -1 \pm i$. Explain this by proving that

$$\sqrt[3]{26 + 15\sqrt{3}} = 2 + \sqrt{3} \quad \sqrt[3]{26 - 15\sqrt{3}} = 2 - \sqrt{3}$$

so that

$$\sqrt[3]{26 + 15\sqrt{3}} + \sqrt[3]{26 - 15\sqrt{3}} = 4.$$

**2.** Let $\zeta_7$ be a primitive $7^{\text{th}}$ root of unity and let $\alpha = \zeta + \zeta^{-1}$.

**(a)** Show that $\zeta_7$ is a root of the quadratic $z^2 - \alpha z + 1$ over $\mathbb{Q}(\alpha)$.

**(b)** Show using the minimal polynomial for $\zeta_7$ that $\alpha$ is a root of the cubic $x^3 + x^2 - 2x - 1$.

**(c)** Use (a) and (b) together with the explicit solution of the cubic in (b) in the text to express $\zeta_7$ in terms of radicals similar to the expressions given earlier for the other roots of unity of small order. (The complicated nature of the expression explains why we did not include $\zeta_7$ earlier in our list of explicit roots of unity.)

**3.** Let $F$ be a field of characteristic $\neq 2$. State and prove a necessary and sufficient condition on $\alpha, \beta \in F$ so that $F(\sqrt{\alpha}) = F(\sqrt{\beta})$. Use this to determine whether $\mathbb{Q}(\sqrt{1-\sqrt{2}}) = \mathbb{Q}(i, \sqrt{2})$.

**4.** Let $K = \mathbb{Q}(\sqrt[n]{a})$, where $a \in \mathbb{Q}, a > 0$ and suppose $[K : \mathbb{Q}] = n$ (i.e., $x^n - a$ is irreducible). Let $E$ be any subfield of $K$ and let $[E : \mathbb{Q}] = d$. Prove that $E = \mathbb{Q}(\sqrt[d]{a})$. [Consider $N_{K/E}(\sqrt[n]{a}) \in E$ .]

**5.** Let $K$ be as in the previous exercise. Prove that if $n$ is odd then $K$ has no nontrivial subfields which are Galois over $\mathbb{Q}$ and if $n$ is even then the only nontrivial subfield of $K$ which is Galois over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{a})$.

**6.** Let $L$ be the Galois closure of $K$ in the previous two exercises (i.e., the splitting field of $x^n - a$). Prove that $[L : \mathbb{Q}] = n\varphi(n)$ or $\frac{1}{2}n\varphi(n)$. [Note that $\mathbb{Q}(\zeta_n) \cap K$ is a Galois extension of $\mathbb{Q}$.]

**7.** (*Kummer Generators for Cyclic Extensions*) Let $F$ be a field of characteristic not dividing $n$ containing the $n^{\text{th}}$ roots of unity and let $K$ be a cyclic extension of degree $d$ dividing $n$. Then $K = F(\sqrt[n]{a})$ for some nonzero $a \in F$. Let $\sigma$ be a generator for the cyclic group $\text{Gal}(K/F)$.

**(a)** Show that $\sigma(\sqrt[n]{a}) = \zeta \sqrt[n]{a}$ for some primitive $d^{\text{th}}$ root of unity $\zeta$.

**(b)** Suppose $K = F(\sqrt[n]{a}) = F(\sqrt[n]{b})$. Use (a) to show that $\dfrac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} = \left(\dfrac{\sigma(\sqrt[n]{b})}{\sqrt[n]{b}}\right)^i$ for some integer $i$ relatively prime to $d$. Conclude that $\sigma$ fixes the element $\dfrac{\sqrt[n]{a}}{(\sqrt[n]{b})^i}$ so this is an element of $F$.

**(c)** Prove that $K = F(\sqrt[n]{a}) = F(\sqrt[n]{b})$ if and only if $a = b^i c^n$ and $b = a^j d^n$ for some $c, d \in F$, i.e., if and only if $a$ and $b$ generate the same subgroup of $F^\times$ modulo $n^{\text{th}}$ powers.

**8.** Let $p, q$ and $r$ be primes in $\mathbb{Z}$ with $q \neq r$. Let $\sqrt[p]{q}$ denote any root of $x^p - q$ and let $\sqrt[p]{r}$ denote any root of $x^p - r$. Prove that $\mathbb{Q}(\sqrt[p]{q}) \neq \mathbb{Q}(\sqrt[p]{r})$.

**9.** (*Artin–Schreier Extensions*) Let $F$ be a field of characteristic $p$ and let $K$ be a cyclic extension of $F$ of degree $p$. Prove that $K = F(\alpha)$ where $\alpha$ is a root of the polynomial $x^p - x - a$ for some $a \in F$. [Note that $\mathrm{Tr}_{K/F}(-1) = 0$ since $F$ is of characteristic $p$ so that $-1 = \alpha - \sigma\alpha$ for some $\alpha \in K$ where $\sigma$ is a generator of $\mathrm{Gal}(K/F)$ by Exercise 26 of Section 2. Show that $a = \alpha^p - \alpha$ is an element of $F$.] Note that since $F$ contains the $p^{\text{th}}$ roots of unity (namely, 1) that this completes the description of all cyclic extensions of prime degree $p$ over fields containing the $p^{\text{th}}$ roots of unity in all characteristics.

**10.** Let $K = \mathbb{Q}(\zeta_p)$ be the cyclotomic field of $p^{\text{th}}$ roots of unity for the prime $p$ and let $G = \text{Gal}(K/\mathbb{Q})$. Let $\zeta$ denote any $p^{\text{th}}$ root of unity. Prove that $\sum_{\sigma \in G} \sigma(\zeta)$ (the trace from $K$ to $\mathbb{Q}$ of $\zeta$) is $-1$ or $p-1$ depending on whether $\zeta$ is or is not a primitive $p^{\text{th}}$ root of unity.

**11.** (*The Classical Gauss Sum*) Let $K = \mathbb{Q}(\zeta_p)$ be the cyclotomic field of $p^{\text{th}}$ roots of unity for the odd prime $p$, viewed as a subfield of $\mathbb{C}$, and let $G = \text{Gal}(K/\mathbb{Q})$. Let $H$ denote the subgroup of index 2 in the cyclic group $G$. Define $\eta_0 = \sum_{\tau \in H} \tau(\zeta_p)$, $\eta_1 = \sum_{\tau \in \sigma H} \tau(\zeta_p)$, where $\sigma$ is a generator of $\text{Gal}(K/\mathbb{Q})$ (the two *periods* of $\zeta_p$ with respect to $H$, i.e., the sum of the conjugates of $\zeta_p$ with respect to the two cosets of $H$ in $G$, cf. Section 5).

**(a)** Prove that $\sigma(\eta_0) = \eta_1$, $\sigma(\eta_1) = \eta_0$ and that

$$\eta_0 = \sum_{a=\text{square}} \zeta_p^a \quad , \quad \eta_1 = \sum_{b \neq \text{square}} \zeta_p^b,$$

where the sums are over the squares and nonsquares (respectively) in $(\mathbb{Z}/p\mathbb{Z})^\times$. [Observe that $H$ is the subgroup of squares in $(\mathbb{Z}/p\mathbb{Z})^\times$.]

**(b)** Prove that $\eta_0 + \eta_1 = (\zeta_p, 1) = -1$ and $\eta_0 - \eta_1 = (\zeta_p, -1)$ where $(\zeta_p, 1)$ and $(\zeta_p, -1)$ are two of the Lagrange resolvents of $\zeta_p$.

**(c)** Let $g = \sum_{i=0}^{p-1} \zeta_p^{i^2}$ (the classical *Gauss sum*). Prove that

$$g = (\zeta_p, -1) = \sum_{i=0}^{p-2} (-1)^i \sigma^i(\zeta_p).$$

**(d)** Prove that $\tau g = g$ if $\tau \in H$ and $\tau g = -g$ if $\tau \notin H$. Conclude in particular that $[\mathbb{Q}(g) : \mathbb{Q}] = 2$. Recall that complex conjugation is the automorphism $\sigma_{-1}$ on $K$ (cf. Exercise 7 of Section 5). Conclude that $\bar{g} = g$ if $-1$ is a square mod $p$ (i.e., if $p \equiv 1 \bmod 4$) and $\bar{g} = -g$ if $-1$ is not a square mod $p$ (i.e., if $p \equiv 3 \bmod 4$) where $\bar{g}$ denotes the complex conjugate of $g$.

**(e)** Prove that $g\bar{g} = p$. [The complex conjugate of a root of unity is its reciprocal. Then $\bar{g} = \sum_{j=0}^{p-2} (-1)^j (\sigma^j(\zeta_p))^{-1}$ gives

$$g\bar{g} = \sum_{i,j=0}^{p-2} (-1)^i (-1)^j \frac{\sigma^i(\zeta_p)}{\sigma^j(\zeta_p)} = \sum_{i,j=0}^{p-2} (-1)^{i-j} \sigma^j \left[ \frac{\sigma^{i-j}(\zeta_p)}{\zeta_p} \right]$$

$$= \sum_{k=0}^{p-2} (-1)^k \sum_{j=0}^{p-2} \sigma^j \left[ \frac{\sigma^k(\zeta_p)}{\zeta_p} \right]$$

where $k = i - j$. If $k = 0$ the element $\dfrac{\sigma^k(\zeta_p)}{\zeta_p}$ is 1, and if $k \neq 0$ then this is a primitive $p^{\text{th}}$ root of unity. Use the previous exercise to conclude that the inner sum is $p - 1$ when $k = 0$ and is $-1$ otherwise.]

**(f)** Conclude that $g^2 = (-1)^{(p-1)/2} p$ and that $\mathbb{Q}(\sqrt{(-1)^{(p-1)/2} p})$ is the unique quadratic subfield of $\mathbb{Q}(\zeta_p)$. (Cf. also Exercise 33 of Section 6.)

**12.** Let $L$ be the Galois closure of the finite extension $\mathbb{Q}(\alpha)$ of $\mathbb{Q}$. For any prime $p$ dividing the order of $\mathrm{Gal}(L/\mathbb{Q})$ prove there is a subfield $F$ of $L$ with $[L : F] = p$ and $L = F(\alpha)$.

**13.** Let $F$ be a subfield of the real numbers $\mathbb{R}$. Let $a$ be an element of $F$ and let $K = F(\sqrt[n]{a})$ where $\sqrt[n]{a}$ denotes a real $n^{\text{th}}$ root of $a$. Prove that if $L$ is any Galois extension of $F$ contained in $K$ then $[L : F] \leq 2$.

**14.** This exercise shows that in general it is necessary to use complex numbers when expressing real roots in terms of radicals and generalizes the *Casus irreducibilis* of cubic equations.

Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial all of whose roots are real. Suppose further that one of the roots, $\alpha$, of $f(x)$ can be expressed in terms of *real* radicals (i.e., there is a root extension of real fields $\mathbb{Q} = K_0 \subset K_1 \subset \ldots \subset K_m \subset \mathbb{R}$ with $K_{i+1} = K_i(\sqrt[n_i]{a_i})$, $i = 1, 2, \ldots, m-1$, for some integers $n_i$ and some $a_i \in K_i$ and $\alpha \in K_m$). Prove that the Galois group of $f(x)$ is a 2-group. Conclude in particular that the degree of $f(x)$ is a power of 2 and that the real roots of such a polynomial can be expressed entirely in terms of real radicals if and only if these roots can be constructed by straightedge and compass. [The argument is similar to the case of cubics. Let $L \in \mathbb{R}$ be the Galois closure of $\mathbb{Q}(\alpha)$ and suppose the order of $\mathrm{Gal}(L/\mathbb{Q})$ is divisible by some odd prime $p$. Let $F$ be a subfield of $L$ with $[L : F] = p$ and $L = F(\alpha)$ (by Exercise 12) and consider the composite fields $K_i' = FK_i$, $i = 0, 1, \ldots, m$. These are again real radical extensions and by the argument in the text for the *Casus irreducibilis*, we may assume each $[K_{i+1}' : K_i']$ is a prime. Since $\alpha \notin F = FK_0$, there is an integer $s$ with $\alpha \notin K_{s-1}'$, $\alpha \in K_s'$. Since the extensions are of prime degree, we have $K_s' = K_{s-1}'(\alpha)$. Since $L = F(\alpha)$ is Galois of degree $p$, $K_s'$ is a Galois extension of $K_{s-1}'$ of degree $p$, contradicting the previous exercise.]

**15.** *('Cardano's Formulas' for a Cubic in Characteristic 2)* Suppose $f(x) = x^3 + px + q$ is an irreducible cubic over a field of characteristic 2. Let $\rho$ be a primitive $3^{\text{rd}}$ root of unity and let $\theta$, $\theta'$ be the roots of the quadratic $x^2 + qx + (p^3 + q^2)$ (cf. Exercise 50 of Section 6). Let $\theta_1$ and $\theta_2$ be cube roots of $\rho q + \theta$ and $\rho q + \theta'$, respectively, where the cube roots are chosen so that $\theta_1\theta_2 = p$. Prove that the roots of $f(x)$ are given by $\alpha = \theta_1 + \theta_2$, $\beta = \rho\alpha + \theta_1$, and $\gamma = \rho\alpha + \theta_2 = \alpha + \beta$.

**16.** Let $a$ be a nonzero rational number.

**(a)** Determine when the extension $\mathbb{Q}(\sqrt{ai})$ ($i^2 = -1$) is of degree 4 over $\mathbb{Q}$.

**(b)** When $K = \mathbb{Q}(\sqrt{ai})$ is of degree 4 over $\mathbb{Q}$ show that $K$ is Galois over $\mathbb{Q}$ with the Klein 4-group as Galois group. In this case determine the quadratic extensions of $\mathbb{Q}$ contained in $K$.

**17.** Let $D \in \mathbb{Z}$ be a squarefree integer and let $a \in \mathbb{Q}$ be a nonzero rational number. Show that $\mathbb{Q}(\sqrt{a\sqrt{D}})$ cannot be a cyclic extension of degree 4 over $\mathbb{Q}$.

$\sqrt{a\sqrt{D}})$ c

**18.** Let $D \in \mathbb{Z}$ be a squarefree integer and let $a \in \mathbb{Q}$ be a nonzero rational number. Prove that if $\mathbb{Q}(\sqrt{a\sqrt{D}})$ is Galois over $\mathbb{Q}$ then $D = -1$.

**19.** Let $D \in \mathbb{Z}$ be a squarefree integer and let $K = \mathbb{Q}(\sqrt{D})$.

**(a)** Prove that if $D = s^2 + t^2$ is the sum of two rational squares then there exists an extension $L/\mathbb{Q}$ containing $K$ which is Galois over $\mathbb{Q}$ with a cyclic Galois group of order 4. [Consider the extension $\mathbb{Q}(\sqrt{D + s\sqrt{D}})$.] (Note also that $D$ is the sum of two rational squares if and only if $D$ is also the sum of two integer squares, so one may assume $s$ and $t$ are integral without loss.)

**(b)** Prove conversely that if $K$ can be embedded in a cyclic extension $L$ of degree 4 as in (a) then $D$ is the sum of two squares. [One approach: (i) observe first that $L$ is quadratic over $K$, so $L = K(\sqrt{a + b\sqrt{D}})$ for some $a, b \in \mathbb{Q}$, (ii) show that $L$ contains the quadratic subfield $\mathbb{Q}(\sqrt{a^2 - b^2 D})$, which must be $\mathbb{Q}(\sqrt{D})$ if $L/\mathbb{Q}$ is cyclic, and use Exercise 7.]

**(c)** Conclude in particular that $\mathbb{Q}(\sqrt{3})$ is not a subfield of any cyclic extension of degree 4 over $\mathbb{Q}$. Similarly conclude that the fields $\mathbb{Q}(\sqrt{D})$ for squarefree integers $D < 0$ are never contained in cyclic extensions of degree 4 over $\mathbb{Q}$ (this gives an alternate proof for Exercise 19, Section 6).

**20.** Let $p$ be a prime. Show that any solvable subgroup of $S_p$ of order divisible by $p$ is contained in the normalizer of a Sylow $p$-subgroup of $S_p$ (a Frobenius group of order $p(p-1)$). Conclude that an irreducible polynomial $f(x) \in \mathbb{Q}[x]$ of degree $p$ is solvable by radicals if and only if its Galois group is contained in the Frobenius group of order $p(p-1)$. [Let $G \leq S_p$ be a solvable subgroup of order divisible by $p$. Then $G$ contains a $p$-cycle, hence is transitive on $\{1, 2, \ldots, p\}$. Let $H < G$ be the stabilizer in $G$ of the element 1, so $H$ has index $p$ in $G$. Show that $H$ contains no nontrivial normal subgroups of $G$ (note that the conjugates of $H$ are the stabilizers of the other points). Let $G^{(n-1)}$ be the last nontrivial subgroup in the derived series for $G$. Show that $H \cap G^{(n-1)} = 1$ and conclude that $|G^{(n-1)}| = p$, so that the Sylow $p$-subgroup of $G$ (which is also a Sylow $p$-subgroup in $S_p$) is normal in $G$.]

**21.** (*Criterion for the Solvability of a Quintic*) By the previous exercise, an irreducible polynomial $f(x)$ in $\mathbb{Q}[x]$ of degree 5 can be solved by radicals if and only if its Galois group (considered as a subgroup of $S_5$) is contained in the Frobenius group of order 20. It is known that this is the case if and only if an associated polynomial $g(x)$ of degree 6 has a rational root (cf. Dummit, *Solving Solvable Quintics*, Math. Comp., 57(1991), pp. 387–401). If the quintic is in the general form (where a translation is performed so that the coefficient of $x^4$ is zero)

$$f(x) = x^5 + px^3 + qx^2 + rx + s \qquad p, q, r, s \in \mathbb{Q}$$

then the associated polynomial of degree 6 is

$$\begin{aligned} g(x) = {} & x^6 + 8rx^5 + (2pq^2 - 6p^2r + 40r^2 - 50qs)\,x^4 \\ & + (-2q^4 + 21pq^2r - 40p^2r^2 + 160r^3 - 15p^2qs - 400qrs + 125ps^2)\,x^3 \\ & + (p^2q^4 - 8q^4r + 9p^4r^2 - 136p^2r^3 + 625q^2s^2 + 400r^4 - 6p^3q^2r \\ & \quad + 76pq^2r^2 - 50pq^3s - 1400qr^2s + 500prs^2 + 90p^2qrs)\,x^2 \\ & + (-108p^5s^2 + 32p^4r^3 - 256p^2r^4 - 3125s^4 + 512r^5 - 2pq^6 + 3q^4r^2 \\ & \quad - 58q^5s + 2750q^2rs^2 - 31p^3q^3s - 500pr^2s^2 + 19p^2q^4r \\ & \quad - 51p^3q^2r^2 + 76pq^2r^3 - 2400qr^3s - 325p^2q^2s^2 + 525p^3rs^2 \\ & \quad + 625pqs^3 + 117p^4qrs + 105pq^3rs + 260p^2qr^2s)\,x \\ & + (q^8 + 256r^6 + 17q^4r^3 - 27p^7s^2 - 4p^6r^3 + 48p^4r^4 - 192p^2r^5 \\ & \quad + 3125p^2s^4 - 9375rs^4 - 1600qr^4s - 99p^5rs^2 - 125pq^4s^2 \\ & \quad - 124q^5rs + 3250q^2r^2s^2 - 2000pr^3s^2 - 13pq^6r + p^5q^2r^2 \\ & \quad + 65p^2q^4r^2 - 128p^3q^2r^3 - 16pq^2r^4 - 4p^5q^3s - 12p^2q^5s \\ & \quad - 150p^4q^2s^2 + 1200p^3r^2s^2 + 18p^6qrs + 12p^3q^3rs + 196p^4qr^2s \\ & \quad + 590pq^3r^2s - 160p^2qr^3s - 725p^2q^2rs^2 - 1250pqrs^3). \end{aligned}$$

In the particular case where $f(x) = x^5 + Ax + B$ this polynomial is simply

$$g(x) = x^6+8Ax^5+40A^2x^4+160A^3x^3+400A^4x^2+(512A^5-3125B^4)x-9375AB^4+256A^6.$$

**(a)** Use this criterion to prove that the Galois group over $\mathbb{Q}$ of the polynomial $x^5-5x+12$ is the dihedral group of order 10. [Show the associated sixth degree polynomial is

$$x^6 - 40x^5 + 1000x^4 - 20000x^3 + 250000x^2 - 66400000x + 976000000$$

and has $x = 40$ as a rational root. Cf. also Exercise 35 in Section 6.]

**(b)** Use this criterion to prove that $x^5 - x - 1$ is not solvable by radicals.

## 14.8. Computation of Galois Groups over Q

**1.** Let $p$ be a prime. Prove that the polynomial $x^4+1$ splits mod $p$ either into two irreducible quadratics or into 4 linear factors using Corollary 41 together with the knowledge that the Galois group of this polynomial is the Klein 4-group.

**2.** (Cf. Exercise 48 of Section 6).

**(a)** Let $K$ be the splitting field of $x^6 - 2x^3 - 2$. Prove that if $[K : \mathbb{Q}] = 12$ then $K = \mathbb{Q}(\sqrt[3]{2}, i, \sqrt{3})$ and $K$ is generated over the biquadratic field $F = \mathbb{Q}(i, \sqrt{3})$ by $\alpha = \sqrt[3]{1+\sqrt{3}}$ and by $\beta = \sqrt[3]{1-\sqrt{3}}$. Show that if this is the case then the elements of order 3 in $\mathrm{Gal}(K/\mathbb{Q})$ lie in $\mathrm{Gal}(K/F)$. Conclude that any element of $\mathrm{Gal}(K/\mathbb{Q})$ of order 3 maps $\alpha$ to another cube root of $1+\sqrt{3}$ and maps $\beta$ to another cube root of $1-\sqrt{3}$ and if it is the identity on $\alpha$ or $\beta$ then it is the identity on all of $K$.

**(b)** Show that the factorization of $f(x)$ into irreducibles over $\mathbb{F}_{13}$ is the polynomial $(x-7)(x-8)(x-11)(x^3+3)$ and use Corollary 41 to show that $[K : \mathbb{Q}] = 36$.

**(c)** Knowing that $G = \mathrm{Gal}(K/\mathbb{Q})$ is of order 36 determine all the elements of $G$ explicitly and in particular show that $G$ is isomorphic to $S_3 \times S_3$.

**3.** Prove that the Galois group of $x^5 + 20x + 16$ is $A_5$.

**4.** Prove that the Galois group of $x^5 + x^4 - 4x^3 - 3x^2 + 3x + 1$ is cyclic of order 5. [Show this is the minimal polynomial of $\zeta_{11} + \zeta_{11}^{-1}$.]

**5.** Prove that the Galois group of $x^5 + 11x + 44$ is the dihedral group $D_{10}$ (cf. Exercise 21 of Section 7).

**6.** Prove that the Galois group of $x^5 + 15x + 12$ is $F_{20}$, the Frobenius group of order 20 (cf. Exercise 21 of Section 7).

**7.** Prove that the Galois group of $x^6 + 24x - 20$ is $A_6$.

**8.** Prove that the Galois group of $x^7 + 7x^4 + 14x + 3$ is $A_7$.

**9.** Determine a polynomial of degree 7 whose Galois group is cyclic of order 7.

**10.** Determine the probable Galois group of $x^7 - 7x + 3$.

## 14.9. Transcendental Extensions, Inseparable Extensions, Infinite Galois Groups

**1.** Prove that every purely inseparable extension is normal.

**2.** Let $p$ be a prime and let $K = \mathbb{F}_p(x, y)$ with $x$ and $y$ independent transcendentals over $\mathbb{F}_p$. Let $F = \mathbb{F}_p(x^p - x, y^p - x)$.

**(a)** Prove that $[K : F] = p^2$ and the separable degree and inseparable degree of $K/F$ are both equal to $p$.

**(b)** Prove that there is a subfield $E$ of $K$ containing $F$ which is purely inseparable over $F$ of degree $p$ (so then $K$ is a separable extension of $E$ of degree $p$). [Let $s = x^p - x \in F$ and $t = y^p - x \in F$ and consider $s - t$.]

**3.** Let $p$ be an odd prime, let $s$ and $t$ be independent transcendentals over $\mathbb{F}_p$, and let $F$ be the field $\mathbb{F}_p(s, t)$. Let $\beta$ be a root of $x^2 - sx + t = 0$ and let $\alpha$ be a root of $x^p - \beta = 0$ (in some algebraic closure of $F$). Set $E = F(\beta)$ and $K = F(\alpha)$.

**(a)** Prove that $E$ is a Galois extension of $F$ of degree 2 and that $K$ is a purely inseparable extension of $E$ of degree $p$.

**(b)** Prove that $K$ is not a normal extension of $F$. [If it were, conjugate $\beta$ over $F$ to show that $K$ would contain a $p^{\text{th}}$ root of $s$ and then also a $p^{\text{th}}$ root of $t$, so $[K : F] \geq p^2$, a contradiction.]

**(c)** Prove that there is no field $K_0$ such that $F \subseteq K_0 \subseteq K$ with $K_0/F$ purely inseparable and $K/K_0$ separable. [If there were such a field, use Exercise 1 and the fact that the composite of two normal extensions is again normal to show that $K$ would be a normal extension of $F$.]

**4.** Under the notation of the previous exercise prove that $\alpha, s$ is a separating transcendence base for $K$ over $\mathbb{F}_p$.

**5.** Let $p$ be a prime, let $t$ be transcendental over $\mathbb{F}_p$ and let $K$ be obtained by adjoining to $\mathbb{F}_p(t)$ all $p$-power roots of $t$. Prove that $K$ has transcendence degree 1 over $\mathbb{F}_p$ and has no separating transcendence base.

**6.** Show that if $t$ is transcendental over $\mathbb{Q}$ then $\mathbb{Q}(t, \sqrt{t^3 - t})$ is not a purely transcendental extension of $\mathbb{Q}$. (This is an example of what is called an *elliptic* function field.)

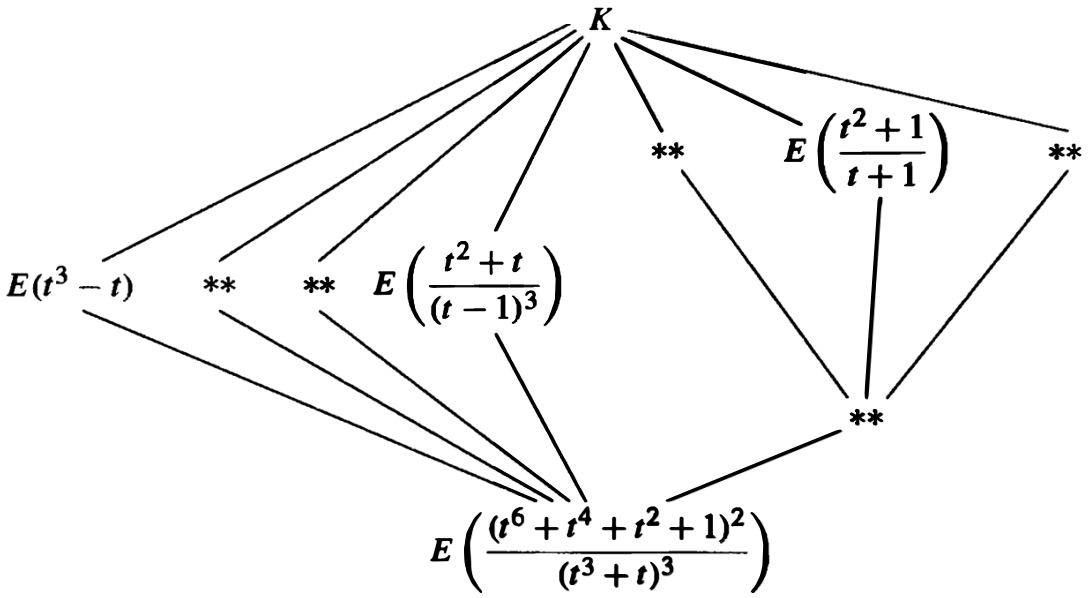
**7.** Let $k$ be the field with 4 elements, $t$ a transcendental over $k$, $F = k(t^4 + t)$ and $K = k(t)$.
- **(a)** Show that $[K : F] = 4$.
- **(b)** Show that $K$ is separable over $F$.
- **(c)** Show that $K$ is Galois over $F$.
- **(d)** Describe the lattice of subgroups of the Galois group and the corresponding lattice of subfields of $K$, giving each subfield in the form $k(r)$, for some rational function $r$.

**8.** Let $p$ be an odd prime, $k$ an algebraically closed field of characteristic $p$ and let $t$ be transcendental over $k$. Suppose $F$ is a degree 2 field extension of $k(t)$. Show that $F$ can be written in the form $k(t, y)$, for some $y \in F$ with $y^2 \in k(t)$ and $y$ transcendental over $k$. If $y^2 = 4t^3 - t - 1$, find $[F : k(y)]$ and describe $k(t) \cap k(y)$ as $k(r)$, for some $r \in k(t)$.

**9.** Let $t$ be transcendental over $\mathbb{F}_3$, let $K = \mathbb{F}_3(t)$, let $G = \mathrm{Aut}(K/\mathbb{F}_3)$ and let $F$ be the fixed field of $G$.

**(a)** Prove $G \cong S_4$ and deduce that there is a unique field $E$ with $F \subseteq E \subseteq K$ and $[E : F] = 2$. [Recall that $G \cong PGL_2(\mathbb{F}_3)$; show that $GL_2(\mathbb{F}_3)$ permutes the 4 lines in a 2-dimensional vector space over $\mathbb{F}_3$ and the kernel of this permutation representation is the scalar matrices.]

**(b)** Complete the description of the lattice of subfields of $K$ containing $E$:




Give each subfield in the form $E(r)$ for some rational function $r$. (The lattice of subgroups of $A_4$ appears in Section 3.5).

**10.** Prove that a purely transcendental proper extension of a field is never algebraically closed.

**11.** Let $S$ be a set of independent transcendentals over a field $F$ and let $\Omega$ be an algebraic closure of $F(S)$. Prove that any permutation on $S$ extends to an element of $\mathrm{Aut}(F(S)/F)$. Prove that any such automorphism of $F(S)$ extends to an automorphism of $\Omega$. Deduce that $\mathbb{C}$ has infinitely many automorphisms.

**12.** Let $K$ be a subfield of $\mathbb{C}$ maximal with respect to the property "$\sqrt{2} \notin K$."

**(a)** Show such a field $K$ exists.

**(b)** Show that $\mathbb{C}$ is algebraic over $K$.

**(c)** Prove that every finite extension of $K$ in $\mathbb{C}$ is Galois with Galois group a cyclic 2-group.

**(d)** Deduce that $[\mathbb{C} : K]$ is countable (and not finite).

**13.** Let $K$ be the fixed field in $\mathbb{C}$ of an automorphism of $\mathbb{C}$. Prove that every finite extension of $K$ in $\mathbb{C}$ is cyclic.

**14.** Let $K_n$ be the splitting field of $(x^2 - p_1)(x^2 - p_2)\cdots(x^2 - p_n)$ over $\mathbb{Q}$, where $p_1, \ldots, p_n$ are the first $n$ primes. Prove that the Galois group of $K_n/\mathbb{Q}$ is an elementary abelian 2-group of order $2^n$.

**15.** Let $K_0 = \mathbb{Q}$ and for $n \geq 0$ define the field $K_{n+1}$ as the extension of $K_n$ obtained by adjoining to $K_n$ all roots of all cubic polynomials over $K_n$. Let $K$ be the union of the subfields $K_n$, $n \geq 0$. Prove that $K$ is a Galois extension of $\mathbb{Q}$. Prove that every cubic polynomial over $K$ splits completely over $K$. Prove that there are nontrivial algebraic extensions of $K$.

**16.** Let $F$ be the composite of all the splitting fields of irreducible cubics over $\mathbb{Q}$. Prove that $F$ does not contain all quadratic extensions of $\mathbb{Q}$.

**17.** Let $K_0 = \mathbb{Q}$ and for $n \geq 0$ define the field $K_{n+1}$ as the extension of $K_n$ obtained by adjoining to $K_n$ all radicals of elements in $K_n$. Let $K$ be the union of the subfields $K_n$, $n \geq 0$. Prove that $K$ is a Galois extension of $\mathbb{Q}$. Prove that there are no nontrivial solvable Galois extensions of $K$. Prove that there are nontrivial Galois extensions of $K$.

**18.** Let $F_0 = \mathbb{Q}$ and for $n \geq 0$ define the field $F_{n+1}$ as the extension of $F_n$ obtained by adjoining to $F_n$ all real radicals of elements in $F_n$. Let $F$ be the union of the subfields $F_n$, $n \geq 0$. Let $K^+$ be the fixed field of complex conjugation restricted to the field $K$ in the previous exercise (the maximal real subfield of $K$). Prove that $F \neq K^+$.

**19.** This exercise proves that if $K/F$ is a Galois extension of fields, then $\mathrm{Gal}(K/F)$ is isomorphic to $\varprojlim \mathrm{Gal}(L/F)$, where the inverse limit is taken over all the finite Galois extensions $L$ of $F$ contained in $K$.

**(a)** Show that $K$ is the union of the fields $L$.

**(b)** Prove that the map $\varphi : \mathrm{Gal}(K/F) \to \varprojlim \mathrm{Gal}(L/F)$ defined by mapping $\sigma$ in $\mathrm{Gal}(K/F)$ to $(\ldots, \sigma|_L, \ldots)$, where $\sigma|_L$ is the restriction of $\sigma$ to $L$, is a homomorphism.

**(c)** Show that $\varphi$ is injective.

**(d)** If $(\ldots, \sigma_L, \ldots) \in \varprojlim \mathrm{Gal}(L/F)$, define $\sigma \in \mathrm{Gal}(K/F)$ by $\sigma(\alpha) = \sigma_L(\alpha)$ if $\alpha \in L$. Prove that $\sigma$ is a well defined automorphism and deduce that $\varphi$ is surjective.

# 15. Commutative Rings and Algebraic Geometry

## 15.1. Noetherian Rings and Affine Algebraic Sets

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**1.** Prove the converse to Hilbert's Basis Theorem: if the polynomial ring $R[x]$ is Noetherian, then $R$ is Noetherian.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**2.** Show that each of the following rings are not Noetherian by exhibiting an explicit infinite increasing chain of ideals:
   **(a)** the ring of continuous real valued functions on $[0, 1]$,
   **(b)** the ring of all functions from any infinite set $X$ to $\mathbb{Z}/2\mathbb{Z}$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**3.** Prove that the field $k(x)$ of rational functions over $k$ in the variable $x$ is not a finitely generated $k$-algebra. (Recall that $k(x)$ is the field of fractions of the polynomial ring $k[x]$. Note that $k(x)$ *is* a finitely generated *field extension* over $k$.)

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**4.** Prove that if $R$ is Noetherian, then so is the ring $R[[x]]$ of formal power series in the variable $x$ with coefficients from $R$ (cf. Exercise 3, Section 7.2). [Mimic the proof of Hilbert's Basis Theorem.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**5.** (*Fitting's Lemma*) Suppose $M$ is a Noetherian $R$-module and $\varphi : M \to M$ is an $R$-module endomorphism of $M$. Prove that $\ker(\varphi^n) \cap \text{image}(\varphi^n) = 0$ for $n$ sufficiently large. Show that if $\varphi$ is surjective, then $\varphi$ is an isomorphism. [Observe that $\ker(\varphi) \subseteq \ker(\varphi^2) \subseteq \ldots$.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**6.** Suppose that $0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$ is an exact sequence of $R$-modules. Prove that $M$ is a Noetherian $R$-module if and only if $M'$ and $M''$ are Noetherian $R$-modules.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**7.** Prove that submodules, quotient modules, and finite direct sums of Noetherian $R$-modules are again Noetherian $R$-modules.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**8.** If $R$ is a Noetherian ring, prove that $M$ is a Noetherian $R$-module if and only if $M$ is a finitely generated $R$-module. (Thus any submodule of a finitely generated module over a Noetherian ring is also finitely generated.)

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**9.** For $k$ a field show that any subring of the polynomial ring $k[x]$ containing $k$ is Noetherian. Give an example to show such subrings need not be U.F.D.s. [If $k \subset R \subseteq k[x]$ and $y \in R - k$ show that $k[x]$ is a finitely generated $k[y]$-module; then use the previous two exercises. For the second, consider $k[x^2, x^3]$.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**10.** Prove that the subring $k[x, x^2y, x^3y^2, \ldots, x^i y^{i-1}, \ldots]$ of the polynomial ring $k[x, y]$ is *not* a Noetherian ring, hence not a finitely generated $k$-algebra. (Thus subrings of Noetherian rings need not be Noetherian and subalgebras of finitely generated $k$-algebras need not be finitely generated.)

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**11.** Suppose $R$ is a commutative ring in which all the prime ideals are finitely generated. This exercise proves that $R$ is Noetherian.

**(a)** Prove that if the collection of ideals of $R$ that are not finitely generated is nonempty, then it contains a maximal element $I$, and that $R/I$ is a Noetherian ring.
**(b)** Prove that there are finitely generated ideals $J_1$ and $J_2$ containing $I$ with $J_1 J_2 \subseteq I$ and that $J_1 J_2$ is finitely generated. [Observe that $I$ is not a prime ideal.]
**(c)** Prove that $I/J_1 J_2$ is a finitely generated $R/I$-submodule of $J_1/J_1 J_2$. [Use Exercise 8.]
**(d)** Show that (c) implies the contradiction that $I$ would be finitely generated over $R$ and deduce that $R$ is Noetherian.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**12.** Suppose $R$ is a Noetherian ring and $S$ is a finitely generated $R$-algebra. If $T \subseteq S$ is an $R$-algebra such that $S$ is a finitely generated $T$-*module*, prove that $T$ is a finitely generated $R$-algebra. [If $s_1, \ldots, s_n$ generate $S$ as an $R$-algebra, and $s_1', \ldots, s_m'$ generate $S$ as a $T$-module, show that the elements $s_i$ and $s_j' s_k'$ can be written as finite $T$-linear combinations of the $s_i'$. If $T_0$ is the $R$-subalgebra generated by the coefficients of these linear combinations, show $S$ (hence $T_0$) is finitely generated (by the $s_i'$) as a $T_0$-module, and conclude that $T$ is finitely generated as an $R$-algebra.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**13.** Verify properties (1) to (10) of the maps $\mathcal{Z}$ and $\mathcal{I}$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**14.** Show that the affine algebraic sets in $\mathbb{A}^1$ over any field $k$ are $\emptyset$, $k$, and finite subsets of $k$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**15.** If $k = \mathbb{F}_2$ and $V = \{(0, 0), (1, 1)\} \subset \mathbb{A}^2$, show that $\mathcal{I}(V)$ is the product ideal $\mathfrak{m}_1\mathfrak{m}_2$ where $\mathfrak{m}_1 = (x, y)$ and $\mathfrak{m}_2 = (x - 1, y - 1)$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**16.** Suppose that $V$ is a finite algebraic set in $\mathbb{A}^n$. If $V$ has $m$ points, prove that $k[V]$ is isomorphic as a $k$-algebra to $k^m$. [Use the Chinese Remainder Theorem.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**17.** If $k$ is a finite field show that every subset of $\mathbb{A}^n$ is an affine algebraic set.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**18.** If $k = \mathbb{F}_q$ is the finite field with $q$ elements show that $\mathcal{I}(\mathbb{A}^1) = (x^q - x) \subset k[x]$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**19.** For each nonconstant $f \in k[x]$ describe $\mathcal{Z}(f) \subseteq \mathbb{A}^1$ in terms of the unique factorization of $f$ in $k[x]$, and then use this to describe $\mathcal{I}(\mathcal{Z}(f))$. Deduce that $\mathcal{I}(\mathcal{Z}(f)) = (f)$ if and only if $f$ is the product of distinct linear factors in $k[x]$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**20.** If $f$ and $g$ are irreducible polynomials in $k[x, y]$ that are not associates (do not divide each other), show that $\mathcal{Z}((f, g))$ is either $\emptyset$ or a finite set in $\mathbb{A}^2$. [If $(f, g) \neq (1)$, show $(f, g)$ contains a nonzero polynomial in $k[x]$ (and similarly a nonzero polynomial in $k[y]$) by letting $R = k[x]$, $F = k(x)$, and applying Gauss's Lemma to show $f$ and $g$ are relatively prime in $F[y]$.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**21.** Identify each $2 \times 2$ matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with entries from $k$ with the point $(a, b, c, d)$ in $\mathbb{A}^4$. Show that the group $SL_2(k)$ of matrices of determinant 1 is an algebraic set in $\mathbb{A}^4$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**22.** Prove that $SL_n(k)$ is an affine algebraic set in $\mathbb{A}^{n^2}$. [Generalize the preceding exercise.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**23.** Let $V$ be any **line** in $\mathbb{R}^2$ (the zero set of any nonzero **linear** polynomial $ax + by - c$). Prove that $\mathbb{R}[V]$ is isomorphic as an $\mathbb{R}$-algebra to the polynomial ring $\mathbb{R}[x]$, and give the corresponding isomorphism from $\mathbb{A}^1$ to $V$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**24.** Let $V = \mathcal{Z}(xy - z) \subseteq \mathbb{A}^3$. Prove that $V$ is isomorphic to $\mathbb{A}^2$ and provide an explicit isomorphism $\varphi$ and associated $k$-algebra isomorphism $\widetilde{\varphi}$ from $k[V]$ to $k[\mathbb{A}^2]$, along with their inverses. Is $V = \mathcal{Z}(xy - z^2)$ isomorphic to $\mathbb{A}^2$?

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**25.** Suppose $V \subseteq \mathbb{A}^n$ is an affine algebraic set and $f \in k[V]$. The *graph* of $f$ is the collection of points $\{(a_1, \ldots, a_n, f(a_1, \ldots, a_n))\}$ in $\mathbb{A}^{n+1}$. Prove that the graph of $f$ is an affine algebraic set isomorphic to $V$. [The morphism in one direction maps $(a_1, \ldots, a_n)$ to $(a_1, \ldots, a_n, f(a_1, \ldots, a_n))$.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**26.** Let $V = \mathcal{Z}(xz - y^2, yz - x^3, z^2 - x^2y) \subseteq \mathbb{A}^3$.

**(a)** Prove that the map $\varphi : \mathbb{A}^1 \to V$ defined by $\varphi(t) = (t^3, t^4, t^5)$ is a surjective morphism. [For the surjectivity, if $(x, y, z) \neq (0, 0, 0)$, let $t = y/x$.]

**(b)** Describe the corresponding $k$-algebra homomorphism $\tilde{\varphi} : k[V] \to k[\mathbb{A}^1]$ explicitly.

**(c)** Prove that $\varphi$ is not an isomorphism.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**27.** Suppose $\varphi : V \to W$ is a morphism of affine algebraic sets. If $W'$ is an affine algebraic subset of $W$ prove that the preimage $V' = \varphi^{-1}(W')$ of $W'$ in $V$ is an affine algebraic subset of $V$. If $W' = \mathcal{Z}(I)$ show that $V' = \mathcal{Z}(\tilde{\varphi}(I))$ for the corresponding morphism $\tilde{\varphi} : k[W] \to k[V]$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**28.** Prove that if $V$ and $W$ are affine algebraic sets, then so is $V \times W$ and $k[V \times W] \cong k[V] \otimes_k k[W]$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following seven exercises introduce the notion of the *associated primes* of an $R$-module $M$. Cf. also Exercises 30–40 in Section 4 and Exercises 25–30 in Section 5.

**Definition.** A prime ideal $P$ of $R$ is said to be *associated* to the $R$-module $M$ (sometimes called an *assassin* for $M$) if $P$ is the annihilator of some element $m$ of $M$, i.e., if $M$ contains a submodule $Rm$ isomorphic to $R/P$. The collection of associated primes for $M$ is denoted $\mathrm{Ass}_R(M)$.

When $M = I$ is an ideal in $R$, it is customary to abuse the terminology and refer instead to the elements of $\mathrm{Ass}_R(R/I)$ (rather than the less interesting collection $\mathrm{Ass}_R(I)$) as the *primes associated to* $I$. (Cf. Exercises 28–29 in Section 5.)

**29.** If $R = \mathbb{Z}$ and $M = \mathbb{Z}/n\mathbb{Z}$, show that $\mathrm{Ass}_R(M)$ consists of the prime ideals $(p)$ for the prime divisors $p$ of $n$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following seven exercises introduce the notion of the *associated primes* of an $R$-module $M$. Cf. also Exercises 30–40 in Section 4 and Exercises 25–30 in Section 5.

**Definition.** A prime ideal $P$ of $R$ is said to be *associated* to the $R$-module $M$ (sometimes called an *assassin* for $M$) if $P$ is the annihilator of some element $m$ of $M$, i.e., if $M$ contains a submodule $Rm$ isomorphic to $R/P$. The collection of associated primes for $M$ is denoted $\mathrm{Ass}_R(M)$.

When $M = I$ is an ideal in $R$, it is customary to abuse the terminology and refer instead to the elements of $\mathrm{Ass}_R(R/I)$ (rather than the less interesting collection $\mathrm{Ass}_R(I)$) as the *primes associated to* $I$. (Cf. Exercises 28–29 in Section 5.)

**30.** If $M$ is the union of some collection of submodules $M_i$, prove that $\mathrm{Ass}_R(M)$ is the union of the collection $\mathrm{Ass}_R(M_i)$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following seven exercises introduce the notion of the *associated primes* of an $R$-module $M$. Cf. also Exercises 30–40 in Section 4 and Exercises 25–30 in Section 5.

**Definition.** A prime ideal $P$ of $R$ is said to be *associated* to the $R$-module $M$ (sometimes called an *assassin* for $M$) if $P$ is the annihilator of some element $m$ of $M$, i.e., if $M$ contains a submodule $Rm$ isomorphic to $R/P$. The collection of associated primes for $M$ is denoted $\mathrm{Ass}_R(M)$.

When $M = I$ is an ideal in $R$, it is customary to abuse the terminology and refer instead to the elements of $\mathrm{Ass}_R(R/I)$ (rather than the less interesting collection $\mathrm{Ass}_R(I)$) as the *primes associated to* $I$. (Cf. Exercises 28–29 in Section 5.)

**31.** Suppose that $\mathrm{Ann}(m) = P$, i.e., that $Rm \cong R/P$. Prove that if $0 \neq m' \in Rm$ then $\mathrm{Ann}(m') = P$. Deduce that $\mathrm{Ass}_R(R/P) = \{P\}$. [Observe that $R/P$ is an integral domain.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following seven exercises introduce the notion of the *associated primes* of an $R$-module $M$. Cf. also Exercises 30–40 in Section 4 and Exercises 25–30 in Section 5.

**Definition.** A prime ideal $P$ of $R$ is said to be *associated* to the $R$-module $M$ (sometimes called an *assassin* for $M$) if $P$ is the annihilator of some element $m$ of $M$, i.e., if $M$ contains a submodule $Rm$ isomorphic to $R/P$. The collection of associated primes for $M$ is denoted $\mathrm{Ass}_R(M)$.

When $M = I$ is an ideal in $R$, it is customary to abuse the terminology and refer instead to the elements of $\mathrm{Ass}_R(R/I)$ (rather than the less interesting collection $\mathrm{Ass}_R(I)$) as the *primes associated to* $I$. (Cf. Exercises 28–29 in Section 5.)

**32.** Suppose that $M$ is an $R$-module and that $P$ is a maximal element in the collection of ideals of the form $\mathrm{Ann}(m)$, for $m \in M$. Prove that $P$ is a prime ideal. [If $P = \mathrm{Ann}(m)$ and $ab \in P$, show that $bm \neq 0$ implies $\mathrm{Ann}(m) \subseteq \mathrm{Ann}(bm)$ and use the maximality of $P$ to deduce that $a \in \mathrm{Ann}(bm) = P$.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following seven exercises introduce the notion of the *associated primes* of an $R$-module $M$. Cf. also Exercises 30–40 in Section 4 and Exercises 25–30 in Section 5.

**Definition.** A prime ideal $P$ of $R$ is said to be *associated* to the $R$-module $M$ (sometimes called an *assassin* for $M$) if $P$ is the annihilator of some element $m$ of $M$, i.e., if $M$ contains a submodule $Rm$ isomorphic to $R/P$. The collection of associated primes for $M$ is denoted $\mathrm{Ass}_R(M)$.

When $M = I$ is an ideal in $R$, it is customary to abuse the terminology and refer instead to the elements of $\mathrm{Ass}_R(R/I)$ (rather than the less interesting collection $\mathrm{Ass}_R(I)$) as the *primes associated to* $I$. (Cf. Exercises 28–29 in Section 5.)

**33.** Suppose $R$ is a Noetherian ring and $M \neq 0$ is an $R$-module. Prove that $\mathrm{Ass}_R(M) \neq \emptyset$. [Use Exercise 32.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following seven exercises introduce the notion of the *associated primes* of an $R$-module $M$. Cf. also Exercises 30–40 in Section 4 and Exercises 25–30 in Section 5.

**Definition.** A prime ideal $P$ of $R$ is said to be *associated* to the $R$-module $M$ (sometimes called an *assassin* for $M$) if $P$ is the annihilator of some element $m$ of $M$, i.e., if $M$ contains a submodule $Rm$ isomorphic to $R/P$. The collection of associated primes for $M$ is denoted $\mathrm{Ass}_R(M)$.

When $M = I$ is an ideal in $R$, it is customary to abuse the terminology and refer instead to the elements of $\mathrm{Ass}_R(R/I)$ (rather than the less interesting collection $\mathrm{Ass}_R(I)$) as the *primes associated to* $I$. (Cf. Exercises 28–29 in Section 5.)

**34.** If $L$ is a submodule of $M$ with quotient $N \cong M/L$, prove that there are containments $\mathrm{Ass}_R(N) \subseteq \mathrm{Ass}_R(M) \subseteq \mathrm{Ass}_R(L) \cup \mathrm{Ass}_R(N)$, and show that both containments can be proper. [If $Rm \cong R/P$, show that $Rm \cap L = 0$ implies $P \in \mathrm{Ass}_R(N)$ and if $Rm \cap L \neq 0$ then $P \in \mathrm{Ass}_R(L)$ (by Exercise 31). For the second statement, consider $n\mathbb{Z} \subset \mathbb{Z}$.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following seven exercises introduce the notion of the *associated primes* of an $R$-module $M$. Cf. also Exercises 30–40 in Section 4 and Exercises 25–30 in Section 5.

**Definition.** A prime ideal $P$ of $R$ is said to be *associated* to the $R$-module $M$ (sometimes called an *assassin* for $M$) if $P$ is the annihilator of some element $m$ of $M$, i.e., if $M$ contains a submodule $Rm$ isomorphic to $R/P$. The collection of associated primes for $M$ is denoted $\mathrm{Ass}_R(M)$.

When $M = I$ is an ideal in $R$, it is customary to abuse the terminology and refer instead to the elements of $\mathrm{Ass}_R(R/I)$ (rather than the less interesting collection $\mathrm{Ass}_R(I)$) as the *primes associated to* $I$. (Cf. Exercises 28–29 in Section 5.)

**35.** Suppose $M$ is an $R$-module and let $\mathcal{S}$ be a subset of the prime ideals in $\mathrm{Ass}_R(M)$. Prove there is a submodule $N$ of $M$ with $\mathrm{Ass}_R(N) = \mathcal{S}$ and $\mathrm{Ass}_R(M/N) = \mathrm{Ass}_R(M) - \mathcal{S}$. [Consider the collection of submodules $N'$ of $M$ with $\mathrm{Ass}_R(N') \subseteq \mathcal{S}$. Use Exercise 30 and Zorn's Lemma to show that there is a maximal submodule $N$ subject to $\mathrm{Ass}_R(N) \subseteq \mathcal{S}$. If $P \in \mathrm{Ass}_R(M/N)$, there is a submodule $M'/N \cong R/P$. Use the previous exercise to show that $\mathrm{Ass}_R(M') \subseteq \mathrm{Ass}_R(N) \cup \{P\}$ and then use maximality of $N$ to show $P \in \mathrm{Ass}_R(M) - \mathcal{S}$, so that $\mathrm{Ass}_R(M/N) \subseteq \mathrm{Ass}_R(M) - \mathcal{S}$ and $\mathrm{Ass}_R(N) \subseteq \mathcal{S}$. Use the previous exercise again to conclude that equality holds in each.]

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

Suppose $M$ is a finitely generated module over the commutative ring $R$ with generators $m_1, \ldots, m_n$. The *Fitting ideal* $\mathcal{F}_R(M)$ (of level 0) of $M$ (also called a *determinant ideal*) is the ideal in $R$ generated by the determinants of all $n \times n$ matrices $A = (r_{ij})$ where $r_{ij} \in R$ and $r_{i1}m_1 + \cdots + r_{in}m_n = 0$ in $M$, i.e., the rows of $A$ consist of the coefficients in $R$ of relations among the generators $m_i$ ($A$ is called an $n \times n$ "relations matrix" for $M$). The following five exercises outline some of the properties of the Fitting ideal.

**36.** **(a)** Show that the Fitting ideal of $M$ is also the ideal in $R$ generated by all the $n \times n$ minors of all $p \times n$ matrices $A = (r_{ij})$ for $p \geq 1$ whose rows consist of the coefficients in $R$ of relations among the generators $m_i$.

**(b)** Let $A$ be a fixed $p \times n$ matrix as in (a) and let $A'$ be a $p \times n$ matrix obtained from $A$ by any elementary row or column operation. Show that the ideal in $R$ generated by all the $n \times n$ minors of $A$ is the same as the ideal in $R$ generated by all the $n \times n$ minors of $A'$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

Suppose $M$ is a finitely generated module over the commutative ring $R$ with generators $m_1, \ldots, m_n$. The *Fitting ideal* $\mathcal{F}_R(M)$ (of level 0) of $M$ (also called a *determinant ideal*) is the ideal in $R$ generated by the determinants of all $n \times n$ matrices $A = (r_{ij})$ where $r_{ij} \in R$ and $r_{i1}m_1 + \cdots + r_{in}m_n = 0$ in $M$, i.e., the rows of $A$ consist of the coefficients in $R$ of relations among the generators $m_i$ ($A$ is called an $n \times n$ "relations matrix" for $M$). The following five exercises outline some of the properties of the Fitting ideal.

**37.** Suppose $m_1, \ldots, m_n$ and $m'_1, \ldots, m'_{n'}$ are two sets of $R$-module generators for $M$. Let $\mathcal{F}$ denote the Fitting ideal for $M$ computed using the generators $m_1, \ldots, m_n$ and let $\mathcal{F}'$ denote the Fitting ideal for $M$ computed using the generators $m_1, \ldots, m_n, m'_1, \ldots, m'_{n'}$.

**(a)** Show that $m'_s = a_{s'1}m_1 + \cdots + a_{s'n}m_n$ for some $a_{s'1}, \ldots, a_{s'n} \in R$, and deduce that $(-a_{s'1}, \ldots, -a_{s'n}, 0, \ldots, 0, 1, 0, \ldots 0)$ is a relation among $m_1, \ldots, m_n, m'_1, \ldots, m'_{n'}$.

**(b)** If $A = (r_{ij})$ is an $n \times n$ matrix whose rows are the coefficients of relations among $m_1, \ldots, m_n$ show that $\det A = \det A'$ where $A'$ is an $(n+n') \times (n+n')$ matrix whose rows are the coefficients of relations among $m_1, \ldots, m_n, m'_1, \ldots, m'_{n'}$. Deduce that $\mathcal{F} \subseteq \mathcal{F}'$. [Use (a) to find a block upper triangular $A'$ having $A$ in the upper left block and the $n' \times n'$ identity matrix in the lower right block.]

**(c)** Prove that $\mathcal{F}' \subseteq \mathcal{F}$ and conclude that $\mathcal{F}' = \mathcal{F}$. [Use the previous exercise.]

**(d)** Deduce from (c) that the Fitting ideal $\mathcal{F}_R(M)$ of $M$ is an invariant of $M$ that does not depend on the choice of generators for $M$ used to compute it.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

Suppose $M$ is a finitely generated module over the commutative ring $R$ with generators $m_1, \ldots, m_n$. The *Fitting ideal* $\mathcal{F}_R(M)$ (of level 0) of $M$ (also called a *determinant ideal*) is the ideal in $R$ generated by the determinants of all $n \times n$ matrices $A = (r_{ij})$ where $r_{ij} \in R$ and $r_{i1}m_1 + \cdots + r_{in}m_n = 0$ in $M$, i.e., the rows of $A$ consist of the coefficients in $R$ of relations among the generators $m_i$ ($A$ is called an $n \times n$ "relations matrix" for $M$). The following five exercises outline some of the properties of the Fitting ideal.

**38.** All modules in this exercise are assumed finitely generated.

- **(a)** If $M$ can be generated by $n$ elements prove that $\text{Ann}(M)^n \subseteq \mathcal{F}_R(M) \subseteq \text{Ann}(M)$, where $\text{Ann}(M)$ is the annihilator of $M$ in $R$. [If $A$ is an $n \times n$ relations matrix for $M$, then $AX = 0$, where $X$ is the column matrix whose entries are $m_1, \ldots, m_n$. Multiply by the adjoint of $A$ to deduce that $\det A$ annihilates $M$.]
- **(b)** If $M = M_1 \times M_2$ is the direct product of the $R$-modules $M_1$ and $M_2$ prove that $\mathcal{F}_R(M) = \mathcal{F}_R(M_1)\mathcal{F}_R(M_2)$.
- **(c)** If $M = (R/I_1) \times \cdots \times (R/I_n)$ is the direct product of cyclic $R$-modules for ideals $I_i$ in $R$ prove that $\mathcal{F}_R(M) = I_1 I_2 \ldots I_n$.
- **(d)** If $R = \mathbb{Z}$ and $M$ is a finitely generated abelian group show that $\mathcal{F}_{\mathbb{Z}}(M) = 0$ if $M$ is infinite and $\mathcal{F}_{\mathbb{Z}}(M) = |M|\mathbb{Z}$ if $M$ is finite.
- **(e)** If $I$ is an ideal in $R$ prove that the image of $\mathcal{F}_R(M)$ in the quotient $R/I$ is $\mathcal{F}_{R/I}(M/IM)$.
- **(f)** Prove that $\mathcal{F}_R(M/IM) \subseteq (\mathcal{F}_R(M), I) \subseteq R$.
- **(g)** If $\varphi : M \to M'$ is a surjective $R$-module homomorphism prove $\mathcal{F}_R(M) \subseteq \mathcal{F}_R(M')$.
- **(h)** If $0 \to L \to M \to N \to 0$ is a short exact sequence of $R$-modules, prove that $\mathcal{F}_R(L)\mathcal{F}_R(N) \subseteq \mathcal{F}_R(M)$.
- **(i)** Suppose $R$ is the polynomial ring $k[x, y, z]$ over the field $k$. Let $M = R/(x, y^2, yz, z^2)$ and let $L$ be the submodule $(x, y, z)/(x, y^2, yz, z^2)$ of $M$. Prove that $\mathcal{F}_R(M)$ is $(x, y^2, yz, z^2)$ and $\mathcal{F}_R(L)$ is $(x, y, z)^2$. (This shows that in general the Fitting ideal of a submodule $L$ of $M$ need not contain the Fitting ideal for $M$.)

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

Suppose $M$ is a finitely generated module over the commutative ring $R$ with generators $m_1, \ldots, m_n$. The *Fitting ideal* $\mathcal{F}_R(M)$ (of level 0) of $M$ (also called a *determinant ideal*) is the ideal in $R$ generated by the determinants of all $n \times n$ matrices $A = (r_{ij})$ where $r_{ij} \in R$ and $r_{i1}m_1 + \cdots + r_{in}m_n = 0$ in $M$, i.e., the rows of $A$ consist of the coefficients in $R$ of relations among the generators $m_i$ ($A$ is called an $n \times n$ "relations matrix" for $M$). The following five exercises outline some of the properties of the Fitting ideal.

**39.** Suppose $M$ is an $R$-module and that $\varphi : R^n \to M$ is a surjective $R$-module homomorphism (i.e., $M$ can be generated by $n$ elements). Let $L = \ker \varphi$. Prove that the image of the

$R$-module homomorphism from $\bigwedge^n(L) \to \bigwedge^n(R^n) \cong R$ induced by the inclusion of $L$ in $R^n$ is the Fitting ideal $\mathcal{F}_R(M)$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

Suppose $M$ is a finitely generated module over the commutative ring $R$ with generators $m_1, \ldots, m_n$. The *Fitting ideal* $\mathcal{F}_R(M)$ (of level 0) of $M$ (also called a *determinant ideal*) is the ideal in $R$ generated by the determinants of all $n \times n$ matrices $A = (r_{ij})$ where $r_{ij} \in R$ and $r_{i1}m_1 + \cdots + r_{in}m_n = 0$ in $M$, i.e., the rows of $A$ consist of the coefficients in $R$ of relations among the generators $m_i$ ($A$ is called an $n \times n$ "relations matrix" for $M$). The following five exercises outline some of the properties of the Fitting ideal.

**40.** Suppose $R$ and $S$ are commutative rings, $\varphi : R \to S$ is a ring homomorphism, $M$ is a finitely generated $R$-module, and $M' = S \otimes_R M$ is the $S$-module obtained by **extending** scalars from $R$ to $S$. Prove that the Fitting ideal $\mathcal{F}_S(M')$ for $M'$ over $S$ is the extension to $S$ of the Fitting ideal $\mathcal{F}_R(M)$ for $M$ over $R$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following two exercises indicate how the remainder in Theorem 23 of Chapter 9 can be used to effect computations in quotients of polynomial rings.

**41.** Suppose $\{g_1, \ldots, g_m\}$ is a Gröbner basis for the ideal $I$ in $k[x_1, \ldots, x_n]$. Prove that the monomials $m$ not divisible by any $LT(g_i)$, $1 \leq i \leq m$, give a $k$-vector space basis for the quotient $k[x_1, \ldots, x_n]/I$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

The following two exercises indicate how the remainder in Theorem 23 of Chapter 9 can be used to effect computations in quotients of polynomial rings.

**42.** Let $I = (x^3y - xy^2 + 1, x^2y^2 - y^3 - 1)$ as in Example 1 following Proposition 9.26.

**(a)** Use the previous exercise to show that $\{1, y, y^2, y^3\}$ is a basis for the $k$-vector space $k[x, y]/I$.

**(b)** Compute the $4 \times 4$ multiplication table for the basis vectors in (a).

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**43.** Suppose $K[x_1, \ldots, x_n]$ is a polynomial ring in $n$ variables over a field $K$ and $k$ is a subfield of $K$. If $I$ is an ideal in $k[x_1, \ldots, x_n]$, let $I'$ be the ideal generated by $I$ in $K[x_1, \ldots, x_n]$.

**(a)** If $G$ is a Gröbner basis for the ideal $I$ in $k[x_1, \ldots, x_n]$ with respect to some monomial ordering, show that $G$ is also a Gröbner basis for the ideal $I'$ in $K[x_1, \ldots, x_n]$ with respect to the same monomial ordering. [Use Buchberger's Criterion.]

**(b)** Prove that the **dimension** of the quotient $k[x_1, \ldots, x_n]/I$ as a vector space over $k$ is the same as the **dimension** of the quotient $K[x_1, \ldots, x_n]/I'$ as a vector space over $K$. [One method: use (a) and Exercise 41.]

**(c)** Prove that $I = k[x_1, \ldots, x_n]$ if and only if $I' = K[x_1, \ldots, x_n]$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**44.** Let $V = \mathcal{Z}(x^3 - x^2z - y^2z)$ and $W = \mathcal{Z}(x^2 + y^2 - z^2)$ in $\mathbb{C}^3$. Then $\mathcal{I}(V) = (x^3 - x^2z - y^2z)$ and $\mathcal{I}(W) = (x^2 + y^2 - z^2)$ in $\mathbb{C}[x, y, z]$ (cf. Exercise 23 in Section 3). Show that $\varphi((a, b, c)) = (a^2c - b^2c, 2abc, -a^3)$ defines a morphism from $V$ to $W$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**45.** Let $V = \mathcal{Z}(x^3 + y^3 + 7z^3) \subset \mathbb{C}^3$. Then $\mathcal{I}(V) = (x^3 + y^3 + 7z^3)$ in $\mathbb{C}[x, y, z]$ (cf. Exercise 24 in Section 3).

**(a)** Show that

$$\tilde{\varphi}(x) = x(y^3 - 7z^3), \qquad \tilde{\varphi}(y) = y(7z^3 - x^3), \qquad \tilde{\varphi}(z) = z(x^3 - y^3)$$

defines a $\mathbb{C}$-algebra homomorphism from $k[V]$ to itself.

**(b)** Let $\varphi : V \to V$ be the morphism corresponding to $\tilde{\varphi}$. Observe that $(-2, 1, 1) \in V$ and compute $\varphi((-2, 1, 1)) \in V$.

**(c)** Prove there are infinitely many points $(a, b, c)$ on $V$ with $a, b, c \in \mathbb{Z}$ and the greatest common divisor of $a$, $b$, and $c$ is 1.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**46.** Let $V = \mathcal{Z}(xz+y^2+z^2, xy-xz+yz-2z^2) \subset \mathbb{C}^3$ and $W = \mathcal{Z}(u^3-uv^2+v^3) \subset \mathbb{C}^2$ as in Example 2 following Corollary 9. Show that the map $\varphi((a, b)) = (-2a^2+ab, ab-b^2, a^2-ab)$ defines a morphism from $W$ to $V$. Show the corresponding $\mathbb{C}$-algebra homomorphism from $k[V]$ to $k[W]$ has a kernel generated by $x^2 - 3y^2 + yz$.

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**47.** Define $\Phi : \mathbb{Q}[u, v, w] \to \mathbb{Q}[x, y]$ by $\Phi(u) = x^2 + y$, $\Phi(v) = x + y^2$, and $\Phi(w) = x - y$. Show that neither $x$ nor $y$ is in the image of $\Phi$. Show that $f = 2x^3 - 4xy - 2y^3 - 4y$ is in the image of $\Phi$ and find a polynomial in $\mathbb{Q}[u, v, w]$ mapping to $f$. Show that $\ker \Phi$ is the ideal generated by

$$u^2 - 2uv - 2uw^2 + 4uw + v^2 - 2vw^2 - 4vw + w^4 + 3w^2.$$

Let $R$ be a commutative ring with $1 \neq 0$ and let $k$ be a field.

**48.** **Suppose $\alpha$ is a root of the irreducible** polynomial $p(x) \in k[x]$ and $\beta = f(\alpha)/g(\alpha)$ with polynomials $f(x), g(x) \in k[x]$ where $g(\alpha) \neq 0$.

**(a)** Show $ag + bp = 1$ for some polynomials $a, b \in k[x]$ and show $\beta = h(\alpha)$ where $h = af$.

**(b)** Show that the ideals $(p, y - h)$ and $(p, gy - f)$ are equal in $k[x, y]$.

**(c)** Conclude that the minimal polynomial for $\beta$ is the monic polynomial in $G \cap k[y]$ where $G$ is the reduced Gröbner basis for the ideal $(p, gy - f)$ in $k[x, y]$ for the lexicographic monomial ordering $x > y$.

**(d)** Find the minimal polynomial over $\mathbb{Q}$ of $(3 - \sqrt[3]{2} + \sqrt[3]{4})/(1 + 3\sqrt[3]{2} - 3\sqrt[3]{4})$.

## 15.2. Radicals and Affine Varieties

**1.** Prove (3) of Corollary 22 directly by considering the collection $\mathcal{S}$ of ideals that do not contain a finite product of prime ideals. [If $I$ is a maximal element in $\mathcal{S}$, show that since $I$ is not prime there are ideals $J$, $K$ properly containing $I$ (hence not in $\mathcal{S}$) with $JK \subseteq I$.]

**2.** Let $I$ and $J$ be ideals in the ring $R$. Prove the following statements:

**(a)** If $I^k \subseteq J$ for some $k \geq 1$ then $\operatorname{rad} I \subseteq \operatorname{rad} J$.

**(b)** If $I^k \subseteq J \subseteq I$ for some $k \geq 1$ then $\operatorname{rad} I = \operatorname{rad} J$.

**(c)** $\operatorname{rad}(IJ) = \operatorname{rad}(I \cap J) = \operatorname{rad} I \cap \operatorname{rad} J$.

**(d)** $\operatorname{rad}(\operatorname{rad} I) = \operatorname{rad} I$.

**(e)** $\operatorname{rad} I + \operatorname{rad} J \subseteq \operatorname{rad}(I + J)$ and $\operatorname{rad}(I + J) = \operatorname{rad}(\operatorname{rad} I + \operatorname{rad} J)$.

**3.** Prove that the intersection of two radical ideals is again a radical ideal.

**4.** Let $I = \mathfrak{m}_1\mathfrak{m}_2$ be the product of the ideals $\mathfrak{m}_1 = (x, y)$ and $\mathfrak{m}_2 = (x-1, y-1)$ in $\mathbb{F}_2[x, y]$. Prove that $I$ is a radical ideal. Prove that the ideal $(x^3 - y^2)$ is a radical ideal in $\mathbb{F}_2[x, y]$.

**5.** If $I = (xy, (x - y)z) \subset k[x, y, z]$ prove that rad $I = (xy, xz, yz)$. For this ideal prove directly that $\mathcal{Z}(I) = \mathcal{Z}(\text{rad } I)$, that $\mathcal{Z}(I)$ is not irreducible, and that rad $I$ is not prime.

**6.** Give an example to show that over a field $k$ that is not algebraically closed the containment $I \subseteq \mathcal{I}(\mathcal{Z}(I))$ can be proper even when $I$ is a radical ideal.

**7.** Suppose $R$ and $S$ are rings and $\varphi : R \to S$ is a ring homomorphism. If $I$ is an ideal of $R$ show that $\varphi(\text{rad } I) \subseteq \text{rad}(\varphi(I))$. If in addition $\varphi$ is surjective and $I$ contains the kernel of $\varphi$ show that $\varphi(\text{rad } I) = \text{rad}(\varphi(I))$.

**8.** Suppose the prime ideal $P$ contains the ideal $I$. Prove that $P$ contains the radical of $I$.

**9.** Prove that for any field $k$ the map $\mathcal{Z}$ in the Nullstellensatz is always surjective and the map $\mathcal{I}$ in the Nullstellensatz is always injective. [Use property (10) of the maps $\mathcal{Z}$ and $\mathcal{I}$ in Section 1.] Give examples (over a field $k$ that is not algebraically closed) where $\mathcal{Z}$ is not injective and $\mathcal{I}$ is not surjective.

**10.** Prove that for $k$ a finite field the Zariski topology is the same as the discrete topology: every subset is closed (and open).

**11.** Let $V$ be a variety in $\mathbb{A}^n$ and let $U_1$ and $U_2$ be two subsets of $\mathbb{A}^n$ that are open in the Zariski topology. Prove that if $V \cap U_1 \neq \emptyset$ and $V \cap U_2 \neq \emptyset$ then $V \cap U_1 \cap U_2 \neq \emptyset$. Conclude that *any* nonempty open subset of a variety is *everywhere dense* in the Zariski topology (i.e., its closure is all of $V$).

**12.** Use the fact that nonempty open sets of an affine variety are everywhere dense to prove that an affine variety is connected in the Zariski topology. (A topological space is *connected* if it is not the union of two disjoint, proper, open subsets.)

**13.** Prove that the affine algebraic set $V$ is connected in the Zariski topology if and only if $k[V]$ is not a direct sum of two nonzero ideals. Deduce from this that a variety is connected in the Zariski topology.

**14.** Prove that if $k$ is an infinite field, then the varieties in $\mathbb{A}^1$ are the empty set, the whole space, and the one point subsets. What are the varieties in $\mathbb{A}^1$ in the case of a finite field $k$?

**15.** Suppose $V$ is a hypersurface in $\mathbb{A}^n$ and $\mathcal{I}(V) = (f)$ for some nonconstant polynomial $f \in k[x_1, x_2, \ldots, x_n]$. Prove that $V$ is a variety if and only if $f$ is irreducible.

**16.** Suppose $V \subseteq \mathbb{A}^n$ is an affine variety and $f \in k[V]$. Prove that the *graph* of $f$ (cf. Exercise 25 in Section 1) is an affine variety.

**17.** Prove that any permutation of the elements of a field $k$ is a continuous map from $\mathbb{A}^1$ to itself in the Zariski topology on $\mathbb{A}^1$. Deduce that if $k$ is an infinite field, there are Zariski continuous maps from $\mathbb{A}^1$ to itself that are not polynomials.

**18.** Let $V$ be an affine algebraic set in $\mathbb{A}^n$ over $k = \mathbb{C}$.

**(a)** Prove that morphisms of algebraic sets over $\mathbb{C}$ are continuous in the Euclidean topology (the topology on $\mathbb{C}^n$ obtained by identifying $\mathbb{C}^n$ with $\mathbb{R}^{2n}$ with its usual Euclidean topology).

**(b)** Prove that $V$ is a closed set in the Euclidean topology on $\mathbb{C}^n$ (so the Zariski closed sets of $\mathbb{A}^n$ over $\mathbb{C}$ are also Euclidean closed).

**(c)** Give an example of a set that is closed in the Euclidean topology but is not closed in the Zariski topology, i.e., is not an affine algebraic set (so the Euclidean topology is "finer" than the Zariski topology).

**19.** Give an example of an injective $k$-algebra homomorphism $\widetilde{\varphi} : k[W] \to k[V]$ whose associated morphism $\varphi : V \to W$ is not surjective.

**20.** Suppose $\varphi : V \to W$ is a surjective morphism of affine algebraic sets. Prove that if $V$ is a variety then $W$ is a variety.

**21.** Let $V$ be an algebraic set in $\mathbb{A}^n$ and let $f \in k[V]$. Define $V_f = \{v \in V \mid f(v) \neq 0\}$.

**(a)** Show that $V_f$ is a Zariski open set in $V$ (called a *principal open set* in $V$).

**(b)** Let $J$ be the ideal in $k[x_1, \ldots, x_n, x_{n+1}]$ generated by $\mathcal{I}(V)$ and $x_{n+1}f - 1$, and let $W = \mathcal{Z}(J) \subseteq \mathbb{A}^{n+1}$. Show that $J = \mathcal{I}(W)$ and that the map $\pi : \mathbb{A}^{n+1} \to \mathbb{A}^n$ by projection onto the first $n$ coordinates is a Zariski continuous bijection from $W$ onto $V_f$ (so the principal *open* set $V_f$ in $V$ may be embedded as a *closed* set in some (larger) affine space).

**(c)** If $U$ is any open set in $V$ show that $U = V_{f_1} \cup \cdots \cup V_{f_m}$ for some $f_1, \ldots, f_m \in k[V]$. (This shows that the principal open sets form a *base* for the Zariski topology.)

**22.** Prove that $GL_n(k)$ is an open affine algebraic set in $\mathbb{A}^{n^2}$ and can be embedded as a closed affine algebraic set in $\mathbb{A}^{n^2+1}$. In particular, deduce that the set $k^\times$ of nonzero elements in $\mathbb{A}^1$ embeds into $\mathbb{A}^2$ as the hyperbola $xy = 1$. [Use the preceding exercise.]

**23.** Show that if $k$ is infinite then $\{(a, a^2, a^3) \mid a \in k\} \subset \mathbb{A}^3$ is an affine algebraic variety. If $k$ is finite show that this set is always reducible.

**24.** Let $V = \mathcal{Z}(xz - y^2, yz - x^3, z^2 - x^2y) \subset \mathbb{A}^3$. Show that if $k$ is infinite then $V$ is an affine variety. [Use Exercise 26 of Section 1 and Exercise 20.]

**25.** Suppose $f(x) = x^3 + ax^2 + bx + c$ is an irreducible cubic in $\mathbb{Q}[x]$ of discriminant $D$. Let $I = (x + y + z + a, xy + xz + yz - b, xyz + c)$ in $\mathbb{Q}[x, y, z]$.

**(a)** Prove that $I$ is a prime ideal if and only if $D$ is not a square in $\mathbb{Q}$, in which case $I$ is a maximal ideal and $\mathbb{Q}[x, y, z]/I$ is a splitting field for $f(x)$ over $\mathbb{Q}$.

**(b)** If $D = r^2$, prove that the primary decomposition of $I$ is $I = Q_+ \cap Q_-$ where $Q_\pm = (I, (x - y)(x - z)(y - z) \pm r)$. Prove $Q_+$ and $Q_-$ are maximal ideals, and $\mathbb{Q}[x, y, z]$ modulo $Q_+$ or $Q_-$ is a splitting field for $f(x)$ over $\mathbb{Q}$.

**26.** A topological space $X$ is called *quasicompact* if whenever any collection of closed subsets $V_i$ of $X$ has empty intersection, then some finite number of these also has empty intersection, i.e.,

$$\text{whenever } \bigcap_i V_i = \emptyset \text{ there exists } V_{i_1}, V_{i_2}, \ldots, V_{i_N} \text{ such that } \bigcap_{t=1}^{N} V_{i_t} = \emptyset.$$

Prove that every affine algebraic set is quasicompact. [Translate the definition into a property of ideals in $k[x_1, \ldots, x_n]$.] (A quasicompact and Hausdorff space is called *compact*.)

**27.** When $k$ is an infinite field prove that the Zariski topology on $k^2$ is not the same as taking the Zariski topology on $k$ and then forming the product topology on $k \times k$. [By Exercise 14 of Section 1, in the product topology on $k \times k$ the Zariski closed sets in $k \times k$ are finite unions of sets of the form $\{a\} \times \{b\}$, $\{a\} \times k$ and $k \times \{b\}$, for any $a, b \in k$.]

**28.** Prove that each of the following rings have infinitely many minimal prime ideals, and that (0) is not the intersection of any finite number of these (so (0) does not have a primary decomposition in these rings):

**(a)** the infinite direct product ring $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \cdots$ (which is a Boolean ring, cf. Exercise 23 in Section 7.4).

**(b)** $k[x_1, x_2, \ldots]/(x_1x_2, x_3x_4, \ldots, x_{2i-1}x_{2i}, \ldots)$, where $x_1, x_2, \ldots$ are independent variables over the field $k$.

**29.** Suppose that $A$ and $B$ are ideals with $AB \subseteq Q$ for a primary ideal $Q$. Prove that if $A \not\subseteq Q$ then $B \subset \operatorname{rad} Q$.

**30.** Let $Q$ be a $P$-primary ideal and suppose $A$ is an ideal not contained in $Q$. Define $A' = \{r \in R \mid rA \subseteq Q\}$ to be the elements of $R$ that when multiplied by elements of $A$ give elements of $Q$. Prove that $A'$ is a $P$-primary ideal.

**31.** Prove that if $Q_1$ and $Q_2$ are primary ideals belonging to the same prime ideal $P$, then $Q_1 \cap Q_2$ is a primary ideal belonging to $P$. Conclude that a finite intersection of $P$-primary ideals is again $P$-primary.

**32.** Prove that if $Q_1$ and $Q_2$ are primary ideals belonging to the same *maximal* ideal $M$, then $Q_1 + Q_2$ and $Q_1 Q_2$ are primary ideals belonging to $M$. Conclude that finite sums and finite products of $M$-primary ideals are again $M$-primary.

**33.** Let $I = (x^2, xy, xz, yz)$ in $k[x, y, z]$. Prove that a primary decomposition of $I$ is $I = (x, y) \cap (x, z) \cap (x, y, z)^2$, determine the isolated and embedded primes of $I$, and find $\operatorname{rad} I$.

**34.** Suppose $\varphi : R \to S$ is a surjective ring homomorphism. Prove that an ideal $Q$ in $R$ containing the kernel of $\varphi$ is primary if and only if $\varphi(Q)$ is primary in $S$, and when this is the case the prime associated to $\varphi(Q)$ is the image $\varphi(P)$ of the prime $P$ associated to $Q$.

**35.** Suppose $\varphi : R \to S$ is a ring homomorphism.

**(a)** Suppose $I$ is an ideal of $R$ containing $\ker \varphi$ with minimal primary decomposition $I = Q_1 \cap \cdots \cap Q_m$ with $\operatorname{rad} Q_i = P_i$. If $\varphi$ is a surjective **homomorphism** prove that $\varphi(I) = \varphi(Q_1) \cap \cdots \cap \varphi(Q_m)$, where $\operatorname{rad} \varphi(Q_i)$ is given by $\varphi(P_i)$, is a minimal primary decomposition of $\varphi(I)$. [Use the previous exercise.]

**(b)** Suppose $I$ is an ideal of $S$ with minimal primary decomposition $I = Q_1 \cap \cdots \cap Q_m$ with $\operatorname{rad} Q_i = P_i$. Prove that $\varphi^{-1}(I) = \varphi^{-1}(Q_1) \cap \cdots \cap \varphi^{-1}(Q_m)$, where $\operatorname{rad} \varphi^{-1}(Q_i)$ is given by $\varphi^{-1}(P_i)$, is a primary decomposition of $\varphi^{-1}(I)$, and is minimal if $\varphi$ is surjective.

**36.** Let $I = (xy, x - yz)$ in $k[x, y, z]$. Prove that $(x, z) \cap (y^2, x - yz)$ is a minimal primary decomposition of $I$. [Consider the ring homomorphism $\varphi : k[x, y, z] \to k[y, z]$ given by mapping $x$ to $yz$, $y$ to $y$, and $z$ to $z$ and use the previous exercise.]

**37.** Prove that a prime ideal $P$ contains the ideal $I$ if and only if $P$ contains one of the associated primes of a minimal primary decomposition of $I$. [Use Exercise 3 and Exercise 11 in Section 7.4.]

**38.** Show that every associated prime ideal for a radical ideal is isolated. [Suppose that $P_2 = \text{rad } Q_2 \subseteq P_1 = \text{rad } Q_1$ in the decomposition of Theorem 21 for the radical ideal $I$. Show that if $a \in Q_2 \cap \cdots \cap Q_m \subseteq P_2$ then $a^n \in I$ for some $n \geq 1$, conclude that $a \in Q_1$ and derive a contradiction to the minimality of the primary decomposition.]

**39.** Fix an element $a$ in the ring $R$. For any ideal $I$ in the ring $R$ let $I_a = \{r \in R \mid ar \in I\}$.

**(a)** Prove that $I_a$ is an ideal and $I_a = R$ if and only if $a \in I$.

**(b)** Prove that $(I \cap J)_a = I_a \cap J_a$ for ideals $I$ and $J$.

**(c)** Suppose that $Q$ is a $P$-primary ideal and that $a \notin Q$. Prove that $Q_a$ is a $P$-primary ideal and that $Q_a = Q$ if $a \notin P$.

**40.** With notation as in the previous exercise, suppose $I = Q_1 \cap \cdots \cap Q_m$ is a minimal primary decomposition of the ideal $I$ and let $P_i$ be the prime ideal associated to $Q_i$.

**(a)** Prove that $I_a = (Q_1)_a \cap \cdots \cap (Q_m)_a$ and that $\operatorname{rad}(I_a) = \operatorname{rad}((Q_1)_a) \cap \cdots \cap \operatorname{rad}((Q_m)_a)$.

**(b)** Prove that $\operatorname{rad}(I_a)$ is the intersection of the prime ideals $P_i$ for which $a \notin Q_i$. [Use the previous exercise.]

**(c)** Prove that if $\operatorname{rad}(I_a)$ is a prime ideal then $\operatorname{rad}(I_a) = P_j$ for some $j$. [Use the fact that prime ideals are irreducible.]

**(d)** For each $i = 1, \ldots, m$, prove that $\operatorname{rad}(I_a) = P_i$ for some $a \in R$. [Show there exists an $a \in R$ with $a \notin Q_i$ but $a \in Q_j$ for all $j \neq i$.]

**(e)** Show from (c) and (d) that the associated primes for a minimal primary decomposition are precisely the collection of prime ideals among the ideals $\operatorname{rad}(I_a)$ for $a \in R$, and conclude that they are uniquely determined by $I$ independent of the minimal primary decomposition.

**41.** Let $P_1, \ldots, P_m$ be the associated prime ideals of the ideal (0) in the Noetherian ring $R$.

**(a)** Show that $P_1 \cap \cdots \cap P_m$ is the collection of nilpotent elements in $R$. [Apply Corollary 22 to $I = (0)$.]

**(b)** Show that $P_1 \cup \cdots \cup P_m$ is the collection of zero divisors in $R$. [Let $I = (0)$ in the previous exercise and show that the set of zero divisors is given by the set $\cup_{a \in R-\{0\}} (0)_a = \cup_{a \in R-\{0\}} \operatorname{rad}((0)_a)$.]

**42.** Suppose $R$ is a Noetherian ring. Prove that $R$ is either an integral domain, has nonzero nilpotent elements, or has at least two minimal prime ideals. [Use the previous exercise.]

**43.** Prove that the ideal $I$ in the Noetherian ring $R$ is radical if and only if the primary components of a minimal primary decomposition are all prime ideals, and conclude that in this case the minimal primary decomposition is unique. [If $I = Q_1 \cap \cdots \cap Q_m$ is radical with $Q_i$ a $P_i$-primary component of a minimal decomposition, show that if $a \in P_1 \cap \cdots \cap P_m$ then some power of $a$ is in $I$, hence $a \in I$ since $I$ is radical. Deduce that $I = P_1 \cap \cdots \cap P_m$ and show that this is also a minimal primary decomposition, i.e., for any $i$ there exists $b$ with $b \notin P_i$, but $b \in P_j$ for $j \neq i$. If $a \in P_i$, show that $ab \in Q_i$, and that $a \in Q_i$. Conclude that $Q_i = P_i$.]

**44.** Prove that a Noetherian integral domain $R$ is a U.F.D. if and only if for every $a \in R$ the isolated primes associated to the principal ideal $(a)$ are principal ideals. [See Example 2 following Corollary 22. To prove $R$ is a U.F.D., show that an irreducible $a \in R$ is prime and then follow the proof of Theorem 14 in Section 8.3.]

**45.** Let $R$ be the ring of all real valued functions on the open interval $(-1, 1)$ that have derivatives of all orders (the ring of $C^\infty$ functions). Let

$$F(x) = \begin{cases} e^{-1/x^4} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

(you may assume $F \in R$ and $F^{(n)}(0) = 0$ for all $n \geq 0$). Let $(F)$ be the principal ideal generated by $F$ and let $A = \text{rad}((F))$. Let $M$ be the (maximal) ideal of all functions in $R$ that are zero at $x = 0$ and let $P = \cap_{n=1}^{\infty} M^n$.

**(a)** Prove that $M = (x)$ is the ideal generated by the function $x$ in $R$ and that $M^n = (x^n)$ consists of the functions whose first $n - 1$ derivatives vanish at the origin.

**(b)** Prove that $R$ is not Noetherian (compare Exercise 33 in Section 7.4). [One approach is the following: Let $G(x)$ be the function that is 0 for $x < 0$ and is equal to $F(x)$ for $x \geq 0$. Let $I_n$ be the ideal of functions in $R$ vanishing for all $x \leq 1/n$. Use translates of $G(x)$ to show that $I_1 \subset I_2 \subset I_3 \subset \cdots$ is an infinite ascending chain.]

**(c)** Prove that $P$ consists of the functions all of whose derivatives are zero at $x = 0$ (i.e., the functions whose associated Taylor series at $x = 0$ is identically zero), and that $P$ is a prime ideal.

**(d)** Prove that $F \in P$ and deduce that $A \subseteq P$.

**(e)** Prove that $A \neq P$. [Let $G(x) = e^{-1/x^2}$ when $x \neq 0$ and $G(0) = 0$. Show that $G \in P$ but $G \notin A$.]

**(f)** Show that there is a prime ideal $Q$ containing $(F)$ with $Q \neq P, M$. Prove that $Q \subset P$ i.e., there are nonzero prime ideals properly contained in $P$.

**46.** Let $\mathcal{A}$ be any ideal in $R = k[x_1, \ldots, x_n, y_1, \ldots, y_m]$.

**(a)** Show that $\operatorname{rad}(\mathcal{A} \cap k[y_1, \ldots, y_m]) = \operatorname{rad} \mathcal{A} \cap k[y_1, \ldots, y_m]$.

**(b)** Suppose $(f_1, \ldots, f_s)$ is an ideal in $k[x_1, \ldots, x_n]$. Let $F_1, \ldots, F_t$ be generators for the radical of $(f_1, \ldots, f_s)$, computed in $k[x_1, \ldots, x_n]$. Suppose $J$ is an ideal in $R$ and let $\mathcal{A} = J + (f_1, \ldots, f_s)$, $\mathcal{B} = J + (F_1, \ldots, F_t)$ as ideals in $R$. Prove that $\operatorname{rad} \mathcal{A} = \operatorname{rad} \mathcal{B}$.

**(c)** Conclude from (a) and (b) that $\mathcal{A} = (y_1 - x_1, \ldots, y_m - x_m, f_1, \ldots, f_s) \cap k[y_1, \ldots, y_m]$ and $\mathcal{B} = (y_1 - x_1, \ldots, y_m - x_m, F_1, \ldots, F_t) \cap k[y_1, \ldots, y_m]$ have the same zero sets over an algebraically closed field $k$. [Use Hilbert's Nullstellensatz.]

**47.** Determine the Zariski closure in $\mathbb{C}^3$ of the points on the curve $\{(a^2, a^3, a^4) \mid a \in \mathbb{C}\}$.

**48.** Show that $\mathcal{Z}(x^3 - xyz + z^2)$ is the smallest algebraic set in $\mathbb{R}^3$ containing the points $\{(st, s + t, s^2t) \mid s, t \in \mathbb{R}\}$.

**49.** Show that $\mathcal{Z}(x^3z^2 - 3xy^2z^2 - y^6 - z^4)$ is the smallest algebraic set in $\mathbb{R}^3$ containing the points $\{(s^2 + t^2, st, s^3) \mid s, t \in \mathbb{R}\}$.

**50.** Find equations defining the Zariski closure of the set of points $\{(s^4, s^3t, st^3, t^4) \mid s, t \in \mathbb{R}\}$.

**51.** Show that $V = \mathcal{Z}(x^2 - y^2 z)$ (the *Whitney umbrella surface*) is the smallest algebraic set in $\mathbb{R}^3$ containing the points $S = \{(st, s, t^2) \mid s, t \in \mathbb{R}\}$. Show that $S$ is not Zariski closed in $V$ (the missing points explain the name for the surface). Do the same over $\mathbb{C}$, but show that in this case $S = V$ is closed.

**52.** Let $V = \mathcal{Z}(xz^2 - w^3, xw^2 - y^4, y^4z^2 - w^5) \subset \mathbb{C}^4$. Determine the Zariski closure of the image of $V$ under the projection $\pi((x, y, z, w)) = (x, y, z)$.

**53.** Let $V = \mathcal{Z}(xy - 1)$ in $\mathbb{A}^2$ and let $S$ be the projection of $V$ onto the $x$-axis in $\mathbb{A}^1$.

**(a)** If $k = \mathbb{R}$, show that $\mathcal{I}(V) = (xy - 1) \subset \mathbb{R}[x, y]$ and that $(u - x, xy - 1) \cap \mathbb{R}[u] = 0$ in $\mathbb{R}[x, y, u]$. Use Propositions 8 and 16 to conclude that the Zariski closure of $S$ is $\mathbb{A}^1$ and show that $S$ is not itself closed.

**(b)** If $k = \mathbb{F}_3$, show that $\mathcal{I}(V) = (xy - 1, x^3 - x, y^3 - y) \subset \mathbb{F}_3[x, y]$ and that $(u - x, xy - 1, x^3 - x, y^3 - y) \cap \mathbb{F}_3[u] = (u^2 - 1)$ in $\mathbb{F}_3[x, y, u]$. Use Propositions 8 and 16 to conclude that $S$ is Zariski closed in $\mathbb{A}^1$.

**54.** Recall the *ideal quotient* $(I : J) = \{r \in R \mid rJ \in I\}$ of two ideals $I$, $J$ in a ring $R$ (cf. Exercise 34 *ff.* in Section 9.6). Clearly $I \subseteq (I : J)$.

**(a)** Show that $\mathcal{Z}(I) - \mathcal{Z}(J)$, the set of elements of $\mathcal{Z}(I)$ not lying in $\mathcal{Z}(J)$, is contained in $\mathcal{Z}((I : J))$ and conclude that the Zariski closure of $\mathcal{Z}(I) - \mathcal{Z}(J)$ is contained in $\mathcal{Z}((I : J))$.

**(b)** Show that if $k$ is algebraically closed and $I$ is a radical ideal then $\mathcal{Z}((I : J))$ is precisely the Zariski closure of $\mathcal{Z}(I) - \mathcal{Z}(J)$.

**(c)** Show that if $V$ and $W$ are affine algebraic sets then $(\mathcal{I}(V) : \mathcal{I}(W)) = \mathcal{I}(V - W)$.

## 15.3. Integral Extensions and Hilbert's Nullstellensatz

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**1.** Use the fact that a U.F.D. is integrally closed to prove that the Gaussian integers, $\mathbb{Z}[i]$, is the ring of integers in $\mathbb{Q}(i)$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**2.** Suppose $k$ is a field and let $t = \bar{x}/\bar{y}$ in the field of fractions of the integral domain $R = k[x, y]/(x^2 - y^3)$. Prove that $K = k(t)$ is the fraction field of $R$ and $k[t]$ is the integral closure of $R$ in $K$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**3.** Suppose $k$ is a field and $i$ and $j$ are relatively prime positive integers. Find the normalization of the integral domain $R = k[x, y]/(x^i - y^j)$ (cf. Exercise 14, Section 9.1).

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**4.** Suppose $k$ is a field and let $P$ be the ideal $(y^2-x^3-x^2)$ in the polynomial ring $k[x, y]$. Prove that $P$ is a prime ideal and find the normalization of the integral domain $R = k[x, y]/P$. [To prove $P$ is prime, show that $y^2 - x^3 - x^2$ is irreducible in the U.F.D. $k[x, y]$. Then consider $t = \bar{y}/\bar{x} \in R$.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**5.** If $R$ is an integral domain with field of fractions $F$, show that $F$ is a finitely generated $R$-module if and only if $R = F$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**6.** For each of the following give specific rings $R \subseteq S$ and explicit ideals in these rings that exhibit the specified relation:

- **(a)** an ideal $I$ of $R$ such that $I \neq SI \cap R$ (so the contraction of the extension of an ideal $I$ need not equal $I$)
- **(b)** a prime ideal $P$ of $R$ such that there is no prime ideal $Q$ of $S$ with $P = Q \cap R$
- **(c)** a maximal ideal $M$ of $S$ such that $M \cap R$ is not maximal in $R$
- **(d)** a prime ideal $P$ of $R$ whose extension $PS$ to $S$ is not a prime ideal in $S$
- **(e)** an ideal $J$ of $S$ such that $J \neq (J \cap R)S$ (so the extension of the contraction of an ideal $J$ need not equal $J$).

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**7.** Let $\mathcal{O}_K$ be the ring of integers in a number field $K$.

**(a)** Suppose that every nonzero ideal $I$ of $\mathcal{O}_K$ can be written as the product of powers of prime ideals. Prove that an ideal $Q$ of $\mathcal{O}_K$ is $P$-primary if and only if $Q = P^m$ for some $m \geq 1$. [Show first that since nonzero primes in $\mathcal{O}_K$ are maximal that ${P_1}^{m_1} \subseteq {P_2}^{m_2}$ for distinct nonzero primes $P_1, P_2$ implies $P_1 = P_2$.]

**(b)** Suppose that an ideal $Q$ of $\mathcal{O}_K$ is $P$-primary if and only if $Q = P^m$ for some $m \geq 1$. Assuming all of Theorem 21, prove that every nonzero ideal $I$ of $\mathcal{O}_K$ can be written uniquely as the product of powers of prime ideals. [Prove that ${P_1}^{m_1}$ and ${P_2}^{m_2}$ are comaximal ideals if $P_1$ and $P_2$ are distinct nonzero prime ideals and use the Chinese Remainder Theorem.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**8.** Prove that if $s_1, \ldots, s_n \in S$ are integral over $R$, then the ring $R[s_1, \ldots, s_n]$ is a finitely generated $R$-module.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**9.** Suppose that $S$ is integral over $R$ and that $P$ is a prime ideal in $R$. Prove that every element $s$ in the ideal $PS$ generated by $P$ in $S$ satisfies an equation $s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0 = 0$ where the coefficients $a_0, a_1, \ldots, a_{n-1}$ are elements of $P$. [If $s = p_1 s_1 + \cdots + p_m s_m \in PS$, show that $T = R[s_1, \ldots, s_m]$ satisfies the hypotheses in Proposition 23(3). Follow the proof in Proposition 23 that $s$ is integral, noting that $s \in PT$ so that the $a_{ij}$ are elements of $P$.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**10.** Prove the following generalization of Proposition 28: Suppose $R$ is an integrally closed integral domain with field of fractions $k$ and $\alpha$ is an element of an extension field $K$ of $k$. Show that $\alpha$ is integral over $R$ if and only if $\alpha$ is algebraic over $k$ and the minimal polynomial $m_{\alpha,k}(x)$ for $\alpha$ over $k$ has coefficients in $R$. [If $\alpha$ is integral prove the conjugates of $\alpha$, i.e., the roots of $m_{\alpha,k}(x)$, are also integral, so the elementary symmetric functions of the conjugates are elements of $k$ that are integral over $R$.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**11.** Suppose $R$ is an integrally closed integral domain with field of fractions $k$ and $p(x) \in R[x]$ is a monic polynomial. Show that if $p(x) = a(x)b(x)$ with monic polynomials $a(x), b(x) \in k[x]$ then $a(x), b(x) \in R[x]$ (compare to Gauss' Lemma, Proposition 5, Section 9.3). [See the previous exercise.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**12.** Suppose $S$ is an integral domain that is integral over a ring $R$ as in the previous exercise. If $P$ is a prime ideal in $R$, let $s$ be any element in the ideal $PS$ generated by $P$ in $S$. Prove that, with the exception of the leading term, the coefficients of the minimal polynomial $m_{s,k}(x)$ for $s$ over $k$ are elements of $P$. [By Exercise 10, $m_{s,k}(x) \in R[x]$. Exercise 9 shows that $s$ is a root of a monic polynomial $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_0, \ldots, a_{n-1} \in P$. Use the previous exercise to show that $p(x) = m_{s,k}(x)b(x)$ with $b(x)$ in $R[x]$, and consider this equation in the integral domain $(R/P)[x]$. ]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

The next two exercises extend Exercise 6 in Section 7.5 by characterizing fields that are not fields of fractions of any of their proper subrings.

**13.** Let $K$ be a field of characteristic 0 and let $A$ be a subring of $K$ maximal with respect to $1/2 \notin A$. (Such $A$ exists by Zorn's Lemma.) Let $F$ be the field of fractions of $A$ in $K$.
- **(a)** Show that $K$ is algebraic over $F$. [If $t$ is transcendental over $F$, show that $1/2 \notin A[t]$.]
- **(b)** Show that $A$ is integrally closed in $K$. [Show that $1/2$ is not in the integral closure of $A$ in $K$.]
- **(c)** Deduce from (a) and (b) that $K = F$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

The next two exercises extend Exercise 6 in Section 7.5 by characterizing fields that are not fields of fractions of any of their proper subrings.

**14.** Show that a field $K$ is the field of fractions of some proper subring of $K$ if and only if $K$ is not a subfield of the algebraic closure of a finite field. [If $K$ contains $t$ transcendental over $\mathbb{F}_p$ argue as in the preceding exercise with $1/t$ in place of $1/2$ to show that $K$ is the quotient field of some proper subring.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

The next exercise gives a "geometric" interpretation of Noether's Normalization Lemma, showing that every affine algebraic set is a *finite covering* of some affine $n$-space.

**15.** Let $V$ be an affine algebraic set over an algebraically closed field $k$. Prove that for some $n$ there is a surjective morphism from $V$ onto $\mathbb{A}^n$ with finite fibers, and that if $V$ is a variety, then $n$ can be taken to be the dimension of $V$. [By Noether's Normalization Lemma the finitely generated $k$-algebra $S = k[V]$ contains a polynomial subalgebra $R = k[x_1, x_2, \ldots, x_n]$ such that $S$ is integral over $R$. Apply Theorem 6 to the inclusion of $R$ in $S$ to obtain a morphism $\varphi$ from $V$ to $\mathbb{A}^n$. To see that $\varphi$ is surjective with finite fibers, apply Corollary 27 to the maximal ideal $(x_1 - a_1, \ldots, x_n - a_n)$ of $R$ corresponding to a point $(a_1, \ldots, a_n)$ of $\mathbb{A}^n$.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**16.** Let $V$ be an affine algebraic set in $\mathbb{C}^n$. Prove that $V$ is compact in the Euclidean topology (i.e., closed and bounded) if and only if it is finite. [Use Exercise 18 in Section 2, the previous exercise, and the behavior of compact sets with respect to continuous functions.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**17.** Let $R$ be a subring of the commutative ring $S$ with $1_S \in R$ and suppose that $S$ is integral over $R$. This exercise proves that $R$ and $S$ have the same *Krull dimension*, cf. Section 16.1.

**(a)** If $P_1 \subset P_2 \subset \cdots \subset P_n$ is a chain of distinct prime ideals in $R$ prove that there is a chain $Q_1 \subset Q_2 \subset \cdots \subset Q_n$ of distinct prime ideals in $S$ with $Q_i \cap R = P_i$.

**(b)** Prove conversely that if $Q_1 \subset Q_2 \subset \cdots \subset Q_n$ is a chain of distinct prime ideals in $S$ and $P_i = Q_i \cap R$ then $P_1 \subset P_2 \subset \cdots \subset P_n$ is a chain of distinct prime ideals in $R$. [To prove the $P_i$ are distinct, pass to a quotient and reduce the problem to showing that if $Q$ is a nonzero prime ideal in the integral domain $S$ then $Q \cap R$ is a nonzero prime ideal in $R$. In this case, if $s \in Q$ is nonzero, show that the constant coefficient of a polynomial of minimal degree in $R[x]$ satisfied by $s$ is a nonzero element in $Q \cap R$.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**18.** Let $V = \mathcal{Z}(I)$ and $W = \mathcal{Z}(J)$ where $I$ is the ideal $(uv + v) \subset \mathbb{C}[u, v]$ and $J$ is the ideal $(-2y - y^2 + 2z + z^2, 2x - yz - z^2) \subset \mathbb{C}[x, y, z]$.

**(a)** Show that $I$ and $J$ are prime ideals. Conclude that $I = \mathcal{I}(V)$ and $J = \mathcal{I}(W)$ and that $V$ and $W$ are varieties.

**(b)** Show that the map $\varphi : V \to W$ defined by $\varphi((a_1, a_2)) = (a_1^2 + a_2, a_1 + a_2, a_1 - a_2)$ is an isomorphism.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**19.** Let $I = (x^3 + y^3 + z^3, x^2 + y^2 + z^2, (x + y + z)^3) \subset k[x, y, z]$. Use Gröbner bases to show that $x, y, z \in \text{rad } I$ if $\text{ch}(k) \neq 2, 3$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**20.** Let $I = (x^3 + y^3 + z^3, xy + xz + yz, xyz) \subset k[x, y, z]$. Use Gröbner bases to show that $x, y, z \in \operatorname{rad} I$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**21.** Let $I = (x^4 + y^4 + z^4, x + y + z) \subset k[x, y, z]$.

**(a)** Use Gröbner bases to show that $xy + xz + yz \in \operatorname{rad} I$ if $\operatorname{ch}(k) \neq 2$ and determine the smallest power of $xy + xz + yz$ contained in $I$. Show that none of $x$, $y$ or $z$ is contained in rad $I$.

**(b)** If $J = (x^4 + y^4 + z^4, x + y + z, xy + xz + yz)$ show that the reduced Gröbner basis of $J$ relative to the lexicographic ordering $x > y > z$ is $\{x + y + z, y^2 + yz + z^2\}$. Deduce that $k[x, y, z]/J \cong k[y, z]/(y^2 + yz + z^2)$ and that $J$ is radical if $\operatorname{ch}(k) \neq 3$.

**(c)** If $\operatorname{ch}(k) \neq 2, 3$, show that $\operatorname{rad} I = J$.

**(d)** If $\operatorname{ch}(k) = 3$, show that $\operatorname{rad} I = (x - y, y - z)$.

**(e)** If $\operatorname{ch}(k) = 2$, show that $I = (x + y + z)$ is a prime, hence radical, ideal.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**22.** Let $I = (x^2y + z^3, x + y^3 - z, 2y^4z - yz^2 - z^3) \subset k[x, y, z]$. Use Gröbner bases to show that $x, y, z \in \operatorname{rad} I$ and conclude that $\operatorname{rad} I = (x, y, z)$. Show that $x^9, y^7, z^9$ are the smallest powers of $x, y, z$, respectively, lying in $I$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**23.** Let $V = \mathcal{Z}(x^3 - x^2z - y^2z)$ and $W = \mathcal{Z}(x^2 + y^2 - z^2)$ in $\mathbb{C}^3$. Show that $\mathcal{I}(V) = (x^3 - x^2z - y^2z)$ and $\mathcal{I}(W) = (x^2 + y^2 - z^2)$ in $\mathbb{C}[x, y, z]$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**24.** Let $V = \mathcal{Z}(x^3 + y^3 + 7z^3) \subset \mathbb{C}^3$. Show that $\mathcal{I}(V) = (x^3 + y^3 + 7z^3)$ in $\mathbb{C}[x, y, z]$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**25.** Let $I = (xz + y^2 + z^2, xy - xz + yz - 2z^2)$ and let $K = I + (x^2 - 3y^2 + yz) \subset \mathbb{C}[x, y, z]$.

**(a)** By Exercise 46 in Section 1, there is an injective $\mathbb{C}$-algebra homomorphism from $\mathbb{C}[x, y, z]/K$ to $\mathbb{C}[u, v]/(u^3 - uv^2 + v^3)$. Use this together with the example preceding Proposition 34 to prove that $K$ is a radical ideal and deduce that rad $I \subseteq K$.

**(b)** Show that rad $I \subseteq (y, z)$.

**(c)** Show that $K \cap (y, z) = I$ and deduce that $I$ is radical, so that $\mathcal{I}(V) = I$ if $V = \mathcal{Z}(I)$.

**(d)** Show that $y(x^2 - 3y^2 + yz)$ and $z(x^2 - 3y^2 + yz)$ are elements of $I$ but none of $y$, $z$, or $x^2 - 3y^2 + yz$ is contained in $I$.

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**26.** Let $I$ be an ideal in $k[x_1, \ldots, x_n]$. Prove that the following are equivalent (an ideal satisfying any of these conditions is called a *zero-dimensional ideal* because of (d)):

**(a)** The quotient $k[x_1, \ldots, x_n]/I$ has finite dimension as a vector space over $k$.

**(b)** $I \cap k[x_i] \neq 0$ for each $i = 1, 2, \ldots, n$.

**(c)** If $G$ is any reduced Gröbner basis for $I$ then for each $i = 1, \ldots, n$, there is a $g_i \in G$ with leading term $x_i^{n_i}$ for some $n_i \geq 1$.

**(d)** The set of common zeros $\mathcal{Z}_{\bar{k}}(I)$ of the polynomials in $I$ in an algebraic closure $\bar{k}$ of $k$ is finite.

[For (a) implies (b) use the injection $k[x_i]/(I \cap k[x_i]) \hookrightarrow k[x_1, \ldots, x_n]/I$. For (b) implies (c) note some $LT(g_i)$ divides the leading term of a generator for $I \cap k[x_i]$. For (c) implies (a) use Exercise 37 in Section 9.6. Show (b) implies (d). For (d) implies (b) show the product $m_{a_1,k}(x_i) \ldots m_{a_N,k}(x_i)$ of the minimal polynomials of the $i^{\text{th}}$ coordinates $a_1, \ldots, a_N$ of the points in $\mathcal{Z}_{\bar{k}}(I)$ is a nonzero polynomial in $\mathcal{I}(\mathcal{Z}_{\bar{k}}(I))$ and apply Corollary 33.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**27.** Let $I$ be a zero-dimensional ideal in $k[x_1, \ldots, x_n]$ and let $I'$ be the ideal generated by $I$ in $\bar{k}[x_1, \ldots, x_n]$ where $\bar{k}$ is the algebraic closure of $k$. Let $\mathcal{Z}(I)$ be the zero set of $I$ in $k^n$ and let $\mathcal{Z}_{\bar{k}}(I)$ be the zero set of $I$ (equivalently, of $I'$) in $\bar{k}^n$.

**(a)** Prove that $|\mathcal{Z}_{\bar{k}}(I)| = \dim_{\bar{k}} \bar{k}[x_1, \ldots, x_n]/\operatorname{rad} I'$. [Show that rad $I'$ is the product of the maximal ideals corresponding to the points in $V_{\bar{k}}$ and use the Chinese Remainder Theorem.]

**(b)** Show $|\mathcal{Z}(I)| \leq \dim_k k[x_1, \ldots, x_n]/I$. [One approach: use Exercise 43 in Section 1 and observe that $\dim_{\bar{k}} \bar{k}[x_1, \ldots, x_n]/\operatorname{rad} I' \leq \dim_{\bar{k}} \bar{k}[x_1, \ldots, x_n]/I'$.]

Let $R$ be a subring of the commutative ring $S$ with $1 \in R$.

**28.** Suppose $I$ is a zero-dimensional ideal in $k[x_1, \ldots, x_n]$, and suppose $I \cap k[x_i]$ is generated by the nonzero polynomial $h_i$ (cf. Exercise 26). Let $r_i$ be the product of the irreducible factors of $h_i$ (the 'squarefree part' of $h_i$).

**(a)** Prove that $I + (r_1, \ldots, r_n) \subseteq \text{rad } I$.

**(b)** *(Radicals of zero-dimensional ideals for perfect fields)* If $k$ is a perfect field, prove that $\text{rad } I = I + (r_1, \ldots, r_n)$. [Use induction on $n$. Write $r_1 = p_1 \ldots p_t$ with distinct irreducibles $p_i$ in $k[x_1]$. If $J = I + (r_1, \ldots, r_n)$ show that $J = J_1 \cap \cdots \cap J_t$ where $J_t = J + (p_t)$. Show for each $i$ that reduction modulo $p_i$ induces an isomorphism $k[x_1, \ldots, x_n]/J_i \cong K[x_2, \ldots, x_n]/J_i'$ where $K$ is the extension field $k[x]/(p_i)$ and $J_i' \subseteq K[x_2, \ldots, x_n]$ is the reduction of the ideal $J_i$ modulo $(p_i)$. Use Exercise 11 of Section 13.5 to show that the image of $r_j$ in $J_i' \cap K[x_j]$ remains a nonzero squarefree polynomial for each $j = 2, \ldots, n$ since $k$ is perfect. Conclude by induction that $J_i'$ is a radical ideal. Deduce that $J_i$ is a radical ideal, and finally that $J$ is a radical ideal.]

**(c)** Find the radicals of $(x^7 + x + y^3, x^4 + y^3 + y)$, $(x^3 - xy^2 + x, x^2y + y^3)$, and $(x^4 + y^3, x^3 - xy + y^2)$ in $\mathbb{Q}[x, y]$ and of $(x^2 + y^2z, x^2y^2 + z^3, y^2 + z^2)$ in $\mathbb{Q}[x, y, z]$.

**(d)** Let $k = \mathbb{F}_p(t)$. Show that $I = (x^p + t, y^p - t)$ is a zero-dimensional ideal in $k[x, y]$ such that both $I \cap k[x]$ and $I \cap k[y]$ contain nonzero squarefree polynomials, but that $I$ is not a radical ideal (so the result in (b) need not hold if $k$ is not perfect). [Show that $x + y \in \text{rad } I$ but $x + y \notin I$.]

## 15.4. Localization

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**1.** Suppose $M$ is a finitely generated $R$-module. Prove that $D^{-1}M = 0$ if and only if $dM = 0$ for some $d \in D$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**2.** Let $I$ be an ideal in $R$, let $D$ be a multiplicatively closed subset of $R$ with ring of fractions $D^{-1}R$, and let ${}^c({}^eI) = R$ be the saturation of $I$ with respect to $D$.

**(a)** Prove that ${}^c({}^eI) = R$ if and only if ${}^eI = D^{-1}R$ if and only if $I \cap D \neq \emptyset$.

**(b)** Prove that $I = {}^c({}^eI)$ is saturated if and only if for every $d \in D$, if $da \in I$ then $a \in I$.

**(c)** Prove that extension and contraction define inverse bijections between the ideals of $R$ saturated with respect to $D$ and the ideals of $D^{-1}R$.

**(d)** Let $I = (2x, 3y) \subset \mathbb{Z}[x, y]$. Show the saturation of $I$ with respect to $\mathbb{Z} - \{0\}$ is $(x, y)$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**3.** If $I$ is an ideal in the commutative ring $R$ let $\varphi : R[x_1, \ldots, x_n] \cong (R/I)[x_1, \ldots, x_n]$ be the ring homomorphism with kernel $I[x_1, \ldots, x_n]$ given by reducing coefficients modulo $I$. If $\overline{A}$ is an ideal in $(R/I)[x_1, \ldots, x_n]$, let $A$ denote the inverse image of $\overline{A}$ under $\varphi$.

**(a)** For any $i \geq 1$ show that the inverse image under $\varphi$ of the subring $(R/I)[x_1, \ldots, x_i]$ is $R[x_1, \ldots, x_i] + I[x_1, \ldots, x_n]$.

**(b)** Prove that $\varphi(A \cap R[x_1, \ldots, x_i]) = \overline{A} \cap (R/I)[x_1, \ldots, x_i]$

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**4.** Let $f = y^5 - z^4$, viewed as a polynomial in $y$ with coefficients in $\mathbb{Q}[z]$.

**(a)** Prove that $f$ has no roots in $\mathbb{Q}[z]$.

**(b)** Suppose $f = (y^2 + ay + b)(y^3 + cy^2 + dy + e)$. Show that $a, b, c, d, e$ satisfy the system of equations

$$a + c = 0, \quad ac + b + d = 0, \quad ad + bc + e = 0, \quad ae + bd = 0, \quad be - z^4 = 0.$$

Deduce that $e^5 = z^{12}$ and conclude that $f$ is irreducible in $\mathbb{Q}[y, z]$. [Use elimination.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**5.** Suppose $R$ is a U.F.D. with field of fractions $F$ and $p \in R[x]$ is a monic polynomial.
**(a)** Show that the ideal $pR[x]$ generated by $p$ in $R[x]$ is prime if and only if the ideal $pF[x]$ generated by $p$ in $F[x]$ is prime. [Use Gauss' Lemma.]
**(b)** Show that $pR[x]$ is saturated, i.e., that $pF[x] \cap R[x] = pR[x]$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**6.** Show that $I = (y^3 - xz, xy^2 - z^2)$ is not a prime ideal in $\mathbb{Q}[x, y, z]$ and find explicit elements $a, b \in \mathbb{Q}[x, y, z]$ with $ab \in I$ but $a \notin I$ and $b \notin I$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**7.** Show that $P = (y^3 - xz, xy^2 - z^2, x^2 - yz)$ is a prime ideal in $\mathbb{Q}[x, y, z]$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**8.** Show that $P = (x^2 - yz, w^2 - x^4z)$ is a prime ideal in $\mathbb{Q}[x, y, z, w]$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**9.** Show that $P = (xz^2 - w^3, xw^2 - y^4, y^4z^2 - w^5)$ is a prime ideal in $\mathbb{Q}[x, y, z, w]$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**10.** Show that $I = (xy - w^3, y^2 - zw)$ is not a prime ideal in $\mathbb{Q}[x, y, z, w]$ and find $a, b$ with $ab \in I$ but $a, b \notin I$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**11.** Let $R_P$ be the localization of $R$ at the prime $P$. Prove that if $Q$ is a $P$-primary ideal of $R$ then $Q = {}^c({}^eQ)$ with respect to the extension and contraction of $Q$ to $R_P$. Show the same result holds if $Q$ is $P'$-primary for some prime $P'$ contained in $P$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**12.** Let $R = \mathbb{R}[x, y, z]/(xy - z^2)$, let $P = (\bar{x}, \bar{z})$ be the prime ideal generated by the images of $x$ and $y$ in $R$, and let $R_P$ be the localization of $R$ at $P$. Prove that $P^2 R_P \cap R = (\bar{x})$ and is strictly larger than $P^2$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**13.** Prove that if $N$ and $N'$ are two $R$-submodules of an $R$-module $M$ with $N_P = N'_P$ in the localization $M_P$ for every prime ideal $P$ of $R$ (or just for every maximal ideal) then $N = N'$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**14.** Suppose $\varphi : M \to N$ is an $R$-module homomorphism. Prove that $\varphi$ is injective (respectively, surjective) if and only if the induced $R_P$-module homomorphism $\varphi : M_P \to N_P$ is injective (respectively, surjective) for every prime ideal $P$ of $R$ (or just for every maximal ideal of $R$).

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**15.** Let $R = \mathbb{Z}[\sqrt{-5}]$ be the ring of integers in the quadratic field $\mathbb{Q}(\sqrt{-5})$ and let $I$ be the prime ideal $(2, 1 + \sqrt{-5})$ of $R$ generated by 2 and $1 + \sqrt{-5}$ (cf. Exercise 5, Section 8.2). Recall that every nonzero prime ideal $P$ of $R$ contains a prime $p \in \mathbb{Z}$.

**(a)** If $P$ is a prime ideal of $R$ not containing 2 prove that $I_P = R_P$.

**(b)** If $P$ is a prime ideal of $R$ containing 2 prove that $P = I$ and that $I_P = (1+\sqrt{-5})R_P$.

**(c)** Prove that $I_P \cong R_P$ as $R_P$-modules for every prime ideal $P$ of $R$ but that $I$ and $R$ are not isomorphic $R$-modules. (This example shows that it is important in Exercise 14 to be *given* the $R$-module homomorphism $\varphi$.) [Observe that $I \cong R$ as $R$-modules if and only if $I$ is a *principal* ideal.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**16.** Prove that localization commutes with tensor products: there is a unique isomorphism of $D^{-1}R$-modules $\varphi : (D^{-1}M) \otimes_{D^{-1}R} (D^{-1}N) \cong D^{-1}(M \otimes_R N)$ with $\varphi((m/d) \otimes (n/d')) = (m \otimes n)/dd'$ for any $R$-modules $M$, $N$, and multiplicatively closed set $D$ in $R$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**17.** Prove that the $R$-module $A$ is a flat $R$-module if and only if $A_P$ is a flat $R_P$-module for every prime ideal $P$ of $R$ (or just for every maximal ideal of $R$). [Use Proposition 41, Exercises 14 and 16, and the exactness properties of localization.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**18.** In the notation of Example 2 following Corollary 37, prove that $R_f \cong R[x]/(fx - 1)$ if $f$ is not nilpotent in $R$. [Show that the map $\varphi : R[x] \to R_f$ defined by $\varphi(r) = r/1$ and $\varphi(x) = 1/f$ gives a surjective ring homomorphism and the universal property in Theorem 36 gives an inverse.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**19.** Prove that if $R$ is an integrally closed integral domain and $D$ is any multiplicatively closed subset of $R$ containing 1, then $D^{-1}R$ is integrally closed.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**20.** Suppose that $R$ is a subring of the ring $S$ with $1 \in R$ and that $S$ is integral over $R$. If $D$ is any multiplicatively closed subset of $R$, prove that $D^{-1}S$ is integral over $D^{-1}R$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**21.** Suppose $\varphi : R \to S$ is a ring homomorphism and $D'$ is a multiplicatively closed subset of $S$. Let $D = \varphi^{-1}(D')$. Prove that $D$ is a multiplicatively closed subset of $R$ and that the map $\varphi' : D^{-1}R \to D'^{-1}S$ given by $\varphi'(r/d) = \varphi(r)/\varphi(d)$ is a ring homomorphism.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**22.** Suppose $P \subseteq Q$ are prime ideals in $R$ and let $R_Q$ be the localization of $R$ at $Q$. Prove that the localization $R_P$ is isomorphic to the localization of $R_Q$ at the prime ideal $PR_Q$ (cf. the preceding exercise).

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**23.** Let $\varphi : A \to B$ be a homomorphism of commutative rings with $\varphi(1_A) = 1_B$, and let $P$ be a prime ideal of $A$. Let contraction and extension of ideals with respect to $\varphi$ be denoted by superscripts $c$ and $e$ respectively. Prove that $P$ is the contraction of a prime ideal in $B$ if and only if $P = (P^e)^c$. [Localize $B$ at $\varphi(A - P)$.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**24.** *(The Going-down Theorem)* Let $S$ be an integral domain, let $R$ be an integrally closed subring of $S$ containing $1_S$, and let $k$ be the field of fractions of $R$. Suppose that $P_2 \subseteq P_1$ are prime ideals in $R$ and that $Q_1$ is a prime ideal in $S$ with $Q_1 \cap R = P_1$. Let $S_{Q_1}$ be the localization of $S$ at $Q_1$.

**(a)** Show that $P_2 \subseteq P_2 S_{Q_1} \cap R$.

**(b)** Suppose that $a \in P_2 S_{Q_1} \cap R$ and write $a = s/d$ with $s \in P_2 S$ and $d \in S$, $d \notin Q_1$. If the minimal polynomial of $s$ over $k$ is $x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ with $a_0, \ldots, a_{n-1} \in P_2$ (cf. Exercise 12 in Section 3) show that the minimal polynomial of $d$ over $k$ is $x^n + b_{n-1}x^{n-1} + \cdots + b_1 x + b_0$ where $b_i = a_i / a^{n-i}$ and conclude that $b_i \in R$. [Use Exercise 10 in Section 3.]

**(c)** Show that $a \in P_2$ and conclude that $P_2 S_{Q_1} \cap R = P_2$. [Show $a \notin P_2$ implies $b_i \in P_2$ for $i = 0, 1, \ldots, n-1$, which would imply $d^n \in P_2 S \subseteq P_1 S \subseteq Q_1$ and so $d \in Q_1$.]

**(d)** Prove that $P_2 S_{Q_1}$ is contained in a prime ideal $P$ of $S_{Q_1}$ with $P \cap R = P_2$. [Use (c) and the previous exercise for $\varphi : R \to S_{Q_1}$.]

**(e)** Let $Q_2 = P \cap S$. Prove that $Q_2 \subseteq Q_1$ and that $Q_2 \cap R = P_2$.

**(f)** Use induction together with the previous result to prove the Going-down Theorem: Theorem 26(4).

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**25.** Let $k$ be an algebraically closed field and let $V = \mathcal{Z}(xz - yw) \subset \mathbb{A}^4$. Prove that the set of points $v$ where $f = \bar{x}/\bar{y} \in k(V)$ is regular is precisely the set of points $(x, y, z, w)$ where $y \neq 0$ or $z \neq 0$. [If $f = \bar{a}/\bar{b}$ show that $ay - bx \in (xz - yw)$ as polynomials in $k[x, y, z, w]$ and conclude that $b \in (y, z)$.] Prove that there is no function $a/b \in k(V)$ with $b(v) \neq 0$ for every $v$ where $f$ is regular.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**26.** (*Differentials of Morphisms*) Let $\varphi : V \to W$ be a morphism of affine varieties over the algebraically closed field $k$ and suppose $\varphi(v) = w$.

**(a)** Show that $\varphi$ induces a linear map from the $k$-vector space $M_w/M_w^2$ to the $k$-vector space $M_v/M_v^2$, and use this to show that $\varphi$ induces a linear map $d\varphi$ (called the *differential* of $\varphi$) from the $k$-vector space $\mathbb{T}_{v,V}$ to the $k$-vector space $\mathbb{T}_{w,W}$.

**(b)** Prove that if $V \subseteq \mathbb{A}^n$, $W \subseteq \mathbb{A}^m$ and $\varphi = (F_1(x_1, \ldots, x_n), \ldots, F_m(x_1, \ldots, x_n))$ then $d\varphi : \mathbb{T}_{v,V} \to \mathbb{T}_{w,W}$ is given explicitly by

$$(d\varphi)(a_1, \ldots, a_n) = (D_v(F_1)(a_1, \ldots, a_n), \ldots, D_v(F_m)(a_1, \ldots, a_n)).$$

[If $g = g(y_1, \ldots, y_m)$ show that the chain rule implies

$$\frac{\partial(g \circ \varphi)}{\partial x_i}(v) = \sum_{j=1}^{m} \frac{\partial g}{\partial y_j}(w) \frac{\partial F_j}{\partial x_i}(v),$$

so that $D_v(g \circ \varphi)(a_1, \ldots, a_n) = D_w(g)(b_1, \ldots, b_m)$ where $b_j = D_v(F_j)(a_1, \ldots, a_n)$. Then use the fact that $g \circ \varphi \in \mathcal{I}(V)$ if $g \in \mathcal{I}(W)$.]

**(c)** If $\psi : U \to V$ is another morphism with $\psi(u) = v$, prove that the associated $d(\varphi \circ \psi) : \mathbb{T}_{u,U} \to \mathbb{T}_{w,W}$ is the same as $d\varphi \circ d\psi$.

**(d)** Prove that if $\varphi$ is an isomorphism then $d\varphi$ is a vector space isomorphism from $\mathbb{T}_{v,V}$ to $\mathbb{T}_{w,W}$ for every $\varphi(v) = w$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**27.** Let $V = \mathbb{A}^1$ and $W = \mathcal{Z}(xz - y^2, yz - x^3, z^2 - x^2y) \subset \mathbb{A}^3$. Let $\varphi : V \to W$ be the surjective morphism $\varphi(t) = (t^3, t^4, t^5)$ (cf. Exercise 26 in Section 1). For each $t \in \mathbb{A}^1$ describe the differential $d\varphi : \mathbb{T}_{t,\mathbb{A}^1} \to \mathbb{T}_{(t^3,t^4,t^5),W}$ in the previous exercise explicitly; in particular prove that $d\varphi$ is an isomorphism of vector spaces for all $t \neq 0$ and is the zero map for $t = 0$. Use this to prove that $V$ and $W$ are not isomorphic.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**28.** If $k$ is a field, the quotient $k[x]/(x^2)$ is called the *ring of dual numbers* over $k$. If $V$ is an affine algebraic set over $k$, show that a $k$-algebra homomorphism from $k[V]$ to $k[x]/(x^2)$ is equivalent to specifying a point $v \in V$ with $\mathcal{O}_{v,V}/\mathfrak{m}_{v,V} = k$ (called a *$k$-rational point* of $V$) together with an element in the tangent space $\mathbb{T}_{v,V}$ of $V$ at $v$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

**29.** (*Computing the dimension of a variety*) Let $P$ be a prime ideal in $k[x_1, \ldots, x_n]$, set $P_0 = 0$ and let $P_i = P \cap k[x_1, \ldots, x_i]$. Define the varieties $V_i = \mathcal{Z}(P_i) \subseteq \mathbb{A}^i$ with $V_0$ the zero dimensional variety consisting of a single point and coordinate ring $k$.

**(a)** Show that $\dim V_{i-1} \leq \dim V_i \leq \dim V_{i-1} + 1$. [First exhibit an injection from $k[V_{i-1}]$ into $k[V_i]$; then show that $k[V_i]$ is a $k$-algebra generated by $k[V_{i-1}]$ and one additional generator.]

**(b)** If the ideal generated by $P_{i-1}$ in $k[x_1, \ldots, x_i]$ equals $P_i$, show that $V_i \cong V_{i-1} \times \mathbb{A}^1$ and deduce that $\dim V_i = \dim V_{i-1} + 1$.

**(c)** If the ideal generated by $P_{i-1}$ in $k[x_1, \ldots, x_i]$ is properly contained in $P_i$, show that $\dim V_i = \dim V_{i-1}$.

**(d)** Show that $\dim V$ equals the number of $i \in \{1, 2, \ldots, n\}$ such that the ideal generated by $P_{i-1}$ in $k[x_1, \ldots, x_i]$ equals the ideal $P_i$. Deduce that if $G$ is the reduced Gröbner basis for $P$ with respect to the lexicographic monomial ordering $x_n > \cdots > x_1$ and $G_i = G \cap k[x_1, \ldots, x_i]$ where $G_0 = \emptyset$, and $N$ is the number of $i$ with $G_i \neq G_{i-1}$ for $1 \leq i \leq n$, then $\dim V = n - N$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\mathrm{Supp}(M)$.

**30.** Prove that $M = 0$ if and only if $\mathrm{Supp}(M) = \emptyset$. [Use Proposition 47.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\mathrm{Supp}(M)$.

**31.** If $0 \to L \to M \to N \to 0$ is an exact sequence of $R$-modules, prove that the localization $M_P$ is nonzero if and only if one of the localizations $N_P$ and $L_P$ is nonzero and deduce that $\mathrm{Supp}(M) = \mathrm{Supp}(L) \cup \mathrm{Supp}(N)$. In particular, if $M = M_1 \oplus \cdots \oplus M_n$ prove that $\mathrm{Supp}(M) = \mathrm{Supp}(M_1) \cup \cdots \cup \mathrm{Supp}(M_n)$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted Supp($M$).

**32.** Suppose $P \subseteq Q$ are prime ideals in $R$ and that $M$ is an $R$-module. Prove that the localization of the $R$-module $M_Q$ at $P$ is the localization $M_P$, i.e., $(M_Q)_P = M_P$. [Argue directly, or use Proposition 41 and the associativity of the tensor product.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\text{Supp}(M)$.

**33.** Suppose $P \subseteq Q$ are prime ideals in $R$ and that $M$ is an $R$-module. Prove that if $P \in \text{Supp}(M)$ then $Q \in \text{Supp}(M)$. [Use the previous exercise.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\text{Supp}(M)$.

**34.** **(a)** Suppose $M = Rm$ is a cyclic $R$-module. Prove that $M_P = 0$ if and only if there is an element $r \in R$, $r \notin P$ with $rm = 0$. Deduce that $P \in \text{Supp}(M)$ if and only if $P$ contains the annihilator of $m$ in $R$ (cf. Exercise 10 in Section 10.1).

**(b)** If $M = Rm_1 + \cdots + Rm_n$ is a finitely generated $R$-module prove that $P \in \text{Supp}(M)$ if and only if $P$ is contained in $\text{Supp}(Rm_i)$ for some $i = 1, \ldots, n$. [Use Proposition 42.] Deduce that $P \in \text{Supp}(M)$ if and only if $P$ contains the annihilator $\text{Ann}(M)$ of $M$ in $R$. [Note $\text{Ann}(M) = \cap_{i=1}^{n} \text{Ann}(Rm_i)$, then use (a) and Exercise 11 of Section 7.4.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted Supp($M$).

**35.** Suppose $P$ is a prime ideal of $R$ with $P \cap D = \emptyset$. Prove that if $P \in \mathrm{Ass}_R(M)$ then $D^{-1}P \in \mathrm{Ass}_{D^{-1}R}(D^{-1}M)$. [Use Proposition 38(3) and Proposition 42.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\mathrm{Supp}(M)$.

**36.** Suppose $D^{-1}P \in \mathrm{Ass}_{D^{-1}R}(D^{-1}M)$ where $P = (a_1, \ldots, a_n)$ is a finitely generated prime ideal in $R$ with $P \cap D = \emptyset$.

**(a)** Suppose $m/d \in D^{-1}M$ has annihilator $D^{-1}P$ in $D^{-1}R$. Show that $d_i a_i m = 0 \in R$ for some $d_1, \ldots, d_n \in D$.

**(b)** Let $d' = d_1 d_2 \ldots d_n$. Show that $P = \mathrm{Ann}(d'm)$ and conclude that $P \in \mathrm{Ass}_R(M)$. [The inclusion $P \subseteq \mathrm{Ann}(d'm)$ is immediate. For the reverse inclusion, show that $b \in \mathrm{Ann}(d'm)$ implies that $b/1$ annihilates $m/d$ in $D^{-1}M$, hence $b/1 \in D^{-1}P$, and conclude $b \in P$.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\mathrm{Supp}(M)$.

**37.** Suppose $M$ is a module over the Noetherian ring $R$. Use the previous two exercises to show that under the bijection of Proposition 38(3) the prime ideals $P$ of $\mathrm{Ass}_R(M)$ with $P \cap D = \emptyset$ correspond bijectively with the prime ideals of $\mathrm{Ass}_{D^{-1}R}(D^{-1}M)$.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\mathrm{Supp}(M)$.

**38.** Suppose $M$ is a module over the Noetherian ring $R$ and $D$ is a multiplicatively closed subset of $R$. Let $\mathcal{S}$ be the subset of prime ideals $P$ in $\mathrm{Ass}_R(M)$ with $P \cap D \neq \emptyset$. This exercise proves that the kernel $N$ of the localization map $M \to D^{-1}M$ is the unique submodule $N$ of $M$ with $\mathrm{Ass}_R(N) = \mathcal{S}$ and $\mathrm{Ass}_R(M/N) = \mathrm{Ass}_R(M) - \mathcal{S}$.

**(a)** If $N'$ is a submodule of $M$ with $\mathrm{Ass}_R(N') = \mathcal{S}$ and $\mathrm{Ass}_R(M/N') = \mathrm{Ass}_R(M) - \mathcal{S}$ as in Exercise 35 in Section 1, prove that the diagram

$$\begin{array}{ccc} M & \xrightarrow{\pi} & M/N' \\ {\scriptstyle\varphi}\downarrow & & \downarrow{\scriptstyle\varphi'} \\ D^{-1}M & \xrightarrow{\pi'} & D^{-1}(M/N') \end{array}$$

is commutative, where $\pi$ and $\pi'$ are the natural projections (cf. Proposition 42(6)) and $\varphi$, $\varphi'$ are the localization homomorphisms.

**(b)** Show that $\mathrm{Ass}_{D^{-1}R}(D^{-1}N') = \emptyset$ and conclude that $D^{-1}N' = 0$ and that $\pi'$ is injective. [Use the previous exercise, the definition of $\mathcal{S}$, and Exercise 34 in Section 1.]

**(c)** If $x$ is the kernel $K$ of $\varphi'$ show that $\mathrm{Ann}(x) \cap D \neq \emptyset$ and that $\mathrm{Ass}_R(K) \subseteq \mathcal{S}$. Show that $\mathrm{Ass}_R(K) \subseteq \mathrm{Ass}_R(M/N')$ implies that $\mathrm{Ass}_R(K) = \emptyset$, and deduce that $K = 0$.

**(d)** Prove $\varphi$ and $\pi$ have the same kernel, i.e., $N = N'$, and this submodule of $M$ is unique.

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\text{Supp}(M)$.

The next two exercises establish a fundamental relation between the sets $\text{Ass}_R(M)$ and $\text{Supp}(M)$ of prime ideals related to the $R$-module $M$.

**39.** Prove that $\text{Ass}_R(M) \subseteq \text{Supp}(M)$. [If $Rm \cong R/P$ use Proposition 42(4) and Proposition 46(1) to show that $0 \neq (Rm)_P \subseteq M_P$.]

As usual $R$ is a commutative ring with 1 and $D$ is a multiplicatively closed set in $R$.

The following eleven exercises introduce the notion of the *support* of an $R$-module $M$ and its relation to the associated primes of $M$. Cf. also Exercises 29 to 35 in Section 1 and Exercises 25 to 30 in Section 5.

**Definition.** If $M$ is an $R$-module, then the set of prime ideals $P$ of $R$ for which the localization $M_P$ is nonzero is called the *support* of $M$, denoted $\text{Supp}(M)$.

The next two exercises establish a fundamental relation between the sets $\text{Ass}_R(M)$ and $\text{Supp}(M)$ of prime ideals related to the $R$-module $M$.

**40.** Suppose that $R$ is Noetherian and $M$ is an $R$-module.

**(a)** If $P \in \text{Supp}(M)$ prove that $P$ contains a prime ideal $Q$ with $Q \in \text{Ass}_R(M)$.

**(b)** If $P$ is a minimal prime in $\text{Supp}(M)$, show that $P \in \text{Ass}_R(M)$. [Use Exercise 33 in Section 1 to show that $\text{Ass}_{R_P}(M_P) \neq \emptyset$ and then use Exercise 37.]

**(c)** Conclude that $\text{Ass}_R(M) \subseteq \text{Supp}(M)$ and that these two sets have the same minimal elements.

## 15.5. The Prime Spectrum of a Ring

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**1.** If $N$ is the nilradical of $R$, prove that Spec $R$ and Spec $R/N$ are homeomorphic. [Show that the natural homomorphism from $R$ to $R/N$ induces a Zariski continuous isomorphism from Spec $R/N$ to Spec $R$.]

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**2.** Let $I$ be an ideal in the ring $R$. Prove that the continuous map from Spec $R/I$ to Spec $R$ induced by the canonical projection homomorphism $R \to R/I$ maps Spec $R/I$ homeomorphically onto the closed set $\mathcal{Z}(I)$ in Spec $R$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**3.** Prove that two elements $f, g \in R$ have the same values at all elements $P$ in Spec $R$ if and only if $f - g$ is contained in the nilradical of $R$. In particular, prove that an element in an affine $k$-algebra is uniquely determined by its values.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**4.** Let $k$ be an arbitrary field, not necessarily algebraically closed. Prove that the prime ideals in $k[x, y]$ (i.e., the elements of Spec $k[x, y]$) are

**(i)** (0),

**(ii)** $(f)$ where $f$ is an irreducible polynomial in $k[x, y]$, and

**(iii)** $(p(x), g(x, y))$ where $p(x)$ is an irreducible polynomial in $k[x]$ and $g(x, y)$ is an irreducible polynomial in $k[x, y]$ that is irreducible modulo $p(x)$, i.e., $g(x, y)$ remains irreducible in the quotient $k[x, y]/(p(x))$.

Prove that mSpec $k[x, y]$ consists of the primes in (iii). [Use Exercise 20 in Section 1.]

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**5.** Let $\mathfrak{m} = (p(x), g(x, y))$ be a maximal ideal in $k[x, y]$ as in the previous exercise. Show that $K = k[x, y]/\mathfrak{m}$ is an algebraic field extension of $k$, so that $k[x, y]$ can also be viewed as a subring of $K[x, y]$. If $x, y$ are mapped to $\alpha, \beta \in K$, respectively, under the canonical homomorphism $k[x, y] \to k[x, y]/\mathfrak{m}$, prove that $\mathfrak{m} = k[x, y] \cap (x - \alpha, y - \beta) \subseteq K[x, y]$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**6.** Describe the elements in Spec $\mathbb{R}[x]$ and Spec $\mathbb{C}[x]$. Describe the elements in Spec $\mathbb{Z}_{(2)}[x]$ where $\mathbb{Z}_{(2)} = \{a/b \in \mathbb{Q} \mid b \text{ is odd}\}$ is the localization of $\mathbb{Z}$ at the prime (2).

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**7.** Let $(f) = (x^5 + x + 1)$ in $\operatorname{Spec} \mathbb{Z}[x]$ viewed as fibered over $\operatorname{Spec} \mathbb{Z}$ as in Example 3 following Proposition 55. Show that there are two closed points in the fiber over (2), three closed points in the fiber over (5), four closed points in the fiber over (19), and five closed points in the fiber over (211).

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**8.** Let $(f) = (x^4 + 1)$ in Spec $\mathbb{Z}[x]$ viewed as fibered over Spec $\mathbb{Z}$ as in Example 3 following Proposition 55. Prove that there is one closed point in the fiber over (2), four closed points in the fiber over $p$ for $p$ odd, $p \equiv 1 \bmod 8$, and two closed points in the fiber over $p$ for all other odd primes $p$ (cf. Corollary 16 in Section 3 of Chapter 14).

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**9.** Prove that the elements in the fiber over $(p)$ of the Zariski continuous map from Spec $\mathbb{Z}[x]$ to Spec $\mathbb{Z}$ are homeomorphic with the elements in Spec$(\mathbb{Z}[x] \otimes_{\mathbb{Z}} \mathbb{F}_p)$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**10.** Let $X = \operatorname{Spec} R$ and let $X_f$ be the principal open set corresponding to $f \in R$. Prove that $X_f \cap X_g = X_{fg}$. Prove that $X_f = X$ if and only if $f$ is a unit in $R$, and that $X_f = \emptyset$ if and only if $f$ is nilpotent.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**11.** If $X_f$ and $X_g$ are principal open sets in $X = \operatorname{Spec} R$, prove that the open set $X_f \cup X_g$ is the complement of the closed set $\mathcal{Z}(I)$ where $I = (f, g)$ is the ideal in $R$ generated by $f$ and $g$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**12.** Prove that a Zariski open subset $U$ of $X = \operatorname{Spec} R$ is quasicompact if and only if $U$ is a finite union of principal open subsets. Give an example of a ring $R$, a Zariski open subset $U$ of $\operatorname{Spec} R$, and a Zariski open covering of $U$ that cannot be reduced to a finite subcovering.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**13.** Let $\varphi : R \to S$ be a homomorphism of rings. Prove that under the induced map $\varphi^*$ from $Y = \operatorname{Spec} S$ to $X = \operatorname{Spec} R$ the full preimage of the principal open set $X_f$ in $X$ is the principal open set $Y_{\varphi(f)}$ in $Y$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**14.** Suppose that $R = R_1 \times R_2$ is the direct product of the rings $R_1$ and $R_2$. Prove that $X = \operatorname{Spec} R$ is the disjoint union of open subspaces $X_1$, $X_2$ (which are therefore also closed), where $X_1$ is homeomorphic to $\operatorname{Spec} R_1$ and $X_2$ is homeomorphic to $\operatorname{Spec} R_2$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**15.** Prove that $X = \operatorname{Spec} R$ is not connected if and only if $R$ is the direct product of two nonzero rings if and only if $R$ contains an idempotent $e$ with $e \neq 0, 1$ (cf. the previous exercise).

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**16.** Prove that $X = \operatorname{Spec} R$ is irreducible (i.e., any two nonempty open subsets have a nontrivial intersection) if and only if $X_f \cap X_g \neq \emptyset$ for any two nonempty principal open sets $X_f$ and $X_g$. Deduce that $X = \operatorname{Spec} R$ is irreducible if and only if the nilradical of $R$ is a prime ideal. [Use Exercise 10.]

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

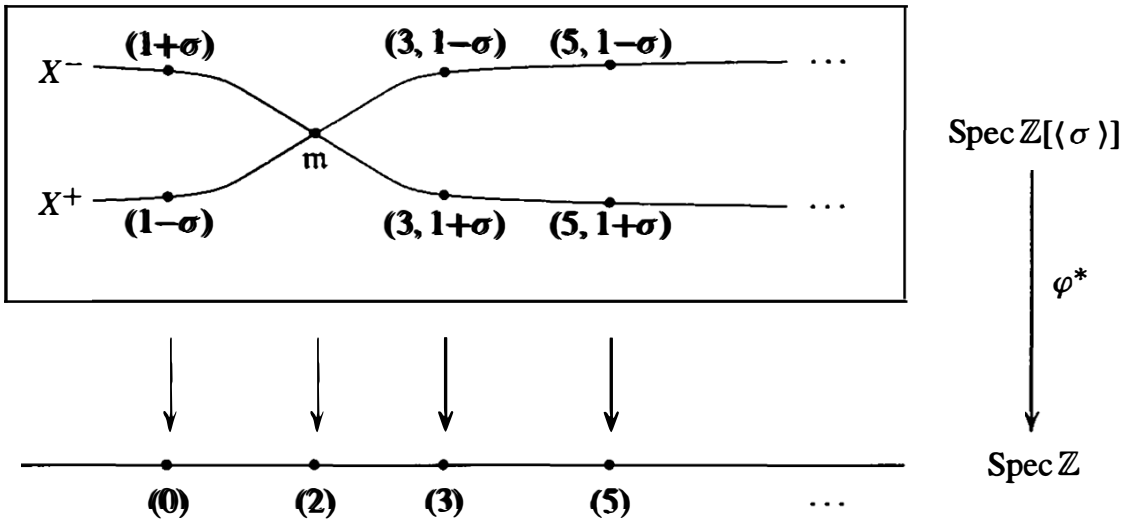
**17.** Let $G = \langle \sigma \rangle$ be a group of order 2, let $R = \mathbb{Z}[G] = \{a + b\sigma \mid a, b \in \mathbb{Z}\}$ be the corresponding group ring, and let $X = \operatorname{Spec} R$.

**(a)** Prove that the nilradical of $R$ is (0) but is not a prime ideal. Prove that $X = X^+ \cup X^-$ where $X^+ = \mathcal{Z}(1 - \sigma)$ and $X^- = \mathcal{Z}(1 + \sigma)$. [Use $(1 + \sigma)(1 - \sigma) = 0$.]

**(b)** Prove that the homomorphism $\mathbb{Z}[G] \to \mathbb{Z}$ defined by mapping $\sigma$ to 1 induces a homeomorphism of $X^+$ with $\operatorname{Spec} \mathbb{Z}$, and the homomorphism mapping $\sigma$ to $-1$ induces a homeomorphism of $X^-$ with $\operatorname{Spec} \mathbb{Z}$.

**(c)** Prove that $X^+ \cap X^-$ consists of the single element $\mathfrak{m} = (1 + \sigma, 1 - \sigma) = (2, 1 - \sigma)$ and that this is a closed point in $X$.

**(d)** Show that $(1 - \sigma)$ and $(1 + \sigma)$ are the unique non-closed points in $X$, with closures $X^+$ and $X^-$, respectively. Describe the closed points, $\operatorname{mSpec} R$, in $X$ and prove that $\operatorname{Spec} \mathbb{Z}[\langle \sigma \rangle]$ can be pictured as follows:



All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**18.** Let $\mathcal{O}$ be the structure sheaf on $X = \operatorname{Spec} R$, let $U$ be an open set in $X$, and suppose $s, t \in \mathcal{O}(U)$. If $s = a/f_1^n$ on $X_{f_1}$ and $t = b/f_2^m$ on $X_{f_2}$, show that

$$st = (abf_1^m f_2^n)/(f_1 f_2)^{n+m} \quad \text{and} \quad s + t = (af_1^m f_2^{m+n} + bf_1^{m+n} f_2^n)/(f_1 f_2)^{n+m}$$

on $X_{f_1 f_2}$. Deduce that $\mathcal{O}(U)$ is a commutative ring with identity.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**19.** Let $\mathcal{O}$ be the structure sheaf on $X = \operatorname{Spec} R$, let $V \subseteq U$ be open sets in $X$, and let $s \in \mathcal{O}(U)$. Suppose $P \in V$ and that $s = a/f^n$ on $X_f \subseteq U$.

**(a)** Show that there is a principal open set $X_{f'} \subseteq V \cap X_f$ containing $P$.

**(b)** Show that $(f')^m = bf$ for some $b \in R$.

**(c)** Show that $s = (ab^n)/(f')^{mn}$ on $X_{f'}$ and conclude that restricting $s$ to $V$ gives a well defined ring homomorphism from $\mathcal{O}(U)$ to $\mathcal{O}(V)$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**20.** Let $\varphi : R \to S$ be a homomorphism of rings, let $X = \operatorname{Spec} R$, $Y = \operatorname{Spec} S$, and let $V \subseteq U$ be Zariski open subsets of $X$. Set $V' = (\varphi^*)^{-1}(V)$ and $U' = (\varphi^*)^{-1}(U)$, the corresponding Zariski open subsets of $Y$ with respect to the continuous map $\varphi^* : Y \to X$ induced by $\varphi$. Prove that the induced map $\varphi^\# : \mathcal{O}_X(U) \to \mathcal{O}_Y(U')$ on sections is a ring homomorphism. Prove that $V' \subseteq U'$ and that $\varphi^\#$ is compatible with restriction i.e., that

the diagram

$$\begin{array}{ccc} \mathcal{O}_X(U) & \xrightarrow{\varphi^\#} & \mathcal{O}_Y(U') \\ \downarrow & & \downarrow \\ \mathcal{O}_X(V) & \xrightarrow{\varphi^\#} & \mathcal{O}_Y(V') \end{array}$$

is commutative, where the vertical maps are the restriction homomorphisms.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**21.** Suppose $D$ is a multiplicatively closed subset of $R$. Show that the localization homomorphism $R \to D^{-1}R$ induces a homeomorphism from $\mathrm{Spec}(D^{-1}R)$ to the collection of prime ideals $P$ of $R$ with $P \cap D = \emptyset$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**22.** Show that $\operatorname{Spec} k[x, y]/(xy)$ is connected but is the union of two proper closed subsets each homeomorphic to $\operatorname{Spec} k[x]$, hence is not irreducible (cf. Exercise 16).

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**23.** For each of the following rings $R$ exhibit the elements of Spec $R$, the open sets $U$ in Spec $R$, the sections $\mathcal{O}(U)$ of the structure sheaf for Spec $R$ for each open $U$, and the stalks $\mathcal{O}_P$ at each point $P \in \text{Spec } R$:
**(a)** $\mathbb{Z}/4\mathbb{Z}$ **(b)** $\mathbb{Z}/6\mathbb{Z}$ **(c)** $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ **(d)** $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**24.** **(a)** If every ideal of $R$ is principal, show every open set in Spec $R$ is a principal open set.
**(b)** Show that if $R = \mathbb{Z}[x]/(4, x^2)$ then $R$ contains a nonprincipal ideal, but every open set in Spec $R$ is a principal open set.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**25.** **(a)** If $M$ is an $R$-module prove that $\text{Supp}(M)$ is a Zariski closed subset of Spec $R$. [Use Exercise 33 of Section 4.]

**(b)** If $M$ is a finitely generated $R$-module prove that $\text{Supp}(M) = \mathcal{Z}(\text{Ann}(M)) \subseteq \text{Spec } R$. [Use Exercise 34 of Section 4.]

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**26.** Suppose $M$ is a finitely generated module over the Noetherian ring $R$.

**(a)** Prove that there are finitely many minimal primes $*P_1, \ldots, P_n$ containing $\text{Ann}(M)$. [Use Corollary 22.]

**(b)** Prove that $\{P_1, \ldots, P_n\}$ is also the set of minimal primes in $\text{Ass}_R(M)$ and that $\text{Supp}(M)$ is the union of the Zariski closed sets $\mathcal{Z}(P_1), \ldots, \mathcal{Z}(P_n)$ in Spec $R$. [Use the previous exercise and Exercise 40 in Section 4.]

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

The previous exercise gives a geometric view of a finitely generated module $M$ over a Noetherian ring $R$: over each point $P$ in Spec $R$ is the localization $M_P$ (the *stalk* over $P$). The stalk is nonzero precisely over the points in the Zariski closed subsets $\mathcal{Z}(P_1), \ldots, \mathcal{Z}(P_n)$ where the $P_i$ are the minimal primes in $\mathrm{Ass}_R(M)$. These ideas lead to the notion of the *(coherent) module sheaf on* Spec $R$ *associated to* $M$ (with a picture similar to that of the structure sheaf following Proposition 58), which is a powerful tool in modern algebraic geometry.

**27.** Let $R = k[x, y]$ and let $M$ be the ideal $(x, y)$ in $R$. Prove that $\mathrm{Supp}(M) = \mathrm{Spec}\, R$ and $\mathrm{Ass}_R(M) = \{0\}$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

The next two exercises show that the associated primes for an ideal $I$ in a Noetherian ring $R$ in the sense of primary decomposition are the associated primes for $I$ in the sense of $\mathrm{Ass}_R(R/I)$.

**28.** This exercise proves that the ideal $Q$ in a Noetherian ring $R$ is **$P$-primary** if and only if $\mathrm{Ass}_R(R/Q) = \{P\}$.

**(a)** Suppose $Q$ is a $P$-primary ideal and let $M$ be the $R$-module $R/Q$. If $0 \neq m \in M$, show that $Q \subseteq \mathrm{Ann}(m) \subseteq P$ and that $\mathrm{rad}\,\mathrm{Ann}(m) = P$. Deduce that if $\mathrm{Ann}(m)$ is a prime ideal then it is equal to $P$ and hence that $\mathrm{Ass}_R(R/Q) = \{P\}$. [Use Exercise 33 in Section 1.]

**(b)** For any ideal $Q$ of $R$, let $0 \neq M \subseteq R/Q$. Prove that the radical of $\mathrm{Ann}(M)$ is the intersection of the prime ideals in $\mathrm{Supp}(M)$. [Use Proposition 12 and Exercise 25.]

**(c)** For $M$ as in (b), prove that the radical of $\mathrm{Ann}M$ is also the intersection of the prime ideals in $\mathrm{Ass}_R(M)$. [Use Exercise 26(b).]

**(d)** If $Q$ is an ideal of $R$ with $\mathrm{Ass}_R(R/Q) = \{P\}$ prove that $\mathrm{rad}\,Q = P$. [Use the fact that $Q = \mathrm{Ann}(R/Q)$ and (c).]

**(e)** If $Q$ is an ideal of $R$ with $\mathrm{Ass}_R(R/Q) = \{P\}$ prove that $Q$ is $P$-primary. [If $ab \in Q$ with $a \notin Q$ consider $0 \neq M = (Ra + Q)/Q \subseteq R/Q$ and show that $b$ is contained in $\mathrm{Ann}M \subseteq \mathrm{rad}\,\mathrm{Ann}(M)$. Use Exercises 33–34 in Section 1, to show that $\mathrm{Ass}_R(M) = \{P\}$, then use (c) to show that $\mathrm{rad}\,\mathrm{Ann}(M) = P$, and conclude finally that $b \in P$.]

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

The next two exercises show that the associated primes for an ideal $I$ in a Noetherian ring $R$ in the sense of primary decomposition are the associated primes for $I$ in the sense of $\mathrm{Ass}_R(R/I)$.

**29.** Suppose $I = Q_1 \cap \cdots \cap Q_n$ is a minimal primary decomposition of the ideal $I$ in the Noetherian ring $R$ with $P_i = \mathrm{rad}\, Q_i$, $i = 1, \ldots, n$. This exercise proves that $\mathrm{Ass}_R(R/I) = \{P_1, \ldots, P_n\}$.

**(a)** Prove that the natural projection homomorphisms induce an injection of $R/I$ into $R/Q_1 \oplus \cdots \oplus R/Q_n$ and deduce that $\mathrm{Ass}_R(R/I) \subseteq \{P_1, \ldots, P_n\}$. [Use Exercise 34 in Section 1 and the previous exercise.]

**(b)** Let $Q_i' = \cap_{j \neq i} Q_j$. Show that the minimality of the decomposition implies that $0 \neq Q_i'/I = (Q_i' + Q_i)/Q_i \subseteq R/Q_i$. Deduce that $\mathrm{Ass}_R(Q_i'/I) = \{P_i\}$. [Use Exercises 33–34 in Section 1 and the previous exercise.] Deduce that $\{P_i\} \in \mathrm{Ass}_R(R/I)$, so that $\mathrm{Ass}_R(R/I) = \{P_1, \ldots, P_n\}$. [Use $Q_i'/I \subseteq R/I$ and Exercise 34 in Section 1.]

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**30.** Let $I$ be the ideal $(x^2, xy, xz, yz)$ in $R = k[x, y, z]$. Prove that $\mathrm{Ass}_R(R/I)$ consists of the primes $\{(x, y), (x, z), (x, y, z)\}$.

All rings are assumed commutative with identity, and all ring homomorphisms are assumed to map identities to identities.

**31.** *(*Spec *for Quadratic Integer Rings)* Let $R$ be the ring of integers in the quadratic field $K = \mathbb{Q}(\sqrt{D})$ where $D$ is a squarefree integer and let $P$ be a nonzero prime ideal in $R$. This exercise shows how the prime ideals in $R$ are determined explicitly from the primes $(p)$ in $\mathbb{Z}$, giving in particular a description of Spec $R$ fibered over Spec $\mathbb{Z}$.

As in the discussion and example following Theorem 29, we have $R = \mathbb{Z}[\omega]$ where $\omega = \sqrt{D}$ if $D \equiv 2, 3 \bmod 4$ (respectively, $\omega = (1+\sqrt{D})/2$ if $D \equiv 1 \bmod 4$), with **minimal** polynomial $m_\omega(x) = x^2 - D$ (respectively, $m_\omega(x) = x^2 - x + (1-D)/4$), and $P \cap \mathbb{Z} = p\mathbb{Z}$ is a nonzero prime ideal of $\mathbb{Z}$.

**(a)** For any prime $p$ in $\mathbb{Z}$ show that $R/pR \cong \mathbb{Z}[x]/(p, m_\omega(x)) \cong \mathbb{F}_p[x]/(\overline{m}_\omega(x))$ as rings, where $\overline{m}_\omega(x)$ is the reduction of $m_\omega(x)$ modulo $p$. Deduce that there is a prime ideal $P$ in $R$ with $P \cap \mathbb{Z} = (p)$ (this gives an alternate proof of Theorem 26(2) in this case).

**(b)** Use the isomorphism in (a) to prove that $P$ is determined explicitly by the factorization of $m_\omega(x)$ modulo $p$:

**(i)** If $\overline{m}_\omega(x) \equiv (x-a)^2 \bmod p$ where $a \in \mathbb{Z}$ then $P = (p, \omega - a)$ and $pR = P^2$. Show that this case occurs only for the finitely many primes $p$ dividing the discriminant of $m_\omega(x)$.

**(ii)** If $\overline{m}_\omega(x) \equiv (x-a)(x-b) \bmod p$ with integers $a, b \in \mathbb{Z}$ that are distinct modulo $p$ then $P$ is either $P_1 = (p, \omega - a)$ or $P_2 = (p, \omega - b)$ and $P_1$, $P_2$ are distinct prime ideals in $R$ with $pR = P_1 P_2$.

**(iii)** If $\overline{m}_\omega(x)$ is irreducible modulo $p$ then $P = pR$.

**(c)** Show that the picture for Spec $R$ over Spec $\mathbb{Z}$ for any $D$ is similar to that for the case $R = \mathbb{Z}[i]$ when $D = -1$: there is precisely one nonclosed point $(0) \in$ Spec $R$ over $(0) \in$ Spec $\mathbb{Z}$, precisely one closed point $P \in$ Spec $R$ over each of the primes $(p)$ in Spec $\mathbb{Z}$ in (i) (called *ramified* primes) and over the primes in (iii) (called *inert* primes), and precisely two closed points over the primes in (ii) (called *split* primes).

# 16. Artinian Rings, Discrete Valuation Rings, and Dedekind Domains

## 16.1. Artinian Rings

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**1.** Suppose $R$ is an Artinian ring and $I$ is an ideal in $R$. Prove that $R/I$ is also Artinian.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**2.** Show that every finite commutative ring with 1 is Artinian.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**3.** Prove that an integral domain of Krull dimension 0 is a field.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**4.** Prove that an Artinian integral domain is a field.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**5.** Suppose $I$ is a nilpotent ideal in $R$ and $M = IM$ for some $R$-module $M$. Prove that $M = 0$.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**6.** Suppose that $0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$ is an exact sequence of $R$-modules. Prove that $M$ is an Artinian $R$-module if and only if $M'$ and $M''$ are Artinian $R$-modules.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**7.** Suppose $R = F$ is a field. Prove that an $R$-module $M$ is Artinian if and only if it is Noetherian if and only if $M$ is a finite dimensional vector space over $F$.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**8.** Let $M$ be a maximal ideal of the ring $R$ and suppose that $M^n = 0$ for some $n \geq 1$. Prove that $R$ is Noetherian if and only if $R$ is Artinian. [Observe the each successive quotient $M^i/M^{i+1}, i = 0, \ldots, n-1$ in the filtration $R \supseteq M \supseteq \cdots \supseteq M^{n-1} \supseteq M^n = 0$ is a module over the field $F = R/M$. Then use the previous two exercises and Exercise 6 of Section 15.1.]

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**9.** Let $M$ be a finitely generated $R$-module. Prove that if $x_1, \ldots, x_n$ are elements of $M$ whose images in $M/\mathcal{J}M$ generate $M/\mathcal{J}M$, then they generate $M$. Deduce that if $R$ is Noetherian and the images of $a_1, \ldots, a_n$ in $\mathcal{J}/\mathcal{J}^2$ generate $\mathcal{J}/\mathcal{J}^2$, then $\mathcal{J} = (a_1, \ldots, a_n)$. [Let $N$ be the submodule generated by $x_1, \ldots, x_n$ and apply Nakayama's Lemma to the module $A = M/N$.]

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**10.** Let $R = \mathbb{Z}_{(2)}$ be the localization of $\mathbb{Z}$ at the prime ideal (2). Prove that Jac $R = (2)$ is the ideal generated by 2. If $M = \mathbb{Q}$, prove that $M/2M$ is a finitely generated $R$-module but that $M$ is not finitely generated over $R$. Why doesn't this contradict the previous exercise? [Note the hypotheses in Nakayama's Lemma.]

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**11.** Let $V$ be an affine variety over a field $k$ and let $R = k[V]$ be its coordinate ring. Let $d_t(R)$ denote the transcendence degree of the field of fractions $k(V)$ over $k$, and let $d_p(R)$ be the Krull dimension of $R$ defined in terms of chains of prime ideals. This exercise shows $d_t(R) = d_p(R)$. By Noether's Normalization Lemma there is a polynomial subring $R_1 = k[y_1, \ldots, y_m]$ of $R$ such that $R$ is integral over $R_1$.

**(a)** Show that $d_t(R_1) = d_t(R) = m$ and that $d_p(R_1) = d_p(R)$. Deduce that we may assume $R = R_1$. [Use the Going-up and Going-down Theorems (cf. Theorem 26, Section 15.3) to prove the second equality.]

**(b)** When $R = R_1$ show that $d_p(R) \geq d_t(R)$ by exhibiting an explicit chain of prime ideals of length $m$.

**(c)** When $R = R_1$ show that any nonzero prime ideal of $R$ contains an element $f$ such that $R(f)$ is transcendental over $R$ of transcendence degree 1. Use induction to show that $d_p(R) \leq d_t(R)$, and deduce that $d_p(R) = d_t(R)$.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**12.** Let $R$ be a Noetherian local ring with maximal ideal $M$.

**(a)** The quotient $M/M^2$ is a module (i.e., vector space) over the field $R/M$. Prove that $d = \dim_{R/M}(M/M^2)$ is finite.

**(b)** Prove that $M$ can be generated as an ideal in $R$ by $d$ elements and by no fewer. [Use Exercise 9.]

**(c)** Let $R = k[x_1, \ldots, x_n]_{(x_1,\ldots,x_n)}$ be the localization of the polynomial ring $k[x_1, \ldots, x_n]$ over the field $k$ at the maximal ideal $(x_1, \ldots, x_n)$, and let $M$ be the maximal ideal in $R$. Prove that $\dim_{R/M}(M/M^2) = n = \dim R$. [Cf. the previous exercise.]

It can be shown that $\dim_{R/M}(M/M^2) \geq \dim R$ for any Noetherian local ring $R$ with maximal ideal $M$. A Noetherian local ring $R$ is called a *regular local ring* if $\dim_{R/M}(M/M^2) = \dim R$. It is a fact that a regular local ring is necessarily an integral domain and is also integrally closed.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**13.** If $R$ is a Noetherian ring, prove that the Zariski topology on Spec $R$ is discrete (i.e., every subset is Zariski open and also Zariski closed) if and only if $R$ is Artinian.

Let $R$ be a commutative ring with 1 and let $\mathcal{J}$ be its Jacobson radical.

**14.** Suppose $I$ is the ideal $(x_1, x_2^2, x_3^3, \dots)$ in the polynomial ring $k[x_1, x_2, x_3, \dots]$ where $k$ is a field and let $R$ be the quotient ring $k[x_1, x_2, x_3, \dots]/I$. Prove that the image of the ideal $(x_1, x_2, x_3, \dots)$ in $R$ is the unique prime ideal in $R$ but is not finitely generated. Deduce that $R$ is a local ring of Krull dimension 0 but is not Artinian.

## 16.2. Discrete Valuation Rings

**1.** Suppose $R$ is a Discrete Valuation Ring with respect to the valuation $\nu$ on the fraction field $K$ of $R$. If $x, y \in K$ with $\nu(x) < \nu(y)$ prove that $\nu(x + y) = \min(\nu(x), \nu(y))$. [Note that $x + y = x(1 + y/x)$.]

**2.** Suppose $R$ is a Discrete Valuation Ring with unique maximal ideal $M$ and quotient $F = R/M$. For any $n \geq 0$ show that $M^n/M^{n+1}$ is a vector space over $F$ and that $\dim_F(M^n/M^{n+1}) = 1$.

**3.** Suppose $R$ is an integral domain that is also a local ring whose unique maximal ideal $M = (t)$ is nonzero and principal, and suppose that $\cap_{n \geq 1}(t^n) = 0$. Prove that $R$ is a Discrete Valuation Ring. [Show that every nonzero element in $R$ can be written in the form $ut^n$ for some unit $u \in R$ and some $n \geq 0$.]

**4.** Suppose $R$ is a Noetherian local ring whose unique maximal ideal $M = (t)$ is principal. Prove that either $R$ is a Discrete Valuation Ring or $t^n = 0$ for some $n \geq 0$. In the latter case show that $R$ is Artinian.

**5.** Suppose that $R$ is a Noetherian integral domain that is also a local ring of Krull dimension 1. Let $M$ be the unique maximal ideal of $R$ and let $F = R/M$, so that $M/M^2$ is a vector space over $F$.

**(a)** Prove that if $\dim_F(M/M^2) = 1$ then $R$ is a Discrete Valuation Ring.

**(b)** If every nonzero ideal of $R$ is a power of $M$ prove that $R$ is a Discrete Valuation Ring.

**6.** Let $R$ be an integral domain with fraction field $K$. Prove that every finitely generated $R$-submodule of $K$ is a fractional ideal of $R$. If $R$ is Noetherian, prove that $A$ is a fractional ideal of $R$ if and only if $R$ is a finitely generated $R$-submodule of $K$.

**7.** If $R$ is an integral domain and $A$ is a fractional ideal of $R$, prove that if $A$ is projective then $A$ is finitely generated. Conclude that every integral domain that is not Noetherian contains an ideal that is not projective.

**8.** Suppose $R$ is a Noetherian integral domain that is also a local ring with nonzero maximal ideal $M$. Prove that $R$ is a D.V.R. if and only if the only $M$-primary ideals in $R$ are the powers of $M$.

**9.** Let $C = \mathcal{Z}(xz - y^2, yz - x^3, z^2 - x^2y) \subset \mathbb{A}^3$ over the algebraically closed field $k$. If $v = (0, 0, 0) \in C$, prove that $\dim_k(\mathfrak{m}_{v,C}/\mathfrak{m}_{v,C}^2) = 3$ so that $v$ is singular on $C$. Conclude that $k[C]$ is not integrally closed in $k(C)$ and determine its integral closure. [cf. Exercise 27, Section 15.4.]

## 16.3. Dedekind Domains

**1.** If $R$ is an integral domain, show that every fractional ideal of $R$ is invertible if and only if every integral ideal of $R$ is invertible.

**2.** Suppose $R$ is an integral domain with fraction field $K$ and $A_1, A_2, \ldots, A_n$ are fractional ideals of $R$ whose product is a nonzero principal fractional ideal: $A_1 A_2 \cdots A_n = Rx$ for some $0 \neq x \in K$. For each $i = 1, \ldots, n$ prove that $A_i$ is an invertible fractional ideal with inverse $(x^{-1})A_1 \cdots A_{i-1} A_{i+1} \cdots A_n$.

**3.** Suppose $R$ is an integral domain with fraction field $K$ and $P$ is a nonzero prime ideal in $R$. Show that the fractional ideals of $R_P$ in $K$ are the $R_P$-modules of the form $AR_P$ where $A$ is a fractional ideal of $R$.

**4.** Suppose $R$ is an integral domain with fraction field $K$ and $A$ is a fractional ideal of $R$ in $K$. Let $A' = \{x \in K \mid xA \subseteq R\}$ as in Proposition 9.

**(a)** For any prime ideal $P$ in $R$ prove that the localization $(A')_P$ of $A'$ at $P$ is a fractional ideal of $R_P$ in $K$.

**(b)** If $A$ is a finitely generated $R$-module, prove that $(A')_P = (A_P)'$ where $(A_P)'$ is the fractional $R_P$ ideal $\{x \in K \mid xA_P \subseteq R_P\}$ corresponding to the localization $A_P$.

**5.** If $Q_1$ is a $P_1$-primary ideal and $Q_2$ is a $P_2$-primary ideal where $P_1$ and $P_2$ are comaximal ideals in a Noetherian ring $R$, prove that $Q_1$ and $Q_2$ are also comaximal. [Use Proposition 14 in Section 15.2.]

**6.** Suppose $R$ is a Dedekind Domain with fraction field $K$.

**(a)** Prove that every nonzero fractional ideal of $R$ in $K$ can be written uniquely as the product of distinct prime powers $P_1^{a_1} \cdots P_n^{a_n}$ where the $a_i$ are nonzero integers, possibly negative.

**(b)** If $0 \neq x \in K$, let $P^{v_P(x)}$ be the power of the prime $P$ in the factorization of the principal ideal $(x)$ as in (a) (where $v_P(x) = 0$ if $P$ is not one of the primes occurring). Prove $v_P$ is a valuation on $K$ with valuation ring $R_P$, the localization of $R$ at $P$.

**7.** Suppose $R$ is a Noetherian integral domain that is not a field. Prove that $R$ is a Dedekind Domain if and only if for every maximal ideal $M$ of $R$ there are no ideals $I$ of $R$ with $M^2 \subset I \subset M$. [Use Exercise 12 in Section 1 and Theorems 7 and 15.]

**8.** Suppose $R$ is a Noetherian integral domain with Krull dimension 1. Prove that every nonzero ideal $I$ in $R$ can be written uniquely as a product of primary ideals whose radicals are all distinct. [Cf. the proof of Theorem 15. Use the uniqueness of the primary components belonging to the isolated primes in a minimal primary decomposition (Theorem 21 in Section 15.2).]

**9.** Suppose $R$ is an integral domain. Prove that $R_P$ is a D.V.R. for every nonzero prime ideal $P$ if and only if $R_M$ is a D.V.R. for every nonzero maximal ideal.

**10.** Suppose $R$ is a Noetherian integral domain that is not a field. Prove that $R$ is a Dedekind Domain if and only if nonzero primes $M$ are maximal and every $M$-primary ideal is a power of $M$.

**11.** If $I$ and $J$ are nonzero ideals in the Dedekind Domain $R$ show there exists an integral ideal $I_1$ in $R$ that is relatively prime to both $I$ and $J$ such that $I_1 I$ is a principal ideal in $R$.

**12.** If $I$ and $J$ are nonzero fractional ideals for the Dedekind Domain $R$ prove there are elements $\alpha, \beta \in K$ such that $\alpha I$ and $\beta J$ are nonzero integral ideals in $R$ are relatively prime.

**13.** Suppose $I$ and $J$ are nonzero ideals in the Dedekind Domain $R$. Prove that there is an ideal $I_1 \cong I$ that is relatively prime to $J$. [Use Corollary 19 to find an ideal $I_2$ with $I_2 I = (a)$ and $(I_2, J) = R$. If $I_2 = P_1^{e_1} \cdots P_n^{e_n}$, choose $b \in R$ with $b \in P_i^{e_i} - P_i^{e_i+1}$ and $b \equiv 1 \bmod P$ for every prime $P$ dividing $J$. Show that $(b) = I_2 I_1$ for some ideal $I_1$ and consider $(a)I_1$ to prove that $I_1 \cong I$.]

**14.** Prove that every finitely generated torsion module over a Dedekind Domain $R$ is isomorphic to a direct sum $R/I_1 \oplus R/I_2 \oplus \cdots \oplus R/I_n$ with unique nonzero ideals $I_1, \ldots, I_n$ of $R$ satisfying $I_1 \subseteq I_2 \subseteq \cdots \subseteq I_n$ (called the *invariant factors* of $M$). [cf. Section 12.1.]

**15.** If $P$ is a nonzero prime ideal in the Dedekind Domain $R$ prove that $R/P^n$ is not a projective $R$-module for any $n \geq 1$. [Consider the exact sequence $0 \to P^n/P^{n+1} \to R/P^{n+1} \to R/P^n \to 0$.] Conclude that if $M \neq 0$ is a finitely generated torsion $R$-module then $M$ is not projective. [cf. Exercise 3, Section 10.5.]

**16.** Prove that the class number of the Dedekind Domain $R$ is 1 if and only if every finitely generated projective $R$-module is free.

**17.** Suppose $R$ is a Dedekind Domain.

**(a)** Show that $I \sim J$ if and only if $I \cong J$ as $R$-modules defines an equivalence relation on the set of nonzero fractional ideals of $R$. Let $C(R)$ be the corresponding set of $R$-module isomorphism classes and let $[I] \in C(R)$ denote the equivalence class containing the fractional ideal $I$ of $R$.

**(b)** Show that the multiplication $[I][J] = [I \oplus J]$ gives a well defined binary operation with respect to which $C(R)$ is an abelian group with identity $1 = [R]$.

**(c)** Prove that the abelian group $C(R)$ in (b) is isomorphic to the class group of $R$.

**18.** If $R$ is a Dedekind Domain and $I$ is any nonzero ideal, prove that $R/I$ contains only finitely many ideals. In particular, show that $R/I$ is an Artinian ring.

**19.** Suppose $I$ is a nonzero fractional ideal in the Dedekind Domain $R$. Explicitly exhibit $I$ as a direct summand of a free $R$-module to show that $I$ is projective. [Consider $I \oplus I^{-1}$ and use Proposition 21.]

**20.** Suppose $I$ and $J$ are two nonzero fractional ideals in the Dedekind Domain $R$ and that $I^n = J^n$ for some $n \neq 0$. Prove that $I = J$.

**21.** Suppose $K$ is an algebraic number field and $\mathcal{O}_K$ is the ring of integers in $K$. If $P$ is a nonzero prime ideal in $\mathcal{O}_K$ prove that $P = (p, \pi)$ for some prime $p \in \mathbb{Z}$ and algebraic integer $\pi \in \mathcal{O}_K$.

**22.** Suppose $K = \mathbb{Q}(\sqrt{D})$ is a quadratic extension of $\mathbb{Q}$ where $D$ is a squarefree integer and $\mathcal{O}_K$ is the ring of integers in $K$.

**(a)** Prove that $|\mathcal{O}_K/(p)| = p^2$. [Observe that $\mathcal{O}_K \cong \mathbb{Z}^2$ as an abelian group.]

**(b)** Use Corollary 16 to show that there are 3 possibilities for the prime ideal factorization of $(p)$ in $\mathcal{O}_K$:

(i) $(p) = P$ is a prime ideal with $|\mathcal{O}_K/P| = p^2$,

(ii) $(p) = P_1P_2$ with distinct prime ideals $P_1$, $P_2$ and $|\mathcal{O}_K/P_1| = |\mathcal{O}_K/P_2| = p$,

(iii) $(p) = P^2$ for some prime ideal $P$ with $|\mathcal{O}_K/P| = p$.

(In cases (i), (ii), and (iii) the prime $p$ is said to be *inert*, *split*, or *ramified* in $\mathcal{O}_K$, respectively. The set of ramified primes is finite: the primes $p$ dividing $D$ if $D \equiv 1, 2 \bmod 4$; $p = 2$ and the primes $p$ dividing $D$ if $D \equiv 3 \bmod 4$. Cf. Exercise 31 in Section 15.5.)

**(c)** Determine the prime ideal factorizations of the primes $p = 2, 3, 5, 7, 11$ in the ring of integers $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ of $K = \mathbb{Q}(\sqrt{-5})$.

**23.** Let $\mathcal{O}$ be the ring of integers in the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$.

**(a)** Show that the infinite sequence of ideals in $\mathcal{O}$ $(2) \subseteq (\sqrt{2}) \subseteq (\sqrt[4]{2}) \subseteq (\sqrt[8]{2}) \subseteq \cdots$ is strictly increasing, and so $\mathcal{O}$ is not Noetherian.

**(b)** Show that $\mathcal{O}$ has Krull dimension 1. [Use Theorem 26 in Section 15.3.]

**(c)** Let $K$ be a number field and let $I$ be any ideal in $\mathcal{O}_K$. Show that there is some finite extension $L$ of $K$ such that $I$ becomes principal when extended to $\mathcal{O}_L$, i.e., the ideal $I\mathcal{O}_L$ is principal (where $L$ depends on $I$)—you may use the theorem that the class group of $K$ is a finite group. [cf. Exercise 20.]

**(d)** Prove that $\mathcal{O}$ is a Bezout Domain (cf. Section 8.1).

**24.** Suppose $F$ and $K$ are algebraic number fields with $\mathbb{Q} \subseteq F \subseteq K$, with rings of integers $\mathcal{O}_F$ and $\mathcal{O}_K$, respectively. Since $\mathcal{O}_F \subseteq \mathcal{O}_K$, the ring $\mathcal{O}_K$ is naturally a module over $\mathcal{O}_F$.

**(a)** Prove $\mathcal{O}_K$ is a torsion free $\mathcal{O}_F$-module of rank $n = [K : F]$. [Compute ranks over $\mathbb{Z}$.] If $\mathcal{O}_K$ is *free* over $\mathcal{O}_F$ then $\mathcal{O}_K$ is said to have a ***relative integral basis*** over $\mathcal{O}_F$.

**(b)** Prove that if $F$ has class number 1 then $\mathcal{O}_K$ has a relative integral basis over $\mathcal{O}_F$.

If $K = \mathbb{Q}(\sqrt{-5}, \sqrt{2})$ then the ring of integers $\mathcal{O}_K$ is given by

$$\mathcal{O}_K = \mathbb{Z} + \mathbb{Z}\sqrt{-5} + \mathbb{Z}\sqrt{-10} + \mathbb{Z}\omega \qquad \text{where } \omega = (\sqrt{-10} + \sqrt{2})/2.$$

**(c)** If $F_1 = \mathbb{Q}(\sqrt{2})$ prove that $\mathcal{O}_K$ has a relative integral basis over $\mathcal{O}_{F_1}$ and find an explicit basis $\{\alpha, \beta\}$: $\mathcal{O}_K = \mathcal{O}_{F_1} \cdot \alpha + \mathcal{O}_{F_1} \cdot \beta$.

**(d)** If $F_2 = \mathbb{Q}(\sqrt{-5})$, show that $P_3 = (3, 1 + \sqrt{-5}) = (3, 5 - \sqrt{-5})$ is a prime ideal of $\mathcal{O}_{F_2}$ that is not principal and that $\mathcal{O}_K = \mathcal{O}_{F_2} \cdot 1 + (1/3)P_3 \cdot \omega$. [Check that $\sqrt{-10} = -(5 - \sqrt{-5})\omega/3$.] Conclude that the Steinitz class of $\mathcal{O}_K$ as a module over $\mathcal{O}_{F_2}$ is the nontrivial class of $P_3$ in the class group of $\mathcal{O}_{F_2}$ and so there is no relative integral basis of $\mathcal{O}_K$ over $\mathcal{O}_{F_2}$.

**(e)** Determine whether $\mathcal{O}_K$ has a relative integral basis over the ring of integers of the remaining quadratic subfield $F_3 = \mathbb{Q}(\sqrt{-10})$ of $K$.

**25.** Suppose $C$ is a nonsingular irreducible affine curve over an algebraically closed field $k$. Prove that the coordinate ring $k[C]$ is a Dedekind Domain.

# 17. Introduction to Homological Algebra and Group Cohomology

## 17.1. Introduction to Homological Algebra—Ext and Tor

**1.** Give the details of the proof of Proposition 1.

**2.** This exercise defines the connecting map $\delta_n$ in the Long Exact Sequence of Theorem 2 and proves it is a homomorphism. In the notation of Theorem 2 let $0 \to \mathcal{A} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\beta} \mathcal{C} \to 0$ be a short exact sequence of cochain complexes, where for simplicity the cochain maps for $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ are all denoted by the same $d$.

**(a)** If $c \in C^n$ represents the class $x \in H^n(\mathcal{C})$ show that there is some $b \in B^n$ with $\beta_n(b) = c$.

**(b)** Show that $d_{n+1}(b) \in \ker \beta_{n+1}$ and conclude that there is a unique $a \in A^{n+1}$ such that $\alpha_{n+1}(a) = d_{n+1}(b)$. [Use $c \in \ker d_{n+1}$ and the commutativity of the diagram.]

**(c)** Show that $d_{n+2}(a) = 0$ and conclude that $a$ defines a class $\bar{a}$ in the quotient group $H^{n+1}(\mathcal{A})$. [Use the fact that $\alpha_{n+2}$ is injective.]

**(d)** Prove that $\bar{a}$ is independent of the choice of $b$, i.e., if $b'$ is another choice and $a'$ is its unique preimage in $A^{n+1}$ then $\bar{a} = \overline{a'}$, and that $\bar{a}$ is also independent of the choice of $c$ representing the class $x$.

**(e)** Define $\delta_n(x) = \bar{a}$ and prove that $\delta_n$ is a group homomorphism from $H^n(\mathcal{C})$ to $H^{n+1}(\mathcal{A})$. [Use the fact that $\delta_n(x)$ is independent of the choices of $c$ and $b$ to compute $\delta_n(x_1 + x_2)$.]

**3.** Suppose

$$\begin{array}{ccccccc} & & A & \xrightarrow{\alpha} & B & \xrightarrow{\beta} & C \longrightarrow 0 \\ & & \downarrow f & & \downarrow g & & \downarrow h \\ 0 & \longrightarrow & A' & \xrightarrow{\alpha'} & B' & \xrightarrow{\beta'} & C' \end{array}$$

is a commutative diagram of $R$-modules with exact rows.

**(a)** If $c \in \ker h$ and $\beta(b) = c$ prove that $g(b) \in \ker \beta'$ and conclude that $g(b) = \alpha'(a')$ for some $a' \in A'$. [Use the commutativity of the diagram.]

**(b)** Show that $\delta(c) = a' \bmod \text{image } f$ is a well defined $R$-module homomorphism from $\ker h$ to the quotient $A'/\text{image } f$.

**(c)** *(The Snake Lemma)* Prove there is an exact sequence

$$\ker f \longrightarrow \ker g \longrightarrow \ker h \xrightarrow{\delta} \operatorname{coker} f \longrightarrow \operatorname{coker} g \longrightarrow \operatorname{coker} h$$

where $\operatorname{coker} f$ (the *cokernel* of $f$) is $A'/(\text{image } f)$ and similarly for $\operatorname{coker} g$ and $\operatorname{coker} h$.

**(d)** Show that if $\alpha$ is injective and $\beta'$ is surjective (i.e., the two rows in the commutative diagram above can be extended to short exact sequences) then the exact sequence in (c) can be extended to the exact sequence

$$0 \longrightarrow \ker f \longrightarrow \ker g \longrightarrow \ker h \xrightarrow{\delta} \operatorname{coker} f \longrightarrow \operatorname{coker} g \longrightarrow \operatorname{coker} h \longrightarrow 0$$

**4.** Let $\mathcal{A} = \{A^n\}$ and $\mathcal{B} = \{B^n\}$ be cochain complexes, where the maps $A^n \to A^{n+1}$ and $B^n \to B^{n+1}$ in both complexes are denoted by $d_{n+1}$ for all $n$. Cochain complex homomorphisms $\alpha$ and $\beta$ from $\mathcal{A}$ to $\mathcal{B}$ are said to be *homotopic* if for all $n$ there are module homomorphisms $s_n : A^{n+1} \to B^n$ such that the maps $\alpha_n - \beta_n$ from $A^n$ to $B^n$ satisfy

$$\alpha_n - \beta_n = d_n s_{n-1} + s_n d_{n+1}.$$

The collection of maps $\{s_n\}$ is called a *cochain homotopy* from $\alpha$ to $\beta$. One may similarly define chain homotopies between chain complexes.

**(a)** Prove that homotopic maps of cochain complexes induce the same maps on cohomology, i.e., if $\alpha$ and $\beta$ are homotopic homomorphisms of cochain complexes then the induced group homomorphisms from $H^n(\mathcal{A})$ to $H^n(\mathcal{B})$ are equal for every $n \geq 0$. (Thus "homotopy" gives a sufficient condition for two maps of complexes to induce the same maps on cohomology or homology; this condition is not in general necessary.) [Use the definition of homotopy to show $(\alpha_n - \beta_n)(z) \in \text{image}\, d_n$ for every $z \in \ker d_{n+1}$.]

**(b)** Prove that the relation $\alpha \sim \beta$ if $\alpha$ and $\beta$ are homotopic is an equivalence relation on any set of cochain complex homomorphisms.

**5.** Establish the first step in the Simultaneous Resolution result of Proposition 7 as follows: assume the first two nonzero rows in diagram (11) are given, except for the map from $P_0 \oplus \overline{P}_0$ to $M$ (where the maps along the row of projective modules are the obvious injection and projection for this split exact sequence). Let $\mu : \overline{P}_0 \to M$ be a lifting to $\overline{P}_0$ of the map $\overline{P}_0 \to N$ (which exists because $\overline{P}_0$ is projective). Let $\lambda$ be the composition $P_0 \to L \to M$ in the diagram. Define

$$\pi : P_0 \oplus \overline{P}_0 \to M \qquad \text{by} \qquad \pi(x, y) = \lambda(x) + \mu(y).$$

Show that with this definition the first two nonzero rows of (11) form a commutative diagram.

**6.** Let $0 \to \mathcal{A} \xrightarrow{\alpha} \mathcal{B} \xrightarrow{\beta} \mathcal{C} \to 0$ be a short exact sequence of cochain complexes. Prove that if any two of $\mathcal{A}, \mathcal{B}, \mathcal{C}$ are exact, then so is the third. [Use Theorem 2.]

**7.** Prove that a finitely generated abelian group $A$ is free if and only if $\mathrm{Ext}^1(A, \mathbb{Z}) = 0$.

**8.** Prove that if $0 \to L \to M \to N \to 0$ is a split short exact sequence of $R$-modules, then for every $n \geq 0$ the sequence $0 \to \mathrm{Ext}^n_R(N, D) \to \mathrm{Ext}^n_R(M, D) \to \mathrm{Ext}^n_R(L, D) \to 0$ is also short exact and split. [Use a splitting homomorphism and Proposition 5.]

**9.** Show that

$$0 \longrightarrow \mathbb{Z}/d\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \xrightarrow{d} \mathbb{Z}/m\mathbb{Z} \xrightarrow{m/d} \mathbb{Z}/m\mathbb{Z} \xrightarrow{d} \mathbb{Z}/m\mathbb{Z} \xrightarrow{m/d} \cdots$$

is an injective resolution of $\mathbb{Z}/d\mathbb{Z}$ as a $\mathbb{Z}/m\mathbb{Z}$-module. [Use Proposition 36 in Section 10.5.] Use this to compute the groups $\mathrm{Ext}^n_{\mathbb{Z}/m\mathbb{Z}}(A, \mathbb{Z}/d\mathbb{Z})$ in terms of the dual group $\mathrm{Hom}_{\mathbb{Z}/m\mathbb{Z}}(A, \mathbb{Z}/m\mathbb{Z})$. In particular, if $m = p^2$ and $d = p$, give another derivation of the result $\mathrm{Ext}^n_{\mathbb{Z}/p^2\mathbb{Z}}(\mathbb{Z}/p\mathbb{Z}, \mathbb{Z}/p\mathbb{Z}) \cong \mathbb{Z}/p\mathbb{Z}$.

**10.** **(a)** Prove that an arbitrary direct sum $\oplus_{i\in I} P_i$ of projective modules $P_i$ is projective and that an arbitrary direct product $\prod_{j\in J} Q_j$ of injective modules $Q_j$ is injective.

**(b)** Prove that an arbitrary direct sum of projective resolutions is again projective and use this to show $\mathrm{Ext}^n_R(\oplus_{i\in I} A_i, B) \cong \prod_{i\in I} \mathrm{Ext}^n_R(A_i, B)$ for any collection of $R$-modules $A_i$ $(i \in I)$. [cf. Exercise 12 in Section 10.5.]

**(c)** Prove that an arbitrary direct product of injective resolutions is an injective resolution and use this to show $\mathrm{Ext}^n_R(A, \prod_{j\in J} B_j) \cong \prod_{j\in J} \mathrm{Ext}^n_R(A, B_j)$ for any collection of $R$-modules $B_j$ $(j \in J)$. [cf. Exercise 12 in Section 10.5.]

**(d)** Prove that $\mathrm{Tor}^R_n(A, \oplus_{j\in J} B_j) \cong \oplus_{j\in J} \mathrm{Tor}^R_n(A, B_j)$ for any collection of $R$-modules $B_j$ $(j \in J)$.

**11.** *(Bass' Characterization of Noetherian Rings)* Suppose $R$ is a commutative ring.

**(a)** If $R$ is Noetherian, and $I$ is any nonzero ideal in $R$ show that the image of any $R$-module homomorphism $f : I \to \oplus_{j \in \mathcal{J}} Q_j$ from $I$ into a direct sum of injective $R$-modules $Q_j$ ($j \in \mathcal{J}$) is contained in some finite direct sum of the $Q_j$.

**(b)** If $R$ is Noetherian, prove that an arbitrary direct sum $\oplus_{j \in \mathcal{J}} Q_j$ of injective $R$-modules is again injective. [Use Baer's Criterion (Proposition 36) and Exercise 4 in Section 10.5 together with (a).]

**(c)** Let $I_1 \subseteq I_2 \subseteq \ldots$ be an ascending chain of ideals of $R$ with union $I$ and let $I/I_i \to Q_i$ for $i = 1, 2, \ldots$ be an injection of the quotient $I/I_i$ into an injective $R$-module $Q_i$ (by Theorem 38 in Section 10.5). Prove that the composition of these injections with the product of the canonical projection maps $I \to I_i$ gives an $R$-module homomorphism $f : I \to \oplus_{i=1,2,\ldots} Q_i$.

**(d)** Prove the converse of (b): if an arbitrary direct sum $\oplus_{j \in \mathcal{J}} Q_j$ of injective $R$-modules is again injective then $R$ is Noetherian. [If the direct sum in (c) is injective, use Baer's Criterion to lift $f$ to a homomorphism $F : R \to \oplus_{i=1,2,\ldots} Q_i$. If the component of $F(1)$ in $Q_i$ is 0 for $i \geq n$ prove that $I = I_n$ and the ascending chain of ideals is finite.]

**12.** Prove Proposition 13: $\mathrm{Tor}_0^R(D, A) \cong D \otimes_R A$. [Follow the proof of Proposition 3.]

**13.** Prove Proposition 16 characterizing flat modules.

**14.** Suppose $0 \to A \to B \to C \to 0$ is a short exact sequence of $R$-modules. Prove that if $C$ is a flat $R$-module, then $A$ is flat if and only if $B$ is also flat. [Use the Tor long exact sequence.] Give an example to show that if $A$ and $B$ are flat then $C$ need not be flat.

**15.** **(a)** If $I$ is an ideal in $R$ and $M$ is an $R$-module, prove that $\mathrm{Tor}_1^R(M, R/I)$ is isomorphic to the kernel of the map $M \otimes_R I \to M$ that maps $m \otimes i$ to $mi$ for $i \in I$ and $m \in M$. [Use the Tor long exact sequence associated to $0 \to I \to R \to R/I \to 0$ noting that $R$ is flat.]

**(b)** *(A Flatness Criterion using* Tor*)* Prove that the $R$-module $M$ is flat if and only if $\mathrm{Tor}_1^R(M, R/I) = 0$ for every finitely generated ideal $I$ of $R$. [Use Exercise 25 in Section 10.5.]

**16.** Suppose $R$ is a P.I.D. and $A$ and $B$ are $R$-modules. If $t(B)$ denotes the torsion submodule of $B$ show that $\mathrm{Tor}_1^R(A, t(B)) \cong \mathrm{Tor}_1^R(A, B)$ and deduce that $\mathrm{Tor}_1^R(A, B)$ is isomorphic to $\mathrm{Tor}_1^R(t(A), t(B))$. [Use Exercise 26 in Section 10.5 to show that $B/t(B)$ is flat over $R$, then use the Tor long exact sequence with $D = A$ applied to the short exact sequence $0 \to t(B) \to B \to B/t(B) \to 0$ and the remarks following Proposition 16.]

**17.** Let $A = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z} \oplus \cdots$. Prove that $\mathrm{Ext}^1(A, B) \cong (B/2B) \times (B/3B) \times (B/4B) \times \cdots$ for any abelian group $A$. [Use Exercise 10.] Prove that $\mathrm{Ext}^1(A, B) = 0$ if and only if $B$ is divisible.

**18.** Prove that $\mathbb{Z}/2\mathbb{Z}$ is a projective $\mathbb{Z}/6\mathbb{Z}$-module and deduce that $\mathrm{Tor}_1^{\mathbb{Z}/6\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) = 0$.

**19.** Suppose $r \neq 0$ is not a zero divisor in the commutative ring $R$.

**(a)** Prove that multiplication by $r$ gives a free resolution $0 \to R \xrightarrow{r} R \to R/rR \to 0$ of the quotient $R/rR$.

**(b)** Prove that $\mathrm{Ext}^0_R(R/rR, B) = {}_rB$ is the set of elements $b \in B$ with $rb = 0$, that $\mathrm{Ext}^1_R(R/rR, B) \cong B/rB$, and that $\mathrm{Ext}^n_R(R/rR, B) = 0$ for $n \geq 2$ for every $R$-module $B$.

**(c)** Prove that $\mathrm{Tor}^R_0(A, R/rR) = A/rA$, that $\mathrm{Tor}^R_1(A, R/rR) = {}_rA$ is the set of elements $a \in A$ with $ra = 0$, and that $\mathrm{Tor}^R_n(A, R/rR) = 0$ for $n \geq 2$ for every $R$-module $A$.

**20.** Prove that $\mathrm{Tor}_0^{\mathbb{Z}/m\mathbb{Z}}(A, \mathbb{Z}/d\mathbb{Z}) \cong A/dA$, that $\mathrm{Tor}_n^{\mathbb{Z}/m\mathbb{Z}}(A, \mathbb{Z}/d\mathbb{Z}) \cong {}_dA/(m/d)A$ for $n$ odd, $n \geq 1$, and that $\mathrm{Tor}_n^{\mathbb{Z}/m\mathbb{Z}}(A, \mathbb{Z}/d\mathbb{Z}) \cong {}_{(m/d)}A/dA$ for $n$ even, $n \geq 2$. [Use the projective resolution in Example 2 following Proposition 3.]

**21.** Let $R = k[x, y]$ where $k$ is a field, and let $I$ be the ideal $(x, y)$ in $R$.

**(a)** Let $\alpha : R \to R^2$ be the map $\alpha(r) = (yr, -xr)$ and let $\beta : R^2 \to R$ be the map $\beta((r_1, r_2)) = r_1x + r_2y$. Show that

$$0 \longrightarrow R \xrightarrow{\alpha} R^2 \xrightarrow{\beta} R \longrightarrow k \longrightarrow 0$$

where the map $R \to R/I = k$ is the canonical projection, gives a free resolution of $k$ as an $R$-module.

**(b)** Use the resolution in (a) to show that $\mathrm{Tor}_2^R(k, k) \cong k$.

**(c)** Prove that $\mathrm{Tor}_1^R(k, I) \cong k$. [Use the long exact sequence corresponding to the short exact sequence $0 \to I \to R \to k \to 0$ and (b).]

**(d)** Conclude from (c) that the torsion free $R$-module $I$ is not flat (compare to Exercise 26 in Section 10.5).

**22.** *(Flat Base Change for* Tor*)* Suppose $R$ and $S$ are commutative rings and $f : R \to S$ is a ring homomorphism making $S$ into an $R$-module as in Example 6 following Corollary 12 in Section 10.4. Prove that if $S$ is flat as an $R$-module, then $\mathrm{Tor}_n^R(A, B) \cong \mathrm{Tor}_n^S(S \otimes_R A, B)$ for all $R$-modules $A$ and all $S$-modules $B$. [Show that since $S$ is flat, tensoring an $R$-module projective resolution for $A$ with $S$ gives an $S$-module projective resolution of $S \otimes_R A$.]

**23.** *(Localization and* Tor*)* Let $D^{-1}R$ be the localization of the commutative ring $R$ with respect to the multiplicative subset $D$ of $R$. Prove that localization commutes with Tor, i.e., $D^{-1}\mathrm{Tor}_n^R(A, B) \cong \mathrm{Tor}_n^{D^{-1}R}(D^{-1}A, D^{-1}B)$ for all $R$-modules $A$ and $B$ and all $n \geq 0$. [Use the previous exercise and the fact that $D^{-1}R$ is flat over $R$, cf. Proposition 42(6) in Section 15.4.]

**24.** ***(Flatness is local)*** Suppose $R$ is a commutative ring. Prove that an $R$-module $M$ is flat if and only if every localization $M_P$ is a flat $R_P$-module for every maximal (hence also for every prime) ideal in $R$. [Use the previous exercise together with the characterization of flatness in terms of Tor.]

**25.** If $R$ is an integral domain with field of fractions $F$, prove that $\mathrm{Tor}_1^R(F/R, B) \cong t(B)$ for any $R$-module $B$, where $t(B)$ denotes the $R$-torsion submodule of $B$.

An $R$-module $M$ is said to be *finitely presented* if there is an exact sequence

$$R^s \longrightarrow R^t \longrightarrow M \longrightarrow 0$$

of $R$-modules for some integers $s$ and $t$. Equivalently, $M$ is finitely generated by $t$ elements and the kernel of the corresponding $R$-module homomorphism $R^t \to M$ can be generated by $s$ elements.

**26.** **(a)** Prove that every finitely generated module over a Noetherian ring $R$ is finitely presented. [Use Exercise 8 in Section 15.1.]

**(b)** Prove that an $R$-module $M$ is finitely presented and projective if and only if $M$ is a direct summand of $R^n$ for some integer $n \geq 1$.

An $R$-module $M$ is said to be *finitely presented* if there is an exact sequence

$$R^s \longrightarrow R^t \longrightarrow M \longrightarrow 0$$

of $R$-modules for some integers $s$ and $t$. Equivalently, $M$ is finitely generated by $t$ elements and the kernel of the corresponding $R$-module homomorphism $R^t \to M$ can be generated by $s$ elements.

**27.** Suppose that $M$ is a finitely presented $R$-module and that $0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} M \to 0$ is an exact sequence of $R$-modules. This exercise proves that if $B$ is a finitely generated $R$-module then $A$ is also a finitely generated $R$-module.

**(a)** Suppose $R^s \xrightarrow{\psi} R^t \xrightarrow{\varphi} M \to 0$ and $e_1, \ldots, e_t$ is an $R$-module basis for $R^t$. Show that there exist $b_1, \ldots, b_t \in B$ so that $\beta(b_i) = \varphi(e_i)$ for $i = 1, \ldots, t$.

**(b)** If $f$ is the $R$-module homomorphism from $R^t$ to $B$ defined by $f(e_i) = b_i$ for $i = 1, \ldots, t$, show that $f(\psi(R^s)) \subseteq \ker \beta$. [Use $\varphi \circ \psi = 0$.] Conclude that there is a commutative diagram

$$\begin{array}{ccccccccc} & & & & R^s & \xrightarrow{\psi} & R^t & \xrightarrow{\varphi} & M & \longrightarrow & 0 \\ & & & & \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} & & \| & & \\ 0 & \longrightarrow & A & \xrightarrow{\alpha} & B & \xrightarrow{\beta} & M & \longrightarrow & 0 & & \end{array}$$

of $R$-modules with exact rows.

**(c)** Prove that $A/\text{image } g \cong B/\text{image } f$ and use this to prove that $A$ is finitely generated. [For the isomorphism, use the Snake Lemma in Exercise 3. Then show that image $g$ and $A/\text{image } g$ are both finitely generated and apply Exercise 7 of Section 10.3.]

**(d)** If $I$ is an ideal of $R$ conclude that $R/I$ is a finitely presented $R$-module if and only if $I$ is a finitely generated ideal.

An $R$-module $M$ is said to be *finitely presented* if there is an exact sequence

$$R^s \longrightarrow R^t \longrightarrow M \longrightarrow 0$$

of $R$-modules for some integers $s$ and $t$. Equivalently, $M$ is finitely generated by $t$ elements and the kernel of the corresponding $R$-module homomorphism $R^t \to M$ can be generated by $s$ elements.

**28.** Suppose $R$ is a local ring with unique maximal ideal $\mathfrak{m}$ and $M$ is a finitely presented $R$-module. Suppose $m_1, \ldots, m_s$ are elements in $M$ whose images in $M/\mathfrak{m}M$ form a basis for $M/\mathfrak{m}M$ as a vector space over the field $R/\mathfrak{m}$.

**(a)** Prove that $m_1, \ldots, m_s$ generate $M$ as an $R$-module. [Use Nakayama's Lemma.]

**(b)** Conclude from (a) that there is an exact sequence $0 \to \ker\varphi \to R^s \xrightarrow{\varphi} M \to 0$ that maps a set of free generators of $R^s$ to the elements $m_1, \ldots, m_s$. Deduce that there is an exact sequence

$$\mathrm{Tor}_1^R(M, R/\mathfrak{m}) \longrightarrow (\ker\varphi)/\mathfrak{m}(\ker\varphi) \longrightarrow 0.$$

[Use the Tor long exact sequence with respect to tensoring with $R/\mathfrak{m}$, using the fact that $N \otimes R/\mathfrak{m} \cong N/\mathfrak{m}N$ for any $R$-module $N$ (Example 8 following Corollary 12 in Section 10.4] and the fact that $\varphi : (R/\mathfrak{m})^s \cong M/\mathfrak{m}M$ is an isomorphism by the choice of $m_1, \ldots, m_s$.]

**(c)** Prove that if $\mathrm{Tor}_1^R(M, R/\mathfrak{m}) = 0$ then $m_1, \ldots, m_s$ are a set of *free* $R$-module generators for $M$. [Use the previous exercise and Nakayama's Lemma to show that $\ker\varphi = 0$.]

An $R$-module $M$ is said to be *finitely presented* if there is an exact sequence

$$R^s \longrightarrow R^t \longrightarrow M \longrightarrow 0$$

of $R$-modules for some integers $s$ and $t$. Equivalently, $M$ is finitely generated by $t$ elements and the kernel of the corresponding $R$-module homomorphism $R^t \to M$ can be generated by $s$ elements.

**29.** Suppose $R$ is a local ring with unique maximal ideal $\mathfrak{m}$. This exercise proves that a finitely generated $R$-module is flat if and only if it is free.

**(a)** Prove that $M = F/K$ is the quotient of a finitely generated free module $F$ by a submodule $K$ with $K \subseteq \mathfrak{m}F$. [Let $F$ be a free module with $F/\mathfrak{m}F \cong M/\mathfrak{m}M$.]

**(b)** Suppose $x \in K$ and write $x = a_1e_1 + \cdots + a_ne_n$ where $e_1, \ldots, e_n$ are an $R$-basis for $F$. Let $I = (a_1, \ldots, a_n)$ be the ideal of $R$ generated by $a_1, \ldots, a_n$). Prove that if $M$ is flat, then $I = \mathfrak{m}I$ and deduce that $K = 0$, so $M$ is free. [Use Exercise 25(d) of Section 10.5 to see that $x \in IK \subseteq \mathfrak{m}IF$ and conclude that $I \subseteq \mathfrak{m}I$. Then apply Nakayama's Lemma to the finitely generated ideal $I$.]

**30.** Suppose $R$ is a local ring with unique maximal ideal $\mathfrak{m}$, $M$ is an $R$-module, and consider the following statements:

**(i)** $M$ is a free $R$-module,

**(ii)** $M$ is a projective $R$-module,

**(iii)** $M$ is a flat $R$-module, and

**(iv)** $\mathrm{Tor}_1^R(M, R/\mathfrak{m}) = 0$.

**(a)** Prove that (i) implies (ii) implies (iii) implies (iv).

**(b)** Prove that (i) , (ii), and (iii) are equivalent if $M$ is finitely generated. (Exercise 34 below shows (iii) need not imply (i) or (ii) if $M$ is finitely generated but $R$ is not local.) [Use the previous exercise.]

**(c)** Prove that (i), (ii), (iii), and (iv) are equivalent if $M$ is finitely presented. (Exercise 35 below shows that (iv) need not imply (i), (ii) or (iii) if $M$ is finitely generated but not finitely presented.) [Use Exercise 28.]

*Remark:* It is a theorem of Kaplansky (cf. *Projective Modules*, Annals of Mathematics, 68(1958), pp. 372-377) that (i) and (ii) are equivalent without the condition that $M$ be finitely generated.

**31.** *(Localization and* Hom *for Finitely Presented Modules)* Suppose $D^{-1}R$ is the localization of the commutative ring $R$ with respect to the multiplicative subset $D$ of $R$, and let $M$ be a finitely presented $R$-module.

**(a)** For any $R$-modules $A$ and $B$ prove there is a unique $D^{-1}R$-module homomorphism from $D^{-1}\text{Hom}_R(A, B)$ to $\text{Hom}_{D^{-1}R}(D^{-1}A, D^{-1}B)$ that maps $\varphi \in \text{Hom}_R(A, B)$ to the homomorphism from $D^{-1}A$ to $D^{-1}B$ induced by $\varphi$.

**(b)** For any $R$-module $N$ and any $m \geq 1$ show that $\text{Hom}_R(R^m, N) \cong N^m$ as $R$-modules and deduce that $D^{-1}\text{Hom}_R(R^m, N) \cong (D^{-1}N)^m$ as $D^{-1}R$-modules.

**(c)** Suppose $R^s \longrightarrow R^t \longrightarrow M \longrightarrow 0$ is exact. Prove there is a commutative diagram

$$\begin{array}{ccccccc}
0 \to & D^{-1}\text{Hom}_R(M, N) & \to & D^{-1}\text{Hom}_R(R^t, N) & \to & D^{-1}\text{Hom}_R(R^s, N) \\
 & \downarrow & & \downarrow & & \downarrow \\
0 \to & \text{Hom}_{D^{-1}R}(D^{-1}M, D^{-1}N) & \to & \text{Hom}_{D^{-1}R}((D^{-1}R)^t, D^{-1}N) & \to & \text{Hom}_{D^{-1}R}((D^{-1}R)^s, D^{-1}N)
\end{array}$$

of $D^{-1}R$-modules with exact rows. [For the first row first take $R$-module homomorphisms from the terms in the presentation for $M$ into $N$ using Theorem 33 of Section 10.5 (noting the first comment in the proof) and then tensor with the flat $R$-module $D^{-1}R$, cf. Propositions 41 and 42(6) in Section 15.4. For the second row first tensor the presentation with $D^{-1}R$ and then take $D^{-1}R$-module homomorphisms into $D^{-1}N$.]

**(d)** Use (b) to prove that localization commutes with taking homomorphisms when $M$ is finitely presented, i.e., $D^{-1}\text{Hom}_R(M, N) \cong \text{Hom}_{D^{-1}R}(D^{-1}M, D^{-1}N)$ as $D^{-1}R$-modules. [Show the second two vertical maps in the diagram above are isomorphisms and deduce that the left vertical map is also an isomorphism.] (This result is not true in general if $M$ is not finitely presented.)

**32.** ***(Localization and* Ext *for Finitely Presented Modules)*** Suppose $D^{-1}R$ is the localization of the commutative ring $R$ with respect to the multiplicative subset $D$ of $R$. Prove that if $M$ is a finitely presented $R$-module then $D^{-1}\text{Ext}^n_R(M, N) \cong \text{Ext}^n_{D^{-1}R}(D^{-1}M, D^{-1}N)$ as $D^{-1}R$-modules for every $R$-module $N$ and every $n \geq 0$. [Use a projective resolution of $N$ and the previous exercise, noting that tensoring the resolution with $D^{-1}R$ gives a projective resolution for the $D^{-1}R$-module $D^{-1}N$.]

**33.** Suppose $R$ is a commutative ring and $M$ is a finitely presented $R$-module (for example a finitely generated module over a Noetherian ring, or a quotient, $R/I$, of $R$ by a finitely generated ideal $I$, cf. Exercises 26 and 27). Prove that the following are equivalent:

**(a)** $M$ is a projective $R$-module,

**(b)** $M$ is a flat $R$-module,

**(c)** $M$ is locally free, i.e., each localization $M_P$ is a free $R_P$-module for every maximal (hence also for every prime) ideal $P$ of $R$.

In particular show that finitely generated projective modules are the same as finitely presented flat modules. [Exercises 24 and 30 show that (b) is equivalent to (c). Use the Ext criterion for projectivity and Exercises 30 and 32 to see that (a) is equivalent to (c).]

**34.** **(a)** Prove that *every* $R$-module for the commutative ring $R$ is flat if and only if every finitely generated ideal $I$ of $R$ is a direct summand of $R$, in which case every finitely generated ideal of $R$ is principal and projective (such a ring is said to be *absolutely flat*). [Use Exercise 15, the previous exercise applied to the finitely presented $R$-module $R/I$, and the remarks following Proposition 16.]

**(b)** Prove that every Boolean ring is absolutely flat. [Use Exercise 24 in Section 7.4, noting that if $I = Rx$ then $x$ is an idempotent so $R = Rx \oplus R(1 - x)$.]

**(c)** Let $R$ be the direct product and $I$ the direct sum of countably many copies of $\mathbb{Z}/2\mathbb{Z}$. Prove that $I$ is an ideal of the Boolean ring $R$ that is not finitely generated and that the cyclic $R$-module $M = R/I$ is flat but not projective (so finitely generated flat modules need not be projective).

**35.** Let $R$ be the local ring obtained by localizing the ring of $C^\infty$ functions on the open interval $(-1, 1)$ at the maximal ideal of functions that are 0 at $x = 0$ (cf. Exercise 45 of Section 15.2), let $\mathfrak{m} = (x)$ be the unique maximal ideal of $R$ and let $P$ be the prime ideal $\cap_{n\geq 1}\mathfrak{m}^n$. Set $M = R/P$.

**(a)** Prove that $\mathrm{Tor}_1^R(M, R/\mathfrak{m}) = 0$. [Use Exercise 19 applied with $r = x$, noting that $R/P$ is an integral domain.]

**(b)** Prove that $M$ is not flat (hence not projective). [Let $F$ be as in Exercise 45 of Section 15.2. Show that the sequence $0 \to R \to R \to R/(F) \to 0$ induced by multiplication by $F$ is exact, but is not exact after tensoring with $M$.]

## 17.2. The Cohomology of Groups

**1.** Let $F_n = \mathbb{Z}G \otimes_{\mathbb{Z}} \mathbb{Z}G \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} \mathbb{Z}G$ ($n+1$ factors) for $n \geq 0$ with $G$-action defined on simple tensors by $g \cdot (g_0 \otimes g_1 \otimes \cdots \otimes g_n) = (gg_0) \otimes g_1 \otimes \cdots \otimes g_n$.

**(a)** Prove that $F_n$ is a free $\mathbb{Z}G$-module of rank $|G|^n$ with $\mathbb{Z}G$ basis $1 \otimes g_1 \otimes g_2 \otimes \cdots \otimes g_n$ with $g_i \in G$.

Denote the basis element $1 \otimes g_1 \otimes g_2 \otimes \cdots \otimes g_n$ in (a) by $(g_1, g_2, \ldots, g_n)$ and define the $G$-module homomorphisms $d_n$ for $n \geq 1$ on these basis elements by $d_1(g_1) = g_1 - 1$ and

$$d_n(g_1, \ldots, g_n) = g_1 \cdot (g_2, \ldots, g_n) + \sum_{i=1}^{n-1} (-1)^i (g_1, \ldots, g_{i-1}, g_i g_{i+1}, g_{i+2}, \ldots, g_n) + (-1)^n (g_1, \ldots, g_{n-1}),$$

for $n \geq 2$. Define the $\mathbb{Z}$-module *contracting homomorphisms*

$$\mathbb{Z} \xrightarrow{s_{-1}} F_0 \xrightarrow{s_0} F_1 \xrightarrow{s_1} F_2 \xrightarrow{s_2} \cdots$$

on a $\mathbb{Z}$ basis by $s_{-1}(1) = 1$ and $s_n(g_0 \otimes \cdots \otimes g_n) = 1 \otimes g_0 \otimes \ldots \otimes g_n$.

**(b)** Prove that

$$\epsilon s_{-1} = 1, \qquad d_1 s_0 + s_{-1}\epsilon = 1, \qquad d_{n+1}s_n + s_{n-1}d_n = 1, \text{ for all } n \geq 1$$

where the map aug : $F_0 \to \mathbb{Z}$ is the augmentation map $\text{aug}(\sum_{g \in G} \alpha_g g) = \sum_{g \in G} \alpha_g$.

**(c)** Prove that the maps $s_n$ are a chain homotopy (cf. Exercise 4 in Section 1) between the identity (chain) map and the zero (chain) map from the chain

$$\cdots \longrightarrow F_n \xrightarrow{d_n} F_{n-1} \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} F_0 \xrightarrow{\text{aug}} \mathbb{Z} \longrightarrow 0 \qquad (*)$$

of $\mathbb{Z}$-modules to itself.

**(d)** Deduce from (c) that all $\mathbb{Z}$-module homology groups of $(*)$ are zero, i.e., $(*)$ is an exact sequence of $\mathbb{Z}$-modules. Conclude that $(*)$ is a projective $G$-module resolution of $\mathbb{Z}$.

**2.** Let $P_n$ denote the free $\mathbb{Z}$-module with basis $(g_0, g_1, g_2, \ldots, g_n)$ with $g_i \in G$ and define an action of $G$ on $P_n$ by $g \cdot (g_0, g_1, \ldots, g_n) = (gg_0, gg_1, \ldots, gg_n)$. For $n \geq 1$ define

$$d_n(g_0, g_1, g_2, \ldots, g_n) = \sum_{i=0}^{n} (-1)^i (g_0, \ldots, \hat{g_i}, \ldots, g_n),$$

where $(g_0, \ldots, \hat{g_i}, \ldots, g_n)$ denotes the term $(g_0, g_1, g_2, \ldots, g_n)$ with $g_i$ deleted.

**(a)** Prove that $P_n$ is a free $\mathbb{Z}G$-module with basis $(1, g_1, g_2, \ldots, g_n)$ where $g_i \in G$.

**(b)** Prove that $d_{n-1} \circ d_n = 0$ for $n \geq 1$. [Show that the term $(g_0, \ldots, \hat{g_j}, \ldots, \hat{g_k}, \ldots, g_n)$ missing the entries $g_j$ and $g_k$ occurs twice in $d_{n-1} \circ d_n(g_0, g_1, g_2, \ldots, g_n)$, with opposite signs.]

**(c)** Prove that $\varphi : P_n \to F_n$ defined by

$$\varphi((g_0, g_1, g_2, \ldots, g_n)) = g_0 \otimes (g_0^{-1} g_1) \otimes (g_1^{-1} g_2) \ldots \otimes (g_{n-1}^{-1} g_n)$$

is a $G$-module isomorphism with inverse $\psi : P_n \to F_n$ given by

$$\psi(g_0 \otimes g_1 \otimes \ldots \otimes g_n) = (g_0, g_0 g_1, g_0 g_1 g_2, \ldots, g_0 g_1 g_2 \cdots g_n).$$

**(d)** Prove that if $\epsilon(g_0) = 1$ for all $g_0 \in G$ then

$$\cdots \longrightarrow P_n \xrightarrow{d_n} P_{n-1} \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_1} P_0 \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0 \qquad (**)$$

is a free $G$-module resolution of $\mathbb{Z}$. [Show that the isomorphisms in (c) take the $G$-module resolutions $(**)$ and $(*)$ of the previous exercise into each other.]

**3.** Let $F_n$ and $P_n$ be as in the previous two exercises and let $A$ be a $G$-module.

**(a)** Prove that $\mathrm{Hom}_{\mathbb{Z}G}(F_n, A)$ can be identified with the collection $C^n(G, A)$ of maps from $G \times G \times \cdots \times G$ ($n$ copies) to $A$ and that under this identification the associated coboundary maps from $C^n(G, A)$ to $C^{n+1}(G, A)$ are given by equation (18).

**(b)** Prove that $\mathrm{Hom}_{\mathbb{Z}G}(P_n, A)$ can be identified with the collection of maps $f$ from $n+1$ copies $G \times G \times \cdots \times G$ to $A$ that satisfy $f(gg_0, gg_1, \ldots, gg_n) = gf(g_0, g_1, \ldots, g_n)$.

The group $C^n(G, A)$ is sometimes called the group of *inhomogeneous n-cochains of G in A*, and the group in (b) of the previous exercise is called the group of *homogeneous n-cochains of G in A*. The inhomogeneous cochains are easier to describe since there is no restriction on the maps from $G^n$ to $A$, but the coboundary map $d_n$ on homogeneous cochains is less complicated (and more naturally suggested in topological contexts) than the coboundary map on inhomogeneous cochains. The results of the previous exercises show that the cohomology groups $H^n(G, A)$ defined using either homogeneous or inhomogeneous cochains are the same and indicate the origin of the coboundary maps $d_n$ used in the text. Historically, $H^n(G, A)$ was originally defined using homogeneous cochains.

**4.** Suppose $H$ is a normal subgroup of the group $G$ and $A$ is a $G$-module. For every $g \in G$ prove that the map $f(a) = ga$ for $a \in A^H$ defines an automorphism of the subgroup $A^H$.

**5.** Suppose the $G$-module $A$ decomposes as a direct sum $A = A_1 \oplus A_2$ of $G$-submodules. Prove that for all $n \geq 0$, $H^n(G, A) \cong H^n(G, A_1) \oplus H^n(G, A_2)$.

**6.** Suppose $0 \to A \to M_1 \to M_2 \to \cdots \to M_k \to C \to 0$ is an exact sequence of $G$-modules where $M_1, M_2, \ldots, M_k$ are cohomologically trivial. Prove that $H^{n+k}(G, A) \cong H^n(G, C)$ for all $n \geq 1$. [Decompose the exact sequence into a succession of short exact sequences and use Corollary 22. For example, if $0 \to A \xrightarrow{\alpha} M_1 \xrightarrow{\beta} M_2 \xrightarrow{\gamma} C \to 0$ is exact, show that $0 \to A \to M_1 \to B \to 0$ and $0 \to B \to M_2 \to C \to 0$ are both exact, where $B = M_1/\text{image}\,\alpha = M_1/\ker\beta \cong \text{image}\,\beta = \ker\gamma$.]

**7.** (*Adjoint Associativity*) Let $R$, $S$ and $T$ be rings with 1, let $P$ be a left $S$-module, let $N$ be a $(T, S)$-bimodule, and let $A$ be a left $T$-module. Prove that

$$\Phi : \mathrm{Hom}_S(P, \mathrm{Hom}_T(N, A)) \longrightarrow \mathrm{Hom}_T(N \otimes_S P, A)$$

defined by $\Phi(f)(n \otimes p) = f(p)(n)$ is an isomorphism of abelian groups. (See also Theorem 43 in Section 10.5).

**8.** Suppose $G$ is cyclic of order $m$ with generator $\sigma$ and let $N = 1+\sigma+\sigma^2+\cdots+\sigma^{m-1} \in \mathbb{Z}G$.

**(a)** Prove that the *augmentation* map $\text{aug}(\sum_{i=0}^{m-1} a_i\sigma^i) = \sum_{i=0}^{m-1} a_i$ is a $G$-module homomorphism from $\mathbb{Z}G$ to $\mathbb{Z}$.

**(b)** Prove that multiplication by $N$ and by $\sigma - 1$ in $\mathbb{Z}G$ define a free $G$-module resolution of $\mathbb{Z}$: $\ldots \xrightarrow{\sigma-1} \mathbb{Z}G \xrightarrow{N} \mathbb{Z}G \xrightarrow{\sigma-1} \ldots \xrightarrow{N} \mathbb{Z}G \xrightarrow{\sigma-1} \mathbb{Z}G \xrightarrow{\text{aug}} \mathbb{Z} \longrightarrow 0$.

**9.** Suppose $G$ is an infinite cyclic group with generator $\sigma$.

**(a)** Prove that multiplication by $\sigma - 1 \in \mathbb{Z}G$ defines a free $G$-module resolution of $\mathbb{Z}$: $0 \longrightarrow \mathbb{Z}G \xrightarrow{\sigma-1} \mathbb{Z}G \longrightarrow \mathbb{Z} \longrightarrow 0$.

**(b)** Show that $H^0(G, A) \cong A^G$, that $H^1(G, A) \cong A/(\sigma-1)A$, and that $H^n(G, A) = 0$ for all $n \geq 2$. Deduce that $H^1(G, \mathbb{Z}G) \cong \mathbb{Z}$ (so free modules need not be cohomologically trivial).

**10.** Suppose $H$ is a subgroup of finite index $m$ in the group $G$ and $A$ is an $H$-module. Let $x_1, \ldots, x_m$ be a set of left coset representatives for $H$ in $G$: $G = x_1 H \cup \cdots \cup x_m H$.

**(a)** Prove that $\mathbb{Z}G = \bigoplus_{i=1}^m x_i \mathbb{Z}H = \bigoplus_{i=1}^m \mathbb{Z}H x_i^{-1}$ and $\mathbb{Z}G \otimes_{\mathbb{Z}H} A = \bigoplus_{i=1}^m (x_i \otimes A)$ as abelian groups.

**(b)** Let $f_{i,a}$ be the function from $\mathbb{Z}G$ to $A$ defined by

$$f_{i,a}(x) = \begin{cases} ha & \text{if } x = hx_i^{-1} \text{ with } h \in H \\ 0 & \text{otherwise.} \end{cases}$$

Prove that $f_{i,a} \in M_H^G(A) = \mathrm{Hom}_{\mathbb{Z}H}(\mathbb{Z}G, A)$, i.e., $f_{i,a}(h'x) = h' f_{i.a}(x)$ for $h' \in H$.

**(c)** Prove that the map $\varphi(f) = \sum_{i=1}^m x_i \otimes f(x_i^{-1})$ from $M_H^G(A)$ to $\mathbb{Z}G \otimes_{\mathbb{Z}H} A$ is a $G$-module homomorphism. [Write $x_i^{-1} g = h_i x_{i'}^{-1}$ for $i = 1, \ldots, m$ and observe that $x_i \otimes f(x_i^{-1} g) = x_i \otimes h_i f(x_{i'}^{-1}) = x_i h_i \otimes f(x_{i'}^{-1}) = g x_{i'} \otimes f(x_{i'}^{-1})$.]

**(d)** Prove that $\varphi$ gives a $G$-module isomorphism $\varphi : M_H^G(A) \cong \mathbb{Z}G \otimes_{\mathbb{Z}H} A$. [For the injectivity observe that an $H$-module homomorphism is 0 if and only if $f(x_i^{-1}) = 0$ for $i = 1, \ldots, m$. For the surjectivity prove that $\varphi(f_{i,a}) = x_i \otimes a$.]

**11.** Prove that the isomorphism $M_H^G(A) \cong \mathbb{Z}G \otimes_{\mathbb{Z}H} A$ in (d) of the previous exercise need not hold if $H$ is not of finite index in $G$. [If $G$ is an infinite cyclic group show that Shapiro's Lemma implies $H^1(G, M_1^G(\mathbb{Z})) = 0$ while $H^1(G, \mathbb{Z}G) \cong \mathbb{Z}$ by Exercise 9.]

**12.** If $H$ is a subgroup of $G$ and $A$ is an abelian group let $M_{G/H}(A)$ denote the abelian group of all maps from the left cosets $gH$ of $H$ in $G$ to $A$.

**(a)** Prove that $M_1^G(A) \cong M_1^H(M_{G/H}(A))$ as $H$-modules. [If $\{g_i\}_{i\in\mathcal{I}}$ is a choice of left coset representatives of $H$ in $G$ define the correspondence between $f \in M_1^G(A)$ and $F : H \to M_{G/H}(A)$ by $F(h)(g_i H) = f(g_i h)$, and check that this is an isomorphism of $H$-modules.]

**(b)** A $G$-module $A$ such that $H^n(H, A) = 0$ for all $n \geq 1$ and all subgroups $H$ of $G$ is called *cohomologically trivial.* Prove that $M_1^G(A)$ is a cohomologically trivial for any abelian group $A$.

**(c)** If $G$ is finite, prove that $\mathbb{Z}G \otimes_{\mathbb{Z}} A$ is cohomologically trivial for all abelian groups $A$.

**13.** Suppose $A$ is a $G$-module and $H$ is a subgroup of $G$. Prove that the group homomorphism from $H^n(G, A)$ to $H^n(G, M_H^G(A))$ for all $n \geq 0$ induced from the $G$-module homomorphism from $A$ to $M_H^G(A)$ in Example 3 following Corollary 22 composed with the isomorphism $H^n(G, M_H^G(A)) \cong H^n(H, A)$ of Shapiro's Lemma is the restriction homomorphism from $H^n(G, A)$ to $H^n(H, A)$.

**14.** Suppose $\varphi : H \to G$ is the inclusion map of the subgroup $H$ of $G$ into $G$. If $A$ is an $H$-module and $M_H^G(A)$ the associated induced $G$-module, define the group homomorphism $\psi : M_H^G(A) \to A$ by mapping $f$ to its value at 1: $\psi(f) = f(1)$.

**(a)** Prove that $\varphi$ and $\psi$ are compatible homomorphisms.

**(b)** Prove that the induced group homomorphism from $H^n(G, M_H^G(A))$ to $H^n(H, A)$ for $n \geq 0$ is the isomorphism in Shapiro's Lemma.

**15.** Suppose $H$ is a normal subgroup of $G$ and $A$ is a $G$-module. For fixed $g \in G$, let $\psi(a) = ga$ and $\varphi(h) = g^{-1}hg$ for $h \in H$.

**(a)** Prove that $\varphi$ and $\psi$ are compatible homomorphisms.

**(b)** For each $n \geq 0$, prove that the homomorphism $\theta_g$ from $H^n(H, A)$ to $H^n(H, A)$ induced by the compatible homomorphisms $\varphi$ and $\psi$ is an automorphism of $H^n(H, A)$. [Observe that both $\varphi$ and $\psi$ have inverses.]

**(c)** Show that $\theta_g$ acting on $H^0(H, A)$ is the automorphism in Exercise 4.

**16.** Let $A$ be a $G$-module and for $g \in G$ let $\theta_g$ denote the automorphism of $H^n(G, A)$ defined in the previous exercise.

**(a)** Prove that $\theta_g$ acting on $H^0(G, A) = A^G$ is the identity map.

**(b)** Prove that $\theta_g$ acting on $H^n(G, A)$ is the identity map for $n \geq 1$. [By induction on $n$ and dimension shifting. For $n = 1$, use the exact sequence in Corollary 22, together with (a) applied to $\theta_g$ on $C^G$. For $n \geq 2$ use the isomorphism $H^{n+1}(G, A) \cong H^n(G, C)$ in Corollary 22.]

**17.** Suppose that $H$ is a normal subgroup of $G$ and $A$ is a $G$-module. For $n \geq 0$ prove that $H^n(H, A)$ is a $G/H$-module where $gH$ acts by the automorphism $\theta_g$ induced by conjugation by $g$ on $H$ and the natural action of $g$ on $A$ as in Exercise 15. [Use the previous exercise to show this action of a coset is well defined.]

**18.** Suppose that $G$ is cyclic of order $m$, that $H$ is a subgroup of $G$ of index $d$, and that $\mathbb{Z}$ is a trivial $G$-module. Use the projective $G$-module resolution in Exercise 8 to prove

**(a)** that Cor : $H^n(H, \mathbb{Z}) \to H^n(G, \mathbb{Z})$ is multiplication by $d$ from $\mathbb{Z}$ to $\mathbb{Z}$ for $n = 0$, from $\mathbb{Z}/(m/d)\mathbb{Z}$ to $\mathbb{Z}/m\mathbb{Z}$ if $n$ is odd, and from 0 to 0 if $n$ is even, $n \geq 2$, and

**(b)** that Res : $H^n(G, \mathbb{Z}) \to H^n(H, \mathbb{Z})$ is the identity map from $\mathbb{Z}$ to $\mathbb{Z}$ for $n = 0$, and is the natural projection map from $\mathbb{Z}/m\mathbb{Z}$ to $\mathbb{Z}/(m/d)\mathbb{Z}$ or from 0 to 0, depending on the parity of $n \geq 1$.

**19.** Let $p$ be a prime and let $P$ be a Sylow $p$-subgroup of the finite group $G$. Show that for any $G$-module $A$ and all $n \geq 0$ the map Res : $H^n(G, A) \to H^n(P, A)$ is injective on the $p$-primary component of $H^1(G, A)$. Deduce that if $|A| = p^a$ then the restriction map is injective on $H^n(G, A)$. [Use Proposition 26.]

**20.** Let $p$ be a prime, let $G = \langle \sigma \rangle$ be cyclic of order $p^m$ and let $W$ be a vector space of dimension $d > 0$ over $\mathbb{F}_p$ on which $\sigma$ acts as a linear transformation. Assume $W$ has a basis such that the matrix of $\sigma$ is a $d \times d$ elementary Jordan block with eigenvalue 1.

**(a)** Prove that $d \leq p^m$. [Use facts about the minimal polynomial of an elementary Jordan block.]

**(b)** Prove that $\dim_{\mathbb{F}_p} W^G = 1$.

**(c)** Prove that $\dim_{\mathbb{F}_p} (\sigma - 1)W = d - 1$.

**(d)** If $N = 1 + \sigma + \cdots + \sigma^{p^m - 1}$ is the usual norm element, prove that $NW$ is of dimension 1 if $d = p^m$ (respectively, of dimension 0 if $d < p^m$) and that the dimension of ${}_N W$ is $d - 1$ (respectively, $d$). [Let $R$ be the group ring $\mathbb{F}_p G$, and show that every nonzero $R$-submodule of $R$ contains $N$. Note that $W$ is a cyclic $R$-module and let $\varphi : R \to W$ be a surjective homomorphism. Conclude that if $\varphi$ is not an isomorphism then $N \in \ker \varphi$.]

**(e)** Deduce that if $d = p^m$ then $H^n(G, W) = 0$, and if $d < p^m$ then $H^n(G, W)$ has order $p$, for all $n \geq 1$ (i.e., these cohomology groups are zero if and only if $W$ is a free $\mathbb{F}_p G$-module).

**21.** Let $p$ be a prime, let $G = \langle \sigma \rangle$ be cyclic of order $p^m$ and let $V$ be a $G$-module of exponent $p$. Let $V = V_1 \oplus V_2 \oplus \cdots \oplus V_k$ be a decomposition of $V$ giving the Jordan Canonical Form of $\sigma$, where each $V_i$ is $\sigma$-invariant and a matrix of $\sigma$ on $V_i$ is an $d_i \times d_i$ elementary Jordan block with eigenvalue 1, $d_i \geq 1$ (cf. Section 12.3). Prove that $|V^G| = p^k$ and $|H^n(G, V)| = p^s$ where $s$ is the number of $V_i$ of dimension less than $p^m$ over $\mathbb{F}_p$, for all $n \geq 1$. [Use the preceding exercise and Exercise 5.]

**22.** Suppose $G$ is a topological group, i.e., there is a topology on $G$ such that the maps $G \times G \to G$ defined by $(g_1, g_2) \mapsto g_1 g_2$ and $G \to G$ defined by $g \mapsto g^{-1}$ are continuous.

**(a)** If $H$ is an open subgroup of $G$ and $g \in G$, prove that the cosets $gH$ and $Hg$ and the subgroup $g^{-1}Hg$ are also open.

**(b)** Prove that any open subgroup is also closed. [The complement is the union of cosets as in (a).]

**(c)** Prove that a closed subgroup of finite index is open.

**(d)** If $G$ is compact prove that every open subgroup $H$ is of finite index.

**23.** Suppose $G$ is a compact topological group. Prove the following are equivalent:

(i) $G$ is profinite, i.e., $G = \varprojlim G_i$ is the inverse limit of finite groups $G_i$.

(ii) There exists a family $\{N_i\}$ $(i \in \mathcal{I})$ of open normal subgroups $N_i$ in $G$ such that $\cap_i N_i = 1$ and in this case $G \cong \varprojlim(G/N_i)$.

(iii) There exists a family $\{H_j\}$ $(j \in \mathcal{J})$ of open subgroups $H_j$ in $G$ such that $\cap_j H_j = 1$.

[To show (iii) implies (ii), let $H$ be open in $G$ and use (d) of the previous exercise to show that $N = \cap_{g \in G} g^{-1} H g$ is a finite intersection and conclude that $N \subseteq H \subseteq G$ and $N$ is open and normal in $G$.]

**24.** Suppose $N$ and $N'$ are open normal subgroups of the profinite group $G$ and $N' \subseteq N$. Prove that the projection homomorphism $\varphi : G/N' \to G/N$ and the injection $\psi : A^N \to A^{N'}$ are compatible homomorphisms and deduce there is an induced homomorphism from $H^n(G/N, A^N)$ to $H^n(G/N', A^{N'})$.

**25.** If $G$ is an infinite **profinite** group show that $G$ does not act continuously on $A = \mathbb{Z}G$. [Show that the stabilizer of $a \in A$ is not always of finite index in $G$.]

## 17.3. Crossed Homomorphisms and $H^1(G, A)$

**1.** Let $G$ be the cyclic group of order 2 and let $A$ be a $G$-module. Compute the isomorphism types of $Z^1(G, A)$, $B^1(G, A)$ and $H^1(G, A)$ for each of the following:

**(a)** $A = \mathbb{Z}/4\mathbb{Z}$ (trivial action),

**(b)** $A = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (trivial action),

**(c)** $A = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (any nontrivial action).

**2.** Let $p$ be a prime and let $P$ be a $p$-group.

**(a)** Show that $H^1(P, \mathbb{F}_p) \cong P/\Phi(P)$, where $\Phi(P)$ is the Frattini subgroup of $P$ (cf. the exercises in Section 6.1).

**(b)** Deduce that the dimension of $H^1(P, \mathbb{F}_p)$ as a vector space over $\mathbb{F}_p$ equals the minimum number of generators of $P$. [Use Exercise 26(c), Section 6.1.]

**3.** If $G$ is the cyclic group of order 2 acting by inversion on $\mathbb{Z}$ show that $|H^1(G, \mathbb{Z})| = 2$. [Show that in $E = \mathbb{Z} \rtimes G$ every element of $E - \mathbb{Z}$ has order 2, and there are two conjugacy classes in this coset.]

**4.** Let $A$ be the Klein 4-group and let $G = \mathrm{Aut}(A) \cong S_3$ act on $A$ in the natural fashion. Prove that $H^1(G, A) = 0$. [Show that in the semidirect product $E = A \rtimes G$, $G$ is the normalizer of a Sylow 3-subgroup of $E$. Apply Sylow's Theorem to show all complements to $A$ in $E$ are conjugate.]

**5.** Let $G$ be the cyclic group of order 2 acting on an elementary abelian 2-group $A$ of order $2^n$. Show that $H^1(G, A) = 0$ if and only if $n = 2k$ and $|A^G| = 2^k$. [In $E = A \rtimes G$ show that $(a, x)$ is an element of order 2 if and only if $a \in A^G$, where $G = \langle x \rangle$. Then compare the number of complements to $A$ with the number of $E$-conjugates of $x$.]

**6.** *(Thompson Transfer Lemma)* Let $G$ be a finite group of even order, let $T$ be a Sylow 2-subgroup of $G$, let $M \leq T$ with $|T : M| = 2$, and let $x$ be an element of order 2 in $G$. Show that if $G$ has no subgroup of index 2 then $M$ contains some $G$-conjugate of $x$ as follows:

**(a)** Let Ver : $G/[G, G] \to T/[T, T]$ be the transfer homomorphism. Show that

$$\text{Ver}(x) = \prod_g g^{-1}xg \bmod [T, T]$$

where the product is over representatives of the cosets $gT$ that are fixed under left multiplication by $x$.

**(b)** Show that under left multiplication $x$ fixes an odd number of left cosets of $T$ in $G$.

**(c)** Show that if $G$ has no subgroup of index 2 then $\text{Ver}(x) \in M/[T, T]$. Deduce that for some $g \in G$ we must have $g^{-1}xg \in M$. [Consider the product $\text{Ver}(x)$ in the group $T/M$ of order 2.]

**7.** Let $H$ be a subgroup of $G$ and let $x \in G$. The transfer $\mathrm{Ver} : G/[G, G] \to H/[H, H]$ may be computed as follows: let $\mathcal{O}_1, \mathcal{O}_2, \ldots, \mathcal{O}_k$ be the distinct orbits of $x$ acting by left multiplication on the left cosets of $H$ in $G$, let $\mathcal{O}_i$ have length $n_i$ and let $g_i H$ be any representative of $\mathcal{O}_i$.

**(a)** Show that $\mathcal{O}_i = \{g_i H, x g_i H, x^2 g_i H, \ldots, x^{n_i - 1} g_i H\}$ and that $g_i^{-1} x^{n_i} g_i \in H$.

**(b)** Show that $\mathrm{Ver}(x) = \prod_{i=1}^{k} g_i^{-1} x^{n_i} g_i \bmod [H, H]$.

**8.** Assume the center, $Z(G)$, of $G$ is of index $m$. Prove that $\mathrm{Ver}(x) = x^m$, for all $x \in G$, where Ver is the transfer homomorphism from $G/[G, G]$ to $Z(G)$. [Use the preceding exercise.]

**9.** Let $p$ be a prime, let $n \geq 3$, and let $V$ be an $n$-dimensional vector space over $\mathbb{F}_p$ with basis $v_1, v_2, \ldots, v_n$. Let $V$ be a module for the symmetric group $S_n$, where each $\pi \in S_n$ permutes the basis in the natural way: $\pi(v_i) = v_{\pi(i)}$.

**(a)** Show that $|H^1(S_n, V)| = \begin{cases} 0, & \text{if } p \neq 2 \\ 2, & \text{if } p = 2 \end{cases}$. [Use Shapiro's Lemma.]

**(b)** Show that $H^1(A_n, V) = 0$ for all primes $p$.

**10.** Let $V$ be the natural permutation module for $S_n$ over $\mathbb{F}_2$, $n \geq 3$, as described in the preceding exercise, and let $W = \{a_1 v_1 + \cdots + a_n v_n \mid a_1 + \cdots + a_n = 0\}$ (the "trace zero" submodule of $V$). Show that if $n$ is even then $H^1(A_n, W) \neq 0$. [Show that in the semidirect product $V \rtimes A_n$ the element $v_1$ induces a nontrivial outer automorphism on $E = W \rtimes A_n$ that stabilizes the series $1 \trianglelefteq W \trianglelefteq E$.]

**11.** Let $F$ be a field of characteristic not dividing $n$ and let $\alpha$ be any nonzero element in $F$. Let $K$ be a Galois extension of $F$ containing the splitting field of $x^n - a$, and let $\sqrt[n]{\alpha}$ be a fixed $n^{\text{th}}$ root of $\alpha$ in $K$.

**(a)** Prove that $\sigma(\sqrt[n]{\alpha})/\sqrt[n]{\alpha}$ is an $n^{\text{th}}$ root of unity.

**(b)** Prove that the function $f(\sigma) = \sigma(\sqrt[n]{\alpha})/\sqrt[n]{\alpha}$ is a 1-cocycle of $G$ with values in the group $\mu_n$ of $n^{\text{th}}$ roots of unity in $K$ (note $\mu_n$ is not assumed to be contained in $F$).

**(c)** Prove that the 1-cocycle obtained by a different choice of $n^{\text{th}}$ root of $\alpha$ in $K$ differs from the 1-cocycle in (b) by a 1-coboundary.

**12.** Let $F$ be a field of characteristic not dividing $n$ that contains the $n^{\text{th}}$ roots of unity, and suppose $L/F$ is a Galois extension with abelian Galois group of exponent dividing $n$. Prove that $L$ is the composite of cyclic extensions of $F$ whose degrees are divisors of $n$ and use this to prove that there is a bijection between the subgroups of the multiplicative group $F^\times/F^{\times n}$ and such extensions $L$.

**13.** The Galois group of the extension $\mathbb{C}/\mathbb{R}$ is the cyclic group $G = \langle \tau \rangle$ of order 2 generated by complex conjugation $\tau$. Prove that $H^2(G, \mathbb{C}^\times) \cong \mathbb{R}^\times/\mathbb{R}^+ \cong \mathbb{Z}/2\mathbb{Z}$ where $\mathbb{R}^+$ denotes the positive real numbers.

**14.** For any group $G$ let $\hat{G} = \text{Hom}(G, \mathbb{Q}/\mathbb{Z})$ denote its dual group.

**(a)** If $\varphi : G_1 \to G_2$ is a group homomorphism prove that composition with $\varphi$ induces a homomorphism $\hat{\varphi} : \hat{G}_2 \to \hat{G}_1$ on their dual groups.

**(b)** For any fixed $g$ in $G$, show that evaluation at $g$ gives a homomorphism $\varphi_g$ from $\hat{G}$ to $\mathbb{Q}/\mathbb{Z}$.

**(c)** Prove that the map taking $g \in G$ to $\varphi_g$ in (b) defines a homomorphism from $G$ to its *double dual* $\widehat{(\hat{G})}$.

**(d)** Prove that if $G$ is a finite abelian group then the homomorphism in (c) is an isomorphism of $G$ with its double dual. (By Exercise 14 in Section 5.2 the group $G$ is (noncanonically) isomorphic to its dual $\hat{G}$. This shows that $G$ is *canonically* isomorphic to its double dual — the isomorphism is independent of any choice of generators for $G$.)

**(e)** If $\psi : \hat{G}_2 \to \hat{G}_1$ is a homomorphism where $G_1$ and $G_2$ are finite abelian groups, then by (a) and (d) there is an induced homomorphism $\varphi : G_1 \to G_2$. Prove that $\varphi(g_1) = g_2$ if $\chi(g_2) = \chi'(g_1)$ for $\chi' = \psi(\chi)$.

**15.** Use Gauss' Lemma in the computation of the transfer map for $\mathbb{F}_p^\times$ to $\{\pm 1\}$ to prove that 2 is a square modulo the odd prime $p$ if and only if $p \equiv \pm 1 \bmod 8$. [Count how many elements in $2, 4, \ldots, p-1$ are greater than $(p-1)/2$.]

## 17.4. Group Extensions, Factor Sets and $H^2(G, A)$

**1.** Let $A = \{1, a, b, c\}$ be the Klein 4-group and let $G = \langle g \rangle$ be the cyclic group of order 2 acting trivially on $A$.

**(a)** Prove that $|C^2(G, A)| = 2^8$.

**(b)** Show that coboundaries are constant functions, and deduce that $|B^2(G, A)| = 4$.

**(c)** Use the cocycle condition to show that $|Z^2(G, A)| \leq 2^4$.

**(d)** If $E = Z_4 \times Z_2 = \langle x \rangle \times \langle y \rangle$, prove that the extensions $1 \to A \xrightarrow{\iota_i} E \xrightarrow{\pi} G \to 1$ defined by $\pi(x) = g$, $\pi(y) = 1$ and $\iota_1(a) = x^2$, $\iota_1(b) = y$ (respectively, $\iota_2(b) = x^2$, $\iota_2(a) = y$, and $\iota_3(c) = x^2$, $\iota_3(a) = y$), together with the split extension $Z_2 \times Z_2 \times Z_2$ give 4 inequivalent extensions of $Z_2$ by the Klein 4-group. Deduce that $H^2(G, A)$ has order 4 by explicitly exhibiting the corresponding cocycles.

**2.** Let $A = \mathbb{Z}/4\mathbb{Z}$ and let $G$ be the cyclic group of order 2 acting trivially on $A$.

**(a)** Prove that $|C^2(G, A)| = 2^8$.

**(b)** Use the coboundary condition to show that $|B^2(G, A)| = 2^3$.

**(c)** Use the cocycle condition to show that $|Z^2(G, A)| \leq 2^4$.

**(d)** Show that $|H^2(G, A)| = 2$ by exhibiting two inequivalent extensions of $G$ by $A$ and their corresponding cocycles.

**3.** Let $A = \mathbb{Z}/4\mathbb{Z}$ and let $G$ be the cyclic group of order 2 acting by inversion on $A$.

**(a)** Show that there are four coboundaries and that only the zero coboundary is normalized.

**(b)** Prove by a direct computation of cocycle and coboundary groups that $|H^2(G, A)| = 2$.

**(c)** Exhibit two distinct cohomology classes and their corresponding extension groups.

**(d)** Show that for a given extension of $G$ by $A$ with extension group isomorphic to $D_8$ there are four normalized sections, all of which have the zero 2-cocycle as their factor set.

**(e)** Show that for a given extension of $G$ by $A$ with extension group isomorphic to $Q_8$ there are sixteen sections, four of which are normalized, and all of the latter have the same factor set.

**4.** For a *non*-normalized 2-cocycle $f$ one defines the extension group $E_f$ on the set $A \times G$ by the same binary operation in equation (34). Verify two of the group axioms in this case by showing that identity is now $(-f(1, 1), 1)$ and inverses are given by

$$(a, x)^{-1} = (-x^{-1} \cdot a - f(x^{-1}, x) - f(1, 1), x^{-1}).$$

(Verification of the associative law is essentially the same as for normalized 2-cocycles.) Prove also that the set $A^{**} = \{(a - f(1, 1), 1) \mid a \in A\}$ is a subgroup of $E_f$ and the map $\iota^{**} : a \mapsto (a - f(1, 1), 1)$ is an isomorphism from $A$ to $A^{**}$. Show that this extension $E_f$, with the injection $\iota^{**}$ and the usual projection map $\pi^*$ onto $G$, is equivalent to an extension derived from a normalized cocycle in the same class as $f$.

**5.** Show that the set of equivalences of a given extension $1 \to A \xrightarrow{\iota} E \xrightarrow{\pi} G \to 1$ with itself form a group under composition, and that this group is isomorphic to the stability group

$\mathrm{Stab}(1 \trianglelefteq \iota(A) \trianglelefteq E)$. (Thus Proposition 31 implies $Z^1(G, A)$ is the group of equivalences of the extension with itself).

**6.** *(Gaschütz's Theorem)* Let $p$ be a prime, let $A$ be an abelian normal $p$-subgroup of a finite group $G$, and let $P$ be a Sylow $p$-subgroup of $G$. Prove that $G$ is a split extension of $G/A$ by $A$ if and only if $P$ is a split extension of $P/A$ by $A$. (Note that $A \leq P$ by Exercise 37 in Section 4.5). [Use Sylow's Theorem to show if $G$ splits over $A$ then so too does $P$. Conversely, show that a normalized 2-cocycle associated to the extension of $P/A$ by $A$ via Theorem 36 is the image of a normalized 2-cocycle in $H^2(G/A, A)$ under the restriction homomorphism Res : $H^2(G/A, A) \to H^2(P/A, A)$. Then use Proposition 26 and the fact that multiplication by $|G : P|$ is an automorphism of $A$.]

**7. (a)** Prove that $H^2(A_4, \mathbb{Z}/2\mathbb{Z}) \neq 0$ by exhibiting a nonsplit extension of $A_4$ by a cyclic group of order 2. [See Exercise 11, Section 4.5.]

**(b)** Prove that $H^2(A_5, \mathbb{Z}/2\mathbb{Z}) \neq 0$ by showing that $SL_2(\mathbb{F}_5)$ is a nonsplit extension of $A_5$ by a cyclic group of order 2. [Use Propositions 21 and 23 in Section 4.5.]

**8.** The *Schur multiplier* of a finite group $G$ is defined as the group $H^2(G, \mathbb{C}^\times)$, where the multiplicative group $\mathbb{C}^\times$ of complex numbers is a trivial $G$-module. Prove that the Schur multiplier is a finite group. [Show that every cohomology class contains a cocycle whose values lie in the $n^{\text{th}}$ roots of unity, where $n = |G|$, as follows: If $f$ is any cocycle then by Corollary 27, $f^n \in B^2(G, \mathbb{C}^\times)$. Define $k \in C^2(G, \mathbb{C}^\times)$ by $k(g_1, g_2) = f(g_1, g_2)^{1/n}$ (take any $n^{\text{th}}$ roots). Show that $k \in B^2(G, \mathbb{C}^\times)$ and $fk^{-1}$ takes values in the group of $n^{\text{th}}$ roots of 1.]

**9.** Use the classification of the extensions of the Klein 4-group by $Z_2$ in the example following Theorem 39 to prove the following (in the notation of that example):

**(a)** There is an (outer) automorphism of $Z_4 \times Z_2$ which interchanges the cosets $Ax$ and $Axy$ and fixes the coset $Ay$.

**(b)** There is an outer automorphism of $D_8$ which interchanges the cosets $As$ and $Asr$ and fixes the coset $Ar$.

**10.** Suppose $\mathbb{F}_q$ is a finite field with $G = \mathrm{Gal}(\mathbb{F}_{q^d}/\mathbb{F}_q) = \langle \sigma_q \rangle$ where $\sigma_q$ is the Frobenius automorphism, and let $N$ be the usual norm element for the cyclic group $G$.

**(a)** Use Hilbert's Theorem 90 to prove that $|_N(\mathbb{F}_{q^d}^\times)| = (q^d - 1)/(q - 1)$, and deduce that the norm map from $\mathbb{F}_{q^d}$ to $\mathbb{F}_q$ is surjective.

**(b)** Prove that $H^n(G, \mathbb{F}_{q^d}^\times) = 0$ for all $n \geq 1$.

# 18. Representation Theory and Character Theory

## 18.1. Linear Actions and Modules over Group Rings

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**1.** Prove that if $\varphi : G \to GL(V)$ is any representation, then $\varphi$ gives a faithful representation of $G/\ker\varphi$.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**2.** Let $\varphi : G \to GL_n(F)$ be a matrix representation. Prove that the map $g \mapsto \det(\varphi(g))$ is a degree 1 representation.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**3.** Prove that the degree 1 representations of $G$ are in bijective correspondence with the degree 1 representations of the abelian group $G/G'$ (where $G'$ is the commutator subgroup of $G$).

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**4.** Let $V$ be a (possibly infinite dimensional) $FG$-module ($G$ is a finite group). Prove that for each $v \in V$ there is an $FG$-submodule containing $v$ of dimension $\leq |G|$.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**5.** Prove that if $|G| > 1$ then every irreducible $FG$-module has dimension $< |G|$.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**6.** Write out the matrices $\varphi(g)$ for every $g \in G$ for each of the following representations that were described in the second set of examples:

**(a)** the representation of $S_3$ described in Example 3 (let $n = 3$ in that example)

**(b)** the representation of $D_8$ described in Example 6 (i.e., let $n = 4$ in that example and write out the values of all the sines and cosines, for all group elements)

**(c)** the representation of $Q_8$ described in Example 7

**(d)** the representation of $Q_8$ described in Example 8.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**7.** Let $V$ be the 4-dimensional permutation module for $S_4$ described in Example 3 of the second set of examples. Let $\pi : D_8 \to S_4$ be the permutation representation of $D_8$ obtained from the action of $D_8$ by left multiplication on the set of left cosets of its subgroup $\langle s \rangle$. Make $V$ into an $FD_8$-module via $\pi$ as described in Example 4 and write out the $4 \times 4$ matrices for $r$ and $s$ given by this representation with respect to the basis $e_1, \ldots, e_4$.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**8.** Let $V$ be the $FS_n$-module described in Examples 3 and 10 in the second set of examples.

**(a)** Prove that if $v$ is any element of $V$ such that $\sigma \cdot v = v$ for all $\sigma \in S_n$ then $v$ is an $F$-multiple of $e_1 + e_2 + \cdots + e_n$.

**(b)** Prove that if $n \geq 3$, then $V$ has a unique 1-dimensional submodule, namely the submodule $N$ consisting of all $F$-multiples of $e_1 + e_2 + \cdots + e_n$.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**9.** Prove that the 4-dimensional representation of $Q_8$ on $\mathbb{H}$ described in Example 8 in the second set of examples is irreducible. [Show that any $Q_8$-stable subspace is a left ideal.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**10.** Prove that $GL_2(\mathbb{R})$ has no subgroup isomorphic to $Q_8$. [This may be done by direct computation using generators and relations for $Q_8$. Simplify these calculations by putting one generator in rational canonical form.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**11.** Let $\varphi : S_n \to GL_n(F)$ be the matrix representation given by the permutation module described in Example 3 in the second set of examples, where the matrices are computed with respect to the basis $e_1, \ldots, e_n$. Prove that $\det \varphi(\sigma) = \epsilon(\sigma)$ for all $\sigma \in S_n$, where $\epsilon(\sigma)$ is the sign of the permutation $\sigma$. [Check this on transpositions.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**12.** Assume the characteristic of $F$ is not 2. Let $H$ be the set of $T \in M_n(F)$ such that $T$ has exactly one nonzero entry in each row and each column and zeros elsewhere, and the nonzero entries are $\pm 1$. Prove that $H$ is a subgroup of $GL_n(F)$ and that $H$ is isomorphic to $E_{2^n} \rtimes S_n$ (semidirect product), where $E_{2^n}$ is the elementary abelian group of order $2^n$.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

The next few exercises explore an important result known as Schur's Lemma and some of its consequences.

**13.** Let $R$ be a ring and let $M$ and $N$ be simple (i.e., irreducible) $R$-modules.

**(a)** Prove that every nonzero $R$-module homomorphism from $M$ to $N$ is an isomorphism. [Consider its kernel and image.]

**(b)** Prove Schur's Lemma: if $M$ is a simple $R$-module then $\mathrm{Hom}_R(M, M)$ is a division ring (recall that $\mathrm{Hom}_R(M, M)$ is the ring of all $R$-module homomorphisms from $M$ to $M$, where multiplication in this ring is composition).

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

The next few exercises explore an important result known as Schur's Lemma and some of its consequences.

**14.** Let $\varphi : G \to GL(V)$ be a representation of $G$. The *centralizer* of $\varphi$ is defined to be the set of all linear transformations, $A$, from $V$ to itself such that $A\varphi(g) = \varphi(g)A$ for all $g \in G$ (i.e., the linear transformations of $V$ which commute with all $\varphi(g)$'s).

**(a)** Prove that a linear transformation $A$ from $V$ to $V$ is in the centralizer of $\varphi$ if and only if it is an $FG$-module homomorphism from $V$ to itself (so the centralizer of $\varphi$ is the same as the *ring* $\text{Hom}_{FG}(V, V)$).

**(b)** Show that if $z$ is in the center of $G$ then $\varphi(z)$ is in the centralizer of $\varphi$.

**(c)** Assume $\varphi$ is an irreducible representation (so $V$ is a simple $FG$-module). Prove that if $H$ is any finite *abelian* subgroup of $GL(V)$ such that $A\varphi(g) = \varphi(g)A$ for all $A \in H$ then $H$ is cyclic (in other words, any finite abelian subgroup of the multiplicative group of units in the ring $\text{Hom}_{FG}(V, V)$ is cyclic). [By the preceding exercise, $\text{Hom}_{FG}(V, V)$ is a division ring, so this reduces to proving that a finite abelian subgroup of the multiplicative group of nonzero elements in a division ring is cyclic. Show that the division subring generated by an abelian subgroup of any division ring is a field and use Proposition 18, Section 9.5.]

**(d)** Show that if $\varphi$ is a faithful irreducible representation then the center of $G$ is cyclic.

**(e)** Deduce from (d) that if $G$ is abelian and $\varphi$ is any irreducible representation then $G/\ker\varphi$ is cyclic.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

The next few exercises explore an important result known as Schur's Lemma and some of its consequences.

**15.** Exhibit all 1-dimensional complex representations of a finite cyclic group; make sure to decide which are inequivalent.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

The next few exercises explore an important result known as Schur's Lemma and some of its consequences.

**16.** Exhibit all 1-dimensional complex representations of a finite abelian group. Deduce that the number of inequivalent degree 1 complex representations of a finite abelian group equals the order of the group. [First decompose the abelian group into a direct product of cyclic groups, then use the preceding exercise.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

The next few exercises explore an important result known as Schur's Lemma and some of its consequences.

**17.** Prove the following variant of Schur's Lemma for complex representations of abelian groups: if $G$ is abelian, any irreducible complex representation, $\varphi$, of $G$ is of degree 1 and $G/\ker\varphi$ is cyclic. [This can be done without recourse to Exercise 14 by using the observation that for any $g \in G$ the eigenspaces of $\varphi(g)$ are $G$-stable. Your proof that $\varphi$ has degree 1 should also work for infinite abelian groups.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

The next few exercises explore an important result known as Schur's Lemma and some of its consequences.

**18.** Prove the following general form of Schur's Lemma for complex representations: if $\varphi : G \to GL_n(\mathbb{C})$ is an irreducible matrix representation and $A$ is an $n \times n$ matrix commuting with $\varphi(g)$ for all $g \in G$, then $A$ is a scalar matrix. Deduce that if $\varphi$ is a faithful, irreducible, complex representation then the center of $G$ is cyclic and $\varphi(z)$ is a scalar matrix for all elements $z$ in the center of $G$. [As in the preceding exercise, the eigenspaces of $A$ are $G$-stable.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**19.** Prove that if $G$ is an abelian group then any finite dimensional complex representation of $G$ is equivalent to a representation into diagonal matrices (i.e., any finite group of commuting matrices over $\mathbb{C}$ can be simultaneously diagonalized). [This can be done without recourse to Maschke's Theorem by looking at eigenspaces.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**20.** Prove that the number of degree 1 complex representations of any finite group $G$ equals $|G : G'|$, where $G'$ is the commutator subgroup of $G$. [Use Exercises 3 and 16.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**21.** Let $G$ be a noncyclic abelian group acting by conjugation on an elementary abelian $p$-group $V$, where $p$ is a prime not dividing the order of $G$.

**(a)** Prove that if $W$ is an irreducible $\mathbb{F}_p G$-submodule of $V$ then there is some nonidentity element $g \in G$ such that $W \leq C_V(g)$ (here $C_V(g)$ is the subgroup of elements of $V$ that are fixed by $g$ under conjugation).

**(b)** Prove that $V$ is generated by the subgroups $C_V(g)$ as $g$ runs over all nonidentity elements of $G$.

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**22.** Let $p$ be a prime, let $P$ be a $p$-group and let $F$ be a field of characteristic $p$. Prove that the only irreducible representation of $P$ over $F$ is the trivial representation. [Do this for a group of order $p$ first using the fact that $F$ contains all $p^{\text{th}}$ roots of 1 (namely 1 itself). If $P$ is not of order $p$, let $z$ be an element of order $p$ in the center of $P$, prove that $z$ is in the kernel of the irreducible representation and apply induction to $P/\langle z \rangle$.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**23.** Let $p$ be a prime, let $P$ be a nontrivial $p$-group and let $F$ be a field of characteristic $p$. Prove that the regular representation is not completely reducible. [Use the preceding exercise.]

Let $F$ be a field, let $G$ be a finite group and let $n \in \mathbb{Z}^+$.

**24.** Let $p$ be a prime, let $P$ be a nontrivial $p$-group and let $F$ be a field of characteristic $p$. Prove that the regular representation is indecomposable.

## 18.2. Wedderburn's Theorem and Some Consequences

Let $G$ be a finite group and let $R$ be a ring with 1.

**1.** Prove that conditions (1) and (2) of Wedderburn's Theorem are equivalent.

Let $G$ be a finite group and let $R$ be a ring with 1.

**2.** Prove that (3) implies (2) in Wedderburn's Theorem. [Let $Q$ be a submodule of an $R$-module $N$. Use Zorn's Lemma to show there is a submodule $M$ maximal with respect to $Q \cap M = 0$. If $Q + M = N$, then (2) holds; otherwise let $M_1$ be the complete preimage in $N$ of some simple module in $N/M$ not contained in $(Q + M)/M$, and argue that $M_1$ contradicts the maximality of $M$.]

Let $G$ be a finite group and let $R$ be a ring with 1.

**3.** Prove that (4) implies (3) in Wedderburn's Theorem. [Let $N$ be a nonzero $R$-module. First show $N$ contains simple submodules by considering a cyclic submodule. Then use Zorn's Lemma applied to the set of direct sums of simple submodules (appropriately ordered) to show that $N$ contains a maximal completely reducible submodule $M$. If $M \neq N$ let $M_1$ be the complete preimage in $N$ of a simple module in $N/M$ and contradict the maximality of $M$.]

Let $G$ be a finite group and let $R$ be a ring with 1.

**4.** Prove that (5) implies (4) in Wedderburn's Theorem. [Use the methods in the proofs of Propositions 6 and 8 to decompose each $R_i$ as a left $R$-module.]

Let $G$ be a finite group and let $R$ be a ring with 1.

The next six exercises establish some general results about rings and modules that imply the remaining implication of Wedderburn's Theorem: (2) implies (5). In these exercises assume $R$ satisfies (2): every $R$-module is injective.

**5.** Show that $R$ has the descending chain condition (D.C.C.) on left ideals. Deduce that $R$ is a finite direct sum of left ideals. [If not, then show that as a left $R$-module $R$ is a direct sum of an infinite number of nonzero submodules. Derive a contradiction by writing the element 1 in this direct sum.]

Let $G$ be a finite group and let $R$ be a ring with 1.

The next six exercises establish some general results about rings and modules that imply the remaining implication of Wedderburn's Theorem: (2) implies (5). In these exercises assume $R$ satisfies (2): every $R$-module is injective.

**6.** Show that $R = R_1 \times R_2 \times \cdots \times R_r$ where $R_j$ is a 2-sided ideal and a simple ring (i.e., has no proper, nonzero 2-sided ideals). Show each $R_j$ has an identity and satisfies D.C.C. on left ideals. [Use the preceding exercise to show $R$ has a minimal 2-sided ideal $R_1$. As a left $R$-module $R = R_1 \oplus R'$ for some left ideal $R'$. Show $R'$ is a right ideal and proceed inductively using D.C.C.]

Let $G$ be a finite group and let $R$ be a ring with 1.

The next six exercises establish some general results about rings and modules that imply the remaining implication of Wedderburn's Theorem: (2) implies (5). In these exercises assume $R$ satisfies (2): every $R$-module is injective.

**7.** Let $S$ be a simple ring with 1 satisfying D.C.C. on left ideals and let $L$ be a minimal left ideal in $S$. Show that $S \cong L^n$ as left $S$-modules, where $L^n = L \oplus \cdots \oplus L$ with $n$ factors. [Argue by simplicity that $LS = S$ so $1 = l_1 s_1 + \cdots + l_n s_n$ for some $l_i \in L$ and $s_i \in S$ with $n$ minimal. Show that the map $(x_1, \ldots, x_n) \mapsto x_1 s_1 + \cdots + x_n s_n$ is a surjective homomorphism of left $S$-modules; use the minimality of $L$ and $n$ to show it is an injection.]

Let $G$ be a finite group and let $R$ be a ring with 1.

The next six exercises establish some general results about rings and modules that imply the remaining implication of Wedderburn's Theorem: (2) implies (5). In these exercises assume $R$ satisfies (2): every $R$-module is injective.

**8.** Let $A$ be any ring with 1, let $L$ be any left $A$-module and let $L^n$ be the direct sum of $n$ copies of $L$ with itself.

**(a)** Prove the ring isomorphism $\mathrm{Hom}_A(L^n, L^n) \cong M_n(D)$, where $D = \mathrm{Hom}_A(L, L)$ (multiplication in the ring $\mathrm{Hom}_A(X, X)$ is function composition, cf. Proposition 2(4) in Section 10.2).

**(b)** Deduce that if $L$ is a simple $A$-module, then $\mathrm{Hom}_A(L^n, L^n)$ is isomorphic to a matrix ring over a division ring. [Use Schur's Lemma and (a).]

**(c)** Prove the ring isomorphism $\mathrm{Hom}_A(A, A) \cong A^{opp}$, where $A^{opp}$ is the opposite ring to $A$ (the elements and addition are the same as in $A$ but the value of the product $x \cdot y$ in $A^{opp}$ is $yx$, computed in $A$), cf. the end of Section 17.4. [Any homomorphism is determined by its value on 1.]

Let $G$ be a finite group and let $R$ be a ring with 1.

The next six exercises establish some general results about rings and modules that imply the remaining implication of Wedderburn's Theorem: (2) implies (5). In these exercises assume $R$ satisfies (2): every $R$-module is injective.

**9.** Prove that if $S$ is a simple ring with 1 satisfying D.C.C. on left ideals then $S \cong M_n(\Delta)$ for some division ring $\Delta$. (This result together with Exercise 6 completes the existence part of the proof that (2) implies (5) in Wedderburn's Theorem). [Use Exercises 7 and 8 to show $S^{opp} \cong \mathrm{Hom}_S(L^n, L^n) \cong M_n(D)$ for some division ring $D$. Then show $S \cong M_n(\Delta)$, where $\Delta$ is the division ring $D^{opp}$.]

Let $G$ be a finite group and let $R$ be a ring with 1.

The next six exercises establish some general results about rings and modules that imply the remaining implication of Wedderburn's Theorem: (2) implies (5). In these exercises assume $R$ satisfies (2): every $R$-module is injective.

**10.** Prove that $\Delta$ and $n$ in the isomorphism $S \cong M_n(\Delta)$ of the previous exercise are uniquely determined by $S$ (proving the uniqueness statement in Wedderburn's Theorem), as follows. Suppose $S = M_n(\Delta) \cong M_{n'}(\Delta')$ as rings, where $\Delta$ and $\Delta'$ are division rings.

**(a)** Prove that $\Delta \cong \mathrm{Hom}_S(L, L)$ where $L$ is a minimal left ideal in $S$. Deduce that $\Delta \cong \Delta'$. [Use Proposition 6(4).]

**(b)** Prove that a finitely generated (left) module over a division ring $\Delta$ has a "basis" (a linearly independent generating set), and that any two bases have the same cardinality. Deduce that $n = n'$. [Mimic the proof of Corollary 4(2) of Section 11.1.]

Let $G$ be a finite group and let $R$ be a ring with 1.

**11.** Prove that if $R$ is a ring with 1 such that every $R$-module is free then $R$ is a division ring.

Let $G$ be a finite group and let $R$ be a ring with 1.

**12.** Let $F$ be a field, let $f(x) \in F[x]$ and let $R = F[x]/(f(x))$. Find necessary and sufficient conditions on the factorization of $f(x)$ in $F[x]$ so that $R$ is a semisimple ring. When $R$ is semisimple, describe its Wedderburn decomposition. [See Proposition 16 in Section 9.5.]

Let $G$ be a finite group and let $R$ be a ring with 1.

**13.** Let $G$ be the cyclic group of order $n$ and let $R = \mathbb{Q}G$. Describe the Wedderburn decomposition of $R$ and find the number and the degrees of the irreducible representations of $G$ over $\mathbb{Q}$. In particular, show that if $n = p$ is a prime then $G$ has exactly one nontrivial irreducible representation over $\mathbb{Q}$ and this representation has degree $p - 1$. [Recall from the first example in Section 1 that $\mathbb{Q}G = \mathbb{Q}[x]/(x^n - 1)$. Use Proposition 16 in Section 9.5 and results from Section 13.6.]

Let $G$ be a finite group and let $R$ be a ring with 1.

**14.** Let $p$ be a prime and let $F = \mathbb{F}_p$ be the field of order $p$. Let $G$ be the cyclic group of order 3 and let $R = FG$. For each of $p = 2$ and $p = 7$ describe the Wedderburn decomposition of $R$ and find the number and the degrees of the irreducible representations of $G$ over $F$.

Let $G$ be a finite group and let $R$ be a ring with 1.

**15.** Prove that if $P$ is a $p$-group for some prime $p$, then $P$ has a faithful irreducible complex representation if and only if $Z(P)$ is cyclic. [Use Exercise 18 in Section 1, Theorem 6.1(2) and Example 3.]

Let $G$ be a finite group and let $R$ be a ring with 1.

**16.** Prove that if $V$ is an irreducible $FG$-module and $F$ is an algebraically closed field then $\mathrm{Hom}_{FG}(V, V)$ is isomorphic to $F$ (as a ring).

Let $G$ be a finite group and let $R$ be a ring with 1.

**17.** Let $F$ be a field, let $R = M_n(F)$ and let $M$ be the unique irreducible $R$-module. Prove that $\mathrm{Hom}_R(M, M)$ is isomorphic to $F$ (as a ring).

Let $G$ be a finite group and let $R$ be a ring with 1.

**18.** Find all 2-sided ideals of $M_n(\mathbb{Z})$.

## 18.3. Character Theory and the Orthogonality Relations

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**1.** Prove that $\operatorname{tr} AB = \operatorname{tr} BA$ for $n \times n$ matrices $A$ and $B$ with entries from any commutative ring.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**2.** In each of (a) to (c) let $\psi$ be the character afforded by the specified representation $\varphi$.

**(a)** Let $\varphi$ be the degree 2 representation of $D_{10}$ described in Example 6 in the second set of examples in Section 1 (here $n = 5$) and show that $||\psi||^2 = 1$ (hence $\varphi$ is irreducible).

**(b)** Let $\varphi$ be the degree 2 representation of $Q_8$ described in Example 7 in the second set of examples in Section 1 and show that $||\psi||^2 = 1$ (hence $\varphi$ is irreducible).

**(c)** Let $\varphi$ be the degree 4 representation of $Q_8$ described in Example 8 in the second set of examples in Section 1 and show that $||\psi||^2 = 4$ (hence even though $\varphi$ is irreducible over $\mathbb{R}$, $\varphi$ decomposes over $\mathbb{C}$ as twice an irreducible representation of degree 2).

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**3.** If $\chi$ is an irreducible character of $G$, prove that the $\chi$-isotypic subspace of a $\mathbb{C}G$-module is unique.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**4.** Prove that if $N$ is any irreducible $\mathbb{C}G$-module and $M = N \oplus N$, then $M$ has infinitely many direct sum decompositions into two copies of $N$.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**5.** Prove that a class function is a character if and only if it is a positive integral linear combination of irreducible characters.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**6.** Let $\varphi : G \to GL(V)$ be a representation with character $\psi$. Let $W$ be the subspace $\{v \in V \mid \varphi(g)(v) = v \text{ for all } g \in G\}$ of $V$ fixed pointwise by all elements of $G$. Prove that $\dim W = (\psi, \chi_1)$, where $\chi_1$ is the principal character of $G$.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**7.** Assume $V$ is a $\mathbb{C}G$-module on which $G$ acts by permuting the basis $\mathcal{B} = \{e_1, \ldots, e_n\}$. Write $\mathcal{B}$ as a disjoint union of the orbits $\mathcal{B}_1, \ldots, \mathcal{B}_t$ of $G$ on $\mathcal{B}$.

**(a)** Prove that $V$ decomposes as a $\mathbb{C}G$-module as $V_1 \oplus \cdots \oplus V_t$, where $V_i$ is the span of $\mathcal{B}_i$.

**(b)** Prove that if $v_i$ is the sum of the vectors in $\mathcal{B}_i$ then the 1-dimensional subspace of $V_i$ spanned by $v_i$ is the unique $\mathbb{C}G$-submodule of $V_i$ affording the trivial representation (in other words, any vector in $V_i$ that is fixed under the action of $G$ is a multiple of $v_i$). [Use the fact that $G$ is transitive on $\mathcal{B}_i$. See also Exercise 8 in Section 1.]

**(c)** Let $W = \{v \in V \mid \varphi(g)(v) = v \text{ for all } g \in G\}$ be the subspace of $V$ fixed pointwise by all elements of $G$. Deduce that $\dim W = t =$ the number of orbits of $G$ on $\mathcal{B}$.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**8.** Prove the following result (sometimes called Burnside's Lemma although its origin is with Frobenius): let $G$ be a subgroup of $S_n$ and for each $\sigma \in G$ let $\mathrm{Fix}(\sigma)$ denote the number of fixed points of $\sigma$ on $\{1, \ldots, n\}$. Let $t$ be the number of orbits of $G$ on $\{1, \ldots, n\}$. Then

$$t|G| = \sum_{g \in G} \mathrm{Fix}(g).$$

[Use the preceding two exercises.]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**9.** Let $G$ be a nontrivial, transitive group of permutations on the finite set $\Omega$ and let $\psi$ be the character afforded by the linear representation over $\mathbb{C}$ obtained from $\Omega$ (cf. Example 4 in Section 1) so $\psi(\sigma)$ is the number of fixed points of $\sigma$ on $\Omega$. Now let $G$ act on the set $\Omega \times \Omega$ by $g \cdot (\omega_1, \omega_2) = (g \cdot \omega_1, g \cdot \omega_2)$ and let $\pi$ be the character afforded by the linear representation obtained from this action.

**(a)** Prove that $\pi = \psi^2$.

**(b)** Prove that the number of orbits of $G$ on $\Omega \times \Omega$ is given by the inner product $(\psi, \psi)$. [By the preceding exercises, the number of orbits on $\Omega \times \Omega$ is equal to $(\pi, \chi_1)$, where $\chi_1$ is the principal character.]

**(c)** Recall that $G$ is said to be *doubly transitive* on $\Omega$ if it has precisely 2 orbits in its action on $\Omega \times \Omega$ (it always has at least 2 orbits since the diagonal, $\{(\omega, \omega) \mid \omega \in \Omega\}$, is one orbit). Prove that if $G$ is doubly transitive on $\Omega$ then $\psi = \chi_1 + \chi_2$, where $\chi_1$ is the principal character and $\chi_2$ is a nonprincipal irreducible character of $G$.

**(d)** Let $\Omega = \{1, 2, \ldots, n\}$ and let $G = S_n$ act on $\Omega$ in the natural fashion. Show that the character of the associated linear representation decomposes as the principal character plus an irreducible character of degree $n - 1$.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**10.** Let $\psi$ be the character of any 2-dimensional representation of a group $G$ and let $x$ be an element of order 2 in $G$. Prove that $\psi(x) = 2, 0$ or $-2$. Generalize this to $n$-dimensional representations.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**11.** Let $\chi$ be an irreducible character of $G$. Prove that for every element $z$ in the center of $G$ we have $\chi(z) = \epsilon\chi(1)$, where $\epsilon$ is some root of 1 in $\mathbb{C}$. [Use Schur's Lemma.]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**12.** Let $\psi$ be the character of some representation $\varphi$ of $G$. Prove that for $g \in G$ the following hold:

**(a)** if $\psi(g) = \psi(1)$ then $g \in \ker \varphi$, and

**(b)** if $|\psi(g)| = \psi(1)$ and $\varphi$ is faithful then $g \in Z(G)$ (where $|\psi(g)|$ is the complex absolute value of $\psi(g)$). [Use the method of proof of Proposition 14.]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**13.** Let $\varphi : G \to GL(V)$ be a representation and let $\chi : G \to \mathbb{C}^\times$ be a degree 1 representation. Prove that $\chi\varphi : G \to GL(V)$ defined by $\chi\varphi(g) = \chi(g)\varphi(g)$ is a representation (note that multiplication of the linear transformation $\varphi(g)$ by the complex number $\chi(g)$ is well defined). Show that $\chi\varphi$ is irreducible if and only if $\varphi$ is irreducible. Show that if $\psi$ is the character afforded by $\varphi$ then $\chi\psi$ is the character afforded by $\chi\varphi$. Deduce that the product of any irreducible character with a character of degree 1 is also an irreducible character.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

**14.** Note that since $F \subseteq \mathbb{C}$, every representation $\varphi : G \to GL_m(F)$ may also be considered as a complex representation. Prove that if $\varphi$ is a representation over $F$ that is irreducible over $F$, then $\varphi$ is also irreducible when considered over the larger field $\mathbb{C}$ (note that this is not true if $F$ is not algebraically closed — cf. Exercise 2(c) above). Show that the set of irreducible characters of $G$ over $F$ is the same as the set of irreducible characters over $\mathbb{C}$ (i.e., these are exactly the same set of class functions on $G$). Deduce that every complex representation is equivalent to a representation over $F$. [Since $F$ is algebraically closed of characteristic 0, the irreducible characters over either $F$ or $\mathbb{C}$ are characterized by the first orthogonality relation.]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

Let $\varphi : G \to GL_m(F)$ be any representation with character $\psi$. Let $\mathbb{Q}(\varphi)$ denote the subfield of $F$ generated by all the entries of the matrices $\varphi(g)$ for all $g \in G$.

**15.** Prove that $\mathbb{Q}(\varphi)$ is a finite extension of $\mathbb{Q}$.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

Let $\varphi : G \to GL_m(F)$ be any representation with character $\psi$. Let $\mathbb{Q}(\varphi)$ denote the subfield of $F$ generated by all the entries of the matrices $\varphi(g)$ for all $g \in G$.

Now let $K$ be any Galois extension of $\mathbb{Q}$ containing $\mathbb{Q}(\varphi)$ and let $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. In fact, since every automorphism of $K$ extends to an automorphism of $F$, we may assume $\sigma$ is any automorphism of $F$. The map $\varphi^\sigma : G \to GL_n(F)$ is defined by letting $\varphi^\sigma(g)$ be the $n \times n$ matrix whose entries are obtained by applying the field automorphism $\sigma$ to the entries of the matrix $\varphi(g)$.

**16.** Prove that $\varphi^\sigma$ is a representation. Prove also that the character of $\varphi^\sigma$ is $\psi^\sigma$, where $\psi^\sigma(g) = \sigma(\psi(g))$.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

Let $\varphi : G \to GL_m(F)$ be any representation with character $\psi$. Let $\mathbb{Q}(\varphi)$ denote the subfield of $F$ generated by all the entries of the matrices $\varphi(g)$ for all $g \in G$.

Now let $K$ be any Galois extension of $\mathbb{Q}$ containing $\mathbb{Q}(\varphi)$ and let $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. In fact, since every automorphism of $K$ extends to an automorphism of $F$, we may assume $\sigma$ is any automorphism of $F$. The map $\varphi^\sigma : G \to GL_n(F)$ is defined by letting $\varphi^\sigma(g)$ be the $n \times n$ matrix whose entries are obtained by applying the field automorphism $\sigma$ to the entries of the matrix $\varphi(g)$.

**17.** Prove that $\varphi$ is irreducible if and only if $\varphi^\sigma$ is irreducible.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

Let $\varphi : G \to GL_m(F)$ be any representation with character $\psi$. Let $\mathbb{Q}(\varphi)$ denote the subfield of $F$ generated by all the entries of the matrices $\varphi(g)$ for all $g \in G$.

Now let $K$ be any Galois extension of $\mathbb{Q}$ containing $\mathbb{Q}(\varphi)$ and let $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. In fact, since every automorphism of $K$ extends to an automorphism of $F$, we may assume $\sigma$ is any automorphism of $F$. The map $\varphi^\sigma : G \to GL_n(F)$ is defined by letting $\varphi^\sigma(g)$ be the $n \times n$ matrix whose entries are obtained by applying the field automorphism $\sigma$ to the entries of the matrix $\varphi(g)$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

**18.** Prove that $\mathbb{Q}(\psi) \subseteq \mathbb{Q}(\epsilon)$. Deduce that $\mathbb{Q}(\psi)$ is a Galois extension of $\mathbb{Q}$ with abelian Galois group. [See Proposition 14.]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

Let $\varphi : G \to GL_m(F)$ be any representation with character $\psi$. Let $\mathbb{Q}(\varphi)$ denote the subfield of $F$ generated by all the entries of the matrices $\varphi(g)$ for all $g \in G$.

Now let $K$ be any Galois extension of $\mathbb{Q}$ containing $\mathbb{Q}(\varphi)$ and let $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. In fact, since every automorphism of $K$ extends to an automorphism of $F$, we may assume $\sigma$ is any automorphism of $F$. The map $\varphi^\sigma : G \to GL_n(F)$ is defined by letting $\varphi^\sigma(g)$ be the $n \times n$ matrix whose entries are obtained by applying the field automorphism $\sigma$ to the entries of the matrix $\varphi(g)$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

Recall from Section 14.5 that $\mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, where the Galois automorphisms are given on the generator $\epsilon$ by $\sigma_a : \epsilon \mapsto \epsilon^a$, where $a$ is an integer relatively prime to $n$.

**19.** Prove that if $\sigma_a \in \mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q})$ is the field automorphism defined above, then for all $g \in G$ we have $\psi^{\sigma_a}(g) = \psi(g^a)$. [Use the method of Proposition 14.]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

Let $\varphi : G \to GL_m(F)$ be any representation with character $\psi$. Let $\mathbb{Q}(\varphi)$ denote the subfield of $F$ generated by all the entries of the matrices $\varphi(g)$ for all $g \in G$.

Now let $K$ be any Galois extension of $\mathbb{Q}$ containing $\mathbb{Q}(\varphi)$ and let $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. In fact, since every automorphism of $K$ extends to an automorphism of $F$, we may assume $\sigma$ is any automorphism of $F$. The map $\varphi^\sigma : G \to GL_n(F)$ is defined by letting $\varphi^\sigma(g)$ be the $n \times n$ matrix whose entries are obtained by applying the field automorphism $\sigma$ to the entries of the matrix $\varphi(g)$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

Recall from Section 14.5 that $\mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, where the Galois automorphisms are given on the generator $\epsilon$ by $\sigma_a : \epsilon \mapsto \epsilon^a$, where $a$ is an integer relatively prime to $n$.

**20.** Prove that if $g$ is an element of $G$ which is conjugate to $g^a$ for all integers $a$ relatively prime to $n$, then $\psi(g) \in \mathbb{Q}$, for every character $\psi$ of $G$. [Use the preceding exercise and the fact that $\mathbb{Q}$ is the field fixed by all $\sigma_a$'s.]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

Let $\varphi : G \to GL_m(F)$ be any representation with character $\psi$. Let $\mathbb{Q}(\varphi)$ denote the subfield of $F$ generated by all the entries of the matrices $\varphi(g)$ for all $g \in G$.

Now let $K$ be any Galois extension of $\mathbb{Q}$ containing $\mathbb{Q}(\varphi)$ and let $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. In fact, since every automorphism of $K$ extends to an automorphism of $F$, we may assume $\sigma$ is any automorphism of $F$. The map $\varphi^\sigma : G \to GL_n(F)$ is defined by letting $\varphi^\sigma(g)$ be the $n \times n$ matrix whose entries are obtained by applying the field automorphism $\sigma$ to the entries of the matrix $\varphi(g)$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

Recall from Section 14.5 that $\mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, where the Galois automorphisms are given on the generator $\epsilon$ by $\sigma_a : \epsilon \mapsto \epsilon^a$, where $a$ is an integer relatively prime to $n$.

**21.** Prove that an element $g \in G$ is conjugate to $g^a$ for all integers $a$ relatively prime to $|G|$ if and only if $g$ is conjugate to $g^{a'}$ for all integers $a'$ relatively prime to $|g|$.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

Let $\varphi : G \to GL_m(F)$ be any representation with character $\psi$. Let $\mathbb{Q}(\varphi)$ denote the subfield of $F$ generated by all the entries of the matrices $\varphi(g)$ for all $g \in G$.

Now let $K$ be any Galois extension of $\mathbb{Q}$ containing $\mathbb{Q}(\varphi)$ and let $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. In fact, since every automorphism of $K$ extends to an automorphism of $F$, we may assume $\sigma$ is any automorphism of $F$. The map $\varphi^\sigma : G \to GL_n(F)$ is defined by letting $\varphi^\sigma(g)$ be the $n \times n$ matrix whose entries are obtained by applying the field automorphism $\sigma$ to the entries of the matrix $\varphi(g)$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

Recall from Section 14.5 that $\mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, where the Galois automorphisms are given on the generator $\epsilon$ by $\sigma_a : \epsilon \mapsto \epsilon^a$, where $a$ is an integer relatively prime to $n$.

**22.** Show for any positive integer $n$ that every character of the symmetric group $S_n$ is rational valued (i.e., $\psi(g) \in \mathbb{Q}$ for all $g \in S_n$ and all characters $\psi$ of $S_n$).

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

The next two exercises establish the converse to Exercise 20.

**23.** Prove that elements $x$ and $y$ are conjugate in a group $G$ if and only if $\chi(x) = \chi(y)$ for all irreducible characters $\chi$ of $G$.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

The next two exercises establish the converse to Exercise 20.

**24.** Let $g \in G$ and assume that every irreducible character of $G$ is rational valued on $g$. Prove that $g$ is conjugate to $g^a$ for every integer $a$ relatively prime to $|G|$. [If $g$ is not conjugate to $g^a$ for some $a$ relatively prime to $|G|$ then by the preceding exercise there is an irreducible character $\chi$ such that $\chi(g) \neq \chi(g^a)$. Derive a contradiction from the hypothesis that $\chi(g) \in \mathbb{Q}$.]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

**25.** Describe which irreducible characters of the cyclic group of order $n$ are algebraically conjugate.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

The next few exercises study the notion of *algebraically conjugate* characters. These exercises may be considered as extensions of Proposition 14 and some consequences of these extensions. In particular we obtain a group-theoretic characterization of the conditions under which all irreducible characters of a group take values in $\mathbb{Q}$.

Let $F$ be the subfield of $\mathbb{C}$ of all elements that are algebraic over $\mathbb{Q}$ (the field of algebraic numbers). Thus $F$ is the algebraic closure of $\mathbb{Q}$ contained in $\mathbb{C}$ and all the results established over $\mathbb{C}$ hold without change over $F$.

The representation $\varphi^\sigma$ (or character $\psi^\sigma$) is called the *algebraic conjugate* of $\varphi$ by $\sigma$ (or of $\psi$, respectively); two representations $\varphi_1$ and $\varphi_2$ (or characters $\psi_1$ and $\psi_2$) are said to be *algebraically conjugate* if there is some automorphism $\sigma$ of $F$ such that $\varphi_1^\sigma = \varphi_2$ (or $\psi_1^\sigma = \psi_2$, respectively). Some care needs to be taken with this (standard) notation since the exponential notation usually denotes a right action whereas automorphisms of $F$ act on the left on representations: $\varphi^{(\sigma\tau)} = (\varphi^\tau)^\sigma$.

Let $\mathbb{Q}(\psi)$ be the subfield of $F$ generated by the numbers $\psi(g)$ for all $g \in G$. Let $|G| = n$ and let $\epsilon$ be a primitive $n^{\text{th}}$ root of 1 in $F$.

**26.** Prove that every irreducible character of both $Q_8$ and $D_8$ is rational valued. Prove that $D_{10}$ has an irreducible character that is not rational valued.

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**27.** Let $G = H \times K$ and let $\varphi : H \to GL(V)$ be an irreducible representation of $H$ with character $\chi$. Then $G \xrightarrow{\pi_H} H \xrightarrow{\varphi} GL(V)$ gives an irreducible representation of $G$, where $\pi_H$ is the natural projection; the character, $\tilde{\chi}$, of this representation is $\tilde{\chi}((h, k)) = \chi(h)$. Likewise any irreducible character $\psi$ of $K$ gives an irreducible character $\tilde{\psi}$ of $G$ with $\tilde{\psi}((h, k)) = \psi(k)$.

**(a)** Prove that the product $\tilde{\chi}\tilde{\psi}$ is an irreducible character of $G$. [Show it has norm 1.]

**(b)** Prove that every irreducible character of $G$ is obtained from such products of irreducible characters of the direct factors. [Use Theorem 10, either (3) or (4).]

Let $G$ be a finite group. Unless stated otherwise all representations and characters are over $\mathbb{C}$.

**28.** *(Finite subgroups of $GL_2(\mathbb{Q})$)* Let $G$ be a finite subgroup of $GL_2(\mathbb{Q})$.

**(a)** Show that $GL_2(\mathbb{Q})$ does not contain an element of order $n$ for $n = 5$, 7, or $n \geq 9$. Deduce that $|G| = 2^a 3^b$. [Use rational canonical forms.]

**(b)** Show that the Klein 4-group is the only noncyclic abelian subgroup of $GL_2(\mathbb{Q})$. Deduce from this and (a) that $|G| \mid 24$.

**(c)** Show that the only finite subgroups of $GL_2(\mathbb{Q})$ are the cyclic groups of order 1, 2, 3, 4, and 6, the Klein 4-group, and the dihedral groups of order 6, 8, and 12. [Use the classifications of groups of small order in Section 4.5 and Exercise 10 of Section 1 to restrict $G$ to this list. Show conversely that each group listed has a 2-dimensional faithful rational representation.]

# 19. Examples and Applications of Character Theory

## 19.1. Characters of Groups of Small Order

**1.** Calculate the character tables of $Z_2 \times Z_2$, $Z_2 \times Z_3$ and $Z_2 \times Z_2 \times Z_2$. Explain why the table of $Z_2 \times Z_3$ contains primitive $6^{\text{th}}$ roots of 1.

**2.** Compute the degrees of the irreducible characters of $D_{16}$.

**3.** Compute the degrees of the irreducible characters of $A_5$. Deduce that the degree 6 irreducible character of $S_5$ is not irreducible when restricted to $A_5$. [The conjugacy classes of $A_5$ are worked out in Section 4.3.]

**4.** Using the character tables in this section, for each of parts (a) to (d) use the first orthogonality relation to write the specified permutation character (cf. Example 3, Section 18.3) as a sum of irreducible characters:

**(a)** the permutation character of the subgroup $A_3$ of $S_3$

**(b)** the permutation character of the subgroup $\langle (1\ 2\ 3\ 4) \rangle$ of $S_4$

**(c)** the permutation character of the subgroup $V_4$ of $S_4$

**(d)** the permutation character of the subgroup $\langle (1\ 2\ 3),\ (1\ 2),\ (4\ 5) \rangle$ of $S_5$ (this subgroup is the normalizer of a Sylow 3-subgroup of $S_5$).

**5.** Assume that for any character $\psi$ of a group, $\psi^2$ is also a character (where $\psi^2(g) = (\psi(g))^2$) — this is a special case of Proposition 17 in Section 18.3. Using the character tables in this section, for each of parts (a) to (e) write out the values of the square, $\chi^2$, of the specified character $\chi$ and use the first orthogonality relation to write $\chi^2$ as a sum of irreducible characters:

**(a)** $\chi = \chi_3$, the degree 2 character in the table of $S_3$

**(b)** $\chi = \chi_5$, the degree 2 character in the table of $Q_8$

**(c)** $\chi = \chi_5$, the last character in the table of $S_4$

**(d)** $\chi = \chi_4$, the second degree 4 character in the table of $S_5$

**(e)** $\chi = \chi_7$, the last character in the table of $S_5$.

**6.** Calculate the character table of $A_5$.

**7.** Show that $S_6$ has an irreducible character of degree 5.

**8.** Calculate the character table of $D_{10}$. (This table contains nonreal entries.)

**9.** Calculate the character table of $D_{12}$.

**10.** Calculate the character table of $S_3 \times S_3$.

**11.** Calculate the character table of $Z_3 \times S_3$.

**12.** Calculate the character table of $Z_2 \times S_4$.

**13.** Calculate the character table of $S_3 \times S_4$.

**14.** Let $n$ be an integer with $n \geq 3$. Show that every irreducible character of $D_{2n}$ has degree 1 or 2 and find the number of irreducible characters of each degree. [The conjugacy classes of $D_{2n}$ were found in Exercises 31 and 32 of Section 4.3 and its commutator subgroup was computed in Section 5.4.]

**15.** Prove that the character table is an invertible matrix. [Use the orthogonality relations.]

**16.** For each of $A_5$ and $D_{10}$ describe which irreducible characters are algebraically conjugate (cf. the exercises in Section 18.3).

**17.** Let $p$ be any prime and let $P$ be a non-abelian group of order $p^3$ (up to isomorphism there are two choices for $P$; for odd $p$ these were constructed when the groups of order $p^3$ were classified in Section 5.5). This exercise determines the character table of $P$ and shows that both isomorphism types have the same character table (the argument includes the $p = 2$ case worked out in this section).

**(a)** Prove that $P$ has $p^2$ characters of degree 1.

**(b)** Prove that $P$ has $p - 1$ irreducible characters of degree $p$ and that these together with the $p^2$ degree 1 characters are all the irreducible characters of $P$. [Use Theorem 10(3) and Theorem 12 in Section 18.2.]

**(c)** Deduce that (regardless of the isomorphism type) the group $P$ has $p^2+p-1$ conjugacy classes, $p$ of which are of size 1 (i.e., are central classes) and $p^2 - 1$ of which each have size $p$. Deduce also that the classes of size $p$ are precisely the nonidentity cosets of the center of $P$ (i.e., if $x \in P - Z(P)$ then the conjugacy class of $x$ is the set of $p$ elements in the coset $xZ(P)$).

**(d)** Prove that if $\chi$ is an irreducible character of degree $p$ then the representation affording $\chi$ is faithful.

**(e)** Fix a generator, $z$, of the center of $P$ and let $\epsilon$ be a fixed primitive $p^{\text{th}}$ root of 1 in $\mathbb{C}$. Prove that if $\chi$ is an irreducible character of degree $p$ then $\chi(z) = p\epsilon^i$ for some $i \in \{1, 2, \ldots, p - 1\}$. Prove further that $\chi(x) = 0$ for all $x \in P - Z(P)$. (Note then that the degree $p$ characters are all algebraically conjugate.) [Use the same reasoning as in the construction of the character table of $Q_8$.]

**(f)** Prove that for each $i \in \{1, 2, \ldots, p - 1\}$ there is a unique irreducible character $\chi_i$ of degree $p$ such that $\chi_i(z) = p\epsilon^i$. Deduce that the character table of $P$ is uniquely determined, and describe it. [Recall from Section 6.1 that regardless of the isomorphism type, $P' = Z(P)$ and $P/P' \cong Z_p \times Z_p$. From this one can write out the degree 1 characters. Part (e) describes the degree $p$ characters.]

## 19.2. Theorems of Burnside and Hall

**1.** Show that every character of the symmetric group $S_n$ is integer valued, for all $n$ (i.e., $\psi(g) \in \mathbb{Z}$ for all $g \in S_n$ and all characters $\psi$ of $S_n$). [See Exercise 22 in Section 18.3.]

**2.** Let $G$ be a finite group with the property that every maximal subgroup has either prime or prime squared index. Prove that $G$ is solvable. (The simple group $GL_3(\mathbb{F}_2)$ has the property that every maximal subgroup has index either 7 or 8, i.e., either prime or prime cubed index — cf. Section 6.2.). [Let $p$ be the largest prime dividing $|G|$ and let $P$ be a Sylow $p$-subgroup of $G$. If $P \trianglelefteq G$, apply induction to $G/P$. Otherwise let $M$ be a maximal subgroup containing $N_G(P)$. Use Exercise 51 in Section 4.5 to show that $p = 3$ and deduce that $|G| = 2^a 3^b$.]

**3.** Assume $G$ is a finite group that possesses an abelian subgroup $H$ whose index is a power of a prime. Prove that $G$ is solvable.

**4.** Repeat the preceding exercise with the word "abelian" replaced by "nilpotent."

**5.** Use the ideas in the proof of Philip Hall's Theorem to prove Burnside's $p^a q^b$ Theorem in the special case when all Sylow subgroups are abelian (without use of character theory.)

## 19.3. Introduction to the Theory of Induced Characters

Throughout the exercises all representations are over the complex numbers.

**1.** Let $G = S_3$, let $H = A_3$ and let $V$ be the 3-dimensional $\mathbb{C}H$-module which affords the natural permutation representation of $A_3$. More explicitly, let $V$ have basis $e_1, e_2, e_3$ and let $\sigma \in A_3$ act on $V$ by $\sigma e_i = e_{\sigma(i)}$. Let 1 and (1 2) be coset representatives for the left cosets of $A_3$ in $S_3$ and write out the explicit matrices described in Theorem 11 for the action of $S_3$ on the induced module $W$, for each of the elements of $S_3$.

Throughout the exercises all representations are over the complex numbers.

**2.** In each of parts (a) to (f) a character $\psi$ of a subgroup $H$ of a particular group $G$ is specified. Compute the values of the induced character $\mathrm{Ind}_H^G(\psi)$ on all the conjugacy classes of $G$ and use the character tables in Section 1 to write $\mathrm{Ind}_H^G(\psi)$ as a sum of irreducible characters:

**(a)** $\psi$ is the unique nonprincipal degree 1 character of the subgroup $\langle (1\ 2) \rangle$ of $S_3$

**(b)** $\psi$ is the degree 1 character of the subgroup $\langle r \rangle$ of $D_8$ defined by $\psi(r) = i$, where $i \in \mathbb{C}$ is a square root of $-1$

**(c)** $\psi$ is the degree 1 character of the subgroup $\langle r \rangle$ of $D_8$ defined by $\psi(r) = -1$

**(d)** $\psi$ is any of the nonprincipal degree 1 characters of the subgroup $V_4 = \langle (1\ 2),\ (3\ 4) \rangle$ of $S_4$

**(e)** $\psi = \chi_4$ is the first of the two characters of degree 3 in the character table of $H = S_4$ in Section 1 and $H$ is a subgroup of $G = S_5$

**(f)** $\psi$ is any of the nonprincipal degree 1 characters of the subgroup $V_4 = \langle (1\ 2),\ (3\ 4) \rangle$ of $S_5$.

Throughout the exercises all representations are over the complex numbers.

**3.** Use Proposition 13 to explicitly write out the character table of each of the following groups:

**(a)** the dihedral group of order 10

**(b)** the non-abelian group of order 57

**(c)** the non-abelian group of order 56 which has a normal, elementary abelian Sylow 2-subgroup.

Throughout the exercises all representations are over the complex numbers.

**4.** Let $H$ be a subgroup of $G$, let $\varphi$ be a representation of $H$ and suppose that $N$ is a normal subgroup of $G$ with $N \leq H$ and $N$ contained in the kernel of $\varphi$. Prove that $N$ is also contained in the kernel of the induced representation of $\varphi$.

Throughout the exercises all representations are over the complex numbers.

**5.** Let $N$ be a normal subgroup of $G$ and let $\psi_1$ be the principal character of $N$. Let $\Psi$ be the induced character $\mathrm{Ind}_N^G(\psi_1)$ so that by the preceding exercise we may consider $\Psi$ as the character of a representation of $G/N$. Prove that $\Psi$ is the character of the regular representation of $G/N$.

Throughout the exercises all representations are over the complex numbers.

**6.** Let $Z$ be any subgroup of the center of $G$, let $|G : Z| = m$ and let $\psi$ be a character of $Z$. Prove that

$$\mathrm{Ind}_Z^G(\psi)(g) = \begin{cases} m\psi(g) & \text{if } g \in Z \\ 0 & \text{if } g \notin Z. \end{cases}$$

Throughout the exercises all representations are over the complex numbers.

**7.** Let $\varphi$ be a matrix representation of the subgroup $H$ of $G$ and define matrices $\Phi(g)$ for every $g \in G$ by the displayed formula in the statement of Theorem 11. Prove directly that $\Phi$ is a representation by showing that $\Phi(xy) = \Phi(x)\Phi(y)$ for all $x, y \in G$.

Throughout the exercises all representations are over the complex numbers.

**8.** Let $G$ be a Frobenius group with Frobenius kernel $Q$. Assume that both $Q$ and $G/Q$ are abelian but $G$ is not abelian (i.e., $G \neq Q$). Let $|Q| = n$ and $|G : Q| = m$.

**(a)** Prove that $G/Q$ is cyclic and show that $G = QC$ for some cyclic subgroup $C$ of $G$ with $C \cap Q = 1$ (i.e., $G$ is a semidirect product of $Q$ and $C$ and $|C| = m$). [Let $q$ be a prime divisor of $n$ and let $G/Q$ act by conjugation on the elementary abelian $q$-group $\{h \in Q \mid h^q = 1\}$. Apply Exercise 14(e) of Section 18.1 and the definition of a Frobenius group to an irreducible constituent of this $\mathbb{F}_q G/Q$-module.]

**(b)** Prove that $n$ and $m$ are relatively prime. [If a prime $p$ divides both the order and index of $Q$, let $P$ be a Sylow $p$-subgroup of $G$. Then $P \cap Q \trianglelefteq P$ and $P \cap Q$ is a Sylow $p$-subgroup of $Q$. Consider the centralizer in $G$ of the subgroup $Z(P) \cap Q$ (this intersection is nontrivial by Theorem 1 of Section 6.1).]

**(c)** Show that $G$ has no elements of order $qp$, where $q$ is any nontrivial divisor of $n$ and $p$ is any nontrivial divisor of $m$. [Argue as in Proposition 13.]

**(d)** Prove that the number of nonidentity conjugacy classes of $G$ contained in $Q$ is $(n-1)/m$ and that each of these classes has size $m$. [Argue as in Proposition 13.]

**(e)** Prove that no two distinct elements of $C$ are conjugate in $G$. Deduce that the nonidentity elements of $C$ are representatives for $m-1$ distinct conjugacy classes of $G$ and that each of these classes has size $n$. Deduce then that every element of $G - Q$ is conjugate to some element of $C$ and that $G$ has $m + (n-1)/m$ conjugacy classes.

**(f)** Prove that $G' = Q$ and deduce that $G$ has $m$ distinct characters of degree 1. [To show $Q \leq G'$ let $C = \langle x \rangle$ and argue that the map $h \mapsto [h, x] = x^{-1}h^{-1}xh$ is a homomorphism from $Q$ to $Q$ whose kernel is trivial, hence this map is surjective.]

**(g)** Show that if $\psi$ is any nonprincipal irreducible character of $Q$, then $\mathrm{Ind}_Q^G(\psi)$ is an **irreducible** character of $G$. Show that every irreducible character of $G$ of degree $> 1$ is equal to $\mathrm{Ind}_Q^G(\psi)$ for some nonprincipal irreducible character $\psi$ of $Q$. Deduce that every irreducible character of $G$ has degree either 1 or $m$ and the number of **irreducible** characters of degree $m$ is $(n-1)/m$. [Check that the proof of Proposition 13(3) establishes this more general result with the appropriate changes to the numbers involved.]

Throughout the exercises all representations are over the complex numbers.

**9.** Use the preceding exercise to explicitly write out the character table of $\langle (1\,2\,3\,4\,5),\ (2\,3\,5\,4) \rangle$, which is the normalizer in $S_5$ of a Sylow 5-subgroup (this group is a Frobenius group of order 20).

Throughout the exercises all representations are over the complex numbers.

**10.** Let $N$ be a normal subgroup of $G$, let $\psi$ be a character of $N$ and let $g \in G$. Define $\psi^g$ by $\psi^g(h) = \psi(ghg^{-1})$ for all $h \in N$.

**(a)** Prove that $\psi^g$ is a character of $N$ ($\psi$ and $\psi^g$ are called *G-conjugate* characters of $N$). Prove that $\psi^g$ is irreducible if and only if $\psi$ is irreducible.

**(b)** Prove that the map $\psi \mapsto \psi^g$ is a right group action of $G$ on the set of characters of $N$ and $N$ is in the kernel of this action.

**(c)** Prove that if $\psi_1$ and $\psi_2$ are $G$-conjugate characters of $N$, then $\text{Ind}_N^G(\psi_1) = \text{Ind}_N^G(\psi_2)$. Prove also that if $\psi_1$ and $\psi_2$ are characters of $N$ that are not $G$-conjugate then $\text{Ind}_N^G(\psi_1) \neq \text{Ind}_N^G(\psi_2)$. [Use the argument in the proof of Proposition 13(3).]

Throughout the exercises all representations are over the complex numbers.

**11.** Show that if $G = A_4$ and $N = V_4$ is its Sylow 2-subgroup then any two nonprincipal irreducible characters of $N$ are $G$-conjugate (cf. the preceding exercise).

Throughout the exercises all representations are over the complex numbers.

**12.** Let $G = D_{2n}$ be presented by its usual generators and relations. Prove that if $\psi$ is any degree 1 character of $H = \langle r \rangle$ such that $\psi \neq \psi^s$, then $\mathrm{Ind}_H^G(\psi)$ is an irreducible character of $D_{2n}$. Show that every irreducible character of $D_{2n}$ is the induced character of some degree 1 character of $\langle r \rangle$.

**Throughout the exercises all representations are over the complex numbers.**

**13.** Prove both parts of Proposition 14.

Throughout the exercises all representations are over the complex numbers.

**14.** Prove the following result known as *Frobenius Reciprocity*: let $H \leq G$, let $\psi$ be any character of $H$ and let $\chi$ be any character of $G$. Then

$$(\psi, \chi|_H)_H = (\mathrm{Ind}_H^G(\psi), \chi)_G.$$

[Expand the right hand side using the formula for the induced character $\mathrm{Ind}_H^G(\psi)$ or follow the proof of Shapiro's Lemma in Section 17.2.]

Throughout the exercises all representations are over the complex numbers.

**15.** Assume $G$ were a simple group of order $3^3 \cdot 7 \cdot 13 \cdot 409$ whose Sylow subgroups and their normalizers are described by properties (1) to (5) in this section. Prove that the permutation character of degree 819 obtained from the action of $G$ on the left cosets of the subgroup $N_4$ decomposes as $\chi_0 + \gamma + \gamma'$, where $\chi_0$ is the principal character of $G$ and $\gamma$ and $\gamma'$ are distinct irreducible characters of $G$ of degree 409. [Use Exercise 9 in Section 18.3 to show that this permutation character $\pi$ has $||\pi||^2 = 3$.]